

**US Army Corps  
of Engineers**  
Louisville District

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# **Solicitation For**

**Base Fire Station  
Youngstown ARS, OH**

**P2: 513392**

**Design-Bid-Build**

**Specifications – Volume 2 of 3  
(Div 07 – 23)  
Certified Final Design**

**13 March 2025  
W912QR25BA009**

**ARIMS: 200A  
Disposition: Maintain for 15yrs after construction**

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P2#: 513392, YARS Fire Station  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## PROFESSIONAL SEALS PAGE

**BASE FIRE STATION**  
**Youngstown Air Reserve Station, Ohio**  
P2#: 513392  
Project No: ZDEL239001

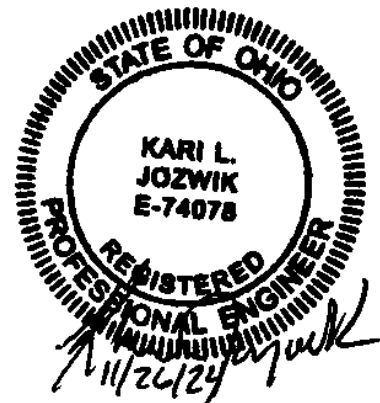
**Design Consultant**     **Tetra Tech, Inc.**  
2000 Warrington Way, Suite 245  
Louisville, Kentucky 40222  
Phone (502) 584-5555 / Fax (502) 584-5696  
Tt Project No.: 213-11568-23001



11/26/2024  
STRUCTURAL



11/26/2024  
ARCHITECTURAL



CIVIL



26 November 2024

FIRE PROTECTION



11/26/2024

MECHANICAL / PLUMBING



ELECTRICAL / TELECOMMUNICATIONS

PROFESSIONAL SEALS PAGE

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P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## PROJECT TABLE OF CONTENTS

### DIVISION 00 - PROCUREMENT AND CONTRACTING REQUIREMENTS

00 80 00.00 06 04/21, CHG 1: 06/23 SPECIAL PROVISIONS

### DIVISION 01 - GENERAL REQUIREMENTS

01 11 00 08/15, CHG 2: 08/21 SUMMARY OF WORK  
01 32 01.00 06 10/23 PROJECT SCHEDULE  
01 33 00 08/18, CHG 4: 02/21 SUBMITTAL PROCEDURES  
01 33 29.00 06 11/17 SUSTAINABILITY REPORTING  
01 35 26.00 06 GOVERNMENT SAFETY REQUIREMENTS  
01 42 00 02/19, CHG 1: 08/23 SOURCES FOR REFERENCE PUBLICATIONS  
01 45 00.15 10 11/23 RESIDENT MANAGEMENT SYSTEM CONTRACTOR  
MODE (RMS CM)  
01 45 04.10 06 CONTRACTOR QUALITY CONTROL  
01 45 35 11/20 SPECIAL INSPECTIONS  
01 46 00.00 06 04/20 TOTAL BUILDING COMMISSIONING  
01 50 00 11/20, CHG 2: 08/22 TEMPORARY CONSTRUCTION FACILITIES AND  
CONTROLS  
01 57 19.00 06 04/20 TEMPORARY ENVIRONMENTAL CONTROLS AND  
PERMITS  
01 74 19 02/19, CHG 3: 11/21 CONSTRUCTION WASTE MANAGEMENT AND  
DISPOSAL  
01 78 23 05/23 OPERATION AND MAINTENANCE DATA

### DIVISION 02 - EXISTING CONDITIONS

02 41 00 08/22 DEMOLITION  
02 61 13 02/10, CHG 1: 02/21 EXCAVATION AND HANDLING OF  
CONTAMINATED MATERIAL  
02 81 00 11/18 TRANSPORTATION AND DISPOSAL OF  
HAZARDOUS MATERIALS  
02 82 00 11/18, CHG 1: 11/19 ASBESTOS REMEDIATION  
02 83 00 11/18 LEAD REMEDIATION  
02 84 16 11/22 HANDLING OF LIGHTING BALLASTS AND  
LAMPS CONTAINING PCBs AND MERCURY

### DIVISION 03 - CONCRETE

03 30 00 02/19, CHG 6: 11/23 CAST-IN-PLACE CONCRETE

### DIVISION 04 - MASONRY

04 20 00 11/15, CHG 2: 05/19 UNIT MASONRY

### DIVISION 05 - METALS

05 05 20 05/22 POST-INSTALLED CONCRETE AND MASONRY  
ANCHORS  
05 12 00 08/18, CHG 2: 05/21 STRUCTURAL STEEL  
05 21 00 05/15, CHG 1: 08/18 STEEL JOIST FRAMING  
05 30 00 05/15, CHG 2: 08/18 STEEL DECKS  
05 40 00 05/15, CHG 1: 08/18 COLD-FORMED METAL FRAMING  
05 50 13 05/17, CHG 1: 08/18 MISCELLANEOUS METAL FABRICATIONS  
05 51 33 02/16, CHG 2: 02/18 METAL LADDERS  
05 52 00 08/23 METAL RAILINGS

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### **DIVISION 06 - WOOD, PLASTICS, AND COMPOSITES**

|                |                     |                                              |
|----------------|---------------------|----------------------------------------------|
| 06 10 00       | 08/16, CHG 2: 11/18 | ROUGH CARPENTRY                              |
| 06 20 00       | 08/16, CHG 2: 11/18 | FINISH CARPENTRY                             |
| 06 41 16.00 10 | 08/10, CHG 1: 11/18 | PLASTIC-LAMINATE-CLAD ARCHITECTURAL CABINETS |
| 06 61 16       | 08/20               | SOLID SURFACING FABRICATIONS                 |

#### **DIVISION 07 - THERMAL AND MOISTURE PROTECTION**

|             |                     |                                     |
|-------------|---------------------|-------------------------------------|
| 07 11 13    | 08/23               | BITUMINOUS DAMPPROOFING             |
| 07 19 00    | 08/23               | WATER REPELLENTS                    |
| 07 21 13    | 08/23               | BOARD AND BLOCK INSULATION          |
| 07 21 16    | 08/23               | BUILDING INSULATION                 |
| 07 22 00    | 08/23               | ROOF AND DECK INSULATION            |
| 07 27 10    | 08/23               | BUILDING AIR BARRIER SYSTEM         |
| 07 27 19.01 | 08/23               | SELF-ADHERING AIR BARRIERS          |
| 07 27 26    | 08/23               | FLUID-APPLIED MEMBRANE AIR BARRIERS |
| 07 41 63    | 11/16               | FABRICATED ROOF PANEL ASSEMBLIES    |
| 07 42 63    | 05/11, CHG 4: 08/20 | FABRICATED WALL PANEL ASSEMBLIES    |
| 07 60 00    | 08/23               | FLASHING AND SHEET METAL            |
| 07 84 00    | 05/10, CHG 1: 08/13 | FIRESTOPPING                        |
| 07 92 00    | 08/23               | JOINT SEALANTS                      |

#### **DIVISION 08 - OPENINGS**

|          |                     |                                           |
|----------|---------------------|-------------------------------------------|
| 08 11 13 | 08/20               | STEEL DOORS AND FRAMES                    |
| 08 14 00 | 08/16, CHG 1: 08/18 | WOOD DOORS                                |
| 08 31 00 | 05/17, CHG 1: 08/18 | ACCESS DOORS AND PANELS                   |
| 08 36 13 | 08/20               | SECTIONAL OVERHEAD DOORS                  |
| 08 41 13 | 08/18, CHG 1: 08/18 | ALUMINUM-FRAMED ENTRANCES AND STOREFRONTS |
| 08 51 13 | 05/19               | ALUMINUM WINDOWS                          |
| 08 60 45 | 08/20               | TRANSLUCENT PANELS                        |
| 08 71 00 | 02/16, CHG 4: 02/22 | DOOR HARDWARE                             |
| 08 81 00 | 05/19               | GLAZING                                   |
| 08 91 00 | 08/20               | METAL WALL LOUVERS                        |

#### **DIVISION 09 - FINISHES**

|          |                     |                                       |
|----------|---------------------|---------------------------------------|
| 09 22 00 | 02/10, CHG 2: 08/18 | SUPPORTS FOR PLASTER AND GYPSUM BOARD |
| 09 29 00 | 08/16, CHG 4: 02/20 | GYPSUM BOARD                          |
| 09 30 10 | 08/20               | CERAMIC TILING                        |
| 09 51 00 | 08/20               | ACOUSTICAL CEILINGS                   |
| 09 65 00 | 08/10, CHG 3: 08/18 | RESILIENT FLOORING                    |
| 09 68 00 | 11/17, CHG 2: 08/20 | CARPETING                             |
| 09 90 00 | 02/21               | PAINTS AND COATINGS                   |
| 09 96 00 | 11/14               | HIGH-PERFORMANCE COATINGS             |

#### **DIVISION 10 - SPECIALTIES**

|             |                     |                          |
|-------------|---------------------|--------------------------|
| 10 11 00    | 08/20               | VISUAL DISPLAY UNITS     |
| 10 14 00.10 | 08/17, CHG 1: 11/18 | EXTERIOR SIGNAGE         |
| 10 14 00.20 | 08/20               | INTERIOR SIGNAGE         |
| 10 21 13    | 08/20               | TOILET COMPARTMENTS      |
| 10 22 13    | 08/16, CHG 1: 08/18 | WIRE MESH PARTITIONS     |
| 10 26 00    | 08/20               | WALL AND DOOR PROTECTION |
| 10 28 13    | 08/20               | TOILET ACCESSORIES       |

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

|          |       |                    |
|----------|-------|--------------------|
| 10 44 16 | 11/19 | FIRE EXTINGUISHERS |
| 10 51 13 | 05/11 | PPE LOCKERS        |
| 10 56 13 | 04/06 | STEEL SHELVING     |

#### **DIVISION 11 - EQUIPMENT**

|          |       |                                   |
|----------|-------|-----------------------------------|
| 11 24 00 | 01/23 | ROOFTOP FALL PROTECTION EQUIPMENT |
|----------|-------|-----------------------------------|

#### **DIVISION 12 - FURNISHINGS**

|          |                     |                                                        |
|----------|---------------------|--------------------------------------------------------|
| 12 24 13 | 08/20               | ROLLER WINDOW SHADES                                   |
| 12 48 13 | 08/23               | ENTRANCE FLOOR MATS AND FRAMES<br>(RECESSED SS GRILLS) |
| 12 93 00 | 08/17, CHG 1: 08/18 | SITE FURNISHINGS                                       |

#### **DIVISION 13 - SPECIAL CONSTRUCTION**

|          |       |                                                |
|----------|-------|------------------------------------------------|
| 13 48 73 | 05/22 | SEISMIC CONTROL FOR MISCELLANEOUS<br>EQUIPMENT |
|----------|-------|------------------------------------------------|

#### **DIVISION 21 - FIRE SUPPRESSION**

|          |       |                                                |
|----------|-------|------------------------------------------------|
| 21 13 13 | 08/20 | WET PIPE SPRINKLER SYSTEMS, FIRE<br>PROTECTION |
|----------|-------|------------------------------------------------|

#### **DIVISION 22 - PLUMBING**

|                |                     |                                                     |
|----------------|---------------------|-----------------------------------------------------|
| 22 00 00       | 11/15, CHG 4: 05/21 | PLUMBING, GENERAL PURPOSE                           |
| 22 05 48.00 20 | 04/06, CHG 1: 05/15 | MECHANICAL SOUND, VIBRATION, AND<br>SEISMIC CONTROL |

#### **DIVISION 23 - HEATING, VENTILATING, AND AIR CONDITIONING (HVAC)**

|                |                     |                                                                              |
|----------------|---------------------|------------------------------------------------------------------------------|
| 23 03 00.00 20 | 08/10, CHG 3: 08/18 | BASIC MECHANICAL MATERIALS AND METHODS                                       |
| 23 05 15       | 05/22               | COMMON PIPING FOR HVAC                                                       |
| 23 05 48.19    | 05/18, CHG 2: 08/20 | BRACING FOR HVAC                                                             |
| 23 05 93       | 11/15               | TESTING, ADJUSTING, AND BALANCING FOR<br>HVAC                                |
| 23 07 00       | 02/13, CHG 7: 05/20 | THERMAL INSULATION FOR MECHANICAL<br>SYSTEMS                                 |
| 23 08 00       | 05/23               | COMMISSIONING OF MECHANICAL AND<br>PLUMBING SYSTEMS                          |
| 23 09 00       | 02/19, CHG 3: 05/21 | INSTRUMENTATION AND CONTROL FOR HVAC                                         |
| 23 09 13       | 11/15, CHG 2: 05/21 | INSTRUMENTATION AND CONTROL DEVICES<br>FOR HVAC                              |
| 23 09 23.02    | 02/19, CHG 1: 02/20 | BACNET DIRECT DIGITAL CONTROL FOR HVAC<br>AND OTHER BUILDING CONTROL SYSTEMS |
| 23 11 20       | 05/20               | FACILITY GAS PIPING                                                          |
| 23 21 13.00 20 | 04/06, CHG 2: 11/19 | LOW TEMPERATURE WATER (LTW) HEATING<br>SYSTEM                                |
| 23 23 00       | 08/21               | REFRIGERANT PIPING                                                           |
| 23 30 00       | 05/20, CHG 1: 02/22 | HVAC AIR DISTRIBUTION                                                        |
| 23 35 16.17 10 | 05/20               | MECHANICAL ENGINE EXHAUST SYSTEMS                                            |
| 23 52 43.00 20 | 05/15, CHG 2: 08/18 | LOW PRESSURE WATER HEATING BOILERS<br>(UNDER 800,000 BTU/HR OUTPUT)          |
| 23 80 20.00 10 | 05/20               | GAS-FIRED HEATING EQUIPMENT                                                  |
| 23 81 00       | 05/18, CHG 1: 02/21 | DECENTRALIZED UNITARY HVAC EQUIPMENT                                         |
| 23 81 23       | 11/20               | COMPUTER ROOM AIR CONDITIONING UNITS                                         |

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### **DIVISION 25 - INTEGRATED AUTOMATION**

|             |                     |                                                                        |
|-------------|---------------------|------------------------------------------------------------------------|
| 25 05 11    | 05/21, CHG 1: 08/23 | CYBERSECURITY FOR FACILITY-RELATED CONTROL SYSTEMS                     |
| 25 05 11.01 | 05/21, CHG 1: 08/23 | CYBERSECURITY FOR FACILITY-RELATED CONTROL SYSTEMS (MECHANICAL BAS)    |
| 25 05 11.02 | 05/21, CHG 1: 08/23 | CYBERSECURITY FOR FACILITY-RELATED CONTROL SYSTEMS (FIRE ALARM)        |
| 25 05 11.03 | 05/21, CHG 1: 08/23 | CYBERSECURITY FOR FACILITY-RELATED CONTROL SYSTEMS (LIGHTING)          |
| 25 05 11.04 | 05/21, CHG 1: 08/23 | CYBERSECURITY FOR FACILITY-RELATED CONTROL SYSTEMS (FIREFIGHTER ALERT) |

#### **DIVISION 26 - ELECTRICAL**

|                |                     |                                                              |
|----------------|---------------------|--------------------------------------------------------------|
| 26 05 73       | 08/23               | POWER SYSTEM STUDIES                                         |
| 26 20 00       | 08/23               | INTERIOR DISTRIBUTION SYSTEM                                 |
| 26 27 13.10 30 | 10/07, CHG 2: 08/18 | ELECTRIC METERS                                              |
| 26 32 15.00    | 05/20               | ENGINE-GENERATOR SET STATIONARY 15-2500 KW, WITH AUXILIARIES |
| 26 33 53       | 05/19               | STATIC UNINTERRUPTIBLE POWER SUPPLY (UPS)                    |
| 26 41 00       | 08/23               | LIGHTNING PROTECTION SYSTEM                                  |
| 26 51 00       | 05/20, CHG 2: 11/21 | INTERIOR LIGHTING                                            |
| 26 56 00       | 08/21               | EXTERIOR LIGHTING                                            |
| 26 56 20       | 02/19               | AIRFIELD AND HELIPORT LIGHTING AND VISUAL NAVIGATION AIDS    |

#### **DIVISION 27 - COMMUNICATIONS**

|             |       |                                            |
|-------------|-------|--------------------------------------------|
| 27 05 13.43 | 05/20 | TELEVISION DISTRIBUTION SYSTEM             |
| 27 10 00    | 08/11 | BUILDING TELECOMMUNICATIONS CABLING SYSTEM |
| 27 20 00    | 07/24 | FIREFIGHTER ALERT SYSTEM                   |
| 27 41 00    | 11/21 | AUDIO-VISUAL SYSTEMS                       |
| 27 51 23    | 11/21 | INTERCOMMUNICATION SYSTEM                  |

#### **DIVISION 28 - ELECTRONIC SAFETY AND SECURITY**

|          |       |                                                               |
|----------|-------|---------------------------------------------------------------|
| 28 31 76 | 08/20 | INTERIOR FIRE ALARM AND MASS NOTIFICATION SYSTEM, ADDRESSABLE |
|----------|-------|---------------------------------------------------------------|

#### **DIVISION 31 - EARTHWORK**

|                |                     |                          |
|----------------|---------------------|--------------------------|
| 31 00 00.00 06 | 07/18               | EARTHWORK                |
| 31 11 00       | 11/18               | CLEARING AND GRUBBING    |
| 31 21 13       | 11/18               | RADON MITIGATION         |
| 31 31 16.13    | 08/22, CHG 1: 11/23 | CHEMICAL TERMITE CONTROL |

#### **DIVISION 32 - EXTERIOR IMPROVEMENTS**

|                |       |                                                                 |
|----------------|-------|-----------------------------------------------------------------|
| 32 01 19.61    | 11/19 | SEALING OF JOINTS IN RIGID PAVEMENT                             |
| 32 05 33       | 08/17 | LANDSCAPE ESTABLISHMENT                                         |
| 32 11 20       | 05/22 | BASE COURSE FOR RIGID AND SUBBASE FOR FLEXIBLE PAVING           |
| 32 12 13       | 05/17 | BITUMINOUS TACK AND PRIME COATS                                 |
| 32 12 16.16    | 11/20 | ROAD-MIX ASPHALT PAVING                                         |
| 32 13 13.06 06 | 06/18 | PORTLAND CEMENT CONCRETE PAVEMENT FOR ROADS AND SITE FACILITIES |



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## SECTION 07 11 13

### BITUMINOUS DAMPPROOFING 08/23

#### PART 1 GENERAL

##### 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

##### ASTM INTERNATIONAL (ASTM)

|                   |                                                                                                             |
|-------------------|-------------------------------------------------------------------------------------------------------------|
| ASTM C208         | (2022) Standard Specification for Cellulosic Fiber Insulating Board                                         |
| ASTM C728         | (2017a; R 2022) Standard Specification for Perlite Thermal Insulation Board                                 |
| ASTM D41/D41M     | (2011; R 2016) Standard Specification for Asphalt Primer Used in Roofing, Dampproofing, and Waterproofing   |
| ASTM D226/D226M   | (2017) Standard Specification for Asphalt-Saturated Organic Felt Used in Roofing and Waterproofing          |
| ASTM D227/D227M   | (2003; R 2018) Standard Specification for Coal-Tar-Saturated Organic Felt Used in Roofing and Waterproofing |
| ASTM D1187/D1187M | (1997; R 2018) Standard Specification for Asphalt-Base Emulsions for Use as Protective Coatings for Metal   |
| ASTM D1227/D1227M | (2013; R 2019; e1) Standard Specification for Emulsified Asphalt Used as a Protective Coating for Roofing   |
| ASTM D4263        | (1983; R 2018) Standard Test Method for Indicating Moisture in Concrete by the Plastic Sheet Method         |

##### 1.2 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-07 Certificates

Materials

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 1.3 DELIVERY AND STORAGE

Deliver materials in sealed containers bearing manufacturer's original labels. Labels must include date of manufacture, contents of each container, performance standards that apply to the contents and recommended shelf life. While in storage, do not allow water based bituminous damproofing to freeze.

## PART 2 PRODUCTS

### 2.1 ASPHALT PRIMER

ASTM D41/D41M.

### 2.2 EMULSION-BASED ASPHALT DAMPPROOFING

#### 2.2.1 Fibrated Emulsion-Based Asphalt

Provide cold-applied fibrated emulsion-based asphalt dampproofing conforming to ASTM D1227/D1227M Type II, Class 1, asbestos-free, manufactured of refined asphalt, emulsifiers and selected clay, fibrated with mineral fibers. For spray or brush application, emulsion must contain a minimum of 59 percent solids by weight, 56 percent solids by volume. For trowel application, emulsion must contain a minimum of 58 percent solids by weight, 55 percent solids by volume.

#### 2.2.2 Non-Fibrated Emulsion-Based Asphalt

Provide cold-applied non-fibrated emulsion-based asphalt dampproofing conforming to ASTM D1187/D1187M Type II or ASTM D1227/D1227M Type III, manufactured of refined asphalt, emulsifiers and selected clay. Asphalt must contain a minimum 58 percent solids by weight, 55 percent solids by volume.

### 2.3 SURFACE PROTECTION

#### 2.3.1 Saturated Felt

ASTM D226/D226M, Asphalt Saturated, Type I, 15 pound; ASTM D227/D227M, Coal-Tar Saturated.

#### 2.3.2 Protection Board

Wood Fiber Board, ASTM C208, or Perlite Board, ASTM C728.

## PART 3 EXECUTION

### 3.1 SURFACE PREPARATION

Remove or cut form ties and repair surface defects as required in Section 03 30 00 CAST-IN-PLACE CONCRETE. Clean concrete and masonry surfaces to receive dampproofing of foreign matter and loose particles. Apply dampproofing to clean dry surfaces. Moisture test in accordance with ASTM D4263. If test indicates moisture, allow a minimum of seven additional days after test completion for curing. If moisture still exists, redo test until substrate is dry.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 3.2 Protection of Surrounding Areas

Before starting the dampproofing work, protect the surrounding areas and surfaces from spillage and migration of dampproofing material onto other work.

### 3.3 APPLICATION

Apply dampproofing after priming coat is dry, but prior to any deterioration of primed surface, and when ambient temperature is above 40 degrees F.

#### 3.3.1 Surface Priming

Prime surfaces to receive asphalt or . Apply primer when ambient temperature is above 40 degrees F and at rate of approximately one gallon per 100 square feet, fully covering entire surface to be dampproofed.

#### 3.3.2 Cold-Application Method

##### 3.3.2.1 Emulsion-Based Asphalt

Do not perform emulsion-based asphalt dampproofing work in temperatures below 40 degrees F. Emulsions must have a smooth and uniform consistency at time of application. Apply dampproofing materials in accordance with manufacturer's published instructions to produce a smooth uniform dry film of not less than 12 mils thick without voids or defects. Recoat any dull or porous spots. Seal dampproofing materials tightly around pipes and other items projecting through dampproofing. Apply materials at rates as follows:

- a. Primer: 1/2 gallon per 100 square feet, cold-applied.
- b. Fibrated Dampproofing: 2 gallons per 100 square feet, cold-applied with spray, brush or trowel.
- c. Non-fibrated Dampproofing: 2 gallons per 100 square feet, cold-applied with spray, brush or trowel.

### 3.4 PROTECTIVE COVERING

Protect dampproofed surfaces against which backfill will be placed with one layer of 15 pound saturated felt. Use asphalt-saturated felt where the dampproofing material is asphalt and use coal-tar-saturated felt where the dampproofing material is coal-tar pitch. Embed felts in the second coating of bitumen and lap edges and ends not less than one inch .

-- End of Section --

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

SECTION 07 19 00

WATER REPELLENTS

**08/23**

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN ARCHITECTURAL MANUFACTURERS ASSOCIATION (AAMA)

AAMA 501.2 (2015) Quality Assurance and Diagnostic Water Leakage Field Check of Installed Storefronts, Curtain Walls and Sloped Glazing Systems

AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS (AASHTO)

AASHTO T 259 (2002; R 2017) Standard Method of Test for Resistance of Concrete to Chloride Ion Penetration

AASHTO T 260 (2021) Standard Method of Test for Sampling and Testing for Chloride Ion in Concrete and Concrete Raw Materials

ASTM INTERNATIONAL (ASTM)

ASTM C140/C140M (2022c) Standard Test Methods for Sampling and Testing Concrete Masonry Units and Related Units

ASTM C642 (2021) Standard Test Method for Density, Absorption, and Voids in Hardened Concrete

ASTM D1653 (2013; R 2021) Standard Test Methods for Water Vapor Transmission of Organic Coating Films

ASTM D2369 (2020) Standard Test Method for Volatile Content of Coatings

ASTM D3278 (2021) Standard Test Methods for Flash Point of Liquids by Small Scale Closed-Cup Apparatus

ASTM E96/E96M (2022a; E 2023) Standard Test Methods for Gravimetric Determination of Water Vapor Transmission Rate of Materials

ASTM E514/E514M (2020) Standard Test Method for Water Penetration and Leakage Through Masonry

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

ASTM G154

(2023) Standard Practice for Operating  
Fluorescent Ultraviolet (UV) Lamp  
Apparatus for Exposure of Materials

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1910.1000

Air Contaminants

## 1.2 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

### SD-03 Product Data

Water Repellents

### SD-06 Test Reports

Water Absorption

Accelerated Weathering

Resistance to Chloride Ion Penetration

Moisture Vapor Transmission

Water Penetration and Leakage

### SD-07 Certificates

Manufacturer's Qualifications

Applicator's Qualifications

Evidence of Acceptable Variation

Warranty

### SD-08 Manufacturer's Instructions

Application Instructions

Manufacturer's Safety Data Sheets

## 1.3 QUALITY ASSURANCE

### 1.3.1 Qualifications

- a. Manufacturer's qualifications: Minimum five years record of successful in-service experience of water repellent treatments manufactured for concrete masonry, application.
- b. Applicator's qualifications: Minimum five years successful experience in projects of similar scope using specified or similar treatment materials and manufacturer's approval for application.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 1.3.2 Performance Requirements

- a. Water absorption: ASTM C140/C140M. Comparison of treated and untreated specimens.
- b. Moisture vapor transmission: ASTM E96/E96M. Comparison of treated and untreated specimens.
- c. Water penetration and leakage through masonry: ASTM E514/E514M.
- d. Resistance to chloride ion penetration: AASHTO T 259 and AASHTO T 260.
- e. Resistance to Accelerated weathering: ASTM G154

### 1.3.3 Evidence of Acceptable Variation

If a product proposed for use does not conform to requirements of the referenced specification, submit for approval to the Contracting Officer, evidence that the proposed product is either equal to or better than the product specified. Include the following:

- a. Identification of the proposed substitution;
- b. Reason why the substitution is necessary;
- c. A comparative analysis of the specified product and the proposed substitution, including tabulations of the composition of pigment and vehicle;
- d. The difference between the specified product and the proposed substitution; and
- e. Other information necessary for an accurate comparison of the proposed substitution and the specified product.

## 1.4 SAMPLE TEST PANEL

The approved Sample Test Panel will serve as the standard of quality for all other water repellent coating work. Do not proceed with application until the sample panel has been approved by the Contracting Officer.

### 1.4.1 Preparation of Sample Test Panel

Prior to commencing work, including bulk purchase and delivery of material, apply water repellent treatment to a minimum 4 feet high by 4 feet long concrete masonry, test-panel specified in. Provide a full height expansion joint at mid-panel length. Prepare and seal joint with materials approved for project use.

#### 1.4.1.1 Testing

Similar to AAMA 501.2 hose testing, provide field water testing of water repellent treated surfaces in the presence of the Contracting Officer and the water repellent treatment manufacturer's representative as follows.

- a. Apply water repellent to left side of mock-up and allow to cure prior to application of treatment to right side.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

- b. Twenty days after completion of application of treatment, test mock-up with 5/8 inch garden hose, with spray nozzle, located 10 feet from wall and aimed upward so water strikes wall at 45-degree downward angle. After water has run continuously for three hours observe back side of mock-up for water penetration and leakage. If leakage is detected, make changes as needed and retest.
- c. Coordinate testing procedures and modify project treatment application as required to pass mock-up tests for water penetration and leakage resistance.

#### 1.4.1.2 Approval

Proceed with water repellent treatment work only after completion of field test application and approval of mock-up and tests by the Contracting Officer.

#### 1.4.2 Pre-Installation Meeting

- a. Attend pre-installation meeting required prior to commencement of concrete masonry, installation.
- b. Review procedures and coordination required between water repellent treatment work and work of other trades which could affect work to be performed under this section of the work.
- c. Convene additional pre-installation meeting prior to water repellent treatment application for coordination with work not previously coordinated including joint sealants.

### 1.5 REGULATORY REQUIREMENTS

#### 1.5.1 Environmental Protection

Provide coating materials that conform to the restrictions of the Local Air Pollution Control jurisdiction. Notify the Contracting Officer of water repellent coating specified herein which fails to conform to the local Air Quality Management District Rules at the location of the Project. In localities where the specified coating is prohibited, the Contracting Officer may direct the substitution of an acceptable coating.

### 1.6 DELIVERY, STORAGE, AND HANDLING

Deliver materials in original sealed containers, clearly marked with the manufacturer's name, brand name, type of material, batch number, percent solids by weight and volume, and date of manufacturer. Store materials off the ground, in a dry area where the temperature will be not less than 50 degrees F nor more than 85 degrees F.

### 1.7 SAFETY METHODS

Apply coating materials using safety methods and equipment.

#### 1.7.1 Toxic Materials

To protect personnel from overexposure to toxic materials, conform to the most stringent guidance of:

- a. The coating manufacturer when using solvents or other chemicals. Use

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

impermeable gloves, chemical goggles or face shield, and other recommended protective clothing and equipment to avoid exposure of skin, eyes, and respiratory system. Conduct work in a manner to minimize exposure of building occupants and the general public.

- b. 29 CFR 1910.1000.
- c. Threshold Limit Values (R) of the American Conference of Governmental Industrial Hygienists.
- d. Manufacturer's Safety Data Sheets.

## 1.8 ENVIRONMENTAL CONDITIONS

### 1.8.1 Weather and Substrate Conditions

Do not proceed with application of water repellents under any of the following conditions, except with written recommendations of manufacturer.

- a. Ambient temperature is less than 40 degrees F.
- b. Substrate faces have cured less than one month.
- c. Rain or temperature below 40 degrees F are predicted for a period of 24 hours before or after treatment.
- d. Earlier than three days after surfaces are wet.
- e. Substrate is frozen or surface temperature is less than 40 degrees F and falling.

### 1.8.2 Moisture Condition

Determine moisture content of substrate meets manufacturer's requirements prior to application of water repellent material.

## 1.9 SEQUENCING AND SCHEDULING

### 1.9.1 Masonry Surfaces

Do not start water repellent coating until all joint tooling, pointing and masonry cleaning operations have been completed. Allow masonry to cure for at least 60 days under normal weather conditions before applying water repellent.

### 1.9.2 Plaster Surfaces

Do not start water repellent coating until all shrinkage and stress cracks are repaired and sound, all surfaces are free of defects and cleaning operations have been completed. Allow plaster to cure for at least 30 days under normal weather conditions before applying water repellent.

### 1.9.3 Concrete Surfaces

Do not start water repellent coating until all patching, pointing and cleaning operations have been completed and concrete has cured a minimum of 30 days under normal weather conditions.



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### 1.9.4 Sealants

Do not apply water repellents until the sealants for joints adjacent to surfaces receiving water repellent treatment have been installed and cured.

- a. Water repellent work may precede sealant application only if sealant adhesion and compatibility have been tested and verified using substrate, water repellent, and sealant materials identical to those used in the work.
- b. Provide manufacturers' test results of compatibility.

#### 1.10 INSPECTIONS

Notify the manufacturer's representative a minimum of 72 hours prior to scheduled application of water repellents for field inspection. Inspect surfaces and obtain approval in writing from the manufacturer's representative prior to application of water repellent coating.

#### 1.11 SURFACES TO BE COATED

Coat all exterior masonry, surfaces where indicated on Drawings. This includes back faces of parapets, top of walls, edges and returns adjacent to windows and door frames and free-standing walls.

#### 1.12 WARRANTY

Provide a warranty, issued jointly by the manufacturer and the applicator of the water repellent treatment against moisture penetration through the treated structurally sound surface for a period of five years. Warranty to provide the material, labor, and equipment necessary to remedy the problem. At the satisfactory completion of the work, complete the warranty sign, notarize, and submit to the Contracting Officer.

### PART 2 PRODUCTS

#### 2.1 MATERIALS

Provide a clear, non-yellowing, deep-penetrating, VOC compliant water repellent solution. Material must not stain or discolor and must produce a mechanical and chemical interlocking bond with the substrate to the depth of the penetration.

#### 2.2 WATER REPELLENTS

##### 2.2.1 Silane, 20 Percent Solids

Penetrating water repellent. A monomeric compound containing approximately 20 percent alkyltrialkoxysilanes with alcohol, mineral spirits, water, and other proprietary solvent carrier.

- a. Composition: Modified alkylalkoxysilane.
- b. Active alkylalkoxysilane content: ASTM D2369 20 percent by weight, plus or minus 1 percent.
- c. Appearance: White, milky liquid.
- d. Average depth of penetration: Up to 3/8 inch depending on substrate.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

- e. VOC content: Less than 350 grams per liter.
- f. Flash point, ASTM D3278.
- g. Specific gravity, at 78 degrees F: 0.96 to 0.98.
- h. Density: .0 to 8.2 pounds per gallon.

#### 2.2.2 Silane, 40 Percent Solids

Penetrating water repellent. A monomeric compound containing approximately 40 percent alkyltrialkoxysilanes with alcohol, mineral spirits, or water.

- a. Composition: Modified alkylalkoxysilane.
- b. Active alkylalkoxysilane content: ASTM D2369 40 percent by weight, plus or minus 1.5 percent.
- c. Appearance: White, milky liquid.
- d. Average depth of penetration: Up to 3/8 inch depending on substrate.
- e. VOC content: Less than 350 grams per liter.
- f. Flash point, ASTM D3278.
- g. Specific gravity, at 78 degrees F: 0.94 to 0.97.
- h. Density: 7.8 to 8.1 pounds per gallon.

#### 2.2.3 Silane, 85 Percent Solids or Greater

Penetrating water repellent. A monomeric compound containing 85 percent or greater alkyltrialkoxysilanes with alcohol, mineral spirits, or water.

- a. Composition: Modified alkylalkoxysilane.
- b. Active alkylalkoxysilane content: ASTM D2369 85 percent by weight, plus or minus 1 percent.
- c. Appearance: White, milky liquid.
- d. Average depth of penetration: Up to 3/8 inch depending on substrate.
- e. VOC content: Less than 350 grams per liter.
- f. Flash point, ASTM D3278.
- g. Specific gravity, at 78 degrees F: 0.96 to 0.98.
- h. Density: 8.0 to 8.2 pounds per gallon.

#### 2.2.4 Siloxanes

Penetrating water repellent. Alkylalkoxysiloxanes that are oligomeric with alcohol, ethanol, mineral spirits, or water.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

- a. Solids by weight: ASTM D2369, 7.5 to 16.0 percent.
- b. Volatile Organic Content (VOC) after blending: Less than 175 grams per liter.
- c. Density, activated: 8.4 pounds per gallon, plus or minus one percent.
- d. Flash point, ASTM D3278: Greater than 212 degrees F.

#### 2.2.5 Low-Solids Acrylic

Water-clear, breathing coating of acrylic resins, water-based, solvent-based, or acrylic emulsions solution containing less than 15 percent solids by volume.

#### 2.2.6 High-Solids Acrylic

Water-clear, breathing coating of acrylic resins, water-based, solvent-based, or acrylic emulsions solution containing 15 percent solids or more by volume.

#### 2.2.7 VOC-Complying Water Repellents

Products certified by the manufacturer that they comply with local regulations controlling use of volatile organic compounds (VOC's).

### 2.3 PERFORMANCE CRITERIA

#### 2.3.1 Silane, 20 Percent Solids

- a. Water absorption test: ASTM C642 and ASTM E514/E514M.
- b. Moisture vapor transmission: ASTM D1653, 28.33 perms or 51.61 percent maximum compared to untreated surfaces.
- c. Resistance to chloride ion penetration: AASHTO T 259 and AASHTO T 260.
- d. Water penetration and leakage through masonry, ASTM E514/E514M percentage reduction of leakage: 97 percent minimum.
- e. Resistance to accelerated weathering, ASTM G154 testing 2,500 hours: No loss in repellency.
- f. Drying time under normal conditions: Four hours per 75 degrees F.

#### 2.3.2 Silane, 40 Percent Solids

- a. Average depth of penetration: 3/8 inches depending on substrate
- b. Resistance to chloride ion penetration, AASHTO T 259 and AASHTO T 260.
- c. Water absorption test, ASTM E514/E514M: 0.42 percent per 48 hours; 1.2 percent per 50 days.
- d. Moisture vapor transmission: ASTM D1653, 28.33 perms or 51.61 percent maximum compared to untreated surfaces.
- e. Resistance to accelerated weathering, ASTM G154. Testing 2,500 hours: No loss in repellency.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

f. Drying time under normal conditions: Four hours per 75 degrees F.

#### 2.3.3 Silane, 85 Percent Solids or Greater

- a. Average depth of penetration: 3/8 inches depending on substrate.
- b. Resistance to chloride ion penetration, AASHTO T 259 and AASHTO T 260.
- c. Water absorption test, ASTM E514/E514M: 0.42 percent per 48 hours; 1.2 percent per 50 days.
- d. Moisture vapor transmission: ASTM D1653, 28.33 perms or 51.61 percent maximum compared to untreated surfaces.
- e. Resistance to accelerated weathering, ASTM G154. Testing 2,500 hours: No loss in repellency.
- f. Drying time under normal conditions: Four hours per 75 degrees F.

#### 2.3.4 Siloxanes

- a. Dry time for recoat, if necessary: One to two hours depending on weather conditions.
- b. Penetration: 3/8 inch, depending on substrate.
- c. Water penetration and leakage through masonry, ASTM E514/E514M, percentage reduction of leakage: 97.0 percent minimum.
- d. Moisture vapor transmission, ASTM E96/E96M: 47.5 perms or 82 percent maximum compared to untreated sample.
- e. Resistance to accelerated weathering, ASTM G154. Testing 2,500 hours: No loss in repellency.
- f. Resistance to chloride ion penetration, AASHTO T 259 and AASHTO T 260.

### PART 3 EXECUTION

#### 3.1 EXAMINATION

Examine concrete, or masonry surfaces to be treated to ensure that:

- a. All visible cracks, voids or holes have been repaired.
- b. All mortar joints in masonry are tight and sound, have not been re-set or misaligned and show no cracks or spalling.
- c. Moisture contents of walls does not exceed 15 percent when measured on an electronic moisture register, calibrated for the appropriate substrate.
- d. Concrete surfaces are free of form release agents, curing compounds and other compounds that would prevent full penetration of the water repellent material.

Do not start water repellent treatment work until all deficiencies have been corrected, examined, and found acceptable to the Contracting Officer

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

and the water repellent treatment manufacturer. Do not apply treatment to damp, dirty, dusty, or otherwise unsuitable surfaces. Comply with the manufacturer's recommendations for suitability of surface.

### 3.2 PREPARATION

#### 3.2.1 Surface Preparation

Prepare substrates in accordance with water repellent treatment manufacturer's recommendation. Clean surfaces of dust, dirt, efflorescence, alkaline, and foreign matter detrimental to proper application of water repellent treatment.

#### 3.2.2 Protection

Provide masking or protective covering for materials which could be damaged by water repellent treatment.

- a. Protect glass, glazed products, and prefinished products from contact with water repellent treatment.
- b. Protect landscape materials with breathing-type drop cloths; plastic covers are not acceptable.

#### 3.2.3 Compatibility

- a. Confirm treatment compatibility with each type of joint sealer within or adjacent to surfaces receiving water repellent treatment in accordance with manufacturer's recommendations.
- b. When recommended by joint sealer manufacturer, apply treatment after application and cure of joint sealers. Coordinate treatment with joint sealers.

### 3.3 MIXING

Mix water repellent material thoroughly in accordance with the manufacturer's recommendations. Mix, in quantities required for that day's work, all containers prior to application. Mix each container the same length of time.

### 3.4 APPLICATION

In strict accordance with the manufacturers written requirements. Do not start application without the manufacturer's representative being present or their written acceptance of the surface to be treated. Submit manufacturer's instructions including preparation, application, recommended equipment to be used, safety measures, and protection of completed application.

#### 3.4.1 Water Repellent Treatment

##### 3.4.1.1 Spray Application

Spray apply water repellent material to exterior concrete, and masonry surfaces using low-pressure airless spray equipment in strict accordance with manufacturer's printed application, instructions, and precautions. Maintain copies at the job site. Apply flood coat in an overlapping pattern allowing approximately 8 to 10 inch rundown on the vertical

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

surface. Maintain a wet edge at all overlaps, both vertical and horizontal. Hold gun maximum 18 inches from wall.

#### 3.4.1.2 Brush or Roller Application

Brush or roller apply water repellent material only at locations where overspray would affect adjacent materials and where not practical for spray applications.

#### 3.4.1.3 Covered Surfaces

Coat all exterior concrete, or masonry surfaces including back faces of parapets, tops of walls, edges and returns adjacent to window and door frames, window sills, and free-standing walls.

#### 3.4.1.4 Rate of Application

Apply materials to exterior surfaces at the coverages recommended by the manufacturer and as determined from sample panel test. Increase or decrease application rates depending upon the surface texture and porosity of the substrate to achieve even appearance and total water repellency.

#### 3.4.1.5 Number of Coats

The results of the sample panel test determines the number of coats required to achieve full coverage and protection.

#### 3.4.1.6 Appearance

If unevenness in appearance, lines of work termination or scaffold lines exist, or detectable changes from the approved sample panel occur, the Contracting Officer may require additional treatment at no additional cost to the Government. Apply required additional treatment to a natural break off point.

### 3.5 CLEANING

Clean all runs, drips, and overspray from adjacent surfaces while the water repellent treatment is still wet in a manner recommended by the manufacturer.

### 3.6 FIELD QUALITY CONTROL

Do not remove drums containing water repellent material from the job site until completion of all water repellent treatment and until so authorized by the Contracting Officer.

#### 3.6.1 Field Testing

At a time not less than twenty days after completion of the water repellent coating application, subject a representative wall area of the building to the Navy Hose Stream Field Test similar to AAMA 501.2 hose test to simulate rainfall for a period of three hours. Use a minimum 5/8 inch diameter hose and a fixed lawn sprinkler spray head which will direct a full flow of water against the wall. Place the sprinkler head so that the water will strike the wall downward at a 45-degree angle to the wall. If the inside of the wall shows traces of moisture during or following the test, apply another coat of water repellent, at the manufacturer's recommended coverage rate to the entire building. Repeat testing and

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

re-coating process until no moisture shows on the inside wall face.  
Accomplish required work retesting and re-coating at no additional cost to the Government.

### 3.6.2 Site Inspection

Inspect treatment in progress by manufacturer's representative to verify compliance with manufacturer instructions and recommendations.

-- End of Section --

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## SECTION 07 21 13

### BOARD AND BLOCK INSULATION 08/23

#### PART 1 GENERAL

##### 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

#### ASTM INTERNATIONAL (ASTM)

|                   |                                                                                                                                               |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM C578         | (2023) Standard Specification for Rigid, Cellular Polystyrene Thermal Insulation                                                              |
| ASTM C930         | (2019) Standard Classification of Potential Health and Safety Concerns Associated with Thermal Insulation Materials and Accessories           |
| ASTM D3833/D3833M | (1996; R 2019) Standard Test Method for Water Vapor Transmission of Pressure-Sensitive Tapes                                                  |
| ASTM E96/E96M     | (2024) Standard Test Methods for Gravimetric Determination of Water Vapor Transmission Rate of Materials                                      |
| ASTM E154/E154M   | (2008a; R 2019) Standard Test Methods for Water Vapor Retarders Used in Contact with Earth Under Concrete Slabs, on Walls, or as Ground Cover |

##### 1.2 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

#### SD-02 Shop Drawings

Manufacturer's Standard Details

#### SD-03 Product Data

Environmental Data for Materials

Block or Board Insulation

Vapor Retarder; FIO

Pressure Sensitive Tape



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

Accessories including sealants

SD-08 Manufacturer's Instructions

Insulation Installation and Handling

SD-11 Closeout Submittals

Draft Guarantee

Final Guarantee

Draft Warranty

Final Warranty

### 1.3 MANUFACTURER'S DETAILS

Submit manufacturer's standard details indicating methods of attachment and spacing, transition and termination details, and installation details. Include verification of existing conditions.

### 1.4 PRODUCT DATA

Include environmental data for materials descriptions, recommendations for product shelf life, requirements for precautions for flammability and toxicity. Include data to verify compatibility of sealants with insulation.

### 1.5 DELIVERY, STORAGE, AND HANDLING

#### 1.5.1 Delivery

Deliver materials to the site in original sealed wrapping bearing manufacturer's name and brand designation, specification number, type, grade, R-value, and class. Store and handle to protect from damage. Do not allow insulation materials to become wet, soiled, crushed, or covered with ice or snow. Comply with manufacturer's recommendations for handling, storing, and protecting of materials before and during installation.

#### 1.5.2 Storage

Inspect materials delivered to the site for damage and store out of weather in manufacturer's original packaging. Store only in dry locations, not subject to open flames or sparks, and easily accessible for inspection and handling. Keep materials wrapped and separated from off-gassing materials (such as drying paints). Do not use materials that have visible moisture or biological growth. Comply with manufacturer's recommendations for handling, storage, and protection of materials before and during installation.

### 1.6 SAFETY PRECAUTIONS

#### 1.6.1 Other Safety Considerations

Comply with the safety requirements of ASTM C930.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## 1.7 SPECIAL WARRANTIES

### 1.7.1 Guarantee

Guarantee insulation installation against failure due to ultraviolet light exposure for a period of three years from the date of Beneficial Occupancy or Substantial Completion. Submit draft guarantee and final guarantee.

### 1.7.2 Warranty

Provide manufacturer's material warranty for all system components for a period of three years from the date of Beneficial Occupancy or Substantial Completion. Submit draft warranty and final warranty.

## PART 2 PRODUCTS

### 2.1 BLOCK OR BOARD INSULATION

Provide thermal insulating materials as recommended by manufacturer for each type of application indicated. Provide insulation with the following physical properties and in accordance with the following standards:

- a. Extruded Preformed Cellular Polystyrene (XPS): ASTM C578 REV A, Type IV

#### 2.1.1 Prohibited Materials

Do not provide materials containing asbestos.

### 2.2 VAPOR RETARDER

#### 2.2.1 Vapor Retarder under Floor Slab

- a. Water vapor permeance: 0.2 Perm or less when tested in accordance with ASTM E96/E96M.
- b. Puncture resistance: Maximum load no less than 40 pounds when tested according to ASTM E154/E154M REV A.

### 2.3 PRESSURE SENSITIVE TAPE

As recommended by manufacturer of vapor retarder(s). Match water vapor permeance rating for each vapor retarder specified. Provide tape in accordance with ASTM D3833/D3833M.

### 2.4 ACCESSORIES

#### 2.4.1 Mechanical Fasteners

Corrosion resistant fasteners as recommended by the insulation manufacturer.

## PART 3 EXECUTION

### 3.1 INSTALLATION

#### 3.1.1 Insulation Installation and Handling

Provide insulation in accordance with the manufacturer's printed installation instructions. Keep material dry and free of extraneous

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

materials.

### 3.1.2 Cold Climate Requirement

Place insulation on the outside of pipes.

### 3.1.3 Continuity of Insulation

Butt tightly against adjoining boards, studs, rafters, joists, sill plates, headers and obstructions. Provide continuity and integrity of insulation at corners, wall to ceiling joint, roof, and floor. Avoid creating thermal bridges and voids. Provide and verify continuity of insulative barrier throughout the building enclosure.

### 3.1.4 Coordination

Verify final installed insulation thicknesses comply with thicknesses indicated and with the approved insulation submittal(s).

## 3.2 PERIMETER AND UNDER SLAB INSULATION

Install perimeter thermal insulation where heated spaces are adjacent to exterior walls, slab edges in slab-on-grade, or floating slab construction.

### 3.2.1 Manufacturer's Instructions

Layout insulation, tape edges, provide vapor retarder and other required accessories to protect against vermin, insects, and damage in accordance with manufacturer's printed instructions.

### 3.2.2 Insulation on Vertical Surfaces

Provide thermal insulation on exterior of foundation walls below grade on edges of slabs-on-grade. Fasten insulation with mechanical fasteners.

### 3.2.3 Insulation Under Slab

Provide insulation horizontally under slab on grade for a distance of two feet from the edge of slab. Install insulation on top of vapor retarder and turn retarder up over the outside edge of insulation to top of slab.

### 3.2.4 Protection of Insulation

Protect insulation from damage during construction and back filling by application of protection board or a coating. Do not leave installed vertical insulation unprotected overnight. Protect installed insulation from weather, including rain and ultraviolet light, from mechanical abuse, compression, and dislocation. Install protection over entire exposed exterior insulation board.

-- End of Section --

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## SECTION 07 21 16

### BUILDING INSULATION

**08/23**

#### PART 1 GENERAL

##### 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

##### ASTM INTERNATIONAL (ASTM)

|                   |                                                                                                                                     |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| ASTM C665         | (2023) Standard Specification for Mineral-Fiber Blanket Thermal Insulation for Light Frame Construction and Manufactured Housing    |
| ASTM C930         | (2019) Standard Classification of Potential Health and Safety Concerns Associated with Thermal Insulation Materials and Accessories |
| ASTM D3833/D3833M | (1996; R 2019) Standard Test Method for Water Vapor Transmission of Pressure-Sensitive Tapes                                        |
| ASTM D5359        | (1998; R 2021) Standard Specification for Glass Cullet Recovered from Waste for Use in Manufacture of Glass Fiber                   |
| ASTM E84          | (2023) Standard Test Method for Surface Burning Characteristics of Building Materials                                               |
| ASTM E96/E96M     | (2022a; E 2023) Standard Test Methods for Gravimetric Determination of Water Vapor Transmission Rate of Materials                   |
| ASTM E136         | (2022) Standard Test Method for Assessing Combustibility of Materials Using a Vertical Tube Furnace at 750 Degrees C                |

##### CALIFORNIA DEPARTMENT OF PUBLIC HEALTH (CDPH)

|                    |                                                                                                                                                            |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CDPH SECTION 01350 | (2017; Version 1.2) Standard Method for the Testing and Evaluation of Volatile Organic Chemical Emissions from Indoor Sources using Environmental Chambers |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|

##### GREEN SEAL (GS)

|       |                                     |
|-------|-------------------------------------|
| GS-36 | (2013) Adhesives for Commercial Use |
|-------|-------------------------------------|

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

|          |                                                                                    |
|----------|------------------------------------------------------------------------------------|
| NFPA 31  | (2020) Standard for the Installation of Oil-Burning Equipment                      |
| NFPA 54  | (2021) National Fuel Gas Code                                                      |
| NFPA 70  | (2023; ERTA 4 2023) National Electrical Code                                       |
| NFPA 211 | (2019) Standard for Chimneys, Fireplaces, Vents, and Solid Fuel-Burning Appliances |

#### SCIENTIFIC CERTIFICATION SYSTEMS (SCS)

|     |                                            |
|-----|--------------------------------------------|
| SCS | SCS Global Services (SCS) Indoor Advantage |
|-----|--------------------------------------------|

#### SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT (SCAQMD)

|                  |                                          |
|------------------|------------------------------------------|
| SCAQMD Rule 1168 | (2017) Adhesive and Sealant Applications |
|------------------|------------------------------------------|

#### TECHNICAL ASSOCIATION OF THE PULP AND PAPER INDUSTRY (TAPPI)

|               |                                         |
|---------------|-----------------------------------------|
| TAPPI T803 OM | (2010) Puncture Test of Container Board |
|---------------|-----------------------------------------|

#### U.S. DEPARTMENT OF ENERGY (DOE)

|             |                                                                     |
|-------------|---------------------------------------------------------------------|
| Energy Star | (1992; R 2006) Energy Star Energy Efficiency Labeling System (FEMP) |
|-------------|---------------------------------------------------------------------|

#### U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

|                 |                        |
|-----------------|------------------------|
| 29 CFR 1910.134 | Respiratory Protection |
|-----------------|------------------------|

#### UNDERWRITERS LABORATORIES (UL)

|         |                                                                                                                 |
|---------|-----------------------------------------------------------------------------------------------------------------|
| UL 2818 | (2022) GREENGUARD Certification Program For Chemical Emissions For Building Materials, Finishes And Furnishings |
|---------|-----------------------------------------------------------------------------------------------------------------|

## 1.2 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

#### SD-03 Product Data

Sound Attenuation Batt Insulation

Rigid Polyisocyanurate Board Insulation

Energy Star Label for Insulation Materials

Spray Polyurethane Foam Insulation - Thermal SPF

Vapor Retarder

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

Pressure Sensitive Tape

Accessories

SD-07 Certificates

Indoor Air Quality for Adhesives

SD-08 Manufacturer's Instructions

Insulation

### 1.3 CERTIFICATIONS

Submit required indoor air quality certifications and validations in one submittal package.

#### 1.3.1 Insulation Products

Provide product certified to meet indoor air quality requirements by UL 2818 (Greenguard) Gold, SCS Global Services Indoor Advantage Gold or provide certification by other third-party programs. Provide current product certification from certification body.

#### 1.3.2 Adhesives and Sealants

Provide products certified to meet indoor air quality requirements by UL 2818 (Greenguard) Gold, SCS Global Services Indoor Advantage Gold or provide certification or validation by other third-party programs that products meet the requirements of this Section. Provide current product certification documentation from certification body. When product does not have certification, provide validation that product meets the indoor air quality product requirements cited herein.

### 1.4 DELIVERY, STORAGE, AND HANDLING

#### 1.4.1 Delivery

Deliver materials to site in original sealed wrapping bearing manufacturer's name and brand designation, specification number, type, grade, R-value, and class. Store and handle to protect from damage. Do not allow insulation materials to become wet, soiled, crushed, or covered with ice or snow. Comply with manufacturer's recommendations for handling, storing, and protecting of materials before and during installation.

#### 1.4.2 Storage

Inspect materials delivered to the site for damage; unload and store out of weather in manufacturer's original packaging. Store only in dry locations, not subject to open flames or sparks, and easily accessible for inspection and handling.

### 1.5 SAFETY PRECAUTIONS

#### 1.5.1 Respirators

Provide installers with dust/mist respirators, training in their use, and

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

protective clothing, all approved by National Institute for Occupational Safety and Health (NIOSH)/Mine Safety and Health Administration (MSHA) in accordance with 29 CFR 1910.134.

#### 1.5.2 Other Safety Concerns

Consider other safety concerns and measures as outlined in ASTM C930.

### PART 2 PRODUCTS

#### 2.1 SOUND ATTENUATION BATT INSULATION

ASTM C665, Type I, blankets without membrane coverings; Class A, membrane-faced surface with a flame spread of 25 or less, except a flame spread rating of 25 or less and a smoke developed rating of 150 or less when tested in accordance with ASTM E84.

Provide insulation with R-Value as indicated on drawings

##### 2.1.1 Recycled Materials

Provide insulation materials containing the following minimum percentage of recycled material content by weight:

- a. Fiberglass: 20 percent glass cullet complying with ASTM D5359
- b. Mineral wool: 75 percent recovered materials content from slag

##### 2.1.2 Energy Star

Provide insulation materials that are Energy Star labeled. Provide data identifying Energy Star label for insulation materials.

##### 2.1.3 Prohibited Materials

Do not provide asbestos-containing materials.

##### 2.1.4 Reduced Volatile Organic Compounds (VOC) for Insulation Materials

Provide certification of indoor air quality for insulation materials.

#### 2.2 RIGID POLYISOCYANURATE BOARD INSULATION - EXTERIOR WALLS

Rigid Polyisocyanurate Board Insulation: Cellular thermal insulation with glass fiber-reinforced Polyisocyanurate closed-cell foam core and aluminum foil facing laminated to both sides: complying with FS HH-I-1972/1, Class 2; aged r-values of 7.2 and 8 at 40 and 75 degrees F (4.4 and 23.9 degrees C), respectively, and as follows:

Surface Burning Characteristics: Maximum values for flame spread and smoke developed of 20 and 150, respectively.

#### 2.3 BLOCKING

Wood, metal, unfaced mineral fiber blankets in accordance with ASTM C665, Type I, or other approved materials. Use only non-combustible materials meeting the requirements of ASTM E136 for blocking around chimneys and heat producing devices.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## 2.4 SPRAY POLYURETHANE FOAM INSULATION - THERMAL SPF

At locations indicated on Drawings.

ASTM C 1029, Type II, closed-cell, with maximum flame-spread and smoke-developed indexes of 75 and 450, respectively, per ASTM E 84.

- a. Minimum density of 1.5 lb/cu. ft., thermal resistivity of 6.2 degrees F x h x sq. ft./Btu x in. at 75 degrees F.
- b. Spray-Applied Insulation: Apply spray-applied insulation according to manufacturer's written instructions. Do not apply insulation unit installation of pipes, ducts, conduits, wiring, and electrical outlets in walls in completed and windows, electrical boxes, and other items not indicated to receive insulation are masked.
- c. Miscellaneous Voids: Install insulation in miscellaneous voids and cavity spaces where required to receive insulation are masked
- d. Thermal Resistance Value (R-Value) of SPF: As indicated on Drawings.

## 2.5 VAPOR RETARDER

- a. Membrane with the following properties:

Water Vapor Permeance: ASTM E96/E96M: 1 perm

Maximum Flame Spread: ASTM E84: 25

Combustion Characteristics: Passing ASTM E136

Puncture Resistance: TAPPI T803 OM: 25

## 2.6 PRESSURE SENSITIVE TAPE

As recommended by the vapor retarder manufacturer and having a water vapor permeance rating of one perm or less when tested in accordance with ASTM D3833/D3833M.

## 2.7 ACCESSORIES

### 2.7.1 Adhesive

As recommended by the insulation manufacturer. Provide non-aerosol adhesive products used on the interior of the building (defined as inside of the weatherproofing system) that meet either emissions requirements of CDPH SECTION 01350 (limit requirements for either office or classroom spaces regardless of space type) or VOC content requirements of SCAQMD Rule 1168. Provide aerosol adhesives used on the interior of the building that meet either emissions requirements of CDPH SECTION 01350 (use the office or classroom requirements, regardless of space type) or VOC content requirements of GS-36. Provide certification or validation of indoor air quality for adhesives.

### 2.7.2 Mechanical Fasteners

Corrosion resistant fasteners as recommended by the insulation manufacturer.



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 2.7.3 Wire Mesh

Corrosion resistant and as recommended by the insulation manufacturer.

## PART 3 EXECUTION

### 3.1 EXISTING CONDITIONS

Before installing insulation, ensure that surfaces that will be in contact with the insulation are dry and free of projections which could cause voids, compressed insulation, or punctured vapor retarders. If moisture or other conditions are found that do not allow the workmanlike installation of the insulation, do not proceed but notify Contracting Officer of such conditions.

### 3.2 PREPARATION

#### 3.2.1 Blocking at Attic Vents and Access Doors

Prior to installation of insulation, install permanent blocking to prevent insulation from slipping over, clogging, or restricting air flow through soffit vents at eaves. Install permanent blocking to maintain accessibility to equipment or controls that require maintenance or adjustment.

#### 3.2.2 Blocking Around Heat Producing Devices

Install non-combustible blocking around heat producing devices to provide the following clearances:

- a. Recessed lighting fixtures, including wiring compartments, ballasts, and other heat producing devices, unless these are certified by the manufacturer for installation surrounded by insulation: 3 inches from outside face of fixtures and devices or as required by NFPA 70 and, if insulation is to be placed above fixture or device, 24 inches above fixture.
- b. Masonry chimneys or masonry enclosing a flue: 2 inches from outside face of masonry. Masonry chimneys for medium and high heat operating appliances: Minimum clearances required by NFPA 211.
- c. Vents and vent connectors used for venting the products of combustion, flues, and chimneys other than masonry chimneys: Minimum clearances as required by NFPA 211.
- d. Gas Fired Appliances: Clearances as required in NFPA 54.
- e. Oil Fired Appliances: Clearances as required in NFPA 31.

Blocking around flues and chimneys is not required when insulation blanket, including any attached vapor retarder, passes ASTM E136, in addition to meeting other requirements stipulated in Part 2. Blocking is also not required if the chimneys are certified by the manufacturer for use in contact with insulating materials.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 3.3 INSTALLATION

#### 3.3.1 Insulation

Install and handle insulation in accordance with manufacturer's instructions. Keep material dry and free of extraneous materials. Any materials that show visual evidence of biological growth due to presence of moisture must not be installed on the building project. Ensure personal protective clothing and respiratory equipment is used as required. Observe safe work practices.

##### 3.3.1.1 Electrical wiring

Do not install insulation in a manner that would sandwich electrical conductors, that are not installed in conduit, between two layers of insulation.

##### 3.3.1.2 Continuity of Insulation

Install blanket insulation to butt tightly against adjoining blankets and to studs, rafters, joists, sill plates, headers and any obstructions. Provide continuity and integrity of insulation at corners, wall to ceiling joints, roof, and floor. Avoid creating thermal bridges.

##### 3.3.1.3 Installation at Bridging and Cross Bracing

Insulate at bridging and cross bracing by splitting blanket vertically at center and packing one half into each opening. Butt insulation at bridging and cross bracing; fill in bridged area with loose or scrap insulation.

##### 3.3.1.4 Sizing of Blankets

Provide only full width blankets when insulating between trusses, joists, or studs. Size width of blankets for a snug fit where trusses, joists or studs are irregularly spaced.

#### 3.3.2 Installation of Separate Vapor Retarder

Apply continuous vapor retarder as indicated. Overlap joints at least 6 inches and seal with pressure sensitive tape. Seal at sill, header, windows, doors and utility penetrations. Repair punctures or tears with pressure sensitive tape.

-- End of Section --

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## SECTION 07 22 00

### ROOF AND DECK INSULATION 08/23

#### PART 1 GENERAL

##### 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

##### ASTM INTERNATIONAL (ASTM)

|                   |                                                                                                    |
|-------------------|----------------------------------------------------------------------------------------------------|
| ASTM C1177/C1177M | (2017) Standard Specification for Glass Mat Gypsum Substrate for Use as Sheathing                  |
| ASTM C1289        | (2023) Standard Specification for Faced Rigid Cellular Polyisocyanurate Thermal Insulation Board   |
| ASTM D226/D226M   | (2017) Standard Specification for Asphalt-Saturated Organic Felt Used in Roofing and Waterproofing |
| ASTM D4601/D4601M | (2004; R 2020) Standard Specification for Asphalt-Coated Glass Fiber Base Sheet Used in Roofing    |
| ASTM E84          | (2023) Standard Test Method for Surface Burning Characteristics of Building Materials              |

##### FM GLOBAL (FM)

|              |                                                                                                               |
|--------------|---------------------------------------------------------------------------------------------------------------|
| FM 4450      | (1989) Approval Standard for Class 1 Insulated Steel Deck Roofs                                               |
| FM APP GUIDE | (updated on-line) Approval Guide<br><a href="http://www.approvalguide.com/">http://www.approvalguide.com/</a> |

##### SCIENTIFIC CERTIFICATION SYSTEMS (SCS)

|     |                                            |
|-----|--------------------------------------------|
| SCS | SCS Global Services (SCS) Indoor Advantage |
|-----|--------------------------------------------|

##### UNDERWRITERS LABORATORIES (UL)

|         |                                                                                                                 |
|---------|-----------------------------------------------------------------------------------------------------------------|
| UL 2818 | (2022) GREENGUARD Certification Program For Chemical Emissions For Building Materials, Finishes And Furnishings |
|---------|-----------------------------------------------------------------------------------------------------------------|

##### 1.2 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Insulation Board Layout and Attachment; G

Verification of Existing Conditions; G

SD-03 Product Data

Insulation; G

Cover Board; G

Fasteners; G

Sheathing Paper; G

Moisture Control; G

Recycled Content For Insulation; S

SD-05 Design Data

Wind Resistance; G

SD-06 Test Reports

Flame Spread Rating; G

SD-07 Certificates

Installer Qualifications; G

Certificates Of Compliance For Felt Materials; G

Indoor Air Quality For Insulation; S

Acceptable Foam Blowing Agents; S

SD-08 Manufacturer's Instructions

Nails and Fasteners; G

Roof Insulation; G

1.3 SHOP DRAWINGS

Submit insulation board layout and attachment indicating methods of attachment and spacing of fasteners on each board, transitions, tapered components, thicknesses of materials, and closure and termination conditions. Show locations of ridges, valleys, crickets, interface with, and slope to through-wall scuppers or, roof drains as indicated. Base shop drawings on verified field measurements and include verification of existing conditions. Show details for drain penetrations, parapet walls, and copings where indicated.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### 1.4 PRODUCT DATA

Include data for material descriptions, recommendations for product shelf life, requirements for cover board or coatings, and precautions for flammability and toxicity. Include data to verify compatibility of sealants with insulation.

#### 1.5 MANUFACTURER'S INSTRUCTIONS

Include field of roof, perimeter, and corner attachment requirements.

Provide a complete description of installation sequencing for each phase of the roofing system. Include weatherproofing procedures.

#### 1.6 QUALITY CONTROL

Provide certification of installer qualifications from the insulation manufacturer confirming the specific installer has the required qualifications for installing the specific roof insulation system(s) indicated.

#### 1.7 WIND RESISTANCE REQUIREMENTS

The complete roof system assembly must be rated and installed to resist wind loads indicated and validated by uplift resistance testing in accordance with applicable site-specific Factory Mutual (FM) test procedures. Coordinate with roof covering attachment requirements and submit wind resistance test certification, attachment patterns for field, perimeter, and corner roof areas along with perimeter and corner boundary dimensions.

#### 1.8 CERTIFICATIONS

Provide products certified to meet indoor air quality requirements by UL 2818(Greenguard) Gold, SCS Global Services Indoor Advantage Gold or provide certification by other third-party programs. Provide current product certification documentation from certification body.

#### 1.9 DELIVERY, STORAGE, AND HANDLING

##### 1.9.1 Delivery

Deliver materials to the project site in manufacturer's unopened and undamaged standard commercial containers bearing the following legible information:

- a. Name of manufacturer
- b. Brand designation
- c. Specification number, type, and class, as applicable, where materials are covered by a referenced specification

Deliver materials in sufficient quantity to allow continuity of the work.

##### 1.9.2 Storage and Handling

Store and handle materials in accordance with manufacturer's printed instructions. Protect from damage, exposure to open flame or other

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

ignition sources, wetting, condensation, and moisture absorption. Keep materials wrapped and separated from off-gassing materials (such as drying paints and adhesives). Do not use materials that have visible moisture or biological growth. Store in an enclosed building or trailer that provides a dry, adequately ventilated environment. Replace damaged material with new material.

#### 1.10 ENVIRONMENTAL CONDITIONS

As per manufacturer's recommendations or Government's instructions, do not install roof insulation during inclement weather or when air temperature is below 40 degrees F and interior humidity is 45 percent or greater, or when there is visible ice, frost, or moisture on the roof deck.

#### 1.11 PROTECTION

##### 1.11.1 Completed Work

Cover completed work with cover board for the duration of construction. Avoid traffic on completed work particularly when ambient temperature is above 80 degrees F. Replace crushed or damaged insulation prior to roof surface installation.

### PART 2 PRODUCTS

#### 2.1 INSULATION

##### 2.1.1 Insulation Types

Provide one, or an assembly of a maximum of three, of the following roof insulation materials. Provide roof insulation that is compatible with attachment methods for the specified insulation and roof membrane.

- a. Polyisocyanurate Board: Provide in accordance with ASTM C1289 REV A Type I, foil faced both sides. For tapered insulation at crickets use Type II, fibrous felt or glass mat membrane both sides, except minimum compressive strength of 25 pounds per square inch (psi).

##### 2.1.2 Recycled Materials

Provide thermal insulation materials containing recycled content. Unless specified otherwise, the minimum required recycled content for listed materials are:

|                                |                              |
|--------------------------------|------------------------------|
| Polyisocyanurate/polyurethane: | 9 percent recovered material |
|--------------------------------|------------------------------|

##### 2.1.3 Insulation Thickness

For insulation over steel decks, satisfy both specified R-value and minimum thickness for width of rib opening recommended in insulation manufacturer's published literature.

##### 2.1.4 Tapered Roof Insulation

One layer of the tapered roof insulation assembly must be factory tapered to a slope of not less than that indicated on the Drawings. Factory fabricate mitered joints from two diagonally cut boards or one board

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

shaped to provide required slopes.

#### 2.1.5 Cants and Tapered Edge Strips

Provide preformed cants and tapered edge strips of the same material as the roof insulation. When unavailable, provide pressure-preservative treated wood, or rigid perlite board cants and edge strips as recommended by the roofing manufacturer for the specific application, unless otherwise indicated. Face of cant strips to incline at 45 degrees with a minimum vertical height of 4 inches. Taper edge strips at a rate of one to 1 1/2 inch per foot down to approximately 1/8 inch thick.

#### 2.2 COVER BOARD

For use as a thermal barrier (underlayment), fire barrier (overlayment), or cover board for hot-mopped, torched-down, or adhesive-applied roofing membrane over roof insulation.

##### 2.2.1 Glass Mat Gypsum Roof Board

ASTM C1177/C1177M, 0 Flame Spread and 0 Smoke Developed when tested in accordance with ASTM E84, 500 psi, Class A, non-combustible, thickness as indicated on Drawings, 4 by 8 feet board size and 4 by 4 feet (1220 by 1220 mm) board size.

#### 2.3 MOISTURE CONTROL

##### 2.3.1 Vapor Retarder

###### 2.3.1.1 Asphalt Saturated Felt Base Sheet for Single Layer Application

Provide in accordance with ASTM D4601/D4601M, weighing not less than 35 pounds per 100 square feet.

##### 2.3.2 Organic Roofing

Provide in accordance with ASTM D226/D226M, Type I.

#### 2.4 FASTENERS

Provide flush-driven fasteners through flat round or hexagonal steel or plastic plates. Provide zinc-coated steel plates, flat round not less than 1 3/8 inch diameter, hexagonal not less than 28 gage. Provide high-density plastic plates, molded thermoplastic with smooth top surface, reinforcing ribs and not less than 3 inches in diameter. Fully recess fastener head into plastic plate after it is driven. Form plates to prevent dishing. Do not use bell or cup shaped plates. Provide fasteners in accordance with insulation manufacturer's recommendations for holding power when driven, or a minimum of 40 pounds each in steel deck, whichever is the higher minimum. Provide fasteners for steel or concrete decks in accordance with FM APP GUIDE (<https://www.approvalguide.com/>) for Class I roof deck construction, and spaced to withstand uplift pressure of 90 pounds per square foot.

##### 2.4.1 Fasteners for Steel Decks

Approved hardened penetrating fasteners or screws in accordance with FM 4450 and listed in FM APP GUIDE for Class I roof deck construction. Quantity and placement to withstand a minimum uplift pressure of 90 psf in

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

accordance with FM APP GUIDE.

## 2.5 FOAM ADHESIVE (OPTION)

Foam adhesive as recommended by the insulation manufacturer for adhering the insulation to concrete surfaces or underlying insulation layers and resist the required wind uplift pressure. Product must be compatible with insulation and air and water barrier materials, and with demonstrated capability to bond insulation securely to substrates without damaging insulation and substrates.

## 2.6 WOOD NAILERS

Pressure-preservative treated wood as specified in Section 06 10 00 ROUGH CARPENTRY.

# PART 3 EXECUTION

## 3.1 EXAMINATION AND PREPARATION

### 3.1.1 Surface Inspection

Ensure surfaces are clean, smooth, and dry prior to application. Check roof deck surfaces, including surfaces sloped to roof drains and outlets, for defects before starting work.

The Contractor must inspect and approve the surfaces immediately before starting installation. Prior to installing insulation, perform the following:

- b. Examine steel decks to ensure that panels are properly secured to structural members and to each other and that surfaces of top flanges are flat or slightly convex.

### 3.1.2 Surface Preparation

To correct defects and inaccuracies in roof deck surface to eliminate poor drainage from hollow or low spots, perform the following:

- a. Cover steel decks with a layer of insulation board of sufficient width to span the width of a deck rib opening, and in accordance with fire safety requirements. Secure with piercing or self-drilling, self-tapping fasteners of quantity and placement in accordance with FM APP GUIDE. Locate insulation joints parallel to ribs of deck on solid bearing surfaces only, not over open ribs.

## 3.2 INSTALLATION OF VAPOR RETARDER

Install vapor retarder in direct contact with roof deck surface. Unless otherwise specified, vapor retarder must consist of two plies of No. 15 asphalt-saturated felt, two plies of asphalt-coated glass felt, or one layer of asphalt-saturated felt base sheet. Lay vapor retarder at right angles to direction of slope. Install first ply of felt or base sheet as specified herein for the specific deck. Apply second ply of 2-ply vapor retarder system using asphalt at rate of 20 to 35 lbs per 100 square feet, applied within plus or minus 25 degrees F of EVT. Do not heat asphalt above asphalt's FBT or 525 degrees F, whichever is less. Use thermometers to check temperatures during heating and application. Completely seal side and end laps. Asphalt must be visible beyond all edges of each ply



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

as it is being installed. Lay plies free of wrinkles, buckles, creases or fishmouths. Do not walk on mopped surfaces while asphalt is sticky. Press out air bubbles to obtain complete adhesion between surfaces. At walls, eaves, rakes, and other vertical surfaces, extend vapor retarder organic felts or separate plies 9 inches, with not less than 9 inches on the substrate, and the extended portion turned back and mopped in over the top of the insulation. At roof penetrations other than walls, eaves and rakes, and vertical surfaces, extend vapor retarder or separate plies 9 inches to form a lap folded back over the edge of the insulation. Provide asphalt roof cement under the vapor retarder for at least 9 inches from walls, eaves, rakes and other penetrations.

### 3.2.1 Vapor Retarder on Steel Decks

Even mop the mechanically secured insulation surface with asphalt before installing vapor retarder. For a two-ply vapor retarder, install each sheet lapping 19 inches over the preceding sheet. Lap ends not less than 4 inches. Stagger the laps a minimum of 12 inches. Cement felts together with solid mopping of asphalt. Apply asphalt moppings at rate of 20 to 35 lbs per 100 square feet. For a vapor retarder consisting of one layer of asphalt base sheet, lap each sheet 4 inches over preceding sheet. Lap ends not less than 4 inches, and stagger laps a minimum of 12 inches. Cement base sheets together with solid mopping of asphalt.

## 3.3 INSULATION INSTALLATION

Apply insulation in two layers with staggered joints when total required thickness of insulation exceeds 1/2 inch. Lay insulation so that continuous longitudinal joints are perpendicular to direction of roofing, as specified in roofing specification section, and end joints of each course are staggered with those of adjoining courses. When using multiple layers of insulation, provide joints of each succeeding layer that are parallel and offset in both directions with respect to the layer below. Keep insulation 1/2 inch clear of vertical surfaces penetrating and projecting from roof surface. Verify required slopes to each roof drain.

### 3.3.1 Installation Using Only Mechanical Fasteners

Secure total thickness of insulation with penetrating type fasteners.

### 3.3.2 Special Precautions for Installation of Foam Insulation

#### 3.3.2.1 Polyisocyanurate Insulation

Where polyisocyanurate foam board insulation is provided, install cover board over top surface of foam board insulation. Stagger joints of insulation with respect to foam board insulation below.

### 3.3.3 Cant Strips

Where indicated, provide cant strips at intersections of roof with walls, parapets, and curbs extending above roof. Wood cant strips must bear on and be anchored to wood blocking. Fit cant strips flush to vertical surfaces. Where possible, nail cant strips to adjoining surfaces. Where cant strips are installed against non-nailable materials, install in an approved adhesive.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 3.3.4 Tapered Edge Strips

Where indicated, provide edge strips in the right angle formed by the juncture of roof and wood nailing strips that extend above the level of the roof. Install edge strips flush to vertical surfaces of wood nailing strips. Where possible, nail edge strips to adjoining surfaces. Where installed against non-nailable materials, install in an approved adhesive.

## 3.4 PROTECTION

### 3.4.1 Protection of Applied Insulation

Completely cover each day's installation of insulation with finished roofing specified in roofing specification section on same day. Phased construction is not permitted. Protect open spaces between insulation and parapets or other walls and spaces at curbs, scuttles, and expansion joints, until permanent roofing and flashing are applied. Storing, walking, wheeling, or trucking directly on insulation or on roofed surfaces is not permitted. Protect exposed edges of insulation with cutoffs at the end of each workday or whenever precipitation is imminent.

Fill all profile voids in cutoffs to prevent trapping moisture below the membrane. Remove cutoffs when work resumes.

### 3.4.2 Damaged Work and Materials

Restore work and materials that become damaged during construction to original condition or replace with new materials.

## 3.5 INSPECTION

Establish and maintain inspection procedures to assure compliance of the installed roof insulation with Contract requirements. Remove, replace, correct in an approved manner, any work found not in compliance. Quality control must include, but is not limited to, the following:

- a. Observation of environmental conditions; number and skill level of insulation workers; start and end time of work.
- b. Verification of certification, listing or label compliance with FM Data Sheets.
- c. Verification of proper storage and handling of insulation and vapor retarder materials before, during, and after installation.
- d. Inspection of vapor retarder application, including edge envelopes and mechanical fastening.
- e. Inspection of mechanical fasteners; type, number, length, and spacing.
- f. Coordination with other materials, cants, sleepers, and nailing strips.
- g. Inspection of insulation joint orientation and laps between layers, joint width and bearing of edges of insulation on deck.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

- h. Installation of cutoffs and proper joining of work on subsequent days.
- i. Continuation of complete roofing system installation to cover insulation installed same day.
- j. Verification of required slope to each roof drain.

-- End of Section --

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## SECTION 07 27 10

### BUILDING AIR BARRIER SYSTEM

**08/23**

#### PART 1 GENERAL

##### 1.1 SUMMARY

This Section specifies the construction and quality control of the installation of an air barrier system. Construct the air barrier system indicated, taking responsibility for the means, methods, and workmanship of the installation of the air barrier system. The air barrier must be contiguous and connected across all surfaces of the enclosed air barrier envelope indicated. The maximum leakage requirements of individual air barrier components and materials are specified in the other specification sections covering these items.

This section also defines the maximum allowable leakage of the final air barrier system. The workmanship must be adequate to meet the maximum allowable leakage requirements of this specification. Test the assembled air barrier system to demonstrate that the building envelope is properly sealed and insulated. Passing the air barrier system leakage test and thermography test will result in system acceptance.

##### 1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referenced within the text by the basic designation only.

#### ASTM INTERNATIONAL (ASTM)

|               |                                                                                                                      |
|---------------|----------------------------------------------------------------------------------------------------------------------|
| ASTM D4541    | (2022) Standard Test Method for Pull-Off Strength of Coatings Using Portable Adhesion Testers                        |
| ASTM E96/E96M | (2022a; E 2023) Standard Test Methods for Gravimetric Determination of Water Vapor Transmission Rate of Materials    |
| ASTM E2178    | (2021a) Standard Test Method for Determining Air Leakage Rate and Calculation of Air Permeance of Building Materials |
| ASTM E2357    | (2018) Standard Test Method for Determining Air Leakage Rate of Air Barrier Assemblies                               |

#### NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

|          |                                                                                                                                                   |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| NFPA 285 | (2023) Standard Fire Test Method for Evaluation of Fire Propagation Characteristics of Exterior Wall Assemblies Containing Combustible Components |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------|

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 1.3 DEFINITIONS

The following terms as they apply to this section:

#### 1.3.1 Air Barrier Accessory

Products designated to maintain air tightness between air barrier materials, air barrier assemblies and air barrier components, to fasten them to the structure of the building, or both (e.g., sealants, tapes, backer rods, transition membranes, fasteners, strapping, primers).

#### 1.3.2 Air Barrier Assembly

The combination of air barrier materials and air barrier accessories that are designated and designed within the environmental separator to act as a continuous barrier to the movement of air through the environmental separator.

#### 1.3.3 Air Barrier Component

Pre-manufactured elements such as windows, doors, dampers and service elements that are installed in the environmental separator.

#### 1.3.4 Air Barrier Envelope

The combination of air barrier assemblies and air barrier components, connected by air barrier accessories that are designed to provide a continuous barrier to the movement of air through an environmental separator. There may be more than one air barrier envelope in a single building. Also known as Air Barrier System.

#### 1.3.5 Air Barrier Material

A building material that is designed, tested or produced to provide the primary resistance to airflow through an air barrier assembly of a building enclosure.

#### 1.3.6 Air Barrier System

Same as AIR BARRIER ENVELOPE.

#### 1.3.7 Air Leakage Rate

The rate of airflow (CFM) driven through a unit surface area (sq.ft.) of an assembly or system by a unit static pressure difference (Pa) across the assembly. (example: 0.25 CFM/sq.ft. @ 75 Pa)

#### 1.3.8 Air Leakage

The total airflow (CFM) driven through the air barrier system by a unit static pressure difference (Pa) across the air barrier envelope. (example: 6500 CFM @ 75 Pa)

#### 1.3.9 Air Permeance

The tested rate of airflow (CFM) through a unit area (sq.ft.) of a material driven by unit static pressure difference (Pa) across the material (example: 0.004 CFM/sq.ft. @ 75 Pa) as established by ASTM E2178.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### 1.3.10 Environmental Separator

The parts of a building that separate the controlled interior environment from the uncontrolled exterior environment, or that separate spaces within a building that have dissimilar environments. Also known as the Control Layer.

#### 1.3.11 Vapor Permeance

Vapor permeance is separated into three classes based on the water vapor permeance of a material as tested via ASTM E96/E96M

Class III Vapor Barrier/Retarder 1.0 perm to 10 perm

#### 1.4 PREPARATORY PHASE OR PRECONSTRUCTION CONFERENCE

Organize pre-construction conferences between the air barrier inspector and the sub-contractors involved in the construction of, or penetration of, the air barrier system to discuss where the work of each sub-contractor begins and ends, the sequence of installation, and each sub-contractor's responsibility to ensure airtight joints, junctures, penetrations and transitions between materials. Discuss the products, and assemblies of products specified in the different sections to be installed by the different sub-contractors.

#### 1.5 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

##### SD-02 Shop Drawings

Air Barrier System Shop Drawings; G, Manufacturer produced warranted air barrier system

##### SD-03 Product Data

Air Barrier System Product Data; G

##### SD-04 Samples

Material Samples For Air Barrier System; G

##### SD-05 Design Data

Design Data And Calculations For The Air Barrier System; G

##### SD-06 Test Reports

Design Review Report; G

Testing and Inspection; G

##### SD-07 Certificates

Air Barrier Inspector; G

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## 1.6 AIR BARRIER INSPECTOR

Employ a designated Air Barrier Inspector on this project. The Air Barrier Inspector performs a Design Review, oversees quality control testing specified in these specifications, performs quality control air barrier inspection as specified, interfaces with the designer and product manufacturer's representatives to assure installation requirements are met, and verifies that the constructed work is in accordance with both the manufacturer's recommendations for products used, the content of this specification and other Contract drawings or documents. Qualifications for the Air Barrier Inspector are as follows:

- a. Training and certification as an Air Barrier Auditor from the Air Barrier Association of America (ABAA) or other third-party air barrier association.
- b. Or, provide documentation in resume format that demonstrates that the individual proposed has the experience, knowledge, skills and abilities to fulfill the above stated duties as the air barrier inspector.
- c. It is acceptable that this individual be employed by the firm performing the building pressurization test or another independent third-party entity, provided they meet the above requirements but must not be a member of the installing Contractor or firm.

Provide copies of Air Barrier Inspector qualifications 30 days after Notice to Proceed.

## 1.7 DESIGN REVIEW

Review the Contract Plans and Specifications and advise the Contracting Officer of any deficiencies that would prevent the construction of an effective air barrier system. Provide a Design Review Report individually listing each deficiency and the corresponding proposed corrective action necessary for a proper air barrier system. Provide copies of the Design Review Report not later than 14 days after approval of the Air Barrier Inspector Qualifications. Submit design data and calculations for the Air Barrier System for a manufacturer produced and warranted air barrier system.

## PART 2 PRODUCTS

### 2.1 AIR BARRIER

Provide air barrier system of compatible parts from one or several manufacturers coordinated by the Contractor or provide a single warranted system provided by a primary manufacturer. The air barrier system as part of a tested exterior wall assembly must meet the conditions of acceptance as tested in accordance with NFPA 285. Materials used for roof assembly air barrier must conform to the appropriate UL and FM wind and fire requirements for the specified roof assemblies.

If a complete air barrier system from a single manufacturer is utilized, whether warranted or not warranted, the air barrier system must conform to ASTM E2357.

Materials in the following categories, as used in the air barrier system

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

or assembly of the exterior wall system, are tested and are required to conform to ASTM E2178: Self-adhered sheet membranes, fluid applied membranes, spray polyurethane foam, mechanically fastened commercial building wrap, factory bonded membranes to sheathing, and adhesive backed commercial building wrap and accessory products.

Other materials used as an air barrier such as concrete, glass, wood, metal or gypsum board may or may not conform to ASTM E2178 but are acceptable provided that when integrated into the air barrier system or assemblies they are not subject to material or environmental induced degradation in their final produced state once incorporated in the permanent construction.

All materials used must be identifiable through manufacturer testing data or literature to be compatible with all the attached or adjoining materials or substrates used in the system.

Provide Air Barrier System Shop Drawings, Material Samples for Air Barrier System and Air Barrier System Product Data.

## PART 3 EXECUTION

### 3.1 QUALITY CONTROL

#### 3.1.1 Documentation and Reporting

Document the entire installation process on daily job site reports. These reports include information on the Installer, substrates, substrate preparation, products used, ambient and substrate temperature, the location of the air barrier installation, the results of the quality control procedures, and testing results.

#### 3.1.2 Quality Control Testing And Inspection

Conduct the following tests and inspections as applicable in the presence of the Contracting Officer during installation of the air barrier system, and submit quality control reports as indicated below.

- a. Provide a Daily Report of Observations with a copy to the Contracting Officer.
- b. Inspect to assure continuity of the air barrier system throughout the building enclosure and that all gaps are covered, the covering is structurally sound, and all penetrations are sealed allowing for no infiltration or exfiltration through the air barrier system.
- c. Inspect to assure structural support of the air barrier system to withstand design air pressures.
- d. Inspect to assure masonry surfaces receiving air barrier materials are smooth, clean, and free of cavities, protrusions and mortar droppings, with mortar joints struck flush or as required by the manufacturer of the air barrier material.
- e. Inspect and test to assure site conditions for application temperature, and dryness of substrates are within guidelines.
- f. Inspect to assure substrate surfaces are properly primed, if applicable, and in accordance with manufacturer's instructions.



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

Priming must extend at least 2 inches beyond the air barrier material to make it obvious that the primer was applied to the substrate before the air barrier material.

- g. Inspect to assure laps in materials are at least a 2-inch minimum, shingled in the correct direction or mastic applied in accordance with manufacturer's recommendations, and with no fishmouths.
- h. Inspect to assure that a roller has been used to enhance adhesion. Identify any defects such as fishmouths, wrinkles, areas of lost adhesion, and improper curing. Note the intended remedy for the deficiencies.
- i. Measure application thickness of liquid applied materials to assure that manufacturer's specifications for the specific substrate are met.
- j. Inspect to assure that the correct materials are installed for compatibility.
- k. Inspect to assure proper transitions for change in direction and structural support at gaps.
- l. Inspect to assure proper connection between assemblies (membrane and sealants) for cleaning, preparation and priming of surfaces, structural support, integrity and continuity of seal.
- m. Perform adhesion tests for fluid-applied and self-adhered air barrier membranes to assure that the manufacturer's specified adhesion strength properties are met. Determine the bond strength of coatings to substrate in accordance with ASTM D4541.
- n. Provide cohesion tests for spray polyurethane foam (SPF).
- o. Provide written test reports of all tests performed.

### 3.2 REPAIR AND PROTECTION

Upon completion of inspection, testing, sample removal and similar services, repair damaged construction and restore substrates, coatings and finishes. Protect construction exposed by or for quality control service activities, and protect repaired construction.

-- End of Section --

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## SECTION 07 27 19.01

### SELF-ADHERING AIR BARRIERS

**08/23**

#### PART 1 GENERAL

##### 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

#### AIR BARRIER ASSOCIATION OF AMERICA (ABAA)

ABAA Accreditation

Accreditation

ABAA QAP

Quality Assurance Program

#### ASTM INTERNATIONAL (ASTM)

ASTM D146/D146M

(2004; E 2012; R 2012) Sampling and Testing Bitumen-Saturated Felts and Woven Fabrics for Roofing and Waterproofing

ASTM D412

(2016; R 2021) Standard Test Methods for Vulcanized Rubber and Thermoplastic Elastomers - Tension

ASTM D570

(1998; E 2010; R 2010) Standard Test Method for Water Absorption of Plastics

ASTM D903

(1998; R 2017) Standard Test Method for Peel or Stripping Strength of Adhesive Bonds

ASTM D1876

(2008; R 2015; E 2015) Standard Test Method for Peel Resistance of Adhesives (T-Peel Test)

ASTM D4263

(1983; R 2018) Standard Test Method for Indicating Moisture in Concrete by the Plastic Sheet Method

ASTM D4541

(2022) Standard Test Method for Pull-Off Strength of Coatings Using Portable Adhesion Testers

ASTM E96/E96M

(2024) Standard Test Methods for Gravimetric Determination of Water Vapor Transmission Rate of Materials

ASTM E154/E154M

(2008a; R 2019) Standard Test Methods for Water Vapor Retarders Used in Contact with Earth Under Concrete Slabs, on Walls, or as Ground Cover

ASTM E283/E283M

(2019) Standard Test Method for

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

Determining Rate of Air Leakage Through  
Exterior Windows, Skylights, Curtain  
Walls, and Doors Under Specified Pressure  
Differences Across the Specimen

ASTM E331 (2000; R 2023) Standard Test Method for  
Water Penetration of Exterior Windows,  
Skylights, Doors, and Curtain Walls by  
Uniform Static Air Pressure Difference

ASTM E2178 (2021a) Standard Test Method for  
Determining Air Leakage Rate and  
Calculation of Air Permeance of Building  
Materials

ASTM E2357 (2018) Standard Test Method for  
Determining Air Leakage Rate of Air  
Barrier Assemblies

#### NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 285 (2023) Standard Fire Test Method for  
Evaluation of Fire Propagation  
Characteristics of Exterior Wall  
Assemblies Containing Combustible  
Components

### 1.2 RELATED REQUIREMENTS

Coordinate the requirements of Section 07 27 10 BUILDING AIR BARRIER SYSTEM and other building enclosure sections to provide a complete building air barrier system. Submit all materials, components, and assemblies of the air barrier system together as one complete submittal package.

### 1.3 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

#### SD-01 Preconstruction Submittals

Qualifications of Manufacturer;

Qualifications of Installer; G

#### SD-02 Shop Drawings

Self-adhering Air Barrier Shop Drawings; G

#### SD-03 Product Data

Self-adhering Air Barrier; G

Primers, Adhesives, and Mastics; G

Safety Data Sheets; G

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### SD-04 Samples

Self-adhering Air Barrier System Mockup; G

#### SD-06 Test Reports

Field Peel Adhesion Test; G

Flame Propagation of Wall Assemblies; G

Site Inspections and Testing; G

#### SD-08 Manufacturer's Instructions

Self-adhering Air Barrier Manufacturer's Printed Instructions; FIO

Primers, Adhesives, and Mastics; FIO

### 1.4 MISCELLANEOUS REQUIREMENTS

For self-adhering air barrier provide the following:

#### 1.4.1 Shop Drawings

Submit self-adhering air barrier shop drawings showing locations and extent of air barrier assemblies and details of all typical conditions, intersections with other building enclosure assemblies and materials, and membrane counterflashings. Show details for bridging of gaps in construction, treatment of inside and outside corners, expansion joints, methods of attachment of materials covering the self-adhered barrier without compromising the barrier. Indicate how miscellaneous penetrations such as conduit, pipes, electric boxes, brick ties, and similar items will be sealed.

#### 1.4.2 Product Data

Submit manufacturer's technical data indicating compliance with performance and environmental requirements in Part 2 of this specification. Also submit self-adhering air barrier manufacturer's printed instructions for evaluating, preparing, and treating substrates, temperature and other limitations of installation conditions, safety requirements for installation, and Safety Data Sheets. Indicate flame and smoke spread ratings for all products.

#### 1.4.3 Mockup

Provide a self-adhering air barrier system mockup. Apply product in an area designated by the Contracting Officer. Apply an area of not less than 54 square feet. Include all components specified as representative of the complete system. Notify the Contracting Officer a minimum of 48 hours prior to the test application. Select a test area representative of conditions to be covered including window or door openings, wall to ceiling transitions, flashings, and penetrations, as applicable.

#### 1.4.4 Test Reports

Submit test reports indicating that field peel-adhesion tests on all materials have been performed and the changes made, if required, in order

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

to achieve successful and lasting adhesion. Submit test reports for flame propagation of wall assemblies tested in accordance with NFPA 285.

## 1.5 DELIVERY, STORAGE, AND HANDLING

### 1.5.1 Delivery

Deliver and store materials in sufficient quantity to allow for uninterrupted flow of work. Inspect materials delivered to the site for damage and store out of weather. Deliver materials to the jobsite in their original unopened packages, clearly marked with the manufacturer's name, brand designation, description of contents, and shelf life of containerized materials. Store and handle to protect from damage.

### 1.5.2 Storage

Inspect materials delivered to the site for damage; unload and store out of weather in manufacturer's original packaging. Store only in dry locations, not subject to open flames or sparks, and easily accessible for inspection and handling. Protect stored materials from direct sunlight. Keep materials sealed and separated from absorptive materials, such as wood and insulation.

## 1.6 AIR BARRIER TESTING

Perform air barrier testing in accordance with.

## 1.7 QUALITY ASSURANCE

### 1.7.1 Qualifications of Manufacturer

Submit documentation verifying that the manufacturer of the self-adhering air barrier is currently accredited by Air Barrier Association of America (ABAA Accreditation <https://www.airbarrier.org/>).

### 1.7.2 Qualifications of Installer

Submit documentation verifying that installers of the self-adhering air barrier are currently certified in accordance with the ABAA QAP Quality Assurance Program (<https://www.airbarrier.org/qap/>).

### 1.7.3 Field Peel Adhesion Test

Perform a field peel-adhesion test on the construction mockup. Test the self-adhering air barrier for adhesion in accordance with ASTM D4541 using a Type II pull tester except use a disk that is 4 inches in diameter and cut through the membrane to separate the material attached to the dish from the surrounding material. Perform test after curing period in accordance with manufacturer's written recommendations. Record mode of failure and area which failed in accordance with ASTM D4541. Compare adhesion values with the manufacturer's established minimum values for the particular combination of material and substrate. Indicate on the inspection report whether the manufacturer's requirement has been met. Where the manufacturer has not declared a minimum adhesion value for their product and substrate combination, the inspector must record actual values.

## 1.8 PRECONSTRUCTION MEETING

Conduct a preconstruction meeting a minimum of two weeks prior to

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

commencing work specified in this specification Section. Agenda must include, at a minimum, construction and testing of mockup, sequence of construction, coordination with substrate preparation, materials approved for use, compatibility of materials, coordination with installation of adjacent and covering materials, and details of construction. Attendance is required by representatives of related trades including covering materials, substrate materials, adjacent materials, and materials and components of the air barrier system.

## 1.9 ENVIRONMENTAL CONDITIONS

### 1.9.1 Temperature

Install air barrier within the range of ambient and substrate temperatures as recommended in writing by the air barrier manufacturer. Verify that the surface to receive self-adhering air barrier is dry for a minimum of 48 hours prior to the installation of the barrier. Do not apply air barrier to damp or wet substrates. Do not apply during inclement weather or when ice, frost, surface moisture, or visible dampness is present on surfaces to be covered, or when precipitation is imminent.

### 1.9.2 Exposure to Weather and Ultraviolet Light

Protect air barrier products from direct exposure to rain, snow, sunlight, mist, and other extreme weather conditions. Replace, at no additional cost to the Government, barrier products that have been exposed to ultraviolet (sun)light longer than allowed by manufacturer's written requirements.

## PART 2 PRODUCTS

### 2.1 SELF-ADHERING AIR BARRIER

Provide minimum 0.040 inch thick self-adhering, vapor permeable, air barrier membrane consisting of a cross-laminated high density polyethylene (HDPE) film, fully coated with rubberized butyl-based or acrylic-based adhesive. Provide membrane in rolls of various widths interleaved with disposable silicone release paper. Self-adhering air barrier must exhibit no visible water leakage when tested in accordance with ASTM E331 and must perform as a liquid water drainage plane flashed to discharge to the exterior incidental condensation or water penetration. Use regular or low temperature formulation depending on site conditions, within temperature ranges specified by manufacturer.

#### 2.1.1 Physical Properties

- a. Air Permeance (ASTM E2178): In accordance with Section 07 27 10 BUILDING AIR BARRIER SYSTEM
- b. Air Leakage (ASTM E2357, ASTM E283/E283M): In accordance with Section 07 27 10 BUILDING AIR BARRIER SYSTEM.
- c. Tensile Strength (ASTM D412 die C modified): Not less than 400 psi.
- d. Tensile Elongation (ASTM D412 die C modified): Not less than 200 percent.
- e. Puncture Resistance (ASTM E154/E154M): Not less than 40 lbs.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

- f. Pliability (ASTM D146/D146M): Unaffected at minus 25 degrees F, 0.063 inch mandrel.
- g. Lap Adhesion (ASTM D1876 modified): Not less than 4.0 lbs per inch.
- h. Peel Adhesion (ASTM D903): Not less than 5.0 lbs per inch.
- i. Water Vapor Permeance (Vapor Permeable Air Barrier) (ASTM E96/E96M, desiccant method B): greater than 10.0 perms.
- j. Water Absorption (ASTM D570): Not to exceed 0.12 percent by weight.
- k. Flame propagation of wall assemblies (NFPA 285): Pass

## 2.2 PRIMERS, ADHESIVES, AND MASTICS

Provide primers, adhesives, mastics and other accessory materials as recommended in writing by the manufacturer of the self-adhering air barrier for adequate bonding to each type of substrate. Adhesive must be butyl- or acrylic-based.

## 2.3 SHEET METAL FLASHING

Provide as specified in Section 07 60 00 FLASHING AND SHEET METAL.

## 2.4 JOINT SEALANTS

Provide as specified in Section 07 92 00 JOINT SEALANTS. Verify compatibility with adjacent products that are or will be in contact with one another.

# PART 3 EXECUTION

## 3.1 EXAMINATION

Before installing air barrier, examine substrates, areas, and conditions under which air barrier assemblies will be applied, with Installer present, for compliance with requirements. Ensure the following conditions are met:

- a. Surfaces are sound, dry, even, and free of oil, grease, dirt, excess mortar or other contaminants.
- b. Concrete surfaces are cured and dry, smooth without large voids, spalled areas or sharp protrusions.
- c. Verify substrate is visibly dry and free of moisture. Test for capillary moisture by plastic sheet method in accordance with ASTM D4263 and take suitable measures until substrate passes moisture test.
- d. Verify sealants used in sheathing are compatible with membrane proposed for use. Perform field peel adhesion test on materials to which sealants are adhered.

## 3.2 PREPARATION

Clean, prepare, and treat substrate in accordance with manufacturer's written instructions. Ensure clean, dust-free, and dry substrate for air

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

barrier application.

- a. Prime masonry and concrete substrates with conditioning primer.
- b. Prime gypsum sheathing an adequate number of coats to achieve required bond, with adequate drying time between coats.
- c. Prime wood, metal, and painted substrates with primer.
- d. Prepare, treat, and seal vertical and horizontal surfaces at terminations and penetrations through air barrier and at protrusions.

### 3.3 INSTALLATION

#### 3.3.1 Installation of Self-adhering Air Barrier

Install materials in accordance with manufacturer's recommendations and the following:

- a. Apply primer at rate recommended by manufacturer prior to membrane installation. Allow primer to dry completely before membrane application. Apply as many coats as necessary for proper adhesion.
- b. When membrane is properly positioned, press into place and roll membrane with roller immediately after placement.
- c. Apply membrane sheets to shed water naturally without interception by a sheet edge, unless that edge is sealed with permanently flexible termination mastic.
- d. Position subsequent sheets of membrane applied above so that membrane overlaps the membrane sheet below by a minimum of 2-1/2 inches, unless greater overlap is recommended by manufacturer. Roll into place with roller.
- e. Make side laps a minimum of 2-1/2 inches and end laps a minimum of 5 inches, unless greater overlap is recommended by manufacturer. Roll seams with roller.
- f. Roll membrane to adhere to substrate. Cover corners and joints with two layers of reinforcement by first applying a 12 inch width of membrane centered along the axis. Flash drains and projections with a second ply of membrane for a distance of 6 inches from the drain or projection.
- g. Seal around all penetrations through the air barrier resulting from pipes, vents, conduit, electrical fixtures, structural members, or other construction passing through it. Seal with termination mastic, extruded silicone sealant, membrane counterflashing or other sealing methods in accordance with manufacturer's written recommendations.
- h. Continuously connect the air barrier between walls, roof, floor and below grade assemblies to form a continuous integrated air barrier system around the entire building enclosure. Extend the air barrier membrane into rough openings such as doors, windows, louvers, and other exterior penetrations. Seal edges of barrier at junctures with rough openings.
- i. At changes in substrate plane, provide transition material (e.g., bead



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

of sealant, mastic, extruded silicone sealant, membrane counterflashing or other material recommended by manufacturer) under membrane to eliminate all sharp 90-degree inside corners and to make a smooth transition from one plane to another.

- j. Provide mechanically fastened non-corrosive metal sheet to span gaps in substrate plane and to make a smooth transition from one plane to the other. Continuously support membrane with substrate.
- k. At deflection and control joints, provide backup for the membrane to accommodate anticipated movement.
- l. At expansion and seismic joints provide transition to the joint assemblies.
- m. Apply a bead or trowel coat of mastic along membrane seams at reverse lapped seams, rough cuts, and as recommended by the manufacturer.
- n. At end of each working day, seal top edge of membrane to substrate with termination mastic.
- o. Do not allow materials to come in contact with chemically incompatible materials.
- p. Counterflash upper edge of thru-wall flashing and air barrier. Counter flashing is specified in Section 07 60 00 FLASHING AND SHEET METAL. See applicable wall-type Sections for thru-wall flashing requirements.

### 3.4 FIELD QUALITY CONTROL

#### 3.4.1 Site Inspections and Testing

Provide site inspections and testing in accordance with the manufacturer's instructions, Section 07 27 10 BUILDING AIR BARRIER SYSTEM and this section.

- a. Conduct inspections and testing at 5, 50, and 95 percent completion of this scope of work. Forward written site inspections and testing reports to the Contracting Officer within five working days of the inspection and test being performed.
- b. If inspections reveal defects, promptly remove and replace defective work at no additional expense to the Government.

#### 3.5 FIELD PEEL ADHESION TEST

Conduct in accordance with test protocol indicated in Part 1, paragraph FIELD PEEL ADHESION TEST.

### 3.6 PROTECTION AND CLEANING

#### 3.6.1 Protection

##### 3.6.1.1 Adjacent Surfaces

Protect exposed adjacent surfaces that could be damaged by primers and adhesives associated with air barrier membrane. Provide protection during application and the remainder of construction in accordance with

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

manufacturer's written instructions.

#### 3.6.1.2 The Air Barrier Assembly

Protect finished portions of the air barrier assembly from damage during ongoing application and throughout the remainder of the construction period in accordance with manufacturer's written instructions. Coordinate timing of installation of materials that will cover the air barrier membrane to ensure the exposure period does not exceed that recommended by the air barrier manufacturer's written installation instructions. Remove and replace, at no additional cost to the Government, membrane products that exceed the manufacturer's allowed exposure limits.

#### 3.6.2 Cleaning

Clean spillage and soiling from adjacent construction using cleaning agents and procedures recommended by manufacturer of affected construction and as acceptable to the primary material manufacturer.

-- End of Section --

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## SECTION 07 27 26

### FLUID-APPLIED MEMBRANE AIR BARRIERS

**08/23**

#### PART 1 GENERAL

##### 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

#### AIR BARRIER ASSOCIATION OF AMERICA (ABAA)

ABAA Accreditation

Accreditation

ABAA QAP

Quality Assurance Program

#### ASTM INTERNATIONAL (ASTM)

ASTM C1305/C1305M

(2016) Standard Test Method for Crack Bridging Ability of Liquid-Applied Waterproofing Membrane

ASTM D412

(2016; R 2021) Standard Test Methods for Vulcanized Rubber and Thermoplastic Elastomers - Tension

ASTM D4263

(1983; R 2018) Standard Test Method for Indicating Moisture in Concrete by the Plastic Sheet Method

ASTM D4541

(2022) Standard Test Method for Pull-Off Strength of Coatings Using Portable Adhesion Testers

ASTM D5590

(2017; R 2021) Standard Test Method for Determining the Resistance of Paint Films and Related Coatings to Fungal Defacement by Accelerated Four-Week Agar Plate Assay

ASTM E96/E96M

(2024) Standard Test Methods for Gravimetric Determination of Water Vapor Transmission Rate of Materials

ASTM E283/E283M

(2019) Standard Test Method for Determining Rate of Air Leakage Through Exterior Windows, Skylights, Curtain Walls, and Doors Under Specified Pressure Differences Across the Specimen

ASTM E331

(2000; R 2023) Standard Test Method for Water Penetration of Exterior Windows, Skylights, Doors, and Curtain Walls by Uniform Static Air Pressure Difference

ASTM E2178

(2021a) Standard Test Method for

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

Determining Air Leakage Rate and  
Calculation of Air Permeance of Building  
Materials

ASTM E2357

(2018) Standard Test Method for  
Determining Air Leakage Rate of Air  
Barrier Assemblies

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 285

(2023) Standard Fire Test Method for  
Evaluation of Fire Propagation  
Characteristics of Exterior Wall  
Assemblies Containing Combustible  
Components

## 1.2 RELATED REQUIREMENTS

Coordinate the requirements of Section 07 27 10 BUILDING AIR BARRIER SYSTEM and other building enclosure sections to provide a complete building air barrier system. Submit all materials, components and assemblies of the air barrier system together as one complete submittal package.

## 1.3 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

### SD-01 Preconstruction Submittals

Qualifications of Manufacturer; G

Qualifications of Installer; G

### SD-02 Shop Drawings

Fluid-Applied Membrane Air Barrier Shop Drawings; G

### SD-03 Product Data

Fluid-Applied Membrane Air Barrier; G

Transition Membrane; G

Primers, Adhesives, and Mastics; G

Reinforcement; G

Safety Data Sheets; G

### SD-04 Samples

Fluid-Applied Membrane Air Barrier Mockup; G

### SD-06 Test Reports

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

Capillary Moisture Test; FIO

Field Peel Adhesion Test; FIO

Flame Propagation of Wall Assemblies; FIO

Site Inspections Reports; G

#### SD-08 Manufacturer's Instructions

Fluid-Applied Membrane Air Barrier Manufacturer's Printed Instructions; G

Primers, Adhesives, and Mastics Printed Manufacturer's Instructions; G

### 1.4 MISCELLANEOUS REQUIREMENTS

For fluid-applied membrane air barriers provide the following:

#### 1.4.1 Shop Drawings

Submit fluid-applied membrane air barrier shop drawings showing locations and extent of barrier assemblies, transition membranes, details of all typical conditions, intersections with other envelope assemblies and materials, and membrane counterflashings. Show details for bridging of gaps in construction, treatment of inside and outside corners, expansion joints, methods of attachment of materials covering the self-adhered barrier without compromising the barrier. Indicate how miscellaneous penetrations such as conduit, pipes, electric boxes, brick ties, and similar items will be sealed.

#### 1.4.2 Product Data

Submit manufacturer's technical data indicating compliance with performance and environmental requirements. Submit fluid-applied membrane air barrier manufacturer's printed instructions for evaluating, preparing, and treating substrates, temperature and other limitations of installation conditions, safety requirements for installation, and Safety Data Sheets. Also, submit primers, adhesives and mastics printed manufacturer's instructions for installation conditions and safety requirements. Indicate flame and smoke spread ratings for all products.

#### 1.4.3 Mockup

Provide a mockup of the fluid-applied membrane air barrier. Apply product in an area designated by the Contracting Officer. Apply an area of not less than 54 square feet. Include all components specified as representative of the complete system. Notify the Contracting Officer a minimum of 48 hours prior to the test application. Select a test area representative of conditions to be covered including window or door openings, wall to ceiling transitions, flashings, and penetrations, as applicable.

#### 1.4.4 Test Reports

Submit test reports indicating that capillary moisture tests and field peel adhesion tests on all substrate materials have been performed and the changes made, if required, in order to achieve successful and lasting

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

adhesion. Submit test reports for flame propagation of wall assemblies tested in accordance with NFPA 285.

## 1.5 DELIVERY, STORAGE, AND HANDLING

### 1.5.1 Delivery

Deliver and store materials in sufficient quantity to allow for uninterrupted flow of work. Inspect materials delivered to the site for damage and store out of weather. Deliver materials to the jobsite in their original unopened packages, clearly marked with the manufacturer's name, brand designation, description of contents, and shelf life of containerized materials. Store and handle to protect from damage.

### 1.5.2 Storage

Inspect materials delivered to the site for damage; unload and store out of weather in manufacturer's original packaging. Store only in dry locations, not subject to open flames or sparks, and easily accessible for inspection and handling. Protect stored materials from direct sunlight.

## 1.6 CAPILLARY MOISTURE TEST

Perform a capillary moisture test by plastic sheet method in accordance with ASTM D4263 on the construction mockup and substrate materials. Perform test after curing period as recommended by the air barrier manufacturer. Record mode of failure and area which failed in accordance with ASTM D4263. Once the air barrier material manufacturer has established a minimum adhesion or moisture level for the product on the particular substrate, indicate on the inspection report whether this requirement has been met. Where the manufacturer has not declared a minimum adhesion or moisture value for their product and substrate combination, the inspector must record actual values.

## 1.7 FIELD PEEL ADHESION TEST

Perform a field peel adhesion test on a construction mockup. Test the applied product for adhesion in accordance with manufacturer's recommendations. Perform test after curing period recommended by the manufacturer. Record mode of failure and area which failed in accordance with ASTM D4541. When the manufacturer has established a minimum adhesion level for the product on the particular substrate, the inspection report must indicate whether this requirement has been met. Where the manufacturer has not declared a minimum adhesion value for their product/substrate combination, the inspector must record actual values.

## 1.8 AIR BARRIER TESTING

Perform air barrier testing in accordance with Section 07 27 10 BUILDING AIR BARRIER SYSTEM.

## 1.9 QUALITY ASSURANCE

### 1.9.1 Qualifications of Manufacturer

Submit documentation verifying that manufacturer of fluid-applied membrane air barrier is currently accredited by the Air Barrier Association of America (ABAA Accreditation <https://www.airbarrier.org/>).

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 1.9.2 Qualifications of Installer

Submit documentation verifying that installers of the fluid-applied membrane air barrier are currently certified in accordance with the ABAA QAP Quality Assurance Program (<https://www.airbarrier.org/qap/>).

### 1.10 PRECONSTRUCTION MEETING

Conduct a preconstruction meeting a minimum of two weeks prior to commencing work specified in this specification Section. Agenda must include, at a minimum, construction and testing of construction mock-up, sequence of construction, coordination with substrate preparation, materials approved for use, compatibility of materials, coordination with installation of adjacent and covering materials, and details of construction. Attendance is required by representatives of related trades including covering materials, substrate materials, adjacent materials, and materials and components of the fluid-applied membrane air barrier.

### 1.11 ENVIRONMENTAL CONDITIONS

#### 1.11.1 Temperature

Install fluid-applied membrane air barrier within the range of ambient and substrate temperatures as recommended in writing by the fluid-applied membrane air barrier manufacturer. Do not apply fluid-applied membrane air barrier to a damp or wet substrate. Do not apply during inclement weather or when ice, frost, surface moisture, or visible dampness is present on surfaces to be covered, or when precipitation is imminent.

#### 1.11.2 Exposure to Weather

Protect fluid-applied membrane air barrier products from direct exposure to rain, snow, sunlight, mist, and other extreme weather conditions. Replace, at no additional cost to the Government, barrier products that have been exposed to ultraviolet (sun)light longer than allowed by manufacturer's written requirements.

## PART 2 PRODUCTS

### 2.1 FLUID-APPLIED MEMBRANE AIR BARRIER

Provide a fluid-applied, vapor permeable, air barrier. This barrier must exhibit no visible water leakage when tested in accordance with ASTM E331 and must perform as a liquid water drainage plane with thru-wall flashing to discharge incidental condensation and water penetration to the exterior of the building enclosure. Provide products suitable for use within temperature ranges specified by manufacturer for the location of the project.

#### 2.1.1 Physical Properties

- a. Air Permeance (ASTM E2178): in accordance with Section 07 27 10 BUILDING AIR BARRIER SYSTEM
- b. Air Leakage (ASTM E2357, ASTM E283/E283M): in accordance with Section 07 27 10 BUILDING AIR BARRIER SYSTEM.
- c. Water Vapor Permeance (Vapor Permeable Membrane) (ASTM E96/E96M, desiccant method A): 10.0 perms.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

- d. Pull Adhesion (ASTM D4541): Not less than 15 psi.
- e. Elongation (ASTM D412): Not less than 300 percent.
- f. Low temperature Flexibility and Crack Bridging (ASTM C1305/C1305M):  
Pass at minus 15 degrees F.
- g. Solids by Volume: minimum 50 percent.
- h. Flame propagation of wall assemblies (NFPA 285): Pass
- i. Resistance to Mold, Mildew and Fungal Growth (ASTM D5590): 0, No growth.

## 2.2 PRIMERS, ADHESIVES, AND MASTICS

Provide primers, adhesives, mastics, sealants and other accessories as recommended by manufacturer of fluid-applied membrane air barrier for a complete installation.

## 2.3 TRANSITION MEMBRANE

Provide as recommended by the air barrier manufacturer.

## 2.4 SHEET METAL FLASHING

Provide as specified in Section 07 60 00 FLASHING AND SHEET METAL.

## 2.5 JOINT SEALANTS

Provide as specified in Section 07 92 00 JOINT SEALANTS.

## 2.6 REINFORCEMENT

Provide fiberglass mesh tape, or fluid-applied air barrier manufacturer's approved comparable equivalent product, reinforcement at seams, edges, projections and penetrations. Reinforce all joints exceeding 1/4 inch with fiberglass mesh.

# PART 3 EXECUTION

## 3.1 EXAMINATION

Before installing fluid-applied membrane air barrier, examine substrates, areas, and conditions under which fluid-applied membrane air barrier assemblies will be applied, with installer present, for compliance with requirements. Ensure the following conditions are met:

- a. Surfaces are sound, dry, even, and free of oil, grease, dirt, excess mortar or other contaminants detrimental to the adhesion of the membranes.
- b. Concrete and masonry surfaces are cured and dry, smooth without large voids, spalled areas or sharp protrusions. Do not proceed with installation until after minimum concrete curing period recommended by fluid-applied membrane air barrier manufacturer.
- c. Fill voids, gaps and spalled areas in substrate to provide an even



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

plane. Strike masonry joints full flush.

- d. Verify substrate is visibly dry and free of moisture. Test for capillary moisture by plastic sheet method in accordance with ASTM D4263 and take suitable measures until substrate passes moisture test.
- e. Verify sealants used in substrates, and in joints between substrates, are compatible with fluid-applied membrane air barrier.

### 3.2 PREPARATION

Clean, prepare, and treat substrate in accordance with manufacturer's written instructions. Ensure clean, dust-free, and dry substrate for fluid-applied membrane air barrier application.

- a. Remove dust, dirt and other contaminants from joints and cracks before coating surfaces.
- b. Prepare, treat, and seal vertical and horizontal surfaces at terminations and penetrations through fluid-applied membrane air barrier.
- c. At changes in substrate plane, provide transition material (bead of sealant, mastic, extruded silicone sealant, membrane counterflashing or other material recommended by manufacturer) under transition membrane to eliminate all sharp 90-degree inside corners and to make a smooth transition from one plane to another.
- d. Provide mechanically fastened non-corrosive metal sheet to span gaps in substrate plane and to make a smooth transition from one plane to the other. Continuously support membrane with substrate.
- e. For exterior sheathing substrates, ensure that exterior sheathing is stabilized, with corners and edges fastened with appropriate screws. Treat all joints in accordance with the air barrier manufacturer's instructions prior to application of air barrier material. Allow sufficient time for joint treatments to fully cure before application of transition membranes and fluid-applied membrane air barrier.
- f. For concrete and masonry substrates, fill all voids and holes, particularly in mortar joints, with non-shrinking grout.
- g. Mask off and cover adjacent surfaces to protect from spillage and overspray.

### 3.3 INSTALLATION

#### 3.3.1 Installation of Transition Membrane

Install transition membrane materials in accordance with the details on the drawings, manufacturer's written recommendations, and the following:

- a. Install transition membrane at all required locations (such as at any voids, gaps, or interstitial spaces that the fluid-applied air barrier cannot adequately protect) prior to installation of the fluid-applied membrane air barrier.
- b. Verify transition membrane is fully adhered to substrate and that its

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

surface is clean, dry and wrinkle free prior to installation of the fluid-applied membrane air barrier.

- c. Verify transition membrane completely covers all transition areas and provides continuity of the finished fluid-applied membrane air barrier without gaps or cracks.

### 3.3.2 Installation of Flashing

Counterflash upper edge of thru-wall flashing and fluid-applied air barrier. Counter flashing is specified in Section 07 60 00 FLASHING AND SHEET METAL. See applicable wall-type Sections for thru-wall flashing requirements.

### 3.3.3 Installation of Fluid-Applied Membrane Air Barrier

Install materials in accordance with manufacturer's recommendations and the following:

- a. Apply fluid-applied membrane air barrier in single or dual coat application by spray or roller. Apply fluid-applied membrane air barrier within manufacturer's recommended temperature range for application.
- b. Apply fluid-applied membrane air barrier in manner and at rate and wet film thickness recommended by manufacturer to yield a finished dry film thickness of not less than 90 mils or as otherwise required by the manufacturer for the application substrate material and surface roughness.
- c. Apply fluid-applied membrane air barrier around all penetrations ensuring a complete and continuous air barrier. Lap fluid-applied membrane air barrier a minimum of 3 inch over transition membrane to seal leading edge.
- d. Seal membrane terminations, heads of mechanical fasteners, masonry tie fasteners, around penetrations, HVAC assemblies, plumbing and electrical assemblies, doors, windows, louvers, and other assemblies penetrating the fluid-applied membrane air barrier with a termination sealant recommended by the fluid-applied membrane air barrier manufacturer.
- e. Notify the Contracting Officer and Testing Agency upon completion of fluid-applied membrane air barrier installation. Air barrier materials and assemblies must remain exposed until tested and inspected in accordance with the ABAA protocol.
- f. Do not allow materials to come in contact with chemically incompatible materials.

### 3.3.4 Installation of Reinforcement

Install reinforcement at projections, corners, joints, and penetrations where applicable.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 3.4 FIELD QUALITY CONTROL

#### 3.4.1 Site Inspections and Testing

Provide site inspections and testing in accordance with the manufacturer's instructions, Section 07 27 10 BUILDING AIR BARRIER SYSTEM and this Section.

- a. Conduct inspections and testing at 5, 50, and 95 percent completion of this scope of work. Forward written inspection reports to the Contracting Officer within five working days of the inspection and test being performed.
- b. If the inspections reveal defects, promptly remove and replace defective work at no additional expense to the Government.

### 3.5 PROTECTION AND CLEANING

#### 3.5.1 Protection

Protect fluid-applied membrane air barrier assemblies from damage during application and remainder of construction in accordance with manufacturer's written instructions.

Coordinate installation, testing, and inspection procedures to ensure exposure period does not exceed that recommended by the product manufacturer. Remove and replace, at no additional cost to the government, membrane products that exceed manufacturer's allowed exposure limits.

#### 3.5.2 Cleaning of Adjacent Surfaces

Clean excess product from adjacent construction using cleaning agents and procedures as recommended in writing by the manufacturer of each type of affected construction and as acceptable to same.

### 3.6 CLEANUP OF SPILLS

Conduct cleanup of uncured product spillage in accordance with manufacturer's written safe handling instructions.

-- End of Section --

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## SECTION 07 41 63

### FABRICATED ROOF PANEL ASSEMBLIES

**11/16**

#### PART 1 GENERAL

##### 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

#### ALUMINUM ASSOCIATION (AA)

AA ADM (2020) Aluminum Design Manual

#### AMERICAN IRON AND STEEL INSTITUTE (AISI)

AISI D100 (2017) Cold-Formed Steel Design Manual

#### AMERICAN SOCIETY OF CIVIL ENGINEERS (ASCE)

ASCE 7-16 (2017; Errata 2018; Supp 1 2018) Minimum Design Loads and Associated Criteria for Buildings and Other Structures

#### AMERICAN WELDING SOCIETY (AWS)

AWS A5.1/A5.1M (2012) Specification for Carbon Steel Electrodes for Shielded Metal Arc Welding

AWS D1.1/D1.1M (2020; Errata 1 2021) Structural Welding Code - Steel

#### ASTM INTERNATIONAL (ASTM)

ASTM A653/A653M (2023) Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process

ASTM A780/A780M (2020) Standard Practice for Repair of Damaged and Uncoated Areas of Hot-Dip Galvanized Coatings

ASTM B117 (2019) Standard Practice for Operating Salt Spray (Fog) Apparatus

ASTM B209 (2014) Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate

ASTM B209M (2014) Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate (Metric)

ASTM B659 (1990; R 2021) Standard Guide for Measuring Thickness of Metallic and Inorganic Coatings

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

|                   |                                                                                                                                   |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| ASTM C553         | (2013; R 2019) Standard Specification for Mineral Fiber Blanket Thermal Insulation for Commercial and Industrial Applications     |
| ASTM C612         | (2014; R 2019) Standard Specification for Mineral Fiber Block and Board Thermal Insulation                                        |
| ASTM C920         | (2018) Standard Specification for Elastomeric Joint Sealants                                                                      |
| ASTM C1396/C1396M | (2017) Standard Specification for Gypsum Board                                                                                    |
| ASTM D522/D522M   | (2017) Mandrel Bend Test of Attached Organic Coatings                                                                             |
| ASTM D523         | (2014; R 2018) Standard Test Method for Specular Gloss                                                                            |
| ASTM D714         | (2002; R 2017) Standard Test Method for Evaluating Degree of Blistering of Paints                                                 |
| ASTM D822         | (2013; R 2018) Filtered Open-Flame Carbon-Arc Exposures of Paint and Related Coatings                                             |
| ASTM D968         | (2022) Standard Test Methods for Abrasion Resistance of Organic Coatings by Falling Abrasive                                      |
| ASTM D1056        | (2020) Standard Specification for Flexible Cellular Materials - Sponge or Expanded Rubber                                         |
| ASTM D1308        | (2002; R 2013) Effect of Household Chemicals on Clear and Pigmented Organic Finishes                                              |
| ASTM D1667        | (2022) Standard Specification for Flexible Cellular Materials - Poly (Vinyl Chloride) Foam (Closed-Cell)                          |
| ASTM D2244        | (2021) Standard Practice for Calculation of Color Tolerances and Color Differences from Instrumentally Measured Color Coordinates |
| ASTM D2247        | (2015; R 2020) Standard Practice for Testing Water Resistance of Coatings in 100% Relative Humidity                               |
| ASTM D2794        | (1993; R 2019) Standard Test Method for Resistance of Organic Coatings to the Effects of Rapid Deformation (Impact)               |
| ASTM D3363        | (2005; E 2011; R 2011; E 2012) Film Hardness by Pencil Test                                                                       |

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

|                                                                   |                                                                                                                                         |
|-------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| ASTM D4214                                                        | (2007; R 2015) Standard Test Method for Evaluating the Degree of Chalking of Exterior Paint Films                                       |
| ASTM E84                                                          | (2023) Standard Test Method for Surface Burning Characteristics of Building Materials                                                   |
| ASTM E119                                                         | (2022) Standard Test Methods for Fire Tests of Building Construction and Materials                                                      |
| ASTM E136                                                         | (2022) Standard Test Method for Assessing Combustibility of Materials Using a Vertical Tube Furnace at 750 Degrees C                    |
| ASTM E1592                                                        | (2017) Standard Test Method for Structural Performance of Sheet Metal Roof and Siding Systems by Uniform Static Air Pressure Difference |
| ASTM E2140                                                        | (2001; R 2023) Standard Test Method for Water Penetration of Metal Roof Panel Systems by Static Water Pressure Head                     |
| ASTM G152                                                         | (2013; R 2021) Standard Practice for Operating Open Flame Carbon Arc Light Apparatus for Exposure of Nonmetallic Materials              |
| ASTM G153                                                         | (2013; R 2021) Standard Practice for Operating Enclosed Carbon Arc Light Apparatus for Exposure of Nonmetallic Materials                |
| FM GLOBAL (FM)                                                    |                                                                                                                                         |
| FM 4471                                                           | (2010) Class I Panel Roofs                                                                                                              |
| FM 4474                                                           | (2014) Evaluating the Simulated Wind Uplift Resistance of Roof Assemblies Using Static Positive and/or Negative Differential Pressures  |
| METAL BUILDING MANUFACTURERS ASSOCIATION (MBMA)                   |                                                                                                                                         |
| MBMA RSDM                                                         | (2012) Metal Roofing Systems Design Manual                                                                                              |
| NATIONAL ASSOCIATION OF ARCHITECTURAL METAL MANUFACTURERS (NAAMM) |                                                                                                                                         |
| NAAMM AMP 500                                                     | (2006) Metal Finishes Manual                                                                                                            |
| NATIONAL ROOFING CONTRACTORS ASSOCIATION (NRCA)                   |                                                                                                                                         |
| NRCA 0429                                                         | (2022) The NRCA Roofing Manual: Architectural Metal Flashing, Condensation and Air Leakage Control and Reroofing                        |

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

NRCA RoofMan (2020) The NRCA Roofing Manual

SHEET METAL AND AIR CONDITIONING CONTRACTORS' NATIONAL ASSOCIATION  
(SMACNA)

SMACNA 1793 (2012) Architectural Sheet Metal Manual,  
7th Edition

SOCIETY FOR PROTECTIVE COATINGS (SSPC)

SSPC PS 9.01 (1982; E 2004) Cold-Applied Asphalt Mastic  
Painting System with Extra-Thick Film

UNDERWRITERS LABORATORIES (UL)

UL 580 (2006; Reprint Mar 2019) UL Standard for  
Safety Tests for Uplift Resistance of Roof  
Assemblies

## 1.2 DEFINITIONS

Fabricated Roof Panel Assembly: Metal roof and liner panels, attachment system components, miscellaneous metal framing, thermal insulation, and accessories shop-fabricated or field-assembled for a complete weathertight roofing system.

## 1.3 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

### SD-01 Preconstruction Submittals

Qualification of Manufacturer; G

Qualification of Installer; G

Qualifications for Welding; G

Work Plan; G

On-Site Inspection and Acceptance Procedure; G

### SD-02 Shop Drawings

Roofing Panels; G

Flashing and Accessories; G

Gutter/Downspout Assembly; G

### SD-03 Product Data

Sustainable Acquisition; G.

Coil Stock; G

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

Factory Color Finish; G  
Sub-girts and Formed Shapes; G  
Closure Materials; G  
Insulation; G  
Pressure-Sensitive Tape; G  
Sealants and Caulking; G  
Rated Wall Assembly; G  
Accessories; G

SD-04 Samples

Coil Stock; G  
Roofing Panels; G  
Fasteners; G  
Metal Closure Strips; G  
Insulation; G  
Manufacturer's Color Charts and Chips; G

SD-05 Design Data

Wind Design Analysis

SD-06 Test Reports

Coatings and Base Metals of Metal Roofing  
Factory Finish and Color Performance Requirements

SD-07 Certificates

Coil Stock  
Fasteners  
Safety Data Sheets  
Coating Physical Properties:

SD-08 Manufacturer's Instructions

Installation of Roof Panel Assemblies

SD-11 Closeout Submittals

Warranty



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

Information Form and Placard; G

Manufacturer's Field Inspection Reports

Application Instructions

Date of Installation Wall-Mounted Placard; G

20-year "No-Dollar-Limit" Warranty for Labor and Materials; G

#### 1.4 QUALITY CONTROL

##### 1.4.1 Preroofing Conference

After submittals are received and approved but before roofing and insulation work, including associated work, is performed, the Contracting Officer will hold a preroofing conference (including the roofing supplier, the erector, the Construction Rep., and the roofing manufacturer's technical representative) within 45 days after contract award to review the following:

a. The drawings and specifications:

- (1) Fabrication and installation drawings for the following items are to indicate completely dimensioned structural frame and erection layouts, openings in the roof, special framing details and construction details at corners, ridges, eaves, building intersections, curbs and flashing, location and type of mastic and metal filler strips, location and erection of flashing and gutter/downspout assembly.
- (2) Installation of roof panel assemblies
- (3) Roofing panels, submit sample 12 inches long by actual panel width
- (4) Flashing and accessories, submit sample 10 inches long of each type
- (5) Gutter/downspout assembly

Prepare written minutes of the pre-roofing conference and submit to the Contracting Officer.

Submit certification from the coil stock manufacturer or supplier that the machinery used will form the provided coil stock without warping, waviness, or rippling that is not a part of the panel profile, and without damage, abrasion or marring of the finish coating, and certification of conformance with the standards specified herein. Submit a sample 12 inches long by the actual panel width.

Submit the manufacturer's standard color charts and chips, approximately 4 by 4 inches, showing the full range of colors, textures and patterns available for roof panels with the factory color finish.

Submit factory finish and color performance requirements verified by an independent testing agency.

Submit a wind design analysis from the manufacturer including wind speed, exposure category, coefficient, importance factor. Designate a type of facility, negative pressures for each zone, methods and requirements of

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

attachment. Include a roof plan delineating dimensions and attachment patterns for each zone. Include a signed and sealed wind design analysis with a Licensed project engineer, in the geographic area where the construction will take place.

(9) Fire Rating Test Report

b. Qualifications including:

(1) Qualification of Manufacturer

(2) Qualification of Installer

(3) Qualifications for Welding

c. Submit an on-site inspection and acceptance procedure of the roofing substrate and pertinent structural details relating to the roofing system, including:

(1) Safety Data Sheets

(2) Sub-girts and Formed Shapes

(3) Closure Materials

(4) Insulation

(5) Pressure-Sensitive Tape

(6) Sealants and Caulking

(7) Accessories

d. Submit a work plan for coordination of the various trades involved in providing the roofing system and other components secured to the roofing.

Include detailed application instructions and standard manufacturer drawings altered as required by these specifications. Explicitly identify in writing the differences between the manufacturer's instructions and the specified requirements.

e. Safety requirements and to establish a mutual understanding of the metal roof system contract requirements.

f. Submit manufacturer's data indicating the percentage of recycle material in roofing panels to verify sustainable acquisition compliance.

#### 1.4.2 Manufacturer's Technical Representative

Ensure the representative has authorization from the manufacturer to approve field changes and is thoroughly familiar with the products and installations in the geographical area where construction will take place. The representative must be on-site during all roof system installation work.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### 1.4.3 Qualification of Manufacturer

Guarantee the metal roof panel system manufacturer possesses the following:

- a. A minimum of five years of experience in manufacturing metal roof system and accessory products.
- b. Engineering services of an authorized engineer; currently licensed in the geographical area where construction will take place, having a minimum of four years of experience as an engineer knowledgeable in roof wind design analysis, protocols and procedures for the MBMA RSDM; ASCE 7-16, UL 580 and FM 4474 FM wind design guide for metal roof systems.
- c. Certified engineering calculations using the products submitted for wind uplift requirements in accordance with FM 4474 and ASCE 7-16.

#### 1.4.4 Qualification of Installation Contractor

Confirm that the installation contractor is approved and certified by the roofing panel manufacturer before installing the metal roofing system.

#### 1.4.5 Qualifications for Welding

Provide certification of welding procedures conforming to AWS A5.1/A5.1M and AWS D1.1/D1.1M

#### 1.4.6 Single-Source

Obtain each type of metal roof and liner panels, clips, closures and other accessories from the standard products of the single-source manufacturer to ensure these items operate as a complete system for the intended use.

### 1.5 DELIVERY, STORAGE, AND HANDLING

Deliver components, sheets, metal roof panels, and other manufactured items, handling them in a manner to prevent damage or deformation; package metal roof panels for protection during transportation and handling.

Unload, store, and erect metal roof panels in a manner to prevent bending, warping, twisting, and surface damage.

Stack metal roof panels on platforms or pallets, covered with a suitable weather-tight and ventilated covering; store metal roof panels to ensure dryness. Do not store metal roof panels in contact with other materials that might cause staining, denting, or other surface damage.

Protect the strippable protective covering on metal roof panels from exposure to sunlight and high humidity, except to extent necessary for the period of metal roof panel installation.

Protect foam-plastic insulation as follows:

- a. Do not expose the foam-plastic insulation to sunlight, except to extent necessary for period of installation and concealment.
- b. Protect the foam-plastic insulation against ignition at all times. Do not deliver foam-plastic insulation materials to the project site before installation time.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

Complete installation and concealment of plastic materials as rapidly as possible.

#### 1.6 PROJECT/SITE CONDITIONS

Weather Limitations: Proceed with installation only when existing and forecasted weather conditions permit metal roof panel work to be performed according to the manufacturer's written instructions and warranty requirements.

Field Measurements: Verify the actual dimensions of construction contiguous with metal roof panels by field measurements before fabrication.

#### 1.7 WARRANTY

Furnish the metal roof panel manufacturer's 20-year no dollar limit roof system materials and installation workmanship warranty, including flashing, insulation, components, trim, and accessories necessary for a watertight roof system construction. Issue the warranty directly to the Government, such that the warranty takes effect at the time of the Government's acceptance of the roof work. Provide a warranty with the following conditions:

- a. If within the warranty period, the metal roof system, as installed for its intended use in the normal climatic and environmental conditions of the facility, becomes nonwater-tight, shows evidence of moisture intrusion within the assembly, displaces, corrodes, perforates, separates at the seams, or shows evidence of excessive weathering due to defective materials or installation workmanship, the repair or replacement of the defective and damaged materials of the metal roof system and correction of defective workmanship is the responsibility of the metal roof panel manufacturer. All costs associated with the repair or replacement work are the responsibility of the metal roof panel manufacturer. Conform galvanized repairs to ASTM A780/A780M.
- b. If the manufacturer or the applicator approved by the manufacturer fail to perform the repairs within 24 hours of notification, emergency temporary repairs performed by others does not void the warranty.

##### 1.7.1 Manufacturer's Finish Warranty

Provide a manufacturer's 20-year "No-Dollar-Limit" Warranty for Labor and Materials for the roofing system. Issue the warranty directly to the Government at the date of Government acceptance, warranting that the factory color finish, under normal atmospheric conditions at the site, will not crack, peel, or delaminate; chalk in excess of a numerical rating of 8 when measured in accordance with ASTM D4214; or fade or change colors in excess of 5 NBS units as measured in accordance with ASTM D2244.

##### 1.7.2 Metal Roof System Installer Warranty

Provide the "Contractors Five 5)-Year No Penalty Sum Warranty for Non-Structural Metal Roof System" attached at the end of this section.

##### 1.7.3 Continuance of Warranty

Approve and accomplish repair or replacement work that becomes necessary within the warranty period to restore the integrity of the roof system

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

assembly and maintain the validity of the metal roof system manufacturer's warranty for the remainder of the manufacturer warranty period.

## PART 2 PRODUCTS

### 2.1 SYSTEM DESCRIPTION

#### 2.1.1 Conformance and Compatibility

Provide an entire roofing and flashing system in accordance with specified and indicated requirements, including wind resistance requirements. Perform any work not specifically addressed, or any deviation from specified requirements in general accordance with recommendations of the MBMA RSDM, NRCA RoofMan, the metal panel manufacturer's published recommendations and details, and compatible with surrounding components and construction. Submit any deviation from specified or indicated requirements to the Contracting Officer for approval before installation.

#### 2.1.2 Performance Requirements

- a. Hydrostatic-Head Resistance: No water penetration when tested according to ASTM E2140. The minimum slope for hydrokinetic roofs is 4:12. In ASHRA Climates Zones 4 and above, which have the potential for ice damming, use a minimum slope of 6:12. Slopes lower than indicated above will require a Hydrostatic roof system with hydrostatic panels (often referred to as structural panels) with a minimum of 2 inch tall seams for a standing seam panel type roof and with no water penetration when tested according to ASTM E2140.
- b. Wind-Uplift Resistance: Wind-Uplift Resistance: Provide portable roll-forming equipment capable of producing sheet metal roofing assemblies that comply with UL 580 for Class 90 wind-uplift resistance.

Maintain UL certification of portable roll-forming equipment for duration of sheet metal roofing work.

Refer to Component and Cladding Wind Pressures Schedule on Structural Drawings.

- c. FMG Listing: Provide FRP roof panels and component materials that comply with the requirements in FM 4471 as part of a panel roofing system. Identify materials with FMG markings.
- d. Structural Performance: Provide roof panel assemblies capable of withstanding the effects of gravity loads and stresses within limits and under conditions indicated, based on testing according to ASTM E1592.

#### 2.1.3 Fire-Resistance

##### 2.1.3.1 Surface-Burning Characteristics

Provide metal roof panels having insulation core material with the following surface-burning characteristics as determined by testing identical products according to ASTM E84 by a qualified testing agency. Identify products with the appropriate markings of an applicable testing agency.

Flame-Spread Index: 25 or less.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

Smoke-Developed Index: 450 or less.

#### 2.1.3.2 Fire-Resistance Ratings

Where indicated, provide metal roof panels identical to those of assemblies tested for fire resistance by a qualified testing agency in accordance with ASTM E119. Identify products with the appropriate markings of the applicable testing agency.

Indicate design designations from UL's "Fire Resistance Directory" or from the listings of another qualified testing agency. Combustion Characteristics: ASTM E136.

### 2.2 FABRICATION

#### 2.2.1 Fabrication

Fabricate and finish metal roof panels and accessories at the factory to the greatest extent possible, using the manufacturer's standard procedures and processes to fulfill the indicated performance requirements. Comply with indicated profiles, and dimensional and structural requirements conforming to AISI D100.

Provide a panel profile, including major ribs and intermediate stiffening ribs, if any, for the full length of panel.

Fabricate metal roof panel side laps with factory-installed captive gaskets or separator strips that provide a tight seal and prevent metal-to-metal contact, in a manner that will seal weather-tight and minimize noise from movements within the panel assembly.

Sheet Metal Accessories: Fabricate flashing and trim to comply with recommendations in SMACNA 1793 that apply to the design, dimensions, metal, and other characteristics of the item indicated.

Form exposed sheet metal accessories without excessive oil canning, buckling, and tool marks, and true to the line and levels indicated, with exposed edges folded back to form hems.

End Seams: Fabricate nonmoving seams with flat-lock seams. Form seams and seal with epoxy seam sealer. Rivet joints for additional strength.

Sealed Joints: Form nonexpansion but movable joints in metal to accommodate elastomeric sealant, compliant with SMACNA 1793.

Conceal fasteners and expansion provisions where possible. Exposed fasteners are not allowed on the faces of accessories exposed to view.

Fabricate cleats and attachment devices of the size and metal thickness recommended by SMACNA 1793 or by the metal roof panel manufacturer for application, but not less than the thickness of the metal being secured.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## 2.2.2 Sheet Metal Flashing and Trim

### 2.2.2.1 Fabrication, General

Custom-fabricate sheet metal flashing and trim to comply with the recommendations in SMACNA 1793 that apply to the design, dimensions, metal, and other characteristics of the items indicated. Shop-fabricate items where practicable. Obtain field measurements for an accurate fit before shop fabrication.

### 2.2.2.2 Roof Drainage Sheet Metal Fabrications

Fabricate gutters to the cross section indicated, with riveted and soldered joints, complete with end pieces, outlet tubes, and other special accessories as required. Fabricate in 96-inch long sections at a minimum. Fabricate expansion joints and accessories from the same metal as the gutters, unless otherwise indicated.

Fabricate rectangular downspouts complete with mitered elbows. Furnish with metal hangers, fabricated from the same material as the downspouts and anchors.

## 2.2.3 Finishes

Comply with NAAMM AMP 500 for recommendations for applying and designating finishes.

Appearance of Finished Work: Ensure that there are no noticeable variations in finish on the same piece. Variations in the appearance of adjoining components are acceptable if they are within the range of approved samples and are assembled or installed to minimize contrast.

## 2.3 COMPONENTS

### 2.3.1 Miscellaneous Metal Framing

#### 2.3.1.1 General

Provide cold-formed metallic-coated steel sheet conforming to AISI D100 and ASTM A653/A653M and specified in Section 05 40 00 COLD-FORMED METAL FRAMING unless otherwise indicated.

#### 2.3.1.2 Fasteners for Miscellaneous Metal Framing

Provide fasteners of a type, material, corrosion resistance, size, and sufficient length to penetrate the supporting member a minimum of 1 inch and possessing the other properties required to fasten miscellaneous metal framing members to substrates in accordance with the roof-panel manufacturer's and ASCE 7-16 requirements.

### 2.3.2 Fasteners

#### 2.3.2.1 General

Provide fasteners of a type, material, corrosion resistance, size, and sufficient length to penetrate the supporting member a minimum of 1 inch and possessing the other properties required to fasten miscellaneous metal framing members to substrates in accordance with the roof-panel manufacturer's and ASCE 7-16 requirements.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### 2.3.2.2 Exposed Fasteners

Provide corrosion-resistant coated steel, aluminum, stainless steel, or nylon-capped, steel-compatible exposed fasteners, with the sheet panel or flashing. Provide exposed fasteners of a type and size recommended by the manufacturer to meet the performance requirements and design loads specified. Provide the manufacturer's standard fasteners for accessories. Provide an integral metal washer that matches the color of material the washer is attached to with a compressible sealing EPDM gasket approximately 3/32 inch thick.

#### 2.3.2.3 Screws

Provide corrosion-resistant coated steel, aluminum or stainless steel screws of the type and size recommended by the manufacturer to meet the performance requirements.

#### 2.3.2.4 Rivets

Provide closed-end rivets, made of corrosion-resistant coated steel, aluminum, or stainless steel where watertight connections are required.

#### 2.3.2.5 Attachment Clips

Provide clips fabricated from steel hot-dipped galvanized in accordance with ASTM A653/A653M G 90 or Series 300 stainless steel. Ensure that the size, shape, thickness and capacity are as required to meet the insulation thickness and design load criteria specified.

### 2.4 MATERIALS

#### 2.4.1 Aluminum Sheet

Roll-form aluminum roof and liner panels to the specified profile, with  $f_y$  equals to a 40 ksi, .040 inch thickness and depth as indicated. Ensure that the material is plumb and true, and within the tolerances listed:

- a. Aluminum Sheet conforming to ASTM B209, ASTM B209M, and AA ADM.
- b. Ensure individual panels have continuous length to that covers the entire length of any unbroken roof slope with no joints or seams, formed without warping, waviness, or ripples that are not part of the panel profile, and free of damage to the finish coating system.
- c. Provide panels with thermal expansion and contraction coefficients consistent with the type of system specified.
  - (1) Provide profile and coverage of minimum height and width based on the manufacturer's standard for the indicated roof slope.
  - (2) Provide a smooth, flat, non-directional surface texture. Color as indicated on Drawings or, if not otherwise indicated as selected by Contracting Officer.

#### 2.4.2 Insulated Panel Construction

Shop-fabricate or field-assemble insulated panel construction with specified exterior and interior aluminum sheet in accordance with



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

manufacturer's printed instructions.

Provide pre-finished interior lath- or board-finished interior surfaces for panel assemblies in accordance with ASTM C1396/C1396M.

Provide glass-fiber insulation conforming to ASTM C553 and ASTM C612 of thickness and density as required for the geographical area where construction will take place. Glass-fiber and mineral-wool-fiber are materials listed in the EPA's Comprehensive Procurement Guidelines (CPG) <https://www.epa.gov/smm/comprehensive-procurement-guidelines-construction-products>  
Submit a sample of insulation approximately 8 inches by 11 inches.

Provide adhesively attached insulation fasteners plate-welded to projecting spindle anchors, capable of holding insulation of thickness indicated, secured in position with self-locking washers and complying with the following requirements:

- a. Plate: Perforated galvanized carbon-steel sheet, 0.030 inch thick by 2 inches square.
- b. Spindle: Copper-coated, low-carbon steel; fully annealed; 0.105 inch in diameter; length to suit depth of insulation indicated.
- c. Insulation-Retaining Washers: Self-locking washers formed from 0.016 inch thick galvanized steel sheet, with a beveled edge for increased stiffness, sized as required to hold insulation in place, but not less than 1 1/2 inches square or in diameter.
- d. Adhesive: Provide an anchor adhesive to bond insulation anchors to the substrates indicated without damaging insulation, fasteners, and substrates.

#### 2.4.3 Finish

Ensure all panels receive a factory-applied Kynar 500/Hylar 5000 finish consisting of a baked-on top-coat and a manufacturer's recommended prime coat with to the following:

- a. Metal Preparation: Prepare all metal surfaces for painting on a continuous process coil coating line by alkali cleaning, hot-water rinsing, application of chemical conversion coating, cold-water rinsing, sealing with acid rinse, and thorough drying.
- b. Prime Coating: Apply a base-coat of epoxy paint, specifically formulated to interact with the top-coat, to the prepared surfaces by roll-coating the paint to a dry film thickness of 0.20 mils plus 0.05 mils. Ensure that the prime coat is oven-cured before the application of finish coat is applied.
- c. Exterior Finish Coating: Apply the finish coating over the primer by roll-coating the finish coating to a dry film thickness of 0.80 plus 5 mils (3.80 plus 0.50 mils for Vinyl Plastisol) for a total dry-film thickness of 1.00 mils plus 0.10 mils (4.00 mils plus 0.10 mils for Vinyl Plastisol). Ensure that the finish coat is oven-cured.
- d. Interior Finish Coating: Apply a wash-coat on the reverse side over the primer by roll-coating to a dry-film thickness of 0.30 mils plus 0.05 mils for a total-dry film thickness of 0.50 mils plus 0.10 mils. Ensure that the wash-coat is oven-cured.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

- e. Color: Ensure that the exterior finish is as chosen from the manufacturer's standard color chart.
- f. Coating Physical Properties: Provide coating conforming to the industry and manufacturer's standard performance criteria as listed by the following certified test reports:
  - (1) Chalking: ASTM D4214
  - (2) Coating Thickness: ASTM B659
  - (3) Color Change and Conformity: ASTM D2244
  - (4) Weatherometer: ASTM G152, ASTM G153 and ASTM D822
  - (5) Humidity: ASTM D2247 and ASTM D714
  - (6) Salt Spray: ASTM B117
  - (7) Chemical Pollution: ASTM D1308
  - (8) Gloss at 60: ASTM D523
  - (9) Pencil Hardness: ASTM D3363
  - (10) Reverse Impact: ASTM D2794
  - (11) Flexibility: ASTM D522/D522M
  - (12) Abrasion: ASTM D968
  - (13) Flame Spread: ASTM E84

## 2.5 ACCESSORIES

### 2.5.1 General

Provide only accessories which are compatible with the metal roof panels. Sheet metal flashing, trim, metal closure strips, caps and similar metal accessories can not be less than the minimum thickness specified for the roof panels. Submit a 10 inches long sample of each type. Ensure the exposed metal accessories and finishes match the panels furnished, except as otherwise indicated. Provide molded-foam rib, ridge and other closure strips which are non-absorbent closed-cell or solid-cell synthetic rubber or pre-molded neoprene to match the configuration of the panels.

### 2.5.2 Rubber Closure Strips

Provide closed-cell, expanded cellular rubber conforming to ASTM D1056 and ASTM D1667; extruded or molded to the configuration of the specified roof panel and in lengths supplied by the roof-panel manufacturer.

### 2.5.3 Metal Closure Strips

Provide factory fabricated aluminum closure strips of the same gauge, color, finish and profile as the specified roof panel.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### 2.5.4 Joint Sealants

##### 2.5.4.1 Sealants

Provide an approved gun-type sealant for use in hand- or air-pressure caulking guns at temperatures above 40 degrees F (or frostfree application at temperatures above 10 degrees F with minimum solid content of 85 percent of the total volume. Provide sealant that has a tough, durable, dry surface skin that permits it to remain soft and pliable underneath, providing a weather-tight joint. No migratory staining is permitted on painted or unpainted metal, stone, glass, vinyl, or wood.

Prime all joints to receive sealants with a compatible one-component or two-component primer as recommended by the roof-panel manufacturer.

- a. Shop-Applied Caulking: Use an approved gun-grade, non-sagging one-component polysulfide or silicone conforming to ASTM C920, Type II, with a curing time to ensure the sealant's plasticity at the time of field erection.
- b. Field Applied Caulking: Use an approved gun-grade, non-sagging one-component polysulfide or two-component polyurethane with an initial maximum Shore A durometer hardness of 25, conforming to ASTM C920, Type II. Match the color to the panel colors.
- c. Tape Sealant: Use a pressure-sensitive, 100 percent solid with a release paper backing, permanently elastic, non-sagging, non-toxic and non-staining as approved by the roof-panel manufacturer.

### PART 3 EXECUTION

#### 3.1 EXAMINATION

The Contracting Officer may request verification and certification testing of coatings and base metals of metal roofing prior to installation. The following areas may be verified:

- a. Examine substrates, areas, and conditions, with the installer present, for compliance with the requirements for installation tolerances, metal roof panel supports, and other conditions affecting performance of the work.
- b. Examine primary and secondary roof framing to verify that rafters, purlins, angles, channels, and other structural panel support members and anchorages have been installed within alignment tolerances required by the metal roof-panel manufacturer, and as required for the geographical area where construction has taken place.
- c. Examine solid roof sheathing to verify that the sheathing joints are supported by framing or blocking and that the installation is within flatness tolerances required by the metal roof-panel manufacturer.
- d. Examine roughing-in for components and systems penetrating the metal roof panels to verify actual locations of penetrations relative to seam locations of metal roof panels before metal roof panel installation.
- e. Submit to the Contracting Officer a written report, endorsed by the installer, listing conditions detrimental to performance of the work.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

- f. Proceed with the installation only after unsatisfactory conditions have been corrected.

### 3.2 PREPARATION

Clean substrates of substances harmful to insulation, remove projections capable of interfering with insulation attachment.

Install sub-purlins, eave angles, furring, and other miscellaneous roof-panel support members and anchorage according to the metal roof-panel manufacturer's written instructions.

### 3.3 INSTALLATION

#### 3.3.1 Workmanship

Ensure lines, arises, and angles are sharp and true. Free exposed surfaces from visible wave, warp, buckle, and tool marks. Fold back exposed edges neatly to form a 1/2 inch hem on the concealed side. Ensure that sheet metal that is exposed to the weather is watertight, with provisions for expansion and contraction.

Ensure surfaces that are to receive sheet metal are plumb and true, clean, even, smooth, dry, and free of defects and projections that might affect the application. Install items not shown in detail or not covered by specifications conform to the applicable requirements of SMACNA 1793. Provide sheet metal flashing in the angles formed where roof decks abut walls, curbs, ventilators, pipes, or other vertical surfaces, and wherever indicated and necessary to make the work watertight.

#### 3.3.2 Roof Panels

Provide metal roof panels of full length from eave to ridge or eave to wall as indicated, unless otherwise indicated or restricted by shipping limitations. Anchor metal roof panels and other components of the work in place, with provisions for thermal and structural movement in accordance with NRCA 0429.

- a. Aluminum Roof Panels: Use aluminum or stainless-steel fasteners for exterior surfaces and aluminum or galvanized steel fasteners for interior surfaces.

- b. Metal Protection: Where dissimilar metals will contact each other or corrosive substrates, protect against galvanic action by painting contact surfaces with bituminous coating conforming to SSPC PS 9.01, by applying rubberized-asphalt underlayment to each contact surface, or with another means to separate the metals and contact surface as recommended by metal roof-panel manufacturer.

- c. Joint Sealers: Install gaskets, joint fillers, and sealants where required for weatherproof performance of metal roof panel assemblies. Provide the types of gaskets, fillers, and sealants indicated; or if not indicated, provide types recommended by the metal roof panel-manufacturer.

Erect the roofing system in accordance with the approved erection drawings, the printed instructions and the safety precautions of the manufacturer.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

Do not overload, abuse, or subject sheets to undue impact. Do not apply bent, chipped, or defective sheets.

Erect sheets true and plumb and in exact alignment with the horizontal and vertical edges of the building, securely anchored, and with the indicated rake, eave, and curb overhang.

Allow for thermal movement of the roofing and movement of the building structure, and provide permanent freedom from noise caused by wind pressure.

Field cutting metal roof panels by torch is not permitted.

Lay roofing sheets with corrugations in the direction of the roof slope. End laps of exterior roofing cannot be less than 8 inches; the side laps of standard exterior corrugated sheets cannot be less than 2-1/2 corrugations.

Do not permit storage, walking, wheeling, and trucking directly on applied roofing materials. Provide temporary walkways, runways, and platforms of smooth clean boards or planks as necessary to avoid damage to the installed roofing materials and to distribute weight to conform to the indicated live-load limits of roof construction.

### 3.3.3 Fasteners

Anchor metal roof panels and other components of the work in place using the manufacturer's approved fasteners according to the manufacturer's written instructions.

### 3.3.4 Flashing, Trim and Closure

#### 3.3.4.1 General Requirements

Comply with performance requirements, the manufacturer's written installation instructions, and SMACNA 1793. Provide concealed fasteners where possible, and set units true to line and level as indicated. Install work with laps, joints, and seams that will be permanently watertight and weather-resistant.

Install sheet metalwork to form weathertight construction without waves, warps, buckles, fastening stresses or distortion, and allow for expansion and contraction. Ensure sheet metal mechanics perform cutting, fitting, drilling, and other operations in connection with sheet metal work required to accommodate the work of other trades.

#### 3.3.4.2 Metal Flashing

Install metal flashing at building corners, rakes and eaves, junctions between metal siding and roofing, valleys and changes of slope or direction in metal roofing, and building expansion joints and gutters.

Provide exposed metal flashing that is the same material, color, and finish as the specified metal roofing.

Fasten flashing at not more than 8 inches on-center for roofs, except where flashing are held in place by the same screws that secure covering sheets.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

Furnish flashing in at least 8 foot lengths. Provide exposed flashing that has 1 inch locked and blind-soldered end joints, and expansion joints at intervals of not more than 16 feet.

Bed exposed flashing and flashing subject to rain penetration in the specified joint sealant.

To prevent electrolytic deterioration, isolate flashing that is in contact with dissimilar metals by means of the specified asphalt-mastic material.

Form drips to the profile indicated, with the edge folded back 1/2 inch to form a reinforced drip edge.

#### 3.3.4.3 Closures

Install metal closure strips at the open ends of corrugated or ribbed pattern roofs, and at the intersection of wall and roof unless open ends are concealed with formed eave flashing, at the rake of the metal roof unless the open end has a formed flashing member, and in other required areas.

Install mastic closure strips at the intersection of the wall with metal roofing, at the top and bottom of metal siding, at the heads of wall openings, and in other required locations.

#### 3.3.5 Information Form and Placard

For each roof, furnish a typewritten information card for facility records and a card laminated in plastic and framed for interior display at the roof access point, or a photoengraved 0.032 inch thick aluminum card for exterior display. Format the card as directed in paragraph FORM ONE.

Provide an information card 8 1/2 inches by 11 inches minimum, identifying the facility name and number, location, contract number, approximate roof area, detailed roof system description, including deck type, roof panel manufacturer and product name, type underlayment, date of completion, installing contractor identification and contact information; manufacturer warranty expiration, warranty reference number, and contact information. Install the card at interior roof top access point and provide a paper copy to the Contracting Officer.

### 3.4 FIELD QUALITY CONTROL

#### 3.4.1 Acceptance Provisions

##### 3.4.1.1 Erection Tolerances

Erect metal roofing straight and true with plumb vertical lines correctly lapped and secured in accordance with the manufacturer's written instructions. Do not vary horizontal lines more than 1/8 inch in 40 feet.

##### 3.4.1.2 Repairs to Finish

Repair scratches, abrasions, and minor surface defects in the finish with the specified repair materials. Ensure repaired finished surfaces are uniform and free from variations of color and surface texture.

Immediately remove and replace repaired metal surfaces that are not

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

acceptable to the project requirements with new material.

#### 3.4.1.3 Paint-Finish Metal Roofing

Test paint-finish metal roofing for color stability by the Contracting Officer during the manufacturer's specified guarantee period.

Remove and replace panels that have visual evidence of color changes, fading, or surface degradation, with new panels at no expense to the Government.

Replaced panels are subject to the specified tests for an additional year from the date of their installation.

#### 3.4.2 Repair of Finish Protection

Provide repair paint for color-finish enameled roofing that is compatible with the paint of the same formula and color as the specified finish furnished by the roofing manufacturer.

### 3.5 ADJUSTING AND CLEANING

Clean all exposed sheet metal work at completion of installation. Remove metal shavings, filings, nails, bolts, and wires from roofs. Remove grease and oil films, excess sealants, handling marks, contamination from steel wool, fittings, and drilling debris and scrub the work clean. Ensure exposed metal surfaces are free of dents, creases, waves, scratch marks, solder or weld marks, and damage to the finish coating.

Collect and place scrap/waste materials in containers. Dispose of demolished materials immediately. Do not allow demolished materials to accumulate on-site; transport demolished materials from government property and legally dispose of them.

### 3.6 SCHEDULES

#### 3.6.1 Form One

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

FORM 1 - PREFORMED ALUMINUM PANEL ROOFING SYSTEM AND COMPONENTS

1. Contract Number:
2. Building Number & Location:
3. NAVFAC Specification Number:
4. Deck/Substrate Type:
5. Slopes of Deck/Roof Structure:
6. Insulation Type & Thickness:
7. Insulation Manufacturer:
8. Vapor Retarder: ( )Yes ( )No
9. Vapor Retarder Type:
10. Preformed Steel Standing Seam Roofing Description:
  - a. Manufacturer (Name, Address, & Phone No.):
  - b. Product Name:
  - c. Width:
  - d. Gage:
  - e. Base Metal:
  - f. Method of Attachment:
11. Repair of Color Coating:
  - a. Coating Manufacturer (Name, Address & Phone No.):
  - b. Product Name:
  - c. Surface Preparation:
  - d. Recoating Formula:
  - e. Application Method:
12. Statement of Compliance or Exception: \_\_\_\_\_  
 \_\_\_\_\_  
 \_\_\_\_\_
13. Date Roof Completed:
14. Warranty Period: From \_\_\_\_\_ To \_\_\_\_\_
15. Roofing Contractor (Name & Address):
16. Prime Contractor (Name & Address):

Contractor's Signature \_\_\_\_\_ Date:

Inspector's Signature \_\_\_\_\_ Date:

3.6.2 Date of Installation Wall-Mounted Placard

For each metal roof panel installation, furnish an exterior "Date of Installation Placard", 0.032 inch thick aluminum, 8-1/2 inches high by 11 inches wide, with mounting accessories, photoengraved to include the following information:



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

Facility Name and Number  
Approximate Roof Area Newly Installed and Date of Completion  
Manufacturer, Type of Roof Panel and Name  
Underlayment and Insulation System, R value  
Installing Contractor and Contact Information  
Warranty Expiration Date  
Warranty Reference Number and Contact Information

Install placard as directed by the Contracting Officer.

#### 3.6.3 Warranty

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

CONTRACTOR'S FIVE (5)  
PENAL SUM WARRANTY

YEAR NO

FOR  
NON-STRUCTURAL METAL ROOF SYSTEM

FACILITY DESCRIPTION\_\_\_\_\_

BUILDING NUMBER:\_\_\_\_\_

CORPS OF ENGINEERS CONTRACT NUMBER:\_\_\_\_\_

CONTRACTOR

CONTRACTOR:\_\_\_\_\_

ADDRESS:\_\_\_\_\_

POINT OF CONTACT:\_\_\_\_\_

TELEPHONE NUMBER:\_\_\_\_\_

OWNER

OWNER:\_\_\_\_\_

ADDRESS:\_\_\_\_\_

POINT OF CONTACT:\_\_\_\_\_

TELEPHONE NUMBER:\_\_\_\_\_

CONSTRUCTION AGENT

CONSTRUCTION AGENT:\_\_\_\_\_

ADDRESS:\_\_\_\_\_

POINT OF CONTACT:\_\_\_\_\_

TELEPHONE NUMBER:\_\_\_\_\_

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

CONTRACTOR'S FIVE (5)  
YEAR NO PENAL SUM WARRANTY  
FOR  
NON-STRUCTURAL METAL ROOF SYSTEM  
(continued)

THE NON-STRUCTURAL METAL ROOF SYSTEM INSTALLED ON THE ABOVE NAMED BUILDING IS WARRANTED BY \_\_\_\_\_ FOR A PERIOD OF FIVE (5) YEARS AGAINST WORKMANSHIP AND MATERIAL DEFICIENCIES, WIND DAMAGE, STRUCTURAL FAILURE, AND LEAKAGE. FOR THE NON-STRUCTURAL METAL ROOFING SYSTEM COVERED UNDER THIS WARRANTY INCLUDE, BUT DO NOT LIMIT TO, THE FOLLOWING: THE ENTIRE ROOFING SYSTEM, MANUFACTURER SUPPLIED FRAMING AND STRUCTURAL MEMBERS, METAL ROOF PANELS, FASTENERS, CONNECTORS, ROOF SECUREMENT COMPONENTS, AND ASSEMBLIES TESTED AND APPROVED IN ACCORDANCE WITH UL 580. IN ADDITION, THE SYSTEM PANEL FINISHES, SLIP SHEET, INSULATION, VAPOR RETARDER, ALL ACCESSORIES, COMPONENTS, AND TRIM AND ALL CONNECTIONS ARE INCLUDED. THIS INCLUDES ROOF PENETRATION ITEMS SUCH AS VENTS, CURBS, SKYLIGHTS; INTERIOR OR EXTERIOR GUTTERS AND DOWNSPOUTS; EAVES, RIDGE, HIP, VALLEY, RAKE, GABLE, WALL, OR OTHER ROOF SYSTEM FLASHING INSTALLED AND ANY OTHER COMPONENTS SPECIFIED WITHIN THIS CONTRACT TO PROVIDE A WEATHERTIGHT ROOF SYSTEM; AND ITEMS SPECIFIED IN OTHER SECTIONS OF THE SPECIFICATIONS THAT ARE PART OF THE NON-STRUCTURAL METAL ROOFING SYSTEM.

REPAIR ALL MATERIAL DEFICIENCIES, WIND DAMAGE, STRUCTURAL FAILURE, AND LEAKAGE ASSOCIATED WITH THE NON-STRUCTURAL METAL ROOF SYSTEM COVERED UNDER THIS WARRANTY AS APPROVED BY THE CONTRACTING OFFICER. IN THIS WARRANTY COVER THE ENTIRE COST OF REPAIR OR REPLACEMENT, INCLUDING ALL MATERIAL, LABOR, AND RELATED MARKUPS. THE ABOVE REFERENCED WARRANTY COMMENCED ON THE DATE OF FINAL ACCEPTANCE ON \_\_\_\_\_ AND WILL REMAIN IN EFFECT FOR STATED DURATION FROM THIS DATE.

SIGNED, DATED, AND NOTARIZED (BY COMPANY PRESIDENT)

\_\_\_\_\_  
(Company President)

\_\_\_\_\_  
(Date)

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

CONTRACTOR'S FIVE (5)

YEAR NO

PENAL SUM WARRANTY  
FOR  
NON-STRUCTURAL METAL ROOFING SYSTEM  
(continued)

ENSURE THE CONTRACTOR SUPPLEMENTS THIS WARRANTY WITH WRITTEN WARRANTIES FROM THE MANUFACTURER AND/OR INSTALLER OF THE NON-STRUCTURAL METAL ROOFING SYSTEM. SUBMIT ALONG WITH THE CONTRACTOR'S WARRANTY. HOWEVER, THE CONTRACTOR IS ULTIMATELY RESPONSIBLE FOR THIS WARRANTY AS OUTLINED IN THE SPECIFICATIONS AND AS INDICATED IN THIS WARRANTY EXAMPLE.

EXCLUSIONS FROM COVERAGE

1. NATURAL DISASTERS, ACTS OF GOD (LIGHTNING, FIRE, EXPLOSIONS, SUSTAINED WIND FORCES IN EXCESS OF THE DESIGN CRITERIA, EARTHQUAKES, AND HAIL).
2. ACTS OF NEGLIGENCE OR ABUSE OR MISUSE BY GOVERNMENT OR OTHER PERSONNEL, INCLUDING ACCIDENTS, VANDALISM, CIVIL DISOBEDIENCE, WAR, OR DAMAGE CAUSED BY FALLING OBJECTS.
3. DAMAGE BY STRUCTURAL FAILURE, SETTLEMENT, MOVEMENT, DISTORTION, WARPAGE, OR DISPLACEMENT OF THE BUILDING STRUCTURE OR ALTERATIONS MADE TO THE BUILDING.
4. CORROSION CAUSED BY EXPOSURE TO CORROSIVE CHEMICALS, ASH OR FUMES GENERATED OR RELEASED INSIDE OR OUTSIDE THE BUILDING FROM CHEMICAL PLANTS, FOUNDRIES, PLATING WORKS, KILNS, FERTILIZER FACTORIES, PAPER PLANTS, AND THE LIKE.
5. FAILURE OF ANY PART OF THE NON-STRUCTURAL METAL ROOF DUE TO ACTIONS BY THE OWNER TO INHIBIT FREE DRAINAGE OF WATER FROM THE ROOF AND GUTTERS AND DOWNSPOUTS OR ALLOW PONDING WATER TO COLLECT ON THE ROOF SURFACE. IN CONTRACTOR'S DESIGN ENSURE FREE DRAINAGE FROM THE ROOF AND DO NOT ALLOW PONDING WATER.
6. THIS WARRANTY APPLIES TO THE NON-STRUCTURAL METAL ROOFING SYSTEM. IT DOES NOT INCLUDE ANY CONSEQUENTIAL DAMAGE TO THE BUILDING INTERIOR OR CONTENTS WHICH IS COVERED BY THE WARRANTY OF CONSTRUCTION CLAUSE INCLUDED IN THIS CONTRACT.
7. THIS WARRANTY CANNOT BE TRANSFERRED TO ANOTHER OWNER WITHOUT WRITTEN CONSENT OF THE CONTRACTOR; AND THIS WARRANTY AND THE CONTRACT PROVISIONS WILL TAKE PRECEDENCE OVER ANY CONFLICTS WITH STATE STATUTES.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

CONTRACTOR'S FIVE (5)

YEAR NO

PENAL SUM WARRANTY  
FOR  
NON-STRUCTURAL METAL ROOF SYSTEM  
(continued)

\*\*RESPOND TO REPORTS OF LEAKS AND ROOF SYSTEM DEFICIENCIES WITHIN 48 HOURS OF RECEIPT OF NOTICE, BY TELEPHONE OR IN WRITING, FROM EITHER THE OWNER OR CONTRACTING OFFICER. INITIATE EMERGENCY REPAIRS TO PREVENT FURTHER ROOF LEAKS IMMEDIATELY; SUBMIT A WRITTEN PLAN FOR APPROVAL TO REPAIR OR REPLACE THIS ROOF SYSTEM WITHIN SEVEN (7) CALENDAR DAYS. COMMENCE ACTUAL WORK FOR PERMANENT REPAIRS OR REPLACEMENT WITHIN 30 DAYS AFTER RECEIPT OF NOTICE, AND COMPLETED WITHIN A REASONABLE TIME FRAME. IF THE CONTRACTOR FAILS TO ADEQUATELY RESPOND TO THE WARRANTY PROVISIONS, AS STATED IN THE CONTRACT AND AS CONTAINED HEREIN, THE CONTRACTING OFFICER MAY HAVE THE NON-STRUCTURAL METAL ROOF SYSTEM REPAIRED OR REPLACED BY OTHERS AND CHARGE THE COST TO THE CONTRACTOR.

UNDER THE CONTRACT DISPUTES ACT, THE CONTRACTOR MAY CHALLENGE THE OWNER'S DEMAND FOR REPAIRS AND/OR REPLACEMENT DIRECTED BY THE OWNER OR CONTRACTING OFFICER EITHER BY REQUESTING A CONTRACTING OFFICER'S DECISION UNDER THE CONTRACT DISPUTES ACT, OR BY REQUESTING THAT AN ARBITRATOR RESOLVE THE ISSUE. MAKE THE REQUEST FOR AN ARBITRATOR WITHIN 48 HOURS OF BEING NOTIFIED OF THE DISPUTED DEFECTS. UPON BEING INVOKED, WITHIN TEN (10) DAYS, ENSURE THE PARTIES JOINTLY REQUEST A LIST OF FIVE (5) ARBITRATORS FROM THE FEDERAL MEDIATION AND CONCILIATION SERVICE. THE PARTIES TEN (10) DAYS AFTER RECEIPT OF THE LIST TO SEEK AGREEMENT ON AN ARBITRATOR TO CONFER. IF THE PARTIES CANNOT AGREE ON AN ARBITRATOR, THE CONTRACTING OFFICER AND THE PRESIDENT OF THE CONTRACTOR'S COMPANY WILL STRIKE ONE (1) NAME FROM THE LIST ALTERNATIVELY UNTIL ONE (1) NAME REMAINS. THE REMAINING PERSON IS THE DULY SELECTED ARBITRATOR. THE COSTS OF THE ARBITRATION, INCLUDING THE ARBITRATOR'S FEE AND EXPENSES, COURT REPORTER, COURTROOM OR SITE SELECTED, ETC., WILL BE BORNE EQUALLY BETWEEN THE PARTIES. EITHER PARTY DESIRING A COPY OF THE TRANSCRIPT PAYS FOR THE TRANSCRIPT. A HEARING WILL BE HELD AS SOON AS THE PARTIES CAN MUTUALLY AGREE. REQUEST A WRITTEN ARBITRATOR'S DECISION NO LATER THAN 30 DAYS FOLLOWING THE HEARING. THE DECISION OF THE ARBITRATOR WILL NOT BE BINDING; HOWEVER, IT WILL BE ADMISSIBLE IN ANY SUBSEQUENT APPEAL UNDER THE CONTRACT DISPUTES ACT.

POST A FRAMED COPY OF THIS WARRANTY IN THE MECHANICAL ROOM OR OTHER APPROVED LOCATION DURING THE ENTIRE WARRANTY PERIOD.

-- End of Section --

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## SECTION 07 42 63

## FABRICATED WALL PANEL ASSEMBLIES

**05/11, CHG 4: 08/20**

## PART 1 GENERAL

## 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

## AMERICAN ARCHITECTURAL MANUFACTURERS ASSOCIATION (AAMA)

AAMA 501.1 (2017) Standard Test Method for Water Penetration of Windows, Curtain Walls and Doors Using Dynamic Pressure

## AMERICAN IRON AND STEEL INSTITUTE (AISI)

AISI S100 (2012) North American Specification for the Design of Cold-Formed Steel Structural Members

AISI SG03-3 (2002; Suppl 2001-2004; R 2008) Cold-Formed Steel Design Manual Set

## AMERICAN SOCIETY OF CIVIL ENGINEERS (ASCE)

ASCE 7-16 (2017; Errata 2018; Supp 1 2018) Minimum Design Loads and Associated Criteria for Buildings and Other Structures

## ASTM INTERNATIONAL (ASTM)

ASTM A36/A36M (2019) Standard Specification for Carbon Structural Steel

ASTM A123/A123M (2017) Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products

ASTM A606/A606M (2023) Standard Specification for Steel Sheet and Strip, High-Strength, Low-Alloy, Hot-Rolled and Cold-Rolled, with Improved Atmospheric Corrosion Resistance

ASTM A653/A653M (2023) Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process

ASTM A1008/A1008M (2024) Standard Specification for Steel, Sheet, Cold-Rolled, Carbon, Structural, High-Strength Low-Alloy, High-Strength Low-Alloy with Improved Formability, Solution Hardened, and Bake Hardenable

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

|                   |                                                                                                                                   |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| ASTM B117         | (2019) Standard Practice for Operating Salt Spray (Fog) Apparatus                                                                 |
| ASTM C273/C273M   | (2020) Standard Test Method for Shear Properties of Sandwich Core Materials                                                       |
| ASTM C553         | (2013; R 2019) Standard Specification for Mineral Fiber Blanket Thermal Insulation for Commercial and Industrial Applications     |
| ASTM C612         | (2014; R 2019) Standard Specification for Mineral Fiber Block and Board Thermal Insulation                                        |
| ASTM C920         | (2018) Standard Specification for Elastomeric Joint Sealants                                                                      |
| ASTM D522/D522M   | (2017) Mandrel Bend Test of Attached Organic Coatings                                                                             |
| ASTM D523         | (2014; R 2018) Standard Test Method for Specular Gloss                                                                            |
| ASTM D714         | (2002; R 2017) Standard Test Method for Evaluating Degree of Blistering of Paints                                                 |
| ASTM D822         | (2013; R 2018) Filtered Open-Flame Carbon-Arc Exposures of Paint and Related Coatings                                             |
| ASTM D968         | (2022) Standard Test Methods for Abrasion Resistance of Organic Coatings by Falling Abrasive                                      |
| ASTM D1056        | (2020) Standard Specification for Flexible Cellular Materials - Sponge or Expanded Rubber                                         |
| ASTM D1308        | (2002; R 2013) Effect of Household Chemicals on Clear and Pigmented Organic Finishes                                              |
| ASTM D1621        | (2016) Standard Test Method for Compressive Properties of Rigid Cellular Plastics                                                 |
| ASTM D1622/D1622M | (2014) Apparent Density of Rigid Cellular Plastics                                                                                |
| ASTM D1667        | (2022) Standard Specification for Flexible Cellular Materials - Poly (Vinyl Chloride) Foam (Closed-Cell)                          |
| ASTM D2244        | (2021) Standard Practice for Calculation of Color Tolerances and Color Differences from Instrumentally Measured Color Coordinates |

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

|            |                                                                                                                                                                                 |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM D2247 | (2015; R 2020) Standard Practice for Testing Water Resistance of Coatings in 100% Relative Humidity                                                                             |
| ASTM D2794 | (1993; R 2019) Standard Test Method for Resistance of Organic Coatings to the Effects of Rapid Deformation (Impact)                                                             |
| ASTM D3363 | (2005; E 2011; R 2011; E 2012) Film Hardness by Pencil Test                                                                                                                     |
| ASTM D4214 | (2007; R 2015) Standard Test Method for Evaluating the Degree of Chalking of Exterior Paint Films                                                                               |
| ASTM D6226 | (2021) Standard Test Method for Open Cell Content of Rigid Cellular Plastics                                                                                                    |
| ASTM E84   | (2023) Standard Test Method for Surface Burning Characteristics of Building Materials                                                                                           |
| ASTM E119  | (2024) Standard Test Methods for Fire Tests of Building Construction and Materials                                                                                              |
| ASTM E136  | (2024b) Standard Test Method for Assessing Combustibility of Materials Using a Vertical Tube Furnace at 750 Degrees C                                                           |
| ASTM E283  | (2019) Standard Test Method for Determining the Rate of Air Leakage Through Exterior Windows, Curtain Walls, and Doors Under Specified Pressure Differences Across the Specimen |
| ASTM E331  | (2000; R 2023) Standard Test Method for Water Penetration of Exterior Windows, Skylights, Doors, and Curtain Walls by Uniform Static Air Pressure Difference                    |
| ASTM E1592 | (2017) Standard Test Method for Structural Performance of Sheet Metal Roof and Siding Systems by Uniform Static Air Pressure Difference                                         |
| ASTM G152  | (2013; R 2021) Standard Practice for Operating Open Flame Carbon Arc Light Apparatus for Exposure of Nonmetallic Materials                                                      |
| ASTM G153  | (2013; R 2021) Standard Practice for Operating Enclosed Carbon Arc Light Apparatus for Exposure of Nonmetallic Materials                                                        |

#### METAL BUILDING MANUFACTURERS ASSOCIATION (MBMA)

|           |                                      |
|-----------|--------------------------------------|
| MBMA MBSM | (2018) Metal Building Systems Manual |
|-----------|--------------------------------------|



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

# NATIONAL ASSOCIATION OF ARCHITECTURAL METAL MANUFACTURERS (NAAMM)

NAAMM AMP 500 (2006) Metal Finishes Manual

## SHEET METAL AND AIR CONDITIONING CONTRACTORS' NATIONAL ASSOCIATION (SMACNA)

SMACNA 1793 (2012) Architectural Sheet Metal Manual, 7th Edition

## UNDERWRITERS LABORATORIES (UL)

UL 580 (2006; Reprint Apr 2024) UL Standard for Safety Tests for Uplift Resistance of Roof Assemblies

UL Bld Mat Dir (updated continuously online) Building Materials Directory

### 1.2 DEFINITIONS

Fabricated Wall Panel Assembly: Metal wall and liner panels, attachment system components, miscellaneous metal framing, thermal insulation, and accessories shop fabricated or field assembled for a complete weather-tight wall system.

### 1.3 DESCRIPTION OF FABRICATED WALL PANEL ASSEMBLY SYSTEM

Factory color finished, galvanized metal wall panel system with concealed fastening attachment. Panel profile must be as shown on drawings.

#### 1.3.1 Metal Wall Panel General Performance

Comply with performance requirements, conforming to AISI S100, without failure due to defective manufacture, fabrication, installation, or other defects in construction. Wall panels and accessory components must conform to the following standards:

ASTM A1008/A1008M  
ASTM A123/A123M  
ASTM A36/A36M  
ASTM A653/A653M  
ASTM A606/A606M  
ASTM C273/C273M  
ASTM D522/D522M for applied coatings  
UL Bld Mat Dir

#### 1.3.2 Structural Performance

Maximum calculated fiber stress must not exceed the allowable value in the AISI or AA manuals; a one third overstress for wind is allowed. Midspan deflection under maximum design loads is limited to L/180. Contract drawings show the design wind loads and the extent and general assembly details of the metal siding. Contractor must provide design for members and connections not shown on the drawings. Siding panels and accessories must be the products of the same manufacturer.

Provide metal wall panel assemblies complying with the load and stress

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

requirements in accordance with ASTM E1592. Wind Load force due to wind action governs the design for panels.

Wall systems and attachments are to resist the wind loads as determined by UL 580 and ASCE 7-16 in the geographic area where the construction will take place, in pounds per square foot.

#### 1.3.3 Air Infiltration

Air leakage must conform to the limits through the wall assembly area when tested according to ASTM E283.

#### 1.3.4 Water Penetration Under Static Pressure

No water penetration when tested according to ASTM E331.

#### 1.3.5 Water Penetration Under Dynamic Pressure

No evidence of water leakage when tested according to AAMA 501.1.

### 1.4 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

#### SD-01 Preconstruction Submittals

Qualification of Manufacturer; G

Qualification of Installer; G

#### SD-02 Shop Drawings

Fabrication and Installation drawings for the following items are to indicate completely dimensioned structural frame and erection layouts, openings in the wall, special framing details, and construction details at corners, building intersections and flashing, location and type of mastic and metal filler strips.

Wall Panel Assemblies; G

Flashing and Accessories

Anchorage Systems

#### SD-03 Product Data

Submit Manufacturer's catalog data for the following items:

Factory Color Finish; G

Sub-girts and Formed Shapes

Closure Materials

Insulation

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

Pressure Sensitive Tape

Sealants and Caulking

Rated Wall Assembly

Galvanizing Repair Paint

Accessories

#### SD-04 Samples

Submit as required each of the following samples:

Wall Panel Assemblies, 12 inches long by actual panel width; G

Fasteners

Metal Closure Strips, 10 inches long of each type

Insulation, approximately 8 by 11 inches

Submit manufacturer's color charts and chips, approximately 4 by 4 inches, showing full range of colors, textures and patterns available for wall panels with factory applied finishes.

#### SD-05 Design Data

Wind Design Analysis; G

#### SD-06 Test Reports

Submit test reports for the following in accordance with the referenced articles in this section.

Leakage Tests

Factory Color Finish Performance Requirements

#### SD-08 Manufacturer's Instructions

Installation of Wall Panels

Include detailed application instructions and standard manufacturer drawings altered as required by these specifications. Explicitly identify in writing, differences between manufacturer's instructions and the specified requirements.

#### SD-11 Closeout Submittals

Instructions To:

Government and/or Contractor Personnel

Safety Data Sheets

Submit 20 year "No-Dollar-Limit" Warranty for labor and materials.;  
G

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## 1.5 QUALITY ASSURANCE

### 1.5.1 Pre-Installation Conference

After submittals are received and approved but before wall panel and insulation work, including associated work, is performed, the Contracting Officer will hold a pre-siding conference to review the following:

- a. The drawings, including Fabrication and Installation drawings, showing complete Wall Panel Assemblies, and specifications.  
Include details for the following for review:

- flashing and accessories
- anchorage systems
- manufacturer's catalog data
- Factory Color Finish

Submit manufacturer's color charts and chips, approximately 4 by 4 inches, showing full range of colors, textures and patterns available for wall panels with factory applied finishes.

- Sub-girts and Formed Shapes
- Closure Materials, including metal closure strips.
- Insulation
- Pressure Sensitive Tape
- Rated Wall Assembly test data
- Accessories
- Fasteners

- b. Finalize construction schedule and verify availability of materials, Installer's personnel, equipment, and facilities needed to make progress and avoid delays.
- c. Methods and procedures related to metal wall panel installation, including manufacturer's written instructions for Installation of Wall panels, and verification of wall system assembly wind load and fire rating classification listings.
- d. Support conditions for compliance with requirements, including alignment between and attachment to structural members. Provide details of wind design analysis including wind speed, exposure category, co-efficient, importance factor, designates type of facility, negative pressures for each zone, methods and requirements of attachment. Wind design analysis to include wall plan delineating dimensions and attachment patterns for each zone. Wind design analysis to be prepared and sealed by Licensed Project Engineer in the geographic area where the construction will take place.
- e. Flashing, special siding details, wall penetrations, openings, and condition of other construction that will affect metal wall panels.
- f. Governing regulations and requirements for insurance, certificates, tests and inspections if applicable. Include certification for wall system assembly wind load and fire rating classification. Safety plan review must include applicable Safety Data Sheets for maintenance/repair materials.
- g. Temporary protection requirements for metal wall panel assembly during

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

and after installation.

- h. Wall panel observation and repair procedures after metal wall panel installation. Include review of sample Galvanizing Repair Paint.
- i. Sample 20 year "No-Dollar-Limit" warranty.

#### 1.5.2 Manufacturer's Technical Representative

The representative must have authorization from manufacturer to approve field changes and be thoroughly familiar with the products and installations in the geographical area where construction will take place.

#### 1.5.3 Qualification of Manufacturer

Metal wall panel system manufacturer must have:

- a. A minimum of five (5) years experience in manufacturing metal wall system and accessory products.
- b. Provide engineering services by an authorized engineer; currently licensed in the geographical area where construction will take place, having a minimum of four (4) years experience as an engineer knowledgeable in wind load design analysis, protocols and procedures for the MBMA MBSM; ASCE 7-16, and ASTM E1592.
- c. Provide certified engineering calculations using the products submitted for:

Wind load requirements in accordance with FM Wind Design Guide and ASCE 7-16.

#### 1.5.4 Qualification of Installer

The installation contractor must be approved and certified by the wall panel manufacturer prior to beginning the installation of the metal wall system.

#### 1.5.5 Single Source

Obtain each type of metal wall and liner panels, clips, closures and other accessories from the standard products of the single source from a single manufacturer to operate as a complete system for the intended use.

#### 1.5.6 Surface-Burning Characteristics

Provide metal wall panels having insulation core material with the following surface-burning characteristics as determined by testing identical products according to ASTM E84 by a qualified testing agency. Identify products with appropriate markings of applicable testing agency.

- a. Flame-Spread Index: 25 or less.
- b. Smoke-Developed Index: 450 or less.

#### 1.5.7 Fire-Resistance Ratings

Where indicated, provide metal wall panels identical to those of assemblies tested for fire resistance per ASTM E119 by a qualified testing agency. Identify products with appropriate markings of applicable testing

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

agency.

Indicate design designations from UL's "Fire Resistance Directory" or from the listings of another qualified testing agency.  
Combustion Characteristics: ASTM E136.

#### 1.5.8 Fabrication

Fabricate and finish metal wall panels and accessories at the factory to greatest extent possible, by manufacturer's standard procedures and processes and as necessary to fulfill indicated performance requirements. Comply with indicated profiles and with dimensional and structural requirements.

Provide panel profile, including major ribs and intermediate stiffening ribs, if any, for full length of panel.

Fabricate metal wall panel side laps with factory-installed captive gaskets or separator strips that provide a tight seal and prevent metal-to-metal contact, in a manner that will seal weather-tight and minimize noise from movements within panel assembly.

##### 1.5.8.1 Sheet Metal Accessories

Fabricate flashing and trim to comply with recommendations in SMACNA 1793 that apply to the design, dimensions, metal, and other characteristics of item indicated:

- a. Form exposed sheet metal accessories that are without excessive oil canning, buckling, and tool marks and that are true to line and levels indicated, with exposed edges folded back to form hems.
- b. End Seams: Fabricate nonmoving seams with flat-lock seams. Form seams and seal with epoxy seam sealer. Rivet joints for additional strength.
- c. Sealed Joints: Form non-expansion but movable joints in metal to accommodate elastomeric sealant to comply with SMACNA standards.
- d. Conceal fasteners and expansion provisions where possible. Exposed fasteners are not allowed on faces of accessories exposed to view.
- e. Fabricate cleats and attachment devices of size and metal thickness recommended by SMACNA or by metal wall panel manufacturer for application, but not less than thickness of metal being secured.

#### 1.5.9 Finishes

Comply with NAAMM AMP 500 for recommendations for applying and designating finishes.

Appearance of Finished Work: Noticeable variations in same piece are not acceptable. Variations in appearance of adjoining components are acceptable if they are within the range of approved Samples and are assembled or installed to minimize contrast.

#### 1.6 DELIVERY, HANDLING, AND STORAGE

Deliver and package components, sheets, metal wall panels, and other

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

manufactured items so as not to be damaged or deformed and protected during transportation and handling.

Unload, store, and erect metal wall panels in a manner to prevent bending, warping, twisting, and surface damage.

Stack and store metal wall panels horizontally on platforms or pallets, covered with suitable weather-tight and ventilated covering to ensure dryness, with positive slope for drainage of water. Do not store metal wall panels in contact with other materials that might cause staining, denting, or other surface damage.

Retain strippable protective covering on metal wall panel for period of metal wall panel installation.

Protect foam-plastic insulation as follows:

- a. Do not expose to sunlight, except to extent necessary for period of installation and concealment.
- b. Protect against ignition at all times. Do not deliver foam-plastic insulation materials to Project site before installation time.

Complete installation and concealment of plastic materials as rapidly as possible in each area of construction.

#### 1.7 PROJECT CONDITIONS

Weather Limitations: Proceed with installation preparation only when existing and forecasted weather conditions permit work to proceed without water entering into existing walling system or building.

Field Measurements: Verify locations of wall framing and opening dimensions by field measurements before metal wall panel fabrication and indicate measurements on Shop Drawings.

#### 1.8 WARRANTY

Furnish manufacturer's no-dollar-limit warranty for the metal wall panel system. The warranty period is to be no less than twenty (20) years from the date of Government acceptance of the work. The warranty is to be issued directly to the Government. The warranty is to provide that if within the warranty period the metal wall panel system shows evidence of corrosion, perforation, rupture or excess weathering due to deterioration of the wall panel system resulting from defective materials and correction of the defective workmanship is to be the responsibility of the metal wall panel system manufacturer. Repairs that become necessary because of defective materials and workmanship while metal wall panel system is under warranty are to be performed within 24 hours after notification, unless additional time is approved by the Contracting Officer. Failure to perform repairs within 24 hours of notification will constitute grounds for having emergency repairs performed by others and not void the warranty.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## PART 2 PRODUCTS

### 2.1 PANEL MATERIALS

#### 2.1.1 Steel Sheet

Roll-form steel wall and liner panels to the specified profile, with  $f_y = 22$  gauge and depth as indicated. Material must be plumb and true, and within the tolerances listed:

- a. Galvanized Steel Sheet conforming to ASTM A653/A653M and AISI SG03-3.
- c. Individual panels to have continuous length to cover the entire length of any unbroken wall area with no joints or seams and formed without warping, waviness, or ripples that are not part of the panel profile and free of damage to the finish coating system.
- d. Provide panels with thermal expansion and contraction consistent with the type of system specified. Provide complete insulated metal wall panel assembly incorporating trim, copings, fasciae, corners, soffits, inside and outside coreners, extrusions, and miscellaneous flahsings. Provide manufacturer's factory-formed concealed clips, shims, flashings, gaskets, lap tapes, closure strips, and caps for a complete installation.
  - 1. Profile as indicated on Drawings - 3 inch thick, concealed-fastener insulated metal wall panels, Color: As indciated on Drawings.
  - 2. Smooth, flat surface texture.

#### 2.1.2 Foam-Insulation Core Wall Panel

Provide factory-formed steel wall panel assembly fabricated from two sheets of metal with modified polyisocyanurate or polyurethane foam insulation core foamed-in-place during fabrication with joints between panels designed to form weather-tight seals. Include accessories required for weather-tight installation.

- a. Closed-Cell Content: 90 percent when tested according to ASTM D6226.
- b. Density: 2.0 to 2.6 lb/cu. ft. when tested according to ASTM D1622/D1622M.
- c. Compressive Strength: Minimum 20 psi when tested according to ASTM D1621.
- d. Shear Strength: 26 psi when tested according to ASTM C273/C273M.

#### 2.1.3 Insulated Panel Construction

Shop fabricate or field assemble insulated panel construction with specified exterior and interior steel sheet in accordance with manufacturer's printed instructions.

Insulation to be glass-fiber conforming to ASTM C553 and ASTM C612 of thickness and density as required for the geographical area where construction will take place. Glass-Fiber and Mineral-Wool-Fiber are materials listed in the EPA's Comprehensive Procurement Guidelines (CPG) (



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

<http://www.epa.gov/cpg/>).

Insulation fasteners to be adhesively attached, plate welded to projecting spindle anchors; capable of holding insulation of thickness indicated, secured in position with self-locking washer and complying with the following requirements:

- a. Plate: Perforated galvanized carbon-steel sheet, 0.030 inch thick by 2 inches square.
- b. Spindle: Copper-coated, low carbon steel; fully annealed; 0.105 inch in diameter; length to suit depth of insulation indicated.
- c. Insulation-Retaining Washers: Self-locking washers formed from 0.016 inch thick galvanized steel sheet, with beveled edge for increased stiffness, sized as required to hold insulation securely in place, but not less than 1-1/2 inches square or in diameter.
- d. Anchor adhesive to be a product with demonstrated capability to bond insulation anchors securely to substrates indicated without damaging insulation, fasteners, and substrates.

#### 2.1.4 Finish

All panels are to receive a factory-applied Kynar 500/Hylar 5000 finish consisting of a baked-on top-coat with a manufacturer's recommended prime coat conforming to the following:

- a. Metal Preparation: All metal is to have the surfaces carefully prepared for painting on a continuous process coil coating line by alkali cleaning, hot water rinsing, application of chemical conversion coating, cold water rinsing, sealing with acid rinse, and thorough drying.
- b. Prime Coating: A base coat of epoxy paint, specifically formulated to interact with the top-coat, is to be applied to the prepared surfaces by roll coating to a dry film thickness of 0.20 plus 0.05 mils. This prime coat must be oven cured prior to application of finish coat.
- c. Exterior Finish Coating: Apply the finish coating over the primer by roll coating to dry film thickness of 0.80 plus 5 mils (3.80 plus 0.50 mils for Vinyl Plastisol) for a total dry film thickness of 1.00 plus 0.10 mils (4.00 plus 0.10 mils for Vinyl Plastisol). This finish coat must be oven-cured.
- d. Interior Finish Coating: Apply a wash-coat on the reverse side over the primer by roll coating to a dry film thickness of 0.30 plus 0.05 mils for a total dry film thickness of 0.50 plus 0.10 mils. The wash-coat must be oven-cured.
- e. Color: The exterior finish chosen from the manufacturer's standard color chart.
- f. Physical Properties: Coating must conform to the industry and manufacturer's standard performance criteria as listed by the following certified test reports:

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

|                              |                                    |
|------------------------------|------------------------------------|
| Chalking:                    | ASTM D4214                         |
| Color Change and Conformity: | ASTM D2244                         |
| Weatherometer:               | ASTM G152, ASTM G153 and ASTM D822 |
| Humidity:                    | ASTM D2247 and ASTM D714           |
| Salt Spray:                  | ASTM B117                          |
| Chemical Pollution:          | ASTM D1308                         |
| Gloss at 60:                 | ASTM D523                          |
| Pencil Hardness:             | ASTM D3363                         |
| Reverse Impact:              | ASTM D2794                         |
| Flexibility:                 | ASTM D522/D522M                    |
| Abrasion:                    | ASTM D968                          |
| Flame Spread:                | ASTM E84                           |

## 2.2 MISCELLANEOUS METAL FRAMING

### 2.2.1 General

Cold-formed metallic-coated steel sheet conforming to ASTM A653/A653M and specified in Division 05 Section 05 40 00 "Cold-Formed Metal Framing" unless otherwise indicated.

## 2.3 FASTENERS

### 2.3.1 General

Type, material, corrosion resistance, size and sufficient length to penetrate the supporting member a minimum of 1 inch with other properties required to fasten miscellaneous metal framing members to substrates in accordance with the wall panel manufacturer's and ASCE 7-16 requirements.

### 2.3.2 Exposed Fasteners

Fasteners for wall panels to be corrosion resistant coated steel, stainless steel, or nylon capped steel compatible with the sheet panel or flashing and of a type and size recommended by the manufacturer to meet the performance requirements and design loads. Fasteners for accessories to be the manufacturer's standard. Provide an integral metal washer matching the color of attached material with compressible sealing EPDM gasket approximately 3/32 inches thick.

### 2.3.3 Screws

Screws to be corrosion resistant coated steel, and/or stainless steel

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

being the type and size recommended by the manufacturer to meet the performance requirements.

#### 2.3.4 Rivets

Rivets to be closed-end type, corrosion resistant coated steel, or stainless steel where watertight connections are required.

#### 2.3.5 Attachment Clips

Fabricate clips from Series 300 stainless steel. Size, shape, thickness and capacity as required meeting the insulation thickness and design load criteria specified.

### 2.4 ACCESSORIES

#### 2.4.1 General

All accessories to be compatible with the metal wall panels. Sheet metal flashing, trim, metal closure strips, caps and similar metal accessories must not be less than the minimum thickness specified for the wall panels. Exposed metal accessories/finishes to match the panels furnished, except as otherwise indicated. Molded foam rib, ridge and other closure strips to be non-absorbent closed-cell or solid-cell synthetic rubber or pre-molded neoprene to match configuration of the panels.

#### 2.4.2 Rubber Closure Strips

Closed-cell, expanded cellular rubber conforming to ASTM D1056 and ASTM D1667; extruded or molded to the configuration of the specified wall panel and in lengths supplied by the wall panel manufacturer.

#### 2.4.3 Metal Closure Strips

Factory fabricated steel closure strips to be the same thickness, color, finish and profile of the specified wall panel.

#### 2.4.4 Joint Sealants

##### 2.4.4.1 Sealants and Caulking

Sealants are to be an approved gun type for use in hand- or air-pressure caulking guns at temperatures above 40 degrees F (or frost-free application at temperatures above 10 degrees F) with minimum solid content of 85 percent of the total volume. Sealant is to dry with a tough, durable surface skin which permits it to remain soft and pliable underneath, providing a weather-tight joint. No migratory staining is permitted on painted or unpainted metal, stone, glass, vinyl, or wood.

Prime all joints to receive sealants with a compatible one-component or two-component primer as recommended by the wall panel manufacturer.

##### 2.4.4.2 Shop-Applied

Sealant for shop-applied caulking must be an approved gun grade, non-sag one component polysulfide or silicone conforming to ASTM C920, Type II, and with a curing time to ensure the sealant's plasticity at the time of field erection.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### 2.4.4.3 Field-Applied

Sealant for field-applied caulking must be an approved gun grade, non-sag one component polysulfide or two-component polyurethane with an initial maximum Shore A durometer hardness of 25, and conforming to ASTM C920, Type II. Color to match panel colors.

#### 2.4.4.4 Tape Sealant

Pressure sensitive, 100 percent solid with a release paper backing; permanently elastic, non-sagging, non-toxic and non-staining as approved by the wall panel manufacturer.

### 2.5 SHEET METAL FLASHING AND TRIM

#### 2.5.1 Fabrication

Shop fabricate sheet metal flashing and trim where practicable to comply with recommendations in SMACNA 1793 that apply to design, dimensions, metal, and other characteristics of item indicated. Obtain field measurements for accurate fit before shop fabrication.

Fabricate sheet metal flashing and trim without excessive oil canning, buckling, and tool marks and true to line and levels indicated, with exposed edges folded back to form hems.

### 2.6 REPAIR OF FINISH PROTECTION

Repair paint for color finish enameled wall panel must be compatible paint of the same formula and color as the specified finish furnished by the wall panel manufacturer.

## PART 3 EXECUTION

### 3.1 EXAMINATION

A. Examine substrates, areas, and conditions, with Installer present, for compliance with requirements for installation tolerances, metal wall panel supports, and other conditions affecting performance of the Work.

B. Examine primary and secondary wall framing to verify that rafters, purlins, angles, channels, and other structural panel support members and anchorages have been installed within alignment tolerances required by metal wall panel manufacturer, UL, ASTM, ASCE 7-16 and as required for the geographical area where construction will take place.

C. Examine solid wall sheathing to verify that sheathing joints are supported by framing or blocking and that installation is within flatness tolerances required by metal wall panel manufacturer.

D. Examine roughing-in for components and systems penetrating metal wall panels to verify actual locations of penetrations relative to seam locations of metal wall panels before metal wall panel installation.

E. Submit to the Contracting Officer a written report, endorsed by Installer, listing conditions detrimental to performance of the Work.

F. Proceed with installation only after unsatisfactory conditions have been corrected.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 3.2 PREPARATION

- A. Clean substrates of substances harmful to insulation, including removing projections capable of interfering with insulation attachment.
- B. Miscellaneous Framing: Install sub-purlins, girts, angles, furring, and other miscellaneous wall panel support members and anchorage according to metal wall panel manufacturer's written instructions.

### 3.3 WALL PANEL INSTALLATION

Provide metal wall panels of full length from sill to eave as indicated, unless otherwise indicated or restricted by shipping limitations. Anchor metal wall panels and other components of the Work securely in place, with provisions for thermal and structural movement in accordance with MBMA Metal Building Systems Manual.

- a. Steel Wall Panels: Use stainless-steel fasteners for exterior surfaces and galvanized steel fasteners for interior surfaces.
- d. Metal Protection: Where dissimilar metals will contact each other or corrosive substrates, protect against galvanic action by painting contact surfaces with bituminous coating, by applying rubberized-asphalt underlayment to each contact surface, or by other permanent separation as recommended by metal wall panel manufacturer.
- e. Joint Sealers: Install gaskets, joint fillers, and sealants where indicated and where required for weatherproof performance of metal wall panel assemblies. Provide types of gaskets, fillers, and sealants indicated or, if not indicated, types recommended by metal wall panel manufacturer.

Erect wall panel system in accordance with the approved erection drawings, the printed instructions and safety precautions of the manufacturer.

Sheets are not to be subjected to overloading, abuse, or undue impact. Bent, chipped, or defective sheets shall not be applied.

Sheets must be erected true and plumb and in exact alignment with the horizontal and vertical edges of the building, securely anchored, and with the indicated eave, and sill.

Work is to allow for thermal movement of the wall panel, movement of the building structure, and to provide permanent freedom from noise due to wind pressure.

Field cutting metal wall panels by torch is not permitted.

### 3.4 FASTENER INSTALLATION

Anchor metal wall panels and other components of the Work securely in place, using manufacturer's approved fasteners according to manufacturer's written instructions.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 3.5 FLASHING, TRIM AND CLOSURE INSTALLATION

#### 3.5.1 General Requirements

Comply with performance requirements, manufacturer's written installation instructions, and SMACNA 1793. Provide concealed fasteners where possible, and set units true to line and level as indicated. Install work with laps, joints, and seams that will be permanently watertight and weather resistant.

Sheet metalwork is to be accomplished to form weather-tight construction without waves, warps, buckles, fastening stresses or distortion, and allow for expansion and contraction. Cutting, fitting, drilling, and other operations in connection with sheet metal required to accommodate the work of other trades is to be performed by sheet metal mechanics.

#### 3.5.2 Metal Flashing

Exposed metal flashing is to be installed at building corners, sills and eaves, junctions between metal siding and walling.

Exposed metal flashing is to be the same material, color, and finish as the specified metal wall panel.

Flashing is to be fastened at not more than 8 inches on center, except where flashing are held in place by the same screws that secure covering sheets.

Flashing is to be furnished in at least 8 foot lengths. Exposed flashing is to have 1 inch locked and blind-soldered end joints, and expansion joints at intervals of not more than 16 feet.

Exposed flashing and flashing subject to rain penetration to be bedded in the specified joint sealant.

Flashing which is in contact with dissimilar metals to be isolated by means of the specified asphalt mastic material to prevent electrolytic deterioration.

Drips to be formed to the profile indicated, with the edge folded back 1/2 inch to form a reinforced drip edge.

#### 3.5.3 Closures

Install metal closure strips at open ends of corrugated or ribbed pattern walls, and at intersection of wall and wall unless open ends are concealed with formed eave flashing; and in other required areas.

Install mastic closure strips at intersection of the wall with metal walling; top and bottom of metal siding; heads of wall openings; and in other required locations.

### 3.6 WORKMANSHIP

Make lines, arises, and angles sharp and true. Free exposed surfaces from visible wave, warp, buckle, and tool marks. Fold back exposed edges neatly to form a 1/2 inch hem on the concealed side. Make sheet metal exposed to the weather watertight with provisions for expansion and contraction.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

Make surfaces to receive sheet metal plumb and true, clean, even, smooth, dry, and free of defects and projections which might affect the application. For installation of items not shown in detail or not covered by specifications conform to the applicable requirements of SMACNA 1793. Provide sheet metal flashing in the angles formed where roof decks abut walls, curbs, ventilators, pipes, or other vertical surfaces and wherever indicated and necessary to make the work watertight.

### 3.7 ACCEPTANCE PROVISIONS

#### 3.7.1 Erection Tolerances

Erect metal wall panels straight and true with plumb vertical lines correctly lapped and secured in accordance with the manufacturer's written instructions. Horizontal lines must not vary more than 1/8 inch in 40 feet.

#### 3.7.2 Leakage Tests

Inspect and test finished application of metal wall panels when directed to do so by the Contracting Officer. Inspection and tests must be conducted without cost to the Government.

Inspection and testing is to be made promptly after erection to permit correction of defects and the removal and replacement of defective materials.

#### 3.7.3 Repairs to Finish

Scratches, abrasions, and minor surface defects of finish may be repaired with the specified repair materials. Finished repaired surfaces must be uniform and free from variations of color and surface texture.

Repaired metal surfaces that are not acceptable to the project requirements are to be immediately removed and replaced with new material.

#### 3.7.4 Paint-Finish Metal Siding

Paint-finish metal siding will be tested for color stability by the Contracting Officer during the manufacturer's specified guarantee period.

Panels that indicate color changes, fading, or surface degradation, determined by visual examination, must be removed and replaced with new panels at no expense to the Government.

New panels will be subject to the specified tests for an additional year from the date of their installation.

### 3.8 CLEAN-UP AND DISPOSAL

Clean all exposed sheet metal work at completion of installation. Remove metal shavings, filings, nails, bolts, and wires from work area. Remove grease and oil films, excess sealants, handling marks, contamination from steel wool, fittings and drilling debris and scrub the work clean. Exposed metal surfaces to be free of dents, creases, waves, scratch marks, solder or weld marks, and damage to the finish coating.

Collect and place scrap/waste materials in containers. Promptly dispose

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

of demolished materials. Do not allow demolished materials to accumulate on-site; transport demolished materials from government property and legally dispose of them.

-- End of Section --



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

SECTION 07 60 00

FLASHING AND SHEET METAL  
**08/23**

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

|                 |                                                                                                                                         |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| ASTM A480/A480M | (2023a) Standard Specification for General Requirements for Flat-Rolled Stainless and Heat-Resisting Steel Plate, Sheet, and Strip      |
| ASTM A653/A653M | (2023) Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process |
| ASTM A792/A792M | (2022) Standard Specification for Steel Sheet, 55% Aluminum-Zinc Alloy-Coated by the Hot-Dip Process                                    |
| ASTM B32        | (2020) Standard Specification for Solder Metal                                                                                          |
| ASTM B69        | (2021) Standard Specification for Rolled Zinc                                                                                           |
| ASTM B209/B209M | (2021a) Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate                                                          |
| ASTM B221       | (2021) Standard Specification for Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes                            |
| ASTM D41/D41M   | (2011; R 2016) Standard Specification for Asphalt Primer Used in Roofing, Dampproofing, and Waterproofing                               |
| ASTM D226/D226M | (2017) Standard Specification for Asphalt-Saturated Organic Felt Used in Roofing and Waterproofing                                      |
| ASTM D1784      | (2020) Standard Specification for Rigid Poly(Vinyl Chloride) (PVC) Compounds and Chlorinated Poly(Vinyl Chloride) (CPVC) Compounds      |
| ASTM D2244      | (2021) Standard Practice for Calculation of Color Tolerances and Color Differences from Instrumentally Measured Color                   |

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### Coordinates

|                   |                                                                                                   |
|-------------------|---------------------------------------------------------------------------------------------------|
| ASTM D4214        | (2007; R 2015) Standard Test Method for Evaluating the Degree of Chalking of Exterior Paint Films |
| ASTM D4586/D4586M | (2007; R 2018) Asphalt Roof Cement, Asbestos-Free                                                 |
| ASTM E2112        | (2023) Standard Practice for Installation of Exterior Windows, Doors and Skylights                |

#### NATIONAL ROOFING CONTRACTORS ASSOCIATION (NRCA)

|           |                                                                                                                  |
|-----------|------------------------------------------------------------------------------------------------------------------|
| NRCA 0429 | (2022) The NRCA Roofing Manual: Architectural Metal Flashing, Condensation and Air Leakage Control and Reroofing |
|-----------|------------------------------------------------------------------------------------------------------------------|

#### SHEET METAL AND AIR CONDITIONING CONTRACTORS' NATIONAL ASSOCIATION (SMACNA)

|             |                                                      |
|-------------|------------------------------------------------------|
| SMACNA 1793 | (2012) Architectural Sheet Metal Manual, 7th Edition |
|-------------|------------------------------------------------------|

#### SINGLE PLY ROOFING INDUSTRY (SPRI)

|                |                                                 |
|----------------|-------------------------------------------------|
| ANSI/SPRI RD-1 | (2019) Performance Standard for Retrofit Drains |
|----------------|-------------------------------------------------|

## 1.2 GENERAL REQUIREMENTS

Finished sheet metal assemblies must form a weathertight enclosure without waves, warps, buckles, fastening stresses or distortion, while allowing for expansion and contraction without damage to the system. The sheet metal installer is responsible for cutting, fitting, drilling, and other operations in connection with sheet metal modifications required to accommodate the work of other trades. Coordinate installation of sheet metal items used in conjunction with roofing with roofing work to permit continuous, uninterrupted roofing operations.

### 1.2.1 General Material Requirements

All materials specified in this Section installed in conjunction with the roofing system must be provided by the roofing system manufacturer, or by a manufacturer approved by the roofing system manufacturer for use in the roofing system, and must form a part of the Warranty as required by the applicable roofing system Section.

## 1.3 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Exposed Sheet Metal Coverings; FIO

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

Expansion Joints; FIO

Flashing for Roof Drains; FIO

Base Flashing; FIO

Counterflashing; FIO

Flashing at Roof Penetrations and Equipment Supports; FIO

Reglets; FIO

Copings; FIO

#### SD-04 Samples

Finish Samples; G

#### SD-07 Certificates

Warranty on Finishes; G

#### SD-08 Manufacturer's Instructions

Instructions for Installation;

#### SD-10 Operation and Maintenance Data

Cleaning and Maintenance; G

### 1.4 MISCELLANEOUS REQUIREMENTS

#### 1.4.1 Product Data

Indicate thicknesses, dimensions, fastenings, anchoring methods, expansion joints, and other provisions necessary for thermal expansion and contraction. Scaled manufacturer's catalog data may be submitted for factory fabricated items.

#### 1.4.2 Finish Samples

Submit two color charts and two finish sample chips from manufacturer's standard color and finish options for each type of finish indicated.

#### 1.4.3 Operation and Maintenance Data

Submit detailed instructions for installation and quality control during installation, cleaning and maintenance, for each type of assembly indicated.

### 1.5 DELIVERY, HANDLING, AND STORAGE

Package and protect materials during shipment. Uncrate and inspect materials for damage, dampness, and wet-storage stains upon delivery to the job site. Remove from the site and replace damaged materials that cannot be restored to like-new condition. Handle sheet metal items to avoid damage to surfaces, edges, and ends. Store materials in dry, weather-tight, ventilated areas until installation.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## PART 2 PRODUCTS

### 2.1 MATERIALS

Use any metal listed by NRCA 0429 or SMACNA 1793 for a particular item, unless otherwise indicated. Provide materials and configurations in accordance with NRCA 0429 or SMACNA 1793 for each material, while also meeting the minimum thickness requirements specified in this Section. Different items need not be of the same metal, except that contact between dissimilar metals must be avoided.

Provide sheet metal items in 8 to 10 foot lengths. Single pieces less than 8 feet long may be used to connect to factory-fabricated inside and outside corners, and at ends of runs. Factory fabricate corner pieces with minimum 12 inch legs. Provide accessories and other items essential to complete the sheet metal installation. Provide accessories made of the same or compatible materials as the items to which they are applied. Provide sheet metal items with mill finish unless specified otherwise.

#### 2.1.1 Exposed Sheet Metal Items

Provide exposed sheet metal items of the same material. Consider the following as exposed sheet metal: cap, valley, steeped, base, and eave flashings and related accessories.

#### 2.1.2 Steel Sheet, Zinc-Coated (Galvanized)

Provide in accordance with ASTM A653/A653M, a minimum of 24 gauge, and a minimum zinc coat weighting of 90 ounces per square feet.

#### 2.1.3 Zinc Sheet and Strip

Provide in accordance with ASTM B69, Type I, a minimum of 0.024 inch thick.

#### 2.1.4 Steel Sheet, Aluminum Zinc-Coated

Provide in accordance with ASTM A792/A792M, a minimum of 24 gauge.

#### 2.1.5 Stainless Steel

Provide in accordance with ASTM A480/A480M, Type 302 or 304, 2D Finish, fully annealed, dead-soft temper, a minimum of 24 gauge.

#### 2.1.6 Aluminum Alloy Sheet and Plate

Provide in accordance with ASTM B209/B209M anodized clear form alloy, and temper appropriate for use. Provide material not less than 0.032-in in thickness.

##### 2.1.6.1 Alclad

When fabricated of aluminum, fabricate the following items with Alclad 3003, Alclad 3004, or Alclad 3005, clad on one side unless otherwise indicated.

##### a. Flashing

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### 2.1.7 Finishes

Provide exposed exterior sheet metal and aluminum with a baked on, factory applied coil fluoropolymer color coating or approved equal siliconized polyester coating. Dry film thickness of coatings must be 0.8 to 1.3 mils. Color to be selected from manufacturer's standard range of color choices. Field applications of color coatings are prohibited and will be rejected.

##### 2.1.7.1 Warranty on Finishes

Provide a manufacturer's warranty to repair, or replace, sheet metal flashing and trim that shows evidence of deterioration of factory-applied finishes within 20 years from date of project substantial completion. Deterioration includes the following:

- a. Color fading more than 5 Delta E units when tested in accordance with ASTM D2244.
- b. Chalking in excess of a No. 8 rating when tested in accordance with ASTM D4214.
- c. Cracking, checking, peeling, or failure of paint to adhere to bare metal.

#### 2.1.8 Aluminum Alloy, Extruded Bars, Rods, Shapes, and Tubes

ASTM B221.

#### 2.1.9 Solder

Provide in accordance with ASTM B32, 95-5 tin-antimony.

#### 2.1.10 Reglets

##### 2.1.10.1 Polyvinyl Chloride Reglets

Provide in accordance with ASTM D1784, Type II, Grade 1, Class 14333-D, 0.075 inch minimum thickness.

##### 2.1.10.2 Metal Reglets

Provide factory fabricated caulked type or friction type reglets with a minimum opening of 1/4 inch and a depth of 1-1/4 inch, as approved.

###### 2.1.10.2.1 Caulked Reglets

Provide with rounded edges, temporary reinforcing cores, and accessories as required for securing to adjacent construction. Provide built-up mitered corner pieces for inside and outside corners.

###### 2.1.10.2.2 Friction Reglets

Provide with flashing receiving slots not less than 5/8 inch deep, one inch jointing tongues, and upper and lower anchoring flanges installed at 24 inch maximum snap-lock type receiver.

#### 2.1.11 Splash Pans

Provide splash pans where downspouts discharge onto roof surfaces and at

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

locations indicated. Unless otherwise indicated, provide pans not less than 24 inches long by 18 inches wide with metal ribs across bottoms of pans. Provide sides of pans with vertical baffles not less than one inch high in the front, and 4 inches high in the back. Ground splash pans to be precast concrete.

#### 2.1.12 Copings

Unless otherwise indicated, provide copings in copper sheets, 8 or 10 feet long, joined by a 3/4 inch locked and soldered seam.

#### 2.1.13 Bituminous Plastic Cement

Provide in accordance with ASTM D4586/D4586M, Type I.

#### 2.1.14 Roofing Felt

Provide in accordance with ASTM D226/D226M Type I.

#### 2.1.15 Asphalt Primer

Provide in accordance with ASTM D41/D41M.

#### 2.1.16 Fasteners

Use the same metal as, or a metal compatible with the item fastened. Confirm compatibility of fasteners and items to be fastened to avoid galvanic corrosion due to dissimilar materials.

### PART 3 EXECUTION

#### 3.1 INSTALLATION

##### 3.1.1 Workmanship

Make lines and angles sharp and true. Free exposed surfaces from visible wave, warp, buckle, and tool marks. Fold back exposed edges neatly to form a 1/2 inch hem on the concealed side. Make sheet metal exposed to the weather watertight with provisions for expansion and contraction.

Make surfaces to receive sheet metal plumb and true, clean, even, smooth, dry, and free of defects and projections. For installation of items not shown in detail or not covered by specifications conform to the applicable requirements of SMACNA 1793, Architectural Sheet Metal Manual. Provide sheet metal flashing in the angles formed where roof decks abut walls, curbs, ventilators, pipes, or other vertical surfaces and wherever indicated and necessary to make the work watertight.

##### 3.1.2 Cleats

Provide cleats for sheet metal 18 inches and over in width. Space cleats evenly not over 12 inches on center unless otherwise specified or indicated. Unless otherwise specified, provide cleats of 2 inches wide by 3 inches long and of the same material and thickness as the sheet metal being installed. Secure one end of the cleat with two nails and the cleat folded back over the nailheads. Lock the other end into the seam. Pre-tin cleats for soldered seams.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 3.1.3 Bolts, Rivets, and Screws

Install bolts, rivets, and screws where indicated or required. Provide compatible washers where required to protect surface of sheet metal and to provide a watertight connection. Provide mechanically formed joints in aluminum sheets 0.040 inches or less in thickness.

### 3.1.4 Seams

Straight and uniform in width and height with no solder showing on the face.

### 3.1.5 Mechanical Fastening

#### 3.1.5.1 Mechanical Fastening of Aluminum

Use No. 12, aluminum alloy, sheet metal screws or other suitable aluminum alloy or stainless steel fasteners. Drive fasteners in holes made with a No. 26 drill in securing side laps, end laps, and flashings. Space fasteners 12 inches maximum on center. Where end lap fasteners are required to improve closure, locate the end lap fasteners not more than 2 inches from the end of the overlapping sheet.

### 3.1.6 Protection from Contact with Dissimilar Materials

#### 3.1.6.1 Copper or Copper-bearing Alloys

Paint with heavy-bodied bituminous paint surfaces in contact with dissimilar metal, or separate the surfaces by means of moistureproof building felts.

#### 3.1.6.2 Aluminum

Do not allow aluminum surfaces in direct contact with other metals except stainless steel, zinc, or zinc coating. Where aluminum contacts another metal, paint the dissimilar metal with a primer followed by two coats of aluminum paint. Where drainage from a dissimilar metal passes over aluminum, paint the dissimilar metal with a non-lead pigmented paint.

#### 3.1.6.3 Metal Surfaces

Paint surfaces in contact with mortar, concrete, or other masonry materials with alkali-resistant coatings such as heavy-bodied bituminous paint.

#### 3.1.6.4 Wood or Other Absorptive Materials

Paint surfaces that may become repeatedly wet and in contact with metal with two coats of aluminum paint or a coat of heavy-bodied bituminous paint.

### 3.1.7 Expansion and Contraction

Provide expansion and contraction joints at not more than 32 foot intervals for aluminum and at not more than 40 foot intervals for other metals. Provide an additional joint where the distance between the last expansion joint and the end of the continuous run is more than half the required interval. Space joints evenly. Join extruded aluminum gravel stops and fascia by expansion and contraction joints spaced not more than

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

12 feet apart.

### 3.1.8 Base Flashing

Extend up vertical surfaces of the flashing not less than 8 inches and not less than 4 inches under the roof covering. Where finish wall coverings form a counterflashing, extend the vertical leg of the flashing up behind the applied wall covering not less than 6 inches. Overlap the flashing strips with the previously laid flashing not less than 3 inches. Fasten the strips at their upper edge to the deck. Horizontal flashing at vertical surfaces must extend vertically above the roof surface and fastened at their upper edge to the deck a minimum of 6 inches on center with a minimum of 2 inch lap of any surface. Solder end laps and provide for expansion and contraction. Extend the metal flashing over crickets at the up-slope side of curbs, and similar vertical surfaces extending through sloping roofs, the metal flashings. Extend the metal flashings onto the roof covering not less than 4.5 inches at the lower side of vertical surfaces extending through the roof decks. Install and fit the flashings so as to be completely weathertight. Provide factory-fabricated base flashing for interior and exterior corners. Do not use metal base flashing on built-up roofing.

### 3.1.9 Counterflashing

Except where indicated or specified otherwise, insert counterflashing in reglets located from 9 to 10 inches above roof decks, extend down vertical surfaces over upturned vertical leg of base flashings not less than 3 inches. Fold the exposed edges of counterflashings 1/2 inch. Where stepped counterflashings are required, they may be installed in short lengths a minimum 8 inches by 8 inches or may be of the preformed single piece type. Provide end laps in counterflashings not less than 3 inches and make it weathertight with plastic cement. Do not make lengths of metal counterflashings exceed 10 feet. Form flashings to the required shapes before installation. Factory form corners not less than 12 inches from the angle. Secure the flashings in the reglets with lead wedges and space not more than 18 inches apart; on chimneys and short runs, place wedges closer together. Fill caulked-type reglets or raked joints which receive counterflashing with caulking compound. Turn up the concealed edge of counterflashings built into masonry or concrete walls not less than 1/4 inch and extend not less than 2 inches into the walls. Install counterflashing to provide a spring action against base flashing. Utilize two piece counterflashing components to minimize damage of counterflashing and reglets during future reroofing activities.

### 3.1.10 Metal Reglets

Keep temporary cores in place during installation. Ensure factory fabricated caulked type or friction type, reglets have a minimum opening of 1/4 inch and a minimum depth of 1-1/4 inch, when installed.

#### 3.1.10.1 Caulked Reglets

Wedge flashing in reglets with lead wedges every 18 inches, caulked full and solid with an approved compound.

#### 3.1.10.2 Friction Reglets

Install flashing snap lock receivers at 24 inches on center maximum. When flashing has been inserted the full depth of the slot, caulk the slot, lock,



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

and fill with sealant.

### 3.1.11 Polyvinyl Chloride Reglets for Temporary Construction

Rigid polyvinyl chloride reglets may be provided in lieu of metal reglets for temporary construction.

### 3.1.12 Gravel Stops and fascia

Prefabricate in the shapes and sizes indicated and in lengths not less than 8 feet. Extend flange at least 4 inches onto roofing. Provide prefabricated, mitered corners internal and external corners. Install gravel stops and fascia after all plies of the roofing membrane have been applied, but before the flood coat of bitumen is applied. Prime roof flange of gravel stops and fascia on both sides with an asphalt primer. After primer has dried, set flange on roofing membrane and strip-in. Nail flange securely to wood nailer with large-head, barbed-shank roofing nails 1.5 inch long spaced not more than 3 inches on center, in two staggered rows.

#### 3.1.12.1 Edge Strip

Hook the lower edge of fascia at least 3/4 inch over a continuous strip of the same material bent outward at an angle not more than 45 degrees to form a drip. Nail hook strip to a wood nailer at 6 inches maximum on center. Where fastening is made to concrete or masonry, use screws spaced 12 inches on center driven in expansion shields set in the concrete or masonry. Where horizontal wood nailers are slotted to provide for insulation venting, install strips to prevent obstruction of vent slots. Where necessary, install strips over 1/16 inch thick compatible spacer or washers.

#### 3.1.12.2 Joints

Leave open the section ends of gravel stops and fascia 1/4 inch and backed with a formed flashing plate, mechanically fastened in place and lapping each section end a minimum of 4 inches set laps in plastic cement. Face nailing is not permitted. Install prefabricated aluminum gravel stops and fascia in accordance with the manufacturer's printed instructions and details.

### 3.1.13 Flashing for Roof Drains

Provide a 30 inches square sheet indicated. Taper insulation to drain from 24 inches out. Set flashing on finished felts in a full bed of asphalt roof cement, ASTM D4586/D4586M. Heavily coat the drain flashing ring with asphalt roof cement. Clamp the roof membrane, flashing sheet, and stripping felt in the drain clamping ring. Secure clamps so that felts and drain flashing are free of wrinkles and folds. Retrofit roof drains must conform to ANSI/SPRI RD-1.

### 3.1.14 Flashing at Wall Openings

Install pan flashing in the rough opening sill at all penetrations in the exterior wall assemblies, such as windows, louvers, storefronts and curtain walls. Pan sill flashing must have end dams at both jambs a minimum of 2 in high and a rear dam of 2 in high. Flashing must comply with ASTM E2112 and SMACNA 1793.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 3.1.15 Expansion Joints

Provide expansion joints for roofs, walls, and floors as indicated.  
Provide.

### 3.1.16 Flashing at Roof Penetrations and Equipment Supports

Provide metal flashing for all pipes, ducts, and conduits projecting through the roof surface and for equipment supports, guy wire anchors, and similar items supported by or attached to the roof deck.

### 3.1.17 Single Pipe Vents

Bend the top of sleeve over and extend down into the vent pipe a minimum of 2 inches. For long runs or long rises above the deck, where it is impractical to cover the vent pipe with lead, use a two-piece formed metal housing. Set metal housing with a metal sleeve having a 4 inches roof flange in bituminous plastic cement and nailed 3 inches on center. Extend sleeve a minimum of 8 inches above the roof deck and lapped a minimum of 3 inches by a metal hood secured to the vent pipe by a draw band. Seal the area of hood in contact with vent pipe with an approved sealant.

### 3.1.18 Copings

Provide copings as indicated in the Drawings.

### 3.1.19 Through Wall Flashing

Provide through wall flashing as required in the applicable wall system Sections.

## 3.2 PAINTING

Touch ups in the field may be applied only after metal substrates have been cleaned and pretreated in accordance with manufacturer's written instructions and products.

Field-paint dissimilar sheet metals in contact to separate and deter galvanic interactions.

### 3.2.1 Aluminum Surfaces

Clean with solvent and apply one coat of zinc-molybdate primer and one coat of aluminum paint.

## 3.3 CLEANING

Clean exposed sheet metal work at completion of installation. Remove grease and oil films, handling marks, contamination from steel wool, fittings and drilling debris, and scrub-clean. Free the exposed metal surfaces of dents, creases, waves, scratch marks, and solder or weld marks.

## 3.4 REPAIRS TO FINISH

Scratches, abrasions, and minor surface defects of finish may be repaired in accordance with the manufacturer's printed instructions and as approved. Repair damaged surfaces caused by scratches, blemishes, and variations of color and surface texture. Replace items which cannot be repaired.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

-- End of Section --

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## SECTION 07 84 00

### FIRESTOPPING

**05/10, CHG 1: 08/13**

#### PART 1 GENERAL

##### 1.1 SUMMARY

Furnish and install tested and listed firestopping systems, combination of materials, or devices to form an effective barrier against the spread of flame, smoke and gases, and maintain the integrity of fire resistance rated walls, partitions, floors, and ceiling-floor assemblies, including through-penetrations and construction joints and gaps.

- a. Through-penetrations include the annular space around pipes, tubes, conduit, wires, cables and vents.
- b. Construction joints include those used to accommodate expansion, contraction, wind, or seismic movement; do not allow firestopping material to interfere with the required movement of the joint.

Gaps requiring firestopping include gaps between the curtain wall and the floor slab and between the top of the fire-rated walls and the roof or floor deck above and at the intersection of shaft assemblies and adjoining fire resistance rated assemblies.

##### 1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

#### ASTM INTERNATIONAL (ASTM)

|                   |                                                                                                                                           |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM E84          | (2023) Standard Test Method for Surface Burning Characteristics of Building Materials                                                     |
| ASTM E119         | (2022) Standard Test Methods for Fire Tests of Building Construction and Materials                                                        |
| ASTM E814         | (2013a; R 2017) Standard Test Method for Fire Tests of Penetration Firestop Systems                                                       |
| ASTM E1399/E1399M | (1997; R 2022) Standard Test Method for Cyclic Movement and Measuring the Minimum and Maximum Joint Widths of Architectural Joint Systems |
| ASTM E1966        | (2015; R 2019) Standard Test Method for Fire-Resistive Joint Systems                                                                      |
| ASTM E2307        | (2020) Standard Test Method for Determining Fire Resistance of Perimeter Fire Barrier Systems Using Intermediate-Scale, Multi-story Test  |

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### Apparatus

#### FM GLOBAL (FM)

FM 4991 (2013) Approval of Firestop Contractors  
FM APP GUIDE (updated on-line) Approval Guide  
<http://www.approvalguide.com/>

#### INTERNATIONAL CODE COUNCIL (ICC)

ICC IBC (2021) International Building Code

#### UNDERWRITERS LABORATORIES (UL)

UL 723 (2020) UL Standard for Safety Test for Surface Burning Characteristics of Building Materials  
UL 1479 (2015; Reprint May 2021) Fire Tests of Through-Penetration Firestops  
UL 2079 (2015; Reprint Jul 2020) Tests for Fire Resistance of Building Joint Systems  
UL Fire Resistance (2014) Fire Resistance Directory

### 1.3 SEQUENCING

Coordinate the specified work with other trades. Apply firestopping materials, at penetrations of pipes and ducts, prior to insulating, unless insulation meets requirements specified for firestopping. Apply firestopping materials, at building joints and construction gaps, prior to completion of enclosing walls or assemblies. Locate cast-in-place firestop devices and install in place before concrete placement. Install pipe, conduit or cable bundles through cast-in-place device after concrete placement but before area is concealed or made inaccessible. Firestop material must be inspected and approved prior to final completion and enclosing of any assemblies that may conceal installed firestop.

### 1.4 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Firestopping System; G

SD-03 Product Data

Firestopping Materials; G

SD-06 Test Reports

Inspection; G

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## SD-07 Certificates

Inspector Qualifications

Firestopping Materials

Installer Qualifications; G

### 1.5 QUALITY ASSURANCE

#### 1.5.1 Installer

Engage an experienced Installer who is:

- a. FM Research approved in accordance with FM 4991, operating as a UL Certified Firestop Contractor, or
- b. Certified, licensed, or otherwise qualified by the firestopping manufacturer as having the necessary staff, training, and a minimum of 3 years experience in the installation of manufacturer's products in accordance with specified requirements. Submit documentation of this experience. A manufacturer's willingness to sell its firestopping products to the Contractor or to an installer engaged by the Contractor does not in itself confer installer qualifications on the buyer. The Installer must be a trained representative of the manufacturer (not distributor or agent) in the proper selection and installation procedures. Obtain and submit installer's written certification of training, and retain proof of certification for duration of firestop installation.

#### 1.5.2 Inspector Qualifications

The inspector must have a minimum of two years experience in construction field inspections of firestopping systems, products, and assemblies. The inspector must be completely independent of, and divested from, the installer, the manufacturer, and the supplier of any material or item being inspected. The inspector must not be a competitor of the installer, the contractor, the manufacturer, or supplier of any material or item being inspected. Include in the qualifications submittal a notarized statement assuring compliance with the requirements stated herein.

### 1.6 DELIVERY, STORAGE, AND HANDLING

Deliver materials in the original unopened packages or containers showing name of the manufacturer and the brand name. Store materials off the ground, protected from damage and exposure to elements and temperatures in accordance with manufacturer requirements. Remove damaged or deteriorated materials from the site. Use materials within their indicated shelf life.

## PART 2 PRODUCTS

### 2.1 FIRESTOPPING SYSTEM

Submit detail drawings including manufacturer's descriptive data, typical details conforming to UL Fire Resistance or other details certified by another nationally recognized testing laboratory, installation instructions or UL listing details for a firestopping assembly in lieu of fire-test data or report. For those firestop applications for which no UL

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

tested system is available through a manufacturer, submit a manufacturer's engineering judgment, derived from similar UL system designs or other tests for review and approval prior to installation. Submittal must indicate the firestopping material to be provided for each type of application. When more than a total of 5 penetrations and/or construction joints are to receive firestopping, provide drawings that indicate location, "F" "T" and "L" ratings, and type of application.

Also, submit a written report indicating locations of and types of penetrations and types of firestopping used at each location; record type by UL list printed numbers.

## 2.2 FIRESTOPPING MATERIALS

Provide firestopping materials, supplied from a single domestic manufacturer, consisting of commercially manufactured, asbestos-free, nontoxic products FM APP GUIDE approved, or UL listed, for use with applicable construction and penetrating items, complying with the following minimum requirements:

### 2.2.1 Fire Hazard Classification

Provide material that has a flame spread of 25 or less, and a smoke developed rating of 50 or less, when tested in accordance with ASTM E84 or UL 723. Provide an approved firestopping material as listed in UL Fire Resistance or by a nationally recognized testing laboratory.

### 2.2.2 Toxicity

Provide material that is nontoxic and carcinogen free to humans at all stages of application or during fire conditions and does not contain hazardous chemicals or require harmful chemicals to clean material or equipment.

### 2.2.3 Fire Resistance Rating

Firestop systems must be UL Fire Resistance listed or FM APP GUIDE approved with "F" rating at least equal to fire-rating of fire wall or floor in which penetrated openings are to be protected. Where required, firestop systems must also have "T" rating at least equal to the fire-rated floor in which the openings are to be protected.

#### 2.2.3.1 Through-Penetrations

Firestopping materials for through-penetrations, as described in paragraph SUMMARY, must provide "F", "T" and "L" fire resistance ratings in accordance with ASTM E814 or UL 1479. Provide fire resistance ratings as follows:

##### 2.2.3.1.1 Penetrations of Fire Resistance Rated Walls and Partitions

F Rating = Rating of wall or partition being penetrated.

##### 2.2.3.1.2 Penetrations of Fire Resistance Rated Floors, Floor-Ceiling Assemblies and the Ceiling Membrane of Roof-Ceiling Assemblies

F Rating Where the penetrating item is outside of a wall cavity the F rating must be equal to the fire resistance rating of the floor penetrated, and the T rating must be in accordance with the requirements

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

of ICC IBC.

#### 2.2.3.2 Construction Joints and Gaps

Fire resistance ratings of construction joints, as described in paragraph SUMMARY, and gaps such as those between floor slabs and curtain walls must be Provide construction joints and gaps with firestopping materials and systems that have been tested in accordance with ASTM E119, ASTM E1966 or UL 2079 to meet the required fire resistance rating. Provide curtain wall joints with firestopping materials and systems that have been tested in accordance with ASTM E2307 to meet the required fire resistance rating. Systems installed at construction joints must meet the cycling requirements of ASTM E1399/E1399M or UL 2079. Provide a minimum class II movement capability for all joints at the intersection of the top of a fire resistance rated wall and the underside of a fire-rated floor, floor ceiling, or roof ceiling assembly.

#### 2.2.4 Material Certification

Submit certificates attesting that firestopping material complies with the specified requirements. Provide certification of compliance with UL 1479 for all intumescent firestop materials used in through penetration systems.

### PART 3 EXECUTION

#### 3.1 PREPARATION

Areas to receive firestopping must be free of dirt, grease, oil, or loose materials which may affect the fitting or fire resistance of the firestopping system. For cast-in-place firestop devices, formwork or metal deck to receive device prior to concrete placement must be sound and capable of supporting device. Prepare surfaces as recommended by the manufacturer.

#### 3.2 INSTALLATION

Completely fill void spaces with firestopping material regardless of geometric configuration, subject to tolerance established by the manufacturer. Firestopping systems for filling floor voids 4 inches or more in any direction must be capable of supporting the same load as the floor is designed to support or be protected by a permanent barrier to prevent loading or traffic in the firestopped area. Install firestopping in accordance with manufacturer's written instructions. Provide tested and listed firestop systems in the following locations, except in floor slabs on grade:

- a. Penetrations of duct, conduit, tubing, cable and pipe through floors and through fire-resistance rated walls, partitions, and ceiling-floor assemblies.
- b. Penetrations of vertical shafts such as pipe chases, elevator shafts, and utility chutes.
- c. Gaps at the intersection of floor slabs and curtain walls, including inside of hollow curtain walls at the floor slab.
- d. Gaps at perimeter of fire-resistance rated walls and partitions, such as between the top of the walls and the bottom of roof decks.



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

- e. Construction joints in floors and fire rated walls and partitions.
- f. Other locations where required to maintain fire resistance rating of the construction.

### 3.2.1 Insulated Pipes and Ducts

Cut and remove thermal insulation where pipes or ducts pass through firestopping, unless insulation meets requirements specified for firestopping. Replace thermal insulation with a material having equal thermal insulating and firestopping characteristics.

### 3.2.2 Fire Dampers

Install and firestop fire dampers in accordance with Division 23. Firestop installed with fire damper must be tested and approved for use in fire damper system. Firestop installed with fire damper must be tested and approved for use in fire damper system.

### 3.2.3 Data and Communication Cabling

Seal cabling for data and communication applications with re-enterable firestopping products and devices.

#### 3.2.3.1 Re-Enterable Devices

Provide firestopping devices that are pre-manufactured modular devices, containing built-in self-sealing intumescent inserts. Allow for cable moves, additions or changes without the need to remove or replace any firestop materials. Devices must be capable of maintaining the fire resistance rating of the penetrated membrane at 0 percent to 100 percent visual fill of penetrants; while maintaining "L" rating of <10 cfm/sf at 0 percent to 100 percent visual fill.

#### 3.2.3.2 Re-Sealable Products

Provide firestopping pre-manufactured modular products, containing self-sealing intumescent inserts. Allow for cable moves, additions or changes. Provide devices capable of maintaining the fire resistance rating of the penetrated membrane at 0 percent to 100 percent visual fill of penetrants.

## 3.3 INSPECTION

For all projects, do not cover or enclose the firestopped areas until inspection is complete and approved by the Contracting Officer. The inspector must inspect the applications initially to ensure adequate preparations (clean surfaces suitable for application, etc.) and periodically during the work to assure that the completed work has been accomplished according to the manufacturer's written instructions and the specified requirements. Submit written reports indicating locations of and types of penetrations and types of firestopping used at each location; record type by UL listed printed numbers.

Install firestopping manufacturer's label(s) at each firestopped penetration. Labels shall be filled out and include, at the minimum, the UL listed number of the firestopping system, date installed, installer's name (the person doing the installation), and the company's name for whom the installer works.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 3.3.1 Inspection Standards

Inspect all firestopping and document inspection results to be submitted.

### 3.3.2 Inspection Reports

Submit inspection report stating that firestopping work has been inspected and found to be applied according to the manufacturer's recommendations and the specified requirements.

-- End of Section --

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

SECTION 07 92 00

JOINT SEALANTS

**08/23**

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

|            |                                                                                                                   |
|------------|-------------------------------------------------------------------------------------------------------------------|
| ASTM C509  | (2006; R 2021) Standard Specification for Elastomeric Cellular Preformed Gasket and Sealing Material              |
| ASTM C734  | (2015; R 2019) Standard Test Method for Low-Temperature Flexibility of Latex Sealants After Artificial Weathering |
| ASTM C834  | (2017; R 2023) Standard Specification for Latex Sealants                                                          |
| ASTM C919  | (2022) Standard Practice for Use of Sealants in Acoustical Applications                                           |
| ASTM C920  | (2018) Standard Specification for Elastomeric Joint Sealants                                                      |
| ASTM C1193 | (2016) Standard Guide for Use of Joint Sealants                                                                   |
| ASTM C1311 | (2022) Standard Specification for Solvent Release Sealants                                                        |
| ASTM C1521 | (2019; R 2020) Standard Practice for Evaluating Adhesion of Installed Weatherproofing Sealant Joints              |
| ASTM D217  | (2021a) Standard Test Methods for Cone Penetration of Lubricating Grease                                          |
| ASTM D1056 | (2020) Standard Specification for Flexible Cellular Materials - Sponge or Expanded Rubber                         |
| ASTM D1667 | (2022) Standard Specification for Flexible Cellular Materials - Poly (Vinyl Chloride) Foam (Closed-Cell)          |
| ASTM D2452 | (2015; R 2019) Standard Test Method for Extrudability of Oil- and Resin-Base Caulking Compounds                   |
| ASTM D2453 | (2015; R 2020; E 2020) Standard Test                                                                              |

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

Method for Shrinkage and Tenacity of Oil-  
and Resin-Base Caulking Compounds

ASTM E84

(2023) Standard Test Method for Surface  
Burning Characteristics of Building  
Materials

CALIFORNIA DEPARTMENT OF PUBLIC HEALTH (CDPH)

CDPH SECTION 01350

(2017; Version 1.2) Standard Method for  
the Testing and Evaluation of Volatile  
Organic Chemical Emissions from Indoor  
Sources using Environmental Chambers

SCIENTIFIC CERTIFICATION SYSTEMS (SCS)

SCS

SCS Global Services (SCS) Indoor Advantage

SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT (SCAQMD)

SCAQMD Rule 1168

(2017) Adhesive and Sealant Applications

UNDERWRITERS LABORATORIES (UL)

UL 2818

(2022) GREENGUARD Certification Program  
For Chemical Emissions For Building  
Materials, Finishes And Furnishings

## 1.2 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

### SD-03 Product Data

Sealants;

Primers;

Bond Breakers;

Backstops;

Caulking;

Cleaning Solvents;

### SD-06 Test Reports

Field Adhesion;

### SD-07 Certificates

Indoor Air Quality For Interior Sealants

Indoor Air Quality For Interior Floor Joint Sealants

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## Indoor Air Quality For Interior Acoustical Sealants

## Indoor Air Quality For Interior Caulking

### 1.3 PRODUCT DATA

Include storage requirements, shelf life, curing time, instructions for mixing and application, and accessories. Provide manufacturer's Safety Data Sheets (SDS) for each solvent, primer and sealant material proposed.

### 1.4 CERTIFICATIONS

#### 1.4.1 Indoor Air Quality Certifications

Submit required indoor air quality certifications in one submittal package.

##### 1.4.1.1 Adhesives and Sealants

Provide products certified to meet indoor air quality requirements by UL 2818 (Greenguard) Gold, SCS Global Services Indoor Advantage Gold or provide certification or validation by other third-party program that products meet the requirements of this Section. Provide current product certification documentation from certification body. When product does not have certification, provide validation that product meets the indoor air quality product requirements cited herein.

### 1.5 ENVIRONMENTAL CONDITIONS

Apply sealant when the ambient temperature is between 40 and 90 degrees F.

### 1.6 DELIVERY AND STORAGE

Deliver materials to the jobsite in unopened manufacturers' sealed shipping containers, with brand name, date of manufacture, and material designation clearly marked thereon. Label elastomeric sealant containers to identify type, class, grade, and use. Handle and store materials in accordance with manufacturer's printed instructions. Prevent exposure to foreign materials or subjection to sustained temperatures exceeding 90 degrees F or lower than 0 degrees F. Keep materials and containers closed and separated from absorptive materials such as wood and insulation.

### 1.7 QUALITY ASSURANCE

#### 1.7.1 Compatibility with Substrate

Verify that each sealant is compatible for use with each joint substrate in accordance with sealant manufacturer's printed recommendations for each application.

#### 1.7.2 Joint Tolerance

Provide joint tolerances in accordance with manufacturer's printed instructions.

#### 1.7.3 Mock-Up

Provide a mock-up of each type of sealant using materials, colors, and techniques approved for use on the project. Approved mock-ups may be

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

incorporated into the Work.

#### 1.7.4 Adhesion

Provide in accordance with ASTM C1193 or ASTM C1521.

### PART 2 PRODUCTS

#### 2.1 SEALANTS

Provide sealant products that have been tested, found suitable, and documented as such by the manufacturer for the particular substrates to which they will be applied.

In areas with ambient temperatures that exceed 110 degrees F, do not use polybutene, bituminous, acrylic-latex, polyvinyl acetate latex sealants, polychloroprene (neoprene), polyvinyl chloride (PVC), and polyurethane foams, and neoprene, PVC, and styrene butadiene rubber extruded seals and closure strips due to these materials having maximum recommended surface temperature ranges from 130 to 180 degrees F.

##### 2.1.1 Interior Sealants

Provide ASTM C834 ASTM C920, Type S or M, Grade NS, Class 12.5, Use NT. Provide sealant products used on the interior of the building (defined as inside of the weatherproofing system) meeting either emissions requirements of CDPH SECTION 01350 (limit requirements for either office or classroom spaces regardless of space type) or VOC content requirements of SCAQMD Rule 1168. Provide certification or validation of indoor air quality for interior sealants. Location(s) and color(s) of sealant for the following (Note, color "as selected" refers to manufacturer's full range of color options):

| LOCATION                                                                                                                                                                 | COLOR                         |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| a. Small voids between walls or partitions and adjacent lockers, casework, shelving, door frames, built-in or surface mounted equipment and fixtures, and similar items. | Match adjacent surface color. |
| b. Perimeter of frames at doors, windows, and access panels which adjoin exposed interior concrete and masonry surfaces.                                                 | Match adjacent surface color. |
| c. Joints of interior masonry walls and partitions which adjoin columns, pilasters, concrete walls, and exterior walls unless otherwise detailed.                        | N.A.                          |
| d. Joints between edge members for acoustical tile and adjoining vertical surfaces.                                                                                      | N.A.                          |
| e. Interior locations, not otherwise indicated or specified, where small voids exist between materials specified to be painted.                                          | White - paintable.            |

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

| LOCATION                                                                                                                                          | COLOR              |
|---------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| f. Joints between bathtubs and ceramic tile; joints between shower receptors and ceramic tile; joints formed where non-planar tile surfaces meet. | Match grout color. |
| g. Joints formed between tile floors and tile base cove; joints between tile and dissimilar materials; joints occurring where substrates change.  | Match grout color. |
| h. Behind escutcheon plates at valve pipe penetrations and showerheads in showers.                                                                | White - paintable. |

### 2.1.2 Exterior Sealants

For joints in vertical surfaces, provide ASTM C920, Type S or M, Grade NS, Class 25, Use NT. For joints in horizontal surfaces, provide ASTM C920, Type S or M, Grade P, Class 25, Use T. Provide location(s) and color(s) of sealant as follows (Note, color "as selected" refers to manufacturer's full range of color options):

| LOCATION                                                                                                                                                                                                                   | COLOR                                                          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| a. Joints and recesses formed where frames and subsills of windows, doors, louvers, and vents adjoin masonry, concrete, or metal frames. Use sealant at both exterior and interior surfaces of exterior wall penetrations. | Match adjacent surface color                                   |
| b. Joints between new and existing exterior masonry walls.                                                                                                                                                                 | N.A.                                                           |
| c. Masonry joints where shelf angles occur.                                                                                                                                                                                | Match adjacent surface color.                                  |
| d. Joints in wash surfaces of stonework.                                                                                                                                                                                   | N.A.                                                           |
| e. Expansion and control joints.                                                                                                                                                                                           | Match grout color in masonry.<br>Match adjacent surface color. |
| f. Interior face of expansion joints in exterior concrete or masonry walls where metal expansion joint covers are not required.                                                                                            | Match adjacent surface color.                                  |
| g. Voids where items pass through exterior walls.                                                                                                                                                                          | Match adjacent surface color.                                  |

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

| LOCATION                                                                                                             | COLOR                              |
|----------------------------------------------------------------------------------------------------------------------|------------------------------------|
| h. Metal reglets, where flashing is inserted into masonry joints, and where flashing is penetrated by coping dowels. | Match grout color.                 |
| i. Metal-to-metal joints where sealant is indicated or specified.                                                    | Match metal color.                 |
| j. Joints between ends of gravel stops, fascia, copings, and adjacent walls.                                         | Match adjacent wall surface color. |

### 2.1.3 Floor Joint Sealants

ASTM C920, Type S or M, Grade P, Class 25, Use T. Provide sealant products used on the interior of the building (defined as inside of the weatherproofing system) meeting either emissions requirements of CDPH SECTION 01350 (limit requirements for either office or classroom spaces regardless of space type) or VOC content requirements of SCAQMD Rule 1168. Provide certification or validation of indoor air quality for interior floor joint sealants. Provide location(s) and color(s) of sealant as follows (Note, color "as selected" refers to manufacturer's full range of color options):

| LOCATION                                                                      | COLOR                         |
|-------------------------------------------------------------------------------|-------------------------------|
| a. Seats of metal thresholds for exterior doors.                              | Gray                          |
| b. Control and expansion joints in floors, slabs, ceramic tile, and walkways. | Match adjacent surface color. |

### 2.1.4 Acoustical Sealants

Rubber or polymer based acoustical sealant in accordance with ASTM C919 to have a flame spread of 25 or less and a smoke developed rating of 50 or less when tested in accordance with ASTM E84. Provide non-staining acoustical sealant with a consistency of 250 to 310 when tested in accordance with ASTM D217. Acoustical sealant must remain flexible and adhesive after 500 hours of accelerated weathering as specified in ASTM C734. Provide sealant products used on the interior of the building (defined as inside of the weatherproofing system) meeting either emissions requirements of CDPH SECTION 01350 (limit requirements for either office or classroom spaces regardless of space type) or VOC content requirements of SCAQMD Rule 1168. Provide certification or validation of indoor air quality for interior acoustical sealants.

### 2.1.5 Preformed Sealants

Provide preformed sealants of polybutylene or isoprene-butylene based pressure sensitive weather resistant tape or bead sealants capable of sealing out moisture, air and dust when installed as recommended by the manufacturer. At temperatures from minus 30 to plus 160 degrees F, sealants must be non-bleeding and have no loss of adhesion.



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### 2.1.5.1 Tape

Tape sealant: Provide cross section dimensions of required width.

#### 2.1.5.2 Bead

Bead sealant: Provide cross section dimensions of required width.

#### 2.1.5.3 Foam Strip

Provide foam strip of polyurethane foam with cross section dimensions of required width. Provide foam strip capable of sealing out moisture, air, and dust when installed and compressed in accordance with manufacturer's printed instructions. Service temperature must be minus 40 to plus 275 degrees F. Install untreated strips with adhesive to hold them in place. Do not allow adhesive to stain or bleed onto adjacent finishes. Saturate treated strips with butylene waterproofing or impregnate with asphalt.

### 2.2 PRIMERS

Non-staining, quick drying type and consistency as recommended by the sealant manufacturer for the particular application. Provide primers for interior applications that meet the indoor air quality requirements of the paragraph INTERIOR SEALANTS above.

### 2.3 BOND BREAKERS

Type and consistency as recommended by the sealant manufacturer to prevent adhesion of the sealant to the backing or to the bottom of the joint. Provide bond breakers for interior applications that meet the indoor air quality requirements of the paragraph INTERIOR SEALANTS above.

### 2.4 BACKSTOPS

Provide glass fiber roving, neoprene, butyl, polyurethane, or polyethylene foams free from oil or other staining elements as recommended by sealant manufacturer. Provide 25 to 33 percent oversized backing for closed cell and 40 to 50 percent oversized backing for open cell material, unless otherwise indicated. Provide backstop material that is compatible with sealant. Do not use oakum or other types of absorptive materials as backstops.

#### 2.4.1 Rubber

Provide in accordance with ASTM D1056, Type 1, open cell, or Type 2, closed cell, Class A typical and Class B where petroleum oil or fuel resistance is required, Grade (as recommended by Manufacturer for intended use), round cross section for cellular rubber sponge backing.

#### 2.4.2 PVC

Provide in accordance with ASTM D1667, Grade V0 12, open-cell foam, round cross section for polyvinyl chloride (PVC) backing.

#### 2.4.3 Synthetic Rubber

Provide in accordance with ASTM C509, Option I, Type I preformed rods or tubes for synthetic rubber backing.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### 2.4.4 Neoprene

Provide in accordance with ASTM D1056, closed cell expanded neoprene cord Type 2, Class C, Grade 2C2 open cell neoprene sponge Type 1, Class C, Grade 1C3 for neoprene backing.

#### 2.4.5 Butyl Rubber Based

Provide in accordance with ASTM C1311, from a single component, with solvent release. color as selected from manufacturer's full range of color choices.

#### 2.4.6 Silicone Rubber Base

Provide in accordance with ASTM C920, from a single component, with solvent release, Non-sag, Type 'S', Grade 'NS', Class 25. Color as selected from manufacturer's full range of color choices.

### 2.5 CAULKING

For interior use and only where there is little or no anticipated joint movement. Provide in accordance with ASTM D2452 and ASTM D2453, Type 'S', for oil and resin-based caulking. Provide products used on the interior of the building (defined as inside of the weatherproofing system) meeting either emissions requirements of CDPH SECTION 01350 (limit requirements for either office or classroom spaces regardless of space type) or VOC content requirements of SCAQMD Rule 1168. Provide certification or validation of indoor air quality for interior caulking.

### 2.6 CLEANING SOLVENTS

Provide type(s) recommended by the sealant manufacturer and in accordance with environmental requirements herein. Protect adjacent aluminum and bronze surfaces from solvents. Provide solvents for interior applications that meet the indoor air quality requirements of the paragraph INTERIOR SEALANTS above.

## PART 3 EXECUTION

### 3.1 FIELD QUALITY CONTROL

Perform a field adhesion test in accordance with manufacturer's instructions and ASTM C1193, Method A or ASTM C1521, Method A, Tail Procedure. Remove sealants that fail adhesion testing; clean substrates, reapply sealants, and re-test. Test sealants adjacent to failed sealants. Submit field adhesion test report indicating tests, locations, dates, results, and remedial actions taken.

### 3.2 SURFACE PREPARATION

Prepare surfaces according to manufacturer's printed installation instructions. Clean surfaces from dirt, frost, moisture, grease, oil, wax, lacquer, paint, or other foreign matter that would destroy or impair adhesion. Remove oil and grease with solvent; thoroughly remove solvents prior to sealant installation. Wipe surfaces dry with clean cloths. When resealing an existing joint, remove existing caulk or sealant prior to applying new sealant. For surface types not listed below, provide in accordance with sealant manufacturer's printed instructions for each specific surface.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 3.2.1 Steel Surfaces

Remove loose mill scale by sandblasting or, if sandblasting is impractical or would damage finished work, scraping and wire brushing. Remove protective coatings by sandblasting or using a residue free solvent. Remove resulting debris and solvent residue prior to sealant installation.

### 3.2.2 Aluminum or Bronze Surfaces

Remove temporary protective coatings from surfaces that will be in contact with sealant. When masking tape is used as a protective coating, remove tape and residual adhesive prior to sealant application. For removing protective coatings and final cleaning, use non-staining solvents recommended by the manufacturer of the item(s) containing aluminum or bronze surfaces.

### 3.2.3 Concrete and Masonry Surfaces

Where surfaces have been treated with curing compounds, oil, or other such materials, remove materials by sandblasting or wire brushing. Remove laitance, efflorescence and loose mortar from the joint cavity. Remove resulting debris prior to sealant installation.

### 3.2.4 Wood Surfaces

Ensure wood surfaces that will be in contact with sealants are free of splinters, sawdust and other loose particles.

### 3.2.5 Removing Existing Hazardous Sealants

For sealants applied prior to 1979, or that have been tested and found to contain polychlorinated biphenyls (PCBs), remove and dispose of these sealants in accordance with Division 02 Existing Conditions.

## 3.3 SEALANT PREPARATION

Do not add liquids, solvents, or powders to sealants. Mix multicomponent elastomeric sealants in accordance with manufacturer's printed instructions.

## 3.4 APPLICATION

### 3.4.1 Joint Width-To-Depth Ratios

Acceptable Ratios:

| JOINT WIDTH                                    | JOINT DEPTH |          |
|------------------------------------------------|-------------|----------|
|                                                | Minimum     | Maximum  |
| For metal, glass, or other nonporous surfaces: |             |          |
| 1/4 inch (minimum)                             | 1/4 inch    | 1/4 inch |

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

| JOINT WIDTH                            | JOINT DEPTH  |                |
|----------------------------------------|--------------|----------------|
|                                        | Minimum      | Maximum        |
| over 1/4 inch                          | 1/2 of width | Equal to width |
| For wood, concrete, masonry, or stone: |              |                |
| 1/4 inch (minimum)                     | 1/4 inch     | 1/4 inch       |
| over 1/4 inch to 1/2 inch              | 1/4 inch     | Equal to width |
| over 1/2 inch to 1 inch                | 1/2 inch     | 5/8 inch       |
| Over 1 inch                            | prohibited   |                |

Unacceptable Ratios: Where joints of acceptable width-to-depth ratios have not been provided, clean out joints to acceptable depths and grind or cut to acceptable widths without damage to the adjoining work. Grinding is prohibited at metal surfaces.

#### 3.4.2 Unacceptable Sealant Use

Do not install sealants in lieu of other required building enclosure weatherproofing components such as flashing, drainage components, and joint closure accessories, or to close gaps between walls, floors, roofs, windows, and doors, that exceed acceptable installation tolerances. Remove sealants that have been used in an unacceptable manner and correct building enclosure deficiencies to comply with contract documents requirements.

#### 3.4.3 Masking Tape

Place masking tape on the finished surface on one or both sides of joint cavities to protect adjacent finished surfaces from primer or sealant smears. Remove masking tape within 10 minutes of joint filling and tooling.

#### 3.4.4 Backstops

Provide backstops dry and free of tears or holes. Tightly pack the back or bottom of joint cavities with backstop material to provide joints in specified depths. Provide backstops where indicated and where backstops are not indicated but joint cavities exceed the acceptable maximum depths specified in JOINT WIDTH-TO-DEPTH RATIOS Table.

#### 3.4.5 Primer

Clean out loose particles from joints immediately prior to application of. Apply primer to joints in concrete masonry units, wood, and other porous surfaces in accordance with sealant manufacturer's printed instructions. Do not apply primer to exposed finished surfaces.

#### 3.4.6 Bond Breaker

Provide bond breakers to surfaces not intended to bond in accordance with, sealant manufacturer's printed instructions for each type of surface and

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

sealant combination specified.

#### 3.4.7 Sealants

Provide sealants compatible with the material(s) to which they are applied. Do not use a sealant that has exceeded its shelf life or has jelled and cannot be discharged in a continuous flow from the sealant gun. Apply sealants in accordance with the manufacturer's printed instructions with a gun having a nozzle that fits the joint width. Work sealant into joints so as to fill the joints solidly without air pockets. Tool sealant after application to ensure adhesion. Apply sealant uniformly smooth and free of wrinkles. Upon completion of sealant application, roughen partially filled or unfilled joints, apply additional sealant, and tool smooth as specified. Apply sealer over sealants in accordance with the sealant manufacturer's printed instructions.

### 3.5 PROTECTION AND CLEANING

#### 3.5.1 Protection

Protect areas adjacent to joints from sealant smears. Masking tape may be used for this purpose if removed 5 to 10 minutes after the joint is filled and no residual tape marks remain.

#### 3.5.2 Final Cleaning

Upon completion of sealant application, remove remaining smears and stains and leave the work in a clean and neat condition.

- a. Masonry and Other Porous Surfaces: Immediately remove fresh sealant that has been smeared on adjacent masonry, rub clean with a solvent, and remove solvent residue, in accordance with sealant manufacturer's printed instructions. Allow excess sealant to cure for 24 hours then remove by wire brushing or sanding. Remove resulting debris.
- b. Metal and Other Non-Porous Surfaces: Remove excess sealant with a solvent moistened cloth. Remove solvent residue in accordance with solvent manufacturer's printed instructions.

-- End of Section --

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## SECTION 08 11 13

### STEEL DOORS AND FRAMES

**08/20**

#### PART 1 GENERAL

##### 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

##### AMERICAN WELDING SOCIETY (AWS)

AWS D1.1/D1.1M (2020; Errata 1 2021) Structural Welding  
Code - Steel

##### ASTM INTERNATIONAL (ASTM)

ASTM A653/A653M (2023) Standard Specification for Steel  
Sheet, Zinc-Coated (Galvanized) or  
Zinc-Iron Alloy-Coated (Galvannealed) by  
the Hot-Dip Process

ASTM A924/A924M (2022a) Standard Specification for General  
Requirements for Steel Sheet,  
Metallic-Coated by the Hot-Dip Process

ASTM C578 (2022) Standard Specification for Rigid,  
Cellular Polystyrene Thermal Insulation

ASTM C591 (2022) Standard Specification for Unfaced  
Preformed Rigid Cellular Polyisocyanurate  
Thermal Insulation

ASTM C612 (2014; R 2019) Standard Specification for  
Mineral Fiber Block and Board Thermal  
Insulation

ASTM D2863 (2019) Standard Test Method for Measuring  
the Minimum Oxygen Concentration to  
Support Candle-Like Combustion of Plastics  
(Oxygen Index)

##### BUILDERS HARDWARE MANUFACTURERS ASSOCIATION (BHMA)

ANSI/BHMA A156.115 (2016) Hardware Preparation in Steel Doors  
and Steel Frames

##### NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 80 (2022) Standard for Fire Doors and Other  
Opening Protectives

NFPA 252 (2022) Standard Methods of Fire Tests of  
Door Assemblies

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### STEEL DOOR INSTITUTE (SDI/DOOR)

|                  |                                                                                                                           |
|------------------|---------------------------------------------------------------------------------------------------------------------------|
| SDI/DOOR 111     | (2009) Recommended Details for Standard Steel Doors, Frames, and Accessories and Related Components                       |
| SDI/DOOR 113     | (2013; R2018) Standard Practice for Determining the Steady-State Thermal Transmittance of Steel Door and Frame Assemblies |
| SDI/DOOR A250.4  | (2018) Test Procedure and Acceptance Criteria for Physical Endurance for Steel Doors, Frames and Frame Anchors            |
| SDI/DOOR A250.6  | (2015) Recommended Practice for Hardware Reinforcing on Standard Steel Doors and Frames                                   |
| SDI/DOOR A250.8  | (2017) Specifications for Standard Steel Doors and Frames                                                                 |
| SDI/DOOR A250.11 | (2012) Recommended Erection Instructions for Steel Frames                                                                 |

#### U.S. DEPARTMENT OF DEFENSE (DOD)

|              |                                                                               |
|--------------|-------------------------------------------------------------------------------|
| UFC 4-010-01 | (2018; with Change 2, 2022) DoD Minimum Antiterrorism Standards for Buildings |
|--------------|-------------------------------------------------------------------------------|

#### UNDERWRITERS LABORATORIES (UL)

|        |                                                                                                 |
|--------|-------------------------------------------------------------------------------------------------|
| UL 10C | (2016; Reprint May 2021) UL Standard for Safety Positive Pressure Fire Tests of Door Assemblies |
|--------|-------------------------------------------------------------------------------------------------|

### 1.2 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

#### SD-02 Shop Drawings

Doors; G  
Frames; G  
Accessories  
Schedule of Doors; G  
Schedule of Frames; G

#### SD-03 Product Data

Doors; G

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

Frames; G

Accessories

SD-04 Samples

Hot-Dip Zinc-Coated and Factory-Primed Finish; G

Submit manufacturer's standard colors and patterns for selection by DOR. Submit not less than two 1'-0" square samples for review.

### 1.3 DELIVERY, STORAGE, AND HANDLING

Deliver doors, frames, and accessories undamaged and with protective wrappings or packaging. Store doors and frames on platforms under cover in clean, dry, ventilated, and accessible locations, with 1/4 inch airspace between doors. Remove damp or wet packaging immediately and wipe affected surfaces dry. Replace damaged materials with new.

## PART 2 PRODUCTS

### 2.1 STANDARD STEEL DOORS

SDI/DOOR A250.8, except as specified otherwise. Prepare doors to receive door hardware as specified in Section 08 71 00 DOOR HARDWARE. Undercut where indicated. Provide exterior doors with top edge closed flush and sealed to prevent water intrusion. Provide doors at 1-3/4 inch thick, unless otherwise indicated. Provide door material that uses a minimum of 25 percent recycled content. Provide data indicating percentage of recycled content for steel door product. Exterior doors must be tested in accordance with Standard 10 to meet requirements of UFC 4-010-01.

Exterior door glazing shall be in accordance with Standard 10 of UFC 4-010-01 "DoD Minimum Antiterrorism Standard for Buildings."

#### 2.1.1 Classification - Level, Performance, Model

##### 2.1.1.1 Heavy Duty Doors

SDI/DOOR A250.8, Level 2, physical performance Level B, Model 1, with core construction as required by the manufacturer for interior doors, of size(s) and design(s) indicated including Schedule STC acoustic ratings. Provide Level 2 typical at all locations.

##### 2.1.1.2 Extra Heavy Duty Doors

SDI/DOOR A250.8, Level 3, physical performance Level A, Model 2 with core construction as required by the manufacturer for interior doors, of size(s) and design(s) indicated. Where vertical stiffener cores are required, the space between the stiffeners must be filled with board insulation.

##### 2.1.1.3 Maximum Duty Doors

SDI/DOOR A250.8, Level 4, physical performance Level A, Model 1 with core construction as required by the manufacturer for indicated exterior doors, of size(s) and design(s) indicated including Schedule "U" insulation value.



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## 2.2 INSULATED STEEL DOOR SYSTEMS

Provide insulated steel doors and frames in accordance with SDI/DOOR 113 at entrances to dwelling units and where indicated. Meet energy requirements including Solar Heat Gain Coefficient (SHGC) and U-factor. Provide insulated steel doors with a core of polyurethane foam; face sheets, edges, and frames of galvanized steel not lighter than 23 gage, 16 gage, and 16 gage respectively; magnetic weatherstripping; nonremovable-pin hinges; thermal-break aluminum threshold; and vinyl door bottom. Provide to doors and frames a phosphate treatment, rust-inhibitive primer, and baked acrylic enamel finish. Test doors in accordance with SDI/DOOR A250.4 and meet the requirements for Level C. Prepare doors to receive specified hardware. Provide doors 1-3/4 inch thick.

## 2.3 ACCESSORIES

### 2.3.1 Astragals

For pairs of exterior steel doors which will not have aluminum astragals or removable mullions, as specified in Section 08 71 00 DOOR HARDWARE provide overlapping steel astragals with the doors.

## 2.4 INSULATION CORES

Provide insulating cores at all exterior doors and other specific doors noted in the door schedule, and provide an apparent U-factor of .48 in accordance with SDI/DOOR 113 and conforming to:

- a. Rigid Cellular Polyisocyanurate Foam: ASTM C591, Type I or II, foamed-in-place or in board form, with oxygen index of not less than 22 percent when tested in accordance with ASTM D2863; or
- b. Rigid Polystyrene Foam Board: ASTM C578, Type I or II; or
- c. Mineral board: ASTM C612, Type I.

## 2.5 STANDARD STEEL FRAMES

SDI/DOOR A250.8, Level 2, except as otherwise specified. Form frames to sizes and shapes indicated, with welded corners. Provide steel frames for doors, transoms, sidelights, mullions, cased openings, and interior glazed panels, unless otherwise indicated. Provide frame product that uses a minimum of 25 percent recycled content.

### 2.5.1 Welded Frames

Continuously weld frame faces at corner joints. Mechanically interlock or continuously weld stops and rabbets. Grind welds smooth.

Weld frames in accordance with the recommended practice of the Structural Welding Code Sections 1 through 6, AWS D1.1/D1.1M and in accordance with the practice specified by the producer of the metal being welded.

### 2.5.2 Mullions and Transom Bars

Provide mullions and transom bars of closed or tubular construction with heads and jambs butt-welded together or knock-down for field assembly. Bottom of door mullions must have adjustable floor anchors and spreader

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

connections.

### 2.5.3 Stops and Beads

Provide form and loose stops and beads from 20 gage steel. Provide for glazed and other openings in standard steel frames. Secure beads to frames with oval-head, countersunk Phillips self-tapping sheet metal screws or concealed clips and fasteners. Space fasteners approximately 12 to 16 inch on center. Miter molded shapes at corners. Butt or miter square or rectangular beads at corners.

### 2.5.4 Terminated Stops

Where indicated, terminate interior door frame stops 6 inch above floor. Do not terminate stops of frames for soundproof, doors.

### 2.5.5 Cased Openings

Fabricate frames for cased openings of same material, gage, and assembly as specified for metal door frames, except omit door stops and preparation for hardware.

### 2.5.6 Anchors

Provide anchors to secure the frame to adjoining construction. Provide steel anchors, zinc-coated not lighter than 18 gage.

#### 2.5.6.1 Wall Anchors

Provide at least three anchors for each jamb. For frames which are more than 7.5 feet in height, provide one additional anchor for each jamb for each additional 2.5 feet or fraction thereof.

- a. Masonry: Provide anchors of corrugated or perforated steel straps or 3/16 inch diameter steel wire, adjustable or T-shaped;
- b. Stud partitions: Weld or otherwise securely fasten anchors to backs of frames. Design anchors to be fastened to closed steel studs with sheet metal screws, and to open steel studs by wiring or welding;
- c. Completed openings: Secure frames to previously placed concrete or masonry with expansion bolts in accordance with SDI/DOOR 111.

#### 2.5.6.2 Floor Anchors

Provide floor anchors drilled for 3/8 inch anchor bolts at bottom of each jamb member. Where floor fill occurs, terminate bottom of frames at the indicated finished floor levels and support by adjustable extension clips resting on and anchored to the structural slabs.

## 2.6 FIRE DOORS AND FRAMES

Where indicated provide fire doors and frames in accordance with NFPA 80 and this specification. Include insulated core materials in fire doors where indicated in the door schedule.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 2.6.1 Labels

Provide fire doors and frames bearing the label of Underwriters Laboratories (UL), Factory Mutual Engineering and Research (FM), or Warnock Hersey International (WHI) attesting to the rating required. Testing must be in accordance with NFPA 252 or UL 10C. Provide labels that are metal with raised letters, bearing the name or file number of the door and frame manufacturer. Labels must be permanently affixed at the factory to frames and to the hinge edge of the door. Do not paint door and labels.

### 2.6.2 Astragal on Fire Doors

On pairs of labeled fire doors, conform to NFPA 80 and UL requirements.

## 2.7 EXTERIOR FRAMES

Provide thermal insulation in all exterior frames. Provide frames of a minimum Level 4, with frames of a minimum thickness of 0.067 inch, 14 gage.

## 2.8 HARDWARE PREPARATION

Drill and tap doors and frames to receive finish hardware. Prepare doors and frames for hardware in accordance with the applicable requirements of SDI/DOOR A250.8 and SDI/DOOR A250.6. For additional requirements refer to ANSI/BHMA A156.115. Drill and tap for surface-applied hardware at the project site. Build additional reinforcing for surface-applied hardware into the door at the factory. Punch door frames, with the exception of frames that will have weatherstripping or soundproof gasketing, to receive a minimum of two rubber or vinyl door silencers on lock side of single doors and one silencer for each leaf at heads of double doors. Set lock strikes out to provide clearance for silencers.

## 2.9 FINISHES

### 2.9.1 Factory-Primed Finish

Thoroughly clean all surfaces of doors and frames then chemically treat and factory prime with a rust inhibiting coating as specified in SDI/DOOR A250.8, or paintable A25 galvanized steel without primer. Where coating is removed by welding, apply touchup of factory primer.

### 2.9.2 Hot-Dip Zinc-Coated and Factory-Primed Finish

Fabricate exterior doors and frames from hot dipped zinc coated steel, alloyed type, that complies with ASTM A924/A924M and ASTM A653/A653M. The coating weight must meet or exceed the minimum requirements for coatings having 0.4 ounces per square foot, total both sides, i.e., A40. Repair damaged zinc-coated surfaces by the application of zinc dust paint. Thoroughly clean and chemically treat to insure maximum paint adhesion. Factory prime as specified in SDI/DOOR A250.8.

## 2.10 FABRICATION AND WORKMANSHIP

Provide finished doors and frames that are strong and rigid, neat in appearance, and free from defects, waves, scratches, cuts, dents, ridges, holes, warp, and buckle. Provide molded members that are clean cut, straight, and true, with joints coped or mitered, well formed, and in true alignment. Dress exposed welded and soldered joints smooth. Design door

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

frame sections for use with the wall construction indicated. Corner joints must be well formed and in true alignment. Conceal fastenings where practicable.

### PART 3 EXECUTION

#### 3.1 INSTALLATION

##### 3.1.1 Frames

Set frames in accordance with SDI/DOOR A250.11. Plumb, align, and brace securely until permanent anchors are set. Anchor bottoms of frames with expansion bolts or powder-actuated fasteners. Build in or secure wall anchors to adjoining construction. Where frames require ceiling struts or overhead bracing, anchor frames to the struts or bracing.

##### 3.1.2 Doors

Hang doors in accordance with clearances specified in SDI/DOOR A250.8. After erection and glazing, clean and adjust hardware.

##### 3.1.3 Fire Doors and Frames

Install fire doors and frames, including hardware, in accordance with NFPA 80.

#### 3.2 PROTECTION

Protect doors and frames from damage. Repair damaged doors and frames prior to completion and acceptance of the project or replace with new, as directed. Wire brush rusted frames until rust is removed. Clean thoroughly. Apply an all-over coat of rust-inhibitive paint of the same type used for shop coat.

#### 3.3 CLEANING

Upon completion, clean exposed surfaces of doors and frames thoroughly. Remove mastic smears and other unsightly marks.

-- End of Section --

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## SECTION 08 14 00

### WOOD DOORS

**08/16, CHG 1: 08/18**

## PART 1 GENERAL

### 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to in the text by the basic designation only.

#### ASTM INTERNATIONAL (ASTM)

ASTM E90 (2009; R2016) Standard Test Method for Laboratory Measurement of Airborne Sound Transmission Loss of Building Partitions and Elements

ASTM E2226 (2015; R 2019b) Standard Practice for Application of Hose Stream

#### NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 80 (2022) Standard for Fire Doors and Other Opening Protectives

NFPA 252 (2022) Standard Methods of Fire Tests of Door Assemblies

#### UNDERWRITERS LABORATORIES (UL)

UL 10B (2008; Reprint May 2020) Fire Tests of Door Assemblies

#### WINDOW AND DOOR MANUFACTURERS ASSOCIATION (WDMA)

ANSI/WDMA I.S.1A (2013) Interior Architectural Wood Flush Doors

ANSI/WDMA I.S.6A (2013) Interior Architectural Stile and Rail Doors

#### WOODWORK INSTITUTE (WI)

NAAWS 3.1 (2017; 2018 Errata Edition) North American Architectural Woodwork Standards

### 1.2 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

Doors; G

Submit drawings or catalog data showing each type of door unit .  
Indicate within drawings and data the door types and construction,  
sizes, thickness, and glazing, .

#### SD-03 Product Data

Doors; G

Accessories

Water-resistant Sealer

Sample Warranty

Sound Transmission Class Rating; G

#### SD-04 Samples

Doors

Prior to the delivery of wood doors, submit a sample section of  
each type of door which shows the stile, rail, veneer, finish, and  
core construction.

Door Finish Colors; G

Submit a minimum of three color selection samples , minimum 3 by 5  
inches in size representing wood stain .

#### SD-06 Test Reports

Cycle-Slam

Hinge Loading Resistance

Submit cycle-slam test report for doors tested in accordance with  
ANSI/WDMA I.S.1A, and hinge loading resistance test report for  
doors tested in accordance with ANSI/WDMA I.S.6A.

#### SD-07 Certificates

Certificates of Grade

#### SD-11 Closeout Submittals

Warranty

### 1.3 CERTIFICATIONS

#### 1.3.1 Certified Wood Grades

Provide certificates of grade from the grading agency on, acoustical doors,  
and fire doors.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### 1.4 DELIVERY, STORAGE, AND HANDLING

Deliver doors to the site in an undamaged condition and protect against damage and dampness. Stack doors flat under cover. Support on blocking, a minimum of 4 inch thick, located at each end and at the midpoint of the door. Store doors in a well-ventilated building so that they will not be exposed to excessive moisture, heat, dryness, direct sunlight, or extreme changes of temperature and humidity. Replace defective or damaged doors with new ones.

#### 1.5 WARRANTY

Warrant doors free of defects as set forth in the door manufacturer's standard door warranty.

### PART 2 PRODUCTS

#### 2.1 DOORS

Provide doors of the types, sizes, and designs indicated free of urea-formaldehyde resins.

##### 2.1.1 Flush Doors

Conform to ANSI/WDMA I.S.1A for flush doors. Provide hollow core doors with lock blocks and 1 inch minimum thickness hinge stile. Hardwood stile edge bands of doors receives a natural finish, compatible with face veneer. Provide mill option for stile edge of doors scheduled to be painted. No visible finger joints will be accepted in stile edge bands. When used, locate finger-joints under hardware.

##### 2.1.1.1 Interior Flush Doors

Provide staved lumber particleboard or agrifiber core, Type II flush doors conforming to ANSI/WDMA I.S.1A with faces of premium grade red oak. Hardwood veneers must be plain sliced.

##### 2.1.2 Acoustical Doors

ANSI/WDMA I.S.1A, solid core, constructed to provide Sound Transmission Class rating of 35 when tested in accordance with ASTM E90.

##### 2.1.3 Fire Doors

Provide doors specified or indicated to have a fire resistance rating conforming to the requirements of UL 10B, ASTM E2226, or NFPA 252 for the class of door indicated. Affix a permanent metal label with raised or incised markings indicating testing agency's name and approved hourly fire rating to hinge edge of each door.

#### 2.2 ACCESSORIES

##### 2.2.1 Door Louvers

Fabricate from wood and of sizes indicated. Provide louvers with a minimum of 35 percent free air. Equip louvers with slat type. Mount louvers in the door with flush wood moldings.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 2.2.2 Door Light Openings

Provide glazed openings with the manufacturer's standard wood moldings. Provide moldings for doors to receive natural finish of the same wood species and color as the wood face veneers. Provide moldings on the exterior doors with sloped surfaces.

### 2.2.3 Additional Hardware Reinforcement

Provide the minimum lock blocks to secure the specified hardware. The measurement of top, bottom, and intermediate rail blocks are a minimum 125 mm 5 inch by full core width. Comply with the manufacturer's labeling requirements for reinforcement blocking, but not mineral material similar to the core.

## 2.3 FABRICATION

### 2.3.1 Marking

Stamp each door with a brand, stamp, or other identifying mark indicating quality and construction of the door.

### 2.3.2 Quality and Construction

Identify the standard on which the construction of the door was based and identify doors having a Type I glue bond.

### 2.3.3 Preservative Treatment

Treat doors scheduled for restrooms, janitor closets and other possible wet locations including exterior doors with a water-repellent preservative treatment and so marketed at the manufacturer's plant.

### 2.3.4 Adhesives and Bonds

ANSI/WDMA I.S.1A. Use Type I bond for exterior doors and Type II bond for interior doors. Provide a nonstaining adhesive on doors with a natural finish.

### 2.3.5 Prefitting

Provide factory prefinished factory prefitted doors for the specified hardware, door frame and door-swing indicated. Machine and size doors at the factory by the door manufacturer in accordance with the standards under which the doors are produced and manufactured. The work includes sizing, beveling edges, mortising, and drilling for hardware and providing necessary beaded openings for glass and louvers. Provide the door manufacturer with the necessary hardware samples, and frame and hardware schedules to coordinate the work.

### 2.3.6 Finishes

#### 2.3.6.1 Field Painting

Factory prime or seal doors, and field paint.

#### 2.3.6.2 Factory Finish

Provide doors finished at the factory by the door manufacturer as follows:



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

The coating is NAAWS 3.1 premium, medium rubbed sheen, closed grain effect. Use stain when required to produce the finish specified for color. Seal edges, cutouts, trim, and wood accessories, and apply two coats of finish compatible with the door face finish. Touch-up finishes that are scratched or marred, or where exposed fastener holes are filled, in accordance with the door manufacturer's instructions. Match color and sheen of factory finish using materials compatible for field application.

#### 2.3.6.3 Color

Provide door finish colors as indicated, or if not indicated as selected by Contracting Officer from Manufacturer's full range of available standard colors.

#### 2.4 SOURCE QUALITY CONTROL

Meet or exceed the following minimum performance criteria of styles of "B" and "C" label fire doors utilizing standard mortise leaf hinges:

- a. Cycle-slam: Heavy Duty Doors: 500,000 cycles with no loose hinge screws or other visible signs of failure when tested in accordance with the requirements of ANSI/WDMA I.S.1A.
- b. Hinge loading resistance: Averages of ten test samples not less than Heavy Duty doors: 475 pounds force when tested for direct screw withdrawal in accordance with ANSI/WDMA I.S.6A using a No. 12, 1-1/4 inch long, steel, fully threaded wood screw. Drill 5/32 inch pilot hole, use 1-1/2 inch opening around screw for bearing surface, and engage screw full, except for last 1/8 inch. Do not use a steel plate to reinforce screw area.

### PART 3 EXECUTION

#### 3.1 INSTALLATION

Do not install building construction materials that show visual evidence of biological growth.

Before installation, seal top and bottom edges of doors with the approved water-resistant sealer. Seal cuts made on the job immediately after cutting using approved water-resistant sealer. Fit, trim, and hang doors with a 1/16 inch minimum, 1/8 inch maximum clearance at sides and top, and a 3/16 inch minimum, 1/4 inch maximum clearance over thresholds. Provide 3/8 inch minimum, 7/16 inch maximum clearance at bottom where no threshold occurs. Bevel edges of doors at the rate of 1/8 inch in 2 inch. Door warp must not exceed 1/4 inch when measured in accordance with ANSI/WDMA I.S.1A.

##### 3.1.1 Fire and Smoke Doors

Install fire doors in accordance with NFPA 80. Do not paint over labels.

-- End of Section --

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## SECTION 08 31 00

### ACCESS DOORS AND PANELS 05/17, CHG 1: 08/18

#### PART 1 GENERAL

##### 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

#### AMERICAN WELDING SOCIETY (AWS)

AWS D1.1/D1.1M (2020; Errata 1 2021) Structural Welding  
Code - Steel

#### ASTM INTERNATIONAL (ASTM)

ASTM A36/A36M (2019) Standard Specification for Carbon  
Structural Steel

ASTM A653/A653M (2023) Standard Specification for Steel  
Sheet, Zinc-Coated (Galvanized) or  
Zinc-Iron Alloy-Coated (Galvannealed) by  
the Hot-Dip Process

ASTM A666 (2023) Standard Specification for Annealed  
or Cold-Worked Austenitic Stainless Steel  
Sheet, Strip, Plate and Flat Bar

ASTM A1008/A1008M (2021a) Standard Specification for Steel,  
Sheet, Cold-Rolled, Carbon, Structural,  
High-Strength Low-Alloy, High-Strength  
Low-Alloy with Improved Formability,  
Solution Hardened, and Bake Hardenable

ASTM E119 (2022) Standard Test Methods for Fire  
Tests of Building Construction and  
Materials

#### MASTER PAINTERS INSTITUTE (MPI)

MPI 79 (2016) Primer, Alkyd, Anti-Corrosive for  
Metal

#### NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 80 (2022) Standard for Fire Doors and Other  
Opening Protectives

NFPA 252 (2022) Standard Methods of Fire Tests of  
Door Assemblies

#### UNDERWRITERS LABORATORIES (UL)

UL 10B (2008; Reprint May 2020) Fire Tests of

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## Door Assemblies

UL 263

(2011; Reprint Aug 2021) UL Standard for  
Safety Fire Tests of Building Construction  
and Materials

### 1.2 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

#### SD-02 Shop Drawings

Access Doors And Panels; G

#### SD-03 Product Data

Access Doors And Panels; G

Hardware Including Locks and Keys; G

Accessories; G

Recycled Content

#### SD-04 Samples

Finishes; G

### 1.3 MISCELLANEOUS REQUIREMENTS

For access doors and panels provide the following:

#### 1.3.1 Shop Drawings

For field assembled access doors and panels, provide plans, elevations, sections, and details for each type of access door and panel assembly. Indicate frame, surface and edge construction, materials, and accessories. Indicate types of finished surfaces and details for panel edge conditions. Provide a door schedule with a unique number for each access door and panel, specific location in the project, location of hinges and hardware for each door.

#### 1.3.2 Product Data

For shop assembled access doors and panels, provide literature indicating sizes, types, frame and edge construction, finishes, hardware, accessories such as gaskets, seals and weatherstripping, and location of each door and panel in the project. Indicate acoustical ratings of assemblies, and fire-ratings of assemblies. Provide details of adjoining work for each condition indicated.

#### 1.3.3 Finish Samples

Submit two color charts from manufacturer's standard color and finish options for each type of frame and panel assembly finish indicated.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### 1.4 PERFORMANCE REQUIREMENTS

##### 1.4.1 Structural Requirements

Provide floor access assemblies to support live loads indicated for floors. Deflection must not exceed 1/180 of span.

##### 1.4.2 Acoustical Requirements

Provide access panels with a minimum sound transmission class (STC) of as indicated on the Drawings. Provide gasketing in accordance with manufacturer's written recommendations.

##### 1.4.3 Fire-Rating Requirements

Provide access panels with a minimum fire-rating of as indicated on the Drawings.

#### 1.5 DELIVERY, STORAGE, AND PROTECTION

Protect from corrosion, deformation, and other types of damage. Store items in an enclosed area free from contact with soil and weather. Remove and replace damaged items with new items.

### PART 2 PRODUCTS

#### 2.1 RECYCLED CONTENT

Provide products with recycled content. Provide data for each product with recycled content, identifying percentage of recycled content.

#### 2.2 MATERIALS

##### 2.2.1 Steel Plates, Shapes, and Bars

Provide in accordance with ASTM A36/A36M.

##### 2.2.2 Sheet Steel

Provide cold rolled steel sheet substrate in accordance with ASTM A1008/A1008M, Commercial Steel (CS), exposed.

##### 2.2.3 Stainless Steel

Provide in accordance with ASTM A666, type 302 or 304.

##### 2.2.4 Metallic Coated Steel Sheet

Provide in accordance with ASTM A653/A653M, Commercial Steel (CS), Type B; with minimum G60 (Z180) or A60 (ZF180) metallic coating.

##### 2.2.5 Hardware

Provide automatic closing devices. Provide latch releases operable from insides of doors.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### 2.2.6 Hinges

Provide concealed spring hinges, 175 degrees of opening, with non-removable hinge pins. Provide hinges of same steel as door and frame or in accordance with manufacturer's written recommendations. If providing non-continuous hinges, provide in numbers required to maintain alignment of door panel with frame. Provide coatings as necessary to permanently protect dissimilar metals from contact with one another; see Part 3 herein for more information.

#### 2.2.7 Locks

Unless otherwise indicated, provide flush tamper proof screws (spanner head locks) for access panels in locations requiring such security.

#### 2.2.8 Accessories

Provide anchors in size, number and location on four sides to secure access door to substrate. Provide anchors in types as recommended by manufacturer's written installation instructions for each substrate indicated. Provide shims, bushings, clips, gaskets, and other devices as necessary for a complete installation.

### 2.3 FABRICATION

#### 2.3.1 Thickness, Size, Edges

Fabricate frames for access doors of steel not lighter than 16 gage with welded joints and anchorage for securing to adjacent construction. Provide doors a minimum of 24 by 24 inches and of not lighter than 16 gage steel, with stiffened edges and welded attachments. Provide with eased (lightly rounded) edges, without burrs, snags or sharpness and exposed welds ground smooth.

#### 2.3.2 Welding

Provide in accordance with AWS D1.1/D1.1M.

### 2.4 ACCESS ASSEMBLY TYPES

Unless indicated otherwise, provide flush-face steel access doors and panels with steel frames and flanges.

#### 2.4.1 Fire-rated Doors

##### 2.4.1.1 Door Construction

Provide ceiling access door construction in accordance with ASTM E119 or UL 263. Provide wall access doors in accordance with NFPA 252 or UL 10B.

##### 2.4.1.2 Labels

Provide class B opening according to UL 10B or test by another nationally recognized laboratory, approved by the Contracting Officer. Provide fire-rating as indicated herein, with a maximum temperature rise of 216 degrees F.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### 2.4.1.3 Door Panel and Frame

Steel sheet, with mineral fiber insulation core, insulated sandwich type construction.

#### 2.4.2 Acoustical Doors

Manufacturer's standard assembly rated in accordance with STC requirements indicated herein. Acoustical insulating materials must have a flame spread rating of no more than 25.

#### 2.4.3 Insulated Doors

Provide access door panels with 25 pounds per square inch density polystyrene, or 5 pound per cubic foot density, chlorofluorocarbon (CFC) free, foamed urethane with a flame spread rating of no more than 25.

Provide ceiling access panels for terminal air blenders as indicated.

### 2.5 FINISHES

Provide steel frames and panel surfaces with a baked enamel or powder coated finish. Provide manufacturer's standard two coat finish system consisting of one coat primer and one thermoset topcoat. Provide dry film thickness in 2 mils minimum. Provide exposed fastenings that approximately match the color and finish of the each material to which fastenings are applied.

## PART 3 EXECUTION

### 3.1 PREPARATION

Field verify all measurements prior to fabrication. Verify access door locations and sizes provide required maintenance access to installed building services components. Protect existing construction and completed work from damage during installation.

### 3.2 GENERAL INSTALLATION REQUIREMENTS

Install items at locations indicated, in accordance with manufacturer's written instructions. Include materials and parts as necessary for a complete installation of each item. Conceal fastenings where practicable. Poor matching of holes to fasteners is cause for rejection of the work.

### 3.3 ACCESS LOCATIONS

Install removable access panels directly below each valve, flow indicator, damper, air splitter or other utility requiring access that is located above ceilings, other than at acoustical panel ceilings, and that would otherwise not be accessible. Install access doors and panels permitting access to service valves, traps, dampers, cleanouts, and other mechanical, electrical and conveyor control items concealed in walls and partitions.

### 3.4 ACCESS LOCATIONS IN WET AREAS

When possible, avoid locating access panels in wet areas. When such locations cannot be avoided, provide moisture resistant assemblies as indicated in Part I herein.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 3.5 RECESSED ACCESS DOORS

Install fire-rated access doors in fire-rated partitions and ceilings in accordance with NFPA 80.

### 3.6 FIELD PAINTING

Field painting primed access doors in accordance with the requirements of Section 09 90 00 PAINTS AND COATINGS.

### 3.7 DISSIMILAR MATERIALS

Where dissimilar metals are in contact, protect surfaces with a coating in accordance with MPI 79 to prevent galvanic or corrosive action.

### 3.8 ADJUSTMENT

Adjust hardware so that door panel opens freely. Adjust door when closed center door panel in frame.

### 3.9 ENVIRONMENTAL CONDITIONS

Do not paint surfaces when damp or exposed to weather, when surface temperature is below 45 degrees F or over 95 degrees F, unless approved by the Contracting Officer.

-- End of Section --

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## SECTION 08 36 13

### SECTIONAL OVERHEAD DOORS 08/20

#### PART 1 GENERAL

##### 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

#### ASTM INTERNATIONAL (ASTM)

|                 |                                                                                                                                                                  |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM A36/A36M   | (2019) Standard Specification for Carbon Structural Steel                                                                                                        |
| ASTM A123/A123M | (2017) Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products                                                                  |
| ASTM A227/A227M | (2017) Standard Specification for Steel Wire, Cold-Drawn for Mechanical Springs                                                                                  |
| ASTM A229/A229M | (2018) Standard Specification for Steel Wire, Quenched and Tempered for Mechanical Springs                                                                       |
| ASTM A653/A653M | (2023) Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process                          |
| ASTM E330/E330M | (2014; R 2021) Standard Test Method for Structural Performance of Exterior Windows, Doors, Skylights and Curtain Walls by Uniform Static Air Pressure Difference |

#### DOOR AND ACCESS SYSTEM MANUFACTURERS ASSOCIATION (DASMA)

|                |                                                                                                                                                                                   |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ANSI/DASMA 102 | (2011) Specifications for Sectional Overhead-Type Doors                                                                                                                           |
| ANSI/DASMA 108 | (2017) Standard Method for Testing Sectional Garage Doors, Rolling Doors and Flexible Doors: Determination of Structural Performance Under Uniform Static Air Pressure Difference |

#### NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

|            |                                                                            |
|------------|----------------------------------------------------------------------------|
| NEMA ICS 1 | (2022) Standard for Industrial Control and Systems: General Requirements   |
| NEMA ICS 2 | (2000; R 2020) Industrial Control and Systems Controllers, Contactors, and |



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

Overload Relays Rated 600 V

NEMA ICS 6 (1993; R 2016) Industrial Control and  
Systems: Enclosures

NEMA MG 1 (2021) Motors and Generators

NEMA ST 20 (2014) Dry-Type Transformers for General  
Applications

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 (2023; ERTA 4 2023) National Electrical  
Code

UNDERWRITERS LABORATORIES (UL)

UL 325 (2017; Reprint Feb 2020) UL Standard for  
Safety Door, Drapery, Gate, Louver, and  
Window Operators and Systems

## 1.2 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Doors; G

SD-03 Product Data

Doors; G

Electric Operators; GSD-08 Manufacturer's Instructions

Doors

SD-10 Operation and Maintenance Data

Doors, Data Package 2; G

### 1.2.1 Shop Drawing Information

Show types, sizes, locations, metal gages including minimum metal decimal thickness, hardware provisions, installation details, and other details of construction. For electrically-operated doors, include supporting brackets for motors, location, type, and ratings of motors, switches, and safety devices.

### 1.2.2 Electric Operators Product Data

For electrically motor-operated doors, submit manufacturer's wiring diagrams for motor and controls.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 1.3 DELIVERY, STORAGE, AND HANDLING

Protect doors and accessories from damage during delivery, storage, and handling. Clearly mark manufacturer's brand name. Store doors in dry locations with adequate ventilation, free from dust and water. Remove damaged items and provide new. Provide easy access for inspection and handling of overhead doors prior to installation.

## PART 2 PRODUCTS

### 2.1 MATERIALS

#### 2.1.1 Hard-Drawn Springwire

ASTM A227/A227M.

#### 2.1.2 Oil-Tempered Springwire

ASTM A229/A229M.

#### 2.1.3 Steel Sheet

ASTM A653/A653M.

#### 2.1.4 Steel Shapes

ASTM A36/A36M.

#### 2.1.5 Glass

Fully tempered, clear float glass inch thick.

### 2.2 DOORS

Doors must comply with ANSI/DASMA 102. Metal doors to have horizontal sections hinged together which operate in a system of tracks to completely close the door opening in the closed position and make the full width and height of the door opening available for use in the open position. Provide a permanent label on the door indicating the name and address of the manufacturer. Provide doors with heavy-duty lift type designed to slide up and back into a horizontal overhead position and requiring a maximum of 16 inch of headroom for 2 inch tracks and 21 inch of headroom for 3 inch tracks. Doors operate by electric power with auxiliary hand chain operation. Provide full vision door panels with heavy-duty tracks.

### 2.3 DESIGN REQUIREMENTS

Design wind load must conform to the site-specific design wind load for the building of door area without damage. Doors must be tested in accordance with either ASTM E330/E330M or ANSI/DASMA 108 and must meet the acceptance criteria of ANSI/DASMA 108. Refer to Design Criteria and Components and Cladding Wind Pressures Schedule on the Structural Drawings (Sheet S-002)

### 2.4 FABRICATION

#### 2.4.1 Steel Overhead Doors

Form door sections of hot-dipped galvanized steel not lighter than 20 gage

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

with longitudinal integral reinforcing ribs. Install sections not less than 2 inch in thickness. Meeting rails to have interlocking joints to ensure a weathertight closure and alignment for full width of the door. Provide sections of the height indicated or the manufacturer's standard. Do not exceed thick 24 inch height for intermediate sections. Bottom sections may be varied to suit door height. Do not exceed 30 inch height for bottom section. Provide glass panels and install panels using manufacturer's standard for rubber gaskets.

#### 2.4.1.1 Insulated Sections

Insulate door sections with plastic foam or other material providing a "U" factor of 0.27 or less. Cover interior of door sections with steel sheets of not lighter than 27 gage to completely enclose the insulating material.

#### 2.4.2 Tracks

Provide galvanized steel tracks not lighter than 14 gage thick for 2 inch tracks and not lighter than 12 gage thick for 3 inch tracks. Provide galvanized steel vertical tracks with continuous galvanized steel angle not lighter than 13 gage for installation to walls. Incline vertical track through use of adjustable brackets to obtain a weathertight closure at jambs. Reinforce horizontal track with galvanized steel angle; support from track ceiling construction with galvanized steel angle and cross bracing to provide a rigid installation.

#### 2.4.3 Hardware

Provide hinges, brackets, rollers, locking devices, and other hardware required for complete installation. Install roller brackets and hinges with 14 gage galvanized steel. Provide rollers with ball bearings and case-hardened races. Provide reinforcing on doors where roller hinges are connected. Provide a positive locking device and cylinder lock with two keys on manually operated doors.

#### 2.4.4 Counterbalancing

Counterbalance doors with an oil-tempered, helical-wound torsional spring mounted on a steel shaft. Provide adjustable spring tension, connect spring to doors with cable through cable drums. Provide cable safety factor of at least 5 to 1.

### 2.5 MANUAL OPERATORS

#### 2.5.1 Pushup Operators

Provide lifting handles on both sides of door. Do not exceed the lesser of 10 percent of the door weight or 25 pounds for the maximum lifting force of required to operate the door. Provide pulldown straps or ropes at bottom of doors over 7 feet high.

#### 2.5.2 Chain Hoist Operators

Provide a galvanized, endless chain operating over a sprocket. Extend chain to within 4 feet of the floor and mount on inside of building. Obtain reduction by use of roller chain and sprocket drive or gearing. Provide chain cleat and pin for securing operator chain. Allow for future installation of power operators to chain hoist operator. Do not exceed the maximum lifting force of 35 pounds required to operate the door.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## 2.6 ELECTRIC OPERATORS

### 2.6.1 Operator Features

Operators must be labeled and listed to the requirements of UL 325. Provide operators of the drawbar type or side mount (jack shaft) type as recommended by the manufacturer. Include operators with electric motor, machine-cut reduction gears, steel chain and sprockets, magnetic brake, brackets, pushbutton radio controls, limit switches, magnetic reversing contactor, a manual chain hoist operator for emergency use, and other accessories necessary for operation. Design electric operator so motor may be removed without disturbing the limit switch timing and without affecting the manual operator. Provide the operator with slipping clutch coupling to prevent stalling the motor. Provide a clutch controlled emergency manual operator so that it may be engaged and disengaged from the floor; do not affect limit switch timing by operation. The manual operator is not required if door can be manual-pushup operated with a force not to exceed 25 pounds. Provide an electrical or mechanical device that disconnects the motor from the operating mechanism when the manual operator is engaged.

### 2.6.2 Motors

NEMA MG 1, high-starting torque, reversible type with sufficient horsepower and torque output to move the door in either direction from any position. Provide a motor to produce a door travel speed of not less than 8 inch or more than one foot per second without exceeding the rated capacity. Motors must be operate on current of the characteristics indicated at not more than 3600 rpm. Motors must be suitable for operation on current of the characteristics indicated. Provide motor enclosures with drip-proof type or NEMA TENV type.

### 2.6.3 Controls

Provide a motor for each door with an enclosed, across-the-line type, magnetic reversing contactor, thermal overload and undervoltage protection, solenoid-operated brake, limit switches, and control switches. Locate control switches at least 5 feet above the floor so the operator will have complete visibility of the door at all times. Provide control equipment to conform to NEMA ICS 1 and NEMA ICS 2. Provide control enclosures with NEMA ICS 6, Type 12 or Type 4, except that contactor enclosures may be Type 1. Provide a three-button type control switch stations with buttons marked "OPEN," "CLOSE," and "STOP." The "OPEN" and "STOP" buttons must require only momentary pressure to operate. The "CLOSE" button must require constant pressure to maintain the closing motion of the door. If the door is in motion and the "STOP" button is pressed or the "CLOSE" button released, the door must stop instantly and remain in the stop position; from the stop position, the door may be operated in either direction by the "OPEN" or "CLOSE" button. Provide full-guarded type pushbuttons to prevent accidental operation. Provide limit switches to automatically stop doors at the fully open and closed positions. Limit switch positions must be readily adjustable.

### 2.6.4 Entrapment Protection Devices

Provide entrapment protection devices for electrically-operated doors in accordance with UL 325. These devices must immediately stop and reverse the door in its closing travel upon sensing an obstruction in the door

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

opening or upon failure of the device or any component of the control system. Any momentary door-closing circuit must be automatically locked out and the door must be operable manually or with constant pressure controls until the failure or damage has been corrected. No entrapment protection device must be used as a limit switch, unless its function is specifically intended to do so.

#### 2.6.5 Control Transformers

NEMA ST 20. Provide transformers in power circuits as necessary to reduce the voltage on the control circuits to 120 volts or less.

#### 2.6.6 Electrical Components

NFPA 70. Provide manual or automatic control and safety devices, including extra flexible Type SO cable and spring-loaded automatic take-up reel or equivalent device, for operation of the doors. Conduit wiring and mounting of controls are specified in the corresponding electrical specification section.

#### 2.7 WEATHER SEALS AND SENSING EDGES

Provide exterior doors with weatherproof joints between sections by means of tongue-and-groove joints, rabbetted joints, shiplap joints, or wool pile, vinyl or rubber weatherstripping; a rubber, or vinyl adjustable weatherstrip at the top and jambs; and a compressible neoprene or rubber weather seal attached to the bottom of the door. On exterior doors that are electrically operated, where a sensing edge is employed, the bottom seal must be combination compressible weather seal and sensing edge for stopping and reversing door movement.

#### 2.8 FINISHES

Hot-dip galvanize concealed metal surfaces and tracks in accordance with ASTM A123/A123M. Hot-dip galvanized and other ferrous metal surfaces, except rollers and lock components, which are galvanized or plated shop primed.

##### 2.8.1 Galvanized and Shop Primed

Provide a zinc coating on specified surfaces, a phosphate treatment, and a shop prime coat of rust-inhibitive paint. Conform to ASTM A653/A653M for galvanized coating, coating designation G60, for steel sheets, and ASTM A123/A123M for assembled steel products. The weight of coatings for assembled products must be as designated in Table I of ASTM A123/A123M for the class of material to be coated. Provide a prime coat especially developed for materials treated by phosphates and adapted to application by dipping or spraying. Repair damaged zinc-coated surfaces with galvanizing repair paint and spot prime. At the Contractor's option, a two-part system including bonderizing, baked-on epoxy primer, and baked-on enamel topcoat may be applied in lieu of prime coat specified.

### PART 3 EXECUTION

#### 3.1 INSTALLATION

NFPA 70. Install doors in accordance with approved shop drawings and manufacturer's written installation instructions. Lubricate and adjust doors to operate freely.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

Provide a weathertight installation and free from warp, twist, or distortion. Adjust and lubricate doors to operate freely.

Provide all items and accessories for a complete installation in every respect.

### 3.2 ELECTRICAL WORK

NFPA 70. Conduit, wiring, and mounting of controls.

### 3.3 TESTING

After installation is complete, operate doors to demonstrate installation and function of operators, safety features, and controls. Correct deficiencies.

-- End of Section --

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## SECTION 08 41 13

### ALUMINUM-FRAMED ENTRANCES AND STOREFRONTS

**08/18, CHG 1: 08/18**

#### PART 1 GENERAL

##### 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

#### AMERICAN ARCHITECTURAL MANUFACTURERS ASSOCIATION (AAMA)

|           |                                                                                                                               |
|-----------|-------------------------------------------------------------------------------------------------------------------------------|
| AAMA 501  | (2015) Methods of Test for Exterior Walls                                                                                     |
| AAMA 611  | (2014) Voluntary Specification for Anodized Architectural Aluminum                                                            |
| AAMA 800  | (2016) Voluntary Specifications and Test Methods for Sealants                                                                 |
| AAMA 1503 | (2009) Voluntary Test Method for Thermal Transmittance and Condensation Resistance of Windows, Doors and Glazed Wall Sections |

#### AMERICAN SOCIETY OF CIVIL ENGINEERS (ASCE)

|           |                                                                                                                  |
|-----------|------------------------------------------------------------------------------------------------------------------|
| ASCE 7-16 | (2017; Errata 2018; Supp 1 2018) Minimum Design Loads and Associated Criteria for Buildings and Other Structures |
|-----------|------------------------------------------------------------------------------------------------------------------|

#### ASTM INTERNATIONAL (ASTM)

|                 |                                                                                                                                                                                 |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM B221       | (2021) Standard Specification for Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes                                                                    |
| ASTM E283       | (2019) Standard Test Method for Determining the Rate of Air Leakage Through Exterior Windows, Curtain Walls, and Doors Under Specified Pressure Differences Across the Specimen |
| ASTM E330/E330M | (2014; R 2021) Standard Test Method for Structural Performance of Exterior Windows, Doors, Skylights and Curtain Walls by Uniform Static Air Pressure Difference                |
| ASTM E331       | (2000; R 2023) Standard Test Method for Water Penetration of Exterior Windows, Skylights, Doors, and Curtain Walls by Uniform Static Air Pressure Difference                    |
| ASTM E783       | (2002; R 2018) Standard Test Method for Field Measurement of Air Leakage Through                                                                                                |

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### Installed Exterior Windows and Doors

|                   |                                                                                                                                                                                                          |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM E1105        | (2015; R 2023) Standard Test Method for Field Determination of Water Penetration of Installed Exterior Windows, Skylights, Doors, and Curtain Walls, by Uniform or Cyclic Static Air Pressure Difference |
| ASTM E1424        | (2022) Standard Test Method for Determining the Rate of Air Leakage Through Exterior Windows, Curtain Walls, and Doors Under Specified Pressure and Temperature Differences Across the Specimen          |
| ASTM E1886        | (2019) Standard Test Method for Performance of Exterior Windows, Curtain Walls, Doors, and Impact Protective Systems Impacted by Missile(s) and Exposed to Cyclic Pressure Differentials                 |
| ASTM E1996        | (2017) Standard Specification for Performance of Exterior Windows, Curtain Walls, Doors, and Impact Protective Systems Impacted by Windborne Debris in Hurricanes                                        |
| ASTM F1642/F1642M | (2017) Standard Test Method for Glazing and Glazing Systems Subject to Airblast Loadings                                                                                                                 |

#### BUILDERS HARDWARE MANUFACTURERS ASSOCIATION (BHMA)

|                   |                                        |
|-------------------|----------------------------------------|
| ANSI/BHMA A156.4  | (2024) Door Controls - Closers         |
| ANSI/BHMA A156.10 | (2024) Power Operated Pedestrian Doors |

#### INTERNATIONAL CODE COUNCIL (ICC)

|         |                                    |
|---------|------------------------------------|
| ICC IBC | (2021) International Building Code |
|---------|------------------------------------|

#### U.S. DEPARTMENT OF DEFENSE (DOD)

|              |                                                                               |
|--------------|-------------------------------------------------------------------------------|
| UFC 4-010-01 | (2018; with Change 3, 2024) DoD Minimum Antiterrorism Standards for Buildings |
|--------------|-------------------------------------------------------------------------------|

#### U.S. GENERAL SERVICES ADMINISTRATION (GSA)

|             |                                                             |
|-------------|-------------------------------------------------------------|
| FS TT-P-645 | (Rev C; Notice 1) Primer, Paint, Zinc-Molybdate, Alkyd Type |
|-------------|-------------------------------------------------------------|

#### UNDERWRITERS LABORATORIES (UL)

|        |                                                                                                               |
|--------|---------------------------------------------------------------------------------------------------------------|
| UL 325 | (2017; Reprint Feb 2020) UL Standard for Safety Door, Drapery, Gate, Louver, and Window Operators and Systems |
|--------|---------------------------------------------------------------------------------------------------------------|



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## 1.2 ADMINISTRATIVE REQUIREMENTS

### 1.2.1 Pre-Installation Meetings

Conduct a meeting before installation begins to verify the project requirements, substrate conditions, manufacturer's installation instructions, and manufacturer's warranty requirements.

Within 30 days of the Contract Award, submit the following for review and approval by the Contracting Officer:

- a. List of product installations
- b. Sample warranty
- c. Finish and color samples
- d. Manufacturer's catalog data

Concurrently submit certified test reports showing compliance with specified performance characteristics and UL 325 for the following:

- a. Wind Load (Resistance) in accordance with AAMA 501
- b. Deflection in accordance with ASTM F1642/F1642M
- c. Condensation Resistance and Thermal Transmittance Performance Requirements in accordance with AAMA 1503
- d. Water Infiltration in accordance with ASTM E331
- e. Structural Requirements in accordance with ASTM F1642/F1642M

## 1.3 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

### SD-01 Preconstruction Submittals

Sample Warranty; G

List of Product Installations; G

### SD-02 Shop Drawings

Installation Drawings; G

Fabrication Drawings; G

### SD-03 Product Data

Manufacturer's Catalog Data; G

Finish; G

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

SD-04 Samples

Finish and Color Samples; G

SD-06 Test Reports

Certified Test Reports; FIO

Deflection

Air Infiltration

Condensation Resistance and Thermal Transmittance

Water Infiltration

SD-08 Manufacturer's Instructions

Manufacturer's Instructions

SD-11 Closeout Submittals

Manufacturer's Product Warranty

1.4 QUALITY CONTROL

1.4.1 Qualifications

1.4.1.1 Installer Qualifications

Provide documentation of the installer's experience in performing the work specified in this section.

Ensure that the installers are specialized in work similar to that required for this project, and that they are acceptable to product manufacturer.

1.4.1.2 Manufacturer Qualifications

Ensure that manufacturers meet the requirements specified in this section and project drawings.

Ensure that the manufacturer is capable of providing field service representation during construction, approving acceptable installers and approving application methods.

1.5 DELIVERY, STORAGE, AND HANDLING

1.5.1 Ordering

To avoid construction delays, comply with the manufacturer's lead-time requirements and instructions for ordering.

1.5.2 Packing, Shipping, Handling and Unloading

Deliver materials in the manufacturer's original, unopened, undamaged containers with identification labels intact.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 1.5.3 Storage and Protection

Store materials in a way that protects them from exposure to harmful weather conditions. Avoid damaging the storefront material and components during handling. Protect storefront material against damage from elements, construction activities, and other hazards before, during, and after storefront installation.

Do not use adhesive papers or sprayed coatings that become firmly bonded when exposed to sunlight. Do not leave coating residue on surfaces.

## 1.6 PROJECT / SITE CONDITIONS

### 1.6.1 Field Measurements

Verify actual measurements or openings by taking field measurements before fabrication; record these measurements on shop drawings. To avoid construction delays, coordinate field measurements, and fabrication schedule with construction progress.

## 1.7 WARRANTY

Provide a written manufacturer's warranty, executed by a company official, warranting against defects in materials and products for 2 years from the date of shipment. Warrant that the door corner construction is for the life of the project. Provide a written installer's warranty, warranting work to be watertight and free from defective materials, defective workmanship, and glass breakage as a result of defective design, and agreeing to replace components that fail within 2 years.

The warranty states the following:

- a. Watertight and airtight system installation is completed within specified tolerances.
- b. The completed installation remains free of rattles, wind whistles and noise caused by thermal movement and wind pressure.
- c. System is structurally sound and free from distortion.
- d. Glass and glazing gaskets will not break or "pop" from frames as a result of design, wind load pressure, movement caused by expansion or contraction, or structural loading.
- e. Glazing sealants and gaskets remain free of abnormal deterioration or dislocation as a result of sunlight, weather, or oxidation.

Provide written warranty stating that the organic coating finish will not fade more than 10 percent or show chalking, yellowing, peeling, cracking, pitting, corroding or variations in color, or gloss deterioration beyond the manufacturer's descriptive standards for Ten (10) years from the shipment date and agreeing to promptly correct defects.

Provide a written thermal integrity warranty for Two (2) years from ship date against thermal barrier system failure resulting from the following:

- a. Longitudinal and transverse thermal barrier shrinkage.
- b. Thermal barrier cracking.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

- c. Structural failure of the thermal barrier material.
- d. Loss of adhesion or loss of prescribed edge pressure on glazing material, resulting in excessive air and water infiltration.

## PART 2 PRODUCTS

### 2.1 SYSTEM DESCRIPTION

Provide aluminum entrances, with glass and glazing, door hardware, and components.

Aluminum entrances include impact resistance entrances; wide vertical face dimension, 1-3/4 inch depth, for interior structural silicone glaze, for high-traffic/impact-resistant applications per UFC 4-010-01 "DOD Anti-Terrorism Standards for Buildings." Provide exterior glazing in accordance with UFC 4-010-01 "DoD Minimum Anti-Terrorism Standards for Buildings" Standard 10 and Standard 12.

#### 2.1.1 Design Requirements for Aluminum (Entrances and Components)

Provide a door portal system designed to withstand the following loads without breakage, loss, failure of seals, product deterioration, or other defects.

- a. Dead and Live Loads: Determined by ASCE 7-16 and calculated in accordance with applicable codes.
- b. Seismic Loads: Design and install the system to comply with the seismic requirements for the project location in accordance with Section 1613 of the International Building Code, ICC IBC.
- c. Wind Loads: Design and install the system so that the effects of wind load acting inward and outward normal to the plane of the wall are in accordance with ASTM E330/E330M.
- d. Thermal Loads And Movement:
  - (1) Ambient Temperature Range: 120 degrees F
  - (2) Material Surfaces Range: 180 degrees F
- e. Water and Air Resistance: Provide weatherstripping, exterior gaskets, sealants, and other accessories to resist water and air penetration.
- f. Impact-Protective Systems Provide an impact-protective system in accordance with ASTM E1886/ASTM E1996.

##### 2.1.1.1 Material Standard

ASTM B221; 6063-T5 alloy and tempered.

Provide wide door stile and rail face dimensions of the entrance doors as follows:

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

| Vertical Stile | Top Rail | Bottom Rail |
|----------------|----------|-------------|
| 5 inches       | 5 inches | 10 inches   |

Provide major portions of the door members at 0.125 inches nominal in thickness and glazing molding at 0.050 inches thick.

#### 2.1.1.2 Sealants

Provide either ethylene propylene diene monomer (EPDM) elastomeric extrusions or thermoplastic elastomer glazing gaskets. Structural silicone sealant is required.

Internal Sealants: Provide sealants that according to the manufacturer will remain permanently elastic, tacky, non-drying, non-migrating, and weather tight.

#### 2.1.1.3 Thermal Barrier

Use a rigid, structural thermal barrier to separate all exterior aluminum from interior aluminum. For purposes of this specification, a structural thermal barrier is defined as a system that transfers shear during bending and, therefore, promotes composite action between the exterior and interior extrusions. Do not use a nonstructural thermal barrier. Ensure that the thermal barrier provides a structural connection between the two sides of the door.

### 2.2 FABRICATION

Provide the following information when submitting fabrication drawings for custom fabrications:

- a. Indicate elevations, detailed design, dimensions, member profiles, joint locations, arrangement of units, and member connections.
- b. Show the following items:
  - (1) Details of special shapes.
  - (2) Reinforcing.
  - (3) Anchorage system.
  - (4) Interfacing with building construction.
  - (5) Provisions for expansion and contraction.
  - (6) Thermal breaks.
- c. Indicate typical glazing details, locations of various types and thickness of glass and internal sealant requirements as recommended by the sealant manufacturer.
- d. Clearly indicate locations of exposed fasteners and joints.
- e. Clearly show where and how the manufacturer's system deviates from Contract drawings and these specifications.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 2.2.1 Entrance System Fabrication

Provide door corner construction consisting of mechanical clip fastening, SIGMA deep penetration plug welds and 1-1/8 inch long fillet welds inside and outside all four corners. Provide a hook-in type exterior glazing stop with EPDM glazing gaskets reinforced with non-stretchable cord. Provide an interior glazing stop that is mechanically fastened to the door member and that incorporates a silicone-compatible spacer used with silicone sealant.

Accurately fit and secure joints and corners. Make joints hairline in appearance. Remove burrs and smooth edges. Prepare components with internal reinforcement for door hardware. Arrange fasteners and attachments so that they are concealed from view.

Separate dissimilar metals with protective coating or pre-formed separators to prevent contact and corrosion.

### 2.2.2 Shop Assembly

Fabricate and assemble units with joints only at the intersection of aluminum members with hairline joints; rigidly secure these units, and seal them in accordance with the manufacturer's recommendations.

#### 2.2.2.1 Welding

Conceal welds on aluminum members in accordance with AWS recommendations or methods recommended by the manufacturer. Members showing welding bloom or discoloration on finish or material distortion will be rejected by the Contacting Officer.

### 2.2.3 Finish

a. Clear Anodized; Conforming to AA-M12C22A31 and AAMA 611.

Select and edit the following items for the appropriate finish; delete types that do not apply.

(1) Architectural Class II

### 2.2.4 Fabrication Tolerance

Fabricate and assemble units with joints only at intersection of aluminum members with hairline joints; rigidly secure these units, and seal them in accordance with the manufacturer's recommendations.

Fabricate aluminum entrances in accordance with the entrance manufacturer's prescribed tolerances.

#### 2.2.4.1 Material Cuts

Square to 1/32 inch off square, over largest dimension; proportionate amount of 1/32 inch on the two dimensions.

#### 2.2.4.2 Maximum Offset at Consecutive Members

1/64 inch in alignment between two consecutive members in line, end to end.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### 2.2.4.3 Maximum Offset at Glazing Pocket Corners

1/64 inch between framing members at glazing pocket corners.

#### 2.2.4.4 Joints

Between adjacent members in same assembly: Joints are hairline and square to the adjacent member.

#### 2.2.4.5 Variation

In squaring diagonals for doors and fabricated assemblies: 1/16 inch.

#### 2.2.4.6 Flatness

For doors and fabricated assemblies: plus/minus 1/16 inch of neutral plane.

### 2.3 MATERIALS

#### 2.3.1 Sealants

Refer to Section 07 92 00 JOINT SEALANTS. Ensure that all sealants conform to AAMA 800.

#### 2.3.2 Glass

Refer to Section 08 81 00 GLAZING.

### 2.4 ACCESSORIES

#### 2.4.1 Fasteners

Provide stainless steel fasteners in areas where the fasteners are exposed.

Use non-corrosive and compatible fasteners with components being fastened. Do not use exposed fasteners, except where unavoidable for application of hardware.

In areas where fasteners are not exposed, use aluminum, non-magnetic stainless steel, or other materials warranted by the manufacturer.

For exposed locations, provide countersunk Phillips head screws when items with a matching finish are fastened. For concealed locations, provide the manufacturer's standard fasteners.

Provide nuts or washers that have been designed with a means to prevent disengagement; do not deform fastener threads.

#### 2.4.2 Perimeter Anchors

When steel anchors are used, provide insulation between steel material and aluminum material to prevent galvanic action.

##### 2.4.2.1 Inserts and Anchorage Devices

Provide manufacturer's standard formed or fabricated assemblies, steel or aluminum, of shapes, plates, bars, or tubes. Shop-coat steel assemblies after fabrication with an alkyd zinc chromate primer complying with FS TT-P-645.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 2.4.3 Standard Entrance Hardware

#### 2.4.3.1 Weatherstripping

Equip meeting stiles on pairs of doors with an adjustable astragal using wool pile with a polymeric fin.

Provide door weatherstripping on a single-acting offset pivot or butt-hung door and frame (single or pairs) consisting of a thermoplastic elastomer weatherstripping on a tubular shape with a semi-rigid polymeric backing.

Provide sill-sweep strips: Provide an EPDM blade gasket sweep strip in an aluminum extrusion applied to the interior exposed surface of the bottom rail with concealed fasteners. (Provide as necessary to meet specified performance tests.)

#### 2.4.3.2 Threshold

Provide an extruded aluminum threshold, one piece per door opening, with ribbed surface.

#### 2.4.3.3 Offset Pivots

Provide the manufacturer's standard top and bottom pivots with one intermediate offset pivot.

#### 2.4.3.4 Panic Device

Provide the manufacturer's recommended standard panic hardware.

#### 2.4.3.5 Closer

Provide a surface closer in accordance with ANSI/BHMA A156.4.

#### 2.4.3.6 Security Lock or Dead Lock

Provide Section 08 71 00 Door Hardware.

## PART 3 EXECUTION

### 3.1 EXAMINATION

#### 3.1.1 Site Verification of Conditions

Verify that the condition of substrate previously installed under other sections is acceptable for product installation in accordance with the manufacturer's instructions.

Verify that openings are sized to receive the storefront system and that the sill plate is level in accordance with the manufacturer's acceptable tolerances.

### 3.2 PREPARATION

Field-verify dimensions before fabricating components for the door portal assembly.

Coordinate requirements for locations of blockouts for anchorage of door



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

portal columns and other embedded components with Section 03 30 00  
CAST-IN-PLACE CONCRETE.

Coordinate the erection of door portal with installation of surrounding glass wall and door assemblies. Ensure that the door portals can provide support and anchorage for assembly components.

Coordinate electrical requirements for electrified door hardware to ensure proper power source, conduit, wiring, and boxes.

### 3.2.1 Adjacent Surfaces Protection

Protect adjacent work areas and finish surfaces from damage during product installation.

### 3.2.2 Aluminum Surface Protection

Protect aluminum surfaces from contact with lime, mortar, cement, acids, and other harmful contaminants.

## 3.3 INSTALLATION

Submit installation drawings for review and approval.

Install the entrance system in accordance with the manufacturer's instructions and the AAMA storefront and entrance guide specifications manual. Attach the entrance system to the structure, allowing it to be adjusted to accommodate construction tolerances and other irregularities. Provide alignment attachments and shims to permanently fasten the system to the building structure. Align the assembly so that it is plumb and level, and free of warp and twist. Maintain assembly dimensional tolerances aligning with adjacent work.

Set thresholds in a bed of mastic and secure the thresholds. Protect aluminum members in contact with masonry, steel, concrete, or dissimilar materials using nylon pads or a bituminous coating. Shim and brace the aluminum system before anchoring the system to the structure. Verify that weep holes are open, and the metal joints are sealed in accordance with the manufacturer's installation instructions. Seal metal-to-metal joints using a sealant recommended by the system manufacturer.

### 3.3.1 Tolerances

Ensure that tolerances for wall thickness and other cross-sectional dimensions of entrance members are nominal and in compliance with Aluminum Standards and Data, published by the Aluminum Association.

### 3.3.2 Adjusting

Adjust operating hardware for smooth operation, and as recommended by the manufacturer.

### 3.3.3 Related Products Installation Requirements

#### 3.3.3.1 Sealants (Perimeter)

Refer to Section 07 92 00 JOINT SEALANTS.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 3.3.3.2 Glass

Refer to Section 08 81 00 GLAZING.

## 3.4 FIELD QUALITY CONTROL

### 3.4.1 Air Infiltration

Test air infiltration in accordance with ASTM E783

Submit certified test reports showing compliance with specified performance characteristics as follows:

- a. For single-acting offset pivot, butt hung, or continuous geared hinge entrances in the closed and locked position, test the specimen in accordance with ANSI/BHMA A156.10, and ASTM E283 at a pressure differential of 1.57 psf for pairs of doors; ensure that maximum infiltration for a pair of 7 foot by 8 foot entrance doors and frame is 1.2 cfm/square foot.
- b. Ensure the maximum allowable infiltration for a completed storefront system does not exceed 0.06 cfm/square foot when tested in accordance with ASTM E1424 at a differential static pressure of 6.24 psf.

### 3.4.2 Wind Loads

Provide a completed storefront system capable of withstanding wind pressure loads, normal to the wall plane indicated on the Componentents and Cladding Wind Pressures Design Criteria Schedule on Structural Drawings.

### 3.4.3 Deflection

Submit certified test reports showing that the maximum allowable deflection in a member when tested in accordance with ASTM E330/E330M with allowable stress is L/175 or 3/4 inches maximum.

### 3.4.4 Condensation Resistance and Thermal Transmittance

Submit certified test reports showing compliance with specified performance characteristics as follows:

- a. U-Value Requirements:
  - (1) Perform test in accordance with the AAMA 1503 procedure and on the configuration specified therein.
  - (2) Thermal Transmittance ("U" Value) maximum 0.65 (6250) BTU/hr/sf/deg F at 15 mph exterior wind.
- b. CRF Class Requirements:
  - (1) Perform a test in accordance with AAMA 1503.
  - (2) Condensation Resistance Factor Requirements (CRF) minimum 15.

### 3.4.5 Water Infiltration

Submit certified test reports showing that the system is designed to

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

provide no uncontrolled water when tested in accordance with ASTM E1105 at a static pressure of 8 psf for storefront and 1.57 psf for exterior doors.

### 3.5 ADJUSTING AND CLEANING

#### 3.5.1 Protection

Protect the installed product's finish surfaces from damage during construction. Protect the aluminum storefront system from damage from grinding and polishing compounds, plaster, lime, acid, cement, or other harmful contaminants.

#### 3.5.2 Cleaning

Repair or replace damaged installed products. Clean installed products in accordance with manufacturer's instructions before acceptance remove excess mastic, mastic smears, and other foreign materials. Remove construction debris from the project site and legally dispose of this debris.

### 3.6 WARRANTY

Submit three signed copies of the manufacturer's product warranty for the entrance system as follows:

- a. Warranty Period: Five years from Date of Substantial Completion of the project, provided that the Limited Warranty begins no later than six months from the date of shipment by the manufacturer. In addition, support welded door corner construction with a limited lifetime warranty for the life of the door under normal use.

Ensure that the Warranty's language is identical to the "As Approved" version of the sample warranty submitted to and returned from the Contracting Officer.

-- End of Section --

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## SECTION 08 51 13

### ALUMINUM WINDOWS

**05/19**

#### PART 1 GENERAL

##### 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

#### ALUMINUM ASSOCIATION (AA)

AA DAF45 (2003; Reaffirmed 2009) Designation System  
for Aluminum Finishes

#### AMERICAN ARCHITECTURAL MANUFACTURERS ASSOCIATION (AAMA)

AAMA 611 (2014) Voluntary Specification for  
Anodized Architectural Aluminum

AAMA 701/702 (2011) Voluntary Specification for Pile  
Weatherstripping and Replaceable  
Fenestration Weatherseals

AAMA 907 (2015) Voluntary Specification for  
Corrosion Resistant Coatings on Carbon  
Steel Components Used in Windows, Doors  
and Skylights

AAMA 1503 (2009) Voluntary Test Method for Thermal  
Transmittance and Condensation Resistance  
of Windows, Doors and Glazed Wall Sections

AAMA/WDMA/CSA 101/I.S.2/A440 (2017) North American Fenestration  
Standard/Specification for Windows, Doors,  
and Skylights

#### AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS (ASHRAE)

ASHRAE 169 (2013) Climate Data for Building Design  
Standards

#### ASTM INTERNATIONAL (ASTM)

ASTM E90 (2009; R2016) Standard Test Method for  
Laboratory Measurement of Airborne Sound  
Transmission Loss of Building Partitions  
and Elements

ASTM E413 (2022) Classification for Rating Sound  
Insulation

ASTM E1300 (2016) Standard Practice for Determining  
Load Resistance of Glass in Buildings

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

|                   |                                                                                                                                                                                          |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM E1332        | (2016) Standard Classification for Rating Outdoor-Indoor Sound Attenuation                                                                                                               |
| ASTM E1886        | (2019) Standard Test Method for Performance of Exterior Windows, Curtain Walls, Doors, and Impact Protective Systems Impacted by Missile(s) and Exposed to Cyclic Pressure Differentials |
| ASTM E1996        | (2017) Standard Specification for Performance of Exterior Windows, Curtain Walls, Doors, and Impact Protective Systems Impacted by Windborne Debris in Hurricanes                        |
| ASTM F1642/F1642M | (2017) Standard Test Method for Glazing and Glazing Systems Subject to Airblast Loadings                                                                                                 |
| ASTM F2248        | (2012) Standard Practice for Specifying an Equivalent 3-Second Duration Design Loading for Blast Resistant Glazing Fabricated with Laminated Glass                                       |
| ASTM F2912        | (2017) Standard Specification for Glazing and Glazing Systems Subject to Airblast Loadings                                                                                               |

#### NATIONAL FENESTRATION RATING COUNCIL (NFRC)

|          |                                                                                                                                 |
|----------|---------------------------------------------------------------------------------------------------------------------------------|
| NFRC 100 | (2020) Procedure for Determining Fenestration Product U-Factors                                                                 |
| NFRC 200 | (2020) Procedure for Determining Fenestration Product Solar Heat Gain Coefficient and Visible Transmittance at Normal Incidence |

#### PASSIVE HOUSE INSTITUTE - US (PHIUS)

|                 |                                               |
|-----------------|-----------------------------------------------|
| PHIUS Certified | Certified Data Program for Window Performance |
|-----------------|-----------------------------------------------|

#### PASSIVE HOUSE INSTITUTE INTERNATIONAL (PHI)

|                     |                                                                                               |
|---------------------|-----------------------------------------------------------------------------------------------|
| Passivhaus Criteria | (2012) Certification Criteria for Certified Passive House Glazings and Transparent Components |
|---------------------|-----------------------------------------------------------------------------------------------|

#### SCREEN MANUFACTURERS ASSOCIATION (SMA)

|          |                                                                                          |
|----------|------------------------------------------------------------------------------------------|
| SMA 1004 | (1987; R 1998) Aluminum Tubular Frame Screens for Windows                                |
| SMA 1201 | (R 2013) Specifications for Insect Screens for Windows, Sliding Doors and Swinging Doors |

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

U.S. AIR FORCE (USAF)

U.S. DEPARTMENT OF DEFENSE (DOD)

UFC 4-010-01 (2018; with Change 2, 2022) DoD Minimum  
Antiterrorism Standards for Buildings

U.S. DEPARTMENT OF ENERGY (DOE)

Energy Star (1992; R 2006) Energy Star Energy  
Efficiency Labeling System (FEMP)

## 1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

### SD-02 Shop Drawings

Windows; G

Fabrication Drawings; G

### SD-03 Product Data

Windows

Hardware

Fasteners

Window Performance

Thermal-Barrier Windows

Mullions

Screens

Weatherstripping

Accessories

Adhesives

Thermal Performance

Energy Star Label For Residential Aluminum Window Products

### SD-04 Samples

Finish Sample

Window Sample

### SD-05 Design Data

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

Structural Calculations for Deflection

Design Analysis

SD-06 Test Reports

Minimum Condensation Resistance Factor

Standard Airblast Test; G

Windborne-Debris-Impact Performance

SD-10 Operation and Maintenance Data

Windows, Data Package 1

Submit in accordance with Section 01 78 23 OPERATION AND  
MAINTENANCE DATA.

Plastic Identification

### 1.3 QUALITY ASSURANCE

#### 1.3.1 Qualification of Manufacturer

Window manufacturer must specialize in designing and manufacturing the type of aluminum windows specified in this section, and have a minimum of 10 years of documented successful experience. Manufacturer must have the facilities capable of meeting contract requirements, single-source responsibility and warranty.

#### 1.3.2 Shop Drawing Requirements

Take field measurements prior to preparation of drawings and fabrications. Provide drawings that indicate elevations of windows, full-size sections, thickness and gages of metal, fastenings, proposed method of anchoring, size and spacing of anchors, details of construction, method of glazing, details of operating hardware, mullion details, method and materials for weatherstripping, method of attaching screens, installation details, and other related items.

#### 1.3.3 Sample Requirements

##### 1.3.3.1 Finish Sample Requirements

Submit color chart of standard factory color coatings when factory-finish color coating is to be provided.

##### 1.3.3.2 Window Sample Requirements

Submit one full-size corner of each window type proposed for use. Where screens or weatherstripping is required, fit sample with such items that are to be used.

#### 1.3.4 Design Data Requirements

Submit calculations to substantiate compliance with deflection requirements. A registered Professional Engineer must provide calculations.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

Submit design analysis with calculations showing that the design of each different size and type of aluminum window unit and its anchorage to the structure meets the requirements of paragraph ANTITERRORISM PERFORMANCE REQUIREMENTS. Calculations verifying the structural performance of each window proposed for use, under the given loads, must be prepared and signed by a registered professional engineer. Reflect the window components and anchorage devices to the structure, as determined by the design analysis, in the shop drawings.

#### 1.3.5 Test Report Requirements

Submit test reports for each type of window attesting that identical windows have been tested and meet the requirements specified herein for conformance to AAMA/WDMA/CSA 101/I.S.2/A440 including test size, and minimum condensation resistance factor (CRF), and, for Antiterrorism windows, in lieu of a Design Analysis, results of a Standard Airblast Test.

#### 1.3.6 Certification

Ensure that construction is performed with products that meet or exceed Energy Star criteria, FEMP Designated criteria, and Passivhaus Criteria and be current in their certification. Provide PHIUS Certified window performance.

Each prime window unit must bear the AAMA Label warranting that the product complies with AAMA/WDMA/CSA 101/I.S.2/A440. Certified test reports attesting that the prime window units meet the requirements of AAMA/WDMA/CSA 101/I.S.2/A440, including test size, will be acceptable in lieu of product labeling.

#### 1.4 DELIVERY AND STORAGE

Deliver windows to project site in an undamaged condition. Use care in handling and hoisting windows during transportation and at the jobsite. Store windows and components out of contact with the ground, under a weathertight covering, so as to prevent bending, warping, or otherwise damaging the windows. Repair damaged windows to an "as new" condition as approved. If windows can not be repaired, provide a new unit.

#### 1.5 PLASTIC IDENTIFICATION

Label plastic products provided to indicate their polymeric composition according to the following list. Where products are not labeled, provide product data indicating polymeric information in Operation and Maintenance Manual.

- a. Type 1: Polyethylene Terephthalate (PET, PETE).
- b. Type 2: High Density Polyethylene (HDPE).
- c. Type 3: Vinyl (Polyvinyl Chloride or PVC).
- d. Type 4: Low Density Polyethylene (LDPE).
- e. Type 5: Polypropylene (PP).
- f. Type 6: Polystyrene (PS).



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

- g. Type 7: Other. Use of this code indicates that the package in question is made with a resin other than the six listed above, or is made of more than one resin listed above, and used in a multi-layer combination.

## 1.6 PERFORMANCE REQUIREMENTS

### 1.6.1 Wind Loading Design Pressure

Design window components, including mullions, hardware, and anchors, to withstand a wind-loading design pressure as indicated on the Design Criteria and Components and Cladding Wind Pressures Schedule on the structural drawings.

### 1.6.2 Tests

Test windows proposed for use in accordance with AAMA/WDMA/CSA 101/I.S.2/A440 for the particular type and quality window specified.

Perform tests by a nationally recognized independent testing laboratory equipped and capable of performing the required tests. Submit the results of the tests as certified laboratory reports required herein.

Minimum design load for a uniform-load structural test must be 50 psf.

## 1.7 DRAWINGS

Submit the Fabrication Drawings for aluminum window units showing complete window assembly including hardware, weatherstripping, and subframe assembly details.

## 1.8 WINDOW PERFORMANCE

Aluminum windows must meet the following performance requirements. Perform testing requirements by an independent testing laboratory or agency.

### 1.8.1 Structural Performance

Structural test pressures on window units must be for positive load (inward) and negative load (outward). After testing, there will be no glass breakage, permanent damage to fasteners, hardware parts, support arms or actuating mechanisms or any other damage which could cause window to be inoperable. There must be no permanent deformation of any main frame, sash or ventilator member in excess of the requirements established by AAMA/WDMA/CSA 101/I.S.2/A440 for the window types and classification specified in this section.

### 1.8.2 Antiterrorism Performance Requirements

Windows must meet the antiterrorism performance criteria as specified in the paragraphs below in accordance with UFC 4-010-01. Conformance to the performance requirements must be validated by one of the following methods.

#### 1.8.2.1 Blast-Resistant System Requirements

Windows and skylights fabricated using laminated glass glazing will be designed in accordance with the requirements below. Provide system

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

designed to meet or exceed the following requirements of the UFC 4-010-01  
"DoD Minimum Antiterrorism Standard for Buildings."

#### Section B-3.1.3.3 Glazing Frame Bite

Refer to ASTM F2248 for minimum glazing frame bite requirements for structurally or non-structurally glazed windows, curtain walls, aluminum doors, steel doors or skylights. For structurally glazed applications, apply the structural silicone bead to both sides of the insulating glass units.

#### 1.8.2.2 Standard Airblast Test Method

As an alternative to the 'Computational Design Analysis Method' and 'Dynamic Design Analysis Method' indicated above, window assembly may be tested for evaluation of hazards generated from airblast loading in accordance with ASTM F1642/F1642M by an independent testing agency regularly engaged in blast testing. For proposed window systems that are of the same type as the tested system but of different size, the test results may be accepted provided the proposed window size is within the range from 25 percent smaller to 10 percent larger in area and aspect ratio of the original qualified tested glazing systems in accordance with ASTM F2912. Proposed window system/assembly of a size outside this range will require testing to evaluate their hazard rating or are certified by the 'Dynamic Design Analysis Method' indicated above. Testing may be by shock tube or arena test. Perform the test on the entire proposed window system/assembly, including, the glazing, its framing/support system, operating devices, and all anchorage devices. Window support system replicate anchorage of the window support system with the method of installation to be used for the project.

#### 1.8.3 Air Infiltration

Air infiltration must not exceed the amount established by AAMA/WDMA/CSA 101/I.S.2/A440 for each window type.

#### 1.8.4 Water Penetration

Water penetration must not exceed the amount established by AAMA/WDMA/CSA 101/I.S.2/A440 for each window type.

#### 1.8.5 Thermal Performance

Windows (including frames and glass) will be independently tested and certified with a Solar Heat Gain Coefficient (SHGC) determined according to NFRC 200 procedures and a whole window U-factor determined in accordance with NFRC 100 within the ranges as indicated below according to the ASHRAE 169 Climate Zone of the project location. Provide visual Transmittance (VT) of 0.5 or greater. Submit documentation supporting compliance with Energy Star, FEMP designated, and Passive House qualifications as applicable. Provide proof of Energy Star label for residential aluminum window products.

##### 1.8.5.1 South-Central Climate

Windows installed within Climate Zone 3 will have a U-Factor of 0.30 BTU/h·ft<sup>2</sup>·°F or less and a SHGC of 0.25 or less.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### 1.8.6 Sound Attenuation

When tested in accordance with AAMA/WDMA/CSA 101/I.S.2/A440 or the following below, provide a minimum Sound Transmission Class (STC) of 35 in accordance with ASTM E90 and as determined by ASTM E413 or Outside-Indoor Transmission Class (OITC) of 25 in accordance with ASTM E1332 and as determined by ASTM E413 with the window glazed with 1/2 inch air space between two pieces of 1/4 inch.

#### 1.8.7 Windborne-Debris-Impact Performance

Exterior window system including glazing must comply with indicated basis or enhanced protection testing requirements in ASTM E1996 for Wind Zone 4 when tested according to ASTM E1886. Test specimens must be no smaller in width and length than glazing indicated for use on Project and must be installed in same manner as glazing indicated for use on Project.

- a. Refer to drawings for classification of window requiring basic or enhanced protection.
- b. Large-Missile Test: For glazing located within 30 feet of grade.
- c. Small-Missile Test: For glazing located more than 30 feet above grade.

#### 1.9 WARRANTY

Provide Manufacturer's standard performance guarantees or warranties that extend beyond a 1 year period.

### PART 2 PRODUCTS

#### 2.1 WINDOWS

Provide prime windows that comply with AAMA/WDMA/CSA 101/I.S.2/A440 and the requirements specified herein. In addition to compliance with AAMA/WDMA/CSA 101/I.S.2/A440, window framing members for each individual light of glass must not deflect to the extent that deflection perpendicular to the glass light exceeds  $L/175$  of the glass edge length when subjected to uniform loads at specified design pressures. Provide Structural calculations for deflection to substantiate compliance with deflection requirements. Provide windows of types, performance classes, performance grades, combinations, and sizes indicated or specified. Design windows to accommodate hardware, glass, weatherstripping, screens, and accessories to be furnished. Each window must be a complete factory assembled unit with or without glass installed. Dimensions shown are minimum. Provide windows with insulating glass and thermal break necessary to achieve a minimum Condensation Resistance Factor (CRF) of 53 when tested in accordance with AAMA 1503. Provide manufacturer's standard hardware fabricated from aluminum, stainless steel, carbon steel complying with AAMA 907, or other corrosion-resistant material compatible with adjacent materials; designed to smoothly operate, tightly close, and securely lock windows, and sized to accommodate sash weight and dimensions.

##### 2.1.1 Fixed Windows (F)

Type F-AW40.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 2.1.2 Glass and Glazing

Materials are specified in Section 08 81 00 GLAZING.

### 2.1.3 Caulking and Sealing

Are specified in Section 07 92 00 JOINT SEALANTS.

### 2.1.4 Weatherstripping

AAMA/WDMA/CSA 101/I.S.2/A440. Provide for all ventilating (operable) sash for all windows. Provide woven wool pile weatherstripping 0.210 inch thick, conforming to AAMA 701/702, or polypropylene multifilament fiber weatherstripping installed in an integral weatherstripping groove in the sash or frame, and flexible polyvinylchloride weatherstripping installed in the sill member.

## 2.2 FABRICATION

Fabrication of window units must comply with AAMA/WDMA/CSA 101/I.S.2/A440.

### 2.2.1 Provisions for Glazing

Design windows and rabbets suitable for glass thickness shown or specified.

For minimum antiterrorism windows, attach glazing to its supporting frame using structural silicone sealant or adhesive glazing tape in accordance with ASTM F2248. Design sash for inside glazing and for securing glass with metal beads.

### 2.2.2 Fasteners

Use window manufacturer's standard for windows, trim, and accessories. Self-tapping sheet-metal screws are not acceptable for material more than 1/16 inch thick.

### 2.2.3 Adhesives

Provide joint sealants as specified in Section 07 92 00 JOINT SEALANTS. For interior application of joint sealants, comply with applicable regulations regarding reduced VOC's, and as specified in Section 07 92 00 JOINT SEALANTS.

### 2.2.4 Drips and Weep Holes

Provide continuous drips over heads of top ventilators. Where fixed windows adjoin ventilators, drips must be continuous across tops of fixed windows. Provide drips and weep holes as required to return water to the outside.

### 2.2.5 Combination Windows

Windows used in combination must be factory assembled of the same class and grade. Where factory assembly of individual windows into larger units is limited by transportation considerations, prefabricate, match mark, transport, and field assemble.

### 2.2.6 Mullions and Transom Bars

Provide mullions between multiple window units to resist two times (2X)

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

glazing resistance in accordance with ASTM F2248 and ASTM E1300. Provide mullions with a thermal break. Secure mullions and transom bars to adjoining construction and window units in such a manner as to permit expansion and contraction and to form a weathertight joint. Provide mullion covers on the interior and exterior to completely close exposed joints and recesses between window units and to present a neat appearance.

Provide special covers over structural support at mullions as indicated.

#### 2.2.7 Accessories

Provide windows complete with necessary hardware, fastenings, clips, fins, anchors, glazing beads, and other appurtenances necessary for complete installation and proper operation. Furnish extruded aluminum subframe receptors and subsill with each window unit.

##### 2.2.7.1 Hardware

AAMA/WDMA/CSA 101/I.S.2/A440. The item, type, and functional characteristics must be the manufacturer's standard for the particular window type. Provide stainless steel hardware of suitable design and of sufficient strength to perform the function for which it is used. Equip all operating ventilators with a lock or latching device which can be secured from the inside.

##### 2.2.7.2 Fasteners

Provide concealed anchors of the type recommended by the window manufacturer for the specific type of construction. Anchors and fasteners must be compatible with the window and the adjoining construction. Provide a minimum of three anchors for each jamb located approximately 6 inches from each end and at midpoint.

##### 2.2.7.3 Window Anchors

Anchoring devices for installing windows must be made of aluminum, cadmium-plated steel, stainless steel, or zinc-plated steel conforming to AAMA/WDMA/CSA 101/I.S.2/A440.

#### 2.2.8 Finishes

Comply with NAAMM's "Metal Finishes Manual" for applying and designating finishes. Exposed aluminum surfaces must be factory finished with an anodic coating. Color must be clear anodized. All windows must have the same finish.

##### 2.2.8.1 Anodic Coating

Clean exposed aluminum surfaces and provide an anodized finish conforming to AA DAF45 and AAMA 611. Finish must be:

- b. Architectural Class I ( 0.7 mil or thicker), designation AA-M10-C22-A41, clear (natural) anodized.

#### 2.2.9 Screens

AAMA/WDMA/CSA 101/I.S.2/A440. Provide one insect screen for each operable exterior sash or ventilator. Design screens to be rewirable, easily removable from inside the building, and to permit easy access to operating hardware. Manufacturers standard aluminum frame complying with SMA 1004

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

or SMA 1201. Fabricate frames with mitered or coped joints or corner extrusion, concealed fasteners and removable PVC spline/anchors concealing edge of frame.

#### 2.2.9.1 Insect Screen

Insect screen mesh to be Aluminum wire fabric, 18x16 mesh of 0.011 inch diameter coated aluminum wire.

### 2.3 THERMAL-BARRIER WINDOWS

Provide thermal-barrier windows, complete with accessories and fittings, where indicated.

Specify material and construction except as follows:

- a. Aluminum alloy must be 6063-T6.
- b. Frame construction, including operable sash, must be factory-assembled and factory-sealed inner and outer aluminum completely separated from metal-to-metal contact. Join assembly by a continuous, concealed, low conductance divider housed in an interlocking extrusion of the inner frame. Metal fasteners, straps, or anchors must not bridge the connection between the inner and outer frame.
- c. Operating hardware for each sash must consist of spring-loaded nylon cushion blocks and pin locks designed to lock in predetermined locations.
- d. Sash must be completely separated from metal-to-metal contact by means of woven-pile weatherstripping, plastic, or elastomeric separation members.
- e. Operating and storm sash must be factory-glazed with the type of glass indicated and of the quality specified in Section 08 81 00 GLAZING.

### 2.4 MULLIONS

Provide mullions between multiple-window units where indicated.

Provide profiles for mullions and mullion covers, reinforced as required for the specified wind loading, and securely anchored to the adjoining construction. Mullion extrusion will include serrations or pockets to receive weatherstripping, sealant, or tape at the point of contact with each window flange.

Mullion assembly must include aluminum window clamps or brackets screwed or bolted to the mullion and the mullion cover.

Mullion cover must be screw-fastened to the mullion unless otherwise indicated.

Mullion reinforcing members must be fabricated of the materials specified in AAMA/WDMA/CSA 101/I.S.2/A440 and meet the specified design loading.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## PART 3 EXECUTION

### 3.1 INSTALLATION

#### 3.1.1 Method of Installation

Install in accordance with the window manufacturer's printed instructions and details. Build in windows as the work progresses or install without forcing into prepared window openings. Set windows at proper elevation, location, and reveal; plumb, square, level, and in alignment; and brace, strut, and stay properly to prevent distortion and misalignment. Protect ventilators and operating parts against accumulation of dirt and building materials by keeping ventilators tightly closed and locked to frame. Bed screws or bolts in sill members, joints at mullions, contacts of windows with sills, built-in fins, and subframes in mastic sealant of a type recommended by the window manufacturer. Install and caulk windows in a manner that will prevent entrance of water and wind. Fasten insect screens securely in place.

Any materials that show visual evidence of biological growth due to the presence of moisture must not be installed on the building project.

#### 3.1.2 Dissimilar Materials

Where aluminum surfaces are in contact with, or fastened to masonry, concrete, wood, or dissimilar metals, except stainless steel or zinc, protect the aluminum surface from dissimilar materials as recommended in the Appendix to AAMA/WDMA/CSA 101/I.S.2/A440. Do not coat surfaces in contact with sealants after installation with any type of protective material. Do not apply coatings or lacquers to surfaces to which caulking and glazing components must adhere.

#### 3.1.3 Anchors and Fastenings

Make provision for securing units to each other, to masonry, and to other adjoining construction. Windows installed in masonry walls must have head and jamb members designed to recess into masonry wall not less than 7/16 inch.

#### 3.1.4 Adjustments After Installation

After installation of windows and completion of glazing and field painting, adjust all ventilators and hardware to operate smoothly and to provide weathertight sealing when ventilators are closed and locked. Lubricate hardware and operating parts as necessary. Verify that products are properly installed, connected, and adjusted.

### 3.2 CLEANING

Clean interior and exterior surfaces of window units of mortar, plaster, paint spattering spots, and other foreign matter to present a neat appearance, to prevent fouling of weathering surfaces and weather-stripping, and to prevent interference with the operation of hardware. Replace all stained, discolored, or abraded windows that cannot be restored to their original condition with new windows.

-- End of Section --

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## SECTION 08 60 45

### TRANSLUCENT PANELS

**08/20**

#### PART 1 GENERAL

##### 1.1 SUMMARY

Provide commercially available which satisfy all requirements contained in this section and have been verified by load testing and independent design analyses (if required) to meet specified design requirements. Provide UV-stabilized, shatterproof and energy efficient skylight systems. Provide light transmitting plastics in the manufacturing of skylights for daylighting applications. Systems must meet requirements of UFC 4-010-01.

##### 1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

#### AMERICAN ARCHITECTURAL MANUFACTURERS ASSOCIATION (AAMA)

|                              |                                                                                                                                                         |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| AAMA 2605                    | (2020) Voluntary Specification, Performance Requirements and Test Procedures for Superior Performing Organic Coatings on Aluminum Extrusions and Panels |
| AAMA/WDMA/CSA 101/I.S.2/A440 | (2017) North American Fenestration Standard/Specification for Windows, Doors, and Skylights                                                             |

#### ASTM INTERNATIONAL (ASTM)

|                 |                                                                                                                                  |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------|
| ASTM C297/C297M | (2016) Flatwise Tensile Strength of Sandwich Constructions                                                                       |
| ASTM D572       | (2004; R 2019) Rubber Deterioration by Heat and Oxygen                                                                           |
| ASTM D635       | (2018) Standard Test Method for Rate of Burning and/or Extent and Time of Burning of Plastics in a Horizontal Position           |
| ASTM D1002      | (2010; R 2019) Apparent Shear Strength of Single-Lap-Joint Adhesively Bonded Metal Specimens by Tension Loading (Metal-to-Metal) |
| ASTM D1003      | (2013) Haze and Luminous Transmittance of Transparent Plastics                                                                   |
| ASTM D1037      | (2012) Evaluating Properties of Wood-Base Fiber and Particle Panel Materials                                                     |
| ASTM D1929      | (2020) Standard Test Method for Determining Ignition Temperature of                                                              |



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### Plastics

|            |                                                                                                                                                                                 |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM D2843 | (2019) Standard Test Method for Density of Smoke from the Burning or Decomposition of Plastics                                                                                  |
| ASTM D3841 | (2016) Standard Specification for Glass Fiber-Reinforced Polyester Plastic Panels                                                                                               |
| ASTM E72   | (2022) Conducting Strength Tests of Panels for Building Construction                                                                                                            |
| ASTM E84   | (2023) Standard Test Method for Surface Burning Characteristics of Building Materials                                                                                           |
| ASTM E108  | (2020a) Standard Test Methods for Fire Tests of Roof Coverings                                                                                                                  |
| ASTM E283  | (2019) Standard Test Method for Determining the Rate of Air Leakage Through Exterior Windows, Curtain Walls, and Doors Under Specified Pressure Differences Across the Specimen |
| ASTM E331  | (2000; R 2023) Standard Test Method for Water Penetration of Exterior Windows, Skylights, Doors, and Curtain Walls by Uniform Static Air Pressure Difference                    |

#### ICC EVALUATION SERVICE, INC. (ICC-ES)

|             |                                                        |
|-------------|--------------------------------------------------------|
| ICC-ES AC04 | (2012; R 2015) Acceptance Criteria for Sandwich Panels |
|-------------|--------------------------------------------------------|

#### INTERNATIONAL CODE COUNCIL (ICC)

|         |                                    |
|---------|------------------------------------|
| ICC IBC | (2021) International Building Code |
|---------|------------------------------------|

#### NATIONAL FENESTRATION RATING COUNCIL (NFRC)

|          |                                                                                                                                 |
|----------|---------------------------------------------------------------------------------------------------------------------------------|
| NFRC 100 | (2020) Procedure for Determining Fenestration Product U-Factors                                                                 |
| NFRC 200 | (2020) Procedure for Determining Fenestration Product Solar Heat Gain Coefficient and Visible Transmittance at Normal Incidence |

#### U.S. DEPARTMENT OF DEFENSE (DOD)

|              |                                                                               |
|--------------|-------------------------------------------------------------------------------|
| UFC 4-010-01 | (2018; with Change 2, 2022) DoD Minimum Antiterrorism Standards for Buildings |
|--------------|-------------------------------------------------------------------------------|

### 1.3 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Shop Drawings; G

SD-03 Product Data

Translucent Panels; G

Warranty

SD-06 Test Reports

Test Reports

SD-07 Certificates

Systems

Qualifications

#### 1.4 QUALITY ASSURANCE

- a. Provide documentation of Qualifications for the following: The manufacturer is a company specializing in the manufacture of the specified products with a minimum of 5 years documented experience. The installer has documented experience of 5 years minimum performing the work specified.
- b. Before fabrication, provide a full service mock-up of complete with glass and AAMA certification label for structural purposes and NFRC temporary and Permanent Label for certification of thermal performance rating for review of skylight construction and quality of hardware operation. Glass and glaze in conformance with the applicable requirements of Section 08 81 00 GLAZING.

#### 1.5 DELIVERY, STORAGE, AND HANDLING

Provide factory assembled system modules to the greatest extent possible. Ship panels to the jobsite in rugged shipping units, ready for erection. Affix conspicuous decals on all skylights warning individuals against sitting or stepping on the units. Store skylight panels on the long edge, several inches above the ground, blocked and under cover to prevent warping. Deliver unit skylights in manufacturer's original containers, dry, undamaged, with seals and labels intact. Deliver, store and protect all products in accordance with manufacturer's recommendations.

#### 1.6 WARRANTY

Provide the manufacturer's complete warranty for materials, workmanship, and installation. The warranty is for 5 years from the time of project completion and with no proration. The warranty must guarantee, but not be limited to, the following:

- a. No change in light transmission and color of the panels after exposure to heat of 300 degrees F for 25 minutes.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

- b. There is no delamination of the panel affecting appearance, performance, weatherability or structural integrity of the panels or the completed system.
- c. There is no fiberbloom on the panel face.
- d. Change in light transmission of no more than 6 percent in accordance with ASTM D1003, and in color (yellowing index) no more than 10 points in comparison to the original specified value over a 10 year period.
- e. Provide a single source warranty for the glazing panels and the framing system. Third party warranty for the glazing panels will not be accepted.

## PART 2 PRODUCTS

### 2.1 TRANSLUCENT PANELS

Fabricate insulated wall panels of panels conforming to the specified requirements and other appropriate lab test specified criteria, weighing not less than 8 ounces/square foot. Submit certified Test Reports from independent testing laboratory for each type and class of panel system. Reports must verify that the material meets specified performance requirements. Previously completed test reports will be acceptable if they are current and indicative of products used on this project. Where a Class A, B or C roof is part of the project, provide a listing certificate for roof covering systems category certifying that the product complies with the safety standards of ASTM E108 and ICC IBC. Size and color of wall panels as indicated.

### 2.2 GLASS-FIBER PANELS

Provide glass-fiber reinforced polyester panels conforming to ASTM D3841, Class A and to the requirements of AAMA/WDMA/CSA 101/I.S.2/A440.

#### 2.2.1 Weatherability

Provide the exposed faces of fiberglass sandwich type panels with a permanent glass veil erosion barrier embedded integrally to provide maximum long term resistance to reinforcing fiber exposure. The exterior face sheet must be uniform in strength and resistant to penetration by pencil point.

#### 2.2.2 Non Combustible Grid Core

Use 6063-T6 aluminum I-beams with provisions for mechanical interlocking of muntin-mullion and perimeter to prevent high and low intersections which do not allow full bonding surface to contact with face material. I-beam width no less than 7/16 inch. Machine I-beam grid to tolerances of not greater than plus or minus 0.002 inch for flat panels. Panels must withstand 1200 degrees F fire for a minimum of one hour without collapse or exterior flaming.

#### 2.2.3 Adhesive

Use heat and pressure resin-type laminate adhesive engineered for structural sandwich panel use; which passes testing requirements specified by the International Conference of Building Officials' "Acceptance Criteria for Sandwich Panel Adhesive". Provide with the following minimum

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

strength:

- a. Tensile Strength of 750 psi in accordance with ASTM C297/C297M after two exposures to six cycles each of the aging conditions prescribed in ASTM D1037.
- b. Shear Strength, after exposure to five separate aging conditions in accordance with ASTM D1002:
  - (1) 540 psi at 50 percent relative humidity and 73 degrees F.
  - (2) 800 psi under accelerated aging in accordance with ASTM D1037 at room temperature.
  - (3) 250 psi under accelerated aging in accordance with ASTM D1037 at 182 degrees F.
  - (4) 1400 psi after 500 hour Oxygen Bomb in accordance with ASTM D572.
  - (5) 100 psi at 182 degrees F.

#### 2.2.4 Panel Construction

Provide panels consisting of fiberglass faces laminated to an aluminum I-beam grid core and deflecting no more than 1.9 inches at 30 psf in 10 feet in accordance with ASTM E72, without a supporting frame. Include manufacturing facilities, sandwich panel components and production sandwich panels in the quality control inspections and required testing, conducted at least once each year, for conformance with ICC-ES AC04 or equivalent.

#### 2.3 THERMOPLASTIC POLYCARBONATE PANELS

Manufacture systems from translucent polycarbonate panels designed for architectural applications. Provide panels consisting of a polycarbonate resin with a permanent, co-extruded, ultra-violet protective layer; co-extruded by the manufacturer during the original extrusion of the panel a permanent part of the exterior and interior layers. Pot-applied coatings or films of dissimilar materials are unacceptable. Provide panel width not to exceed 2 feet to ensure best performance for wind uplift, vibration, oil canning and visual appearance. Meet the following manufacturing requirements:

- a. Extruded in one single formable length. Transverse sections are unacceptable. Manufacture the panels with upstands which are integral to the unit, and with the upstands 90 degrees to the panel face (standing seam dry glazed concept). Welding or gluing of upstands or standing seam is unacceptable.
- b. Provide dry glazed profiles mullions, using no sealant, welding, adhesives or gaskets; thermally break mullions continuous for panel length.
- c. For structural performance, the use of adhesives, plastic or sonic welding or sealant is not allowed.
- d. For longevity, the minimum ratio of panel weight to thickness must be 0.5 psf for 0.4 inch thick panel.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

- e. Extruded panel includes integral extruded multi-cells, and truss-like structural core for resistance to buckling. Interconnect the panel's exterior skins and space apart by supporting ribs, perpendicular to the skins, at a spacing not to exceed 0.16 inches (truss-like construction). In addition, divide the space between the two exterior skins in a cross section by multiple parallel intermediate surfaces, at a spacing not to exceed 0.16 inches.
- f. Interior flame spread classification is Class I in accordance with ASTM E84.
- g. Smoke density no greater than 70 in accordance with ASTM D2843.
- h. The exterior and interior faces must be an approved light transmitting panel with a CC1 fire rating classification in accordance with ASTM D635.
- i. Self-ignition greater than 1058 degrees F in accordance with ASTM D1929.
- j. Fire rated roof assembly translucent panels must be successfully evaluated for fire from exterior exposure in accordance with ASTM E108 to meet Class A rating. Provide panel listed by an independent recognized listing laboratory.

## 2.4 COMMON PANEL REQUIREMENTS

### 2.4.1 Appearance

Provide face sheets uniform in color to prevent splotchy appearance and completely free of ridges and wrinkles which prevent proper surface contact. Clusters of air bubbles/pinholes which collect moisture and dirt are not acceptable.

### 2.4.2 Panel Fabrication

Panel construction must meet the following requirements:

- a. Light transmission 20 percent; color White.
- b. Assembled panel thickness 2-3/4 inches.
- c. Grid size as indicated on Drawings. If not indicated 12 inch x 24 inch; Shoji.

### 2.4.3 Thermal Performance

Provide non-residential insulated wall panels (including frames and glass) certified by the National Fenestration Rating Council with a whole-unit Solar Heat Gain Coefficient (SHGC) maximum of .028 determined according to NFRC 200 procedures and a U-factor maximum of 0.23 Btu/hr-ft<sup>2</sup>-F in accordance with NFRC 100.

### 2.4.4 Condensation Index Rating

The condensation index rating must be 45 as determined using National Fenestration Rating Council approved software THERM.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## 2.5 TRANSLUCENT PANEL SYSTEMS

Submit manufacturer's certificate that the systems meet or exceed specified requirements. Provide systems evaluated and listed (the whole translucent panel as a unit, not just a glazing material in the unit) by the recognized building code authorities: ICC and SBCCI-Public Safety Testing and Evaluation Services Inc. Product ratings determined using NFRC 100 and NFRC 200 must be authorized for certification and properly labeled by the manufacturer. Provide translucent panel systems meeting the following requirements:

- a. Integral perimeter framing system assembly by the manufacturer.
- b. Exterior panel faces white in color. Interior panel faces white in color.
- c. Air infiltration at 1.57 psf less than 0.04 cfm/ft<sup>2</sup> and at 6.24 psf less than 0.07 cfm/ft<sup>2</sup> in accordance with ASTM E283.
- d. Water penetration at test pressure of 15 psf equals zero in accordance with ASTM E331.
- e. Manufacturer is responsible for maximum system deflection, in accordance with the applicable building code, and without damage to system performance. Calculate deflection in accordance with engineering principles.
- f. Incorporate weepage elements within the perimeter framework of the glazing system for drainage of any condensation or water penetration.
- g. System must accommodate movement within the system; movement between the system and perimeter framing components; dynamic loading and release of loads; and deflection of supporting members. Achieve this without damage to system or components, deterioration of weather seals and fenestration properties specified.
- h. Exposed aluminum color must be a shade selected from the manufacturer's standard range. Provide corrosion resistant superior-performance organic finish in accordance with AAMA 2605 .
- i. Provide a system requiring no scheduled recoating to maintain its performance or for UV resistance.
- j. Design criteria:
  - (1) Wind Load refer to Design Criteria and Components and Cladding Wind Pressures Schedule on Structural Drawings; snow load .
- k. Use 6063-T6 and 6063-T5 extruded aluminum; all fasteners of stainless steel or plated steel.

## 2.6 FLEXIBLE SEALING TAPE

Provide manufacturer's standard pre-applied sealing tape to closure system at the factory under controlled conditions.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### PART 3 EXECUTION

#### 3.1 EXAMINATION

Field verify all submitted opening sizes, dimensions and tolerances; preparation of openings includes isolating dissimilar materials from aluminum system to avoid damage by electrolysis. The installer must examine area of installation to verify readiness of site conditions and to notify the Contractor about any defects requiring correction. Verify when structural support is ready to receive all specified work and to convene a pre-installation conference, if approved by the Quality Control Manager, including the Contractor, wall panel installer and all parties directly affecting and affected by the specified work. Do not install any materials that show visual evidence of biological growth due to the presence of moisture. Do not commence work until conditions are satisfactory.

#### 3.2 ERECTION

Erect translucent wall panel system in accordance with the approved shop drawings supplied by the manufacturer. Submit drawings showing fabrication details, materials, dimensions, installation methods, anchors, and relationship to adjacent construction. Fasten and seal in accordance with the manufacturer's shop drawings. Remove all panel, after other trades have completed work on adjacent materials. Carefully inspect and adjust panel installation as necessary to ensure proper installation and weather-tight conditions. provide all staging, lifts and hoists required for the complete installation and field measuring. Install system clean of dirt, debris or staining and thoroughly examined for removal of all protective material prior to final inspection of the designated work area.

-- End of Section --

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## SECTION 08 71 00

### DOOR HARDWARE 02/16, CHG 4: 02/22

#### PART 1 GENERAL

##### 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

#### ASTM INTERNATIONAL (ASTM)

|           |                                                                                                                                                                                 |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM E283 | (2019) Standard Test Method for Determining the Rate of Air Leakage Through Exterior Windows, Curtain Walls, and Doors Under Specified Pressure Differences Across the Specimen |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

#### BUILDERS HARDWARE MANUFACTURERS ASSOCIATION (BHMA)

|                   |                                                                             |
|-------------------|-----------------------------------------------------------------------------|
| ANSI/BHMA A156.1  | (2021) Butts and Hinges                                                     |
| ANSI/BHMA A156.3  | (2020) Exit Devices                                                         |
| ANSI/BHMA A156.4  | (2013) Door Controls - Closers                                              |
| ANSI/BHMA A156.5  | (2020) Cylinder and Input Devices for Locks                                 |
| ANSI/BHMA A156.6  | (2021) Architectural Door Trim                                              |
| ANSI/BHMA A156.7  | (2016) Template Hinge Dimensions                                            |
| ANSI/BHMA A156.8  | (2021) Door Controls - Overhead Stops and Holders                           |
| ANSI/BHMA A156.13 | (2022) Mortise Locks & Latches Series 1000                                  |
| ANSI/BHMA A156.15 | (2021) Release Devices Closer Holder, Electromagnetic and Electromechanical |
| ANSI/BHMA A156.16 | (2023) Auxiliary Hardware                                                   |
| ANSI/BHMA A156.18 | (2020) Materials and Finishes                                               |
| ANSI/BHMA A156.19 | (2013) Power Assist & Low Energy Power Operated Doors                       |
| ANSI/BHMA A156.21 | (2019) Thresholds                                                           |
| ANSI/BHMA A156.22 | (2021) Gasketing                                                            |
| ANSI/BHMA A156.23 | (2010) Electromagnetic Locks                                                |
| ANSI/BHMA A156.24 | (2012) Delayed Egress Locking Systems                                       |



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

|                   |                                                     |
|-------------------|-----------------------------------------------------|
| ANSI/BHMA A156.25 | (2013) Electrified Locking Devices                  |
| ANSI/BHMA A156.26 | (2012) Continuous Hinges                            |
| ANSI/BHMA A156.31 | (2013) Electric Strikes and Frame Mounted Actuators |
| ANSI/BHMA A156.36 | (2010) Auxiliary Locks                              |

#### NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

|          |                                                              |
|----------|--------------------------------------------------------------|
| NFPA 70  | (2023; ERTA 4 2023) National Electrical Code                 |
| NFPA 72  | (2022; ERTA 22-1) National Fire Alarm and Signaling Code     |
| NFPA 80  | (2022) Standard for Fire Doors and Other Opening Protectives |
| NFPA 101 | (2021; TIA 21-1) Life Safety Code                            |
| NFPA 252 | (2022) Standard Methods of Fire Tests of Door Assemblies     |

#### STEEL DOOR INSTITUTE (SDI/DOOR)

|                 |                                                           |
|-----------------|-----------------------------------------------------------|
| SDI/DOOR A250.8 | (2023) Specifications for Standard Steel Doors and Frames |
|-----------------|-----------------------------------------------------------|

#### U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

|             |                                                                                                                                                        |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| 36 CFR 1191 | Americans with Disabilities Act (ADA) Accessibility Guidelines for Buildings and Facilities; Architectural Barriers Act (ABA) Accessibility Guidelines |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|

#### UNDERWRITERS LABORATORIES (UL)

|                |                                                            |
|----------------|------------------------------------------------------------|
| UL Bld Mat Dir | (updated continuously online) Building Materials Directory |
|----------------|------------------------------------------------------------|

## 1.2 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

#### SD-02 Shop Drawings

- Manufacturer's Detail Drawings; G
- Verification of Existing Conditions; G
- Hardware Schedule; G
- Keying System; G

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

SD-03 Product Data

Hardware Items; G

SD-08 Manufacturer's Instructions

Installation

SD-10 Operation and Maintenance Data

Hardware Schedule Items, Data Package 1; G

SD-11 Closeout Submittals

Key Bitting

1.3 SHOP DRAWINGS

Submit manufacturer's detail drawings indicating all hardware assembly components and interface with adjacent construction. Indicate power components and wiring coordination for electrified hardware. Base shop drawings on verified field measurements and include verification of existing conditions.

1.4 PRODUCT DATA

Indicate fire-ratings at applicable components. Provide documentation of ABA/ADA accessibility compliance of applicable components, as required by 36 CFR 1191 Appendix D - Technical.

1.5 HARDWARE SCHEDULE

Provide Hardware Item List and Hardware Schedule containing the following information, and additional information as needed to identify the complete make up of each hardware set and its application to each opening:

1.5.1 Hardware Item List:

- a. Hardware Type
- b. Item Number
- c. Quantity
- d. Size(s)
- e. Reference Publication / Type Number
- f. Manufacturer's Name / Catalog Number
- g. Key Control Symbols
- h. UL Mark (If fire rated and listed)
- i. BHMA Finish(es)
- j. Remarks

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### 1.5.2 Hardware Schedule

- a. Hardware Set Number
- b. Opening Number(s)
- c. Opening Description (single/double leaf, hand, size, door/frame material)
- d. Fire Rating
- e. Sound Rating
- f. Hardware Items
- g. Quantity
- h. Size
- i. BHMA Finish
- j. Remarks

In addition, submit hardware schedule data package 1 in accordance with Section 01 78 23 OPERATION AND MAINTENANCE DATA.

#### 1.6 KEY BITTING CHART REQUIREMENTS

##### 1.6.1 Requirements

Submit key bitting charts to the Contracting Officer prior to completion of the work. Include:

- a. Complete listing of all keys (e.g. AA1 and AA2).
- b. Complete listing of all key cuts (AA1-123456, AA2-123458).
- c. Tabulation showing which key fits which door.
- d. Copy of floor plan showing doors and door numbers.
- e. Listing of 20 percent more key cuts than are presently required in each master system.

#### 1.7 QUALITY ASSURANCE

##### 1.7.1 Hardware Manufacturers and Modifications

Provide, as far as feasible, locks, hinges, and closers of one lock, hinge, or closer manufacturer's make. Modify hardware as necessary to provide features indicated or specified.

##### 1.7.2 Key Shop Drawings Coordination Meeting

Prior to the submission of the key shop drawing, the Contracting Officer, Contractor, Door Hardware Subcontractor, using Activity and Base Locksmith must meet to discuss and coordinate key requirements for the facility.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## 1.8 DELIVERY, STORAGE, AND HANDLING

Deliver hardware in original individual containers, complete with necessary appurtenances including fasteners and instructions. Mark each individual container with item number as shown on hardware schedule. Deliver permanent keys and removable cores to the Contracting Officer, either directly or by certified mail. Deliver construction master keys with the locks.

## PART 2 PRODUCTS

### 2.1 TEMPLATE HARDWARE

Hardware applied to metal or to prefinished doors must be manufactured using a template. Provide templates to door and frame manufacturers in accordance with ANSI/BHMA A156.7 for template hinges. Coordinate hardware items to prevent interference with other hardware.

### 2.2 HARDWARE FOR FIRE DOORS AND EXIT DOORS

Provide all hardware necessary to meet the requirements of NFPA 72 for door alarms, NFPA 80 for fire doors, NFPA 101 for exit doors, NFPA 252 for fire tests of door assemblies, ABA/ADA accessibility requirements, and all other requirements indicated, even if such hardware is not specifically mentioned in paragraph HARDWARE SCHEDULE. Provide Underwriters Laboratories, Inc. labels for such hardware in accordance with UL Bld Mat Dir or equivalent labels in accordance with another testing laboratory approved in writing by the Contracting Officer.

### 2.3 HARDWARE ITEMS

Clearly and permanently mark with the manufacturer's name or trademark, hinges, pivots, locks, latches, exit devices, bolts and closers where the identifying mark is visible after the item is installed. For closers with covers, the name or trademark may be beneath the cover. Coordinate electrified door hardware components with corresponding components specified in Division 28 ELECTRONIC SECURITY SYSTEMS (ESS).

#### 2.3.1 Hinges

Provide in accordance with ANSI/BHMA A156.1. Provide hinges that are 4-1/2 by 4-1/2 inch unless otherwise indicated. Construct loose pin hinges for interior doors and reverse-bevel exterior doors so that pins are non-removable when door is closed. Other anti-friction bearing hinges may be provided in lieu of ball bearing hinges.

#### 2.3.2 Continuous Hinges

Where continuous hinges are required, provide in accordance with ANSI/BHMA A156.26.

#### 2.3.3 Locks and Latches

- a. At exterior locations provide locksets of full stainless steel type 302 or 304 construction including fronts, strike, escutcheons, knobs, bolts and all interior working parts. Marine Grade I, fully non-ferrous.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### 2.3.3.1 Mortise Locks and Latches

Provide in accordance with ANSI/BHMA A156.13, Series 1000, Operational Grade 1, Security Grade 2. Provide knobs and roses of mortise locks with screwless shanks and no exposed screws.

#### 2.3.3.2 Auxiliary Locks

Provide in accordance with ANSI/BHMA A156.36, Grade 1.

#### 2.3.3.3 Combination Locks

Heavy-duty, mechanical combination lockset with five push buttons, standard sized knobs, 3/4 inch deadlocking latch, 2-3/4 inch backset. Locks to operate by pressing two or more of the buttons in unison or individually in the proper sequence. Inside knob operates the latch. Provide a keyed cylinder on the interior to permit setting the combination. Provide a keyed removable core cylinder on the exterior to permit bypassing the combination.

#### 2.3.4 Exit Devices

Provide in accordance with ANSI/BHMA A156.3, Grade 1. Provide adjustable strikes for rim type and vertical rod devices. Provide open back strikes for pairs of doors with mortise and vertical rod devices. Provide touch bars in lieu of conventional crossbars and arms.

#### 2.3.5 Cylinders and Cores

Provide cylinders and cores for new locks, including locks provided under other sections of this specification. Provide cylinders and cores with seven pin tumblers fully compatible with the Stanley Best Access Systems 1E series. Cores shall be solid brass construction with individually capped barrels. Provide cylinders from the products of one manufacturer, and provide cores from the products of one manufacturer. Rim cylinders, mortise cylinders, and knobs of bored locksets have interchangeable cores which are removable by special control keys. Stamp each interchangeable core with a key control symbol in a concealed place on the core.

#### 2.3.6 Push Button Mechanisms

Provide in accordance with ANSI/BHMA A156.5, Grade 1.

#### 2.3.7 Electrified Hardware

Comply with the requirements of NFPA 70 for wiring of electrified hardware.

##### 2.3.7.1 Electric Strikes and Frame Mounted Actuators

Provide in accordance with ANSI/BHMA A156.31, Grade 1. Provide electric strikes and actuators as required to meet operational requirements. Provide electric strikes that remain secure during power failure. Provide a separate power supply for electric strikes, other locking devices and ancillary parts. Provide strikes and actuators with a minimum opening force of 2300 pounds.

Provide facility interface devices that use direct current (dc) power to energize the solenoids. Provide electric strikes and actuators that incorporate end-of-line resistors to facilitate line supervision by the

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

system. If not incorporated into the electric strike or local controller, provide metal oxide resistors (MOVs) to protect the controller from reverse current surges.

#### 2.3.7.1.1 Solenoid

Provide actuating solenoid for strikes and actuators that are rated for continuous duty, cannot dissipate more than 12 Watts and must operate on 12 or 24 Volts dc. Inrush current cannot exceed 1 ampere and the holding current cannot be greater than 500 milliamperes. Actuating solenoid must move from fully secure to fully open positions in less than 500 milliseconds.

#### 2.3.7.1.2 Signal Switches

Provide strikes and actuators with signal switches to indicate to the system when the bolt is not engaged or the strike mechanism is unlocked. Signal switches must report a forced entry to the system.

#### 2.3.7.1.3 Tamper Resistance

Provide strike guards that prevent tampering with the latch bolt of the locking hardware or the latch bolt keeper of the electric strike. Strike guards to bolt through the door using tamper resistant screws. Provide strike guards made of 1/8 inch thick brass and that are 11-1/4 inch high by 1-5/8 inch wide, with a minimum 5/32 inch wide offset.

#### 2.3.7.1.4 Coordination

Provide electric strikes and actuators of a size, weight and profile compatible with each specified door frame. Field verify installation clearances prior to procurement.

#### 2.3.7.1.5 Mounting Method

Provide electric strikes and actuators suitable for use with single and double doors, with mortise or rim type hardware specified, and for right or left hand mounting as specified. In double door installations, locate the lock in the active leaf and monitor the fixed leaf.

#### 2.3.7.2 Electrified Mortise Locks

Provide in accordance with ANSI/BHMA A156.25, Grade 1. Provide electrified mortise locks that remain secure during power failure. Provide facility interface devices that use dc power to energize solenoids. Provide solenoids, resistors, and signal switches in accordance with paragraph ELECTRIC STRIKES AND FRAME MOUNTED ACTUATORS.

##### 2.3.7.2.1 Power Transfer Hinges

Provide power transfer hinges with each electrified lock that route power and monitoring signals from the lockset to the door frame. Coordinate power transfer hinges with door frames.

#### 2.3.7.3 Card Readers and Keypad Access Control Hardware

Coordinate access control hardware with corresponding devices and systems specified in Division 28 ELECTRONIC SECURITY SYSTEMS (ESS).

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### 2.3.7.4 Release Devices

In accordance with ANSI/BHMA A156.15, Grade 1.

##### 2.3.7.4.1 Release Devices

Provide wall mounted Electromagnetic release devices connected to fire and smoke detecting devices.

#### 2.3.7.5 Power Assist and Low Energy Power Operated Doors

Provide in accordance with ANSI/BHMA A156.19, Grade 1.

#### 2.3.7.6 Electromagnetic Locks

Provide in accordance with ANSI/BHMA A156.23, Grade 1. Provide electromagnetic locks that do not contain any moving parts and depend solely upon electromagnetism to secure a portal by generating at least 1200 pounds of holding force. The lock must interface with the local processors without external, internal or functional alteration of the local processor. The electromagnetic lock must incorporate an end of line resistor to facilitate line supervision by the system. Provide metal-oxide resistors (MOVs) to protect controllers from reverse current surges, if not incorporated into the electromagnetic lock or local controller.

##### 2.3.7.6.1 Armature

Provide electromagnetic locks with internal circuitry to eliminate residual magnetism and inductive kickback. Provide actuating armature that operates on 12 or 24 Volts dc and cannot dissipate more than 12 Watts. Holding current must be less than 500 milliamperes. Actuating armature must take less than 300 milliseconds to change the status of the lock from fully secure to fully open or fully open to fully secure.

##### 2.3.7.6.2 Tamper Resistance

Provide lock mechanism encased in hardened guard barriers to deter forced entry.

##### 2.3.7.6.3 Mounting Method

Provide electromagnetic lock suitable for use with single and double door with mortise or rim type hardware and compatible with right or left hand mounting.

#### 2.3.7.7 Delayed Egress Locking System

Provide in accordance with ANSI/BHMA A156.24, Grade 1.

#### 2.3.8 Keying System

Provide a grand master keying system Provide construction interchangeable cores.

#### 2.3.9 Lock Trim

Provide cast, forged, or heavy wrought construction and commercial plain design for lock trim.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### 2.3.9.1 Lever Handles

Provide lever handles where indicated in the Hardware Schedule. Provide in accordance with ANSI/BHMA A156.3 for mortise locks of lever handles for exit devices. Provide lever handle locks with a breakaway feature (such as a weakened spindle or a shear key) to prevent irreparable damage to the lock when force in excess of that specified in ANSI/BHMA A156.13 is applied to the lever handle. Provide lever handles return to within 1/2 inch of the door face.

#### 2.3.10 Keys

Furnish one file key, one duplicate key, and one working key for each key change and for each master and grand master keying system. Furnish one additional working key for each lock of each keyed-alike group. Furnish 4 great grand master keys, 12 master keys, 15 construction master keys, and 4 control keys for removable cores. Stamp each key with appropriate key control symbol and "U.S. property - do not duplicate." Do not place room number on keys.

#### 2.3.11 Door Bolts

Provide in accordance with ANSI/BHMA A156.16. Provide dustproof strikes for bottom bolts, except at doors having metal thresholds. Provide automatic latching flush bolts in accordance with ANSI/BHMA A156.3, Type 25.

#### 2.3.12 Closers

Provide in accordance with ANSI/BHMA A156.4, Series C02000, Grade 1, with PT 4C. Provide with brackets, arms, mounting devices, fasteners, and other features necessary for the particular application. Size closers in accordance with manufacturer's printed recommendations, or provide multi-size closers, Sizes 1 through 6, and list sizes in the Hardware Schedule. Provide manufacturer's 10 year warranty.

##### 2.3.12.1 Identification Marking

Engrave each closer with manufacturer's name or trademark, date of manufacture, and manufacturer's size designation in locations that will be visible after installation.

#### 2.3.13 Overhead Holders

Provide in accordance with ANSI/BHMA A156.8.

#### 2.3.14 Door Protection Plates

Provide in accordance with ANSI/BHMA A156.6.

##### 2.3.14.1 Sizes of Armor Mop and Kick Plates

2 inch less than door width for single doors; 1 inch less than door width for pairs of doors. Provide 10 inch kick plates for flush doors. Provide a minimum 36 inch armor plates for flush doors and completely cover lower panels of panel doors, except UL listed armor plates on fire doors. Provide 10 inch mop plates.



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### 2.3.15 Door Stops and Silencers

Provide in accordance with ANSI/BHMA A156.16. Silencers Type L03011. Provide three silencers for each single door, two for each pair.

#### 2.3.16 Thresholds

Provide in accordance with ANSI/BHMA A156.21. Use J35100, with vinyl or silicone rubber insert in face of stop, for exterior doors opening out, unless specified otherwise.

#### 2.3.17 Weatherstripping Gasketing

Provide in accordance with ANSI/BHMA A156.22. Provide the type and function designation where specified in paragraph HARDWARE SCHEDULE. Provide a set to include head and jamb seals, sweep strips, and, for pairs of doors, astragals. Air leakage of weatherstripped doors not to exceed 0.5 cubic feet per minute of air per square foot of door area when tested in accordance with ASTM E283. Provide weatherstripping with one of the following: Silicone rubber, or polyurethane inserts. Provide clear anodized aluminum.

##### 2.3.17.1 Extruded Aluminum Retainers

Extruded aluminum retainers not less than 0.050 inch wall thickness with vinyl, neoprene, silicone rubber, or polyurethane inserts. Provide clear (natural) anodized aluminum.

##### 2.3.17.2 Interlocking Type

Zinc or bronze not less than 0.018 inch thick.

##### 2.3.17.3 Spring Tension Type

Spring bronze or stainless steel not less than 0.008 inch thick.

#### 2.3.18 Lightproofing and Soundproofing Gasketing

Provide in accordance with ANSI/BHMA A156.22. Provide adjustable doorstops at heads, jams and automatic door bottoms in accordance with the hardware set, of extruded aluminum, clear (natural) anodized, surface applied, with vinyl fin seals between plunger and housing. Provide doorstops with solid neoprene tube, silicone rubber, or closed cell sponge gasket. Provide door bottoms with adjustable operating rod and silicone rubber or closed cell sponge neoprene gasket. Provide doorstops that are mitered at corners. Provide type and function designation where specified in paragraph HARDWARE SETS.

#### 2.3.19 Rain Drips

Provide in accordance with ANSI/BHMA A156.22. Provide extruded aluminum rain drips, not less than 0.08 inch thick, clear anodized finish. Provide the manufacturer's full range of color choices to the Contracting Officer for color selection. Provide rain drips with a 4 inch overlap on each side of each exterior door that is not protected by an awning, roof, eave or other horizontal projection. Set drips in sealant and fasten with stainless steel screws.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### 2.3.19.1 Door Rain Drips

Approximately 1-1/2 inch high by 5/8 inch projection. Align bottom with bottom edge of door.

#### 2.3.19.2 Overhead Rain Drips

Approximately 1-1/2 inch high by 2-1/2 inch projection. Align bottom with door frame rabbet.

#### 2.3.20 Auxiliary Hardware (Other than locks)

Provide in accordance with ANSI/BHMA A156.16, Grade 1.

#### 2.3.21 Special Tools

Provide special tools, such as spanner and socket wrenches and dogging keys, as required to service and adjust hardware items.

### 2.4 FASTENERS

Provide fasteners of type, quality, size, and quantity appropriate to the specific application. Fastener finish to match hardware. Provide stainless steel or nonferrous metal fasteners in locations exposed to weather. Verify metals in contact with one another are compatible and will avoid galvanic corrosion when exposed to weather.

### 2.5 FINISHES

Provide in accordance with ANSI/BHMA A156.18. Provide hardware in BHMA 630 finish (satin stainless steel), unless specified otherwise. Provide items not manufactured in stainless steel in BHMA 626 finish (satin chromium plated) over brass or bronze, except aluminum paint finish for surface door closers, and except BHMA 652 finish (satin chromium plated) for steel hinges. Provide hinges for exterior doors in stainless steel with BHMA 630 finish or chromium plated brass or bronze with BHMA 626 finish. Furnish exit devices in BHMA 626 finish in lieu of BHMA 630 finish except where BHMA 630 is specified under paragraph HARDWARE SETS. Match exposed parts of concealed closers to lock and door trim. Match hardware finish for aluminum doors to the doors.

### 2.6 KEY CABINET AND CONTROL SYSTEM

Provide in accordance with ANSI/BHMA A156.5, Type required to yield a capacity (number of hooks) 50 percent greater than the number of key changes used for door locks.

## PART 3 EXECUTION

### 3.1 INSTALLATION

Provide hardware in accordance with manufacturers' printed installation instructions. Fasten hardware to wood surfaces with full-threaded wood screws or sheet metal screws. Provide machine screws set in expansion shields for fastening hardware to solid concrete and masonry surfaces. Provide toggle bolts where required for fastening to hollow core construction. Provide through bolts where necessary for satisfactory installation.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 3.1.1 Weatherstripping Installation

Provide full contact, weathertight seals that allow operation of doors without binding the weatherstripping.

#### 3.1.1.1 Stop Applied Weatherstripping

Fasten in place with color matched sheet metal screws not more than 9 inch on center after doors and frames have been finish painted.

#### 3.1.1.2 Interlocking Type Weatherstripping

Provide interlocking, self adjusting type on heads and jambs and flexible hook type at sills. Nail weatherstripping to door 1 inch on center and to heads and jambs at 4 inch on center.

#### 3.1.1.3 Spring Tension Type Weatherstripping

Provide spring tension type on heads and jambs. Provide bronze nails with bronze. Provide stainless steel nails with stainless steel. Space nails not more than 1-1/2 inch on center.

### 3.1.2 Lightproofing and Soundproofing Installation

Provide as specified for stop applied weatherstripping.

### 3.1.3 Threshold Installation

Extend thresholds the full width of the opening and notch end for jamb stops. Set thresholds in a full bed of sealant and anchor to floor with cadmium-plated, countersunk, steel screws in expansion sleeves. For aluminum thresholds placed on top of concrete surfaces, coat the underside surfaces that are in contact with the concrete with fluid applied waterproofing as a separation measure prior to placement.

## 3.2 FIRE DOORS AND EXIT DOORS

Provide hardware in accordance with NFPA 72 for door alarms, NFPA 80 for fire doors, NFPA 101 for exit doors, and NFPA 252 for fire tests of door assemblies.

## 3.3 HARDWARE LOCATIONS

Provide in accordance with SDI/DOOR A250.8, unless indicated or specified otherwise.

- a. Kick and Armor Plates: Push side of single-acting doors. Both sides of double-acting doors.
- b. Mop Plates: Bottom flush with bottom of door.

## 3.4 KEY CABINET AND CONTROL SYSTEM

Locate where directed. Tag one set of file keys and one set of duplicate keys. Place other keys in appropriately marked envelopes, or tag each key. Provide complete instructions for setup and use of key control system. On tags and envelopes, indicate door and room numbers or master or grand master key.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 3.5 FIELD QUALITY CONTROL

After installation, protect hardware from paint, stains, blemishes, and other damage until acceptance of work. Submit notice of testing 15 days before scheduled, so that testing can be witnessed by the Contracting Officer. Adjust hinges, locks, latches, bolts, holders, closers, and other items to operate properly. Demonstrate that permanent keys operate respective locks, and give keys to the Contracting Officer. Correct, repair, and finish, errors in cutting and fitting and damage to adjoining work.

### 3.6 HARDWARE SETS

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

Provide hardware for aluminum doors under this section. Deliver Hardware templates and hardware, except field applied hardware, to the aluminum door and frame manufacturer for use in fabricating doors and frames.

## Set #1

|   |                         |                                 |       |  |
|---|-------------------------|---------------------------------|-------|--|
| 2 | Continuous Hinge        | 662HD                           | 628   |  |
| 1 | Semi-Auto Flushbolt     | 3820 X 3810                     | 630   |  |
| 1 | Dust Proof Strike       | 3911                            | 630   |  |
| 1 | Push/Pull Classroom Set | J 6656 AM                       | 630AM |  |
| 1 | Mortise Cylinder        | 1E-74 PATD                      | 626   |  |
| 2 | Door Closer             | C72061 PT4C/D/F/G/H X 1.5" BORE | 689   |  |
| 2 | Armor Plate             | J103 34"                        | 630   |  |
| 2 | Frame Protector         | FG8000-36                       | 630   |  |
| 1 | Gasketing               | 2525 C                          |       |  |

NOTE: Astragal by door manufacturer.

## Set #2

|   |                         |                                 |        |  |
|---|-------------------------|---------------------------------|--------|--|
| 1 | Continuous Hinge        | 662HD                           | 628    |  |
| 1 | Push/Pull Classroom Set | J 6656 AM                       | 630AM  |  |
| 1 | Mortise Cylinder        | 1E-74 PATD                      | 626    |  |
| 1 | Door Closer             | C72061 PT4C/D/F/G/H X 1.5" BORE | 689    |  |
| 1 | Armor Plate             | J103 34"                        | 630    |  |
| 2 | Frame Protector         | FG8000-36                       | 630    |  |
| 1 | Gasketing               |                                 | 2525 C |  |

## Set #3

|   |              |                                 |       |  |
|---|--------------|---------------------------------|-------|--|
| 3 | Hinge        | CB1900R 4.5" x 4.5"             | 652   |  |
| 1 | Exit Device  | FL TYPE 1 FUNC. 8               | 630AM |  |
| 1 | Rim Cylinder | 12E-72 PATD                     | 626   |  |
| 1 | Door Closer  | C72011 PT4C/D/F/G/H X 1.5" BORE | 689   |  |
| 1 | Kick Plate   | J102 8"                         | 630   |  |
| 1 | Gasketing    | 2525 C                          |       |  |

## Set #4

|   |                     |                                 |       |  |
|---|---------------------|---------------------------------|-------|--|
| 6 | Hinge               | CB1900R 4.5" x 4.5"             | 652   |  |
| 1 | Semi-Auto Flushbolt | 3820 X 3810                     | 630   |  |
| 1 | Dust Proof Strike   | 3911                            | 630   |  |
| 1 | Lockset             | F07 (45H-7D14N)                 | 630AM |  |
| 2 | Door Closer         | C72011 PT4C/D/F/G/H X 1.5" BORE | 689   |  |
| 2 | Kick Plate          | J102 8"                         | 630   |  |
| 1 | Gasketing           | 2525 C                          |       |  |

NOTE: Astragal by door manufacturer.

## Set #5 - Push Button Entry

|   |                      |                                 |       |  |
|---|----------------------|---------------------------------|-------|--|
| 1 | Continuous Hinge     | 662HD                           | 628   |  |
| 1 | Keypad Lock          | 45HZ-7DV14KP PATD               | 626AM |  |
| 1 | Door Closer          | C72061 PT4C/D/F/G/H X 1.5" BORE | 689   |  |
| 1 | Armor Plate          | J103 34"                        | 630   |  |
| 2 | Frame Protector      | FG8000-36                       | 630   |  |
| 1 | Door Position Switch | HSS SERIES, CONCEALED           |       |  |
| 1 | Weatherstrip         | 700 ES                          |       |  |
| 1 | Door Sweep           | C627 A                          |       |  |
| 1 | Threshold            | 896 S                           | 628   |  |

## Set #6 - Exterior, Deadlock

|   |                  |              |     |  |
|---|------------------|--------------|-----|--|
| 1 | Continuous Hinge | 662HD        | 628 |  |
| 1 | Deadlock         | F29 (48H-7R) | 626 |  |
| 1 | Flush Pull       | 1111C        | 630 |  |

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

1 Door Closer C72011 PT4C/D/F/G/H X 1.5" BORE 689  
1 Kick Plate J102 8" 630  
1 Weatherstrip 700 ES  
1 Door Sweep C627 A  
1 Threshold 896 S 628

Set #7 - Exterior, Exit Only

1 Continuous Hinge 662HD 628  
1 Exit Device TYPE 3 FUNC. 1 630AM  
1 Door Closer C72061 PT4C/D/F/G/H X 1.5" BORE 689  
1 Kick Plate J102 8" 630  
1 Door Position Switch HSS SERIES, CONCEALED  
1 Weatherstrip 700 ES  
1 Door Sweep C627 A  
1 Threshold 896 S 628

Set #8 - Overhead

1 Padlock 41B-722T PATD M1 606  
2 Door Position Switch AS REQUIRED

Set #9

1 Continuous Hinge 662HD 628  
1 Push/Pull Classroom Set J 6656 AM 630  
1 Mortise Cylinder 1E-74 PATD 626  
1 Door Closer C72011 PT4C/D/F/G/H X 1.5" BORE 689  
1 Kick Plate J102 8" 630  
1 Mop Plate J101 4" 630  
1 Gasketing 2525 C

Set #10 - Card Reader, Exterior

2 Continuous Hinge 662HD 628  
1 Exit Device ELR TYPE 6 FUNC. 2 CD DS 630AM  
1 Exit Device ELR TYPE 6 FUNC. 3 CD DS 630AM  
2 Mortise Cylinder 1E-74 PATD 626  
1 Rim Cylinder 12E-72 PATD 626  
2 Door Pull AP311 630AM  
2 Door Closer C72011 PT4C/D/F/H X 1.5" BORE 689  
2 Overhead Stop N1020 A 630  
2 Door Position Switch HSS SERIES, CONCEALED  
2 Power Transfer PT1000 689  
1 Power Supply ELR152  
1 Card Reader BY SECURITY  
2 Door Sweep C627 A  
1 Threshold 896 S 628

Set #11 - Card Reader, Remote Push Button

2 Continuous Hinge 662HD 628  
2 Exit Device ELR TYPE 8 FUNC. 8 D CD DS LBR 630AM  
2 Mortise Cylinder 1E-74 PATD 626  
2 Rim Cylinder 12E-72 PATD 626  
2 Door Closer C72011 PT4C/D/F/G/H X 1.5" BORE 689  
2 Kick Plate J102 8" 630  
2 Door Position Switch HSS SERIES, CONCEALED  
2 Power Transfer PT1000 689  
1 Power Supply ELR152  
1 Card Reader BY SECURITY  
2 Gasketing 2525 C

Set #12 - Exterior

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

2 Continuous Hinge 662HD 628  
1 Semi-Auto Flushbolt 3820 X 3810 630  
1 Dust Proof Strike 3911 630  
1 Exit Device TYPE 3 FUNC. 3 630AM  
1 Mortise Cylinder 1E-74 PATD 626  
1 Flush Pull 1111C 630AM  
2 Door Closer C72061 PT4C/D/F/G/H X 1.5" BORE 689  
2 Kick Plate J102 8" 630  
1 Weatherstrip 700 ES  
1 Door Sweep C627 A  
1 Astragal 158 SA  
1 Threshold 896 S 628

Set #13 - Card Reader, Exterior

1 Continuous Hinge 662HD 628  
1 Exit Device ELR TYPE 3 FUNC. 3 CD DS 630AM  
2 Mortise Cylinder 1E-74 PATD 626  
1 Flush Pull 1111C 630AM  
1 Door Closer C72061 PT4C/D/F/G/H X 1.5" BORE 689  
1 Kick Plate J102 8" 630  
1 Door Position Switch HSS SERIES, CONCEALED  
1 Power Transfer PT1000 689  
1 Power Supply ELR151  
1 Card Reader BY SECURITY  
1 Weatherstrip 700 ES  
1 Door Sweep C627 A  
1 Threshold 896 S 628

Set #14

1 Continuous Hinge 662HD 628  
1 Exit Device FL TYPE 3 FUNC. 8 630AM  
1 Mortise Cylinder 1E-74 PATD 626  
1 Door Closer C72011 PT4C/D/F/G/H X 1.5" BORE 689  
1 Kick Plate J102 8" 630  
1 Mop Plate J101 4" 630  
1 Gasketing 2525 C  
1 Auto Door Bottom 320 S

Set #15 - Push Button Lock, STC

3 Hinge CB1900R 4.5" x 4.5" 652  
1 Keypad Lock 45HZ-7DV14KP PATD 626AM  
1 Door Closer C72011 PT4C/D/F/H X 1.5" BORE 689  
1 Kick Plate J102 8" 630  
1 Wall Bumper L02101 626  
1 Coat Hook AM801 630  
1 Weatherstrip 700 ES  
1 Auto Door Bottom 320 S

Set #16

3 Hinge CB1900R 4.5" x 4.5" 652  
1 Privacy Set F19 (F19 (45H-0L14M VIT)) 630AM  
1 Kick Plate J102 8" 630  
1 Mop Plate J101 4" 630  
1 Wall Bumper L02101 626  
1 Gasketing 2525 C  
1 Auto Door Bottom 320 S

Set #17 - STC

3 Hinge CB1900R 4.5" x 4.5" 652

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

1 Lockset F20 (45H-7AB14N) 630AM  
1 Door Closer C72011 PT4C/D/F/H X 1.5" BORE 689  
1 Kick Plate J102 8" 630  
1 Wall Bumper L02101 626  
1 Coat Hook AM801 630  
1 Gasketing 2525 C  
1 Auto Door Bottom 320 S

## Set #18

3 Hinge CB1900R 4.5" x 4.5" 652  
1 Lockset F05 (45H-7R14N) 630AM  
1 Door Closer C72011 PT4C/D/F/H X 1.5" BORE 689  
1 Kick Plate J102 8" 630  
1 Mop Plate J101 4" 630  
1 Wall Bumper L02101 626  
1 Gasketing 2525 C

## Set #19

1 Continuous Hinge 662HD 628  
1 Push/Pull Passage Set J 6610 AM 630  
1 Door Pull AP311 630  
1 Door Closer C72011 PT4C/D/F/G/H X 1.5" BORE 689  
1 Kick Plate J102 8" 630  
1 Gasketing 2525 C  
1 Auto Door Bottom 320 S

## Set #20

3 Hinge CB1900R 4.5" x 4.5" 652  
1 Lockset F05 (45H-7R14N) 630AM  
1 Overhead Stop N 4023 630  
3 Silencer L03011 GREY

## Set #21

1 Continuous Hinge 662HD 628  
1 Push/Pull Passage Set J 6610 AM 630  
1 Door Closer C72011 PT4C/D/F/H X 1.5" BORE 689  
1 Kick Plate J102 8" 630  
1 Mop Plate J101 4" 630  
1 Wall Bumper L02101 626  
1 Gasketing 2525 C

## Set #22

3 Hinge CB1900R 4.5" x 4.5" 652  
1 Privacy Set F19 (F19 (45H-0L14M VIT)) 630AM  
1 Door Closer C72011 PT4C/D/F/H X 1.5" BORE 689  
1 Kick Plate J102 8" 630  
1 Mop Plate J101 4" 630  
1 Wall Bumper L02101 626  
1 Gasketing 2525 C

## Set #23 - STC

3 Hinge CB1900R 4.5" x 4.5" 652  
1 Passage Set F01 (45H-0N14M) 630AM  
1 Door Closer C72011 PT4C/D/F/H X 1.5" BORE 689  
1 Kick Plate J102 8" 630  
1 Mop Plate J101 4" 630  
1 Wall Bumper L02101 626  
1 Weatherstrip 700 ES  
1 Auto Door Bottom 4447 S



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

NOTE: Provide gasketing to meet STC ratings as required by Wall partition or Door Schedule. Typical.

## Set #24

3 Hinge CB1900R 4.5" x 4.5" 652  
1 Privacy Set F19 (F19 (45H-0L14M VIT)) 630AM  
1 Kick Plate J102 8" 630  
1 Mop Plate J101 4" 630  
1 Wall Bumper L02101 626  
1 Coat Hook AM801 630  
3 Silencer L03011 GREY

## Set #25

3 Hinge CB1900R 4.5" x 4.5" 652  
1 Lockset F05 (45H-7R14N) 630AM  
1 Door Closer C72011 PT4C/D/F/G/H X 1.5" BORE 689  
1 Kick Plate J102 8" 630  
1 Gasketing 2525 C

## Set #26 - Vestibule

1 Continuous Hinge 662HD 628  
1 Dummy Touchbar N673DR-3 630  
1 Flush Pull 1111C 630  
1 Door Closer C72061 PT4C/D/F/G/H X 1.5" BORE 689  
1 Kick Plate J102 8" 630  
1 Gasketing 2525 C  
1 Auto Door Bottom 320 S

## Set #27 - Card Reader

1 Continuous Hinge 662HD 628  
1 Exit Device ELR TYPE 3 FUNC. 3 CD DS 630AM  
1 Mortise Cylinder 1E-74 PATD 626  
1 Flush Pull 1111C 630  
1 Door Closer C72011 PT4C/D/F/G/H X 1.5" BORE 689  
1 Kick Plate J102 8" 630  
1 Door Position Switch HSS SERIES, CONCEALED  
1 Power Transfer PT1000 689  
1 Power Supply ELR152  
1 Card Reader BY SECURITY  
1 Weatherstrip 700 ES  
1 Door Sweep C627 A  
1 Threshold 896 S 628

## Set #28 - STC

3 Hinge CB1900R 4.5" x 4.5" 652  
1 Lockset F05 (45H-7R14N) 630AM  
1 Door Closer C72011 PT4C/D/F/G/H X 1.5" BORE 689  
1 Kick Plate J102 8" 630  
1 Gasketing 2525 C  
1 Weatherstrip 700 ES  
1 Auto Door Bottom 4447 S

## Set #29

3 Hinge CB1900R 4.5" x 4.5" 652  
1 Lockset F05 (45H-7R14N) 630AM  
1 Wall Bumper L02101 626  
3 Silencer L03011 GREY

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

Set #30 - Miscellaneous

Provide one 24 slot SAM-24 Asset Key Manager, Keystone Software,  
transportable key binder with Velcro hooks and all necessary key rings and  
tools.

-- End of Section --

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

SECTION 08 81 00

GLAZING  
05/19

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN ARCHITECTURAL MANUFACTURERS ASSOCIATION (AAMA)

AAMA 800 (2016) Voluntary Specifications and Test  
Methods for Sealants

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI Z97.1 (2015) Safety Glazing Materials Used in  
Buildings - Safety Performance  
Specifications and Methods of Test

ASTM INTERNATIONAL (ASTM)

ASTM C509 (2006; R 2021) Standard Specification for  
Elastomeric Cellular Preformed Gasket and  
Sealing Material

ASTM C864 (2005; R 2015) Dense Elastomeric  
Compression Seal Gaskets, Setting Blocks,  
and Spacers

ASTM C920 (2018) Standard Specification for  
Elastomeric Joint Sealants

ASTM C1021 (2008; R 2014) Standard Practice for  
Laboratories Engaged in Testing of  
Building Sealants

ASTM C1036 (2021) Standard Specification for Flat  
Glass

ASTM C1048 (2018) Standard Specification for  
Heat-Strengthened and Fully Tempered Flat  
Glass

ASTM C1087 (2016) Standard Test Method for  
Determining Compatibility of  
Liquid-Applied Sealants with Accessories  
Used in Structural Glazing Systems

ASTM C1172 (2019) Standard Specification for  
Laminated Architectural Flat Glass

ASTM C1184 (2023) Standard Specification for  
Structural Silicone Sealants

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

|            |                                                                                          |
|------------|------------------------------------------------------------------------------------------|
| ASTM C1281 | (2016) Standard Specification for<br>Preformed Tape Sealants for Glazing<br>Applications |
| ASTM D395  | (2016; E 2017) Standard Test Methods for<br>Rubber Property - Compression Set            |
| ASTM D2287 | (2019) Nonrigid Vinyl Chloride Polymer and<br>Copolymer Molding and Extrusion Compounds  |
| ASTM E1300 | (2016) Standard Practice for Determining<br>Load Resistance of Glass in Buildings        |
| ASTM E2190 | (2010) Standard Specification for<br>Insulating Glass Unit Performance and<br>Evaluation |

#### GLASS ASSOCIATION OF NORTH AMERICA (GANA)

|                       |                                     |
|-----------------------|-------------------------------------|
| GANA Glazing Manual   | (2008) Glazing Manual               |
| GANA Sealant Manual   | (2008) Sealant Manual               |
| GANA Standards Manual | (2008) Engineering Standards Manual |

#### INSULATING GLASS MANUFACTURERS ALLIANCE (IGMA)

|              |                                                                                                                           |
|--------------|---------------------------------------------------------------------------------------------------------------------------|
| IGMA TB-1200 | (1983; R 2016) Guidelines for Insulating<br>Glass Dimensional Tolerances                                                  |
| IGMA TB-3001 | (2001) Guidelines for Sloped Glazing                                                                                      |
| IGMA TM-3000 | (1990; R 2016) North American Glazing<br>Guidelines for Sealed Insulating Glass<br>Units for Commercial & Residential Use |

#### NATIONAL FENESTRATION RATING COUNCIL (NFRC)

|          |                                                                                                                                          |
|----------|------------------------------------------------------------------------------------------------------------------------------------------|
| NFRC 100 | (2020) Procedure for Determining<br>Fenestration Product U-Factors                                                                       |
| NFRC 200 | (2020) Procedure for Determining<br>Fenestration Product Solar Heat Gain<br>Coefficient and Visible Transmittance at<br>Normal Incidence |

#### NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

|          |                                                                       |
|----------|-----------------------------------------------------------------------|
| NFPA 80  | (2022) Standard for Fire Doors and Other<br>Opening Protectives       |
| NFPA 252 | (2022) Standard Methods of Fire Tests of<br>Door Assemblies           |
| NFPA 257 | (2022) Standard on Fire Test for Window<br>and Glass Block Assemblies |

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

16 CFR 1201

Safety Standard for Architectural Glazing  
Materials

## 1.2 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

### SD-03 Product Data

Insulating Glass

Glazing Accessories

Sealants

Joint Backer

### SD-04 Samples

Insulating Glass

Glazing Compound

Glazing Tape

Sealing Tapes

### SD-07 Certificates

Insulating Glass

### SD-08 Manufacturer's Instructions

Setting and Sealing Materials

Glass Setting

### SD-11 Closeout Submittals

Warranty for Insulated Glass Units

Warranty for Polycarbonate Sheet

## 1.3 SYSTEM DESCRIPTION

Fabricate and install watertight and airtight glazing systems to withstand thermal movement and wind loading without glass breakage, gasket failure, deterioration of glazing accessories, or defects in the work. Glazed panels must comply with the safety standards, in accordance with ANSI Z97.1, and comply with indicated wind/snow loading in accordance with ASTM E1300.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### 1.4 QUALITY CONTROL

Submit two 8 by 10 inch samples of each of the following: tinted glass, patterned glass, heat-absorbing glass, and insulating glass units.

Submit three samples of each other material. Samples of plastic sheets must be minimum 5 by 7 inches.

#### 1.5 DELIVERY, STORAGE, AND HANDLING

Deliver products to the site in unopened containers, labeled plainly with manufacturers' names and brands. Store glass and setting materials in safe, enclosed dry locations and do not unpack until needed for installation. Handle and install materials in a manner that will protect them from damage.

#### 1.6 ENVIRONMENTAL REQUIREMENTS

Do not start glazing work until the outdoor temperature is above 40 degrees F and rising, unless procedures recommended by the glass manufacturer and approved by the Contracting Officer are made to warm the glass and rabbet surfaces. Provide ventilation to prevent condensation of moisture on glazing work during installation. Do not perform glazing work during damp or rainy weather.

#### 1.7 WARRANTY

##### 1.7.1 Warranty for Insulated Glass Units

Warranty insulating glass units against development of material obstruction to vision (such as dust, fogging, or film formation on the inner glass surfaces) caused by failure of the hermetic seal, other than through glass breakage, for a 10-year period following acceptance of the work. Provide new units for any units failing to comply with terms of this warranty within 45 working days after receipt of notice from the Government.

##### 1.7.2 Warranty for Polycarbonate Sheet

For a 5-year period following acceptance of the work:

- a. Warranty Type I, Class A (UV stabilized) sheets against breakage;
- b. Warranty Type III (coated, mar-resistant) sheets against breakage and against coating delamination;
- c. Warranty Type IV (coated sheet) against breakage and against yellowing;
- d. Warranty extruded polycarbonate profile sheet against breakage.

For a 10-year period following acceptance of the work, warranty Type IV against yellowing and loss of light transmission.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## PART 2 PRODUCTS

### 2.1 PRODUCT SUSTAINABILITY CRITERIA

### 2.2 GLASS

ASTM C1036, unless specified otherwise. In doors and sidelights, provide safety glazing material conforming to 16 CFR 1201.

#### 2.2.1 Clear Glass

For interior glazing (i.e., pass and observation windows), 1/4 inch thick glass should be used.

Type I, Class 1 (clear), Quality q4 (A). Provide for glazing openings not indicated or specified otherwise. Use double-strength sheet glass or 1/8 inch float glass for openings up to and including 15 square feet, 3/16 inch for glazing openings over 15 square feet but not over 30 square feet, and 1/4 inch for glazing openings over 30 square feet but not over 45 square feet.

#### 2.2.2 Annealed Glass

Annealed glass must be Type I transparent flat type, Class 1 - clear, Quality q3 - glazing select, conforming to ASTM C1036.

#### 2.2.3 Laminated Glass

ASTM C1172, Laminated glass fabricated from two nominal 1/8 inch pieces of Type I, Class 1, Quality Q3, flat annealed ; clear glass conforming to ASTM C1036. Flat glass to be laminated together with a minimum of 0.030 inch inch thick, clear polyvinyl butyral laminate, conforming to requirements of 16 CFR 1201 and ASTM C1172. The total thickness of nominally 1/4 inches. Color to be clear.

#### 2.2.4 Mirrors

##### 2.2.4.1 Glass Mirrors

Glass for mirrors must be Type I transparent flat type, Class 1-clear, Glazing Quality q1 1/4 inch thick conforming to ASTM C1036. Glass must be coated on one surface with silver coating, copper protective coating, and mirror backing paint. Silver coating must be highly adhesive pure silver coating of a thickness which must provide reflectivity of 83 percent or more of incident light when viewed through 1/4 inch thick glass, and must be free of pinholes or other defects. Copper protective coating must be pure bright reflective copper, homogeneous without sludge, pinholes or other defects, and must be of proper thickness to prevent "adhesion pull" by mirror backing paint. Mirror backing paint must consist of two coats of special scratch and abrasion-resistant paint, and must be baked in uniform thickness to provide a protection for silver and copper coatings which will permit normal cutting and edge fabrication.

#### 2.2.5 Tempered Glass

ASTM C1048, Kind FT (fully tempered), Condition A (uncoated), Type I, Class 1 (transparent), Quality q3, 1/4 inch thick, 89 percent light transmittance, percent shading coefficient conforming to ASTM C1048 and GANA Standards Manual. Color must be clear.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 2.2.6 Heat-Strengthened Glass

ASTM C1048, Kind HS (heat strengthened), Condition A (uncoated), Type I, Class 1 (clear), Quality q3, 1/4 inch thick.

### 2.2.7 Fire/Safety Rated Glass

#### 2.2.7.1 Fire Protection Rated Glass

Clear tempered and meet 16 CFR 1201 Category I (under 9 square feet) or II (over 9 square feet) impact safety standard. Glass to make 20 minute rating when tested in accordance with NFPA 257 and NFPA 252. Glass to be permanently labeled with appropriate markings.

### 2.3 INSULATING GLASS UNITS

Two panes of glass separated by a dehydrated airspace, filled with argon gas and hermetically sealed, conforming to ASTM E2190. Submit performance and compliance documentation for each type of insulating glass.

Insulated glass units must have a Solar Heat Gain Coefficient (SHGC) maximum of 0.29 determined according to NFRC 200 and a U-factor maximum of 0.27 Btu per square foot by hr by degree F in accordance with NFRC 100.

Dimensional tolerances must be as specified in IGMA TB-1200. Spacer must be black, roll-formed, thermally broken aluminum, with bent or tightly welded or keyed and sealed joints to completely seal the spacer periphery and eliminate moisture and hydrocarbon vapor transmission into airspace through the corners. Primary seal must be compressed polyisobutylene and the secondary seal must be a specially formulated silicone.

The inner light must be ASTM C1172, clear annealed flat glass Type I, Class I, Quality q3. The intermediate light must be ASTM C1172, clear annealed flat glass Type I, Class I, Quality q3. The outer light must be ASTM C1036, Type I, Class 1 (transparent), 2 (solar-reflective), Quality q4, 1/4 inch thick.

#### 2.3.1 Low Emissivity Coatings

Interior and exterior glass panes for Low-E insulating units must be Type I annealed flat glass, Class 1-clear with anti-reflective low-emissivity coating or heat-strengthened or fully tempered glass complying with ASTM C1048, Condition C on No. 2 surface (inside surface of exterior pane), Quality q3 - glazing select, conforming to ASTM C1036. Glass performance must be U value maximum of 0.29 Btu/hr-ft<sup>2</sup>-F, Solar Heat Gain Coefficient (SHGC) maximum of 0.38. Color must be Kawneer Solar Bronze.

### 2.4 SETTING AND SEALING MATERIALS

Provide as specified in the GANA Glazing Manual, IGMA TM-3000, IGMA TB-3001, and manufacturer's recommendations, unless specified otherwise herein. Do not use metal sash putty, nonskinning compounds, nonresilient preformed sealers, or impregnated preformed gaskets. Materials exposed to view and unpainted must be gray or neutral color. Sealant testing must be performed by a testing agency qualified according to ASTM C1021.

Submit glass manufacturer's recommendations for setting and sealing materials and for installation of each type of glazing material specified.



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### 2.4.1 Putty and Glazing Compound

Provide glazing compound as recommended by manufacturer for face-glazing metal sash. Putty must be linseed oil type. Do not use putty and glazing compounds with insulating glass or laminated glass.

#### 2.4.2 Glazing Compound

Use for face glazing metal sash. Do not use with insulating glass units or laminated glass.

#### 2.4.3 Sealants

Provide elastomeric sealants.

##### 2.4.3.1 Elastomeric Sealant

ASTM C920, Type S, Grade NS, Class 12.5, Use G. Use for channel or stop glazing metal sash. Sealants must be chemically compatible with setting blocks, edge blocks, and sealing tapes, with sealants used in manufacture of insulating glass units. Color of sealant must be white.

##### 2.4.3.2 Structural Sealant

ASTM C1184, Type S.

#### 2.4.4 Joint Backer

Joint backer must have a diameter size at least 25 percent larger than joint width; type and material as recommended in writing by glass and sealant manufacturer.

#### 2.4.5 Glazing Tapes

##### 2.4.5.1 Back-Bedding Mastic Glazing Tapes

Preformed, butyl-based, 100 percent solids elastomeric tape; nonstaining and nonmigrating in contact with nonporous surfaces; with or without spacer rod as recommended in writing by tape and glass manufacturers for application indicated; and complying with ASTM C1281 and AAMA 800 for products indicated below:

- a. AAMA 804.3 tape, where indicated.
- b. AAMA 806.3 tape, for glazing applications in which tape is subject to continuous pressure.
- c. AAMA 807.3 tape, for glazing applications in which tape is not subject to continuous pressure.

##### 2.4.5.2 Expanded Cellular Glazing Tapes

Closed-cell, PVC foam tapes; factory coated with adhesive on both surfaces; and complying with AAMA 800 for the following types:

- a. AAMA 810.1, Type 1, for glazing applications in which tape acts as the primary sealant.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

- b. AAMA 810.1, Type 2, for glazing applications in which tape is used in combination with a full bead of liquid sealant.

#### 2.4.6 Sealing Tapes

Preformed, semisolid, PVC-based material of proper size and compressibility for the particular condition, complying with ASTM D2287. Use only where glazing rabbet is designed for tape and tape is recommended by the glass or sealant manufacturer. Provide spacer shims for use with compressible tapes. Tapes must be chemically compatible with the product being set.

#### 2.4.7 Setting Blocks and Edge Blocks

Closed-cell neoprene setting blocks must be dense extruded type conforming to ASTM C509 and ASTM D395, Method B, Shore A durometer between 70 and 90. Edge blocking must be Shore A durometer of 50 (plus or minus 5). Provide silicone setting blocks when blocks are in contact with silicone sealant. Profiles, lengths and locations must be as required and recommended in writing by glass manufacturer. Block color must be black.

#### 2.4.8 Glazing Gaskets

Glazing gaskets must be extruded with continuous integral locking projection designed to engage into metal glass holding members to provide a watertight seal during dynamic loading, building movements and thermal movements. Glazing gaskets for a single glazed opening must be continuous one-piece units with factory-fabricated injection-molded corners free of flashing and burrs. Glazing gaskets must be in lengths or units recommended by manufacturer to ensure against pull-back at corners. Provide glazing gasket profiles as recommended by the manufacturer for the intended application.

##### 2.4.8.1 Fixed Glazing Gaskets

Fixed glazing gaskets must be closed-cell (sponge) smooth extruded compression gaskets of cured elastomeric virgin neoprene compounds conforming to ASTM C509, Type 2, Option 1.

##### 2.4.8.2 Wedge Glazing Gaskets

Wedge glazing gaskets must be high-quality extrusions of cured elastomeric virgin neoprene compounds, ozone resistant, conforming to ASTM C864, Option 1, Shore A durometer between 65 and 75.

##### 2.4.8.3 Aluminum Framing Glazing Gaskets

Glazing gaskets for aluminum framing must be permanent, elastic, non-shrinking, non-migrating, watertight and weathertight.

#### 2.4.9 Accessories

Provide as required for a complete installation, including glazing points, clips, shims, angles, beads, and spacer strips. Provide noncorroding metal accessories. Provide primer-sealers and cleaners as recommended by the glass and sealant manufacturers. Use ASTM C1087 to determine whether priming and other specific joint preparation techniques are required to obtain rapid, optimum adhesion of glazing sealants to surface.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## 2.5 MIRROR ACCESSORIES

### 2.5.1 Mastic

Mastic for setting mirrors must be a polymer type mirror mastic resistant to water, shock, cracking, vibration and thermal expansion. Provide mastic compatible with mirror backing paint, and as approved by mirror manufacturer.

### 2.5.2 Mirror Frames

Provide mirrors with mirror frames (J-mold channels) fabricated of one-piece roll-formed Type 304 stainless steel with No. 4 brushed satin finish and concealed fasteners which will keep mirrors snug to wall. Frames must be 1-1/4 by 1/4 by 1/4 inch continuous at top and bottom of mirrors. Concealed fasteners of type to suit wall construction material must be provided with mirror frames.

### 2.5.3 Mirror Clips

Provide clips with concealed fasteners of type to suit wall construction material.

## PART 3 EXECUTION

Any materials that show visual evidence of biological growth due to the presence of moisture must not be installed on the building project.

### 3.1 PREPARATION

Preparation, unless otherwise specified or approved, must conform to applicable recommendations in the GANA Glazing Manual, GANA Sealant Manual, IGMA TB-3001, IGMA TM-3000, and manufacturer's recommendations. Determine the sizes to provide the required edge clearances by measuring the actual opening to receive the glass. Grind smooth in the shop glass edges that will be exposed in finish work. Leave labels in place until the installation is approved, except remove applied labels on heat-absorbing glass and on insulating glass units as soon as glass is installed. Securely fix movable items or keep in a closed and locked position until glazing compound has thoroughly set.

### 3.2 GLASS SETTING

Shop glaze or field glaze items to be glazed using glass of the quality and thickness specified or indicated. Glazing, unless otherwise specified or approved, must conform to applicable recommendations in the GANA Glazing Manual, GANA Sealant Manual, IGMA TB-3001, IGMA TM-3000, and manufacturer's recommendations. Aluminum windows, wood doors, and wood windows may be glazed in conformance with one of the glazing methods described in the standards under which they are produced, except that face puttying with no bedding will not be permitted. Handle and install glazing materials in accordance with manufacturer's instructions. Use beads or stops which are furnished with items to be glazed to secure the glass in place. Verify products are properly installed, connected, and adjusted.

#### 3.2.1 Sheet Glass

Cut and set with the visible lines or waves horizontal.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 3.2.2 Patterned Glass

Set glass with one patterned surface with smooth surface on the weather side. When used for interior partitions, place the patterned surface in same direction in all openings.

### 3.2.3 Insulating Glass Units

Do not grind, nip, or cut edges or corners of units after the units have left the factory. Springing, forcing, or twisting of units during setting will not be permitted. Handle units so as not to strike frames or other objects. Installation must conform to applicable recommendations of IGMA TB-3001 and IGMA TM-3000.

### 3.2.4 Installation of Wire Glass

Install glass for fire doors in accordance with installation requirements of NFPA 80.

### 3.2.5 Installation of Heat-Absorbing Glass

Provide glass with clean-cut, factory-fabricated edges. Field cutting will not be permitted.

### 3.2.6 Installation of Laminated Glass

Sashes which are to receive laminated glass must be weeped to the outside to allow water drainage into the channel.

## 3.3 CLEANING

Clean glass surfaces and remove labels, paint spots, putty, and other defacement as required to prevent staining. Glass must be clean at the time the work is accepted.

## 3.4 PROTECTION

Protect glass work immediately after installation. Identify glazed openings with suitable warning tapes, cloth or paper flags, attached with non-staining adhesives. Protect reflective glass with a protective material to eliminate any contamination of the reflective coating. Place protective material far enough away from the coated glass to allow air to circulate to reduce heat buildup and moisture accumulation on the glass. Upon removal, separate protective materials for reuse or recycling. Remove and replace glass units which are broken, chipped, cracked, abraded, or otherwise damaged during construction activities with new units.

-- End of Section --

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

SECTION 08 91 00

METAL WALL LOUVERS  
**08/20**

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AIR MOVEMENT AND CONTROL ASSOCIATION INTERNATIONAL, INC. (AMCA)

AMCA 500-L (2015) Laboratory Methods of Testing  
Louvers for Rating

AMCA 511 (2010; R 2016) Certified Ratings Program  
for Air Control Devices

AMERICAN ARCHITECTURAL MANUFACTURERS ASSOCIATION (AAMA)

AAMA 2603 (2020) Voluntary Specification,  
Performance Requirements and Test  
Procedures for Pigmented Organic Coatings  
on Aluminum Extrusions and Panels

ASTM INTERNATIONAL (ASTM)

ASTM A123/A123M (2017) Standard Specification for Zinc  
(Hot-Dip Galvanized) Coatings on Iron and  
Steel Products

ASTM A653/A653M (2023) Standard Specification for Steel  
Sheet, Zinc-Coated (Galvanized) or  
Zinc-Iron Alloy-Coated (Galvannealed) by  
the Hot-Dip Process

ASTM A780/A780M (2020) Standard Practice for Repair of  
Damaged and Uncoated Areas of Hot-Dip  
Galvanized Coatings

ASTM A1008/A1008M (2021a) Standard Specification for Steel,  
Sheet, Cold-Rolled, Carbon, Structural,  
High-Strength Low-Alloy, High-Strength  
Low-Alloy with Improved Formability,  
Solution Hardened, and Bake Hardenable

ASTM B209 (2014) Standard Specification for Aluminum  
and Aluminum-Alloy Sheet and Plate

ASTM B221 (2021) Standard Specification for Aluminum  
and Aluminum-Alloy Extruded Bars, Rods,  
Wire, Profiles, and Tubes

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## 1.2 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

### SD-02 Shop Drawings

Wall Louvers

### SD-03 Product Data

Metal Wall Louvers

Door Louvers

### SD-04 Samples

Wall Louver Samples; G

Door Louver Samples; G

## 1.3 DELIVERY, STORAGE, AND PROTECTION

Deliver materials to the site in an undamaged condition. Carefully store materials off the ground to provide proper ventilation, drainage, and protection against dampness. Louvers must be free from nicks, scratches, and blemishes. Replace defective or damaged materials with new.

## 1.4 DETAIL DRAWINGS

Show all information necessary for fabrication and installation of wall louvers. Indicate materials, sizes, thicknesses, fastenings, and profiles.

## 1.5 COLOR SAMPLES

Colors of finishes for wall louver samples must closely approximate colors indicated. Where color is not indicated, submit the manufacturer's standard colors to the Contracting Officer for selection.

## PART 2 PRODUCTS

### 2.1 MATERIALS

#### 2.1.1 Galvanized Steel Sheet

ASTM A653/A653M, coating designation G90.

#### 2.1.2 Aluminum Sheet

ASTM B209, alloy 3003 or 5005 with temper as required for forming.

#### 2.1.3 Extruded Aluminum

ASTM B221, alloy 6063-T5 or -T52.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### 2.1.4 Cold Rolled Steel Sheet

ASTM A1008/A1008M, Class 1, with matte finish. Use for interior louvers only.

### 2.2 METAL WALL LOUVERS

Wind driven rain resistant type, with bird screens and made to withstand a wind load of not less than 30 pounds per square foot. Wall louvers must bear the AMCA certified ratings program seal for air performance and water penetration in accordance with AMCA 500-L and AMCA 511. The rating must show a water penetration of 0.20 or less ounce per square foot of free area at a free velocity of 800 feet per minute.

#### 2.2.1 Extruded Aluminum Louvers

Fabricated of extruded 6063-T5 or -T52 aluminum with a wall thickness of not less than 0.081 inch.

#### 2.2.2 Formed Metal Louvers

Formed of zinc-coated steel sheet not thinner than 16 U.S. gage, or aluminum sheet not less than 0.08 inch thick.

#### 2.2.3 Mullions and Mullion Covers

Same material and finish as louvers. Provide mullions where indicated and for all louvers more than 5 feet in width at not more than 5 feet on centers. Provide mullion covers on both faces of joints between louvers.

#### 2.2.4 Screens and Frames

For aluminum louvers, provide 1/2 inch square mesh, 14 or 16 gage aluminum or 1/4 inch square mesh, 16 gage aluminum bird screening. For steel louvers, provide 1/2 inch square mesh, 12 or 16 gage zinc-coated steel; 1/2 inch square mesh, 16 gage copper; or 1/4 inch square mesh, 16 gage zinc-coated steel or copper bird screening. Mount screens in removable, rewirable frames of same material and finish as the louvers.

### 2.3 FASTENERS AND ACCESSORIES

Provide stainless steel screws and fasteners for aluminum louvers and zinc-coated or stainless steel screws and fasteners for steel louvers. Provide other accessories as required for complete and proper installation.

### 2.4 FINISHES

#### 2.4.1 Aluminum

Exposed aluminum surfaces must be factory finished with an organic coating. Color must be as indicated. All louvers must have the same finish.

##### 2.4.1.1 Organic Coating

Clean and prime exposed aluminum surfaces. Provide a baked enamel finish conforming to AAMA 2603, with total dry film thickness not less than 0.8 mil, color as indicated.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### 2.4.2 Steel

Surfaces specified must have a zinc coating, a phosphate treatment, and a shop prime coat of rust-inhibitive paint. The galvanized coating must conform to ASTM A653/A653M, coating designation Z275 (G90), except that louvers located in conditioned spaces on interior of the building may be Z180 (G60). The weight of zinc coatings must be as designated in Table I of ASTM A123/A123M for the thickness of base metal to be coated. The prime coat must be a type especially developed for materials treated by phosphates and adapted to application by dipping or spraying. Repair damaged zinc-coated surfaces by the materials and methods conforming to ASTM A780/A780M and spot prime. At the option of the Contractor, a two-part system including bonderizing, baked-on epoxy primer, and baked-on enamel top coat may be applied before forming, in lieu of prime coat specified.

### PART 3 EXECUTION

#### 3.1 INSTALLATION

##### 3.1.1 Wall Louvers

Install using stops or moldings, flanges, strap anchors, or jamb fasteners as appropriate for the wall construction and in accordance with manufacturer's recommendations.

##### 3.1.2 Screens and Frames

Attach frames to louvers with screws or bolts.

#### 3.2 PROTECTION FROM CONTACT OF DISSIMILAR MATERIALS

##### 3.2.1 Copper or Copper-Bearing Alloys

Paint copper or copper-bearing alloys in contact with dissimilar metal with heavy-bodied bituminous paint or separate with inert membrane.

##### 3.2.2 Aluminum

Where aluminum contacts metal other than zinc, paint the dissimilar metal with a primer and two coats of aluminum paint.

##### 3.2.3 Metal

Paint metal in contact with mortar, concrete, or other masonry materials with alkali-resistant coatings such as heavy-bodied bituminous paint.

##### 3.2.4 Wood

Paint wood or other absorptive materials that may become repeatedly wet and in contact with metal with two coats of aluminum paint or a coat of heavy-bodied bituminous paint.

-- End of Section --



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## SECTION 09 22 00

### SUPPORTS FOR PLASTER AND GYPSUM BOARD 02/10, CHG 2: 08/18

#### PART 1 GENERAL

##### 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

##### ASTM INTERNATIONAL (ASTM)

|                 |                                                                                                                                         |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| ASTM A463/A463M | (2022) Standard Specification for Steel Sheet, Aluminum-Coated, by the Hot-Dip Process                                                  |
| ASTM A653/A653M | (2023) Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process |
| ASTM C645       | (2014; E 2015) Nonstructural Steel Framing Members                                                                                      |
| ASTM C754       | (2020) Standard Specification for Installation of Steel Framing Members to Receive Screw-Attached Gypsum Panel Products                 |
| ASTM C841       | (2023) Installation of Interior Lathing and Furring                                                                                     |
| ASTM C847       | (2014a) Standard Specification for Metal Lath                                                                                           |

##### NATIONAL ASSOCIATION OF ARCHITECTURAL METAL MANUFACTURERS (NAAMM)

|                |                                                           |
|----------------|-----------------------------------------------------------|
| NAAMM EMLA 920 | (2009) Guide Specifications for Metal Lathing and Furring |
|----------------|-----------------------------------------------------------|

##### UNDERWRITERS LABORATORIES (UL)

|                    |                                  |
|--------------------|----------------------------------|
| UL Fire Resistance | (2014) Fire Resistance Directory |
|--------------------|----------------------------------|

##### 1.2 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Metal Support Systems

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

Submit for the erection of metal framing, furring, and ceiling suspension systems. Indicate materials, sizes, thicknesses, and fastenings.

#### SD-03 Product Data

##### Metal Support Systems

##### Recycled Content for Metal Support Systems

### 1.3 DELIVERY, STORAGE, AND HANDLING

Deliver materials to the job site and store in ventilated dry locations permitting easy access for inspection and handling. If materials are stored outdoors, stack materials off the ground, supported on a level platform, and fully protected from the weather. Handle materials carefully to prevent damage. Remove damaged items and provide new items.

## PART 2 PRODUCTS

### 2.1 MATERIALS

Provide steel materials for metal support systems with galvanized coating ASTM A653/A653M, G-60; aluminum coating ASTM A463/A463M, T1-25; or a 55-percent aluminum-zinc coating.

Provide metal support systems containing a minimum of 20 percent recycled content. Provide data identifying percentage of recycled content for metal support systems.

#### 2.1.1 Materials for Attachment of Lath

##### 2.1.1.1 Suspended and Furred Ceiling Systems and Wall Furring

ASTM C841, and ASTM C847.

##### 2.1.1.2 Non-load Bearing Wall Framing

NAAMM EMLA 920.

#### 2.1.2 Materials for Attachment of Gypsum Wallboard

##### 2.1.2.1 Suspended and Furred Ceiling Systems

ASTM C645.

##### 2.1.2.2 Non-load Bearing Wall Framing and Furring

ASTM C645, but not thinner than 0.0179 inch thickness, with 0.0329 inch minimum thickness supporting wall hung items such as cabinetwork, equipment and fixtures.

##### 2.1.2.3 Furring Structural Steel Columns

ASTM C645. Steel (furring) clips and support angles listed in UL Fire Resistance may be provided in lieu of steel studs for erection of gypsum wallboard around structural steel columns.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### 2.1.2.4 Z-Furring Channels with Wall Insulation

Not lighter than 26 gage galvanized steel, Z-shaped, with 1-1/4 inch and 3/4 inch flanges and depth as required by the insulation thickness provided.

### PART 3 EXECUTION

#### 3.1 INSTALLATION

##### 3.1.1 Systems for Attachment of Lath

###### 3.1.1.1 Suspended and Furred Ceiling Systems and Wall Furring

ASTM C841, except as indicated otherwise.

###### 3.1.1.2 Non-load Bearing Wall Framing

NAAMM EMLA 920, except provide framing members 16 inches o.c. unless indicated otherwise.

##### 3.1.2 Systems for Attachment of Gypsum Wallboard

###### 3.1.2.1 Suspended and Furred Ceiling Systems

ASTM C754, except provide framing members 16 inches o.c. unless indicated otherwise.

###### 3.1.2.2 Non-load Bearing Wall Framing and Furring

ASTM C754, except as indicated otherwise.

###### 3.1.2.3 Furring Structural Steel Columns

Install studs or galvanized steel clips and support angles for erection of gypsum wallboard around structural steel columns in accordance with the UL Fire Resistance, design number(s) indicated .

###### 3.1.2.4 Z-Furring Channels with Wall Insulation

Install Z-furring channels vertically spaced not more than 24 inches o.c. Locate Z-furring channels at interior and exterior corners in accordance with manufacturer's printed erection instructions. Fasten furring channels to masonry and concrete walls with powder-driven fasteners or hardened concrete steel nails through narrow flange of channel. Space fasteners not more than 24 inches o.c.

#### 3.2 ERECTION TOLERANCES

Provide framing members which will be covered by finish materials such as wallboard, plaster, or ceramic tile set in a mortar setting bed, within the following limits:

- a. Layout of walls and partitions: 1/4 inch from intended position;
- b. Plates and runners: 1/4 inch in 8 feet from a straight line;
- c. Studs: 1/4 inch in 8 feet out of plumb, not cumulative; and
- d. Face of framing members: 1/4 inch in 8 feet from a true plane.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

Provide framing members which will be covered by ceramic tile set in dry-set mortar, latex-portland cement mortar, or organic adhesive within the following limits:

- a. Layout of walls and partitions: 1/4 inch from intended position;
- b. Plates and runners: 1/8 inch in 8 feet from a straight line;
- c. Studs: 1/8 inch in 8 feet out of plumb, not cumulative; and
- d. Face of framing members: 1/8 inch in 8 feet from a true plane.

-- End of Section --

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

SECTION 09 29 00

GYPSUM BOARD  
**08/16, CHG 4: 02/20**

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

|                   |                                                                                                                                                                                                      |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM C475/C475M   | (2017; R 2022) Standard Specification for Joint Compound and Joint Tape for Finishing Gypsum Board                                                                                                   |
| ASTM C840         | (2020) Standard Specification for Application and Finishing of Gypsum Board                                                                                                                          |
| ASTM C954         | (2022) Standard Specification for Steel Drill Screws for the Application of Gypsum Panel Products or Metal Plaster Bases to Steel Studs from 0.033 in. (0.84 mm) to 0.112 in. (2.84 mm) in Thickness |
| ASTM C1002        | (2022) Standard Specification for Steel Self-Piercing Tapping Screws for the Application of Gypsum Panel Products or Metal Plaster Bases to Wood Studs or Steel Studs                                |
| ASTM C1047        | (2019) Standard Specification for Accessories for Gypsum Wallboard and Gypsum Veneer Base                                                                                                            |
| ASTM C1178/C1178M | (2013) Standard Specification for Glass Mat Water-Resistant Gypsum Backing Panel                                                                                                                     |
| ASTM C1396/C1396M | (2017) Standard Specification for Gypsum Board                                                                                                                                                       |
| ASTM C1766        | (2015; R2019) Standard Specification for Factory-Laminated Gypsum Panel Products                                                                                                                     |
| ASTM D226/D226M   | (2017) Standard Specification for Asphalt-Saturated Organic Felt Used in Roofing and Waterproofing                                                                                                   |
| ASTM D3273        | (2021) Standard Test Method for Resistance to Growth of Mold on the Surface of Interior Coatings in an Environmental Chamber                                                                         |
| ASTM E336         | (2023) Standard Test Method for Measurement of Airborne Sound Attenuation                                                                                                                            |

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

between Rooms in Buildings

#### CALIFORNIA DEPARTMENT OF PUBLIC HEALTH (CDPH)

CDPH SECTION 01350 (2017; Version 1.2) Standard Method for  
the Testing and Evaluation of Volatile  
Organic Chemical Emissions from Indoor  
Sources using Environmental Chambers

#### FM GLOBAL (FM)

FM APP GUIDE (updated on-line) Approval Guide  
<http://www.approvalguide.com/>

#### GYPSUM ASSOCIATION (GA)

GA 214 (2010) Recommended Levels of Gypsum Board  
Finish

GA 216 (2016) Application and Finishing of Gypsum  
Panel Products

GA 600 (2015) Fire Resistance Design Manual Sound  
Control

#### SCIENTIFIC CERTIFICATION SYSTEMS (SCS)

SCS SCS Global Services (SCS) Indoor Advantage

#### UNDERWRITERS LABORATORIES (UL)

UL 2818 (2022) GREENGUARD Certification Program  
For Chemical Emissions For Building  
Materials, Finishes And Furnishings

UL Fire Resistance (2014) Fire Resistance Directory

### 1.2 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

#### SD-03 Product Data

Cementitious Backer Units

Glass Mat Water-Resistant Gypsum Tile Backing Board

Water-Resistant Gypsum Backing Board

Accessories

Submit for each type of gypsum board and for cementitious backer units.

Gypsum Board

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

Recycled Content for Gypsum Board

Recycled Content for Paper Facing and Gypsum Cores

VOC Content of Joint Compound

#### SD-06 Test Reports

ASTM E90 Factory Test Report;

ASTM E336 Field Test Report;

#### SD-07 Certificates

Asbestos Free Materials;

Certify that gypsum board types, gypsum backing board types, cementitious backer units, and joint treating materials do not contain asbestos.

Indoor Air Quality for Gypsum Board

#### SD-08 Manufacturer's Instructions

#### SD-10 Operation and Maintenance Data

Manufacturer Maintenance Instructions

### 1.3 CERTIFICATIONS

#### 1.3.1 Indoor Air Quality Certifications

Submit required indoor air quality certifications in one submittal package.

##### 1.3.1.1 Ceiling and Wall Systems

Provide products certified to meet indoor air quality requirements by UL 2818 (Greenguard) Gold, SCS Global Services Indoor Advantage Gold or provide certification or validation by other third-party program that products meet the requirements of this Section. Provide current product certification documentation from certification body. When product does not have certification, provide validation that product meets the indoor air quality product requirements cited herein.

##### 1.3.1.2 Adhesives and Sealants

Provide products certified to meet indoor air quality requirements by UL 2818 (Greenguard) Gold, SCS Global Services Indoor Advantage Gold or provide certification or validation by other third-party program that products meet the requirements of this Section. Provide current product certification documentation from certification body. When product does not have certification, provide validation that product meets the indoor air quality product requirements cited herein.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### 1.4 DELIVERY, STORAGE, AND HANDLING

##### 1.4.1 Delivery

Deliver materials in the original packages, containers, or bundles with each bearing the brand name, applicable standard designation, and name of manufacturer, or supplier.

##### 1.4.2 Storage

Keep materials dry by storing inside a sheltered building. Where necessary to store gypsum board and cementitious backer units outside, store off the ground, properly supported on a level platform, and protected from direct exposure to rain, snow, sunlight, and other extreme weather conditions. Provide adequate ventilation to prevent condensation. Store per manufacturer's recommendations for allowable temperature and humidity range.

Do not store gypsum wallboard with materials which have high emissions of volatile organic compounds (VOCs) or other contaminants, including. Do not store panels near materials that may offgas or emit harmful fumes, such as kerosene heaters, fresh paint, or adhesives. Do not use materials that have visible moisture or biological growth.

##### 1.4.3 Handling

Neatly stack gypsum board and cementitious backer units flat to prevent sagging or damage to the edges, ends, and surfaces.

#### 1.5 QUALIFICATIONS

Furnish type of gypsum board work specialized by the installer with a minimum of 3 years of documented successful experience.

#### 1.6 SCHEDULING

The gypsum wallboard must be taped, finished and primed before the installation of the highly-emitting materials, or installed after the installation and ventilation period of the highly-emitting materials.

Commence application only after the area scheduled for gypsum board work is completely weathertight. The heating, ventilating, and air-conditioning systems must be complete and in operation prior to application of the gypsum board. If the mechanical system cannot be activated before gypsum board is begun, the gypsum board work may proceed in accordance with an approved plan to maintain the environmental conditions specified below. Apply gypsum board prior to the installation of finish flooring and acoustic ceiling.

#### 1.7 ENVIRONMENTAL REQUIREMENTS

Do not expose the gypsum board to excessive sunlight prior to gypsum board application. Maintain a continuous uniform temperature of not less than 50 degrees F and not more than 80 degrees F for at least one week prior to the application of gypsum board work, while the gypsum board application is being done, and for at least one week after the gypsum board is set. Shield air supply and distribution devices to prevent any uneven flow of air across the plastered surfaces. Provide ventilation to exhaust moist air to the outside during gypsum board application, set, and until gypsum



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

board jointing is dry. In glazed areas, keep windows open top and bottom or side to side 3 to 4 inches. Reduce openings in cold weather to prevent freezing of joint compound when applied. For enclosed areas lacking natural ventilation, provide temporary mechanical means for ventilation. In unglazed areas subjected to hot, dry winds or temperature differentials from day to night of 20 degrees F or more, screen openings with cheesecloth or similar materials. Avoid rapid drying. During periods of low indoor humidity, provide minimum air circulation following gypsum boarding and until gypsum board jointing complete and is dry.

## 1.8 FIRE RESISTIVE CONSTRUCTION

Comply with specified fire-rated assemblies for design numbers indicated per UL Fire Resistance or FM APP GUIDE.

## PART 2 PRODUCTS

### 2.1 MATERIALS

Conform to specifications, standards and requirements specified. Provide gypsum board types, gypsum backing board types, cementitious backing units, and joint treating materials manufactured from asbestos free materials only. Submit Safety Data Sheets and manufacturer maintenance instructions for gypsum materials including adhesives.

#### 2.1.1 Gypsum Board

ASTM C1396/C1396M. Gypsum board must contain a minimum of 5 percent post-consumer recycled content, or a minimum of 20 percent post-industrial recycled content. Provide data identifying percentage of recycled content for gypsum board. Paper facings must contain a minimum of 100 percent recycled paper content. Gypsum cores must contain a minimum of 95 percent post-industrial recycled gypsum content. Provide data identifying percentage of recycled content for paper facing and gypsum cores. Provide gypsum wall board and panels meeting the emissions requirements of CDPH SECTION 01350 (limit requirements for either office or classroom spaces regardless of space type). Provide certification or validation of indoor air quality for gypsum board.

##### 2.1.1.1 Regular

48 inch wide, 1/2, 5/8 inch thick, tapered edges.

##### 2.1.1.2 Type X (Special Fire-Resistant)

48 inch wide, 1/2, 5/8 inch thick, tapered edges.

##### 2.1.1.3 Mold Resistant / Anti-Microbial Gypsum

ASTM D3273. 48 inch wide, 1/2, 5/8 inch thick, tapered edges. This product must be used on the interior face of all exterior metal stud walls.

#### 2.1.2 Gypsum Backing Board

ASTM C1396/C1396M, gypsum backing board must be used as a base in a multilayer system.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

2.1.2.1 Regular

48 inch wide, 1/2, 5/8 inch thick, square edges.

2.1.2.2 Foil-Backed

48 inch wide, 1/2, 5/8 inch thick, square edges.

2.1.2.3 Type X (Special Fire-Resistant)

48 inch wide, 1/2, 5/8 inch thick, square edges.

2.1.3 Regular Water-Resistant Gypsum Backing Board

ASTM C1396/C1396M

2.1.3.1 Regular

48 inch wide, 1/2, 5/8 inch thick, tapered edges.

2.1.3.2 Type X (Special Fire-Resistant)

48 inch wide, 1/2, 5/8 inch thick, tapered edges.

2.1.4 Glass Mat Water-Resistant Gypsum Tile Backing Board

ASTM C1178/C1178M

2.1.4.1 Regular

48 inch wide, 1/2, 5/8 inch thick, square edges.

2.1.4.2 Type X (Special Fire-Resistant)

48 inch wide, 1/2, 5/8 inch thick, square edges.

2.1.5 Factory-Laminated Gypsum Board

ASTM C1766, regular or Type X, 48 inch wide 1/2, 5/8 inch thick, sound dampening gypsum panel products composed of factory-laminated gypsum panels laminated into a composite panel.

2.1.6 Joint Treatment Materials

ASTM C475/C475M. Product must be low emitting VOC types with VOC limits not exceeding 50 g/L. Provide data identifying VOC content of joint compound. Use all purpose joint and texturing compound containing inert fillers and natural binders, including lime compound. Pre-mixed compounds must be free of antifreeze, vinyl adhesives, preservatives, biocides and other slow releasing compounds.

2.1.6.1 Embedding Compound

Specifically formulated and manufactured for use in embedding tape at gypsum board joints and compatible with tape, substrate and fasteners.

2.1.6.2 Finishing or Topping Compound

Specifically formulated and manufactured for use as a finishing compound.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### 2.1.6.3 All-Purpose Compound

Specifically formulated and manufactured to serve as both a taping and a finishing compound and compatible with tape, substrate and fasteners.

#### 2.1.6.4 Setting or Hardening Type Compound

Specifically formulated and manufactured for use with fiber glass mesh tape.

#### 2.1.6.5 Joint Tape

Use cross-laminated, tapered edge, reinforced paper, or fiber glass mesh tape recommended by the manufacturer.

#### 2.1.7 Fasteners

##### 2.1.7.1 Screws

ASTM C1002, Type "G", Type "S" or Type "W" steel drill screws for fastening gypsum board to gypsum board, wood framing members and steel framing members less than 0.033 inch thick. ASTM C954 steel drill screws for fastening gypsum board to steel framing members 0.033 to 0.112 inch thick. Provide cementitious backer unit screws with a polymer coating.

#### 2.1.8 Accessories

ASTM C1047. Fabricate from corrosion protected steel or plastic designed for intended use. Accessories manufactured with paper flanges are not acceptable. Flanges must be free of dirt, grease, and other materials that may adversely affect bond of joint treatment. Provide prefinished or job decorated materials.

#### 2.1.9 Asphalt Impregnated Building Felt

Provide a 15 lb asphalt moisture barrier over glass mat covered or reinforced gypsum sheathing. Conforming to ASTM D226/D226M Type 1 (No. 15) for asphalt impregnated building felt.

#### 2.1.10 Water

Provide clean, fresh, and potable water.

### PART 3 EXECUTION

#### 3.1 EXAMINATION

##### 3.1.1 Framing and Furring

Verify that framing and furring are securely attached and of sizes and spacing to provide a suitable substrate to receive gypsum board and cementitious backer units. Verify that all blocking, headers and supports are in place to support plumbing fixtures and to receive soap dishes, grab bars, towel racks, and similar items. Do not proceed with work until framing and furring are acceptable for application of gypsum board and cementitious backer units.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 3.1.2 Building Construction Materials

Do not install building construction materials that show visual evidence of biological growth.

## 3.2 APPLICATION OF GYPSUM BOARD

Apply gypsum board to framing and furring members in accordance with ASTM C840 or GA 216 and the requirements specified. Apply gypsum board with separate panels in moderate contact; do not force in place. Stagger end joints of adjoining panels. Neatly fit abutting end and edge joints. Use gypsum board of maximum practical length; select panel sizes to minimize waste. Cut out gypsum board to make neat, close, and tight joints around openings. In vertical application of gypsum board, provide panels in lengths required to reach full height of vertical surfaces in one continuous piece. Lay out panels to minimize waste; reuse cutoffs whenever feasible. Surfaces of gypsum board and substrate members may be bonded together with an adhesive, except where prohibited by fire rating(s). Treat edges of cutouts for plumbing pipes, screwheads, and joints with water-resistant compound as recommended by the gypsum board manufacturer. Minimize framing by floating corners with single studs and drywall clips. Install 5/8 inch gypsum or 1/2 inch ceiling board over framing at 24 inch on center. Provide type of gypsum board for use in each system specified herein as indicated.

### 3.2.1 Application of Gypsum Board to Steel Framing and Furring

Apply in accordance with ASTM C840, System VIII or GA 216.

### 3.2.2 Arches and Bending Radii

Apply gypsum board in accordance with ASTM C840, System IX or GA 216.

### 3.2.3 Gypsum Board for Wall Tile or Tile Base Applied with Adhesive

In dry areas (areas other than tubs, shower enclosures, saunas, steam rooms, gang shower rooms), apply glass mat water-resistant gypsum tile backing board or water-resistant gypsum backing board in accordance with ASTM C840, System X or GA 216.

### 3.2.4 Exterior Application

Apply exterior gypsum board (such as at soffits) in accordance with ASTM C840, System XI or GA 216.

### 3.2.5 Floating Interior Angles

Minimize framing by floating corners with single studs and drywall clips. Locate the attachment fasteners adjacent to ceiling and wall intersections in accordance with ASTM C840, System XII or GA 216, for single-ply and two-ply applications of gypsum board to wood framing.

### 3.2.6 Control Joints

Install expansion and contraction joints in ceilings and walls in accordance with ASTM C840, System XIII or GA 216. Fill control joints between studs in fire-rated construction with firesafing insulation to match the fire-rating of construction.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 3.3 FINISHING OF GYPSUM BOARD

Tape and finish gypsum board in accordance with ASTM C840, GA 214 and GA 216. Finish plenum areas above ceilings to Level 1 in accordance with GA 214. Finish water resistant gypsum backing board, ASTM C1396/C1396M, to receive ceramic tile to Level 2 in accordance with GA 214. Finish walls and ceilings to receive a heavy-grade wall covering or heavy textured finish before painting to Level 3 in accordance with GA 214. Finish walls and ceilings without critical lighting to receive flat paints, light textures, or wall coverings to Level 4 in accordance with GA 214. Unless otherwise specified, finish all gypsum board walls, partitions and ceilings to Level 5 in accordance with GA 214. Provide joint, fastener depression, and corner treatment. Tool joints as smoothly as possible to minimize sanding and dust. Do not use self-adhering fiber glass mesh tape with conventional drying type joint compounds; use setting or hardening type compounds only. Provide treatment for water-resistant gypsum board as recommended by the gypsum board manufacturer. Protect workers, building occupants, and HVAC systems from gypsum dust.

#### 3.3.1 Uniform Surface

Wherever gypsum board is to receive eggshell, semigloss or gloss paint finish, or where severe, up or down lighting conditions occur, finish gypsum wall surface in accordance to GA 214 Level 5. In accordance with GA 214 Level 5, apply a thin skim coat of joint compound to the entire gypsum board surface, after the two-coat joint and fastener treatment is complete and dry.

### 3.4 SEALING

Seal openings around pipes, fixtures, and other items projecting through gypsum board and cementitious backer units as specified in Section 07 92 00 JOINT SEALANTS. Apply material with exposed surface flush with gypsum board or cementitious backer units.

#### 3.4.1 Sealing for Glass Mat or Reinforced Gypsum Board Sheathing

Apply silicone sealant in a 3/8 inch bead to all joints and trowel flat. Apply enough of the same sealant to all fasteners penetrating through the glass mat gypsum board surface to completely cover the penetration when troweled flat.

### 3.5 FIRE-RESISTANT ASSEMBLIES

Wherever fire-rated construction is indicated, provide materials and application methods, including types and spacing of fasteners, wall and ceiling framing in accordance with the specifications contained in UL Fire Resistance for the Design Number(s) indicated, or GA 600 for the File Number(s) indicated. Joints of fire-rated gypsum board enclosures must be closed and sealed in accordance with UL test requirements or GA requirements. Seal penetrations through rated partitions and ceilings tight in accordance with tested systems.

### 3.6 SOUND RATED ASSEMBLIES

When sound rated assemblies are required, provide materials and application methods, including panels, insulation, types and spacing of fasteners, wall and ceiling framing in accordance with the contract document and the description of the assembly in the ASTM E90 Factory Test

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

Report. Seal partitions continuously with acoustical foam or sealant (both sides) and finished to match wall wherever it abuts another element such as the floor, ceiling, wall, column, mullion, or another system or assembly.

### 3.7 PATCHING

Patch surface defects in gypsum board to a smooth, uniform appearance, ready to receive finishes.

### 3.8 SHAFTWALL FRAMING

Install the shaftwall system in accordance with the system manufacturer's published instructions. Coordinate bucks, anchors, blocking and other items placed in or behind shaftwall framing with electrical and mechanical work. Patch or replace fireproofing materials which are damaged or removed during shaftwall construction.

### 3.9 SOUND RATED ASSEMBLY FIELD TESTING

Provide third party testing of sound rated assemblies tested in accordance with ASTM E336. Provide the ASTM E336 Field Test Report verifying that the installed assemblies perform no less than five ASTC rating points below the ASTM E90 Factory Test Report. Examine, modify adjust, and retest any installation not meeting the STC Rating until compliance is obtained.

-- End of Section --

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## SECTION 09 30 10

### CERAMIC TILING

**08/20**

#### PART 1 GENERAL

##### 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

#### AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI A108/A118/A136.1 (2019) American National Standard  
Specifications for the Installation of  
Ceramic Tile

ANSI A137.1 (2019) American National Standards  
Specifications for Ceramic Tile

#### ASTM INTERNATIONAL (ASTM)

ASTM C33/C33M (2023) Standard Specification for Concrete  
Aggregates

ASTM C144 (2018) Standard Specification for  
Aggregate for Masonry Mortar

ASTM C150/C150M (2022) Standard Specification for Portland  
Cement

ASTM C206 (2014; R 2022) Standard Specification for  
Finishing Hydrated Lime

ASTM C207 (2018) Standard Specification for Hydrated  
Lime for Masonry Purposes

ASTM C648 (2020) Standard Test Method for Breaking  
Strength of Ceramic Tile

ASTM C1026 (2013; R 2018) Standard Test Method for  
Measuring the Resistance of Ceramic and  
Glass Tile to Freeze-Thaw Cycling

ASTM C1027 (2009; R 2017) Standard Test Method for  
Determining Visible Abrasion Resistance of  
Glazed Ceramic Tile

ASTM C1178/C1178M (2013) Standard Specification for Glass  
Mat Water-Resistant Gypsum Backing Panel

#### CALIFORNIA DEPARTMENT OF PUBLIC HEALTH (CDPH)

CDPH SECTION 01350 (2017; Version 1.2) Standard Method for  
the Testing and Evaluation of Volatile  
Organic Chemical Emissions from Indoor

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### Sources using Environmental Chambers

##### GREEN SEAL (GS)

GS-36 (2013) Adhesives for Commercial Use

##### SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT (SCAQMD)

SCAQMD Rule 1168 (2017) Adhesive and Sealant Applications

##### TILE COUNCIL OF NORTH AMERICA (TCNA)

TCNA Hdbk (2017) Handbook for Ceramic, Glass, and  
Stone Tile Installation

## 1.2 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

### SD-02 Shop Drawings

#### SD-03 Product Data

Porcelain Tile

Glazed Ceramic Wall Tile

Transition Strips

Setting-Bed

Mortar, Grout, and Adhesive

Glass-Mat Gypsum Water-Resistant Backing Board

Waterproof Membrane

Crack Isolation Membrane

#### SD-04 Samples

Tile

Accessories

Transition Strips

Grout

#### SD-07 Certificates

Indoor Air Quality for Adhesives

Indoor Air Quality for Sealants

#### SD-08 Manufacturer's Instructions



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### Manufacturer's Approved Cleaning Instructions

#### SD-10 Operation and Maintenance Data

Porcelain Tile; G

Glazed Ceramic Wall Tile; G

Transition Strips; G

#### 1.3 CERTIFICATIONS

#### 1.4 QUALITY ASSURANCE

Provide installers having a minimum of two years of experience with a company specializing in performing the type of work described. Each type and color of tile to be provided from a single source. Each type and color of mortar, adhesive, and grout to be provided from the same source.

#### 1.5 DELIVERY, STORAGE, AND HANDLING

Ship tiles in sealed packages and clearly marked with the grade, type of tile, producer identification, and country of origin. Deliver materials to the project site in manufacturer's original unopened containers with seals unbroken and labels and hallmarks intact. Protect materials from weather, and store them under cover in accordance with manufacturer's printed instructions. Store and handle tiles per manufacturer's instructions for gauged porcelain tile and gauged porcelain tile panels/slabs.

#### 1.6 ENVIRONMENTAL REQUIREMENTS

Do not perform ceramic tile work unless the substrate and ambient temperature is at least 50 degrees F and rising. Maintain temperature above 50 degrees F while the work is being performed and for at least 7 days after completion of the work. When temporary heaters are used, ventilate the area to the outside to avoid carbon dioxide damage to new tilework.

#### 1.7 WARRANTY

Provide manufacturer's warranty to repair or replace defective tiling materials and workmanship, including tile, mortar and grout products and installation as a system, for a period of one year from date of final acceptance of the work.

#### 1.8 EXTRA MATERIALS

Supply an extra 2 percent of each type tile used in clean and marked cartons.

### PART 2 PRODUCTS

#### 2.1 TILE

Provide tiles that comply with ANSI A137.1 and are standard grade tiles. Provide a minimum breaking strength of 125 lbs. for wall tile and 250 lbs. for floor tile in accordance with ASTM C648. Provide exterior building

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

tile for cold climate projects that is approved by the manufacturer for exterior use when tested in accordance with ASTM C1026. Provide floor tiles with a minimum wet dynamic coefficient of friction (DCOF) value of 0.42 when tested in accordance with ANSI A137.1 requirements. Provide glazed floor tile with a Class IV-Commercial classification as rated by the manufacturer when tested in accordance with ASTM C1027 for visible abrasion resistance as related to foot traffic. For materials like tile, accessories, and transition strips submit samples of sufficient size to show color range, pattern, type and joints.

Submit manufacturers' descriptive product data for each type of ceramic, quarry and glass tiling indicated. Include manufacturers' literature, finishes, profiles and thicknesses of materials.

Submit manufacturers' operations and maintenance data for each type of ceramic, quarry and glass tiling indicated in accordance with Section 01 78 23 OPERATIONS AND MAINTENANCE DATA.

#### 2.1.1 Porcelain Tile

Provide unglazed through body (surface color and pattern go all the way through the tile body), porcelain tile and cove base and trim pieces. Provide nominal tile size(s) of as indicated on Drawings.

#### 2.1.2 Glazed Ceramic Wall Tile

Provide glazed ceramic wall tile as indicated on Drawings.

### 2.2 SETTING-BED

Submit manufacturer's catalog data. Compose the setting-bed of the following materials:

#### 2.2.1 Aggregate for Concrete Fill

Conform to ASTM C33/C33M for aggregate fill. Do not exceed one-half the thickness of concrete fill for maximum size of coarse aggregate.

#### 2.2.2 Portland Cement

Conform to ASTM C150/C150M for cement, Type I, white for wall mortar and gray for other uses.

#### 2.2.3 Sand

Conform to ASTM C144 for sand.

#### 2.2.4 Hydrated Lime

Conform to ASTM C206 for hydrated lime, Type S or ASTM C207, Type S.

### 2.3 WATER

Provide potable water.

### 2.4 MORTAR, GROUT, AND ADHESIVE

Provide non-aerosol adhesive products used on the interior of the building (defined as inside of the weatherproofing system) meeting either emissions

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

requirements of CDPH SECTION 01350 (limit requirements for either office or classroom spaces regardless of space type) or VOC content requirements of SCAQMD Rule 1168. Provide aerosol adhesives used on the interior of the building meeting either emissions requirements of CDPH SECTION 01350 (limit requirements for either office or classroom spaces regardless of space type) or VOC content requirements of GS-36. For products located on the interior of the building (inside of the weatherproofing system, provide certification or validation of indoor air quality for adhesives. Provide bond coat, mortar, and grout supplied from the same manufacturer.

#### 2.4.1 Latex-Portland Cement Mortar

TCNA Hdbk.

#### 2.4.2 Ceramic Tile Grout

TCNA Hdbk; petroleum-free and plastic-free high-performance cement grout (latex-portland cement grout).

#### 2.4.3 Sealants

Comply with applicable regulations regarding toxic and hazardous materials and as specified. Provide sealant that does not change the color or alter the appearance of the grout. Refer to Section 07 92 00 JOINT SEALANTS.

Provide sealants used on the interior of the building (defined as inside of the weatherproofing system) meeting either emissions requirements of CDPH SECTION 01350 (limit requirements for either office or classroom spaces regardless of space type) or VOC content requirements of SCAQMD Rule 1168. For products located on the interior of the building (inside of the weatherproofing system), provide certification or validation of indoor air quality for sealants.

### 2.5 SUBSTRATES

#### 2.5.1 Glass-Mat Gypsum Water-Resistant Backing Board

Provide glass-mat water-resistant backing board, for use as tile substrate as indicated, in accordance with ASTM C1178/C1178M. Provide 1/2 inch thick glass-mat water-resistant backing board.

### 2.6 MISCELLANEOUS TRIMS

#### 2.6.1 Transition Strips

Provide as indicated on Drawings.

### 2.7 WATERPROOF MEMBRANE

#### 2.7.1 General

Manufacturer's standard product that complies with ANSI A108/A118/A136.1 and is recommended by the manufacturer for the application indicated. Include reinforcement and accessories recommended by manufacturer.

#### 2.7.2 Chlorinated-Polyethylene Shower Waterproof Membrane

Nonplasticized, chlorinated polyethylene faced on both sides with nonwoven polyester fabric; 0.040 inch nominal thickness.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## 2.8 COLOR, TEXTURE, AND PATTERN

Provide color, pattern and texture as indicated; colors listed are not intended to limit the selection of equal colors from other manufacturers..

## PART 3 EXECUTION

### 3.1 PREPARATORY WORK AND WORKMANSHIP

Inspect surface to receive tile in conformance to the requirements of TCNA Hdbk for surface conditions for the type setting bed specified and for workmanship. Provide variations of tiled surfaces that fall within maximum values shown below:

| TYPE                         | WALLS             | FLOORS             |
|------------------------------|-------------------|--------------------|
| Dry-Set Mortar               | 1/8 inch in 8 ft. | 1/8 inch in 10 ft. |
| Organic Adhesives            | 1/8 inch in 8 ft. | 1/16 inch in 3 ft. |
| Latex-Portland Cement Mortar | 1/8 inch in 8 ft. | 1/8 inch in 10 ft. |
| Epoxy                        | 1/8 inch in 8 ft. | 1/8 inch in 10 ft. |

### 3.2 GENERAL INSTALLATION REQUIREMENTS

Do not start tile work until roughing in for mechanical and electrical work has been completed and tested, and built-in items requiring membrane waterproofing have been installed and tested. Close space, in which tile is being set, to traffic and other work. Keep closed until tile is firmly set. Do not start floor tile installation in spaces requiring wall tile until after wall tile has been installed. Apply tile in colors and patterns indicated in the area shown on the drawings. Install tile with the respective surfaces in true even planes to the elevations and grades shown. Provide special shapes as required for sills, jambs, recesses, offsets, external corners, and other conditions to provide a complete and neatly finished installation. Solidly back tile bases and coves with mortar. Do not walk or work on newly tiled floors without using kneeling boards or equivalent protection of the tiled surface. Keep traffic off horizontal portland cement mortar installations for at least 72 hours. Keep all traffic off epoxy installed floors for at least 40 hours after grouting, and heavy traffic off for at least 7 days, unless otherwise specifically authorized by manufacturer. Dimension and draw detail drawings at a minimum scale of 1/4 inch = 1 foot. Include drawings of pattern at inside corners, outside corners, termination points and location of all equipment items such as thermostats, switch plates, mirrors and toilet accessories mounted on surface. Submit drawings showing ceramic tile pattern elevations and floor plans. Submit manufacturer's preprinted installation instructions.

Do not install building construction materials that show visual evidence of biological growth.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 3.3 INSTALLATION OF SUBSTRATES

#### 3.3.1 Glass-Mat Water-Resistant Backing Board

Install in accordance with manufacturer's written instructions.

### 3.4 INSTALLATION OF WALL TILE

Install wall tile in accordance with the TCNA Hdbk, method and with grout joints as recommended by the manufacturer for the type of tile.

#### 3.4.1 Ceramic Tile Grout

Prepare and install ceramic tile grout in accordance with TCNA Hdbk. Provide and apply manufacturer's standard product for sealing grout joints in accordance with manufacturer's recommendations.

### 3.5 INSTALLATION OF MISCELLANEOUS TRIMS

#### 3.5.1 Transition Strips

Install transition strips where indicated, in a manner similar to that of the ceramic tile floor and as recommended by the manufacturer. Provide thresholds full width of the opening. Install head joints at ends not exceeding 1/4 inch in width and grouted full.

#### 3.5.2 Metal Trims

Install trim where indicated. Embed anchoring leg in setting mortar in accordance with manufacturer's instructions. During grouting of tile joints, immediately wipe grout from finish surface.

### 3.6 EXPANSION JOINTS

Form and seal joints as specified in Section 07 92 00 JOINT SEALANTS.

#### 3.6.1 Walls

Provide expansion joints at control joints in backing material. Wherever backing material changes, install an expansion joint to separate the different materials.

### 3.7 CLEANING AND PROTECTING

Upon completion, thoroughly clean tile surfaces in accordance with manufacturer's approved cleaning instructions. Do not use acid for cleaning glazed tile. Clean floor tile with resinous grout or with factory mixed grout in accordance with printed instructions of the grout manufacturer. After the grout has set, provide a protective coat of a noncorrosive soap or other approved method of protection for tile wall surfaces. Cover tiled floor areas with building paper before foot traffic is permitted over the finished tile floors. Provide board walkways on tiled floors that are to be continuously used as passageways by workmen. Replace damaged or defective tiles.

-- End of Section --

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## SECTION 09 51 00

### ACOUSTICAL CEILINGS

**08/20**

#### PART 1 GENERAL

##### 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

#### ASTM INTERNATIONAL (ASTM)

|                   |                                                                                                                                                             |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM A641/A641M   | (2019) Standard Specification for Zinc-Coated (Galvanized) Carbon Steel Wire                                                                                |
| ASTM C423         | (2023) Sound Absorption and Sound Absorption Coefficients by the Reverberation Room Method                                                                  |
| ASTM C635/C635M   | (2022) Standard Specification for Manufacture, Performance, and Testing of Metal Suspension Systems for Acoustical Tile and Lay-In Panel Ceilings           |
| ASTM C636/C636M   | (2013) Standard Practice for Installation of Metal Ceiling Suspension Systems for Acoustical Tile and Lay-In Panels                                         |
| ASTM C834         | (2017; R 2023) Standard Specification for Latex Sealants                                                                                                    |
| ASTM E413         | (2022) Classification for Rating Sound Insulation                                                                                                           |
| ASTM E580/E580M   | (2022) Standard Practice for Installation of Ceiling Suspension Systems for Acoustical Tile and Lay-in Panels in Areas Subject to Earthquake Ground Motions |
| ASTM E795         | (2023) Standard Practices for Mounting Test Specimens During Sound Absorption Tests                                                                         |
| ASTM E1111/E1111M | (2014; R 2022) Standard Test Method for Measuring the Interzone Attenuation of Open Office Components                                                       |
| ASTM E1264        | (2023) Standard Classification for Acoustical Ceiling Products                                                                                              |
| ASTM E1414/E1414M | (2021a) Standard Test Method for Airborne Sound Attenuation Between Rooms Sharing a Common Ceiling Plenum                                                   |
| ASTM E1477        | (1998; R 2022a) Standard Test Method for                                                                                                                    |

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

Luminous Reflectance Factor of Acoustical  
Materials by Use of Integrating-Sphere  
Reflectometers

U.S. DEPARTMENT OF DEFENSE (DOD)

UFC 3-301-01

(2023) Structural Engineering

## 1.2 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

### SD-02 Shop Drawings

Drawings

### SD-03 Product Data

Acoustical Ceiling Systems

Acoustical Performance

### SD-04 Samples

Acoustical Units

Acoustical Ceiling Tiles

## 1.3 DELIVERY, STORAGE. AND HANDLING

Deliver materials to the site in the manufacturer's original unopened containers with brand name and type clearly marked. Carefully handle and store materials in dry, watertight enclosures. Immediately before installation, store acoustical units for not less than 24 hours at the same temperature and relative humidity as the space where they will be installed in order to assure proper temperature and moisture acclimation.

## 1.4 ENVIRONMENTAL REQUIREMENTS

Maintain a uniform temperature of not less than 60 degrees F nor more than 85 degrees F and a relative humidity of not more than 70 percent for 24 hours before, during, and 24 hours after installation of acoustical units.

## 1.5 SCHEDULING

Complete and dry interior finish work such as plastering, concrete and terrazzo work before ceiling installation. Complete mechanical, electrical, and other work above the ceiling line; install and start operating heating, ventilating, and air conditioning systems in order to maintain temperature and humidity requirements.

## 1.6 WARRANTY

Provide manufacturer's warranty to repair or replace defective materials

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

and workmanship including but not limited to, sagging and warping of panels and rusting and of grid systems, for a period of one (1) year from date of final acceptance of the work.

#### 1.7 EXTRA MATERIALS

Furnish spare tiles, from the same lot as those installed, of each color at the rate of 5 tiles for each 1000 tiles installed.

### PART 2 PRODUCTS

#### 2.1 SYSTEM DESCRIPTION

Provide sound controlling units mechanically mounted on a ceiling suspension system for acoustical treatment. Provide the unit size, texture, finish, and color as specified. The Contractor has the option to substitute inch-pound (I-P) Recessed Light Fixtures (RLF) for metric RLF. If the Contractor opts to provide I-P RLF, then provide I-P products for other ceiling elements like acoustical ceiling tiles, air diffusers, air registers and grills. Coordinate the entire ceiling system with other details, like the location of access panels and ceiling penetrations, for instance, shown on the drawings. Submit drawings showing the the location, extent and details of acoustical treatment including suspension system, method of anchoring and fastening, and reflected ceiling plan. Coordinate with paragraph RECLAMATION PROCEDURES for reclamation of mineral fiber acoustical ceiling panels to be removed from the job site.

##### 2.1.1 Acoustical Performance

###### 2.1.1.1 Ceiling Sound Transmission

Provide ceiling systems with the specified Ceiling Attenuation Class (CAC) ratings as determined in accordance with ASTM E1414/E1414M and ASTM E413. Provide sound attenuators over light fixtures, air terminals and other ceiling penetrations, provide acoustical blanket insulation on top of the ceiling or adjacent to partitions to provide lightweight acoustical plenum barriers above partitions as required to achieve the specified CAC ratings. Provide test ceiling continuous at the partition and assembled in the suspension system in the same manner that the ceiling will be installed on the project.

###### 2.1.1.2 Ceiling Sound Absorption

Determine the Noise Reduction Coefficient (NRC) in accordance with ASTM C423. Determine Articulation Class (AC) in accordance with ASTM E1111/E1111M.

###### 2.1.2 Light Reflectance

Determine light reflectance factor in accordance with ASTM E1477 test method.

#### 2.2 ACOUSTICAL UNITS

Submit samples of each type of acoustical unit and each type of suspension grid tee section showing texture, finish, and color. Conform acoustical units to ASTM E1264, Class A, and the following requirements:



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## 2.2.1 Humidity Resistant Composition Units

### 2.2.1.1 Type

Non-asbestos mineral or glass fibers bonded with ceramic, moisture resistant thermo-setting resin, or other moisture resistant material and having a factory applied white paint finish. Provide panels that do not sag or warp under conditions of heat, high humidity or chemical fumes.

### 2.2.1.2 Flame Spread

Class: A, 25 or less

### 2.2.1.3 Pattern

As indicated on Drawings.

### 2.2.1.4 Minimum NRC

0.75 when tested on Mounting Type E-400 of ASTM E795.

### 2.2.1.5 Minimum Light Reflectance Coefficient

LR-1, 0.75 or greater

### 2.2.1.6 Nominal Size

24 by 24 inch

### 2.2.1.7 Edge Detail

Square

### 2.2.1.8 Finish

Factory-applied standard finish. See paragraph COLORS AND PATTERNS.

## 2.3 SUSPENSION SYSTEM

Provide as indicated suspension system conforming to ASTM C635/C635M for intermediate-duty systems. Provide surfaces exposed to view of aluminum or steel with a factory-applied white baked-enamel finish. Provide wall molding having a flange of not less than 15/16 inch. Provide mitered corners. Provide a suspension system with a maximum deflection of 1/360 of the span length capable of supporting the finished ceiling, light fixtures, air diffusers, and accessories, as shown. Conform seismic details to the guidance in UFC 3-301-01, ASTM E580/E580M and as indicated Drawings.

Provide Suspension System containing a minimum of 15 percent recycled content. Provide data identifying percentage of recycled content for suspension systems.

## 2.4 HANGERS

Provide hangers and attachment capable of supporting a minimum 300 pound ultimate vertical load without failure of supporting material or attachment.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### 2.4.1 Wires

Conform wires to ASTM A641/A641M, Class 1, 0.11 in diameter.

#### 2.5 FINISHES

Use manufacturer's standard textures, patterns and finishes as specified for acoustical units and suspension system members. Treat ceiling suspension system components to inhibit corrosion.

#### 2.6 COLORS AND PATTERNS

Use colors and patterns for acoustical units and suspension system components as indicated; colors listed are not intended to limit the selection of equal colors from other manufacturers.

#### 2.7 ACOUSTICAL SEALANT

Conform acoustical sealant to ASTM C834, nonstaining.

### PART 3 EXECUTION

#### 3.1 INSTALLATION

Do not install building construction materials that show visual evidence of biological growth.

Examine surfaces to receive directly attached acoustical units for unevenness, irregularities, and dampness that would affect quality and execution of the work. Rid areas, where acoustical units will be cemented, of oils, form residue, or other materials that reduce bonding capabilities of the adhesive. Complete and dry interior finish work such as plastering, concrete, and terrazzo work before installation. Complete and approve mechanical, electrical, and other work above the ceiling line prior to the start of acoustical ceiling installation. Provide acoustical work complete with necessary fastenings, clips, and other accessories required for a complete installation. Do not expose mechanical fastenings in the finished work. Lay out hangers for each individual room or space. Provide hangers to support framing around beams, ducts, columns, grilles, and other penetrations through ceilings. Keep main runners and carrying channels clear of abutting walls and partitions. Provide at least two main runners for each ceiling span. Wherever required to bypass an object with the hanger wires, install a subsuspension system so that all hanger wires will be plumb.

##### 3.1.1 Suspension System

Install suspension system in accordance with ASTM C636/C636M and as specified herein. Do not suspend hanger wires or other loads from underside of steel decking.

##### 3.1.1.1 Plumb Hangers

Install hangers plumb and not pressing against insulation covering ducts and pipes. Where lighting fixtures are supported from the suspended ceiling system, provide hangers at a minimum of four hangers per fixture and located not more than 6 inch from each corner of each fixture.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### 3.1.1.2 Splayed Hangers

Splay (slope or slant) hangers around obstructions, offsetting the resulting horizontal force by bracing, countersplaying, or other acceptable means.

#### 3.1.2 Wall Molding

Provide wall molding where ceilings abut vertical surfaces. Miter corners where wall moldings intersect or install corner caps. Secure wall molding not more than 3 inch from ends of each length and not more than 16 inch on centers between end fastenings. Provide wall molding springs at each acoustical unit in semi-exposed or concealed systems.

#### 3.1.3 Acoustical Units

Install acoustical units in accordance with the approved installation instructions of the manufacturer. Ensure that edges of acoustical units are in close contact with metal supports, with each other, and in true alignment. Arrange acoustical units so that units less than one-half width are minimized. Hold units in exposed-grid system in place with manufacturer's standard hold-down clips, if units weigh less than 1 psf or if required for fire resistance rating.

#### 3.1.4 Acoustical Sealant

Seal all joints around pipes, ducts or electrical outlets penetrating the ceiling. Apply a continuous ribbon of acoustical sealant on vertical web of wall or edge moldings.

### 3.2 CEILING ACCESS PANELS

Locate ceiling access panels directly under the items which require access.

### 3.3 CLEANING

Following installation, clean dirty or discolored surfaces of acoustical units and leave them free from defects. Remove units that are damaged or improperly installed and provide new units as directed.

-- End of Section --

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## SECTION 09 65 00

RESILIENT FLOORING  
**08/10, CHG 3: 08/18**

## PART 1 GENERAL

## 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

## ASTM INTERNATIONAL (ASTM)

|            |                                                                                                                              |
|------------|------------------------------------------------------------------------------------------------------------------------------|
| ASTM D4078 | (2002; R 2015) Water Emulsion Floor Polish                                                                                   |
| ASTM F710  | (2022) Standard Practice for Preparing Concrete Floors to Receive Resilient Flooring                                         |
| ASTM F1344 | (2021a) Standard Specification for Rubber Floor Tile                                                                         |
| ASTM F1482 | (2021) Standard Practice for Installation and Preparation of Panel Type Underlayments to Receive Resilient Flooring          |
| ASTM F1861 | (2021) Standard Specification for Resilient Wall Base                                                                        |
| ASTM F1869 | (2022) Standard Test Method for Measuring Moisture Vapor Emission Rate of Concrete Subfloor Using Anhydrous Calcium Chloride |
| ASTM F2170 | (2019a) Standard Test Method for Determining Relative Humidity in Concrete Floor Slabs Using in situ Probes                  |

## CALIFORNIA DEPARTMENT OF PUBLIC HEALTH (CDPH)

|                    |                                                                                                                                                            |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CDPH SECTION 01350 | (2017; Version 1.2) Standard Method for the Testing and Evaluation of Volatile Organic Chemical Emissions from Indoor Sources using Environmental Chambers |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|

## GREEN SEAL (GS)

|       |                                     |
|-------|-------------------------------------|
| GS-36 | (2013) Adhesives for Commercial Use |
|-------|-------------------------------------|

## SCIENTIFIC CERTIFICATION SYSTEMS (SCS)

|     |                                            |
|-----|--------------------------------------------|
| SCS | SCS Global Services (SCS) Indoor Advantage |
|-----|--------------------------------------------|

## SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT (SCAQMD)

|                  |                                          |
|------------------|------------------------------------------|
| SCAQMD Rule 1168 | (2017) Adhesive and Sealant Applications |
|------------------|------------------------------------------|

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

UNDERWRITERS LABORATORIES (UL)

UL 2818

(2022) GREENGUARD Certification Program  
For Chemical Emissions For Building  
Materials, Finishes And Furnishings

1.2 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Resilient Flooring and Accessories; G

SD-03 Product Data

Resilient Flooring and Accessories; G

Adhesives

Rubber Tile  
Wall Base

SD-04 Samples

Resilient Flooring and Accessories; G

SD-06 Test Reports

Moisture, Alkalinity and Bond Tests; G

SD-07 Certificates

Indoor Air Quality for Rubber Tile  
Indoor Air Quality for Adhesives

SD-08 Manufacturer's Instructions

Surface Preparation

Installation

SD-10 Operation and Maintenance Data

Resilient Flooring and Accessories; G

1.3 CERTIFICATES

1.3.1 Indoor Air Quality

Submit required indoor air quality certifications and validations in one

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

submittal package.

#### 1.3.1.1 Floor Covering Materials

Provide Rubber Tile, and wall base products certified to meet indoor air quality requirements by FLOORSCORE, UL 2818 (Greenguard) Gold, SCS Global Services Indoor Advantage Gold or provide certification by other third-party programs. Provide current product certification documentation from certification body.

#### 1.3.1.2 Adhesives, Caulking and Sealants

Provide products certified to meet indoor air quality requirements by UL 2818 (Greenguard) Gold, SCS Global Services Indoor Advantage Gold or provide certification or validation by other third-party programs that products meet the requirements of this Section. Provide current product certification documentation from certification body. When product does not have certification, provide validation that product meets the indoor air quality product requirements cited herein.

#### 1.4 DELIVERY, STORAGE, AND HANDLING

Deliver materials to the building site in original unopened containers bearing the manufacturer's name, style name, pattern color name and number, production run, project identification, and handling instructions. Store materials in a clean, dry, secure, and well-ventilated area free from strong contaminant sources and residues with ambient air temperature maintained above 68 degrees F and below 85 degrees F, stacked according to manufacturer's recommendations. Remove resilient flooring products from packaging to allow ventilation prior to installation. Protect materials from the direct flow of heat from hot-air registers, radiators and other heating fixtures and appliances. Observe ventilation and safety procedures specified in the MSDS. Do not store rubber surface products with materials that have a high capacity to adsorb volatile organic compound (VOC) emissions, including. Do not store exposed rubber surface materials in occupied spaces.

#### 1.5 ENVIRONMENTAL REQUIREMENTS

Maintain areas to receive resilient flooring at a temperature above 68 degrees F and below 85 degrees F for 3 days before application, during application and 2 days after application, unless otherwise directed by the flooring manufacturer for the flooring being installed. Maintain a minimum temperature of 55 degrees F thereafter. Provide adequate ventilation to remove moisture from area and to comply with regulations limiting concentrations of hazardous vapors.

#### 1.6 SCHEDULING

Schedule resilient flooring application after the completion of other work which would damage the finished surface of the flooring.

#### 1.7 WARRANTY

Provide manufacturer's standard performance guarantees or warranties that extend beyond a one year period.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## 1.8 EXTRA MATERIALS

Provide extra flooring material of each color and pattern at the rate of 5 tiles for each 1000 tiles and 5 square feet for each 1000 square feet of sheet flooring installed. Provide extra wall base material composed of 20 linear feet of each type, color and pattern. Package all extra materials in original properly marked containers bearing the manufacturer's name, brand name, pattern color name and number, production run, and handling instructions. Provide extra materials from the same lot as those installed. Leave extra stock at the site in location assigned by Contracting Officer.

## PART 2 PRODUCTS

### 2.1 RUBBER TILE

Conform to ASTM F1344 Class 1 homogeneous, Type A (solid color), size as indicated on Drawings. Provide smooth stud profile. Provide 0.125 inch overall thickness.

Provide certification of indoor air quality for Rubber Tile.

### 2.2 WALL BASE

Conform to ASTM F1861, Type TS (vulcanized thermoset rubber) or Type TP (thermoplastic rubber), or Type TV (thermoplastic vinyl), Style A (straight - installed with carpet), and Style B (coved - installed with resilient flooring). Provide 4 inch high and a minimum 1/8 inch thick wall base. Provide preformed corners in matching height, shape, and color.

### 2.3 MOULDING

Provide tapered mouldings of vinyl and types as recommended by flooring manufacturer for both edges and transitions of flooring materials specified. Provide vertical lip on moulding of maximum 1/4 inch. Provide bevel change in level between 1/4 and 1/2 inch with a slope no greater than 1:2.

### 2.4 ADHESIVES

Provide adhesives for flooring, base and accessories as recommended by the manufacturer and comply with local indoor air quality standards. Submit manufacturer's descriptive data, documentation stating physical characteristics, and mildew and germicidal characteristics.

Provide non-aerosol adhesive products used on the interior of the building (defined as inside of the weatherproofing system) that meet either emissions requirements of CDPH SECTION 01350 (limit requirements for either office or classroom spaces regardless of space type) or VOC content requirements of SCAQMD Rule 1168. Provide aerosol adhesives used on the interior of the building that meet either emissions requirements of CDPH SECTION 01350 (use the office or classroom requirements, regardless of space type) or VOC content requirements of GS-36. Provide certification or validation of indoor air quality for adhesives.

### 2.5 POLISH/FINISH

Provide polish finish as recommended by the manufacturer and conform to

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

ASTM D4078 for polish.

## 2.6 CAULKING AND SEALANTS

Provide caulking and sealants in accordance with Section 07 92 00 JOINT SEALANTS.

## 2.7 MANUFACTURER'S COLOR, PATTERN AND TEXTURE

Provide color, pattern and texture for resilient flooring and accessories as indicated on the drawings . Provide floor patterns as specified . Provide flooring in any one continuous area or replacement of damaged flooring in continuous area from same production run with same shade and pattern. Submit scaled drawings indicating patterns (including location of patterns and colors) and dimensions. Submit manufacturer's descriptive data and three samples of each indicated color and type of flooring, base, mouldings, and accessories sized a minimum 2-1/2 by 4 inch.

# PART 3 EXECUTION

## 3.1 EXAMINATION

Examine and verify that site conditions are in agreement with the design package. Report all conditions that will prevent a proper installation. Do not take any corrective action without written permission from the Government. Work will proceed only when conditions have been corrected and accepted by the installer. Submit manufacturer's printed installation instructions for all flooring materials and accessories, including preparation of substrate, seaming techniques, and recommended adhesives.

## 3.2 SURFACE PREPARATION

Provide a smooth, true, level plane for surface preparation of the flooring, except where indicated as sloped. Floor to be flat to within 3/16 inch in 10 feet. Prepare subfloor in accordance with flooring manufacturer's recommended instructions. Prepare the surfaces of lightweight concrete slabs (as defined by the flooring manufacturer) as recommended by the flooring manufacturer. Comply with ASTM F710 for concrete subfloor preparation. Floor fills or toppings may be required as recommended by the flooring manufacturer. Install underlayments, when required by the flooring manufacturer, in accordance with manufacturer's recommended printed installation instructions. Comply with ASTM F1482 for panel type underlayments. Before any work under this section is begun, correct all defects such as rough or scaling concrete, chalk and dust, cracks, low spots, high spots, and uneven surfaces. Repair all damaged portions of concrete slabs as recommended by the flooring manufacturer. Remove concrete curing and sealer compounds from the slabs, other than the type that does not adversely affect adhesion. Remove paint, varnish, oils, release agents, sealers, waxes, and adhesives, as required by the flooring product in accordance with manufacturer's printed installation instructions.

## 3.3 MOISTURE, ALKALINITY AND BOND TESTS

Determine the suitability of the concrete subfloor for receiving the resilient flooring with regard to moisture content and pH level by moisture and alkalinity tests. Conduct moisture testing in accordance with ASTM F1869 or ASTM F2170, unless otherwise recommended by the flooring manufacturer. Conduct alkalinity testing as recommended by the



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

flooring manufacturer. Determine the compatibility of the resilient flooring adhesives to the concrete floors by a bond test in accordance with the flooring manufacturer's recommendations. Submit copy of test reports for moisture and alkalinity content of concrete slab, and bond test stating date of test, person conducting the test, and the area tested.

### 3.4 GENERAL INSTALLATION

Do not install building construction materials that show visual evidence of biological growth.

### 3.5 PLACING MOULDING

Provide moulding where flooring termination is higher than the adjacent finished flooring and at transitions between different flooring materials. When required, locate moulding under door centerline. Moulding is not required at doorways where thresholds are provided. Secure moulding with adhesive as recommended by the manufacturer. Prepare and apply adhesives in accordance with manufacturer's printed directions.

### 3.6 PLACING WALL BASE

Install wall base in accordance with manufacturer's printed installation instructions. Prepare and apply adhesives in accordance with manufacturer's printed directions. Tighten base joints and make even with adjacent resilient flooring. Fill voids along the top edge of base at masonry walls with caulk. Roll entire vertical surface of base with hand roller, and press toe of base with a straight piece of wood to ensure proper alignment. Avoid excess adhesive in corners.

### 3.7 CLEANING

Immediately upon completion of installation of flooring in a room or an area, dry and clean the flooring and adjacent surfaces to remove all surplus adhesive. Clean flooring as recommended in accordance with manufacturer's printed maintenance instructions and within the recommended time frame. As required by the manufacturer, apply the recommended number of coats and type of polish and finish in accordance with manufacturer's written instructions.

### 3.8 PROTECTION

From the time of installation until acceptance, protect flooring from damage as recommended by the flooring manufacturer. Remove and replace flooring which becomes damaged, loose, broken, or curled and wall base which is not tight to wall or securely adhered.

-- End of Section --

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

SECTION 09 68 00

CARPETING

**11/17, CHG 2: 08/20**

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

|           |                                                                               |
|-----------|-------------------------------------------------------------------------------|
| AATCC 16  | (2004; E 2008; E 2010) Colorfastness to Light                                 |
| AATCC 107 | (2013) Colorfastness to Water                                                 |
| AATCC 134 | (2016) Electrostatic Propensity of Carpets                                    |
| AATCC 165 | (2013) Colorfastness to Crocking: Textile Floor Coverings - Crockmeter Method |
| AATCC 174 | (2016) Antimicrobial Activity Assessment of New Carpets                       |

ASTM INTERNATIONAL (ASTM)

|            |                                                                                                                                 |
|------------|---------------------------------------------------------------------------------------------------------------------------------|
| ASTM D1335 | (2017; E 2018) Standard Test Method for Tuft Bind of Pile Yarn Floor Coverings                                                  |
| ASTM D2859 | (2016) Standard Test Method for Ignition Characteristics of Finished Textile Floor Covering Materials                           |
| ASTM D3278 | (2021) Standard Test Methods for Flash Point of Liquids by Small Scale Closed-Cup Apparatus                                     |
| ASTM D5793 | (2018) Standard Test Method for Binding Sites Per Unit Length or Width of Pile Yarn Floor Coverings                             |
| ASTM D5848 | (2020) Standard Test Method for Mass Per Unit Area of Pile Yarn Floor Coverings                                                 |
| ASTM D6859 | (2011) Standard Test Method for Pile Thickness of Finished Level Pile Yarn Floor Coverings                                      |
| ASTM D7330 | (2015) Standard Test Method for Assessment of Surface Appearance Change in Pile Floor Coverings Using Standard Reference Scales |
| ASTM E648  | (2023) Standard Test Method for Critical Radiant Flux of Floor-Covering Systems                                                 |

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### Using a Radiant Heat Energy Source

#### CALIFORNIA DEPARTMENT OF PUBLIC HEALTH (CDPH)

|                    |                                                                                                                                                            |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CDPH SECTION 01350 | (2017; Version 1.2) Standard Method for the Testing and Evaluation of Volatile Organic Chemical Emissions from Indoor Sources using Environmental Chambers |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|

#### CARPET AND RUG INSTITUTE (CRI)

|                     |                                                                                                            |
|---------------------|------------------------------------------------------------------------------------------------------------|
| CRI 104             | (2015) Carpet Installation Standard for Commercial Carpet                                                  |
| CRI 105             | (2015) Carpet Installation Standard for Residential Carpet                                                 |
| CRI GLP QM          | (2017) Green Label Plus Quality Manual                                                                     |
| CRI Test Method 103 | (2015) Standard Test Method for the Evaluation of Texture Appearance Retention of Carpet Standards Program |

#### GREEN SEAL (GS)

|       |                                     |
|-------|-------------------------------------|
| GS-36 | (2013) Adhesives for Commercial Use |
|-------|-------------------------------------|

#### INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

|          |                                                                                                                                                                                                  |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ISO 2551 | (2020) Textile Floor Coverings and Textile Floor Coverings in Tile Form- Determination of Dimensional Changes Due to the Effects of Varied Water and Heat Conditions and Distortion Out of Plane |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

#### SCIENTIFIC CERTIFICATION SYSTEMS (SCS)

|     |                                            |
|-----|--------------------------------------------|
| SCS | SCS Global Services (SCS) Indoor Advantage |
|-----|--------------------------------------------|

#### SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT (SCAQMD)

|                  |                                          |
|------------------|------------------------------------------|
| SCAQMD Rule 1113 | (2016) Architectural Coatings            |
| SCAQMD Rule 1168 | (2022) Adhesive and Sealant Applications |

#### U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

|             |                                                                     |
|-------------|---------------------------------------------------------------------|
| 16 CFR 1630 | Standard for the Surface Flammability of Carpets and Rugs (FF 1-70) |
|-------------|---------------------------------------------------------------------|

#### UNDERWRITERS LABORATORIES (UL)

|         |                                                                                                                 |
|---------|-----------------------------------------------------------------------------------------------------------------|
| UL 2818 | (2022) GREENGUARD Certification Program For Chemical Emissions For Building Materials, Finishes And Furnishings |
|---------|-----------------------------------------------------------------------------------------------------------------|

## 1.2 SUBMITTALS

Government approval is required for submittals with a "G" classification.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Installation Drawings; G

SD-03 Product Data

Carpet; G

Carpet Cushion; G

Moldings; G

Indoor Air Quality for Non-Aerosol Adhesives

Indoor Air Quality for Concrete Primer

SD-04 Samples

Carpet; G

Moldings; G

Carpet Cushion; G

SD-06 Test Reports

Moisture and Alkalinity Tests; FIO

SD-08 Manufacturer's Instructions

Surface Preparation

SD-10 Operation and Maintenance Data

Cleaning and Protection

Maintenance Service

SD-11 Closeout Submittals

Warranty

1.3 CERTIFICATIONS

1.3.1 Indoor Air Quality Certifications

1.3.1.1 Floor Covering Materials

Provide carpet and cushion products certified to meet indoor air quality requirements by UL 2818 (GreenGuard) Gold, SCS Global Services Indoor Advantage Gold, CRI GLP QM or provide certification or validation by other third-party program that products meet the requirements of this Section. Provide current product certification documentation from certification body. When product does not have certification, provide validation that

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

product meets the indoor air quality product requirements cited herein.

#### 1.4 DELIVERY, STORAGE, AND HANDLING

Deliver materials to the site in the manufacturer's original wrappings and packages clearly labeled with the manufacturer's name, brand name, size, dye lot number, and related information. Remove materials from packaging and store them in a clean, dry, well ventilated area (100 percent outside air supply, minimum of 1.5 air changes per hour, and no recirculation), protected from damage, soiling, and moisture, and strong contaminant sources and residues, and maintain at a temperature above 60 degrees F for 2 days prior to installation. Do not store carpet or carpet tiles with materials which have high emissions of volatile organic compounds (VOCs) or other contaminants, including paints and adhesives. Do not store carpet near materials that may off gas or emit harmful fumes, such as kerosene heaters, fresh paint, or adhesives.

#### 1.5 AMBIENT CONDITIONS

Maintain areas in which carpeting is to be installed at a temperature above 60 degrees F and below 90 degrees F for 2 days before installation, during installation, and for 2 days after installation. Provide temporary ventilation during work of this section. Maintain a minimum temperature of 55 degrees F thereafter for the duration of the contract.

#### 1.6 WARRANTY

Provide manufacturer's standard performance guarantees or warranties including minimum ten year wear warranty, two year material and workmanship and ten year tuft bind and delamination.

### PART 2 PRODUCTS

#### 2.1 CARPET

Furnish first quality carpet that is free of visual blemishes, streaks, poorly dyed areas, fuzzing of pile yarn, spots or stains, and other physical and manufacturing defects. Provide carpet materials and treatments as reasonably nonallergenic and free of other recognized health hazards. Provide a static control construction on all grade carpets which gives adequate durability and performance. Submit manufacturer's catalog data and printed documentation stating physical characteristics, durability, resistance to fading, and flame resistance characteristics for each type of carpet material and installation accessory. Submit manufacturer's Product Data for 1) Carpet, 2) Moldings, and 3) Carpet Cushion. Also, submit Samples of the following:

- a. Carpet Tile: Two "Production Quality" samples of selected products, showing quality, pattern, and color specified. Refer to Finish Plan and Schedule for product information.

##### 2.1.1 Indoor Air Quality Requirements

Products must meet emissions requirements of CDPH SECTION 01350.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## 2.1.2 Physical Characteristics for Modular Tile and Entrance Tile Carpets.

### 2.1.2.1 Carpet Construction

Carpet Tile: Woven,  
Entrance Tile: Multilevel Patter Loop

### 2.1.2.2 Type

Modular tile 24 by 24 inch square with 0.15 percent growth/shrink rate in accordance with ISO 2551. Entrance 24 by 24 inch square.

### 2.1.2.3 Pile Types

Carpet Tile: Multilevel loop,  
Entrance Tile: Multilevel loop.

### 2.1.2.4 Pile Fiber

Commercial 100 percent branded nylon continuous filament.

### 2.1.2.5 Gauge or Pitch

Carpet Tile: Minimum 8 per inch,  
Entrance Tile: Minimum 10 per suare inch - Each in accordance with ASTM D5793.

### 2.1.2.6 Stitches or Rows/Wires

Carpet Tile: Minimum 8 per square inch,  
Entrance Tile: Minimum 10 pewr square inch.

### 2.1.2.7 Surface Pile Weight

Carpet Tile: Minimum 28 ounces per square yard,  
Entrance Tile: Minimum 49.3 ounces per square yard. This does not include weight of backings. Determine weight in accordance with ASTM D5848.

### 2.1.2.8 Pile Thickness

Carpet Tile: Minimum 0.174 inch,  
Entrance Tile: Minimum 0.362 inches - Each in accordance with ASTM D6859.

### 2.1.2.9 Pile Density

Carpet Tile: Minimum 5793 ounces per square yard,  
Entrance Tile: Minimum 6477 ounces per square yard.

### 2.1.2.10 Dye Method

100 percent Solution dyed.

### 2.1.2.11 Backing Materials

Provide primary backing materials like those customarily used and accepted by the trade for each type of carpet. Provide secondary backing to suit project requirements of those customarily used and accepted by the trade for each type of carpet.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## 2.2 PERFORMANCE REQUIREMENTS

### 2.2.1 Texture Appearance Retention Rating (TARR)

Provide carpet with a greater than or equal to 3.0 (Heavy) for Carpet Tile and 3.5 (Severe) for Entrance Tile - TARR traffic level classification in accordance with ASTM D7330 or CRI Test Method 103.

### 2.2.2 Static Control

Provide static control to permanently regulate static buildup to less than 3.5 kV when tested at 20 percent relative humidity and 70 degrees F in accordance with AATCC 134.

### 2.2.3 Flammability and Critical Radiant Flux Requirements

Comply with 16 CFR 1630 or ASTM D2859. Provide carpet in corridors and exits with a minimum average critical radiant flux of 0.45 watts per square centimeter when tested in accordance with ASTM E648.

### 2.2.4 Tuft Bind

Comply with ASTM D1335 for tuft bind force required to pull a tuft or loop free from carpet backing with a minimum 8 pound average force for modular carpet tile.

### 2.2.5 Colorfastness to Crocking

Comply dry and wet crocking with AATCC 165 and with a Class 4 minimum rating on the AATCC Color Transference Chart for all colors.

### 2.2.6 Colorfastness to Light

Comply colorfastness to light with AATCC 16, Test Option E "Water-Cooled Xenon-Arc Lamp, Continuous Light" and with a minimum 4 grey scale rating after 40 hours.

### 2.2.7 Colorfastness to Water

Comply colorfastness to water with AATCC 107 and with a minimum 4.0 gray scale rating and a minimum 4.0 transfer scale rating.

### 2.2.8 Delamination Strength

Provide delamination strength for tufted carpet with a secondary back of minimum 2.5 lbs/inch.

### 2.2.9 Antimicrobial

Nontoxic antimicrobial treatment in accordance with AATCC 174 Part I (qualitative), guaranteed by the carpet manufacturer to last the life of the carpet.

## 2.3 ADHESIVES AND CONCRETE PRIMER

Comply with applicable regulations regarding toxic and hazardous materials. Provide water resistant, mildew resistant, nonflammable, and nonstaining adhesives and concrete primers for carpet installation as required by the carpet manufacturer. Provide release adhesive for modular

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

tile carpet as recommended by the carpet manufacturer. Provide adhesives flashpoint of minimum 140 degrees F in accordance with ASTM D3278. Non-aerosol adhesive products used on the interior of the building (defined as inside of the weatherproofing system) must meet either emissions requirements of CDPH SECTION 01350 (limit requirements for either office or classroom spaces regardless of space type) or VOC content requirements of SCAQMD Rule 1168. Aerosol adhesive products used on the interior of the building (defined as inside of the weatherproofing system) must meet either emissions requirements of CDPH SECTION 01350 (limit requirements for either office or classroom spaces regardless of space type) or VOC content requirements of GS-36. Provide validation of indoor air quality for aerosol adhesives. Provide validation of indoor air quality for non-aerosol adhesives. Concrete primer products used on the interior of the building (defined as inside of the weatherproofing system) must meet either emissions requirements of CDPH SECTION 01350 (limit requirements for either office or classroom spaces regardless of space type) or VOC content requirements of SCAQMD Rule 1113. Provide validation of indoor air quality for concrete primer.

#### 2.4 MOLDINGS

Provide carpet moldings where floor covering material changes or carpet edge does not abut a vertical surface. Provide a heavy-duty vinyl or rubber molding designed for the type of carpet being installed. Provide floor flange of a minimum 1 1/2 inches wide. Provide color to match resilient base.

#### 2.5 COLOR, TEXTURE, AND PATTERN

Provide color, texture, and pattern in accordance with the drawings .

### PART 3 EXECUTION

#### 3.1 SURFACE PREPARATION

Do not install carpet on surfaces that are unsuitable and will prevent a proper installation. Prepare subfloor in accordance with flooring manufacturer's recommended instructions. Repair holes, cracks, depressions, or rough areas using material recommended by the carpet or adhesive manufacturer. Free floor of any foreign materials and sweep clean. Before beginning work, test subfloor with glue and carpet to determine "open time" and bond. Submit three copies of the manufacturer's printed Installation instructions for the carpet, including Surface Preparation, seaming techniques, and recommended adhesives and tapes.

#### 3.2 MOISTURE AND ALKALINITY TESTS

Test concrete slab for moisture content and excessive alkalinity in accordance with CRI 104/CRI 105. Submit three copies of reports of Moisture and Alkalinity Tests including content of concrete slab stating date of test, person conducting the test, and the area tested.

#### 3.3 PREPARATION OF CONCRETE SUBFLOOR

Do not commence installation of the carpeting until concrete substrate is at least 90 days old. Prepare the concrete surfaces in accordance with the carpet manufacturer's instructions. Match carpet, when required, and adhesives to prevent off-gassing to a type of curing compounds, leveling agents, and concrete sealer.



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 3.4 INSTALLATION

Isolate area of installation from rest of building. Perform all work by manufacturer's approved installers. Conduct installation in accordance with the manufacturer's printed instructions and CRI 104/CRI 105. Protect edges of carpet meeting hard surface flooring with molding and install in accordance with the molding manufacturer's printed instructions. Use autofoam mothproofing system for wool carpets. Follow ventilation, personal protection, and other safety precautions recommended by the adhesive manufacturer. Continue ventilation during installation and for at least 72 hours following installation. Do not permit traffic or movement of furniture or equipment in carpeted area for 24 hours after installation. Complete other work which would damage the carpet prior to installation of carpet. Submit three copies of Installation Drawings for Carpet and Moldings indicating areas receiving carpet, carpet types, patterns, direction of pile, location of seams, and locations of edge molding.

Do not install building construction materials that show visual evidence of biological growth.

#### 3.4.1 Modular Tile Installation

Install modular tiles with adhesive and snug joints. Use monolithic pattern for Entrance Tile and brick pattern for Carpet Tile installation method. Comply with manufacturer installation instructions for required drying time of releasable adhesive so it sets up properly. Provide accessibility to the subfloor where required.

### 3.5 CLEANING AND PROTECTION

Submit three copies of carpet manufacturer's maintenance instructions describing recommended type of cleaning equipment and material, spotting and cleaning methods, and cleaning cycles.

#### 3.5.1 Cleaning

As specified in Section 01 78 23 OPERATION AND MAINTENANCE DATA. After installation of the carpet, remove debris, scraps, and other foreign matter. Remove soiled spots and adhesive from the face of the carpet with appropriate spot remover. Cut off and remove protruding face yarn. Vacuum carpet clean with a high-efficiency particulate air (HEPA) filtration vacuum.

#### 3.5.2 Protection

Protect the installed carpet from soiling and damage with heavy, reinforced, nonstaining kraft paper, plywood, or hardboard sheets. Lap and secure edges of kraft paper protection to provide a continuous cover. Restrict traffic for at least 48 hours. Remove protective covering when directed by the Contracting Officer.

### 3.6 MAINTENANCE

#### 3.6.1 Extra Materials

Provide extra material from same dye lot consisting of uncut carpet tiles for future maintenance. Provide a minimum of three percent of total square yards of each carpet type, pattern, and color.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 3.6.2 Maintenance Service

Collect information from the manufacturer about maintenance agreement and options, and submit to Contracting Officer. Service must reclaim materials for recycling and/or reuse. Service must not landfill or burn reclaimed materials. When such a service is not available, seek local recyclers to reclaim the materials. Submit documentation of manufacturer's maintenance agreement, take-back program, and green lease for carpet. Include contact information, summary of procedures, and the limitations and conditions applicable to the project. Indicate manufacturer's commitment to reclaim materials for recycling and reuse.

-- End of Section --

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

SECTION 09 90 00

PAINTS AND COATINGS  
02/21

PART 1 GENERAL

1.1 RELATED REQUIREMENTS

1.1.1 Painting Included

Where a space or surface is indicated to be painted, include the following unless indicated otherwise.

- a. Surfaces behind portable objects and surface mounted articles readily detachable by removal of fasteners, such as screws and bolts.
- b. New factory finished surfaces that require identification or color coding and factory finished surfaces that are damaged during performance of the work.
- c. Existing coated surfaces that are damaged during performance of the work.

1.1.1.1 Exterior Painting

Includes new surfaces of the building and appurtenances.

1.1.1.2 Interior Painting

Includes new surfaces of the building and appurtenances as indicated. Where a space or surface is indicated to be painted, include the following items, unless indicated otherwise.

- a. Exposed columns, girders, beams, joists, and metal deck; and
- b. Other contiguous surfaces.

1.1.2 Painting Excluded

Do not paint the following unless indicated otherwise.

- a. Surfaces concealed and made inaccessible by panelboards, fixed ductwork, machinery, and equipment fixed in place.
- b. Surfaces in concealed spaces. Concealed spaces are defined as enclosed spaces above suspended ceilings, furred spaces, attic spaces, crawl spaces, elevator shafts and chases.
- c. Steel to be embedded in concrete.
- d. Copper, stainless steel, aluminum, anodized aluminum, brass, and lead except existing coated surfaces.
- e. Hardware, fittings, and other factory finished items.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 1.1.3 Mechanical and Electrical Painting

Includes field coating of interior new surfaces.

- a. Where a space or surface is indicated to be painted, include the following items unless indicated otherwise.

- (1) Exposed piping, conduit, and ductwork;
- (2) Supports, hangers, air grilles, and registers;
- (3) Miscellaneous metalwork and insulation coverings.

- b. Do not paint the following, unless indicated otherwise:

- (1) New zinc-coated, aluminum, and copper surfaces under insulation
- (2) New aluminum jacket on piping
- (3) New interior ferrous piping under insulation.

#### 1.1.3.1 Fire Extinguishing Sprinkler Systems

Clean, pretreat, prime, and paint new fire extinguishing sprinkler systems including valves, piping, conduit, hangers, supports, miscellaneous metalwork, and accessories. Apply coatings to clean, dry surfaces, using clean brushes.

### 1.1.4 Exterior Painting of Site Work Items

Field coat the following items:

|    | New Surfaces             | Existing Surfaces |
|----|--------------------------|-------------------|
| a. | Steel Doors and Frames   | N.A.              |
| b. | Exposed Structural Steel | N.A.              |

## 1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

#### AMERICAN CONFERENCE OF GOVERNMENTAL INDUSTRIAL HYGIENISTS (ACGIH)

ACGIH 0100 (2017; Suppl 2020) Documentation of the Threshold Limit Values and Biological Exposure Indices

#### AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME A13.1 (2020) Scheme for the Identification of Piping Systems

#### ASTM INTERNATIONAL (ASTM)

ASTM D523 (2014; R 2018) Standard Test Method for Specular Gloss

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

|            |                                                                                                                                         |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| ASTM D4263 | (1983; R 2018) Standard Test Method for Indicating Moisture in Concrete by the Plastic Sheet Method                                     |
| ASTM D4444 | (2013; R 2018) Standard Test Method for Laboratory Standardization and Calibration of Hand-Held Moisture Meters                         |
| ASTM D6386 | (2016a) Standard Practice for Preparation of Zinc (Hot-Dip Galvanized) Coated Iron and Steel Product and Hardware Surfaces for Painting |
| ASTM F1869 | (2023) Standard Test Method for Measuring Moisture Vapor Emission Rate of Concrete Subfloor Using Anhydrous Calcium Chloride            |

#### CENTERS FOR DISEASE CONTROL AND PREVENTION (CDC)

|                          |                                                                 |
|--------------------------|-----------------------------------------------------------------|
| Intelligence Bulletin 65 | (2013) Occupational Exposure to Carbon Nanotubes and Nanofibers |
|--------------------------|-----------------------------------------------------------------|

#### MASTER PAINTERS INSTITUTE (MPI)

|        |                                                                            |
|--------|----------------------------------------------------------------------------|
| MPI 3  | (2016) Primer, Alkali Resistant, Water Based                               |
| MPI 4  | (2016) Interior/Exterior Latex Block Filler                                |
| MPI 9  | (2016) Alkyd, Exterior Gloss (MPI Gloss Level 6)                           |
| MPI 10 | (2016) Latex, Exterior Flat (MPI Gloss Level 1)                            |
| MPI 11 | (2016) Latex, Exterior Semi-Gloss, MPI Gloss Level 5                       |
| MPI 44 | (2016) Latex, Interior, (MPI Gloss Level 2)                                |
| MPI 50 | (2015) Primer Sealer, Latex, Interior                                      |
| MPI 52 | (2016) Latex, Interior, (MPI Gloss Level 3)                                |
| MPI 54 | (2016) Latex, Interior, Semi-Gloss (MPI Gloss Level 5)                     |
| MPI 72 | (2016) Polyurethane, Two-Component, Pigmented, Gloss (MPI Gloss Level 6-7) |
| MPI 77 | (2015) Epoxy, Gloss                                                        |
| MPI 79 | (2016) Primer, Alkyd, Anti-Corrosive for Metal                             |
| MPI 94 | (2016) Alkyd, Exterior, Semi-Gloss (MPI Gloss Level 5)                     |

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

|              |                                                                                        |
|--------------|----------------------------------------------------------------------------------------|
| MPI 95       | (2015) Primer, Quick Dry, for Aluminum                                                 |
| MPI 101      | (2016) Primer, Epoxy, Anti-Corrosive, for Metal                                        |
| MPI 107      | (2016) Primer, Rust-Inhibitive, Water Based                                            |
| MPI 108      | (2015) Epoxy, High Build, Low Gloss                                                    |
| MPI 119      | (2016) Latex, Exterior, Gloss (MPI Gloss Level 6)                                      |
| MPI 134      | (2015) Primer, Galvanized, Water Based                                                 |
| MPI 138      | (2016) Latex, Interior, High Performance Architectural, (MPI Gloss Level 2)            |
| MPI 139      | (2016) Latex, Interior, High Performance Architectural, (MPI Gloss Level 3)            |
| MPI 141      | (2016) Latex, Interior, High Performance Architectural, Semi-Gloss (MPI Gloss Level 5) |
| MPI 144      | (2016) Latex, Interior, Institutional Low Odor/VOC, (MPI Gloss Level 2)                |
| MPI 145      | (2016) Latex, Interior, Institutional Low Odor/VOC, ( MPI Gloss Level 3)               |
| MPI 146      | (2016) Latex, Interior, Institutional Low Odor/VOC, (MPI Gloss Level 4)                |
| MPI 147      | (May 2016) Latex, Interior, Institutional Low Odor/VOC, Semi-Gloss (MPI Gloss Level 5) |
| MPI 149      | (2016) Primer Sealer, Interior, Institutional Low Odor/VOC                             |
| MPI 161      | (2016) Light Industrial Coating, Exterior, Water Based ( MPI Gloss Level 3)            |
| MPI 163      | (2016) Light Industrial Coating, Exterior, Water Based, Semi-Gloss (MPI Gloss Level 5) |
| MPI 164      | (2016) Light Industrial Coating, Exterior, Water Based, Gloss (MPI Gloss Level 6)      |
| MPI 214      | (2016) Latex, Exterior (MPI Gloss Level 2)                                             |
| MPI ASM      | (2019) Architectural Painting Specification Manual                                     |
| MPI GPS-1-14 | (2014) Green Performance Standard GPS-1-14                                             |
| MPI GPS-2-14 | (2014) Green Performance Standard GPS-2-14                                             |
| MPI MRM      | (2015) Maintenance Repainting Manual                                                   |

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

# SOCIETY FOR PROTECTIVE COATINGS (SSPC)

|                        |                                                                                                                                                                                             |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SSPC 7/NACE No.4       | (2007) Brush-Off Blast Cleaning                                                                                                                                                             |
| SSPC Glossary          | (2011) SSPC Protective Coatings Glossary                                                                                                                                                    |
| SSPC Guide 6           | (2021) Guide for Containing Surface Preparation Debris Generated During Paint Removal Operations                                                                                            |
| SSPC Guide 7           | (2015) Guide to the Disposal of Lead-Contaminated Surface Preparation Debris                                                                                                                |
| SSPC PA 1              | (2016) Shop, Field, and Maintenance Coating of Metals                                                                                                                                       |
| SSPC QP 1              | (2019) Standard Procedure for Evaluating the Qualifications of Industrial/Marine Painting Contractors (Field Application to Complex Industrial Steel Structures and Other Metal Components) |
| SSPC SP 1              | (2015) Solvent Cleaning                                                                                                                                                                     |
| SSPC SP 2              | (2018) Hand Tool Cleaning                                                                                                                                                                   |
| SSPC SP 3              | (2018) Power Tool Cleaning                                                                                                                                                                  |
| SSPC SP 6/NACE No.3    | (2007) Commercial Blast Cleaning                                                                                                                                                            |
| SSPC SP 10/NACE No. 2  | (2015) Near-White Blast Cleaning                                                                                                                                                            |
| SSPC VIS 1             | (2002; E 2004) Guide and Reference Photographs for Steel Surfaces Prepared by Dry Abrasive Blast Cleaning                                                                                   |
| SSPC VIS 3             | (2004) Guide and Reference Photographs for Steel Surfaces Prepared by Hand and Power Tool Cleaning                                                                                          |
| SSPC VIS 4/NACE VIS 7  | (1998; E 2000; E 2004) Guide and Reference Photographs for Steel Surfaces Prepared by Waterjetting                                                                                          |
| SSPC-SP WJ-1/NACE WJ-1 | (2012) Clean to Bare Substrate, Waterjet Cleaning of Metals                                                                                                                                 |
| SSPC-SP WJ-2/NACE WJ-2 | (2012) Very Thorough Cleaning, Waterjet Cleaning of Metals                                                                                                                                  |
| SSPC-SP WJ-3/NACE WJ-3 | (2012) Thorough Cleaning, Waterjet Cleaning of Metals                                                                                                                                       |
| SSPC-SP WJ-4/NACE WJ-4 | (2012) Light Cleaning, Waterjet Cleaning of Metals                                                                                                                                          |

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

U.S. DEPARTMENT OF DEFENSE (DOD)

MIL-STD-101 (2014; Rev C) Color Code for Pipelines and  
for Compressed Gas Cylinders

U.S. ENVIRONMENTAL PROTECTION AGENCY (EPA)

EPA Method 24 (2000) Determination of Volatile Matter  
Content, Water Content, Density, Volume  
Solids, and Weight Solids of Surface  
Coatings

U.S. GENERAL SERVICES ADMINISTRATION (GSA)

FED-STD-313 (2018) Material Safety Data,  
Transportation Data and Disposal Data for  
Hazardous Materials Furnished to  
Government Activities

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1910.1000 Air Contaminants

29 CFR 1910.1025 Lead

29 CFR 1926.62 Lead

### 1.3 DEFINITIONS

#### 1.3.1 Qualification Testing

Qualification testing is the performance of all test requirements listed in the product specification. This testing is accomplished by MPI to qualify each product for the MPI Approved Product List, and may also be accomplished by Contractor's third-party testing lab if an alternative to Batch Quality Conformance Testing by MPI is desired.

#### 1.3.2 Batch Quality Conformance Testing

Batch quality conformance testing determines that the product provided is the same as the product qualified to the appropriate product specification. This testing must be accomplished by an MPI testing lab.

#### 1.3.3 Coating

SSPC Glossary; (1) A liquid, liquefiable, or mastic composition that is converted to a solid protective, decorative, or functional adherent film after application as a thin layer; (2) Generic term for paint, lacquer, enamel.

#### 1.3.4 DFT or dft

Dry film thickness, the film thickness of the fully cured, dry paint or coating.

#### 1.3.5 DSD

Degree of Surface Degradation, the MPI system of defining degree of surface degradation. Five levels are generically defined under the



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

Assessment sections in the MPI MRM, MPI Maintenance Repainting Manual.

#### 1.3.6 EXT

MPI short term designation for an exterior coating system.

#### 1.3.7 INT

MPI short term designation for an interior coating system.

#### 1.3.8 Loose Paint

Paint or coating that can be removed with a dull putty knife.

#### 1.3.9 mil / mils

The English measurement for 0.001 in or one one-thousandth of an inch.

#### 1.3.10 MPI Gloss Levels

MPI system of defining gloss. Seven gloss levels (G1 to G7) are generically defined under the Evaluation sections of the MPI Manuals. Traditionally, Flat refers to G1/G2, Eggshell refers to G3, Semigloss refers to G5, and Gloss refers to G6.

Gloss levels are defined by MPI as follows:

| Gloss Level | Description   | Units at 60 degree angle | Units at 80 degree angle |
|-------------|---------------|--------------------------|--------------------------|
| G1          | Matte or Flat | 0 to 5                   | 10 max                   |
| G2          | Velvet        | 0 to 10                  | 10 to 35                 |
| G3          | Eggshell      | 10 to 25                 | 10 to 35                 |
| G4          | Satin         | 20 to 35                 | 35 min                   |
| G5          | Semi-Gloss    | 35 to 70                 |                          |
| G6          | Gloss         | 70 to 85                 |                          |
| G7          | High Gloss    |                          |                          |

Gloss is tested in accordance with ASTM D523. Historically, the Government has used Flat (G1 / G2), Eggshell (G3), Semi-Gloss (G5), and Gloss (G6).

#### 1.3.11 MPI System Number

The MPI coating system number in each MPI Division found in either the MPI Architectural Painting Specification Manual or the Maintenance Repainting Manual and defined as an exterior (EXT/REX) or interior system (INT/RIN).

#### 1.3.12 Paint

SSPC Glossary; (1) Any pigmented liquid, liquefiable, or mastic composition designed for application to a substrate in a thin layer that is converted to an opaque solid film after application. Used for

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

protection, decoration, identification, or to serve some other functional purposes; (2) Application of a coating material.

#### 1.3.13 REX

MPI short term designation for an exterior coating system used in repainting projects or over existing coating systems.

#### 1.3.14 RIN

MPI short term designation for an interior coating system used in repainting projects or over existing coating systems.

### 1.4 SCHEDULING

Allow paint, polyurethane, varnish, and wood stain installations to cure prior to the installation of materials that adsorb VOCs, including carpets, textiles, unprimed gypsum wallboard, acoustical ceiling panels, .

### 1.5 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

Samples of specified materials may be taken and tested for compliance with specification requirements.

#### SD-02 Shop Drawings

Piping Identification

#### SD-03 Product Data

Coating; G

Product Data Sheets

#### SD-04 Samples

Color; G

#### SD-07 Certificates

Qualification Testing laboratory for coatings

Indoor Air Quality for Paints and Primers

Indoor Air Quality for Consolidated Latex Paints

#### SD-08 Manufacturer's Instructions

Application Instructions

Mixing

Manufacturer's Safety Data Sheets

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## SD-10 Operation and Maintenance Data

### Coatings, Data Package 1; G

#### 1.6 QUALITY ASSURANCE

##### 1.6.1 Regulatory Requirements

###### 1.6.1.1 Environmental Protection

In addition to requirements specified elsewhere for environmental protection, provide coating materials that conform to the restrictions of the local Air Pollution Control District and regional jurisdiction. Notify Contracting Officer of any paint specified herein which fails to conform.

###### 1.6.1.2 Lead Content

Do not use coatings having a lead content over 0.06 percent by weight of nonvolatile content.

###### 1.6.1.3 Chromate Content

Do not use coatings containing zinc-chromate or strontium-chromate.

###### 1.6.1.4 Asbestos Content

Provide asbestos-free materials.

###### 1.6.1.5 Mercury Content

Provide materials free of mercury or mercury compounds.

###### 1.6.1.6 Silica

Provide abrasive blast media containing no free crystalline silica.

###### 1.6.1.7 Human Carcinogens

Provide materials that do not contain ACGIH 0100 confirmed human carcinogens (A1) or suspected human carcinogens (A2).

###### 1.6.1.8 Carbon Based Fibers / Tubes

Materials must not contain carbon based fibers such as carbon nanotubes or carbon nanofibers. Intelligence Bulletin 65 ranks toxicity of carbon nanotubes on a par with asbestos.

##### 1.6.2 Coating Contractor's Qualification

Submit the name, address, telephone number, and e-mail address of the Contractor that will be performing all surface preparation and coating application. Submit evidence that key personnel have successfully performed surface preparation and application of coatings on a minimum of three similar projects within the past three years. List information by individual and include the following:

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

- a. Name of individual and proposed position for this work.
- b. Information about each previous assignment including:

Position or responsibility

Employer (if other than the Contractor)

Name of facility owner

Mailing address and telephone number of facility owner

Name of individual in facility owner's organization who can be contacted as a reference

Location, size and description of structure

Dates work was carried out

Description of work carried out on structure

#### 1.6.3 SSPC QP 1 Certification

Contractors that perform surface preparation or coating application on steel substrates must be certified by the Society for Protective Coatings (formerly Steel Structures Painting Council) (SSPC) to the requirements of SSPC QP 1 prior to Contract award, and must remain certified while accomplishing any surface preparation or coating application. If a Contractor's certification expires, the firm will not be allowed to perform any work until the certification is reissued. Requests for extension of time for any delay to the completion of the project due to an inactive certification will not be considered. Notify the Contracting Officer of any change in Contractor certification status. Notify the Contracting Officer of all scheduled and unannounced on-site audits from SSPC and furnish a copy of all audit reports.

#### 1.6.4 Approved Products List

The current MPI, "Approved Product List" which lists paint by brand, label, product name and product code as of the date of Contract award, will be used to determine compliance with the submittal requirements of this specification. The Contractor may choose to use a subsequent MPI "Approved Product List", however, only one list may be used for the entire Contract and each coating system is to be from a single manufacturer. Provide all coats on a particular substrate from a single manufacturer. No variation from the MPI Approved Products List is acceptable.

#### 1.6.5 Paints and Coatings Indoor Air Quality Certifications

Provide paint and coating products certified to meet indoor air quality requirements by MPI GPS-1-14, MPI GPS-2-14 or provide certification by other third-party programs. Provide current product certification documentation from certification body.

Provide certification of Indoor Air Quality for Paints and Primers.  
Provide certification of Indoor Air Quality for Consolidated Latex Paints.  
Submit required indoor air quality certifications in one submittal package.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### 1.6.6 Field Samples and Tests

The Contracting Officer may choose up to two coatings that have been delivered to the site to be tested at no cost to the Government. Take samples of each chosen product as specified in the paragraph SAMPLING PROCEDURE. Test each chosen product as specified in the paragraph TESTING PROCEDURE. Remove products from the job site which do not conform, and replace with new products that conform to the referenced specification. Test replacement products that failed initial testing as specified in the paragraph TESTING PROCEDURE at no cost to the Government.

##### 1.6.6.1 Sampling Procedure

Select paint at random from the products that have been delivered to the job site for sample testing. The Contractor must provide one quart samples of the selected paint materials. Take samples in the presence of the Contracting Officer, and label, and identify each sample. Provide labels in accordance with the paragraph PACKAGING, LABELING, AND STORAGE.

##### 1.6.6.2 Testing Procedure

Provide Batch Quality Conformance Testing for specified products, as defined by and performed by MPI. As an alternative to Batch Quality Conformance Testing, the Contractor may provide Qualification Testing for specified products above to the appropriate MPI product specification, using the third-party laboratory approved under the paragraph QUALIFICATION TESTING laboratory for coatings. Include the backup data and summary of the test results within the qualification testing lab report. Provide a summary listing of all the reference specification requirements and the result of each test. Clearly indicate in the summary whether the tested paint meets each test requirement. Note that Qualification Testing may take 4 to 6 weeks to perform, due to the extent of testing required.

Submit name, address, telephone number, FAX number, and e-mail address of the independent third party laboratory selected to perform testing of coating samples for compliance with specification requirements. Submit documentation that laboratory is regularly engaged in testing of paint samples for conformance with specifications, and that employees performing testing are qualified. If MPI is chosen to perform the Batch Quality Conformance testing, the above submittal information is not required, only a letter is required from the Contractor stating that MPI will perform the testing.

#### 1.7 PACKAGING, LABELING, AND STORAGE

Provide paints in sealed containers that legibly show the Contract specification number, designation name, formula or specification number, batch number, color, quantity, date of manufacture, manufacturer's formulation number, manufacturer's directions including any warnings and special precautions, and name and address of manufacturer. Furnish pigmented paints in containers not larger than 5 gallons. Store paints and thinners in accordance with the manufacturer's written directions, and as a minimum, stored off the ground, under cover, with sufficient ventilation to prevent the buildup of flammable vapors, and at temperatures between 40 to 95 degrees F. Do not store paint, polyurethane, varnish, or wood stain products with materials that have a high capacity to absorb VOC emissions. Do not store paint, polyurethane,

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

varnish, or wood stain products in occupied spaces.

## 1.8 SAFETY AND HEALTH

Comply with applicable Federal, State, and local laws and regulations. Include in the Activity Hazard Analysis the potential impact of painting operations on painting personnel and on others involved in and adjacent to the work zone.

### 1.8.1 Toxic Materials

To protect personnel from overexposure to toxic materials, conform to the most stringent guidance of:

- a. The applicable manufacturer's Safety Data Sheets (SDS) or local regulation.
- b. 29 CFR 1910.1000.
- c. ACGIH 0100, threshold limit values.
- d. The appropriate OSHA standard in 29 CFR 1910.1025 and 29 CFR 1926.62 for surface preparation on painted surfaces containing lead. Removal and disposal of coatings which contain lead is specified in Section 02 83 00 LEAD REMEDIATION. Additional guidance is given in SSPC Guide 6 and SSPC Guide 7. Refer to drawings for list of hazardous materials located on this project. Coordinate paint preparation activities with this specification section.

Submit manufacturer's Safety Data Sheets for coatings, solvents, and other potentially hazardous materials, as defined in FED-STD-313.

## 1.9 ENVIRONMENTAL REQUIREMENTS

Comply, at minimum, with manufacturer recommendations for space ventilation during and after installation. Isolate area of application from rest of building when applying high-emission paints or coatings.

### 1.9.1 Coatings

Do not apply coating when air or substrate conditions are:

- a. Less than 5 degrees F above dew point;
- b. Below 50 degrees F or over 95 degrees F, unless specifically pre-approved by the Contracting Officer and the product manufacturer. Do not, under any circumstances, violate the manufacturer's application recommendations.

### 1.9.2 Post-Application

Vacate space for as long as possible after application. Wait a minimum of 48 hours before occupying freshly painted rooms. Maintain one of the following ventilation conditions during the curing period, or for 72 hours after application:

- a. Supply 100 percent outside air 24 hours a day.
- b. Supply airflow at a rate of 6 air changes per hour, when outside

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

temperatures are between 55 degrees F and 85 degrees F and humidity is between 30 percent and 60 percent.

- c. Supply airflow at a rate of 1.5 air changes per hour, when outside air conditions are not within the range stipulated above.

## PART 2 PRODUCTS

### 2.1 MATERIALS

Conform to the coating specifications and standards referenced in PART 3. Submit Product Data Sheets for specified coatings and solvents. Provide preprinted cleaning and maintenance instructions for all coating systems. Submit Manufacturer's Instructions on Mixing: Detailed mixing instructions, minimum and maximum application temperature and humidity, pot life, and curing and drying times between coats.

### 2.2 COLOR SELECTION OF FINISH COATS

Provide colors of finish coats as indicated or specified. Allow Contracting Officer to select colors not indicated or specified. Manufacturers' names and color identification are used for the purpose of color identification only. Named products are acceptable for use only if they conform to specified requirements. Products of other manufacturers are acceptable if the colors are approximately the colors indicated and the product conforms to specified requirements.

Provide color, texture, and pattern of wall coating systems as indicated. Submit manufacturer's samples of paint colors. Cross reference color samples to color scheme as indicated. Submit color stencil codes. Tint each coat progressively darker to enable confirmation of the number of coats.

### 2.3 CONCRETE FLOOR SURFACE SEALER

Clear concrete water based, wet look, floor sealer system in conformance with MPI ASM, No. 99.

## PART 3 EXECUTION

### 3.1 PROTECTION OF AREAS AND SPACES NOT TO BE PAINTED

Prior to surface preparation and coating applications, remove, mask, or otherwise protect hardware, hardware accessories, machined surfaces, radiator covers, plates, lighting fixtures, public and private property, and other such items not to be coated that are in contact with surfaces to be coated. Following completion of painting, reinstall removed items by workmen skilled in the trades. Restore surfaces contaminated by coating materials, to original condition and repair damaged items.

### 3.2 SURFACE PREPARATION

Remove dirt, splinters, loose particles, grease, oil, disintegrated coatings, and other foreign matter and substances deleterious to coating performance as specified for each substrate before application of paint or surface treatments. Remove oil and grease prior to mechanical cleaning. Schedule cleaning so that dust and other contaminants will not fall on wet, newly painted surfaces. Spot-prime exposed ferrous metals such as nail heads on or in contact with surfaces to be painted with water-thinned

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

paints, with a suitable corrosion-inhibitive primer capable of preventing flash rusting and compatible with the coating specified for the adjacent areas. Refer to MPI ASM and MPI MRM for additional more specific substrate preparation requirements.

### 3.2.1 Substrate Repair

- a. Repair substrate surface damaged during coating removal;
- b. Sand edges of adjacent soundly-adhered existing coatings so they are tapered as smooth as practical to areas involved with coating removal; and
- c. Clean and prime the substrate as specified.

## 3.3 PREPARATION OF METAL SURFACES

### 3.3.1 New Ferrous Surfaces

- a. Ferrous Surfaces including Shop-coated Surfaces and Small Areas That Contain Rust, Mill Scale and Other Foreign Substances: Solvent clean or detergent wash in accordance with SSPC SP 1 to remove oil and grease. Where shop coat is missing or damaged, clean according to SSPC SP 2, SSPC SP 3, SSPC SP 6/NACE No.3, or SSPC SP 10/NACE No. 2. Brush-off blast remaining surface in accordance with SSPC 7/NACE No.4; Water jetting to SSPC-SP WJ-4/NACE WJ-4 may be used to remove loose coating and other loose materials. Use inhibitor as recommended by coating manufacturer to prevent premature rusting. Protect shop-coated ferrous surfaces from corrosion by treating and touching up corroded areas immediately upon detection.
- b. Surfaces With More Than 20 Percent Rust, Mill Scale, and Other Foreign Substances: Clean entire surface in accordance with SSPC SP 6/NACE No.3 / SSPC-SP WJ-3/NACE WJ-3SSPC SP 10/NACE No. 2 / SSPC-SP WJ-2/NACE WJ-2.

### 3.3.2 Final Ferrous Surface Condition:

#### 3.3.2.1 Tool Cleaned Surfaces

Comply with SSPC SP 2 and SSPC SP 3. Use as a visual reference, photographs in SSPC VIS 3 for the appearance of cleaned surfaces.

#### 3.3.2.2 Abrasive Blast Cleaned Surfaces

Comply with SSPC 7/NACE No.4, SSPC SP 6/NACE No.3, and SSPC SP 10/NACE No. 2. Use as a visual reference, photographs in SSPC VIS 1 for the appearance of cleaned surfaces.

#### 3.3.2.3 Waterjet Cleaned Surfaces

Comply with SSPC-SP WJ-1/NACE WJ-1, SSPC-SP WJ-2/NACE WJ-2, SSPC-SP WJ-3/NACE WJ-3 or SSPC-SP WJ-4/NACE WJ-4. Use as a visual reference, photographs in SSPC VIS 4/NACE VIS 7 for the appearance of cleaned surfaces.

### 3.3.3 Galvanized Surfaces

- a. New or Existing Galvanized Surfaces With Only Dirt and Zinc Oxidation



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

Products: Clean with solvent, or non-alkaline detergent solution in accordance with SSPC SP 1. Completely remove coating by brush-off abrasive blast if the galvanized metal has been passivated or stabilized. Do not "passivate" or "stabilize" new galvanized steel to be coated. If the absence of hexavalent stain inhibitors is not documented, test as described in ASTM D6386, Appendix X2, and remove by one of the methods described therein.

- b. Galvanized with Slight Coating Deterioration or with Little or No Rusting: Water jetting to SSPC-SP WJ-3/NACE WJ-3 to remove loose coating from surfaces with less than 20 percent coating deterioration and no blistering, peeling, or cracking. Use inhibitor as recommended by the coating manufacturer to prevent rusting.
- c. Galvanized With Severe Deteriorated Coating or Severe Rusting: Water jet to SSPC-SP WJ-3/NACE WJ-3 degree of cleanliness. Spot abrasive blast rusted areas as described for steel in SSPC SP 6/NACE No.3, and waterjet to SSPC-SP WJ-3/NACE WJ-3 to remove existing coating.

### 3.3.4 Non-Ferrous Metallic Surfaces

Aluminum and aluminum-alloy, lead, copper, and other nonferrous metal surfaces.

Surface Cleaning: Solvent clean in accordance with SSPC SP 1 and wash with mild non-alkaline detergent to remove dirt and water soluble contaminants.

## 3.4 PREPARATION OF CONCRETE AND CEMENTITIOUS SURFACE

### 3.4.1 Concrete and Masonry

- a. Curing: Allow concrete, stucco and masonry surfaces to cure at least 30 days before painting, and concrete slab on grade to cure at least 90 days before painting.
- b. Surface Cleaning: Remove the following deleterious substances.
  - (1) Dirt, Chalking, Grease, and Oil: Wash new surfaces with a solution composed of 1/2 cup trisodium phosphate, 1/4 cup household detergent, and 4 quarts of warm water. Then rinse thoroughly with fresh water. For large areas, water blasting may be used.
  - (2) Fungus and Mold: Wash new surfaces with a solution composed of 1/2 cup trisodium phosphate, 1/4 cup household detergent, one quart 5 percent sodium hypochlorite solution and 3 quarts of warm water. Rinse thoroughly with fresh water.
  - (3) Paint and Loose Particles: Remove by wire brushing.
  - (4) Efflorescence: Remove by scraping or wire brushing followed by washing with a 5 to 10 percent by weight aqueous solution of hydrochloric (muriatic) acid. Do not allow acid to remain on the surface for more than five minutes before rinsing with fresh water. Do not acid clean more than 4 square feet of surface, per workman, at one time.
- c. Cosmetic Repair of Minor Defects: Repair or fill mortar joints and

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

minor defects, including but not limited to spalls, in accordance with manufacturer's recommendations and prior to coating application.

- d. Allowable Moisture Content: Latex coatings may be applied to damp surfaces, but not to surfaces with droplets of water. Do not apply epoxies to damp vertical surfaces as determined by ASTM D4263 or horizontal surfaces that exceed 3 lbs of moisture per 1000 square feet in 24 hours as determined by ASTM F1869. In all cases follow manufacturer's recommendations. Allow surfaces to cure a minimum of 30 days before painting.

### 3.4.2 Gypsum Board, Plaster, and Stucco

#### 3.4.2.1 Surface Cleaning

Verify that plaster and stucco surfaces are free from loose matter and that gypsum board is dry. Remove loose dirt and dust by brushing with a soft brush, rubbing with a dry cloth, or vacuum-cleaning prior to application of the first coat material. A damp cloth or sponge may be used if paint is water-based.

#### 3.4.2.2 Repair of Minor Defects

Prior to painting, repair joints, cracks, holes, surface irregularities, and other minor defects with patching plaster or spackling compound and sand smooth.

#### 3.4.2.3 Allowable Moisture Content

Latex coatings may be applied to damp surfaces, but not surfaces with droplets of water. Do not apply epoxies to damp surfaces as determined by ASTM D4263. Verify that new plaster to be coated has a maximum moisture content of 8 percent, when measured in accordance with ASTM D4444, Method A, unless otherwise authorized. In addition to moisture content requirements, allow new plaster to age a minimum of 30 days before preparation for painting.

### 3.5 PREPARATION OF WOOD AND PLYWOOD SURFACES

#### 3.5.1 New Plywood and Wood Surfaces, Except Floors:

- a. Surface Cleaning: Clean wood surfaces of foreign matter. Verify that surfaces are free from dust and other deleterious substances and in a condition approved by the Contracting Officer prior to receiving paint or other finish. Do not use water to clean uncoated wood. Scrape to remove loose coatings. Lightly sand to roughen the entire area of previously enamel-coated wood surfaces.

#### 3.6 APPLICATION

#### 3.6.1 Coating Application

- a. Comply with applicable federal, state and local laws enacted to ensure compliance with Federal Clean Air Standards. Apply coating materials in accordance with SSPC PA 1. SSPC PA 1 methods are applicable to all substrates, except as modified herein.
- b. At the time of application, paint must show no signs of deterioration. Maintain uniform suspension of pigments during application.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

- c. Unless otherwise specified or recommended by the paint manufacturer, paint may be applied by brush, roller, or spray. Use trigger operated spray nozzles for water hoses. Use rollers for applying paints and enamels of a type designed for the coating to be applied and the surface to be coated. Wear protective clothing and respirators when applying oil-based paints or using spray equipment with any paints.
- d. Only apply paints, except water-thinned types, to surfaces that are completely free of moisture as determined by sight or touch.
- e. Thoroughly work coating materials into joints, crevices, and open spaces. Pay special attention to ensure that all edges, corners, crevices, welds, and rivets receive a film thickness equal to that of adjacent painted surfaces.
- f. Apply each coat of paint so that dry film is of uniform thickness and free from runs, drops, ridges, waves, pinholes or other voids, laps, brush marks, and variations in color, texture, and finish. Completely hide all blemishes.
- g. Touch up damaged coatings before applying subsequent coats. Broom clean and clear dust from interior areas before and during the application of coating material.
- g. Piping in Unfinished Areas: Provide primed surfaces with one coat of red alkyd gloss enamel (MPI 9) applied to a minimum dry film thickness of 1.0 mil in attic spaces, spaces above suspended ceilings, crawl spaces, pipe chases, mechanical equipment room, and spaces where walls or ceiling are not painted or not constructed of a prefinished material.h. Piping in Finished Areas: Provide primed surfaces with two coats of paint to match adjacent surfaces, except provide valves and operating accessories with one coat of red alkyd gloss enamel (MPI 9) applied to a minimum dry film thickness of 1.0 mil or two component gloss polyurethane (MPI 72) in exterior applications.i. Provide labeling on the surfaces of all feed and cross mains to show the pipe function such as "Sprinkler System", "Fire Department Connection", "Standpipe". For pipe sizes 4-inch and larger provide white painted stenciled letters and arrows, a minimum of 2 in in height and visible from at least two sides when viewed from the floor. For pipe sizes less than 4-inch, provide white painted stenciled letters and arrows, a minimum of 0.75 in in height and visible from the floor.j. All fire suppression system valves must be marked with permanent tags indicating normally open or normally closed.
- k. Drying Time: Allow time between coats, as recommended by the coating manufacturer, to permit thorough drying, but not to present topcoat adhesion problems. Provide each coat in specified condition to receive next coat.
- l. Primers, and Intermediate Coats: Do not allow primers or intermediate coats to dry more than 30 days, or longer than recommended by manufacturer, before applying subsequent coats. Follow manufacturer's recommendations for surface preparation if primers or intermediate coats are allowed to dry longer than recommended by manufacturers of subsequent coatings. Cover each preceding coat or surface completely by ensuring visually perceptible difference in shades of successive coats.
- m. Finished Surfaces: Provide finished surfaces free from runs, drops, ridges, waves, laps, brush marks, and variations in colors.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

- n. Thermosetting Paints: Apply topcoats over thermosetting paints (epoxies and urethanes) within the overcoat window recommended by the manufacturer.
- o. Floors: For nonslip surfacing on level floors, as the intermediate coat is applied, cover wet surface completely with almandite garnet, Grit No. 36, with maximum passing U.S. Standard Sieve No. 40 less than 0.5 percent. When the coating is dry, use a soft bristle broom to sweep up excess grit, which may be reused, and vacuum up remaining residue before application of the topcoat.

### 3.6.2 Mixing and Thinning of Paints

Reduce paints to proper consistency by adding fresh paint, except when thinning is mandatory to suit surface, temperature, weather conditions, application methods, or for the type of paint being used. Obtain written permission from the Contracting Officer to use thinners. Verify that the written permission includes quantities and types of thinners to use.

When thinning is allowed, thin paints immediately prior to application with not more than one pint of suitable thinner per gallon. The use of thinner does not relieve the Contractor from obtaining complete hiding, full film thickness, or required gloss. Thinning cannot cause the paint to exceed limits on volatile organic compounds. Do not mix paints of different manufacturers.

### 3.6.3 Two-Component Systems

Mix two-component systems in accordance with manufacturer's instructions. Follow recommendation by the manufacturer for any thinning of the first coat to ensure proper penetration and sealing for each type of substrate.

### 3.6.4 Coating Systems

- a. Systems by Substrates: Apply coatings that conform to the respective specifications listed in the following Tables:

| Table for Exterior Applications |                                                                        |
|---------------------------------|------------------------------------------------------------------------|
| MPI Division                    | Substrate Application                                                  |
| MPI Division 3                  | Refer to Section 07 19 00 Water Repellents                             |
| MPI Division 4                  | Refer to Section 07 19 00 Water Repellents                             |
| MPI Division 5                  | Exterior Metal, Ferrous and Non-Ferrous Paint Table                    |
| MPI Division 6                  | Exterior Wood; Dressed Lumber, Paneling, Decking, Shingles Paint Table |
| MPI Division 9                  | Exterior Stucco Paint Table                                            |

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

| Table for Exterior Applications |                                                                     |
|---------------------------------|---------------------------------------------------------------------|
| MPI Division 10                 | Exterior Cloth Coverings and Bituminous Coated Surfaces Paint Table |
| Table for Interior Applications |                                                                     |
| MPI Division                    | Substrate Application                                               |
| MPI Division 3                  | Interior Concrete Paint Table                                       |
|                                 |                                                                     |
| MPI Division 5                  | Interior Metal, Ferrous and Non-Ferrous Paint Table                 |
| MPI Division 6                  | Interior Wood Paint Table                                           |
|                                 |                                                                     |
| MPI Division 9                  | Gypsum Board, Paint Table                                           |

- b. Minimum Dry Film Thickness (DFT): Apply paints, primers, varnishes, enamels, undercoats, and other coatings to a minimum dry film thickness of 1.5 mil each coat unless specified otherwise in the Tables. Coating thickness, where specified, refers to the minimum dry film thickness.
- c. Coatings for Surfaces Not Specified Otherwise: Coat unspecified surfaces the same as surfaces having similar conditions of exposure.

### 3.7 COATING SYSTEMS FOR METAL

Apply coatings of Tables in MPI Division 5 for Exterior and Interior.

- a. Apply specified ferrous metal primer to steel surfaces on the same day that surface is cleaned, to surfaces that meet all specified surface preparation requirements at time of application.
- b. Inaccessible Surfaces: Prior to erection, use one coat of specified primer on metal surfaces that will be inaccessible after erection.
- c. Shop-primed Surfaces: Touch up exposed substrates and damaged coatings to protect from rusting prior to applying field primer.
- d. Pipes and Tubing: The semitransparent film applied to some pipes and tubing at the mill is not to be considered a shop coat. Overcoat these items with the specified ferrous-metal primer prior to application of finish coats.
- e. Exposed Nails, Screws, Fasteners, and Miscellaneous Ferrous Surfaces. On surfaces to be coated with water thinned coatings, spot prime exposed nails and other ferrous metal with latex primer MPI 107.

### 3.8 COATING SYSTEMS FOR CONCRETE AND CEMENTITIOUS SUBSTRATES

Apply coatings of Tables in MPI Division 3, 4 and 9 for Exterior and Interior.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 3.9 COATING SYSTEMS FOR WOOD AND PLYWOOD

- a. Apply coatings of Tables in MPI Division 6 for Exterior and Interior.
- b. Prior to erection, apply two coats of specified primer to treat and prime wood and plywood surfaces which will be inaccessible after erection.
- c. Apply stains in accordance with manufacturer's printed instructions.

### 3.10 PIPING IDENTIFICATION

Piping Identification, Including Surfaces In Concealed Spaces: Provide in accordance with MIL-STD-101ASME A13.1. Place stenciling in clearly visible locations. On piping not covered by MIL-STD-101ASME A13.1, stencil approved names or code letters, in letters a minimum of 1/2 inch high for piping and a minimum of 2 inches high elsewhere. Stencil arrow-shaped markings on piping to indicate direction of flow using black stencil paint.

### 3.11 FIRE WALL MARKING AND IDENTIFICATION

For other than existing assemblies where there is an accessible concealed floor/ceiling or attic space, fire walls, fire barriers, smoke barriers, and smoke partitions must be permanently identified with an ink roller system within 15 feet of end of each wall, and at intervals not exceeding 30 feet measured horizontally along the wall. Include lettering not less than 3 inches in height with a minimum of 3/8 inch stroke in contrasting color. Identify the fire resistive rating as applicable.

### 3.12 INSPECTION AND ACCEPTANCE

In addition to meeting previously specified requirements, demonstrate mobility of moving components, including swinging and sliding doors, cabinets, and windows with operable sash, for inspection by the Contracting Officer. Perform this demonstration after appropriate curing and drying times of coatings have elapsed and prior to invoicing for final payment.

### 3.13 WASTE MANAGEMENT

As specified in the Waste Management Plan and as follows. Do not use kerosene or any such organic solvents to clean up water based paints. Properly dispose of paints or solvents in designated containers. Close and seal partially used containers of paint to maintain quality as necessary for reuse. Store in protected, well-ventilated, fire-safe area at moderate temperature. Place materials defined as hazardous or toxic waste in designated containers. Coordinate with manufacturer for take-back program. Set aside scrap to be returned to manufacturer for recycling into new product. When such a service is not available, contact local recyclers to reclaim the materials. Set aside extra paint for future color matches or reuse by the Government. Where local options exist for leftover paint recycling, collect all waste paint by type and provide for delivery to recycling or collection facility for reuse by local organizations.

### 3.14 PAINT TABLES

All DFT's are minimum values. Use only materials with a MPI GPS-1-14

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

green check mark having a minimum MPI "Environmentally Friendly" E2 or E3 rating based on VOC (EPA Method 24) content levels. Acceptable products are listed in the MPI Green Approved Products List, available at <http://www.specifygreen.com/APL/ProductIdxByMPInum.asp>.

### 3.14.1 Exterior Paint Tables

#### 3.14.1.1 MPI Division 3: Exterior Concrete Paint Table

A. Concrete; No Table Refer to Section 07 19 00 Water Repellents.

(1) New concrete; vertical surfaces, including undersides of balconies and soffits but excluding tops of slabs

| Latex                                                                                  |                              |        |              |         |            |
|----------------------------------------------------------------------------------------|------------------------------|--------|--------------|---------|------------|
| New and uncoated existing                                                              | Existing, previously painted | Primer | Intermediate | Topcoat | System DFT |
| MPI EXT 3.1A-G1 (Flat)                                                                 | MPI REX 3.1A-G1 (Flat)       | MPI 3  | MPI 10       | MPI 10  | 3.5 mils   |
| MPI EXT 3.1A-G2 (Velvet)                                                               | MPI REX 3.1A-G2 (Velvet)     | MPI 3  | MPI 214      | MPI 214 | 3.5 mils   |
| MPI EXT 3.1A-G5 (Semigloss)                                                            | MPI REX 3.1A-G5 (Semigloss)  | MPI 3  | MPI 11       | MPI 11  | 3.5 mils   |
| MPI EXT 3.1A-G6 (Gloss)                                                                | MPI REX 3.1A-G6 (Gloss)      | MPI 3  | MPI 119      | MPI 119 | 3.5 mils   |
| Primer as recommended by manufacturer.<br>Topcoat: Coating to match adjacent surfaces. |                              |        |              |         |            |

#### 3.14.1.2 MPI Division 5: Exterior Metal, Ferrous and Non-Ferrous Paint Table

A. Steel / Ferrous Surfaces

(1) New Steel that has been blast-cleaned to SSPC SP 6/NACE No.3

| Option 1: Alkyd             |                             |        |              |         |            |
|-----------------------------|-----------------------------|--------|--------------|---------|------------|
| New                         | Existing, uncoated          | Primer | Intermediate | Topcoat | System DFT |
| MPI EXT 5.1D-G5 (Semigloss) | MPI REX 5.1D-G5 (Semigloss) | MPI 79 | MPI 94       | MPI 94  | 5.25 mils  |
| MPI EXT 5.1D-G6 (Gloss)     | MPI REX 5.1D-G6 (Gloss)     | MPI 79 | MPI 9        | MPI 9   | 5.25 mils  |

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

Topcoat: Coating to match adjacent surfaces.

| Option 2: Pigmented Polyurethane             |         |              |         |            |
|----------------------------------------------|---------|--------------|---------|------------|
| Existing, previously coated with epoxy       | Primer  | Intermediate | Topcoat | System DFT |
| MPI REX 5.1H-G6 (Gloss)                      | MPI 101 | MPI 108      | MPI 72  | 8.5 mils   |
| Topcoat: Coating to match adjacent surfaces. |         |              |         |            |

B. Exterior Galvanized Surfaces (only where designated as "painted" on Drawings.

(1) New Galvanized surfaces

| Waterborne Primer / Latex                    |         |              |         |            |
|----------------------------------------------|---------|--------------|---------|------------|
| New Galvanized Surfaces                      | Primer  | Intermediate | Topcoat | System DFT |
| MPI EXT 5.3H-G1 (Flat)                       | MPI 134 | MPI 10       | MPI 10  | 4.5 mils   |
| EXT 5.3H-G5 (Semigloss)                      | MPI 134 | MPI 11       | MPI 11  | 4.5 mils   |
| MPI EXT 5.3H-G6 (Gloss)                      | MPI 134 | MPI 119      | MPI 119 | 4.5 mils   |
| Topcoat: Coating to match adjacent surfaces. |         |              |         |            |

| Waterborne Primer / Waterborne Light Industrial Coating |         |              |         |            |
|---------------------------------------------------------|---------|--------------|---------|------------|
| New Galvanized Surfaces                                 | Primer  | Intermediate | Topcoat | System DFT |
| MPI EXT 5.3J-G5 (Semigloss)                             | MPI 134 | MPI 163      | MPI 163 | 4.5 mils   |
| MPI EXT 5.3J-G6 (Gloss)                                 | MPI 134 | MPI 164      | MPI 164 | 4.5 mils   |
| Topcoat: Coating to match adjacent surfaces.            |         |              |         |            |

| Epoxy Primer / Waterborne Light Industrial Coating |  |  |  |  |
|----------------------------------------------------|--|--|--|--|
|----------------------------------------------------|--|--|--|--|



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

| New Galvanized Surfaces                      | Primer  | Intermediate | Topcoat | System DFT |
|----------------------------------------------|---------|--------------|---------|------------|
| MPI EXT 5.3K-G5 (Semigloss)                  | MPI 101 | MPI 163      | MPI 163 | 5 mils     |
| MPI EXT 5.3K-G6 (Gloss)                      | MPI 101 | MPI 164      | MPI 164 | 5 mils     |
| Topcoat: Coating to match adjacent surfaces. |         |              |         |            |

| Option 1: Pigmented Polyurethane             |         |              |         |            |
|----------------------------------------------|---------|--------------|---------|------------|
| New Galvanized Surfaces                      | Primer  | Intermediate | Topcoat | System DFT |
| MPI EXT 5.3L-G6 (Gloss)                      | MPI 101 | N/A          | MPI 72  | 5 mils     |
| Topcoat: Coating to match adjacent surfaces. |         |              |         |            |

C. Exterior Surfaces, Other Metals (Non-Ferrous) - Only where designated as "Painted" on Drawings.

(1) Aluminum, aluminum alloy and other miscellaneous non-ferrous metal items not otherwise specified except hot metal surfaces, roof surfaces, and new prefinished equipment

| Waterborne Light Industrial Coating          |        |              |         |            |
|----------------------------------------------|--------|--------------|---------|------------|
| New Galvanized Surfaces                      | Primer | Intermediate | Topcoat | System DFT |
|                                              |        |              |         |            |
| MPI EXT 5.4F-G5 (Semigloss)                  | MPI 95 | MPI 163      | MPI 163 | 5 mils     |
| MPI EXT 5.4F-G6 (Gloss)                      | MPI 95 | MPI 164      | MPI 164 | 5 mils     |
| Topcoat: Coating to match adjacent surfaces. |        |              |         |            |

(2) Surfaces adjacent to painted surfaces; Mechanical, Electrical, Fire extinguishing sprinkler systems including valves, conduit, hangers, supports, exposed copper piping, and miscellaneous metal items not otherwise specified except floors, hot metal surfaces, and new prefinished equipment

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

| Waterborne Light Industrial Coating                                                    |        |              |         |            |
|----------------------------------------------------------------------------------------|--------|--------------|---------|------------|
| New                                                                                    | Primer | Intermediate | Topcoat | System DFT |
| MPI EXT 5.1C-G3(Eggshell)                                                              | MPI 79 | MPI 161      | MPI 161 | 5 mils     |
| MPI EXT 5.1C-G5(Semigloss)                                                             | MPI 79 | MPI 163      | MPI 163 | 5 mils     |
| MPI EXT 5.1C-G6(Gloss)                                                                 | MPI 79 | MPI 164      | MPI 164 | 5 mils     |
| Primer as recommended by manufacturer.<br>Topcoat: Coating to match adjacent surfaces. |        |              |         |            |

### 3.14.2 Interior Paint Tables

#### 3.14.2.1 MPI Division 3: Interior Concrete Paint Table

##### A. New Concrete, vertical surfaces, not specified otherwise

| Institutional Low Odor / Low VOC Latex       |                                 |         |              |         |            |
|----------------------------------------------|---------------------------------|---------|--------------|---------|------------|
| New, uncoated<br>Existing                    | Existing,<br>previously painted | Primer  | Intermediate | Topcoat | System DFT |
| MPI INT 3.1M-G2<br>(Flat)                    | MPI RIN 3.1L-G2<br>(Flat)       | MPI 149 | MPI 144      | MPI 144 | 4 mils     |
| MPI INT 3.1M-G3<br>(Eggshell)                | MPI RIN 3.1L-G3<br>(Eggshell)   | MPI 149 | MPI 145      | MPI 145 | 4 mils     |
| MPI INT 3.1M-G4<br>(satin)                   | MPI RIN 3.1L-G4                 | MPI 149 | MPI 146      | MPI 146 | 4 mils     |
| MPI INT 3.1M-G5<br>(Semigloss)               | MPI RIN 3.1L-G5<br>(Semigloss)  | MPI 149 | MPI 147      | MPI 147 | 4 mils     |
| Topcoat: Coating to match adjacent surfaces. |                                 |         |              |         |            |

##### E. New concrete floors:

| Epoxy                      |                                 |        |              |         |            |
|----------------------------|---------------------------------|--------|--------------|---------|------------|
| New, uncoated<br>Existing  | Existing,<br>previously painted | Primer | Intermediate | Topcoat | System DFT |
| MPI INT 3.2C-G6<br>(Gloss) | MPI RIN 3.2C-G6<br>(Gloss)      | MPI 77 | MPI 77       | MPI 77  | 5 mils     |

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

Note: Primer may be reduced for penetration per manufacturer's instructions.

### 3.14.2.2 MPI Division 4: Interior Concrete Masonry Units Paint Table

#### A. New Concrete Masonry

| Institutional Low Odor / Low VOC Latex |        |        |              |         |            |
|----------------------------------------|--------|--------|--------------|---------|------------|
| New, uncoated<br>Existing              | Filler | Primer | Intermediate | Topcoat | System DFT |
| MPI INT 4.2E-G2<br>(Flat)              | MPI 4  | N/A    | MPI 144      | MPI 144 | 4 mils     |
| MPI INT 4.2E-G3<br>(Eggshell)          | MPI 4  | N/A    | MPI 145      | MPI 145 | 4 mils     |
| MPI INT 4.2E-G4<br>(Satin)             | MPI 4  | N/A    | MPI 146      | MPI 146 | 4 mils     |
| MPI INT 4.2E-G5<br>(Semigloss)         | MPI 4  | N/A    | MPI 147      | MPI 147 | 4 mils     |
| Fill all holes in masonry surface      |        |        |              |         |            |

### 3.14.2.3 MPI Division 9: Interior Plaster, Gypsum Board, Textured Surfaces Paint Table - Only where designated as "Painted" on Drawings.

#### A. Interior New and Existing, previously painted not otherwise specified

| Latex                                        |                                 |        |              |         |            |
|----------------------------------------------|---------------------------------|--------|--------------|---------|------------|
| New                                          | Existing,<br>previously painted | Primer | Intermediate | Topcoat | System DFT |
| MPI INT 9.2A-G2<br>(Flat)                    | RIN 9.2A-G2 (Flat)              | MPI 50 | MPI 44       | MPI 44  | 4 mils     |
| MPI INT 9.2A-G3<br>(Eggshell)                | RIN 9.2A-G3<br>(Eggshell)       | MPI 50 | MPI 52       | MPI 52  | 4 mils     |
| MPI INT 9.2A-G5<br>(Semigloss)               | RIN 9.2A-G5<br>(Semigloss)      | MPI 50 | MPI 54       | MPI 54  | 4 mils     |
| Topcoat: Coating to match adjacent surfaces. |                                 |        |              |         |            |

High Performance Architectural Latex - High Traffic Areas

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

| New                                          | Existing,<br>previously painted | Primer | Intermediate | Topcoat | System DFT |
|----------------------------------------------|---------------------------------|--------|--------------|---------|------------|
| MPI INT 9.2B-G2<br>(Flat)                    | MPI RIN 9.2B-G2<br>(Flat)       | MPI 50 | MPI 138      | MPI 138 | 4 mils     |
| MPI INT 9.2B-G3<br>(Eggshell)                | MPI RIN 9.2B-G3<br>(Eggshell)   | MPI 50 | MPI 139      | MPI 139 | 4 mils     |
| MPI INT 9.2B-G5<br>(Semigloss)               | MPI RIN 9.2B-G5<br>(Semigloss)  | MPI 50 | MPI 141      | MPI 141 | 4 mils     |
| Topcoat: Coating to match adjacent surfaces. |                                 |        |              |         |            |

-- End of Section --

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## SECTION 09 96 00

## HIGH-PERFORMANCE COATINGS

**11/14**

## PART 1 GENERAL

## 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

## MASTER PAINTERS INSTITUTE (MPI)

MPI ASM (2019) Architectural Painting  
Specification Manual

## SOCIETY FOR PROTECTIVE COATINGS (SSPC)

SSPC 7/NACE No.4 (2007) Brush-Off Blast Cleaning

## 1.2 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

## SD-01 Preconstruction Submittals

Equipment List; G

## SD-03 Product Data

Epoxy Coatings; FIO

Polyurethane Coatings; FIO

## SD-04 Samples

Color Chips; G

## SD-07 Certificates

Epoxy Coatings; G

Polyurethane Coatings; G

Manufacturer's Printed Instructions; FIO

## 1.3 QUALITY CONTROL

Comply with Master Painters Institute (MPI) Standards indicated and listed in "MPI Approved Products List." Comply with the requirements in "MPI Architectural Painting Specification Manual" before any project is started.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

Submit an equipment list consisting of a list of proposed equipment to be used in performance of construction work.

Submit three color chips 3-inch by 4-inch or manufacture's pull-down of each finish color and gloss as scheduled.

#### 1.4 DELIVERY, STORAGE, AND HANDLING

Deliver special coating materials to the project in their original containers bearing manufacturer's name, descriptive label, and coating formulations. Provide new and unopened containers.

Store special coating materials in tightly closed containers in a covered, well-ventilated area where they are not exposed to excessive heat, fumes, sparks, flame, or direct sunlight. Protect water-based coatings against freezing.

Store solvents, thinners, and equipment cleaners with the same care as the coating materials with ambient temperatures continuously maintained at a minimum 45 degrees F.

### PART 2 PRODUCTS

#### 2.1 PERFORMANCE REQUIREMENTS

Submit manufacturer's catalog data including manufacturer's name and identification. Include detailed data analysis of each special coating material required for the project, with all the coating constituents measured as percentages of the total weight of the coating. Also provide manufacturer's data concerning application, thinning, and average coverage per gallon

#### 2.2 MATERIALS

##### 2.2.1 Epoxy Coatings

Conform to MPI ASM, No. 116 for epoxy coatings and epoxy block filler, as modified.

Resins for finish coats are based on a polyamide-cured, epoxy-resin material. Apply finish coats with a dry-film thickness of not less than 4 mils per coat. Finish color and gloss are as indicated.

##### 2.2.1.1 Concrete Surface Coatings

Apply a epoxy coating system in conformance with MPI ASM, No. 77 for vertical concrete surfaces. Apply an epoxy slip-resistant deck coating system in conformance with MPI ASM, No. 82. Apply a prime coat to fill concrete surface pores with a total dry-film thickness of not less than 2 mils.

##### 2.2.1.2 Masonry Surfaces Coatings

Apply a Water-Based Epoxy Coating System in conformance with MPI ASM, No. 115 . Apply a block filler to fill surface pores with a total dry-film thickness of not less than 7 mils.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### 2.2.1.3 Ferrous and Galvanized Metal Surface Coatings

Coatings on ferrous and galvanized metal surfaces consist of a prime coat and not less than two finish coats. Comply with MPI ASM, No. 101 for an epoxy zinc primer with a metallic-zinc pigment for the substrate to be coated and the end use of the coated surface. Ensure resin solids and zinc pigment are not less than 80 percent of the total weight of the coating material. Apply prime coat with a total dry-film thickness of not less than 4 mils. Provide an epoxy-based finished coat as specified.

#### 2.2.2 Polyurethane Coatings

Ensure polyurethane coatings conform to MPI ASM for each substrate indicated.

Resins for finish coats are based on a two-part, prepolymer, catalytic-cured, polyurethane material. Apply catalytic-cured coatings with a total dry-film thickness of not less than 10 mils per coat. Indicate finish color and gloss on the schedules.

##### 2.2.2.1 Concrete Surface Coatings

Apply a polyurethane, pigmented coating system in conformance with MPI ASM, No. 72 and MPI ASM, No. 80. Ensure the prime coat fills surface pores with a total dry-film thickness of not less than 2 mils. Finish coats are polyurethane-based material as specified.

##### 2.2.2.2 Masonry Surface Coatings

Apply a polyurethane, clear, two-component coating system in conformance with MPI ASM, No. 78. Apply block filler to fill surface pores with a total dry-film thickness of not less than 7 mils. Finish coats are polyurethane-based material as specified.

Clear concrete floor sealer system in conformance with MPI ASM, No. 99.

##### 2.2.2.3 Ferrous and Galvanized Metal Surface Coatings

Apply a polyurethane, pigmented coating system in conformance with MPI ASM, No. 72, MPI ASM, No. 77, and MPI ASM, No. 101. Apply a prime coat with a dry-film thickness of not less than 2 mils. Finish coats are polyurethane-based material as specified.

### PART 3 EXECUTION

#### 3.1 PREPARATION

##### 3.1.1 Surface Preparation

Protect adjacent materials and equipment against damage from spillage, dripping, and spatter of coating materials. Leave clean building materials and equipment with all damaged surfaces corrected. Provide "WET PAINT" signs to indicate newly painted surfaces.

Protect work of other trades against damage from coating operation. Correct damage by cleaning, repairing, replacing, and recoating, as approved by the Contracting Officer, and leave in an undamaged condition. At completion of construction activities of other trades, touch up and restore damaged or defaced coated surfaces.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

Provide forced ventilation for interior spaces during application and drying of coatings to prevent the buildup of toxic or explosive concentrations of solvent vapors.

Provide fire extinguishers of the required quantity and correct type to combat flammable liquid fires.

Dispose of rags that are used to wipe up coating materials, solvents, and thinners by drenching with water and placing them in a covered metal container

### 3.1.2 Cleaning

At end of each workday, remove rubbish, empty cans, rags, and other discarded materials from Project site.

After completing coating application, clean spattered surfaces. Remove spattered coatings by washing, scraping, or other methods. Do not scratch or damage adjacent finished surfaces.

Clean application equipment promptly and thoroughly with a suitable solvent after each use and stored in a clean, covered, well-ventilated container.

#### 3.1.2.1 Protection

- A. PROTECTION: Protect work of other trades, whether to be coated or not, against damage by coating work. Correct any damage by cleaning, repairing or replacing, and refinishing, as acceptable to Architect. Provide signs, barricades, etc. as required to protect new Textured Acrylic Coating work from damage by others (see Section 1.07-B of this Division for protection times)
- B. AFTER COMPLETION of work of all other trades, remove all protection materials (including other trades' if provided by them) and clean/touch up as necessary to restore coating work to new and unblemished condition.

### 3.1.3 Concrete Surfaces

Conform to MPI ASM for substrates indicated. Remove plates, machined surfaces, and similar items already in place that are not to be coated. Provide surface-applied protection before surface preparation and coating where removal is impractical or impossible. After completing coating operations, reinstall items that were removed.

Clean dirt, oil, grease, and incompatible paints from substrates to ensure bonding. Coordination of shop-applied prime coats with high-performance coatings is critical. Remove incompatible primers. Reprime substrate with compatible primers as required to produce coating systems indicated.

#### 3.1.3.1 Concrete Substrates

Remove release agents, curing compounds, efflorescence, and chalk. Maximum allowable moisture content of concrete is 12 percent. Measure moisture content with an electronic moisture meter.



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

Clean surfaces with pressurized water. Use pressure range of 1500 to 4000 psi at 6 inch to 12 inch.

Comply with SSPC 7/NACE No.4 (NACE No. 4), "Brush-Off Blast Cleaning" for abrasive cleaning.

#### 3.1.3.2 Clay Masonry Substrates

Remove efflorescence and chalk. Do not coat surfaces if moisture content or alkalinity of surfaces exceeds that permitted in manufacturer's written instructions.

Clean surfaces with pressurized water. Use pressure range of 100 to 600 psi at 6 inch to 12 inch.

#### 3.1.3.3 Steel Substrates

Remove rust and loose mill scale. Clean using methods recommended in writing by coating manufacturer. Conform to SSPC 7/NACE No.4 for blast cleaning.

#### 3.1.3.4 Galvanized-Metal Substrates

Remove grease and oil residue from galvanized sheet metal fabricated from coil stock by mechanical methods to produce clean, lightly etched surfaces that promote adhesion of subsequently applied coatings.

#### 3.1.4 Coating Material Preparation

Mix and prepare coating materials in accordance with the coating manufacturer's printed instructions for applying the particular material and coat. Keep materials which are not in actual use in closed containers.

Coating materials that have been mixed with an automatic shaker are allowed to stand to let air bubbles escape, then given a final hand mixing before application. Stir materials so as to produce a mixture of uniform density. Stir at frequent intervals during application to prevent skinning. Do not stir film which may form on the surface of the material. Remove film and strain, if necessary.

##### 3.1.4.1 Thinning

Thinning is done in accordance with coating manufacturer's printed directions for the particular material and coat.

##### 3.1.4.2 Tinting

Ensure prime and intermediate coats of paint are slightly different tints from the finish coat to facilitate identification of each coat. Tinting is done by the coating manufacturer and clearly identified as to color and coat.

#### 3.2 APPLICATION

Do not perform exterior painting in damp or rainy weather. Interior painting is not allowed until the building is enclosed and has thoroughly dried out. Painting is not allowed below 50 degrees F or above 95 degrees F. Apply paint in accordance with the coating manufacturer's recommendations, and as specified.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

Ensure coating application is done by skilled applicators. Apply coatings to clean and properly prepared surfaces. Apply coatings with clean, high-quality application equipment. Allow sufficient time between coats to ensure complete drying and curing. Sand and dust surfaces between coatings, as required, to produce a surface free of visible defects. Lightly sand high gloss coatings and clear finishes between coats to ensure bond of following coats.

Apply coats to the surfaces in an even film. Cloudiness, spotting, holidays, laps, application marks, runs, sags, and other similar surface imperfections are not acceptable. Remove defective coating applications and re-coat as directed.

Ensure coating lines such as wainscots are sharp, true, and well-defined. Tape may be used to establish coating lines, providing tape is removed before ragging or sawtooth edges form.

Ensure surfaces, including edges, corners, crevices, welds, and other similar changes in surface plane, meet the dry-film thickness not less than specified.

### 3.2.1 Brush Application

Use clean, proper size brushes for high-quality application of the specified coating materials. Brush out slow-dry coatings. Brush out quick-dry coatings only enough to spread out evenly.

### 3.2.2 Roller Application

Use clean roller covers of the proper nap length, nap texture, and material for high-quality application of the specified coating materials.

Ensure roller application is equivalent in all respects to the same coats applied by high-quality brush application.

### 3.2.3 Spray Application

Do not allow spray application of coatings. Spray application equipment is limited to airless-spray equipment and electrostatic-spray equipment. Ensure equipment is clean and operated by workmen skilled in high quality application of coating materials.

Cover surfaces adjacent to sprayed areas to prevent damage from overspray, coating rebound, and spray drift.

### APPLICATION:

- A. After proper preparation of various wall assemblies as described in section 3.1. Apply coating materials in accordance with the manufacturer's instructions and recommendations as required to achieve the appearance of the approved samples and achieve performance standards as specified herein. Coating materials shall be used as a system that may include primers or under coatings as required by the manufacturer's installation directions.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 3.3 FIELD QUALITY CONTROL

#### 3.3.1 Field Test

Government may take dry-film tests from time to time on finished surfaces. Apply additional coatings to surfaces where there is less than the minimum specified dry-film thickness.

#### 3.3.2 Repairing

Remove damaged and unacceptable portions of completed work and replace with new work to match adjacent surfaces at no additional cost to the Government.

-- End of Section --

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## SECTION 10 11 00

### VISUAL DISPLAY UNITS

08/20

#### PART 1 GENERAL

##### 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

##### ASTM INTERNATIONAL (ASTM)

|           |                                                                                                              |
|-----------|--------------------------------------------------------------------------------------------------------------|
| ASTM B221 | (2021) Standard Specification for Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes |
|-----------|--------------------------------------------------------------------------------------------------------------|

##### 1.2 DEFINITIONS

The term visual display unit when used herein includes presentation boards, markerboards, tackboards, board cases, display track systems, horizontal sliding units, copyboards, and interactive whiteboards; submit manufacturer's descriptive data and catalog cuts plus manufacturer's installation instructions, and cleaning and maintenance instructions. Provide visual display units from manufacturer's standard product line. Submit certificate of compliance signed by Contractor attesting that visual display units conform to the requirements specified.

##### 1.3 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Placement Schedule; G

SD-03 Product Data

Visual Display Unit; G

SD-04 Samples

Aluminum; G

Hardwood; G

Porcelain Enamel; G

SD-08 Manufacturer's Instructions

Manufacturer's Cleaning Instructions

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## Manufacturer's Printed Installation Instructions

### SD-10 Operation and Maintenance Data

#### Visual Display Units

#### 1.4 DELIVERY, STORAGE, AND HANDLING

Deliver materials to the building site in the manufacturer's original unopened containers and store them in a clean dry area with temperature maintained above 50 degrees F. Stack materials according to manufacturer's recommendations. Allow visual display units to acclimate to the building temperature for 24 hours prior to installation.

#### 1.5 WARRANTY

Provide manufacturer's warranty to repair or replace defective materials and workmanship for period of one year from date of final acceptance of the work.

### PART 2 PRODUCTS

#### 2.1 MATERIALS

For each type, submit a section of core material and backing showing the lamination of porcelain enamel coating on steel, colored cork, natural cork, woven fabric, non-woven fabric, or vinyl wall covering, as applicable. Submit a sample of hardwood, plastic laminate finish, or glass type, as applicable. Provide minimum 4 by 4 inch samples, or larger, showing range of color.

Submit manufacturers' descriptive product data for visual display unit indicated. Include manufacturers' literature, finishes, profiles and thicknesses of materials.

Submit manufacturers' operations and maintenance data for visual display unit in accordance with Section 01 78 23 OPERATIONS AND MAINTENANCE DATA.

##### 2.1.1 Porcelain Enamel

Provide markerboard writing surface composed of porcelain enamel fused to a nominal 28 gauge thick steel, laminated to a minimum 1/4 inch thick core material with a steel or foil backing sheet. Writing surface must be capable of supporting paper by means of magnets. Markerboard surface for display track system may be a powder paint dry erase surface adhered to a nominal 18 gauge thick steel.

##### 2.1.2 Aluminum

Provide a minimum 0.06 inch thick, 6063-T5 or 6063-T6 aluminum alloy frame extrusion conforming to ASTM B221. Exposed aluminum must have anodized, satin finish. Use straight, single lengths wherever possible and keep joints to a minimum. Provide mitered corners with a hairline closure. Submit sections of frame, map rail, and marker rail, and two map hooks.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## 2.2 MARKERBOARD

### 2.2.1 Porcelain Markerboard

Provide a factory assembled markerboard with a porcelain enamel, magnetic writing surface. Unit to be comprised of one piece, without joints whenever possible. When markerboard dimensions require delivery in separate sections, components must be prefit at the factory, disassembled for delivery and jointed at the site. Provide aluminum frame with marker rail and extending the full length of the markerboard inches long. The markerboard includes a map rail with a tackable insert extending the full length of the markerboard, map hooks and clips for holding sheets of paper. Provide two map hooks for each 4 feet of map rail. Dry erase markings must be removable with a felt eraser or dry cloth without ghosting. Supply each unit with an eraser and four different color compatible dry erase markers.

## PART 3 EXECUTION

### 3.1 PLACEMENT SCHEDULE

Location, size and mounting height of visual display units as shown on the drawings. Boards shall be installed 6'-0" above finish floor to top of display board frame. Mounting height is defined as distance from finished floor to top of the visual display unit frame.

### 3.2 INSTALLATION

Do not install items that show visual evidence of biological growth. Perform installation and assembly in accordance with manufacturer's printed installation instructions. Use concealed fasteners. Attach visual display units to the walls with suitable devices to anchor each unit. Furnish and install trim items, accessories and miscellaneous items in total, including but not limited to hardware, grounds, clips, backing materials, adhesives, brackets, and anchorages incidental to or necessary for a sound, secure, complete and finished installation. Do not initiate installation until completion of room painting and finishing operations. Install visual display units in locations and at mounting heights indicated. Install visual display units level and plumb, and if applicable align doors and adjust hardware. Repair or replace damaged units as directed by the Contracting Officer.

### 3.3 CLEANING

Clean writing surfaces in accordance with manufacturer's cleaning instructions.

-- End of Section --

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## SECTION 10 14 00.10

### EXTERIOR SIGNAGE 08/17, CHG 1: 11/18

#### PART 1 GENERAL

##### 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

#### AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI Z97.1 (2015) Safety Glazing Materials Used in Buildings - Safety Performance Specifications and Methods of Test

#### AMERICAN WELDING SOCIETY (AWS)

AWS C1.1M/C1.1 (2019) Recommended Practices for Resistance Welding

AWS D1.1/D1.1M (2020; Errata 1 2021) Structural Welding Code - Steel

AWS D1.2/D1.2M (2014; Errata 1 2014; Errata 2 2020) Structural Welding Code - Aluminum

#### ASTM INTERNATIONAL (ASTM)

ASTM A36/A36M (2019) Standard Specification for Carbon Structural Steel

ASTM A123/A123M (2017) Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products

ASTM A653/A653M (2023) Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process

ASTM A924/A924M (2022a) Standard Specification for General Requirements for Steel Sheet, Metallic-Coated by the Hot-Dip Process

ASTM A1011/A1011M (2023) Standard Specification for Steel Sheet and Strip, Hot-Rolled, Carbon, Structural, High-Strength Low-Alloy, High-Strength Low-Alloy with Improved Formability, and Ultra-High Strength

ASTM B26/B26M (2018; E 2018) Standard Specification for Aluminum-Alloy Sand Castings

ASTM B108/B108M (2019) Standard Specification for

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### Aluminum-Alloy Permanent Mold Castings

|            |                                                                                                              |
|------------|--------------------------------------------------------------------------------------------------------------|
| ASTM B209  | (2014) Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate                                |
| ASTM B221  | (2021) Standard Specification for Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes |
| ASTM D3841 | (2016) Standard Specification for Glass Fiber-Reinforced Polyester Plastic Panels                            |
| ASTM E84   | (2023) Standard Test Method for Surface Burning Characteristics of Building Materials                        |

#### NATIONAL ASSOCIATION OF ARCHITECTURAL METAL MANUFACTURERS (NAAMM)

|               |                              |
|---------------|------------------------------|
| NAAMM AMP 500 | (2006) Metal Finishes Manual |
|---------------|------------------------------|

#### SOCIETY OF AUTOMOTIVE ENGINEERS INTERNATIONAL (SAE)

|             |                                                                                 |
|-------------|---------------------------------------------------------------------------------|
| SAE AMS3611 | (2011; Rev E; Stabilized (S) 2011) Plastic Sheet, Polycarbonate General Purpose |
|-------------|---------------------------------------------------------------------------------|

## 1.2 GENERAL REQUIREMENTS

All exterior signage must be provided by a single manufacturer. Exterior signage must be of the design, detail, sizes, types, and message content shown on the drawings, must conform to the requirements specified, and must be provided at the locations indicated. Submit exterior signage schedule in electronic media with spread sheet format. Spread sheet must include sign location, sign type, and message. Signs must be complete with lettering, framing as detailed, and related components for a complete installation. Each sample must consist of a complete sign panel with letters and symbols. Samples may be installed in the work, provided each sample is identified and location recorded. Submit three color samples for each material requiring color and 12 inch square sample of sign face color sample.

### 1.2.1 Wind Load Requirements

Exterior signage must be designed to withstand windloads indicated on Structural Drawings. Submit design analysis and supporting calculations performed in support of specified signage.

### 1.2.2 Character Proportions and Heights

Letters and numbers on indicated signs for handicapped-accessible buildings must have a width-to-height ratio between 3:5 and 1:1 and a stroke-width-to-height ratio between 1:5 and 1:10. Characters and numbers on indicated signs must be sized according to the viewing distance from which they are to be read. The minimum height is measured using an upper case letter "X". Lower case characters are permitted.

## 1.3 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Approved Detail Drawings; G

SD-03 Product Data

Modular Exterior Signage System

Installation

Exterior Signage; G

Wind Load Requirements

SD-04 Samples

Exterior Signage; G

SD-10 Operation and Maintenance Data

Protection and Cleaning; G

1.4 QUALIFICATIONS

Signs, plaques, and dimensional letters must be the standard product of a manufacturer regularly engaged in the manufacture of the products. Items of equipment must essentially duplicate equipment that has been in satisfactory use at least 2 years prior to bid opening.

1.5 DELIVERY AND STORAGE

Materials must be wrapped for shipment and storage, delivered to the jobsite in manufacturer's original packaging, and stored in a clean, dry area in accordance with manufacturer's instructions.

1.6 WARRANTY

Manufacturer's standard performance guarantees or warranties that extend beyond a one year period must be provided.

PART 2 PRODUCTS

2.1 MODULAR EXTERIOR SIGNAGE SYSTEM

Exterior signage must consist of a system of coordinated directional, identification, and regulatory type signs located where shown. Dimensions, details, materials, message content, and design of signage must be as shown. Submit manufacturer's descriptive data and catalog cuts.

2.1.1 Free-Standing Base Mount Pylon/Monolith Type Signs

2.1.1.1 Framing

Interior framing must consist of aluminum or galvanized steel tube columns welded to companion plates. Perimeter framing must consist of aluminum or

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

steel angle framing welded to the post and plate system as designed. Framing members must be designed to permit. Mounting must be provided as shown. Framing members of steel must be finished with semi-gloss baked enamel or two-component acrylic polyurethane. Openings must be sealed from moisture and made tamper-proof.

#### 2.1.1.2 Exterior Sheeting Panels

Modular panels must be provided in sizes shown on drawings. Panels must be fabricated a minimum of 0.090 inch thick aluminum or steel or 0.125 inch thick fiberglass reinforced plastic (FRP). Top and end panels must be removable and must be secured by 3/16 inch socket head jack nuts. Finish for metal panels must be two-component acrylic polyurethane.

#### 2.1.1.3 Mounting

Mount by securing to concrete foundation as indicated.

#### 2.1.1.4 Finishes

Base finish must be semi-gloss baked enamel or two-component acrylic polyurethane. Metal panel system finish must be baked enamel or two-component acrylic polyurethane.

### 2.1.2 Panel And Post/Panel Type Signs

#### 2.1.2.1 Posts

One-piece aluminum or galvanized steel posts must be provided with minimum 0.125 inch wall thickness. Posts must be designed to accept panel framing system described. The post must be designed to permit attachment of panel framing system without exposed fasteners. Caps must be provided for each post.

#### 2.1.2.2 Panel Framing System

Panel framing consisting of aluminum sections and interlocking track components must be designed to interlock with posts with concealed fasteners.

#### 2.1.2.3 Panels

Modular message panels must be provided in sizes shown on drawings. Panels must be fabricated a minimum of 0.080 inch aluminum or 0.125 inch acrylic or 0.125 inch fiberglass reinforced plastic (FRP). Panels must be designed to be interchangeable.

#### 2.1.2.4 Finishes

Post finish must be semi-gloss baked enamel. Metal panel system finish must be baked enamel or two-component acrylic polyurethane.

#### 2.1.2.5 Mounting

Provide removable mounting by a steel or an aluminum sleeve or flange embedded in concrete as indicated.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## 2.2 METAL PLAQUES

Design and location of plaques must be as indicated.

### 2.2.1 Cast Metal Plaques

#### 2.2.1.1 Fabrication

Cast metal plaques must have the logo, emblem and artwork cast in the flat relief technique. Plaques must be fabricated from prime aluminum.

#### 2.2.1.2 Size

Plaque size must be 48 inches in diameter.

#### 2.2.1.3 Border

Border must be flat band.

#### 2.2.1.4 Background

Background texture must be fine pebble.

#### 2.2.1.5 Mounting

Mounting must be concealed threaded aluminum bolts secured to solid backing mounted on exterior face of wall studs.

#### 2.2.1.6 Finish

Finishes must consist of aluminum with background sprayed dark gunmetal colored lacquer. Letters must be satin polished and entire plaque sprayed with two coats clear lacquer.

## 2.3 DIMENSIONAL BUILDING LETTERS

### 2.3.1 Fabrication

Letters must be fabricated from cast aluminum 0.125 inch aluminum sheet or extruded aluminum. Letters must be cleaned by chemical etching or cleaned ultrasonically in a special degreasing bath. Letters must be packaged for protection until installation.

### 2.3.2 Typeface

Typeface must be ARIAL or as indicated.

### 2.3.3 Size

Letter size must be 12" or as indicated.

### 2.3.4 Finish

Baked enamel or two-component acrylic polyurethane finish must be provided.

### 2.3.5 Mounting

Threaded studs of number and size as recommended by manufacturer, must be

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

used for concealed anchorage. Letters, studs, and sleeves must be of the same material. Supply templates for mounting. Provide solid backup behind wall panels on outside face of metal studs.

## 2.4 ALUMINUM ALLOY PRODUCTS

Aluminum alloy products must conform to ASTM B209 for sheet or plate, ASTM B221 for extrusions and ASTM B26/B26M or ASTM B108/B108M for castings. Aluminum extrusions must be provided at least 1/8 inch thick and aluminum plate or sheet at least 16 gauge thick. Welding for aluminum products must conform to AWS C1.1M/C1.1.

## 2.5 ORGANIC COATING

Clean, prime and give surfaces a semi-gloss baked enamel or two-component acrylic polyurethane finish in accordance with NAAMM AMP 500, AMP 505, with total dry film thickness not less than 1.2 mils.

## 2.6 STEEL PRODUCTS

Structural steel products must conform to ASTM A36/A36M. Sheet and strip steel products must conform to ASTM A1011/A1011M. Welding for steel products must conform to AWS D1.2/D1.2M.

## 2.7 VINYL SHEETING FOR GRAPHICS

Vinyl sheeting must be 5 to 7 year premium type and must be in accordance with the flammability requirements of ASTM E84 and must be a minimum 0.003 inch film thickness. Film must include a precoated pressure sensitive adhesive backing, Class 1, or positionable pressure sensitive adhesive backing, Class 3.

## 2.8 FIBER-REINFORCED POLYESTER (FRP) PANELS

Fiber-reinforced polyester (FRP) must be in accordance with ASTM D3841, Type II, Grade 1, Class 124.

## 2.9 ACRYLIC SHEET

Acrylic sheet must be in accordance with the flammability requirements of ASTM E84 and must conform to ANSI Z97.1.

## 2.10 POLYCARBONATE SHEET

Polycarbonate sheet must conform to SAE AMS3611.

## 2.11 ANCHORS AND FASTENERS

Exposed anchor and fastener materials must be compatible with metal to which applied and must match in color and finish and must be non-rusting, non-corroding, and non-staining. Exposed fasteners must be tamper-proof.

## 2.12 SHOP FABRICATION AND MANUFACTURE

### 2.12.1 Factory Workmanship

Work must be assembled in the shop, as far as practical, ready for installation at the site. Work that cannot be shop assembled must be given a trial fit in the shop to ensure proper field assembly. Holes for

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

bolts and screws must be drilled or punched. Drilling and punching must produce clean, true lines and surfaces. Welding to or on structural steel must be in accordance with AWS D1.1/D1.1M. Welding must be continuous along the entire area of contact. Exposed welds must be ground smooth. Exposed surfaces of work must have a smooth finish and exposed riveting must be flush. Fastenings must be concealed where practical. Items specified to be galvanized must be by hot-dip process after fabrication if practical. Galvanization must be in accordance with ASTM A123/A123M and ASTM A653/A653M, as applicable. Other metallic coatings of steel sheet must be in accordance with ASTM A924/A924M. Joints exposed to the weather must be formed to exclude water. Drainage and weep holes must be included as required to prevent condensation buildup.

#### 2.12.2 Dissimilar Materials

Where dissimilar metals are in contact, or where aluminum is in contact with concrete, mortar, masonry, wet or pressure-treated wood, or absorptive materials subject to wetting, the surfaces must be protected with a coat of asphalt varnish or a coat of zinc-molybdate primer to prevent galvanic or corrosive action.

#### 2.12.3 Shop Painting

Surfaces of miscellaneous metal work, except nonferrous metal, corrosion resisting steel, and zinc-coated work, must be given one coat of zinc-molybdate primer or an approved rust-resisting treatment and metallic primer in accordance with manufacturer's standard practice. Surfaces of items to be embedded in concrete must not be painted. Upon completion of work, damaged surfaces must be recoated.

#### 2.13 COLOR, FINISH, AND CONTRAST

Color must be as indicated on the drawings. Color listed is not intended to limit the selection of equal colors from other manufacturers. For buildings required to be handicapped-accessible, the characters and background of signs must be eggshell, matte, or other non-glare finish. Characters and symbols must contrast with their background - either light characters on a dark background or dark characters on a light background.

### PART 3 EXECUTION

#### 3.1 INSTALLATION

Signs, plaques, or dimensional letters must be installed in accordance with approved manufacturer's instructions at locations shown on the approved detail drawings; submit drawings showing elevations of each type of sign; dimensions, details, and methods of mounting or anchoring; shape and thickness of materials; and details of construction. A schedule showing the location, each sign type, and message must be included. Circuits installed underground must conform to the requirements of Section 33 71 02 UNDERGROUND ELECTRICAL DISTRIBUTION. Steel conduits installed underground and illuminated signage mounted directly on buildings must be in conformance with the requirements of Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Signs must be installed plumb and true at mounting heights indicated, and by method shown or specified. Signs mounted on other surfaces must not be installed until finishes on such surfaces have been completed. Submit manufacturer's installation instructions and cleaning instructions.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 3.1.1 Anchorage

Anchorage and fastener materials must be in accordance with approved manufacturer's instructions for the indicated substrate. Anchorage not otherwise specified or indicated must include slotted inserts, expansion shields, and powder-driven fasteners when approved for concrete; toggle bolts and through bolts for masonry; machine carriage bolts for steel; lag bolts and screws for wood.

### 3.1.2 Protection and Cleaning

The work must be protected against damage during construction. Hardware and electrical equipment must be adjusted for proper operation. Glass, frames, and other sign surfaces must be cleaned in accordance with manufacturer's instructions. After signs are completed and inspected, cover all project identification, directional, and other signs which may mislead the public. Covering must be maintained until instructed to be removed by the Contracting Officer or until the facility is to be opened for business. Submit six copies of maintenance instructions listing routine maintenance procedures, possible breakdowns and repairs, and troubleshooting guides. The instructions must include simplified diagrams for the equipment as installed. Signs must be cleaned, as required, at time of cover removal.

## 3.2 FIELD PAINTED FINISH

Miscellaneous metals and frames must be field painted in accordance with Section 09 90 00 PAINTS AND COATINGS. Anodized metals, masonry, and glass must be protected from paint. Finish must be free of scratches or other blemishes.

-- End of Section --

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## SECTION 10 14 00.20

## INTERIOR SIGNAGE

**08/20**

## PART 1 GENERAL

## 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to in the text by basic designation only.

## ALUMINUM ASSOCIATION (AA)

AA DAF45 (2003; Reaffirmed 2009) Designation System for Aluminum Finishes

AA PK-1 (2015) Pink Sheets: Designations and Chemical Composition Limits for Aluminum Alloys in the Form of Castings & Ingot

## AMERICAN WELDING SOCIETY (AWS)

AWS D1.2/D1.2M (2014; Errata 1 2014; Errata 2 2020) Structural Welding Code - Aluminum

## ASTM INTERNATIONAL (ASTM)

ASTM B209 (2014) Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate

ASTM B221 (2021) Standard Specification for Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes

## NATIONAL INSTITUTE OF STANDARDS AND TECHNOLOGY (NIST)

NIST SP 800-82 (2015; Rev 2) Guide to Industrial Control Systems (ICS) Security

## U.S. DEPARTMENT OF DEFENSE (DOD)

DOD 8510.01 (2022) Risk Management Framework (RMF) for DoD Systems

DODI 8500.01 (2014) Cybersecurity

UFC 4-010-06 (2016; with Change 1, 2017) Cybersecurity of Facility-Related Control Systems

## U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

36 CFR 1191 Americans with Disabilities Act (ADA) Accessibility Guidelines for Buildings and Facilities; Architectural Barriers Act (ABA) Accessibility Guidelines

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## 1.2 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

### SD-02 Shop Drawings

Detail Drawings; G

### SD-03 Product Data

Room Identification And Directional Signage System; G

### SD-04 Samples

Interior Signage; G

Room Identification And Directional Signage System; G

### SD-10 Operation and Maintenance Data

Approved Manufacturer's Instructions; G

Protection and Cleaning; G

## 1.3 QUALITY ASSURANCE

### 1.3.1 Samples

Submit interior signage samples of each of the following sign types showing typical quality, workmanship and color: all sign types included in project , , , , , , , Pressure Sensitive Letters. Approved samples may be installed in the work, provided each sample is identified and location recorded.

### 1.3.2 Detail Drawings

Submit detail drawings showing elevations of each type of sign, dimensions, details and methods of mounting or anchoring, mounting height, shape and thickness of materials, and details of construction. Include a schedule showing the location, each sign type, and message.

### 1.3.3 Sign Fabricator

Sign Fabricator to follow room number strategies created by designer. The room numbering system to be reviewed and approved by the Contracting Officer and command end users during the shop drawing phase, and prior to fabrication.

### 1.3.4 Cybersecurity

- a. The Risk Management Framework (RMF) is the process by which information systems are accredited for operation by a designated official from the Using Military Department. It is the standard process under which all DoD information systems achieve and maintain their Authority To Operate. The cybersecurity process is documented



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

in DOD 8510.01 and NIST SP 800-82. Refer to UFC 4-010-06 and DODI 8500.01 for additional requirements.

- b. All systems that are IP addressable or interface with the Assured Network required certification to operate. Coordinate with the Government to initiate and complete the accreditation process.
- c. Cybersecurity requires input from the system vendor or provider and support from the local IMD. The local IMD-IA office is the point of contact for all Cyber Security requirements. The local CMIO is the point of contact for all clinical and functional system requirements.

#### 1.4 DELIVERY, STORAGE, AND HANDLING

Package materials to prevent damage and deterioration during shipment, handling, storage and installation. Deliver products to the jobsite in manufacturer's original packaging and store in a clean, dry area in accordance with manufacturer's instructions.

#### 1.5 WARRANTY

Provide manufacturer's warranty to repair or replace defective interior signage materials and workmanship for a period of 5 years from date of final acceptance of the work.

### PART 2 PRODUCTS

#### 2.1 ROOM IDENTIFICATION AND DIRECTIONAL SIGNAGE SYSTEM

Provide signs, plaques, directories, and dimensional building letters that are standard products of manufacturers regularly engaged in the manufacture of such products that essentially duplicate signs that have been in satisfactory use at least 2 years prior to bid opening. Obtain signage from a single manufacturer with edges and corners of finished letter forms and graphics true and clean.

##### 2.1.1 Standard Room Signs

Provide signs that include tactile letters, symbols and Braille for interior rooms or spaces where the sign is not likely to change over time. Tactile text descriptions are required for pictograms that are provided to identify a permanent room. Examples include interior signs that label restrooms, stairs, room numbers or letters, and room names. These permanent room signs can include paper inserts for updatable information.

##### 2.1.1.1 Tactile Letters, Symbols and Braille

Provide ADA compliant material per 36 CFR 1191 which is raised 1/32 inch from the first surface, has a minimum 5/8 inch in height and is an ADA acceptable font. The color of the tactile letters is required to contrast with the sign face color per ADA standards. The ADA required Braille has a minimum durometer reading of 90. All raised letters, numbers and symbols are to comply.

##### 2.1.2 Directional Signs

Directional signs provide arrows with messages which point to critical destinations such as departments, offices, or other pertinent

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

destinations. These can be a panel sign system with a series of permanently attached messages or a modular system with updatable inserts. Directional signs have header panels with applied or direct print messages.

#### 2.1.3 Message Inserts

Provide updatable message inserts covered with a clear matte 0.015 inch vinyl protective overlay. The insert is typeset message laser printed on paper card stock. Provide software with message template for creating text and symbols for computers identified for Government production of paper inserts after project completion. Manufacturer is required to offer online ordering capabilities to facilitate and expedite ordering packages of replacement, color-coated paper inserts. Provide sliding inserts that slide horizontally exposing different graphic information as identified on the drawings.

#### 2.1.4 Type of Mounting for Signs

Provide surface mounted signs mounted with concealed mechanical fastening through the holders. Secure inserts in holders with flexible plastic clips. Mount framed plaques with manufacturer's standard (1/6 inch) 1.59 mm thick closed cell vinyl foam with adhesive backing. Adhesive must be transparent, long aging, high tech formulation on two sides of the vinyl foam. Double-faced tape consisting of acrylic adhesive on polyurethane foam used in conjunction with silicone adhesive. Provide signs with aluminum ceiling/projecting mount attachment extrusion to secure to ceiling or wall surface, along with matting ceiling/projecting mount track extrusion for hanging, projecting, and double-sided signs. Provide mounting for ceiling/projecting mount attachment extrusion by mechanical fasteners, selected based on wall or ceiling conditions. Mount track extrusion hinges over width of mount attachment and secured with 3.5 by 0.06 mm (6-32 inch) by 6 mm (1/4 inch) cone point stainless steel set screws.

#### 2.1.5 Character Proportions and Heights

Letters and numbers on signs conform to 36 CFR 1191.

### 2.2 DIMENSIONAL BUILDING LETTERS

#### 2.2.1 Fabrication

Letters are cast. Package letters for protection until installation.

#### 2.2.2 Size

Letter size is as indicated. Provide letter thickness that is manufacturer's standard for the size of letter.

#### 2.2.3 Finish

Provide clear anodized finish.

#### 2.2.4 Mounting

Threaded studs of number and size recommended by manufacturer; concealed anchorage. Letters which project from the mounting surface have stud spacer sleeves. Letters, studs, and sleeves are of the same material. Supply templates for mounting.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## 2.3 PRESSURE SENSITIVE LETTERS

### 2.3.1 Fabrication

Ensure that vinyl letter edges and corners of finished letterforms and graphics are true and clean. Do not use letterforms and graphics with rounded positive or negative corners, nicked, cut, or ragged edges.

### 2.3.2 Size

Letter size: as indicated.

## 2.4 MATERIALS

### 2.4.1 Aluminum Alloy Products

Aluminum extrusions are at least 1/8 inch thick, and aluminum plate or sheet are at least 0.0508 inch thick. Extrusions conform to ASTM B221; plate and sheet conforms to ASTM B209. Where anodic coatings are specified, alloy conforms to AA PK-1 alloy designation 514.0. Exposed anodized aluminum finishes are as shown. Welding for aluminum products conforms to AWS D1.2/D1.2M.

### 2.4.2 Anodic Coating

Anodized finish conforming to AA DAF45

### 2.4.3 Fabrication and Manufacture

#### 2.4.3.1 Factory Workmanship

Holes for bolts and screws are drilled or punched. Drilling and punching produces clean, true lines and surfaces. Exposed surfaces of work have a smooth finish; exposed riveting is flush. Conceal fastenings where practicable.

#### 2.4.3.2 Dissimilar Materials

Where dissimilar metals are in contact, protect surfaces prevent galvanic or corrosive action.

### 2.4.4 Typeface

ADA-ABA compliant font for Room Signs.

## 2.5 GRAPHICS

Provide signage graphics for modular signs to the following:

### 2.5.1 Photopolymer

Integral graphics and Braille achieved by photomechanical stratification processes. Provide photopolymer used for ADA compliant graphics of the type that has a minimum durometer reading of 90. Tactile graphics are raised 1/32 inch from the first surface of plaque by photomechanical stratification process.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## 2.6 COLOR, FINISH, AND CONTRAST

Provide color as indicated; colors listed are not intended to limit the selection of equal colors from other manufacturers. Finish of eggshell, matte, or other non-glare finish for all signs as required in handicapped-accessible buildings.

## PART 3 EXECUTION

### 3.1 INSTALLATION

Install signs plumb and true and in accordance with approved manufacturer's instructions at locations shown on the detail drawings. Submit operating instructions outlining the step-by-step procedures required for system operation. The instructions include simplified diagrams for the system as installed, the manufacturer's name, model number, service manual, parts list, and brief description of all equipment and their basic operating features. Provide each set permanently bound with a hard cover. The following identification must be inscribed on the covers: "OPERATING AND MAINTENANCE INSTRUCTIONS", name and location of the facility, name of the Contractor, and contract number. Submit in accordance with Section 01 78 23 OPERATING AND MAINTENANCE DATA. Mounting height and mounting location complies with 36 CFR 1191. Install required blocking. Do not install signs on doors or other surfaces until finishes on such surfaces have been installed. Signs installed on glass surfaces are installed with matching blank back-up plates in accordance with manufacturer's instructions.

Do not install items that show visual evidence of biological growth.

#### 3.1.1 Anchorage

Provide anchorage in accordance with approved manufacturer's instructions. Anchorage not otherwise specified or shown includes slotted inserts, expansion shields, and powder-driven fasteners when approved for concrete; toggle bolts and through bolts for masonry; machine carriage bolts for steel; lag bolts and screws for wood. Provide exposed anchor and fastener materials compatible with metal to which applied with matching color and finish.

- a. Signs mounted to painted gypsum board surfaces must be removable for painting maintenance.
- b. Mount signs to lay-in ceiling grids with clip connections to ceiling tees.
- c. Install signs mounted on metal surfaces with magnetic tape.
- d. Install signs mounted on fabric surfaces with hook and loop tape or pin mount.
- e. Install signs to workstation panels with panel clips.

#### 3.1.2 Protection and Cleaning

Protect the work against damage during construction. Adjust hardware and electrical equipment for proper operation. Clean glass, frames, and other sign surfaces at completion of signage installation in accordance with the manufacturer's written instructions.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

-- End of Section --

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## SECTION 10 21 13

### TOILET COMPARTMENTS

**08/20**

#### PART 1 GENERAL

##### 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

##### ALUMINUM ASSOCIATION (AA)

AA DAF45 (2003; Reaffirmed 2009) Designation System  
for Aluminum Finishes

##### ASTM INTERNATIONAL (ASTM)

ASTM A123/A123M (2017) Standard Specification for Zinc  
(Hot-Dip Galvanized) Coatings on Iron and  
Steel Products

ASTM A167 (2011) Standard Specification for  
Stainless and Heat-Resisting  
Chromium-Nickel Steel Plate, Sheet, and  
Strip

ASTM A336/A336M (2023) Standard Specification for Alloy  
Steel Forgings for Pressure and  
High-Temperature Parts

ASTM A385/A385M (2022) Standard Practice for Providing  
High-Quality Zinc Coatings (Hot-Dip)

ASTM B86 (2023) Standard Specification for Zinc and  
Zinc-Aluminum (ZA) Alloy Foundry and Die  
Castings

ASTM B221 (2021) Standard Specification for Aluminum  
and Aluminum-Alloy Extruded Bars, Rods,  
Wire, Profiles, and Tubes

ASTM D7611/D7611M (2013; E 2014) Standard Practice for  
Coding Plastic Manufactured Articles for  
Resin Identification

##### INTERNATIONAL CODE COUNCIL (ICC)

ICC A117.1 (2017) Standard And Commentary Accessible  
and Usable Buildings and Facilities

##### U.S. GENERAL SERVICES ADMINISTRATION (GSA)

CID A-A-60003 (Basic; Notice 1) Partitions, Toilet,  
Complete

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

36 CFR 1191

Americans with Disabilities Act (ADA)  
Accessibility Guidelines for Buildings and  
Facilities; Architectural Barriers Act  
(ABA) Accessibility Guidelines

1.2 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Fabrication Drawings

Installation Drawings; FIO

SD-03 Product Data

Cleaning and Maintenance Instructions

Colors And Finishes

Anchoring Devices and Fasteners

Hardware and Fittings

Brackets

Door Hardware

Toilet Enclosures

Urinal Screens

Pilaster Shoes

Finishes; G

SD-04 Samples

Colors and Finishes; G

Hardware and Fittings

Anchoring Devices and Fasteners

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## SD-07 Certificates

### Warranty

## SD-10 Operation and Maintenance Data

### Plastic Identification; G

#### 1.3 REGULATORY REQUIREMENTS

Comply with to ICC A117.1 code for access for the handicapped operation of toilet compartment door and hardware.

#### 1.4 DELIVERY, STORAGE, AND HANDLING

Deliver materials in the manufacturer's original unopened packages with the brand, item identification, and project reference clearly marked. Store components in a dry location that is adequately ventilated; free from dust, water, other contaminants, and damage during delivery, storage, and construction.

#### 1.5 WARRANTY

Provide manufacturer's warranty to repair or replace defective materials and workmanship for a period of one year from date of final acceptance of the work.

## PART 2 PRODUCTS

### 2.1 SYSTEM REQUIREMENTS

Provide a complete and usable toilet partition system, including toilet enclosures, room entrance screens, urinal screens, system of panels, hardware, and support components. Furnish the partition system from a single manufacturer, with a standard product as shown in the most recent catalog data. Submit Fabrication Drawings for toilet partitions and urinal screens consisting of fabrication and assembly details to be performed in the factory. Submit manufacturer's Cleaning and Maintenance Instructions in accordance with Section 01 78 23 OPERATIONS AND MAINTENANCE DATA.

#### 2.1.1 Plastic Identification

Verify that plastic products to be incorporated into the project are labeled in accordance with ASTM D7611/D7611M. Where products are not labeled, provide product data indicating polymeric information in the Operation and Maintenance Manual.

|        |                                  |
|--------|----------------------------------|
| Type 2 | High Density Polyethylene (HDPE) |
|--------|----------------------------------|



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## 2.2 MATERIALS

### 2.2.1 Solid Polyethylene Panels (Finish 5)

Provide high density polyethylene (HDPE) suitable for exposed application. Waterproof, non-absorbent and graffiti resistant textured surface with a Flame Spread Index of 75 or less, and a Smoke Developed Index of 450 or less.

### 2.2.2 Anchoring Devices and Fasteners

Provide steel anchoring devices and fasteners hot-dipped galvanized after fabrication, in conformance with ASTM A385/A385M and ASTM A123/A123M. Conceal all galvanized anchoring devices.

### 2.2.3 Brackets

Provide two-ear panel wall brackets, T-style, 1 inch stock. Provide stirrup style panel-to-pilaster brackets.

### 2.2.4 Hardware and Fittings

#### 2.2.4.1 General Requirements

Provide hardware for the toilet partition system that complies with CID A-A-60003 for the specified type and style of partitions. Provide hardware finish highly resistant to alkalis, urine, and other common toilet room acids. Comply with 36 CFR 1191 of latching devices and hinges for handicap compartments; provide chrome-plated steel or stainless steel devices and hinges with door latches that operate without either tight grasping or twisting of the wrist of the operator.

| Material                  | Conformance Standard                |
|---------------------------|-------------------------------------|
| Cold-rolled sheet steel   | ASTM A336/A336M, commercial quality |
| Zinc-base alloy           | ASTM B86, Alloy AC41-A              |
|                           |                                     |
| Aluminum                  | ASTM B221                           |
| Corrosion-resistant steel | ASTM A167, Type 302                 |

#### 2.2.4.2 Finishes

- Provide aluminum with clear anodic coating that complies with AA DAF45.
- Provide corrosion-resistant steel with a No. 4 finish.
- Provide stainless steel with a No. 4 finish.
- Provide exposed fasteners that match the hardware and fittings.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## 2.2.5 Door Hardware

### 2.2.5.1 Hinges

Provide adjustable hinges to hold in-swinging doors open at any angle up to 90 degrees and outswinging doors up to 10 degrees. Provide self-lubricating hinges with the indicated swing. Provide hinges that wrap-around / bolted surface-mounted type.

### 2.2.5.2 Coat Hooks

Provide coat hooks that are combination units with hooks and rubber tipped pins.

## 2.3 PARTITION PANELS AND DOORS

Fabricate partition panels, and pilasters of materials and construction listed:

Provide . plastic (HDPE) partition panels, doors and pilasters not less than 1 inch thick homogenous partition panels, and doors and pilasters no less than 1 inch thick.

. . .

### 2.3.1 Toilet Enclosures

Provide toilet enclosures that comply with CID A-A-60003, Type I, Style C, overhead braced . Furnish width, length, and height of toilet enclosures as shown. Finish surface of panels are solid polyethylene (Finish 5); water resistant; graffiti resistant; non-absorbent radius beveled edges. Provide panels indicated to receive toilet paper holders or grab bars for mounting of the items required, and provide cut outs for through partition toilet accessories. Provide grab bars to withstand a bending stress, shear stress, shear force, and a tensile force induced by 250 lbf. Grab bars cannot rotate within their fittings.

### 2.3.2 Urinal Screens

Provide urinal screens that comply with CID A-A-60003, Type III, Style F, wall hung. Provide finish for surface of screens as solid polyethylene (Finish 5); water resistant; graffiti resistant; non-absorbent with radius beveled edges; with manufacturer's standard post design of materials matching the thickness and construction of pilasters. Furnish width and height of urinal screens as shown. Provide thickness to match toilet compartment panel construction. Secure wall hung urinal screens with continuous brackets. Fabricate screens from the same types of panels and pilasters as the toilet partitions. Use corrosion-resistant steel fittings and fasteners.

## 2.4 CEILING-HUNG PARTITIONS

Provide pilasters in size indicated that are manufacturer's standard corrosion resistant anchoring assemblies complete with leveling adjustment nuts at pilasters for connection to structural support above finished ceiling. Design anchoring device to transmit the strain and loading on the pilaster directly to the structural support above without putting strain or loading on the finished ceiling. Provide sleeves or caps at tops of pilasters to conceal anchorage.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## 2.5 OVERHEAD-BRACED PARTITIONS

Provide pilasters in sizes indicated that are manufacturer's standard corrosion-resistant supports, leveling mechanism, and anchors at pilasters to suit floor conditions. Provide shoes at pilasters to conceal supports and leveling mechanism. Provide anchoring device at the bottom of the pilaster consisting of a channel-shaped floor stirrup fabricated from not less than 0.0635 inch thick material and a leveling bolt. Secure the stirrup to the pilaster with not less than a 3/16 inch bolt and nut after the pilaster is leveled. Secure the stirrup to the floor with not less than two lead expansion shields and sheetmetal screws. Fabricate overhead brace from a continuous extruded aluminum tube not less than 1 inch wide by 1-1/2 inch high, 0.125 inch wall thickness. Finish is AA-C22A31 in accordance with AA DAF45. Set and secure brace into the top of each pilaster. Provide shoes at pilasters to conceal supports and leveling mechanism.

## 2.6 PILASTER SHOES

Provide shoes at pilasters to conceal floor-mounted anchorage. Provide stainless steel pilaster shoes. Height is a minimum 3 inches.

## 2.7 HARDWARE

Provide hardware for the toilet partition system that complies with CID A-A-60003 for the specified type and style of partitions. Provide hardware pre-drilled by manufacturer. Use a hardware finish that is highly resistant to alkalis, urine, and other common toilet room acids. Hardware includes: gravity type, adjustable for door close positioning; nylon bearings; or aluminum door latch; door strike and keeper with rubber bumper; and cast alloy chrome plated coat hook and bumper. Provide latching devices and hinges for handicap compartments complying with 36 CFR 1191 and stainless steel door latches that operate without either tight grasping or twisting of the wrist of the operator. Use stainless steel, tamper proof type screws and bolts. Wall mounting brackets are continuous, full height, stainless steel, in accordance with toilet compartment manufacturer's instructions.

## 2.8 COLORS AND FINISHES

### 2.8.1 Colors

Provide color as indicated; colors listed are not intended to limit the selection of equal colors from other manufacturers.

Submit three samples showing color and a finished edge on two adjacent sides and core construction, each not less than 12 inch square.

### 2.8.2 Finishes

#### 2.8.2.1 Finishes No. 5

Provide manufacturer's standard (Finish 5) formed under high pressure rendering a single component section not less than 1 inch thick. Colors extend throughout the panel thickness.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### PART 3 EXECUTION

#### 3.1 PREPARATION

Take field measurements prior to the preparation of drawing and fabrication to ensure proper fits. Verify that field measurements, surfaces, substrates and conditions are as required, and ready to receive work. Verify correct spacing of plumbing fixtures. Verify correct location of built in framing, anchorage, and bracing. Report in writing to Contracting Officer prevailing conditions that adversely affect satisfactory execution of the work of this section. Do not proceed with work until unsatisfactory conditions have been corrected.

#### 3.2 INSTALLATION

Do not install items that show visual evidence of biological growth. Install partitions rigid, straight, plumb, and level, with the panels centered between the fixtures. Provide a panel clearance of not more than 1/2 inch and secure the panels to walls and pilasters with continuous full height wall brackets. Locate wall brackets so that holes for wall bolts occur in masonry or tile joints. Secure panels to pilasters with brackets matching the wall brackets. Provide for adjustment due to minor floor variations. Locate head rail joints at pilaster center lines. Install adjacent components for consistency of line and plane. Equip each door with hinges, one door latch, and one coat hook and bumper. Align hardware to uniform clearance at vertical edges of doors.

- a. Secure panels to ceramic tile on metal stud walls or hollow concrete-masonry walls with toggle bolts using not less than 1/4-20 screws of the length required for the wall thickness. Provide toggle bolts with a load-carrying strength of not less than 600 pounds per anchor.
- b. Submit Installation Drawings for toilet partitions, room entrance screens, and urinal screens showing plans, elevations, details of construction, hardware, reinforcing and blocking, fittings, mountings and escutcheons. Indicate on drawings the type of partition, location, mounting height, cutouts, and reinforcement required for toilet-room accessories.

#### 3.3 OVERHEAD-BRACED PARTITIONS

Secure pilasters to the floor with the anchorage device specified. Make all leveling devices readily accessible for leveling, plumbing, and tightening the installation. Secure overhead brace to the pilaster face with not less than two fasteners per face. Expansion shields have a minimum 2 inch penetration into the concrete slab. Make tops of doors parallel with the overhead brace when doors are in a closed position.

#### 3.4 FINAL ADJUSTMENT

After completion of the installation, make final adjustments to the pilaster-leveling devices, door hardware, and other working parts of the partition assembly. Doors have a uniform vertical edge clearance of approximately 3/16 inch and rest open at approximately 30 degrees when unlatched.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 3.5 CLEANING

Clean all surfaces and adjacent surfaces soiled as a result of the work, in an approved manner compliant with the manufacturer's recommended cleaning and protection from damage procedures until accepted. Remove all equipment, tools, surplus materials, and work debris from the site.

-- End of Section --

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## SECTION 10 22 13

### WIRE MESH PARTITIONS 08/16, CHG 1: 08/18

#### PART 1 GENERAL

##### 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

##### AMERICAN IRON AND STEEL INSTITUTE (AISI)

AISI SG03-3 (2002; Suppl 2001-2004; R 2008)  
Cold-Formed Steel Design Manual Set

##### ASTM INTERNATIONAL (ASTM)

ASTM A36/A36M (2019) Standard Specification for Carbon  
Structural Steel

##### 1.2 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

##### SD-02 Shop Drawings

##### Wire Mesh Partitions

Show layout, details, materials, dimensions, finishes, and all information necessary for fabrication and installation.

##### SD-03 Product Data

##### Wire Mesh Partitions

Submit for each type of partition, door, and window.

##### 1.3 DELIVERY, STORAGE, AND HANDLING

Deliver materials in manufacturer's original, unopened containers or packaging with labels intact and legible. Deliver, store, and handle materials so as to prevent damage. Replace damaged or defective materials with new.

##### 1.4 DESCRIPTION OF WORK

Wire mesh partitions must be all wire type, heavy duty for extra heavy industrial use. Provide partitions complete with fasteners, capping bars, adjustable floor sockets, bracing, doors, hardware, and other items necessary for a complete, useable, and rigid installation.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## PART 2 PRODUCTS

### 2.1 MATERIALS

#### 2.1.1 Steel Shapes, Plates, and Bars

ASTM A36/A36M.

#### 2.1.2 Cold-Formed Steel

AISI SG03-3.

#### 2.1.3 Wire Mesh

Carbon steel wire, woven diamond mesh, intermediate crimped.

#### 2.1.4 Floor Sockets

Cast or forged steel or ductile iron, adjustable, approximately 2-1/2 inches high.

### 2.2 HEAVY DUTY PARTITIONS

#### 2.2.1 Wire Mesh

6 gage wire, 2 inch mesh.

#### 2.2.2 Panel Frames

1-1/2 by 3/4 by 1/8 inch steel channels.

#### 2.2.3 Center Reinforcing Bar

One 1-1/2 by 3/4 by 1/8 inch channel with all wires woven through, or two 1-1/4 by 3/8 by 1/8 inch channels bolted together with mesh in between.

#### 2.2.4 Capping Bar

Structural steel channel, 3 inch by 4.1 pounds.

#### 2.2.5 Corner Posts

Structural steel angles, 1-3/4 by 1-3/4 by 1/8 inch.

#### 2.2.6 Line Posts

Unless otherwise indicated, provide partitions with flat bar line posts bolted between vertical frame channels. Sizes of posts must be as follows:

| Partition Height   | Size of Posts                       |
|--------------------|-------------------------------------|
| 7 feet to 12 feet  | 2-1/2 by 5/16 inch                  |
| 12 feet to 16 feet | 3 by 5/16 inch or 2-1/2 by 3/8 inch |
| 16 feet to 20 feet | 3-1/2 by 5/16 inch                  |

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 2.2.7 Hinged Doors

Frames must be 1-1/2 by 3/4 by 1/8 inch channels with 1-1/2 by 1/8 inch flat bar cover on top and bottom rails and on hinge stile and a 1-5/8 by 7/8 by 1/8 inch angle riveted to the lock stile. Provide 1-1/2 pairs of heavyweight, wrought steel, non-removable pin, butt hinges riveted or welded to the door and the door opening frame for each door.

### 2.3 DOOR OPENING FRAMES

Provide frames the same size and shape as the vertical frames for the mesh panels.

### 2.4 LOCKS

Provide each door with a mortise type lock with a six-pin tumbler lock cylinder on the outside and a recessed knob on the inside.

### 2.5 FABRICATION

#### 2.5.1 Standard Panels

Wire must be woven into diamond mesh, intermediate crimped, and securely clinched to frames. Joints must be mortised and tenoned. Wire must be continuous at center reinforcing bars, either woven through a single channel or bolted between two channels. Panel vertical frames must have 3/8 inch bolt holes 18 inches o.c. for heavy duty partitions.

#### 2.5.2 Doors

Construction must be similar to that specified for panels. Wire mesh must be the same as that used in the adjacent partition panels.

#### 2.5.3 Finish

Thoroughly clean ferrous metal, treat with phosphate, and paint with gray enamel in the shop.

## PART 3 EXECUTION

### 3.1 INSTALLATION

#### 3.1.1 Wire Mesh Partitions

Install plumb, level, and true to line, within a tolerance of 1/8 inch in 10 feet or the height or run of the partition, if less than 10 feet. Anchor floor sockets to the floor with expansion bolts. Bolt vertical frames and posts together with 3/8 inch bolts 18 inches o.c. for heavy duty partitions. Secure top frames to a continuous capping bar with 1/4 inch diameter U bolts not more than 28 inches o.c.

#### 3.1.2 Doors

Install in accordance with the manufacturers' recommendations. Adjust as required so that doors and hardware operate freely and properly.

#### 3.1.3 Bracing

Brace free standing partitions more than 20 feet in length, at intervals



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

not greater than 20 feet with a steel channel brace connected to the capping bar and anchored to the building wall or framing member .

#### 3.1.4 Touch-Up

Clean and paint scratches, abrasions, and other damage to shop painted surfaces to match the shop-applied finish.

Repair minor surface rust areas. Clean and prime with rust inhibitive primer paint. Apply final paint to match shop-applied finishes.

-- End of Section --

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## SECTION 10 26 00

### WALL AND DOOR PROTECTION 08/20

#### PART 1 GENERAL

##### 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

##### ASTM INTERNATIONAL (ASTM)

|           |                                                                                                                        |
|-----------|------------------------------------------------------------------------------------------------------------------------|
| ASTM A167 | (2011) Standard Specification for Stainless and Heat-Resisting Chromium-Nickel Steel Plate, Sheet, and Strip           |
| ASTM B221 | (2021) Standard Specification for Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes           |
| ASTM D256 | (2010; R 2018) Standard Test Methods for Determining the Izod Pendulum Impact Resistance of Plastics                   |
| ASTM D543 | (2020) Standard Practices for Evaluating the Resistance of Plastics to Chemical Reagents                               |
| ASTM D635 | (2018) Standard Test Method for Rate of Burning and/or Extent and Time of Burning of Plastics in a Horizontal Position |
| ASTM E84  | (2023) Standard Test Method for Surface Burning Characteristics of Building Materials                                  |
| ASTM G21  | (2015; R 2021; E 2021) Standard Practice for Determining Resistance of Synthetic Polymeric Materials to Fungi          |

##### CALIFORNIA DEPARTMENT OF PUBLIC HEALTH (CDPH)

|                    |                                                                                                                                                            |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CDPH SECTION 01350 | (2017; Version 1.2) Standard Method for the Testing and Evaluation of Volatile Organic Chemical Emissions from Indoor Sources using Environmental Chambers |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|

##### GREEN SEAL (GS)

|       |                                     |
|-------|-------------------------------------|
| GS-36 | (2013) Adhesives for Commercial Use |
|-------|-------------------------------------|

##### SCIENTIFIC CERTIFICATION SYSTEMS (SCS)

|     |                                            |
|-----|--------------------------------------------|
| SCS | SCS Global Services (SCS) Indoor Advantage |
|-----|--------------------------------------------|

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

SOCIETY OF AUTOMOTIVE ENGINEERS INTERNATIONAL (SAE)

SAE J1545 (2005; R 2021) Instrumental Color  
Difference Measurement for Exterior  
Finishes, Textiles and Colored Trim

SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT (SCAQMD)

SCAQMD Rule 1168 (2017) Adhesive and Sealant Applications

UNDERWRITERS LABORATORIES (UL)

UL 2818 (2022) GREENGUARD Certification Program  
For Chemical Emissions For Building  
Materials, Finishes And Furnishings

## 1.2 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

### SD-02 Shop Drawings

Corner Guards

Wall Guards

Wall Covering and Panels

### SD-03 Product Data

Corner Guards

Wall Guards

Wall Covering and Panels

### SD-04 Samples

Corner Guards

Wall Guards

Wall Covering and Panels

### SD-06 Test Reports

Fire Resistance Rating

### SD-07 Certificates

### SD-10 Operation and Maintenance Data

Corner Guards, Data Package 1; G

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

Wall Guards, Data Package 1; G

Wall Covering and Panels, Data Package 1; G

### 1.3 CERTIFICATIONS

#### 1.3.1 Indoor Air Quality

##### 1.3.1.1 Wall Covering and Panels

Provide sheet and high impact resistant resilient materials certified to meet indoor air quality requirements by UL 2818 (Greenguard) Gold, SCS Global Services Indoor Advantage Gold or provide certification or validation by other third-party program that products meet the requirements of this section.

##### 1.3.1.2 Adhesives and Sealants

Provide products certified to meet indoor air quality requirements by UL 2818 (Greenguard) Gold, SCS Global Services Indoor Advantage Gold or provide certification or validation by other third-party program that products meet the requirements of this section. Provide current product certification documentation from certification body. When product does not have certification, provide validation that product meets the indoor air quality product requirements cited herein.

### 1.4 DELIVERY, STORAGE, AND HANDLING

Deliver materials to the project site in manufacturer's original unopened containers with seals unbroken and labels and trademarks intact. Keep materials dry, protected from weather and damage, and stored under cover. Store materials at approximately 70 degrees F for at least 48 hours prior to installation.

### 1.5 WARRANTY

Provide manufacturer's warranty to repair or replace defective materials and workmanship for a 1 year period of one year from date of final acceptance of the work.

## PART 2 PRODUCTS

### 2.1 STANDARD PRODUCTS

To the maximum extent possible, provide wall and door protection items that are standard products of a single manufacturer and furnished as detailed. Drawings show general configuration of products required.

Submit detailed shop drawings of each wall and door protection item indicated. Include elevations, dimensions, clearances, details of construction and anchorage, and details of joints and connections.

Submit manufacturers' descriptive product data for each wall and door protection item indicated. Include manufacturers' literature, finishes, profiles and thicknesses of materials.

Submit manufacturers' operations and maintenance data for each wall and door protection item indicated in accordance with Section 01 78 23 OPERATIONS AND MAINTENANCE DATA.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 2.1.1 Resilient Material

Provide resilient material consisting of high impact resistant extruded PVC free conforming to the following:

#### 2.1.1.1 Minimum Impact Resistance

Minimum impact resistance must be 18 ft-lbs/sq. inch when tested in accordance with ASTM D256, (Izod impact, ft-lbs per sq inch notched).

#### 2.1.1.2 Fire Resistance Rating

Provide the following surface burning characteristics when tested and labeled in accordance with ASTM E84 by a qualified testing agency: maximum flame spread of 25 and a smoke developed rating of 450 or less. Provide material rated as self extinguishing when tested in accordance with ASTM D635. Provide resilient material used for protection on fire rated doors and frames listed by the qualified testing agency performing the tests. Provide resilient material installed on fire rated wood/steel door and frame assemblies tested on similar type assemblies. Test results of material tested on any other combination of door/frame assembly are not acceptable.

#### 2.1.1.3 Integral Color

Provide colored components having integral color and matched in accordance with SAE J1545 to within plus or minus 1.0 on the CIE-LCH scales.

#### 2.1.1.4 Chemical and Stain Resistance

Provide materials resistant to chemicals and stains reagents in accordance with ASTM D543.

#### 2.1.1.5 Fungal and Bacterial Resistance

Provide materials resistant to fungi and bacteria in accordance with ASTM G21, as applicable.

### 2.2 CORNER GUARDS

#### 2.2.1 Resilient Corner Guards

Provide as indicated on Drawings.

#### 2.2.2 Stainless Steel Corner Guards

Provide stainless steel base material that contains a minimum of 60 percent recycled content. Fabricate stainless steel base material of 16 gauge thick material conforming to ASTM A167, type 430 or 304. Provide corner guards that extend from floor to ceiling. Form corner guard to dimensions shown.

### 2.3 WALL GUARDS

Provide product with prefabricated end closure caps, inside and outside corners, concealed splices, cushions, mounting hardware and other accessories standard with the manufacturer. Extruded continuous aluminum retainers must conform to ASTM B221, alloy 6063, temper T5 or T6. Field

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

adjust all end caps and corners to assure close alignment.

## 2.4 WALL COVERING AND PANELS

Provide wall covering and panels consisting of high impact PVC free resilient material. Panel sizes are 4 by 8 feet. Provide wall covering material used on the interior of the building (defined as inside of the weatherproofing system) that meets either emissions requirements of CDPH SECTION 01350 (limit requirements for either office or classroom spaces regardless of space type) the VOC content requirements of SCAQMD Rule 1168, or VOC content requirements of GS-36.

### 2.4.1 High Impact Wall Panels

Provide wall panel face and edge thickness that are 0.040 inch. Factory bond panel face to a 0.375 inch thick fiberboard core. Laminate the backside of the panel with a moisture resistant vapor barrier.

## 2.5 TRIM, FASTENERS AND ANCHORS

Provide PVC free trim, fasteners and anchors for each specific installation as indicated.

## 2.6 FINISH

Submit samples indicating color and texture of materials requiring color and finish.

### 2.6.1 Stainless Steel Finish

Provide stainless steel finish in accordance with ASTM A167, Type 302 or 304, finish number 4.

### 2.6.2 Resilient Material Finish

Provide resilient material finish of high gloss vinyl texture with colors in accordance with SAE J1545.

## 2.7 ADHESIVES

Provide adhesive for resilient material in accordance with manufacturers recommendations. Provide sealants and non-aerosol adhesive products used on the interior of the building (defined as inside of the weatherproofing system) that meet either emissions requirements of CDPH SECTION 01350 (limit requirements for either office or classroom spaces regardless of space type) the VOC content requirements of SCAQMD Rule 1168, or VOC content requirements of GS-36.

## 2.8 COLOR

Provide color as indicated; colors listed are not intended to limit the selection of equal colors from other manufacturers.

## PART 3 EXECUTION

### 3.1 INSTALLATION

Do not install items that show visual evidence of biological growth. Install items on surfaces that are clean, smooth, and free of obstructions.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 3.1.1 Corner Guards and Wall Guards

- a. Mount guards as indicated and in accordance with manufacturer's written installation instructions.
- b. For wall guards, space brackets at no more than 3 feet on centers and anchor to the wall in accordance with the manufacturer's written installation instructions.

#### 3.1.1.1 Stainless Steel Guards

- a. Mount guards as indicated and in accordance with manufacturer's recommendations.
- b. Where corner guards are installed on walls, partitions or columns finished with plaster or ceramic tile, provide continuous 16 gauge thick, perforated, galvanized z-shape steel anchors welded to back edges of corner guards and wired to metal studs. Coat back surfaces of corner guards, where shown, with a non-flammable, sound deadening material. Overlap corner guards on finish plaster surfaces.
- c. Where corner guards are installed on gypsum board, clean surfaces and anchor guards with a neoprene solvent-type contact adhesive specifically manufactured for use on gypsum board construction. Remove excess adhesive from the guard edges and allow to cure undisturbed for 24 hours.
- d. For wall guards, space brackets at no more than 3 feet on center and anchor to the wall in accordance with the manufacturer's installation instructions.

### 3.1.2 Wall Coverings and Panels

Install as indicated in accordance with manufacturer's written instructions.

-- End of Section --

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## SECTION 10 28 13

### TOILET ACCESSORIES 08/20

#### PART 1 GENERAL

##### 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

##### U.S. DEPARTMENT OF DEFENSE (DOD)

|              |                                                                                                   |
|--------------|---------------------------------------------------------------------------------------------------|
| MIL-STD-1691 | (1994; Rev F) Construction and Material<br>Schedule for Military Medical and Dental<br>Facilities |
|--------------|---------------------------------------------------------------------------------------------------|

##### 1.2 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

##### SD-02 Shop Drawings

##### Product Schedule; FIO

Submit product Schedule indicating types, quantities, sizes, and installation locations by room for each toilet accessory item required. Identify locations using room designations indicated on the drawings.

##### SD-03 Product Data

Item A5030 Bench, Stall, Shower, Built In; FIO

Item A5080 Dispenser, Paper Towel, SS, Surface Mounted; FIO

Item A5109 Grab Bar, 1-1/4 inch Dia., SS, 2 Wall, W/C Accessible; FIO

Item A5110 Grab Bar, 1-1/4 inch Dia., SS, 2 Wall, Shower Use; FIO

Item A5135 Shelf, Utility W/ Mop/Broom Holders, SS, Surf Mntd; FIO

Item A5140 Hook, Garment, Security; FIO

Item A5170 Rod, Shower Curtain, 1 inch Diameter, W/Curtain & Hooks; FIO

Item A5195 Dispenser, Toilet Tissue, SS, 1-Roll, Surface Mntd; FIO

Item A5202 Soap Dispenser, wall-mount; FIO



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

Item A5205 Mirror, SS, Surface Mounted; FIO

Item A5209 Toilet Seat Cover Dispenser, Surface Mntd; FIO

Soap Dispenser, Wall-mount vertical type, SS, 1200ml minimum capacity, corrosion-resistant valve; FIO

Mirror, Electro-copper plated plate glass, SS frame, 24" x 36"; FIO

Combination Paper Towel Dispenser / Waste Receptacle, 400 C-fold - single fold or quarter fold, key lock for towel door; FIO

Toilet Seat Cover Dispenser, Surface-mount, SS, 250 capacity; FIO

Submit catalog numbers, literature, data sheets, construction details, profiles, anchoring and mounting requirements and other pertinent data for each toilet accessory item to evaluate function, materials, dimensions and appearance.

Accessory Items

#### SD-07 Certificates

Baby Changing Stations

#### SD-10 Operation and Maintenance Data

Item A5030 Bench, Stall, Shower, Built In; G

Item A5080 Dispenser, Paper Towel, SS, Surface Mounted; G

Item A5109 Grab Bar, 1-1/4 inch Dia., SS, 2 Wall, W/C Accessible; G

Item A5110 Grab Bar, 1-1/4 inch Dia., SS, 2 Wall, Shower Use; G

Item A5135 Shelf, Utility W/ Mop/Broom Holders, SS, Surf Mntd; G

Item A5140 Hook, Garment, Security; G

Item A5170 Rod, Shower Curtain, 1 inch Diameter, W/Curtain & Hooks; G

Item A5195 Dispenser, Toilet Tissue, SS, 1-Roll, Surface Mntd; G

Item A5207 Combination Towel Bar, SS, Surface Mntd; G

Item L1200 Cabinet, Specimen, Pass Thru, CRS; G

Soap Dispenser, Wall-mount vertical type, SS, 1200ml minimum capacity, corrosion-resistant valve;

Mirror, Electro-copper plated plate glass, SS frame, 24" x 36";

Combination Paper Towel Dispenser / Waste Receptacle, 400 C-fold - single fold or quarter fold, key lock for towel door;

Toilet Seat Cover Dispenser, Surface-mount, SS, 250 capacity; G

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

Submit Data Package 1 for each toilet accessory item in accordance with Section 01 78 23 OPERATIONS AND MAINTENANCE DATA.

### 1.3 DELIVERY, STORAGE, AND HANDLING

Wrap toilet accessories for shipment and storage, then deliver to the jobsite in manufacturer's original packaging, and store in a clean, dry area protected from construction damage and vandalism.

### 1.4 WARRANTY

Provide manufacturer's warranty to repair or replace defective materials and workmanship for a period of one year from date of final acceptance of the work..

## PART 2 PRODUCTS

### 2.1 ACCESSORY ITEMS

Provide toilet accessories where indicated in accordance with Contractor-provided product schedule. Conform to the requirements for accessory items specified herein which are based on MIL-STD-1691 Joint Schedule Numbers (JSN). Refer to Restroom Accessories Schedule on Drawings for Accessory Abbreviations.

Provide stainless steel products listed herein manufactured from materials containing a minimum of 50 percent recycled content.

#### 2.1.1 Anchors and Fasteners

Provide corrosion-resistant anchors and fasteners capable of developing a restraining force commensurate with the strength of the accessory to be mounted and suited for use with the supporting construction. Provide tamperproof design exposed fasteners with finish to match the accessory. Provide fasteners proposed for use for each type of wall construction and mounting.

#### 2.1.2 Finishes

Except where noted otherwise, provide the following finishes on metal:

| Metal                                 | Finish                  |
|---------------------------------------|-------------------------|
| Stainless steel                       | No. 4 satin finish      |
| Carbon steel, copper alloy, and brass | Chromium plated, bright |

#### 2.1.3 Item A5030 Bench, Stall, Shower, Built-In

Wall mounted shower seat. Frame made of 18 gauge stainless steel with satin finish. Seat made of one piece of 1/2 inch thick nonporous solid phenolic with slots to permit water to drain, secured to frame with stainless steel carriage bolts and acorn nuts. Mounting hardware included. Seat to support a minimum static load of 250 pounds. Hinge seat to fold up when not in use. Seat complies with ADA guidelines.

Approximate size: 34 inches wide by 22 inches deep by 1/2 inch thick.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

2.1.4 Item A5080 Dispenser, Paper Towel, SS, Surface Mounted

Surface mounted unit constructed of stainless steel with satin finish, welded construction, and have full length piano hinge, tumbler lock, refill indicator. Unit has smooth corners, free of burrs and sharp edges. Unit has a capacity of 400 single fold paper towels.

Approximate size: 11 inches wide by 8 inches high by 6 inches deep.

2.1.5 Item A5109 Grab Bar, 1-1/4 Inch Diameter, SS, 2 Wall, W/C Accessible

Grab bar of 1-1/4 inch diameter satin finish stainless steel with peened gripping surface for use in toilet stall/room. Snap-on flange covers for concealed mounting are stainless steel and equipped with two screw holes for attachment to wall. Grab bars designed to meet and exceed ADA requirements for structural strength. Grab bars designed to withstand loads of 900 pounds when properly installed. Clearance from wall to grab bar is 1-1/2 inches to meet ADA and ANSI codes.

2.1.6 Item A5110 Grab Bar, 1-1/4 Inch Diameter, SS, 2 Wall, Shower Use

Grab bar of 1-1/4 inch diameter satin finish stainless steel with peened gripping surface. Snap-on flange covers for concealed mounting stainless steel. Bent ends of tubing pass through the flanges and are Heliarc welded for maximum strength. Grab bars designed to meet and exceed ADA requirements for structural strength. Grab bars designed to withstand loads of 900 pounds when properly installed. Clearance from wall to grab bar is 1-1/2 inches to meet ADA and ANSI codes.

2.1.7 Item A5135 Shelf, Utility W/ Mop/Broom Holders, SS, Surf Mounted

Surface mounted mop/broom holder with shelf made of 18 gauge stainless steel with all exposed surfaces in satin finish. Unit has shelf 8 inches deep with shelf support brackets of satin finish stainless steel welded to mounting base, and a minimum of 3 hooks/3 holders. Mop holders have spring-loaded rubber cams and hold mop or broom handle with a diameter between 5/8 inch and 1 inch.

Approximate size: 36 inches wide by 8 inches deep.

2.1.8 Item A5140 Hook, Garment, Security

Surface mounted safety hook made of stainless steel and secured to wall with tamper resistant mounting screws, exposed mounting. Mounting hardware to be included. Hook designed to snap down when it exceeds load limit.

2.1.9 Item A5170 Rod, Shower Curtain, 1 Inch Diameter, W/Curtain & Hooks

Shower Curtain Rod with concealed mounting. Shower curtain rod made of satin finish stainless steel, 1 inch diameter, with flanges included, and have white vinyl shower curtain, 72 inches high, and stainless steel curtain hooks. Shower curtain has corrosion resistant grommets, reinforced heading, and treated with antibacterial and flame retardant agents. Shower hooks are stainless steel. Length as indicated on drawings.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

2.1.10 Item A5195 Dispenser, Toilet Tissue, SS, 1-Roll, Surface Mounted

Concealed surface mounted single roll toilet tissue dispenser of satin finish stainless steel. Spindle to be free-spinning for non-controlled delivery, chrome-plated high impact resistant plastic and equipped with heavy-duty internal spring. Unit accommodates standard core toilet paper roll up to 5-1/2 inches diameter. Mounting hardware included.

Approximate size: 7-3/4 inches wide by 2 inches high by 4 inches deep.

2.1.11 Item A5202 Soap Dispenser, wall-mount

Vertical type, SS, 1200ml minimum capacity, corrosion-resistant valve.

Approximate size of shelf: 18 inches wide by 5 inches deep.

2.1.12 Item A5205 Mirror

Electro-copper plated plate glass, SS frame, 24" x 36".

2.1.13 Item A5207 Combination Towel Dispenser / Waste Receptacle, SS, Surface Mounted,

400 C-fold - single fold or quarter fold, key lock for towel door.

2.1.14 Item A5209 Toilet Seat Cover Dispenser, Surface Mounted

SS Dispenser with 250 capacity.

Approximate size: 13 inches wide by 12 inches high by 6 inches deep.

PART 3 EXECUTION

3.1 INSTALLATION

Do not install items that show visual evidence of biological growth. Provide the same finish for the surfaces of fastening devices exposed after installation as the attached accessory. Provide oval exposed screw heads. Install accessories at the location and height indicated. Protect exposed surfaces of accessories with strippable plastic or by other means until the installation is accepted. After acceptance of accessories, remove and dispose of strippable plastic protection. Coordinate accessory manufacturer's mounting details with other trades as their work progresses. Use sealants for brackets, plates, anchoring devices and similar items in showers (a silicone sealant specified in Section 07 92 00 JOINT SEALANTS) as they are set to provide a watertight installation. After installation, thoroughly clean exposed surfaces and restore damaged work to its original condition or replace with new work.

3.1.1 Recessed Accessories

Fasten accessories with wood screws to studs, blocking or rough frame in wood construction. Set anchors in mortar in masonry construction. Fasten to metal studs or framing with sheet metal screws in metal construction.

3.1.2 Surface Mounted Accessories

Mount on concealed backplates, unless specified otherwise. Conceal fasteners on accessories without backplates. Install accessories with

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

corrosion-resistant fasteners as required by the construction. Install backplates in the same manner, or provide with lugs or anchors set in mortar, as required by the construction. Fasten accessories mounted on gypsum board and plaster walls without solid backing into the metal or wood studs, or to backplates secured to metal studs.

### 3.2 CLEANING

Clean material in accordance with manufacturer's recommendations. Do not use alkaline or abrasive agents. Take precautions to avoid scratching or marring exposed surfaces.

-- End of Section --

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## SECTION 10 44 16

### FIRE EXTINGUISHERS 11/19

#### PART 1 GENERAL

##### 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

#### NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

|          |                                                                                                                                               |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| NFPA 1   | (2021) Fire Code                                                                                                                              |
| NFPA 10  | (2022; ERTA 1 2021) Standard for Portable Fire Extinguishers                                                                                  |
| NFPA 101 | (2021; TIA 21-1) Life Safety Code                                                                                                             |
| NFPA 505 | (2018) Fire Safety Standard for Powered Industrial Trucks Including Type Designations, Areas of Use, Conversions, Maintenance, and Operations |

#### U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

|                 |                                    |
|-----------------|------------------------------------|
| 29 CFR 1910.157 | (2003) Portable Fire Extinguishers |
|-----------------|------------------------------------|

#### UNDERWRITERS LABORATORIES (UL)

|        |                                                  |
|--------|--------------------------------------------------|
| UL 299 | (2012; May 2021) Dry Chemical Fire Extinguishers |
|--------|--------------------------------------------------|

##### 1.2 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

#### SD-02 Shop Drawings

Fire Extinguishers

Accessories

Cabinets

Wall Brackets

Schedule

#### SD-03 Product Data

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

Fire Extinguishers

Accessories

Cabinets

Wall Brackets

Replacement Parts List

SD-07 Certificates

Fire Extinguishers Certifications

Manufacturer's Warranty with Inspection Tag

### 1.3 DELIVERY, STORAGE, AND HANDLING

Protect materials from weather, soil, and damage during delivery, storage, and construction.

Deliver materials in their original packages, containers, or bundles bearing the brand name and the name and type of the material.

Provide portable fire extinguishers in compliance with NFPA 505 for all ancillary vehicles where Fire Safety Standard for Powered Industrial Trucks, including type designations, special conditions relating to areas of use, conversions, maintenance, or specific operations apply.

### 1.4 WARRANTY

Guarantee that Fire Extinguishers are free of defects in materials, fabrication, finish, and installation and that they will remain so for a period of not less than five (5) years after completion.

Submit the manufacturer's warranty with inspection tag.

### 1.5 PROJECT SCHEDULE

For fire extinguishers. Coordinate final fire extinguisher schedule with fire protection cabinet schedule to ensure proper fit and function. Use same designations indicated on Drawings.

## PART 2 PRODUCTS

Submit fabrication drawings consisting of fabrication and assembly details performed in the factory and product data for the following items: Fire Extinguishers; Accessories, cabinets, and Wall Brackets.

### 2.1 SYSTEM DESCRIPTION

#### 2.1.1 Types

Submit fire extinguishers certifications showing compliance with local codes and regulations.

Provide fire extinguishers conforming to NFPA 10. Provide quantity and placement in compliance with the applicable sections of NFPA 1, NFPA 101, and 29 CFR 1910.157.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

Provide dry chemical type fire extinguishers compliant with UL 299.

#### 2.1.2 Material

Provide corrosion-resistant steel extinguisher shell.

#### 2.1.3 Size

Dimension cabinets to accommodate the specified fire extinguisher.

#### 2.1.4 Accessories

### 2.2 EQUIPMENT

#### 2.2.1 Cabinets

##### 2.2.1.1 Material

Provide corrosion-resistant steel cabinets.

##### 2.2.1.2 Type

Provided cabinet type for walls as scheduled on Drawings.

##### 2.2.1.3 Size

Dimension cabinets to accommodate the specified fire extinguishers.

#### 2.2.2 Wall Brackets

Provide spring-clip fire extinguisher wall brackets.

Provide wall bracket and accessories as approved.

##### 2.2.2.1 Identification

Provide lettering complying with authorities having jurisdiction for letter style, size, spacing, and location. Locate as indicated by the drawings.

Identify bracket-mounted fire extinguishers with the words "FIRE EXTINGUISHER" in red letter decals applied to mounting surface.

Orientation: Vertical.

## PART 3 EXECUTION

### 3.1 INSTALLATION

Install Fire Extinguishers where indicated on the drawings. Verify exact locations prior to installation.

Provide extinguishers which are fully charged and ready for operation upon installation. Provide extinguishers complete with Manufacturer's Warranty with Inspection Tag attached.

Install fire extinguishers in locations indicated and in compliance with requirements of authorities having jurisdiction.



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

Comply with the manufacturer's recommendations for all installations.

### 3.2 PROTECTION

#### 3.2.1 Repairing

Remove and replace damaged and unacceptable portions of completed work with new work at no additional cost to the Government.

Submit replacement parts list indicating specified items replacement part, replacement cost, and name, address and contact for replacement parts distributor.

#### 3.2.2 Cleaning

Clean all surfaces of the work, and adjacent surfaces which are soiled as a result of the work. Remove from the site all construction equipment, tools, surplus materials and rubbish resulting from the work.

-- End of Section --

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## SECTION 10 51 13

### PPE LOCKERS 05/11

#### PART 1 GENERAL

##### 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

##### ASTM INTERNATIONAL (ASTM)

|                   |                                                                                                                                                                                                     |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM A1008/A1008M | (2023) Standard Specification for Steel, Sheet, Cold-Rolled, Carbon, Structural, High-Strength Low-Alloy, High-Strength Low-Alloy with Improved Formability, Solution Hardened, and Bake Hardenable |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|           |                                                                                                                                  |
|-----------|----------------------------------------------------------------------------------------------------------------------------------|
| ASTM B456 | (2017; R 2022) Standard Specification for Electrodeposited Coatings of Copper Plus Nickel Plus Chromium and Nickel Plus Chromium |
|-----------|----------------------------------------------------------------------------------------------------------------------------------|

##### U.S. GENERAL SERVICES ADMINISTRATION (GSA)

|               |                                  |
|---------------|----------------------------------|
| FS AA-L-00486 | (Rev J) Lockers, Clothing, Steel |
|---------------|----------------------------------|

##### 1.2 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

##### SD-02 Shop Drawings

Types; G

Location; G

Installation

##### SD-03 Product Data

Material

Finish

Locker components

Assembly instructions

##### SD-04 Samples

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

Color chips; G

### 1.3 DELIVERY, HANDLING, AND STORAGE

Deliver lockers and associated materials in their original packages, containers, or bundles bearing the manufacturer's name and the name of the material. Protect from weather, soil, and damage during delivery, storage, and construction.

### 1.4 FIELD MEASUREMENTS

To ensure proper fits, make field measurements prior to the preparation of drawings and fabrication. Verify correct location

### 1.5 QUALITY ASSURANCE

#### 1.5.1 Color Chips

Provide a minimum of three color chips, not less than 3 inches square, of each color indicated.

Government may request performance-characteristic tests on assembled lockers. Tests and results must conform to FS AA-L-00486. Lockers not conforming will be rejected.

## PART 2 PRODUCTS

### 2.1 TYPES

Locker must have the following type and size in the location and quantities indicated. Locker finish colors will be as scheduled.

#### 2.1.1 Double-Tier

Double-tier lockers must be as follows:

Double-tier locker 24 inches wide, 20 inches deep, and 72 inches high, attached to 6-inch high legs

### 2.2 MATERIAL

#### 2.2.1 Steel Sheet (Perforated Sides and Doors)

ASTM A1008/A1008M, commercial quality, minimized spangle material. Prepare material surfaces for baked enamel finishing in accordance with FS AA-L-00486. Fabricate locker bodies from not less than 0.0239-inch thick steel sheet.

#### 2.2.2 Chromium Coating

Nickel and chromium electrodeposited on the specified base metal. Conform to ASTM B456, SC-3, as applicable to the base metal.

#### 2.2.3 Finish

FS AA-L-00486.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### 2.2.3.1 Color

Double-Tier PPE Lockers: Red.

### 2.3 COMPONENTS

#### 2.3.1 Coat Hooks

FS AA-L-00486, chromium plated.

#### 2.3.2 Shelves

FS AA-L-00486. Fabricate from not less than 0.0598 inch thick steel sheet.

#### 2.3.3 Base Panels

FS AA-L-00486.

#### 2.3.4 Fastening Devices

Provide bolts, nuts, and rivets as specified in FS AA-L-00486.

## PART 3 EXECUTION

### 3.1 ASSEMBLY AND INSTALLATION

Assemble lockers according to the locker manufacturer's instructions. Align lockers horizontally and vertically. Secure lockers to wall and base with screws as indicated. Bolt adjacent lockers together. Adjust doors to operate freely without sticking or binding and to ensure they close tightly.

### 3.2 FIELD QUALITY CONTROL

#### 3.2.1 Testing

Government may request performance-characteristic tests on assembled lockers in accordance with FS AA-L-00486. Lockers not conforming will be rejected.

#### 3.2.2 Repairing

Remove and replace damaged and unacceptable portions of completed work with new.

#### 3.2.3 Cleaning

Clean surfaces of the work, and adjacent surfaces soiled as a result of the work, in an approved manner. Remove equipment, surplus materials, and rubbish from the site.

-- End of Section --

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## SECTION 10 56 13

### STEEL SHELVING 04/06

#### PART 1 GENERAL

##### 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

#### MATERIAL HANDLING INDUSTRY OF AMERICA (MHI)

MHI MH28.1 (1997) Specification: Industrial Steel  
Grade Shelving

##### 1.2 DEFINITIONS

For the purposes of this specification the shelf category, "medium weight," "heavy weight," will be as follows. Load is given per shelf in pounds for evenly distributed load. This does not limit the shelf size, only the shelving category.

| Minimum Evenly Distributed Load Per Shelf in Pounds |                     |                    |
|-----------------------------------------------------|---------------------|--------------------|
| Shelf Size                                          | Type<br>Medium Duty | Type<br>Heavy Duty |
| 18 by 36 in.                                        | 700                 | 1300               |
| 18 by 48 in.                                        | 500                 | 900                |

##### 1.3 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

#### SD-01 Preconstruction Submittals

Shelving Units

#### SD-03 Product Data

Shelving Units

Accessories

Installation instructions

#### SD-04 Samples

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

Finish

SD-06 Test Reports

Shelving Units

Finish

#### 1.4 DELIVERY, STORAGE, AND HANDLING

Deliver materials in original packages, containers or bundles bearing the brand name and identification of the manufacturer. Store inside under cover. Protect surfaces from damage.

### PART 2 PRODUCTS

#### 2.1 MANUFACTURED UNITS

MHI MH28.1. Provide shelving units indicated . Provide shelving units designed for full dead and live load, designated heavy duty.

#### 2.2 ACCESSORIES

- a. Drawers, 400 pound capacity, and mounting brackets
- b. Partitions and dividers
- c. Label holder 2 1/4 by 3/4 inches.

#### 2.3 FINISH

Provide the shelving units in the manufacturer's standard colors as indicated. Clean metal by multiple stage phosphatizing and sealing process, for rust resistance and paint adhesion. Provide electrostatically applied enamel finish coats, baked hard for a minimum of 30 minutes at 300 degrees F.

#### 2.4 SOURCE QUALITY CONTROL

- a. MHI MH28.1, for tests of shelf capacity, lateral stability and shelf connections.

### PART 3 EXECUTION

#### 3.1 EXAMINATION

Before installation, examine shelving units for dents and scratches. Replace damaged shelving.

#### 3.2 INSTALLATION

Install shelving according to manufacturer's installation instructions.

#### 3.3 PROTECTION

Cover and protect shelving from damage during the completion of construction. Remove prior to acceptance of project.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

-- End of Section --

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## SECTION 11 24 00

### ROOFTOP FALL PROTECTION EQUIPMENT 01/23

#### PART 1 GENERAL

##### 1.1 RELATED DOCUMENTS

Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

##### 1.2 SUMMARY

Roof mounted fall protection system.

##### 1.3 COORDINATION

Coordinate layout and installation of fall protection system with roof structure

##### 1.4 REFERENCES

###### AMERICAN INSTITUTE OF STEEL CONSTRUCTION (AISC)

AISC 325 (2017) Steel Construction Manual

###### AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI Z359 (2020) Fall Protection Code

###### AMERICAN SOCIETY OF SAFETY PROFESSIONALS (ASSP)

ASSP Z359.6 (2016) Specifications and Design  
Requirements for Active Fall Protection  
Systems

###### ASTM INTERNATIONAL (ASTM)

ASTM A666 (2015) Standard Specification for Annealed  
or Cold-Worked Austenitic Stainless Steel  
Sheet, Strip, Plate and Flat Bar

###### U.S. DEPARTMENT OF DEFENSE (DOD)

UFC 3-301-01 (2019, with Change 1, 2022) Structural  
Engineering

###### OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION REGULATIONS

OSHA 29 CFR Standards - 29 CFR 1910.140 Personal  
Fall Protection Systems

The publications listed above form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## 1.5 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

### SD-01 Preconstruction Submittals

#### Manufacturer's Qualifications

### SD-02 Shop Drawings

#### Fall Protection System; G

Include plans, elevations, keyed details, and attachments to other work. Indicate dimensions, loadings, and special conditions. Distinguish between plant- and field-assembled work. Indicate size and location (refer to drawings for general locations), method of attachment to structure, coordination with other systems and required clearances.

### SD-03 Product Data

#### Fall Protection System; G

Submit product data for system and all accessories. Include construction details, material descriptions, dimensions of individual components and profiles, and finishes. Provide capacities, performance, standard use, and applied forces to system.

### SD-04 Design Data

#### Delegated-Design Submittal

#### Color Chip Samples; G

### SD-05 Operation and Maintenance Data

Fall Protection System, all components. Provide Manufacturer's assembly, operation, and maintenance instruction manual, preventive maintenance practices with included assembly drawings.

### SD-06 Closeout Submittals

#### Record Drawings; G

Include Training for Government Employees  
and Preventive Maintenance Practices

### SD-07 Certificates

Submit Manufacturer Warranty and Ensure That Forms Have Been Completed In Owner's Name And Registered With Manufacturer.

#### Qualification Of Professional Engineer

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### Qualification of Installer

Warranty; G

## 1.6 QUALITY ASSURANCE

Standard system shall be designed, fabricated, and installed in accordance with ANSI Z359 Specifications and Design Requirements for Active Fall Protection Systems, OSHA 29 CFR 1910.140(c), and AISC 325.

FALL PROTECTION SYSTEM IS TO BE USED ON LOW-SLOPE ROOF AREAS WHERE ROOF PERIMETER HAS NO GUARDRAIL SYSTEM OR NO PARAPET REACHING 42 INCHES ABOVE THE LEVEL OF THE ROOF.

### 1.6.1 MANUFACTURER'S QUALIFICATIONS

An ISO 9001 registered company with more than 20 years of experience successfully designing and manufacturing fall protection solutions.

### 1.6.2 Qualification of Installer

A company that is acceptable to the manufacturer and meets OSHA 1910.140(c) requirements for a Qualified Person assembling and installing fall protection systems for multiple applications.

## 1.7 DELEGATED-DESIGN SUBMITTAL

Validate structure, mounting, and securing complies with the performance requirements and design criteria, including analysis data signed and sealed by the qualified U.S licensed professional engineer responsible for their preparation.

Design, furnish, install, and certify a complete and useable permanent fall protection system, inclusive of all components, appurtenances, fasteners, end stops, carriages, and equipment necessary for safety of maintenance workers. Salient characteristics which will be reviewed under best value source selection include:

1. Compliance with all Occupational Health and Safety regulations.
2. Design system to accommodate 3 users at one time.
3. Design must be performed, signed, and sealed by a Professional Engineer from the manufacturer experienced in the design of horizontal lifeline fall arrest systems. Submit qualifications of Professional Engineer.
4. Design must provide for access from the roof hatch (where provided), allowing for continuous worker tie off in areas where there is a fall risk. For buildings without a roof hatch, design must provide for access from a roof ladder (at an acceptable location), allowing for continuous worker tie off in areas where there is a fall risk.
5. Low profile and architecturally pleasing systems are preferred. Powder coated or anodized aluminum alloy are preferred. Concealed fasteners are preferred.
6. After fall protection system is installed, the manufacturer's authorized representative shall inspect and operate the system and make any final adjustments. The manufacturer's authorized representative shall

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

issue a certificate attesting to the system's design and installation, and formally submit to the USG.

7. Fall protection system shall be installed under the direction of manufacturer's authorized trained personnel.

8. Train designated US Government personnel on-site regarding proper use of system.

9. Submit horizontal life line and anchorage connector systems for approval by the designer of record and the installation public works group's designated fall protection expert. The chosen fall protection system must be compatible with the Base Exterior Architectural Plan, low-maintenance, and compliant with ANSI/ASSE Z359 requirements. The following proprietary systems would meet government requirements, but other options may be proposed by the contractor, subject to final approval by the government:

- a. Linear Access Solutions - Rooftop Anchor, Fall Protection Systems
- b. Miller ShockFusion Horizontal Lifeline Roof System (BASIS OF DESIGN).
- c. Kee Safety KeeLine Engineered Horizontal Lifeline Solution
- d. 3M Capital Safety Roofsafe Horizontal Rail System -  
(3M SYSTEM FOR METAL STANDING SEAM ROOFS ONLY).

10. The systems shall be designed to be supported by and integral with the roofing system.

11. Fall protection systems shall be designed for three concurrent users. Provide layout as shown on drawings, unless approved otherwise.

12. Design all fall protection systems to safely resist the dynamically applied loads while maintaining a safety factor of two against failure.

13. Design of fall protection systems shall be based on the understanding that the underlying structural steel supports are provided as part of the structure and are adequate to support the imposed loads. Fall Protection loads shall be furnished to the building design structural engineer of record.

14. Design Engineer shall prepare a fall clearance analysis verifying adequate fall distance to safely stop the worker in the event of a fall.

15. System shall be designed for three workers falling at a time with full body harness and shock absorbing lifeline with a maximum arresting force of 900 pounds per worker.

16. Design Engineer shall coordinate Fall Protection System with mechanical and lightning protection systems, and other components of the roofing.

17. Color must comply with the Base Exterior Architectural Plan. Submit color chip samples from manufacturer's full line of standard colors for approval.

18. All fall protection anchorages must be designed for a minimum 3,100 lb nominal live load capacity in accordance with UFC 3-301-01 chapter 2-4.2 section 1604.11. Anchorages, including any supporting structure, must be designed to allow future replacement of the horizontal lifeline system or anchorage connectors regardless of the required capacity of the

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

chosen horizontal lifeline system or anchorage connectors. Anchorages that support two or more independent horizontal life line spans at a single point must be designed for the combined load effects of the horizontal lifeline systems in accordance with ANSI/ASSP Z359.6.

## 1.8 CONDITIONS / DELIVERY, STORAGE, AND HANDLING

### 1.8.1 FIELD MEASUREMENTS

Verify field measurements and indicate measurements on Shop Drawings to ensure required fit.

### 1.8.2 PROJECT CONDITIONS

Manufacturer shall consider extreme environmental factors present at project location when selecting materials and products. Products will be exposed to salt air in the tropical marine environment.

Do not install products under environmental conditions outside manufacturer's absolute limits.

### 1.8.3 DELIVERY, STORAGE, AND HANDLING

- a. Store products in manufacturer's packaging until ready for installation.
- b. Store and dispose of solvent-based materials in accordance with requirements of local authorities.

## 1.9 WARRANTY

### 1.9.1 MANUFACTURER'S WARRANTY

Included on manufacturer's standard form and outlines the manufacturer's agreement to repair or replace assemblies and components that fail in materials and/or execution within warranty period from date of substantial completion.

1. Warranty covers all components to be free from defects in material and workmanship for a period of ten (10) years.

## PART 2 PRODUCTS

### 2.1 SYSTEM DESCRIPTION: SAFETY LINE FALL PROTECTION SYSTEM

Allow users to enter the roof surface near the ridge and walk uninterrupted the entire area of the roof and provide secure anchorage to arrest a fall. The Safety Line System is to allow attachment at any point along the cable and enables freedom of movement along the stainless steel cable as it passes by intermediate anchors. Maximum span of 50 feet (15 m) between anchors and provides continuous hands free access for the user of the roof fall protection system.

#### 2.1.1 ACCESSORIES

Two (2) sets of Self-Retracting Lanyards and Harnesses.

#### 2.1.2 SOURCE LIMITATION

1. Obtain fall protection system and components from a single

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

manufacturer.

## 2.2 PERFORMANCE REQUIREMENTS

Provide fall protection anchors as indicated on drawings For one user, the system shall be designed based on a worst-case placement of the 5000-pound maximum arresting force.

## 2.3 Materials

Primary cable assembly components: Stainless Steel: ASTM A666, Type 316  
Aluminum: 6061 or 6082 aluminum alloy  
Connectors: Comply with OSHA regulation 1910.140 Personal Fall Protection Systems and 1926.502 Fall Protection Systems Criteria and Practices.

## PART 3 EXECUTION

### 3.1 PREPARATION

DO NOT start installation until support structures are properly prepared. Check materials to ensure all parts are present.

Prepare substrates using the methods recommended by the manufacturer for achieving the best result for the substrate under the project conditions.

#### 3.1.1 COORDINATION

Contractor to coordinate layout of fall protection system and clip Connections prior to starting construction.

### 3.2 INSTALLATION

Units and accessories should be installed in accordance with manufacturer's Assembly and Operation Instructions Manual. Exposed work shall be true to line and level with accurate angles, surfaces and with straight square edges. Coordinate anchorage system with supporting structure.

Do not load or stress system until materials and fasteners are properly installed and ready for service.

DO NOT USE UNTIL TRAINED IN THE USE OF THE SYSTEM.

### 3.3 FIELD QUALITY CONTROL

Provide manufacturer's certified installer to inspect installed fall protection system. Provide Record Drawings for existing conditions at closeout.

### 3.4 CLOSEOUT ACTIVITIES

#### 3.4.1 Demonstration

Demonstrate operation of system to Owner's personnel. Briefly describe function, operation, and maintenance of each component.

#### 3.4.2 Training

Train Owner's personnel on operation and maintenance of system.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

1. Use operation and maintenance manual as training reference, supplemented with additional training materials as required.
2. Provide minimum of two hours of training.
3. Provide training at the lifeline installation site.
4. Training to take place at the completion of the installation.

-- End of Section --

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## SECTION 12 24 13

### ROLLER WINDOW SHADES

**08/20**

#### PART 1 GENERAL

##### 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to in the text by basic designation only.

##### AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/WCMA A100.1 (2018) American National Standard for  
Safety of Window Covering Products

##### ASTM INTERNATIONAL (ASTM)

ASTM G21 (2015; R 2021; E 2021) Standard Practice  
for Determining Resistance of Synthetic  
Polymeric Materials to Fungi

##### NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 701 (2023; ERTA 1 2023) Standard Methods of  
Fire Tests for Flame Propagation of  
Textiles and Films

##### UNDERWRITERS LABORATORIES (UL)

UL 325 (2017; Reprint Feb 2020) UL Standard for  
Safety Door, Drapery, Gate, Louver, and  
Window Operators and Systems

##### 1.2 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES

##### SD-02 Shop Drawings

Detailed Drawings; G

Location Schedule; G

##### SD-03 Product Data

Window Shades; G

##### SD-04 Samples

Window Shades; G

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### SD-06 Test Reports

##### Flammability Requirements

#### SD-10 Operation and Maintenance Data

##### Window Shades, Data Package 1; G

#### SD-11 Closeout Submittals

Submit Data Package 1 for roller window shades, and Data Package 2 for electrical operators, in accordance with Section 01 78 23 OPERATIONS AND MAINTENANCE DATA.

### 1.3 QUALITY ASSURANCE

#### 1.3.1 Qualifications

##### 1.3.1.1 Installer's Qualifications

Installer trained and certified by the manufacturer with a minimum of ten years of experience in installing products comparable to those specified in this section.

#### 1.3.2 Flammability Requirements

Passes in accordance with NFPA 701 small and large-scale vertical burn. Materials tested are identical to products proposed for use.

#### 1.3.3 Electrical Requirements

NFPA Article 100 listed and labeled in accordance with UL 325 or other testing agency acceptable to authorities having jurisdiction, marked for intended use, and tested as a system. Individual testing of components is not acceptable in lieu of system testing.

#### 1.3.4 Anti-Microbial Requirements

'No Growth' per ASTM G21 results for fungi ATCC9642, ATCC 9644, ATCC9645.

### 1.4 DELIVERY, STORAGE, AND HANDLING

Deliver components to the jobsite in the manufacturer's original packaging with the brand or company name, item identification, and project reference clearly marked. Store components in a dry location that is adequately ventilated and free from dust, water, or other contaminants and has easy access for inspection and handling. Store materials flat in a clean dry area with temperature maintained above 50 degrees F. Do not open containers until needed for installation unless verification inspection is required. Handle and store shades in accordance with manufacturer's recommendations.

### 1.5 WARRANTY

Provide manufacturer's warranty to repair or replace defective materials and workmanship for a period of 10 years from date of final acceptance of the work.



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## PART 2 PRODUCTS

### 2.1 WINDOW SHADES

Submit drawings showing plans, elevations, sections, product details, installation details, operational clearances, wiring diagrams and relationship to work. Submit a location schedule showing location, size and quantity of shades. Include the use of same room designations as indicated on the drawings.

Provide product data composed of catalog cuts, brochures, and operating and maintenance instructions on each product to be used. Include styles, profiles and features.

Furnish samples of each type and color of roller shade fabric and roller shade channel. Provide shade material minimum 6 by 6 inches in size. Mark face of material to indicate interior faces.

Mock up: Install shade in area designated by Contracting Officer. Do not proceed with remaining work until the Contracting Officer approves workmanship and operation. Rework mock up as required to produce acceptable work. The approved shade can be used in the installation.

Submit fire resistance data, flame spread and smoke contribution data.

Provide roller tube that operates smoothly and of sufficient diameter and thickness to prevent excessive deflection. Provide brackets that are appropriate for inside mount. Provide shade cloth meeting the performance described in NFPA 701, small scale test. Treat steel features for corrosion resistance.

#### 2.1.1 Manufacturer's Qualifications

Obtain motor-controlled roller shades through one source from a single manufacturer with a minimum of twenty years of experience and minimum of three projects of similar scope and size in manufacturing products comparable to those specified in this section. Furnish manual and motorized shades produced by the same manufacturer to provide matching appearance.

#### 2.1.2 Manually Operated Shades with Single Rollers

##### 2.1.2.1 Chain-and-Clutch Operating Mechanisms

Provide continuous-loop bead chain and clutch that stops shade movement when bead chain is released; shade to be permanently adjusted and lubricated.

##### 2.1.2.2 Bead Chains

Provide bead chain from #10 stainless steel rated to 90 lb. minimum breaking strength with pull chain tensioning device complying with ANSI/WCMA A100.1

- a. Loop Length: Full length of roller shade.
- b. Limit Stops: Allows shade to stop when chain is released. Provide

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

limit stops to prevent shade from being raised or lowered too far.

- c. Chain-Retainer Type: Chain tensioner, jamb mounted.

#### 2.1.2.3 Rollers

Provide corrosion-resistant steel or extruded-aluminum tubes of diameters and wall thicknesses required to accommodate operating mechanisms and weights and widths of shade bands indicated without deflection. Provide with permanently lubricated drive-end assemblies and idle-end assemblies designed to facilitate removal of shade cloth for service.

- a. Roller Drive-End Location: Right side of interior face of shade.
- b. Direction of Shade cloth Roll: Regular, from back (exterior face) of roller.
- c. Shade cloth-to-Roller Attachment: Removable spline fitting into integral channel in tube. Adhesive attachment is not acceptable.

#### 2.1.2.4 Mounting Hardware

Provide corrosion resistant brackets or endcaps compatible with roller assembly, operating mechanism, installation accessories, and mounting location and conditions indicated. Provide hardware that allows for field adjustment or removal of shade roller tube and other operable hardware component without removal of brackets and end or center supports.

#### 2.1.2.5 Shade Cloth

- a. Shade Material: Light-filtering fabric: Openness 5 percent.
- b. Shade Cloth Bottom (Hem) Bar: Steel or extruded aluminum. Provide shade bar enclosed in sealed pocket of shade band material.

#### 2.1.2.6 Installation Accessories

- a. Front Fascia: L-shaped aluminum extrusion to conceal shade roller and hardware that snaps onto end caps without requiring exposed fasteners of any kind. Fascia can be mounted continuously across two or more shade bands. Provide manufacturers standard height fascia as required to conceal roller and shade band assembly when shade is fully open.
- b. Exposed Headbox: Rectangular, extruded-aluminum enclosure including front fascia, top and back covers, endcaps, and removable bottom closure. Provide manufacturers standard height fascia as required to conceal roller and shade band assembly when shade is fully open.
- c. Endcaps: Extruded aluminum with universal design suitable for mounting to window mullions. Provide size compatible with roller size. Provide end cap covers matching fascia/headbox finish.
- e. Closure Panel and Wall Clip: Removable aluminum panel designed for installation at bottom of site-constructed ceiling recess or pocket and for snap-in attachment to wall clip without fasteners.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## 2.2 COLOR

Provide color, pattern and texture for metal trim and shade fabric as indicated; colors listed are not intended to limit the selection of equal colors from other manufacturers.

## PART 3 EXECUTION

### 3.1 FIELD MEASUREMENTS

After becoming familiar with details of the work, verify all dimensions in the field, and advise the Contracting Officer of any discrepancy before performing the work.

### 3.2 INSTALLATION

Do not install building construction materials that show visual evidence of biological growth.

Provide roller window shades, complete with necessary brackets, fittings, and hardware as indicated.

Perform installation in accordance with the approved detailed drawings and manufacturer's installation instructions. Install units level, plumb, secure, and at proper height and location relative to window units. Provide and install supplementary or miscellaneous items in total, including clips, brackets, or anchorages incidental to or necessary for a sound, secure, and complete installation. Do not start installation until completion of room painting and finishing operations.

### 3.3 CLEAN-UP

Upon completion of the installation, clean window treatments and exposed components as recommended by manufacturer. Adjust window treatment for form and appearance and proper operating condition. Repair or replace damaged units as directed by the Contracting Officer. Isolate metal parts from direct contact with concrete, mortar, or dissimilar metals. Ensure shades installed in recessed pockets can be removed without disturbing the pocket. The entire shade, when retracted, is contained inside the pocket. For shades installed outside the jambs and mullions, overlap each jamb and mullion 0.75 inch or more when the jamb and mullion sizes permit. Include all hardware, brackets, anchors, fasteners, and accessories necessary for a complete, finished installation.

-- End of Section --

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## SECTION 12 48 13

### ENTRANCE FLOOR MATS AND FRAMES (RECESSED SS GRILLS)

**08/23**

#### PART 1 GENERAL

##### 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

##### ASTM INTERNATIONAL (ASTM)

ASTM D2047 (2017) Standard Test Method for Static Coefficient of Friction of Polish-Coated Floor Surfaces as Measured by the James Machine

ASTM E648 (2023) Standard Test Method for Critical Radiant Flux of Floor-Covering Systems Using a Radiant Heat Energy Source

##### U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

36 CFR 1191 Americans with Disabilities Act (ADA) Accessibility Guidelines for Buildings and Facilities; Architectural Barriers Act (ABA) Accessibility Guidelines

##### 1.2 SUSTAINABILITY REPORTING

Materials in this technical specification may increase contract compliance with sustainability requirements.

##### 1.3 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

##### SD-02 Shop Drawings

Installation Drawings; G

Detail Drawings; G

##### SD-03 Product Data

Entrance Floor Mats and Frames; G

Adhesives and Concrete Primers; G

##### SD-04 Samples

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

Entrance Floor Mats and Frames; G

SD-08 Manufacturer's Instructions

Manufacturer's Instructions

SD-10 Operation and Maintenance Data

Protection, Maintenance, and Repair Information

#### 1.4 QUALITY CONTROL

Comply with 36 CFR 1191 Americans with Disabilities Act (ADA) Accessibility Guidelines for Buildings and Facilities; Architectural Barriers Act (ABA) Accessibility Guidelines for installed entrance floor mats and frames. Ensure that entrance floor mats and frames are slip-resistant in accordance with ASTM D2047, with a minimum 0.60 coefficient of friction, for accessible routes and are structurally capable of withstanding a uniform floor load of 300 lb/sq ft. Ensure that flammability is in accordance with ASTM E648, Class 1, Critical Radiant Flux, minimum 0.45 watts/square centimeter.

#### 1.5 DELIVERY, STORAGE, AND HANDLING

Deliver materials to the project site in their original packages or containers bearing labels clearly identifying the manufacturer, brand name, and quality or grade.

Store materials in their original unbroken packages or containers in the area in which they will be installed. Unwrap, inspect, and place mats at indicated locations. Remove all excess packing materials.

### PART 2 PRODUCTS

#### 2.1 MANUFACTURED UNITS

##### 2.1.1 Entrance Floor Mats and Frames

Submit the manufacturer's catalog data. Submit samples of assembled sections of floor mats showing corners, intersections, and other details of construction. Submit samples of custom graphics, exposed floor mats, frame finishes and accessories.

##### 2.1.1.1 Recessed Floor Grills

Provide manufacturer's standard floor-grille assemblies consisting of treads of type and profile indicated, interlocked or joined together by cross members, and with support legs (if any) and other components needed to produce a complete installation. Basis of Design Product: C/S Construction Specialties, Pedisystems - Gridline G6.

- a. Structural Performance: Provide floor grilles and frames capable of withstanding a wheel load of 500 lb per wheel.
- b. Comply with applicable provisions in the U.S. Architectural & Transportation Barriers Compliance Board's ADA-ABA Accessibility Guidelines for Buildings and Facilities and ICC A117.1 COMM.
- c. Grill shall be manufactured from corrosion resistant Type 304 stainless steel complying with ASTM A276/A276 in

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

- 1 1/8" depth. Wires to be 0.090" (2.28mm) x 0.150" (3.81mm) electronically welded and spaced 0.145 (3.68mm) apart. Unit must withstand 500 lb./wheel loads (load applied to a solid 5" x 2" wide polyurethane wheel, 1000 passes without damage).
- d. Stainless Steel Angle Frame shall be Type 304 Stainless Steel with 1/8" exposed surface with hidden lockdown made from Type 304 Stainless Steel.

#### 2.1.1.2 Recessed Frames

Provide manufacturer's standard frames of size and style for grille type, for permanent recessed installation in subfloor, complete with installation anchorages and accessories. Unless otherwise indicated, fabricate frame of same material and finish as grilles.

- a. Support System: Level Bed Applications - Provide manufacturer's standard, vinyl cushion support system.
- a. Drainage Pit Applications: Provide manufacturer's special deep-pit frame and support extrusion system with intermediate support beams, sized and spaced as recommended by manufacturer for indicated spans and equipped with vinyl support cushions.

#### 2.1.2 Adhesives and Concrete Primers

Provide adhesives and concrete primers, where required, according to the manufacturer's recommendations.

#### 2.1.3 Color and Size

Ensure that color is in accordance with the drawings. Ensure that the size of mat is as indicated.

### PART 3 EXECUTION

#### 3.1 EXAMINATION

Comply with the manufacturer's requirements for substrates and floor conditions affecting installation of floor mats and frames. Ensure that all unsatisfactory conditions have been corrected before installation.

#### 3.2 INSTALLATION

Submit detail drawings, and custom graphics drawings as required. Provide installation drawings. Provide the manufacturer's protection, maintenance, and repair information.

Install floor mats and frames according to manufacturer's instructions. Set mat tops at the height recommended by the manufacturer for the most effective cleaning action. Provide clearance between bottoms of doors and tops of mats.

-- End of Section --

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

SECTION 12 93 00

SITE FURNISHINGS

**08/17, CHG 1: 08/18**

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN INSTITUTE OF STEEL CONSTRUCTION (AISC)

AISC 303 (2016) Code of Standard Practice for Steel Buildings and Bridges

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B18.2.1 (2012; R 2021) Square, Hex, Heavy Hex, and Askew Head Bolts and Hex, Heavy Hex, Hex Flange, Lobed Head, and Lag Screws (Inch Series)

ASME B18.2.2 (2022) Nuts for General Applications: Machine Screw Nuts, and Hex, Square, Hex Flange, and Coupling Nuts (Inch Series)

ASME B18.6.2 (2020) Square Head Set Screws and Slotted Headless Set Screws (Inch Series)

ASME B18.6.3 (2024) Machine Screws, Tapping Screws, and Metallic Drive Screws (Inch Series)

ASME B18.21.1 (2009; R 2016) Washers: Helical Spring-Lock, Tooth Lock, and Plain Washers (Inch Series)

AMERICAN SOCIETY OF SAFETY PROFESSIONALS (ASSP)

ASSP A10.3 (2020) Safety Requirements for Powder-Actuated Fastening Systems American National Standard for Construction and Demolition Operations

AMERICAN WELDING SOCIETY (AWS)

AWS D1.1/D1.1M (2020; Errata 1 2021) Structural Welding Code - Steel

ASTM INTERNATIONAL (ASTM)

ASTM A36/A36M (2019) Standard Specification for Carbon Structural Steel

ASTM A47/A47M (1999; R 2022; E 2022) Standard Specification for Ferritic Malleable Iron

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### Castings

|                   |                                                                                                                                         |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| ASTM A48/A48M     | (2022) Standard Specification for Gray Iron Castings                                                                                    |
| ASTM A53/A53M     | (2024) Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless                                   |
| ASTM A123/A123M   | (2017) Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products                                         |
| ASTM A153/A153M   | (2023) Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware                                                     |
| ASTM A307         | (2021) Standard Specification for Carbon Steel Bolts, Studs, and Threaded Rod 60 000 PSI Tensile Strength                               |
| ASTM A500/A500M   | (2023) Standard Specification for Cold-Formed Welded and Seamless Carbon Steel Structural Tubing in Rounds and Shapes                   |
| ASTM A501/A501M   | (2021) Standard Specification for Hot-Formed Welded and Seamless Carbon Steel Structural Tubing                                         |
| ASTM A653/A653M   | (2023) Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process |
| ASTM B26/B26M     | (2018; E 2018) Standard Specification for Aluminum-Alloy Sand Castings                                                                  |
| ASTM B62          | (2017) Standard Specification for Composition Bronze or Ounce Metal Castings                                                            |
| ASTM B108/B108M   | (2019) Standard Specification for Aluminum-Alloy Permanent Mold Castings                                                                |
| ASTM B209         | (2014) Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate                                                           |
| ASTM B221         | (2021) Standard Specification for Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes                            |
| ASTM D1187/D1187M | (1997; R 2018) Standard Specification for Asphalt-Base Emulsions for Use as Protective Coatings for Metal                               |
| ASTM D3451        | (2006; R 2017) Standard Guide for Testing Coating Powders and Powder Coatings                                                           |



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### FOREST STEWARDSHIP COUNCIL (FSC)

FSC STD 01 001 (2015) Principles and Criteria for Forest Stewardship

#### SOCIETY FOR PROTECTIVE COATINGS (SSPC)

SSPC Paint 25 (1997; E 2004) Zinc Oxide, Alkyd, Linseed Oil Primer for Use Over Hand Cleaned Steel, Type I and Type II

### 1.2 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

#### SD-02 Shop Drawings

Bollards; FIO

Assembly Instruction Drawings

#### SD-03 Product Data

Bollards

### 1.3 CERTIFICATIONS

#### 1.3.1 Certified Sustainably Harvested Wood

Provide wood certified as sustainably harvested by FSC STD 01 001. Provide a letter of Certification of Sustainably Harvested Wood signed by the wood supplier. Identify certifying organization and their third party program name and indicate compliance with chain-of-custody program requirements. Submit sustainable wood certification data; identify each certified product on a line item basis. Provide current product certification documentation from certification body. Submit copies of invoices bearing certification numbers.

### 1.4 QUALITY ASSURANCE

Qualify welders in accordance with AWS D1.1/D1.1M using procedures, materials, and equipment of the type required for the work.

#### 1.4.1 Fabrication Drawings

Submit fabrication drawings showing layout(s), connections to structural system, and anchoring details as specified in AISC 303.

#### 1.4.2 Installation Drawings

Submit templates, erection and installation drawings indicating thickness, type, grade, class of metal, and dimensions. Show construction details, reinforcement, anchorage, and installation.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### 1.4.3 Assembly Instruction Drawings

Submit assembly instruction drawings showing layout(s), connections, bolting and anchoring details in accordance with manufacturer's standards. Submit drawings showing scaled details of proposed site furnishings, elevations for each type of site furnishing; dimensions, details, and methods of mounting or anchoring; shape and thickness of materials; and details of construction.

#### 1.4.4 Primer Certificate

Submit a certificate from the manufacturer stating that the primer conforms to requirements of SSPC Paint 25.

#### 1.4.5 Powder Coatings Certificate

Submit a certificate from the manufacturer stating that the powder coat conforms to ASTM D3451.

### 1.5 DELIVERY, STORAGE, AND HANDLING

Ship items knocked-down (KD) ready for site assembly. Packaged components must be complete including all accessories and hardware. Materials must be delivered, handled, and stored in accordance with the manufacturer's recommendations. Site furnishings must be inspected upon arrival at the job site for conformity to specifications and quality in accordance with paragraph MATERIALS. Protect from corrosion, staining, and other types of damage. Store items in designated area free from contact with soil and weather. Remove and replace damaged items with new items.

## PART 2 PRODUCTS

### 2.1 MATERIALS

Provide materials which are the standard products of a manufacturer regularly engaged in the manufacture of such products. Provide materials with proven satisfactory usage for at least 2 years.

#### 2.1.1 Metals

Metallic materials and products must conform to Section 08 31 00 ACCESS DOORS AND PANELS. Furnish metal components with factory drilled holes and free of excess weld and spatter. Metal components with holes that will not be filled by hardware or hidden by other components will be rejected. Structural steel products must conform to ASTM A36/A36M, ASTM A500/A500M and ASTM A501/A501M. Provide Steel Components with a minimum of 70 percent recycled content. Provide data identifying percentage of recycled content for steel components.

#### 2.1.2 Structural Tubing

ASTM A500/A500M

#### 2.1.3 Steel Pipe and Fittings

Steel pipe must conform to ASTM A53/A53M, Type E or S, Grade B; standard malleable iron fittings must conform to ASTM A47/A47M.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### 2.1.4 Gray Cast Iron

Gray cast iron must conform to ASTM A48/A48M Class 35 or better. Provide castings manufactured true to pattern and component parts that fit together in a satisfactory manner. Castings must be of uniform quality, free from blowholes, porosity, hard spots, shrinkage, distortion, or other defects. Smooth castings must be well-cleaned by sand or shot blasting.

#### 2.1.5 Aluminum Products

Provide Aluminum Components with a minimum of 50 percent total recycled content. Provide data identifying percentage of recycled content for aluminum components

#### 2.1.6 Cast Aluminum

Cast aluminum must conform to ASTM B26/B26M and ASTM B108/B108M. Provide castings manufactured true to pattern and component parts that fit together in a satisfactory manner. Provide castings of uniform quality, free from blowholes, porosity, hard spots, shrinkage, distortion, or other defects. Smooth castings must be well-cleaned by sand or shot blasting.

#### 2.1.7 Aluminum Alloy Products

Conform to ASTM B209 for sheet plate, ASTM B221 for extrusions and ASTM B26/B26M or ASTM B108/B108M for castings, as applicable. Provide aluminum extrusions at least 1/8 inch thick and aluminum plate or sheet at least 0.050 inch thick.

#### 2.1.8 Anchors and Hardware

Provide anchors, where necessary, for fastening site furnishings securely in place and in accordance with approved manufacturer's instructions. Anchoring devices that may be used, when no anchors are otherwise specified or indicated, include anchor bolts, slotted inserts, expansion shields for concrete; toggle bolts and through bolts for masonry; machine carriage bolts for steel; and lag bolts and screws for wood. Anchor bolts must conform to ASTM A307. Hardware must be stainless steel or galvanized steel in accordance with ASTM A153/A153M and compatible with the material to which applied. All exposed hardware must match in color and finish. Mounting hardware must be concealed, recessed, and plugged.

##### 2.1.8.1 Lag Screws and Bolts

ASME B18.2.1, type and grade best suited for the purpose.

##### 2.1.8.2 Toggle Bolts

ASME B18.2.1.

##### 2.1.8.3 Bolts, Nuts, Studs and Rivets

ASME B18.2.2 or ASTM A307.

##### 2.1.8.4 Power Driven Fasteners

Follow safety provisions of ASSP A10.3.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### 2.1.8.5 Screws

ASME B18.2.1, ASME B18.6.2, and ASME B18.6.3.

#### 2.1.8.6 Washers

Provide plain washers to conform to ASME B18.21.1. Provide beveled washers for American Standard beams and channels, square or rectangular, tapered in thickness, and smooth. Provide lock washers to conform to ASME B18.21.1.

#### 2.1.9 Ounce Metals

Bronze, copper, and other ounce metals must conform to ASTM B62.

#### 2.1.10 Concrete

Cast-in-place concrete materials and products must conform to Section 03 30 00 CAST-IN-PLACE CONCRETE.

### 2.2 COATINGS AND FINISHES

#### 2.2.1 Galvanizing

Hot-dip galvanize items specified to be zinc-coated, after fabrication where practicable. Galvanizing must conform to ASTM A123/A123M, ASTM A153/A153M or ASTM A653/A653M, as applicable. Tailings and sharp protrusions formed as a result of the hot-dip process must be removed and exposed edges burnished. Galvanize anchor bolts, grating fasteners, washers and parts or devices necessary for proper installation, unless otherwise indicated.

#### 2.2.2 Polyester Powder

Powder-coated surfaces must receive electrostatic zinc coating prior to painting. Powder coating must be electrostatically applied and oven cured. Polyester powder coating must be resistant to ultraviolet (UV) light.

#### 2.2.3 Polyvinyl-Chloride (PVC)

PVC coating must be primed with a clear acrylic thermosetting solution. The primed parts must be preheated prior to dipping. The liquid polyvinyl chloride must be ultraviolet (UV) stabilized and mold-resistant. The coated parts must be cured. The coating must be a minimum 2/25 inches thick plus or minus 0.020 inches and must have an 85 durometer hardness with a slip-resistant finish.

#### 2.2.4 Finish

Finish must be as specified by the manufacturer or as indicated. Exposed surfaces and edges must be rounded, polished, or sanded. Finish must be non-toxic, non-glare, and resistant to corrosion. Exposed surfaces must be smooth and splinter-free exposed surfaces. Submit two sets of color data for each furnishing displaying manufacturer's color selections and finishes, and identifying those colors and finishes proposed for use.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### 2.2.4.1 Paint

Paint must be factory applied with a minimum of 2 coats. Paint must be weather-resistant and resistant to cracking, peeling and fading.

#### 2.2.4.2 Color

Color of site furnishing components must be as indicated on Drawings.

### 2.3 BOLLARDS

Provide reinforced concrete bollards 16 inch round, height as indicated, suitable for ground mount installation. Provide exposed aggregate or sandblast finish as indicated; manufacturer's standard clear acrylic sealer. Submit manufacturer's descriptive data and catalog cuts.

#### 2.3.1 Concrete Strength

5000 psi, 28 day minimum compressive strength

## PART 3 EXECUTION

### 3.1 INSTALLATION

Verify that finished grades and other operations affecting mounting surfaces have been completed prior to the installation of site furnishings. Site furnishings must be installed plumb and true, at locations indicated, in accordance with the approved manufacturer's instructions.

#### 3.1.1 Assembly and Erection of Components

New parts must be acquired from the manufacturer; substitute parts will not be accepted unless approved by the manufacturer. When the inspection of parts has been completed, the site furnishings must be assembled and anchored according to manufacturer's instructions or as indicated. When site furnishings are assembled at the site, assembly must not interfere with other operations or pedestrian and vehicular circulation.

#### 3.1.2 Anchorage, Fastenings, and Connections

Furnish metal work, mounting bolts or hardware in ample time for securing into concrete or masonry as the work progresses. Provide anchorage where necessary for fastening furniture or furnishings securely in place. Provide, for anchorage not otherwise specified or indicated, slotted inserts, expansion shields, and power-driven fasteners, when approved for concrete; toggle bolts and through bolts for masonry; machine and carriage bolts for steel; through bolts, lag bolts, and screws for wood. Do not use wood plugs in any material. Provide non-ferrous attachments for non-ferrous metal. Make exposed fastenings of compatible materials, generally matching in color and finish the fastenings to which they are applied. Conceal fastenings where practicable.

### 3.2 FINISHES

#### 3.2.1 Field Finishes

Where indicated, field finishes must be applied in accordance with Section 09 90 00 PAINTS AND COATINGS. Where dissimilar metals are in contact,

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

protect surfaces with a coat conforming to SSPC Paint 25 to prevent galvanic or corrosive action. Where aluminum is in contact with concrete, mortar, masonry, wood, or absorptive materials subject to wetting, protect with ASTM D1187/D1187M, asphalt-base emulsion.

### 3.3 BOLLARDS

Install in accordance with construction details in Drawings.

### 3.4 RESTORATION AND CLEAN UP

When the installation has been completed, clean up and protect the site. Existing areas that have been damaged from the installation operation must be restored to original condition at Contractor's expense.

#### 3.4.1 Clean Up

The site must be cleaned of all materials associated with the installation. Site furnishing surfaces must be cleaned of dirt, stains, filings, and other blemishes occurring from shipment and installation. Cleaning methods and agents must be according to manufacturer's instructions or as indicated.

#### 3.4.2 Protection

The area must be protected as required or directed by providing barricades and signage. Signage must be in accordance with Section 10 14 00.10 EXTERIOR SIGNAGE.

#### 3.4.3 Disposal of Materials

Excess and waste material must be removed and disposed off Government property.

### 3.5 RE-INSTALLATION

Where re-installation is required, the following must be accomplished:

- a. Re-install the product as specified. Material acquisition of replacement parts is the responsibility of the Contractor. Provide replacement materials that are new and supplied by the original manufacturer to match.
- b. Damage caused by the failed installation must be repaired.

-- End of Section --

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## SECTION 13 48 73

### SEISMIC CONTROL FOR MISCELLANEOUS EQUIPMENT 05/22

#### PART 1 GENERAL

##### 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

#### AMERICAN CONCRETE INSTITUTE (ACI)

- |           |                                                                                                |
|-----------|------------------------------------------------------------------------------------------------|
| ACI 355.2 | (2007) Qualification of Post-Installed Mechanical Anchors in Concrete and Commentary           |
| ACI 355.4 | (2011) Qualification of Post-Installed Adhesive Anchors in Concrete (ACI 355.4) and Commentary |

#### AMERICAN INSTITUTE OF STEEL CONSTRUCTION (AISC)

- |          |                                  |
|----------|----------------------------------|
| AISC 325 | (2017) Steel Construction Manual |
|----------|----------------------------------|

#### AMERICAN SOCIETY OF CIVIL ENGINEERS (ASCE)

- |           |                                                                                                                  |
|-----------|------------------------------------------------------------------------------------------------------------------|
| ASCE 7-16 | (2017; Errata 2018; Supp 1 2018) Minimum Design Loads and Associated Criteria for Buildings and Other Structures |
|-----------|------------------------------------------------------------------------------------------------------------------|

#### ASTM INTERNATIONAL (ASTM)

- |                 |                                                                                                                       |
|-----------------|-----------------------------------------------------------------------------------------------------------------------|
| ASTM A36/A36M   | (2019) Standard Specification for Carbon Structural Steel                                                             |
| ASTM A53/A53M   | (2022) Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless                 |
| ASTM A153/A153M | (2023) Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware                                   |
| ASTM A325       | (2014) Standard Specification for Structural Bolts, Steel, Heat Treated, 120/105 ksi Minimum Tensile Strength         |
| ASTM A490       | (2014a) Standard Specification for Structural Bolts, Alloy Steel, Heat Treated, 150 ksi Minimum Tensile Strength      |
| ASTM A500/A500M | (2023) Standard Specification for Cold-Formed Welded and Seamless Carbon Steel Structural Tubing in Rounds and Shapes |

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

|                 |                                                                                           |
|-----------------|-------------------------------------------------------------------------------------------|
| ASTM A563       | (2021; E 2022a) Standard Specification for Carbon and Alloy Steel Nuts                    |
| ASTM A603       | (2019) Standard Specification for Zinc-Coated Steel Structural Wire Rope                  |
| ASTM E488/E488M | (2022) Standard Test Methods for Strength of Anchors in Concrete Elements                 |
| ASTM F1554      | (2020) Standard Specification for Anchor Bolts, Steel, 36, 55, and 105-ksi Yield Strength |

#### ICC EVALUATION SERVICE, INC. (ICC-ES)

|              |                                                                                                                                                                                                  |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ICC ES AC156 | (2012) Acceptable Criteria for Seismic Certification by Shake-Table Testing of Nonstructural Components                                                                                          |
| ICC-ES AC23  | (2012; R 2016) Acceptance Criteria for Sprayed Fire-resistant Materials (SFRMs), Intumescent Fire-resistant Coatings and Mastic Fire-resistant Coatings Used to Protect Structural Steel Members |

#### INTERNATIONAL CODE COUNCIL (ICC)

|         |                                    |
|---------|------------------------------------|
| ICC IBC | (2021) International Building Code |
|---------|------------------------------------|

#### METAL FRAMING MANUFACTURERS ASSOCIATION (MFMA)

|        |                                            |
|--------|--------------------------------------------|
| MFMA-4 | (2004) Metal Framing Standards Publication |
|--------|--------------------------------------------|

#### U.S. DEPARTMENT OF DEFENSE (DOD)

|              |                                                                               |
|--------------|-------------------------------------------------------------------------------|
| UFC 3-301-01 | (2023) Structural Engineering                                                 |
| UFC 4-010-01 | (2018; with Change 2, 2022) DoD Minimum Antiterrorism Standards for Buildings |

#### VIBRATION ISOLATION AND SEISMIC CONTROL MANUFACTURERS ASSOCIATION (VISCMA)

|            |                                                               |
|------------|---------------------------------------------------------------|
| VISCMA 412 | (2014) Installing Seismic Restraints for Mechanical Equipment |
|------------|---------------------------------------------------------------|

## 1.2 SYSTEM DESCRIPTION

### 1.2.1 General Requirements

Apply the requirements for seismic protection measures, described in this section and on the drawings, of the miscellaneous equipment and systems listed below, in accordance with UFC 3-301-01 and additional data furnished by the Contracting Officer. Provide seismic protection measures in addition to any other requirements called for in other sections of these specifications. Where there is a conflict between the specifications and the drawings, the specifications will take precedence. Accomplish resistance to lateral forces induced by earthquakes without



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

consideration of friction resulting from gravity loads.

#### 1.2.2 Miscellaneous Equipment and Systems

Provide bracing and attachment for the following miscellaneous equipment and components developed by the Contractor in accordance with the requirements of this specification:

Equipment/Components with  $I_p = 1.0$   
Storage cabinets  
Ornamentations  
Storage Racks  
Shelving  
Partitions  
Stacks  
Pole or frame supported equipment  
Storage tanks for water and oil

#### 1.2.3 Contractor Designed Bracing

Submit copies of the design calculations with the drawings. Calculations must be approved, certified, stamped and signed by a registered Professional Structural Engineer. Calculations must verify the capability of structural members to which bracing is attached for carrying the load from the brace. Design the bracing in accordance with UFC 3-301-01, UFC 4-010-01 and additional data furnished by the Contracting Officer. Resistance to lateral forces induced by earthquakes must be accomplished without consideration of friction resulting from gravity loads. UFC 3-301-01 uses parameters for the building, not for the equipment in the building; therefore, corresponding adjustments to the formulas must be required. Loadings determined using UFC 3-301-01 are based on strength design; therefore, AISC 325 Specifications must be used for the design. The bracing for the equipment designated in paragraph 1.2.2 must be developed by the Contractor.

### 1.3 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

#### SD-02 Shop Drawings

Bracing; G

Resilient Vibration Isolation Devices; G

Equipment Requirements; G

#### SD-03 Product Data

Bracing; G

Equipment Requirements; G

Anchor Bolts; G

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

Vibration Isolators; G

Snubbers; G

SD-05 Design Data

Design Calculations; G

SD-06 Test Reports

Anchor Bolts; G

SD-07 Certificates

ICC ES AC156 Shake Table Test; G

## PART 2 PRODUCTS

### 2.1 EQUIPMENT REQUIREMENTS

Submit detail drawings of bracing along with calculations, catalog cuts, templates, and erection and installation details, as appropriate, for the items listed in Paragraph 1.2.2. Indicate thickness, type, grade, class of metal, and dimensions; and show construction details, reinforcement, anchorage, and installation with relation to the building construction. Provide calculations and drawings that are stamped by a registered structural engineer, and that verify the capability of structural members to which bracing is attached for carrying the load from the brace. Design must be based on actual equipment and system layout. Design must include calculated dead loads, static seismic loads and capacity of materials utilized for the connection of the equipment or system to the structure. Analysis must detail anchoring methods.

Include drawing for Designated Seismic System Equipment indicating the equipment location in the facility sufficient to be used for the installation. Equipment must be rigidly or flexibly mounted as indicated in the specifications and/or drawings depending on vibration isolation requirements as follows below. Roof mounted equipment both vibration isolated and nonisolated, must have support members designed and anchored to building structural steel or concrete as required for seismic restraint and wind loads.

#### 2.1.1 Rigidly (Base and Suspended) Mounted Equipment

Equipment furnished under this contract must be rigidly mounted using cast-in-place anchor bolts to anchor them or post-installed anchors that are qualified for earthquake loading in accordance with ACI 355.2 and ACI 355.4. Anchor bolts must conform to ASTM F1554-07ae1. For any rigid equipment which is rigidly anchored, provide flexible joints for piping, electrical conduit, etc., that are capable of accommodating displacements equal to the full width of the joint in both orthogonal directions. Suspended equipment bracing attachments should be located just above the center of gravity to minimize swinging.

#### 2.1.2 Nonrigid or Flexibly-Mounted Equipment

Select vibration isolation devices so that the maximum movement of equipment from the static deflection point is 1/4 inch. Equipment

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

flexibly mounted on vibration isolators must have a bumper restraint or snubber in each horizontal direction and vertical restraints must be provided where require to resist overturning. Isolator housing and restraints must be constructed of ductile materials. A viscoelastic pad or similar material of appropriate thickness must be used between the bumper and components to limit the impact load. Restraints must be designed to resist the calculated horizontal lateral and vertical forces.

Spring vibration isolators must be seismically rated, restrained isolators for equipment subject to load variations and large external forces. The seismically rated housing must be sized to meet or exceed the force requirements applicable to the project and meet the required isolation criteria. Spring vibration isolator manufacturer's will be a member of VISCMA. Design force,  $F_p$ , must be doubled for vibration isolators with an air gap greater than 0.25 inches as specified in ASCE 7-16, Chapter 13.

## 2.2 BOLTS AND NUTS

Hex head bolts, and heavy hexagon nuts must be ASTM A325 or ASTM A490 bolts and ASTM A563 nuts. Provide bolts and nuts galvanized in accordance with ASTM A153/A153M when used underground or exposed to weather.

## 2.3 SWAY BRACING

Material used for members listed in this section and on the drawings, must be structural steel conforming with the following:

- a. Plates, rods, and rolled shapes, ASTM A36/A36M.
- b. Wire rope, ASTM A603.
- c. Tubes, ASTM A500/A500M, Grade B.
- d. Pipes, ASTM A53/A53M, Grade B.
- e. Angles, ASTM A36/A36M.
- f. Channels (Struts) with in-turned lips and associated hardware for fastening to channels at random points conforming to MFMA-4.

## PART 3 EXECUTION

### 3.1 BRACING

Provide bracing conforming to the arrangements shown. Install cables at a 45-degree slope. Where interference is present, the slope may be minimum of 30 degrees or a maximum of 60 degrees per VISCMA 412.

### 3.2 BUILDING DRIFT

Do not attach sway braces for equipment to two dissimilar structural elements of a building that may respond differentially during an earthquake unless a flexible joint is provided. Bracing must be capable of accommodating building drift due to seismic displacements.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 3.3 ANCHOR BOLTS

#### 3.3.1 General

Submit copies of test results to verify the adequacy of the specific anchor and application, as specified.

Ensure housekeeping pads have adequate space to mount equipment and seismic restraint devices allowing adequate edge distance and embedment depth for restraint anchor bolts. Identify position of reinforcing steel and other embedded items prior to drilling holes for anchors. Do not drill holes in concrete or masonry until concrete, mortar, or grout has achieved full design strength. Install neoprene grommet washers or till the gap with epoxy on equipment anchor bolts where clearance between anchor and equipment support hole exceeds 0.125 inches.

#### 3.3.2 Cast-In-Place

Use templates to locate cast-in-place bolts accurately and securely in formwork. Provide anchor bolts with an embedded straight length equal to at least 12 times nominal diameter of the bolt. Anchor bolts that exceed the normal depth of equipment foundation piers or pads must either extend into concrete floor or the foundation or be increased in depth to accommodate bolt lengths. Use templates to locate cast-in-place bolts accurately and securely in formwork.

#### 3.3.3 Drilled-In Anchor Bolts

Drill holes with rotary impact hammer drills. Drill bits must be of diameters as specified by the anchor manufacturer. Unless otherwise shown on the drawings, all holes must be drilled perpendicular to the concrete surface. Where anchors are permitted to be installed in cored holes, use core bits with matched tolerances as specified by the manufacturer. Properly clean cored hole per manufacturer's instructions. Identify position of reinforcing steel and other embedded items prior to drilling holes for anchors. Exercise care in coring or drilling to avoid damaging existing reinforcing or embedded items. Notify the COR if reinforcing steel or other embedded items are encountered during drilling. Take precautions as necessary to avoid damaging prestressing tendons, electrical and telecommunications conduit, and gas lines. Unless otherwise specified, do not drill holes in concrete or masonry until concrete, mortar, or grout has achieved full design strength. Perform anchor installation in accordance with manufacturer instructions. For Wedge Anchors, Heavy-Duty Sleeve Anchors, and Undercut Anchors, protect threads from damage during anchor installation. Heavy-duty sleeve anchors must be installed with sleeve fully engaged in part to be fastened. Set anchors to manufacturer's recommended torque, using a torque wrench. Following attainment of 10 percent of the specified torque, 100 percent of the specified torque must be reached within 7 or fewer complete turns of the nut. If the specified torque is not achieved within the required number of turns, the anchor must be removed and replaced unless otherwise directed by the Engineer.

For Cartridge Injection Adhesive Anchors where approved for seismic application, clean all holes per manufacturer instructions to remove loose material and drilling dust prior to installation of adhesive. Inject adhesive into holes proceeding from the bottom of the hole and progressing toward the surface in such a manner as to avoid introduction of air pockets in the adhesive. Follow manufacturer recommendations to ensure

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

proper mixing of adhesive components. Sufficient adhesive must be injected in the hole to ensure that the annular gap is filled to the surface. Remove excess adhesive from the surface. Shim anchors with suitable device to center the anchor in the hole. Do not disturb or load anchors before manufacturer specified cure time has elapsed. For Capsule Anchors where approved for seismic application, perform drilling and setting operations in accordance with manufacturer instructions. Clean all holes to remove loose material and drilling dust prior to installation of adhesive. Remove water from drilled holes in such a manner as to achieve a surface dry condition. Capsule anchors must be installed with equipment conforming to manufacturer recommendations. Do not disturb or load anchors before manufacturer specified cure time has elapsed. Observe manufacturer recommendations with respect to installation temperatures for cartridge injection adhesive anchors and capsule anchors.

### 3.3.4 Anchor Bolt Testing

Test in place expansion and chemically bonded anchors not more than in accordance with manufacturer approval reports hours after installation of the anchor, conducted by an independent testing agency; testing must be performed on random anchor bolts as described below.

#### 3.3.4.1 Torque Wrench Testing

Perform torque wrench testing on not less than 50 percent of the total installed expansion anchors and at least one anchor for every piece of equipment containing more than two anchors. The test torque must equal the minimum required installation torque as required by the bolt manufacturer. Calibrate torque wrenches at the beginning of each day the torque tests are performed. Recalibrate torque wrenches for each bolt diameter whenever tests are run on bolts of various diameters. Apply torque between 20 and 100 percent of wrench capacity. Reach the test torque within one half turn of the nut, except for 3/8 inch sleeve anchors which must reach their torque by one quarter turn of the nut. If any anchor fails the test, test similar anchors not previously tested until 20 consecutive anchors pass. Failed anchors must be retightened and retested to the specified torque; if the anchor still fails the test it must be replaced.

#### 3.3.4.2 Pullout Testing

Test expansion and chemically bonded anchors by applying a pullout load using a hydraulic ram attached to the anchor bolt. Testing must be done in accordance with ASTM E488/E488M or ICC-ES AC23. At least 5 percent of the anchors per day must be tested. Apply the load to the anchor without removing the nut; when that is not possible, the nut must be removed and a threaded coupler must be installed of the same tightness as the original nut. Check the test setup to verify that the anchor is not restrained from withdrawing by the baseplate, the test fixture, or any other fixtures. The support for the testing apparatus must be at least 1.5 times the embedment length away from the bolt being tested. Load each tested anchor to 1 times the design tension value for the anchor. The anchor must have no observable movement at the test load. If any anchor fails the test, similar anchors not previously tested must be tested until 10 consecutive anchors pass. Failed anchors must be retightened and retested to the specified load; if the anchor still fails the test it must be replaced.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 3.4 RESILIENT VIBRATION ISOLATION DEVICES

Where the need for these devices is determined, based on the magnitude of the design seismic forces, select anchor bolts for vibration isolation devices and/or snubbers for equipment base and foundations that follow the same procedure as in paragraph ANCHOR BOLTS, except use an equipment weight equal to five times the actual equipment weight.

#### 3.4.1 Spring-Type Vibration Devices

Select vibration isolation devices so that the maximum movement of equipment from the static deflection point is 1/2 inch. Equipment flexibly mounted on vibration isolators must have a bumper restraint or snubber in each horizontal direction and vertical restraints must be provided where required to resist overturning. Isolator housing and restraints must be constructed of ductile materials. A viscoelastic pad or similar material of appropriate thickness must be used between the bumper and components to limit the impact load. Restraints must be designed to resist the calculated horizontal lateral and vertical forces.

Spring vibration isolators must be seismically rated, restrained isolators for equipment subject to load variations and large external forces. The seismically rated housing must be sized to meet or exceed the force requirements applicable to the project and meet the required isolation criteria. Spring vibration isolator manufacturers will be a member of VISCMA.

#### 3.4.2 Multidirectional Seismic Snubbers

Install multidirectional seismic snubbers employing elastomeric pads on floor- or slab-mounted equipment. Use snubbers that provide 1/4 inch free vertical and horizontal movement from the static deflection point. Provide snubber medium consisting of multiple pads of cotton duct and neoprene or other suitable materials arranged around a flanged steel trunnion so both horizontal and vertical forces are resisted by the snubber medium.

### 3.5 EQUIPMENT SWAY BRACING

#### 3.5.1 Suspended Equipment

Provide equipment sway bracing for items supported from floor, overhead floor or roof structural systems. Provide braces that consist of angles, rods, wire rope, bars, channels (struts) or pipes arranged as shown in bracing submittals and secured at both ends with not less than 1/2 inch bolts. Provide sufficient braces for equipment to resist a horizontal force as specified in UFC 3-301-01 without exceeding safe working stress of bracing components. Provide, for approval, specific force calculations in accordance with UFC 3-301-01 for the equipment in the project. Submit details of equipment bracing for acceptance. In lieu of bracing with vertical supports, these items may be supported with hangers inclined at 45 degrees directed up and radially away from equipment and oriented symmetrically in 90-degree intervals on the horizontal plane, bisecting the angles of each corner of the equipment, provided that supporting members are properly sized to support operating weight of equipment when hangers are inclined at a 45-degree angle.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 3.5.2 Floor or Pad Mounted Equipment

#### 3.5.2.1 Shear Resistance

Bolt to the floor, floor mounted equipment. Provide the number and installation of bolts to resist shear forces in accordance with paragraph ANCHOR BOLTS.

#### 3.5.2.2 Overturning Resistance

Use the ratio of the overturning moment from seismic forces to the resisting moment due to gravity loads to determine if overturning forces need to be considered in the sizing of anchor bolts. Provide calculations to verify the adequacy of the anchor bolts for combined shear and overturning.

### 3.6 SPECIAL TESTING FOR SEISMIC-RESISTING EQUIPMENT

Equipment and components designated as Designated Seismic Systems required to remain operational after an earthquake will be seismic qualified by shake table testing conforming to ICC ES AC156 Shake Table Test procedures. The manufacturer is to provide a certification by a fully qualified testing agency for the specific equipment and/or components. Prequalified certifications are acceptable unless noted otherwise.

Miscellaneous components that are required to be certified must bear permanent marking or nameplates constructed of a durable heat and water resistant material. Nameplates must be mechanically attached to such nonstructural components and placed on each component for clear identification. The nameplate must not be less than 5 inches x 7 inches with red letters 1 inch in height on a white background stating "Certified Equipment." The following statement must be on the nameplate: "This equipment/component is certified. No modifications are allowed unless authorized in advance and documented in the Equipment Certification Documentation file." The nameplate must also contain the component identification number in accordance with the drawings/specifications and the O&M manuals.

### 3.7 SPECIAL INSPECTION FOR SEISMIC-RESISTING SYSTEMS AND EQUIPMENT

Perform special inspections for seismic-resisting systems, equipment and components designated seismic systems and equipment per ICC IBC 1705.12.4; and storage racks per ICC IBC 1705.12.7. Periodic special inspections will be conducted on miscellaneous equipment as required by Section 1705.12 of the International Building Code and paragraph 2-2.4.3 of UFC 3-301-01. Provide a Statement of Special Inspections and Final Report in accordance with paragraph 2-2.4.3 of UFC 3-301-01.

-- End of Section --

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## SECTION 21 13 13

### WET PIPE SPRINKLER SYSTEMS, FIRE PROTECTION 08/20

#### PART 1 GENERAL

##### 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

#### AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

|             |                                                                             |
|-------------|-----------------------------------------------------------------------------|
| ASME B16.1  | (2020) Gray Iron Pipe Flanges and Flanged Fittings Classes 25, 125, and 250 |
| ASME B16.3  | (2021) Malleable Iron Threaded Fittings, Classes 150 and 300                |
| ASME B16.4  | (2021) Gray Iron Threaded Fittings; Classes 125 and 250                     |
| ASME B16.21 | (2021) Nonmetallic Flat Gaskets for Pipe Flanges                            |

#### AMERICAN SOCIETY OF SANITARY ENGINEERING (ASSE)

|           |                                                                                 |
|-----------|---------------------------------------------------------------------------------|
| ASSE 1015 | (2021) Performance Requirements for Double Check Backflow Prevention Assemblies |
|-----------|---------------------------------------------------------------------------------|

#### AMERICAN WATER WORKS ASSOCIATION (AWWA)

|          |                                                                                          |
|----------|------------------------------------------------------------------------------------------|
| AWWA M14 | (2015) Manual: Recommended Practice for Backflow Prevention and Cross-Connection Control |
|----------|------------------------------------------------------------------------------------------|

#### ASTM INTERNATIONAL (ASTM)

|                 |                                                                                                       |
|-----------------|-------------------------------------------------------------------------------------------------------|
| ASTM A47/A47M   | (1999; R 2022; E 2022) Standard Specification for Ferritic Malleable Iron Castings                    |
| ASTM A53/A53M   | (2022) Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless |
| ASTM A135/A135M | (2021) Standard Specification for Electric-Resistance-Welded Steel Pipe                               |
| ASTM A153/A153M | (2023) Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware                   |
| ASTM A183       | (2014; R 2020) Standard Specification for Carbon Steel Track Bolts and Nuts                           |



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

ASTM A536 (1984; R 2019; E 2019) Standard  
Specification for Ductile Iron Castings

FM GLOBAL (FM)

FM APP GUIDE (updated on-line) Approval Guide  
<http://www.approvalguide.com/>

MANUFACTURERS STANDARDIZATION SOCIETY OF THE VALVE AND FITTINGS  
INDUSTRY (MSS)

MSS SP-71 (2018) Gray Iron Swing Check Valves,  
Flanged and Threaded Ends

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 13 (2022) Standard for the Installation of  
Sprinkler Systems

NFPA 24 (2022) Standard for the Installation of  
Private Fire Service Mains and Their  
Appurtenances

NFPA 291 (2022) Recommended Practice for Fire Flow  
Testing and Marking of Hydrants

NATIONAL INSTITUTE FOR CERTIFICATION IN ENGINEERING TECHNOLOGIES  
(NICET)

NICET 1014-7 (2012) Program Detail Manual for  
Certification in the Field of Fire  
Protection Engineering Technology (Field  
Code 003) Subfield of Automatic Sprinkler  
System Layout

U.S. DEPARTMENT OF DEFENSE (DOD)

UFC 3-600-01 (2016; with Change 6, 2021) Fire  
Protection Engineering for Facilities

UNDERWRITERS LABORATORIES (UL)

UL 199 (2020) UL Standard for Safety Automatic  
Sprinklers for Fire-Protection Service

UL 312 (2022) UL Standard for Safety Check Valves  
for Fire-Protection Service

UL 405 (2013; Bul. 2020) UL Standard for Safety  
Fire Department Connection Devices

UL 668 (2004; Reprint Oct 2021) UL Standard for  
Safety Hose Valves for Fire-Protection  
Service

UL 1626 (2008; Bul. 2018) UL Standard for Safety

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## Residential Sprinklers for Fire-Protection Service

UL Fire Prot Dir

(2012) Fire Protection Equipment Directory

### 1.2 SYSTEM DESCRIPTION

Provide wet pipe sprinkler system in all areas of the building. Except as modified herein, the system must meet the requirements of UFC 3-600-01, NFPA 13, and NFPA 13R for residential systems. Pipe sizes which are not indicated on the Contract drawings must be determined by hydraulic calculations.

#### 1.2.1 Hydraulic Design

##### 1.2.1.1 Basis for Calculations

A waterflow test was performed on October 17, 2023 at Youngstown Air Reserve Station in Ohio. Testing Hydrants 39 & 40 resulted in a static pressure of 55 psi with a residual pressure of 49 psi while flowing 1,130 gpm. The Contractor shall be responsible for performing a fire hydrant flow test within 6 months prior to shop drawing submittal in accordance with NFPA 291. Results must include hydrant elevations relative to the building and hydrant number/identifiers for the tested hydrants, including which were flowed, which had a gauge. This information must be presented in a tabular form if multiple hydrants were flowed. The results must be included with the hydraulic calculations. Hydraulic calculations must be based on flow test noted in this paragraph, unless approved by the Architect/Engineer. Hydraulic calculations must be based upon the Hazen-Williams formula with a "C" value noted in NFPA 13 for piping, and 100 for existing underground piping. The minimum residual pressure in a service lateral (lead-in) at the design flow rate must be 20 psi at the inlet to the backflow preventer.

##### 1.2.1.2 Hydraulic Calculations

- a. Water supply curves and system requirements must be plotted on semi-logarithmic graph ( $N^{1.85}$ ) paper so as to present a summary of the complete hydraulic calculation.
- b. Provide a summary sheet listing sprinklers in the design area and their respective hydraulic reference points, elevations, minimum discharge pressures and minimum flows. Elevations of hydraulic reference points (nodes) must be indicated.
- c. Documentation must identify each pipe individually and the nodes connected thereto. Indicate the diameter, length, flow, velocity, friction loss, number and type fittings, total friction loss in the pipe, equivalent pipe length and Hazen-Williams coefficient for each pipe.
- d. Where the sprinkler system is supplied by interconnected risers, the sprinkler system must be hydraulically calculated using the hydraulically most demanding single riser. The calculations must not assume the simultaneous use of more than one riser.
- e. All calculations must include the backflow preventer manufacturer's stated friction loss at the design flow or 8 psi for double check

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

backflow preventer, whichever is greater.

- f. All calculations must be performed back to the actual location of the flow test, taking into account the direction of flow in the service main at the test location.
- g. For gridded systems, calculations must show peaking of demand area friction loss to verify that the hydraulically most demanding area is being used. A flow diagram indicating the quantity and direction of flows must be included.

#### 1.2.1.3 Design Criteria

Hydraulically design the system to discharge a minimum density as indicated on the drawings. Hydraulic calculations must be in accordance with the Area/Density Method of NFPA 13. Add an allowance for exterior hose streams of 250 gpm to the sprinkler system demand at the point of connection to the existing water system. The design area reduction in NFPA 13 for quick response sprinklers is not permitted. Hydraulic calculations shall include a minimum 10 psi safety factor on system pressure.

#### 1.2.2 Sprinkler Coverage

Sprinklers must be uniformly spaced on branch lines. Provide coverage throughout 100 percent of the building. This includes, but is not limited to, telephone rooms, electrical equipment rooms (regardless of the fire resistance rating of the enclosure), boiler rooms, switchgear rooms, transformer rooms, attached electrical vaults and other electrical and mechanical spaces. Coverage per sprinkler must be in accordance with NFPA 13. Provide sprinklers below all obstructions in accordance with NFPA 13. Exceptions are as follows:

- a. Facilities that are designed in accordance with NFPA 13R.

#### 1.2.3 Qualified Fire Protection Engineer (QFPE)

The Contractor must provide the services of a QFPE, an individual who is a licensed professional engineer (P.E.) who has passed the fire protection engineering written examination administered by the National Council of Examiners for Engineering and Surveying (NCEES) and has relevant fire protection engineering experience. The QFPE shall be the same individual responsible for Division 28 31 76. Services of the Contractor's QFPE must include:

- a. Reviewing SD-02, SD-03, and SD-05 submittal packages for completeness and compliance with the provisions of this specification. Working (shop) drawings and calculations must be prepared by, or prepared under the immediate supervision of, the QFPE. The QFPE must affix their professional engineering stamp with signature to the shop drawings, calculations, and material data sheets, indicating approval prior to submitting the shop drawings to the DFPE.
- b. Provide a letter documenting that the SD-02, SD-03, and SD-05 submittal package has been reviewed and noting all outstanding comments.
- c. Performing in-progress construction surveillance prior to installation of ceilings (rough-in inspection).

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

- d. Witnessing pre-Government and final Government functional performance testing and performing a final installation review.
- e. Signing applicable certificates under SD-07.

### 1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Partial submittals and submittals not fully complying with NFPA 13 and this specification section must be returned disapproved without review. SD-02, SD-03 and SD-05 must be submitted simultaneously.

Shop drawings (SD-02), product data (SD-03) and calculations (SD-05) must be prepared by the Contractor and combined and submitted as one complete package. The QFPE must review the SD-02/SD-03/SD-05 submittal package for completeness and compliance with the Contract provisions prior to submission to the Government. The QFPE must provide a Letter of Confirmation that they have reviewed the submittal package for compliance with the contract provisions. This letter must include their professional engineer stamp and signature. Partial submittals and submittals not reviewed by the QFPE must be returned disapproved without review.

Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

#### SD-01 Preconstruction Submittals

Qualified Fire Protection Engineer (QFPE); G

Sprinkler System Designer; G

Sprinkler System Installer; G

#### SD-02 Shop Drawings

Shop Drawing; G

#### SD-03 Product Data

Pipe; G

Fittings; G

Valves, including gate, check, butterfly, and globe; G

Alarm Valves; G

Relief Valves; G

Sprinklers ; G

Pipe Hangers and Supports ; G

Sprinkler Alarm Switch; G

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

Valve Supervisory (Tamper) Switch; G

Fire Department Connection; G

Backflow Prevention Assembly; G

Air Vent; G

Hose Valve; G

Nameplates; G

Waterflow Alarm Bell

#### SD-05 Design Data

Hydraulic Calculations; G

#### SD-06 Test Reports

Test Procedures; G

#### SD-07 Certificates

Verification of Compliant Installation; G

Request for Government Final Test; G

#### SD-10 Operation and Maintenance Data

Operating and Maintenance (O&M) Instructions; G

Spare Parts Data; G

#### SD-11 Closeout Submittals

As-built drawings

### 1.4 QUALITY ASSURANCE

#### 1.4.1 Preconstruction Submittals

Within 36 days of contract award but no less than 14 days prior to commencing work on site, the prime Contractor must submit the following for review and approval. SD-02, SD-03 and SD-05 submittals received prior to the review and approval of the qualifications will be returned Disapproved Without Review.

##### 1.4.1.1 Shop Drawing

Three (3) copies of the shop drawings, no later than 28 days prior to the start of system installation. Working drawings conforming to the requirements prescribed in NFPA 13 and must be no smaller than the Contract Drawings. Each set of drawings must include the following:

- a. A descriptive index with drawings listed in sequence by number. A legend sheet identifying device symbols, nomenclature, and conventions used in the package.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

- b. Floor plans drawn to a scale not less than 1/8-inch equals 1-foot clearly showing locations of devices, equipment, risers, and other details required to clearly describe the proposed arrangement.
- c. Actual center-to-center dimensions between sprinklers on branch lines and between branch lines; from end sprinklers to adjacent walls; from walls to branch lines; from sprinkler feed mains, cross mains and branch lines to finished floor and roof or ceiling. A detail must show the dimension from the sprinkler and sprinkler deflector to the ceiling in finished areas.
- d. Longitudinal and transverse building sections showing typical branch line and cross main pipe routing, elevation of each typical sprinkler above finished floor and elevation of "cloud" or false ceilings in relation to the building ceilings.
- e. Plan and elevation views which establish that the equipment will fit the allotted spaces with clearance for installation and maintenance.
- f. Riser layout drawings drawn to a scale of not less than 1/2-inch equals 1-foot to show details of each system component, clearances between each other and from other equipment and construction in the room.
- g. Details of each type of riser assembly, pipe hanger, and restraint of underground water main at point-of-entry into the building, and electrical devices and interconnecting wiring. The dimension from the edge of vertical piping to the nearest adjacent wall(s) must be indicated on the drawings when vertical piping is located in stairs or other portions of the means of egress.
- h. Details of each type of pipe hanger and related components.

#### 1.4.1.2 Product Data

Three (3) copies of annotated catalog data to show the specific model, type, and size of each item. Catalog cuts must also indicate the NRTL listing. The data must be highlighted to show model, size, options, and other pertinent information, that are intended for consideration. Data must be adequate to demonstrate compliance with all contract requirements. Product data for all equipment must be combined into a single submittal.

#### 1.4.1.3 Hydraulic Calculations

Calculations must be as outlined in NFPA 13 except that calculations must be performed by computer using software intended specifically for fire protection system design using the design data shown on the drawings. Calculations must include isometric diagram indicating hydraulic nodes and pipe segments. Include fire pump curve with submittal.

#### 1.4.1.4 Operating and Maintenance (O&M) Instructions

Submit in accordance with Division 01 Specifications as supplemented and modified by this specification section.

Provide six manuals and one pdf version on electronic media. The manuals must include the manufacturer's name, model number, parts list, list of parts and tools that should be kept in stock by the owner for routine

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

maintenance, troubleshooting guide, and recommended service organization (including address and telephone number) for each item of equipment. Each service organization submitted must be capable of providing 4-hour on-site response to a service call on an emergency basis.

Submit spare parts data for each different item of material and equipment specified. The data must include a complete list of parts and supplies, and a list of parts recommended by the manufacturer to be replaced after 1-year and 3 years of service. Include a list of special tools and test equipment required for maintenance and testing of the products supplied.

#### 1.4.2 Qualifications

##### 1.4.2.1 Sprinkler System Designer

The sprinkler system designer must be certified as a minimum Level III Technician by National Institute for Certification in Engineering Technologies (NICET) in the Water-Based Systems Layout subfield of Fire Protection Engineering Technology in accordance with NICET 1014-7.

##### 1.4.2.2 Sprinkler System Installer

The sprinkler system installer must be regularly engaged in the installation of the type and complexity of system specified in the contract documents, and must have served in a similar capacity for at least three systems that have performed in the manner intended for a period of not less than 6 months.

#### 1.4.3 Regulatory Requirements

Equipment and material must be listed or approved. Listed or approved, as used in this Section, means listed, labeled or approved by a Nationally Recognized Testing Laboratory (NRTL) such as UL Fire Prot Dir or FM APP GUIDE. The omission of these terms under the description of an item or equipment described must not be construed as waiving this requirement. All listings or approvals by testing laboratories must be from an existing ANSI or UL published standard. The recommended practices stated in the manufacturer's literature or documentation are mandatory requirements.

#### 1.5 DELIVERY, STORAGE, AND HANDLING

Protect all equipment delivered and placed in storage from the weather, excessive humidity and temperature variations, dirt and dust, or other contaminants. All pipes must be either capped or plugged until installed.

#### 1.6 EXTRA MATERIALS

Spare sprinklers and wrench(es) must be provided as spare parts in accordance with NFPA 13.

### PART 2 PRODUCTS

#### 2.1 MATERIALS AND EQUIPMENT

##### 2.1.1 Standard Products

Provide materials, equipment, and devices listed for fire protection service when so required by NFPA 13 or this specification. Select

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

material from one manufacturer, where possible, and not a combination of manufacturers, for a classification of material. Material and equipment must be standard products of a manufacturer regularly engaged in the manufacture of the products for at least 2 years prior to bid.

#### 2.1.2 Nameplates

Major components of equipment must have the manufacturer's name, address, type or style, model or serial number, catalog number, date of installation, installing Contractor's name and address, and the contract number provided on a new name plate permanently affixed to the item or equipment. Nameplates must be etched metal or plastic, permanently attached by screws to control units, panels or adjacent walls.

#### 2.1.3 Identification and Marking

Pipe and fitting markings must include name or identifying symbol of manufacturer and nominal size. Pipe must be marked with ASTM designation. Valves and equipment markings must have name or identifying symbol of manufacturer, specific model number, nominal size, name of device, arrow indicating direction of flow, and position of installation (horizontal or vertical), except if valve can be installed in either position. Markings must be included on the body casting or on an etched or stamped metal nameplate permanently on the valve or cover plate.

#### 2.1.4 Pressure Ratings

Valves, fittings, couplings, alarm switches, and similar devices must be rated for the maximum working pressures that can be experienced in the system, but in no case less than 175 psi.

### 2.2 UNDERGROUND PIPING COMPONENTS

#### 2.2.1 Pipe

Pipe must comply with NFPA 24. Minimum pipe size is 6 inches. Piping more than 5 feet outside the building walls must comply with Division 33 specifications. A continuous section of welded stainless steel fire water service piping from a point outside the building perimeter to a flanged fitting at least 1-foot above the finished floor within the building is acceptable.

#### 2.2.2 Fittings and Gaskets

Fittings must be ductile-iron conforming to AWWA C110/A21.10 with cement mortar lining conforming to AWWA C104/A21.4. Gaskets must be suitable in design and size for the pipe with which such gaskets are to be used. Gaskets for ductile-iron pipe joints must conform to AWWA C111/A21.11.

#### 2.2.3 Buried Utility Warning and Identification Tape

Provide detectable aluminum foil plastic backed tape or detectable magnetic plastic tape manufactured specifically for warning and identification of buried piping. Tape must be detectable by an electronic detection instrument. Provide tape, 3 inches minimum width, color coded for the utility involved with warning and identification imprinted in bold block letters continuously and repeatedly over the entire tape length. Warning and identification must read "CAUTION BURIED WATER PIPING BELOW" or similar wording. Use permanent code and letter coloring unaffected by



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

moisture and other substances contained in trench backfill material.

## 2.3 ABOVEGROUND PIPING COMPONENTS

### 2.3.1 Steel Piping Components

#### 2.3.1.1 Steel Pipe

Except as modified herein, steel pipe must be black as permitted by NFPA 13 and conform to the applicable provisions of ASTM A53/A53M, ASTM A135/A135M or ASTM A153/A153M.

Steel pipe must be minimum Schedule 40 for sizes 2 inches and less; and minimum Schedule 10 for sizes larger than 2 inches. Steel piping with wall thickness less than Schedule 40 must not be threaded.

#### 2.3.1.2 Fittings

Fittings must be welded, threaded, or grooved-end type. Threaded fittings must be cast-iron conforming to ASME B16.4, malleable-iron conforming to ASME B16.3 or ductile-iron conforming to ASTM A536. Plain-end fittings with mechanical couplings, fittings that use steel gripping devices to bite into the pipe, steel press fittings and field welded fittings are not permitted. Fittings, mechanical couplings, and rubber gaskets must be supplied by the same manufacturer. Threaded fittings must use Teflon tape or manufacturer's approved joint compound. Mechanical/Saddle tees using rubber gasketed fittings are not permitted. Reducing couplings are not permitted except as allowed by NFPA 13.

#### 2.3.1.3 Grooved Mechanical Joints and Fittings

Joints and fittings must be designed for not less than 175 psi service and the product of the same manufacturer. Field welded fittings must not be used. Fitting and coupling housing must be malleable-iron conforming to ASTM A47/A47M, Grade 32510; ductile-iron conforming to ASTM A536, Grade 65-45-12. Rubber gasketed grooved-end pipe and fittings with mechanical couplings are permitted in pipe sizes 2 inches and larger. Gasket must be the flush type that fills the entire cavity between the fitting and the pipe. Nuts and bolts must be heat-treated steel conforming to ASTM A183 and must be cadmium-plated or zinc-electroplated.

#### 2.3.1.4 Flanges

Flanges must conform to NFPA 13 and ASME B16.1. Gaskets must be non-asbestos compressed material in accordance with ASME B16.21, 1/16-inch thick, and full face or self-centering flat ring type.

### 2.3.2 Flexible Sprinkler Hose

The use of flexible hose is not permitted.

### 2.3.3 Pipe Hangers and Supports

Provide galvanized pipe hangers and supports in accordance with NFPA 13. Provide Clevis-type hangers for piping 4 inches and larger.

### 2.3.4 Valves

Provide valves of types approved for fire service. Valves must open by

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

counterclockwise rotation.

#### 2.3.4.1 Control Valve

Manually operated sprinkler control/gate valve must be outside stem and yoke (OS&Y) type or butterfly type as indicated on the drawings and must be listed.

#### 2.3.4.2 Check Valves

Check valves must comply with UL 312. Check valves 4 inches and larger must be of the swing type, have a clear waterway and meet the requirements of MSS SP-71, for Type 3 or 4. Inspection plate must be provided on valves larger than 6 inches.

#### 2.3.4.3 Hose Valve

Valve must comply with UL 668. Provide hose valves for testing of the backflow preventer.

#### 2.3.5 Riser Check Valves

Provide riser check valve, pressure gauges and main drain.

### 2.4 ALARM INITIATING AND SUPERVISORY DEVICES

#### 2.4.1 Sprinkler Alarm Switch

Vane-type flow switch(es). Connection of switch must be by the fire alarm installer. Vane type alarm actuating devices must have mechanical diaphragm controlled retard device adjustable from 10 to 60 seconds and must instantly recycle.

#### 2.4.2 Valve Supervisory (Tamper) Switch

Switch must be integral to the control valve or suitable for mounting to the type of control valve to be supervised open. The switch must be tamper resistant and contain SPDT (Form C) contacts arranged to transfer upon removal of the housing cover or closure of the valve of more than two rotations of the valve stem.

### 2.5 BACKFLOW PREVENTION ASSEMBLY

Double-check valve assembly backflow preventer complying with ASSE 1013, ASSE 1015 and AWWA M14. Each check valve must have a drain. Backflow prevention assemblies must have current "Certificate of Approval from the Foundation for Cross-Connection Control and Hydraulic Research, FCCCHR List" and be listed for fire protection use. Listing of the specific make, model, design, and size in the FCCCHR List is acceptable as the required documentation. Where indicated on the drawings, the backflow prevention assembly shall be listed for use in the vertical position.

#### 2.5.1 Backflow Preventer Test Connection

Test connection must consist of a series of listed hose valves with 2 1/2-inch National Standard male hose threads with cap and chain. Provide one hose valve per 250 gpm of sprinkler system demand (including hose stream allowance) or fraction thereof.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## 2.6 FIRE DEPARTMENT CONNECTION

Fire department connection must be freestanding, projecting or flush type as indicated on the drawings, with cast-brass body, matching wall escutcheon lettered "Auto Spkr" with a chromium-plated finish. The connection must have individual self-closing clappers, caps with drip drains and chains. Female inlets must have 2 1/2-inch diameter American National Fire Hose Connection Screw Threads (NH) per NFPA 1963. Comply with UL 405.

## 2.7 WATERFLOW ALARM

Electrically operated, exterior-mounted, weatherproof waterflow alarm bell shall be provided and installed above the fire department connection in accordance with NFPA 13. The waterflow alarm bell shall be rated 24 VDC and shall be connected to the Fire Alarm Control Unit (FACU/FACP) in accordance with Section 28 31 76 INTERIOR FIRE ALARM AND MASS NOTIFICATION SYSTEM, ADDRESSABLE.

## 2.8 SPRINKLERS

Sprinklers must comply with UL 199 and NFPA 13. Sprinklers with internal O-rings are not acceptable. Sprinklers in high heat areas including attic spaces or in close proximity to unit heaters must have temperature classification in accordance with NFPA 13. Extended coverage sprinklers are not permitted. Listed residential sprinklers are permitted to be used at their listed/approved spacing and coverage.

### 2.8.1 Pendent Sprinkler

Pendent sprinkler must be recessed, quick-response type with nominal K-factor as indicated on the drawings. Pendent sprinklers must have a polished chrome finish. Assembly must include an integral escutcheon.

### 2.8.2 Upright Sprinkler

Upright sprinkler must be brass, quick-response type and have a nominal K-factor as indicated on the drawings.

### 2.8.3 Sidewall Sprinkler

Sidewall sprinkler must be the quick-response type. Sidewall sprinkler must have a nominal K-factor as indicated on the drawings. Sidewall sprinkler must have a brass finish.

### 2.8.4 Concealed Sprinkler

Concealed sprinkler must be chrome-plated, quick-response type and have a nominal K-factor as indicated on the drawings. Coverplate must be white.

### 2.8.5 Residential Sprinkler

Residential sprinkler must be recessed pendent, or sidewall type with nominal K-factor as indicated on the drawings. The minimum k-factor for residential sprinklers must be 5.6. Residential sprinkler must have a polished-chrome finish. Sprinkler must comply with UL 1626.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### 2.8.6 Corrosion-Resistant Sprinkler

Corrosion-resistant sprinkler must be the upright or pendent type installed in locations as indicated. Corrosion-resistant coatings must be factory-applied by the sprinkler manufacturer.

#### 2.8.7 Dry Sprinkler Assembly

Dry sprinkler assembly must be of the pendent or sidewall type as indicated on the drawings. Assembly must include an integral escutcheon. Maximum length must not exceed maximum indicated in its listing. The minimum exposed length of the barrel of the dry sprinkler shall be in accordance with NFPA 13. Sprinkler must have a polished chrome finish.

### 2.9 ACCESSORIES

#### 2.9.1 Sprinkler Cabinet

Provide spare sprinklers in accordance with NFPA 13 and must be placed in a suitable metal cabinet of sufficient size to accommodate all the spare sprinklers and wrenches in designated locations. Spare sprinklers must be representative of, and in proportion to, the number of each type and temperature rating of the sprinklers installed as required by NFPA 13. At least one wrench of each type required must be provided.

#### 2.9.2 Pendent Sprinkler Escutcheon

Escutcheon must be one-piece metallic type with a depth of less than 3/4-inch and suitable for installation on pendent sprinklers. The escutcheon must have a factory finish that matches the pendent sprinkler.

#### 2.9.3 Pipe Escutcheon

Provide split hinge metal plates for piping entering walls, floors, and ceilings in exposed spaces. Provide polished stainless steel plates or chromium-plated finish on copper alloy plates in finished spaces. Provide paint finish on metal plates in unfinished spaces.

#### 2.9.4 Sprinkler Guard

Listed guard must be a steel wire cage designed to encase the sprinkler and protect it from mechanical damage. Guards shall be listed for use with the specific sprinkler (manufacturer and model) to be protected. Guards must be provided on sprinklers located in all electrical and communication rooms without ceilings regardless of elevation, sprinklers located beneath obstructions such as ductwork, sprinklers within 7 feet of the floor, and where otherwise subject to mechanical damage.

#### 2.9.5 Relief Valve

Relief valves must be listed and installed at the riser in accordance with NFPA 13.

#### 2.9.6 Air Vent

Air vents must be of the automatic type and piped to drain to the building exterior. The automatic air release valve shall consist of a 40 mesh "Y" type strainer connected to an automatic air vent valve. The output of the air vent shall be 1/2-inch NPT male connections which allows a drain

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

attachment for safely draining inadvertent discharge of water that is inherent in the operation of the automatic air vent. The automatic air vent assembly shall be field replaceable without disabling the sprinkler system by the installation of a 1/2-inch ball valve installed before the "Y" type strainer for isolation purposes. The automatic air release valve shall be mounted in a vertical position and shall require a minimum of 8-inches of clearance above the sprinkler main or branch line piping. Install the automatic air release valve(s) at the high point(s) of the fire sprinkler system piping to ensure evacuation of air from the system during and after filling.

#### 2.9.7 Identification Sign

Valve identification sign must be minimum 6 inches wide by 2 inches high with enamel baked finish on minimum 18 gage steel or 0.024-inch aluminum with red letters on a white background or white letters on red background. Wording of sign must include, but not be limited to "main drain", "auxiliary drain", "inspector's test", "alarm test", "alarm line", and similar wording as required to identify operational components. Where there is more than one sprinkler system, signage must include specific details as to the respective system.

### PART 3 EXECUTION

#### 3.1 VERIFYING ACTUAL FIELD CONDITIONS

Before commencing work, examine all adjoining work on which the Contractor's work that is dependent for perfect workmanship according to the intent of this specification section, and report to the Contracting Officer's Representative a condition that prevents performance of first class work. No "waiver of responsibility" for incomplete, inadequate or defective adjoining work will be considered unless notice has been filed before submittal of a proposal.

#### 3.2 INSTALLATION

The installation must be in accordance with the applicable provisions of NFPA 13, NFPA 24 and publications referenced therein. Locate sprinklers in a consistent pattern with ceiling grid, lights, and air supply diffusers. Install sprinkler system over and under ducts, piping and platforms when such equipment can negatively affect or disrupt the sprinkler discharge pattern and coverage.

- a. Piping offsets, fittings, and other accessories required must be furnished to provide a complete installation and to eliminate interference with other construction.
- b. Wherever the Contractor's work interconnects with work of other trades the Contractor must coordinate with other Contractors to insure all Contractors have the information necessary so that they may properly install all necessary connections and equipment. Identify all work items needing access (dampers and similar equipment) that are concealed above hung ceilings by permanent color coded pins/tabs in the ceiling directly below the item.
- c. Provide required supports and hangers for piping, conduit, and equipment so that loading will not exceed allowable loadings of structure. Submittal of a bid must be a deemed representation that the Contractor submitting such bid has ascertained allowable loadings

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

and has included in his estimates the costs associated in furnishing required supports.

### 3.2.1 Waste Removal

At the conclusion of each day's work, clean up and stockpile on site all waste, debris, and trash which may have accumulated during the day as a result of work by the Contractor and of his presence on the job. Sidewalks and streets adjoining the property must be kept broom clean and free of waste, debris, trash and obstructions caused by work of the Contractor, which will affect the condition and safety of streets, walks, utilities, and property.

### 3.3 UNDERGROUND PIPING INSTALLATION

The fire protection water main must be laid, and joints anchored, in accordance with NFPA 24. Minimum depth of cover must be 3 feet or the frost line, whichever is deeper. The supply line must terminate inside the building with a flanged piece, the bottom of which must be set not less than 1-foot above the finished floor. A blind flange must be installed temporarily on top of the flanged piece to prevent the entrance of foreign matter into the supply line. The service main shall be restrained against movement at changes in pipe direction in accordance with NFPA 24 by one of the following methods:

1. A concrete thrust block provided the soil conditions are stable and capable of resisting the anticipated thrust forces.
2. Listed restrained joint systems such as locking mechanical or push-on joints, mechanical joints utilizing setscrew retainer glands, bell joint restraints, bolted flange joints, or pipe clamps and tie rods.
3. Threaded connections, grooved connections, welded connections, heat-fused connections, or chemical or solvent cemented connections provided that such joints can pass the NFPA 24 hydrostatic test without shifting of piping.

In addition, joints must be anchored in accordance with NFPA 24. Buried steel components must be provided with a corrosion protective coating in accordance with AWWA C203. Piping more than 5 feet outside the building walls must meet the requirements of Division 33 Specifications.

### 3.4 ABOVEGROUND PIPING INSTALLATION

The methods of fabrication and installation of the aboveground piping must fully comply with the requirements and recommended practices of NFPA 13 and this specification section.

#### 3.4.1 Protection of Piping Against Earthquake Damage

Seismic restraint is not required.

#### 3.4.2 Piping in Exposed Areas

Install exposed piping without diminishing exit access widths, corridors or equipment access. Exposed horizontal piping, including drain piping, must be installed to provide maximum headroom, but no less than 7.5 feet above the finished floor, except where specifically noted on the drawings. Provide a minimum clearance of 36-inches access to and in front of all

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

equipment and a minimum 6-inches behind equipment (control valves, backflow preventer, check valves, waterflow switches, etc.). Pipe installation shall not encroach upon the Dedicated Equipment Space required by NFPA 70 for electrical panels and equipment. Piping located directly above such electrical equipment must be located at least 6 feet above the top of electrical equipment and provided with leak protection.

#### 3.4.3 Piping in Finished Areas

In areas with suspended or dropped ceilings and in areas with concealed spaces above the ceiling, piping must be concealed above ceilings. Piping must be inspected, hydrostatically tested and approved before being concealed. Risers and similar vertical runs of piping in finished areas must be concealed.

#### 3.4.4 Pendent Sprinklers

- a. Drop nipples to pendent sprinklers must consist of minimum 1-inch pipe with a reducing coupling into which the sprinkler must be threaded.
- b. Where sprinklers are installed below suspended or dropped ceilings, drop nipples must be cut such that sprinkler ceiling plates or escutcheons are of a uniform depth throughout the finished space. The outlet of the reducing coupling must not extend below the underside of the ceiling.
- c. Recessed pendent sprinklers must be installed such that the distance from the sprinkler deflector to the underside of the ceiling must not exceed the manufacturer's listed range and must be of uniform depth throughout the finished area.
- d. Pendent sprinklers in suspended ceilings must be located in the center of the tile (plus or minus 2 inches).

#### 3.4.5 Upright Sprinklers

Riser nipples or "sprigs" to upright sprinklers must contain no fittings between the branch line tee and the reducing coupling at the sprinkler. Riser nipples exceeding 30 inches in length shall be individually supported.

#### 3.4.6 Pipe Joints

Pipe joints must conform to NFPA 13, except as modified herein. Not more than four threads must show after joint is made up. Welded joints will be permitted, only if welding operations are performed as required by NFPA 13 at the Contractor's fabrication shop, not at the project construction site. Flanged joints must be provided where indicated or required by NFPA 13. Grooved pipe and fittings must be prepared in accordance with the manufacturer's latest published specification according to pipe material, wall thickness and size. Grooved couplings, fittings and grooving tools must be products of the same manufacturer. The diameter of grooves made in the field must be measured using a "go/no-go" gauge, vernier or dial caliper, narrow-land micrometer, or other method specifically approved by the coupling manufacturer for the intended application. Groove width and dimension of groove from end of pipe must be measured and recorded for each change in grooving tool setup to verify compliance with coupling manufacturer's tolerances.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### 3.4.7 Reducers

Reductions in pipe sizes must be made with one-piece tapered reducing fittings. When standard fittings of the required size are not manufactured, single bushings of the face or hex type will be permitted. Where used, face bushings must be installed with the outer face flush with the face of the fitting opening being reduced. Bushings cannot be used in elbow fittings, in more than one outlet of a tee, in more than two outlets of a cross, or where the reduction in size is less than 1/2-inch.

#### 3.4.8 Pipe Penetrations

- a. Cutting structural members for passage of pipes or for pipe-hanger fastenings will not be permitted. Pipes that must penetrate concrete or masonry walls or concrete floors must be core-drilled and provided with pipe sleeves. Each sleeve must be Schedule 40 galvanized steel, ductile-iron or cast-iron pipe and extend through its respective wall or floor and be cut flush with each wall surface. Sleeves must provide required clearance between the pipe and the sleeve per NFPA 13. The space between the sleeve and the pipe must be firmly packed with mineral wool insulation.
- b. Where pipes and sleeves penetrate fire walls, fire partitions, or floors, pipes/sleeves must be firestopped in accordance with Section 07 84 00 FIRESTOPPING.
- c. In penetrations that are not fire-rated or not a floor penetration, the space between the sleeve and the pipe must be sealed at both ends with plastic waterproof cement that will dry to a firm but pliable mass or with a mechanically adjustable segmented elastomer seal.

#### 3.4.9 Escutcheons

Escutcheons must be provided for pipe penetration in finished areas of ceilings, floors and walls. Escutcheons must be securely fastened to the pipe at surfaces through which piping passes.

#### 3.4.10 Inspector's Test Connection

Unless otherwise indicated, the test connection must consist of 1-inch pipe connected at the riser as a combination test and drain valve; a test valve located approximately 7 feet above the floor; a smooth bore brass outlet equivalent to the smallest orifice sprinkler used in the system; and a painted metal identification sign affixed to the valve with the words "Inspector's Test". All test connection piping must be inside of the building and penetrate the exterior wall at the location of the discharge orifice only. The discharge orifice must be located outside the building wall no more than 2 feet above finished grade, directed so as not to cause damage to adjacent construction or landscaping during full flow discharge, or to the sanitary sewer. Discharge to the exterior must not interfere with exiting from the facility. Water discharge or runoff must not cross the path of egress from the building. Do not discharge to the roof. Discharge to floor drains, janitor sinks or similar fixtures is not permitted.

Provide concrete splash blocks at all drain and inspector's test connection discharge locations if not discharging to a concrete surface. Splash blocks must be large enough to mitigate erosion and not become dislodged during a full flow of the drain. Ensure all discharged water



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

drains away from the facility and does not cause property damage.

#### 3.4.11 Backflow Preventer

Locate within the building.

Install backflow preventers so that the bottom of the assembly is a minimum of 6 inches above the finished floor/grade. Install horizontal backflow preventers so that the bottom of the assembly is no greater than 24 inches above the finished floor/grade. Clearance around control valve handles must be minimum 6 inches above grade/finished floor and away from walls.

##### 3.4.11.1 Test Connection

Provide downstream of the backflow prevention assembly UL 668 hose valves with 2.5-inch National Standard male hose threads with cap and chain. Provide one valve for each 250 gpm of system demand or fraction thereof. Provide a permanent sign in accordance with paragraph entitled "Identification Signs" which reads, "Test Valve". Indicate location of test header. If an exterior connection, provide a control valve inside a heated mechanical room to prevent freezing.

#### 3.4.12 Drains

- a. Main drain piping must be provided to discharge at a safe point outside the building, no more than 2 feet above finished grade. Provide a concrete splash block at drain outlet. Discharge to the exterior must not interfere with exiting from the facility. Water discharge or runoff must not cross the path of egress from the building.
- b. Auxiliary drains must be provided as required by NFPA 13. Auxiliary drains are permitted to discharge to a floor drain if the drain is sized to accommodate full flow (min 40 gpm). Discharge to service sinks or similar plumbing fixtures is not permitted.

#### 3.4.13 Installation of Fire Department Connection

Connection must be as indicated on the drawings, freestanding or be mounted on the exterior wall approximately 2 feet above finished grade. The piping between the connection and the check valve must be provided with an automatic drip in accordance with NFPA 13 and piped to drain to the outside.

#### 3.4.14 Identification Signs

Signs must be affixed to each control valve, inspector test valve, main drain, auxiliary drain, test valve, and similar valves as appropriate or as required by NFPA 13. Main drain test results must be etched into main drain identification sign. Hydraulic design data must be etched into the nameplates and permanently affixed to each sprinkler riser as specified in NFPA 13. Provide labeling on the surfaces of all feed and cross mains to show the pipe function (e.g., "Sprinkler System", "Fire Department Connection") and normal valve position (e.g. "Normally Open", "Normally Closed"). For pipe sizes 4-inch and larger provide white painted stenciled letters and arrows, a minimum of 2 inches in height and visible from at least two sides when viewed from the floor. For pipe sizes less than 4-inch, provide white painted stenciled letters and arrows, a minimum

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

of 0.75-inch in height and visible from the floor.

### 3.5 ELECTRICAL

Except as modified herein, electric equipment and wiring must be in accordance with Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Alarm signal wiring connected to the building fire alarm control system must be by the fire alarm installer.

### 3.6 PAINTING

Color code mark piping as specified in Division 09 Specifications and modified by this specification section.

Clean, pretreat, prime, and paint new fire extinguishing sprinkler systems including valves, piping, conduit, hangers, supports, miscellaneous metalwork, and accessories. The paint manufacturer's instructions shall be followed in the preparation of the surface and in the method of application. At a minimum, apply coatings to clean, dry surfaces using clean brushes. Clean the surfaces to remove dust, dirt, rust, and loose millscale. Immediately after cleaning, provide the metal surfaces with one coat primer. Shield sprinklers with protective covering while painting is in progress. Upon completion of painting, remove protective covering from sprinklers. Remove sprinklers which have been painted and replace with new sprinklers. Adjacent walls, ceilings, lighting, ductwork, equipment, etc., shall also be provided with protective covering as necessary while painting is in progress.

- a. Piping in Unfinished Areas: Provide primed surfaces with one coat of red alkyd gloss enamel applied to a minimum dry film thickness of 1.0 mil in attic spaces, spaces above suspended ceilings, crawl spaces, pipe chases, mechanical equipment room, and spaces where walls or ceilings are not painted or not constructed of a prefinished material.
- b. Piping in Finished Areas: Provide primed surfaces with two coats of paint to match adjacent surfaces, except provide valves and operating accessories with one coat of red alkyd gloss enamel applied to a minimum dry film thickness of 1.0 mil. Provide piping with 2-inch wide red enamel bands or self-adhering red plastic bands spaced at maximum 20 foot intervals throughout the piping system.

### 3.7 FIELD QUALITY CONTROL

#### 3.7.1 Test Procedures

Submit detailed test procedures, prepared and signed by the NICET Level III or IV Fire Sprinkler Technician, and the representative of the installing company, and reviewed by the QFPE 60 days prior to performing system tests. Detailed test procedures must list all components of the installed system. Test procedures must include sequence of testing, time estimate for each test, and sample test data forms. The test data forms must be in a check-off format (pass/fail with space to add applicable test data; similar to the forms in NFPA 13). The test procedures and accompanying test data forms must be used for the pre-Government testing and the Government final testing.

- a. Provide space to identify the date and time of each test. Provide space to identify the names and signatures of the individuals conducting and witnessing each test.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 3.7.2 Pre-Government Testing

#### 3.7.2.1 Verification of Compliant Installation

Conduct inspections and tests to ensure that equipment is functioning properly. Tests must meet the requirements of paragraph entitled "Minimum System Tests" and "System Acceptance" as noted in NFPA 13. The Contractor and QFPE must be in attendance at the pre-Government testing to make necessary adjustments. After inspection and testing is complete, provide a signed Verification of Compliant Installation letter by the QFPE that the installation is complete, compliant with the specification and fully operable. The letter must include the names and titles of the witnesses to the pre-Government tests. Provide all completion documentation as required by NFPA 13 and the test reports noted below.

- a. NFPA 13 Aboveground Material and Test Certificate
- b. NFPA 13 Underground Material and Test Certificate

#### 3.7.2.2 Request for Government Final Test

When the verification of compliant installation has been completed, submit a formal request for Government final test to the Contracting Officers Designated Representative (COR). Government final testing will not be scheduled until the COR has received copies of the request for Government final testing and Verification of Compliant Installation letter with all required reports. Government final testing will not be performed until after the connections to the building fire alarm system have been completed and tested to confirm communications are fully functional. Submit request for test at least 15 calendar days prior to the requested test date.

#### 3.7.3 Correction of Deficiencies

If equipment was found to be defective or non-compliant with contract requirements, perform corrective actions and repeat the tests. Tests must be conducted and repeated if necessary until the system has been demonstrated to comply with all contract requirements.

#### 3.7.4 Government Final Tests

The tests must be performed in accordance with the approved test procedures in the presence of the COR. Furnish instruments and personnel required for the tests. The following must be provided at the job site for Government Final Testing:

- a. The manufacturer's technical representative.
- b. The Contractor's Qualified Fire Protection Engineer (QFPE).
- c. Marked-up red line drawings of the system as actually installed.

Government Final Tests will be witnessed by the Contracting Officer, the Contractor's Qualified Fire Protection Engineer (QFPE), and the Designer of Record (DOR). At this time, all required tests noted in the paragraph "Minimum System Tests" must be repeated at their discretion.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 3.8 MINIMUM SYSTEM TESTS

The system, including the underground water mains, and the aboveground piping and system components, must be tested to ensure that equipment and components function as intended. The underground and aboveground interior piping systems and attached appurtenances subjected to system working pressure must be tested in accordance with NFPA 13 and NFPA 24.

#### 3.8.1 Underground Piping

##### 3.8.1.1 Flushing

Underground piping must be flushed at a minimum of 10 fps in accordance with NFPA 24.

##### 3.8.1.2 Hydrostatic Test

New underground piping must be hydrostatically tested in accordance with NFPA 24.

#### 3.8.2 Aboveground Piping

##### 3.8.2.1 Hydrostatic Test

Aboveground piping must be hydrostatically tested in accordance with NFPA 13. There must be no drop in gauge pressure or visible leakage when the system is subjected to the hydrostatic test. The test pressure must be read from a gauge located at the low elevation point of the system or portion being tested.

##### 3.8.2.2 Backflow Prevention Assembly Forward Flow Test

Each backflow prevention assembly must be tested at system flow demand, including all applicable hose streams, as specified in NFPA 13. The Contractor must provide all equipment and instruments necessary to conduct a complete forward flow test, including 2.5-inch diameter hoses, playpipe nozzles or flow diffusers, calibrated pressure gauges, and pitot tube gauge. The Contractor must provide all necessary supports to safely secure hoses and nozzles during the test. At the system demand flow, the pressure readings and pressure drop (friction loss) across the assembly must be recorded. A metal placard must be provided on the backflow prevention assembly that lists the pressure readings both upstream and downstream of the assembly, total pressure drop, and the system test flow rate determined during the preliminary testing. The pressure drop must be compared to the manufacturer's data and the readings observed during the final inspections and tests.

##### 3.8.3 Main Drain Flow Test

Following flushing of the underground piping, a main drain test must be made to verify the adequacy of the water supply. Static and residual pressures must be recorded on the certificate specified in paragraph SUBMITTALS.

### 3.9 SYSTEM ACCEPTANCE

Following acceptance of the system, as-built drawings and O&M manuals must be delivered to the Contracting Officer for review and acceptance. Submit six sets of detailed as-built drawings. The drawings must show the system

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

as installed, including deviations from both the project drawings and the approved shop drawings. These drawings must be submitted within two weeks after the final acceptance test of the system. At least one set of as-built (marked-up) drawings must be provided at the time of, or prior to the final acceptance test.

- a. Provide one set of full size paper as-built drawings and schematics. The drawings must be prepared electronically and sized no less than the contract drawings. Furnish one set of CDs or DVDs containing software back-up and CAD based drawings in latest version of AutoCAD, DXF and portable document formats of as-built drawings and schematics.
- b. Provide operating and maintenance (O&M) instructions.

### 3.10 ONSITE TRAINING

Conduct a training course for the responding fire department and operating and maintenance personnel as designated by the Contracting Officer. Training must be performed on two separate days (to accommodate different shifts of Fire Department personnel) for a period of 4 hours of normal working time and must start after the system is functionally complete and after the final acceptance test. The on-site training must cover all of the items contained in the approved Operating and Maintenance Instructions.

-- End of Section --

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## SECTION 22 00 00

### PLUMBING, GENERAL PURPOSE 11/15, CHG 4: 05/21

#### PART 1 GENERAL

##### 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

#### AIR-CONDITIONING, HEATING AND REFRIGERATION INSTITUTE (AHRI)

AHRI 1010 (2002) Self-Contained, Mechanically Refrigerated Drinking-Water Coolers

#### AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI Z21.10.1/CSA 4.1 (2019) Gas Water Heaters Vol. I, Storage Water Heaters with Input Ratings of 75,000 Btu Per Hour or Less

ANSI Z21.10.3/CSA 4.3 (2019) Gas-Fired Water Heaters Vol.III, Storage Water Heaters With Input Ratings Above 75,000 Btu Per Hour, Circulating and Instantaneous

ANSI Z21.22/CSA 4.4 (2015; R 2020) Relief Valves for Hot Water Supply Systems

#### AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS (ASHRAE)

ASHRAE 90.1 - IP (2019) Energy Standard for Buildings Except Low-Rise Residential Buildings

ASHRAE 146 (2020) Method of Testing and Rating Pool Heaters

#### AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME A112.1.2 (2012; R 2017; R 2022) Air Gaps in Plumbing Systems (For Plumbing Fixtures and Water-Connected Receptors)

ASME A112.6.1M (1997; R 2017) Floor Affixed Supports for Off-the-Floor Plumbing Fixtures for Public Use

ASME A112.6.3 (2019) Standard for Floor and Trench Drains

ASME A112.6.4 (2003; R 2012) Roof, Deck and Balcony Drains

ASME A112.14.1 (2003; R 2017; R 2022) Backwater Valves

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

|                          |                                                                                                                          |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------|
| ASME A112.19.2/CSA B45.1 | (2018; ERTA 2018) Standard for Vitreous China Plumbing Fixtures and Hydraulic Requirements for Water Closets and Urinals |
| ASME A112.19.3/CSA B45.4 | (2022) Stainless Steel Plumbing Fixtures                                                                                 |
| ASME A112.36.2M          | (1991; R 2017) Cleanouts                                                                                                 |
| ASME B1.20.1             | (2013; R 2018) Pipe Threads, General Purpose (Inch)                                                                      |
| ASME B16.15              | (2018) Cast Copper Alloy Threaded Fittings Classes 125 and 250                                                           |
| ASME B16.18              | (2021) Cast Copper Alloy Solder Joint Pressure Fittings                                                                  |
| ASME B16.22              | (2021) Wrought Copper and Copper Alloy Solder Joint Pressure Fittings                                                    |
| ASME B16.23              | (2021) Cast Copper Alloy Solder Joint Drainage Fittings - DWV                                                            |
| ASME B16.29              | (2022) Wrought Copper and Wrought Copper Alloy Solder-Joint Drainage Fittings - DWV                                      |
| ASME B16.34              | (2021) Valves - Flanged, Threaded and Welding End                                                                        |
| ASME B16.50              | (2021) Wrought Copper and Copper Alloy Braze-Joint Pressure Fittings                                                     |
| ASME B16.51              | (2013) Copper and Copper Alloy Press-Connect Pressure Fittings                                                           |
| ASME B31.1               | (2022) Power Piping                                                                                                      |
| ASME B31.5               | (2022) Refrigeration Piping and Heat Transfer Components                                                                 |
| ASME B40.100             | (2022) Pressure Gauges and Gauge Attachments                                                                             |
| ASME BPVC SEC IV         | (2017) BPVC Section IV-Rules for Construction of Heating Boilers                                                         |
| ASME BPVC SEC VIII D1    | (2019) BPVC Section VIII-Rules for Construction of Pressure Vessels Division 1                                           |
| ASME CSD-1               | (2021) Control and Safety Devices for Automatically Fired Boilers                                                        |

#### AMERICAN SOCIETY OF SANITARY ENGINEERING (ASSE)

|           |                                                                                 |
|-----------|---------------------------------------------------------------------------------|
| ASSE 1001 | (2021) Performance Requirements for Atmospheric Type Vacuum Breakers            |
| ASSE 1003 | (2020) Performance Requirements for Water Pressure Reducing Valves for Domestic |

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### Water Distribution Systems - (ANSI approved 2010)

|           |                                                                                                           |
|-----------|-----------------------------------------------------------------------------------------------------------|
| ASSE 1010 | (2021) Performance Requirements for Water Hammer Arresters                                                |
| ASSE 1011 | (2017) Performance Requirements for Hose Connection Vacuum Breakers                                       |
| ASSE 1012 | (2023) Performance Requirements for Backflow Preventer with an Intermediate Atmospheric Vent              |
| ASSE 1013 | (2021) Performance Requirements for Reduced Pressure Principle Backflow Prevention Assemblies             |
| ASSE 1018 | (2023) Performance Requirements for Trap Seal Primer Valves - Potable Water Supplied (ANSI Approved 2002) |
| ASSE 1019 | (2011; R 2016) Performance Requirements for Wall Hydrant with Backflow Protection and Freeze Resistance   |
| ASSE 1020 | (2020) Performance Requirements for Pressure Vacuum Breaker Assemblies                                    |

#### AMERICAN WATER WORKS ASSOCIATION (AWWA)

|            |                                                                                                           |
|------------|-----------------------------------------------------------------------------------------------------------|
| AWWA 10084 | (2017) Standard Methods for the Examination of Water and Wastewater                                       |
| AWWA B300  | (2018) Hypochlorites                                                                                      |
| AWWA B301  | (2018) Liquid Chlorine                                                                                    |
| AWWA C203  | (2020) Coal-Tar Protective Coatings and Linings for Steel Water Pipelines - Enamel and Tape - Hot-Applied |
| AWWA C606  | (2022) Grooved and Shouldered Joints                                                                      |
| AWWA C651  | (2014) Standard for Disinfecting Water Mains                                                              |
| AWWA C652  | (2019) Disinfection of Water-Storage Facilities                                                           |
| AWWA D100  | (2021) Welded Steel Tanks for Water Storage                                                               |

#### AMERICAN WELDING SOCIETY (AWS)

|                |                                                                          |
|----------------|--------------------------------------------------------------------------|
| AWS A5.8/A5.8M | (2019) Specification for Filler Metals for Brazing and Braze Welding     |
| AWS B2.2/B2.2M | (2016) Specification for Brazing Procedure and Performance Qualification |



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## ASTM INTERNATIONAL (ASTM)

|                 |                                                                                                                                               |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM A74        | (2021) Standard Specification for Cast Iron Soil Pipe and Fittings                                                                            |
| ASTM A888       | (2021a) Standard Specification for Hubless Cast Iron Soil Pipe and Fittings for Sanitary and Storm Drain, Waste, and Vent Piping Applications |
| ASTM B32        | (2020) Standard Specification for Solder Metal                                                                                                |
| ASTM B42        | (2020) Standard Specification for Seamless Copper Pipe, Standard Sizes                                                                        |
| ASTM B75/B75M   | (2020) Standard Specification for Seamless Copper Tube                                                                                        |
| ASTM B88        | (2022) Standard Specification for Seamless Copper Water Tube                                                                                  |
| ASTM B88M       | (2020) Standard Specification for Seamless Copper Water Tube (Metric)                                                                         |
| ASTM B117       | (2019) Standard Practice for Operating Salt Spray (Fog) Apparatus                                                                             |
| ASTM B152/B152M | (2019) Standard Specification for Copper Sheet, Strip, Plate, and Rolled Bar                                                                  |
| ASTM B306       | (2020) Standard Specification for Copper Drainage Tube (DWV)                                                                                  |
| ASTM B370       | (2022) Standard Specification for Copper Sheet and Strip for Building Construction                                                            |
| ASTM B813       | (2016) Standard Specification for Liquid and Paste Fluxes for Soldering of Copper and Copper Alloy Tube                                       |
| ASTM B828       | (2016) Standard Practice for Making Capillary Joints by Soldering of Copper and Copper Alloy Tube and Fittings                                |
| ASTM C564       | (2020a) Standard Specification for Rubber Gaskets for Cast Iron Soil Pipe and Fittings                                                        |
| ASTM C920       | (2018) Standard Specification for Elastomeric Joint Sealants                                                                                  |
| ASTM D2564      | (2020) Standard Specification for Solvent Cements for Poly(Vinyl Chloride) (PVC) Plastic Piping Systems                                       |
| ASTM D2657      | (2007; R 2015) Heat Fusion Joining Polyolefin Pipe and Fittings                                                                               |

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

|                   |                                                                                                                                               |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM D2661        | (2014; E 2018) Standard Specification for Acrylonitrile-Butadiene-Styrene (ABS) Schedule 40, Plastic Drain, Waste, and Vent Pipe and Fittings |
| ASTM D2665        | (2014) Standard Specification for Poly(Vinyl Chloride) (PVC) Plastic Drain, Waste, and Vent Pipe and Fittings                                 |
| ASTM D2822/D2822M | (2005; R 2011; E 2011) Standard Specification for Asphalt Roof Cement, Asbestos-Containing                                                    |
| ASTM D2855        | (2015) Standard Practice for Making Solvent-Cemented Joints with Poly(Vinyl Chloride) (PVC) Pipe and Fittings                                 |
| ASTM D3139        | (2019) Joints for Plastic Pressure Pipes Using Flexible Elastomeric Seals                                                                     |
| ASTM D3212        | (2020) Standard Specification for Joints for Drain and Sewer Plastic Pipes Using Flexible Elastomeric Seals                                   |
| ASTM D3311        | (2017) Standard Specification for Drain, Waste, and Vent (DWV) Plastic Fittings Patterns                                                      |
| ASTM D4551        | (2017) Standard Specification for Poly(Vinyl Chloride) (PVC) Plastic Flexible Concealed Water-Containment Membrane                            |
| ASTM E1           | (2014) Standard Specification for ASTM Liquid-in-Glass Thermometers                                                                           |
| ASTM F477         | (2014; R 2021) Standard Specification for Elastomeric Seals (Gaskets) for Joining Plastic Pipe                                                |
| ASTM F628         | (2023) Standard Specification for Acrylonitrile-Butadiene-Styrene (ABS) Schedule 40 Plastic Drain, Waste, and Vent Pipe with a Cellular Core  |
| ASTM F877         | (2023) Standard Specification for Crosslinked Polyethylene (PEX) Plastic Hot- and Cold-Water Distribution Systems                             |
| ASTM F891         | (2016) Standard Specification for Coextruded Poly (Vinyl Chloride) (PVC) Plastic Pipe with a Cellular Core                                    |
| ASTM F1290        | (2019) Standard Practice for Electrofusion Joining Polyolefin Pipe and Fittings                                                               |
| ASTM F1760        | (2016; R 2020) Standard Specification for Coextruded Poly(Vinyl Chloride) (PVC) Non-Pressure Plastic Pipe Having                              |

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

# Reprocessed-Recycled Content

ASTM F2389 (2023) Standard Specification for  
Pressure-rated Polypropylene (PP) Piping  
Systems

## CAST IRON SOIL PIPE INSTITUTE (CISPI)

CISPI 301 (2018) Hubless Cast Iron Soil Pipe and  
Fittings for Sanitary and Storm Drain,  
Waste, and Vent Piping Applications

CISPI 310 (2012) Coupling for Use in Connection with  
Hubless Cast Iron Soil Pipe and Fittings  
for Sanitary and Storm Drain, Waste, and  
Vent Piping Applications

## COPPER DEVELOPMENT ASSOCIATION (CDA)

CDA A4015 (2016; 14/17) Copper Tube Handbook

## INTERNATIONAL ASSOCIATION OF PLUMBING AND MECHANICAL OFFICIALS (IAPMO)

IAPMO PS 117 (2005b) Press Type Or Plain End Rub  
Gasketed W/ Nail CU & CU Alloy Fittings 4  
Install On CU Tubing

IAPMO UPC (2003) Uniform Plumbing Code

## INTERNATIONAL CODE COUNCIL (ICC)

ICC A117.1 (2017) Standard And Commentary Accessible  
and Usable Buildings and Facilities

ICC IPC (2021) International Plumbing Code

## INTERNATIONAL SAFETY EQUIPMENT ASSOCIATION (ISEA)

ANSI/ISEA Z358.1 (2014; R 2020) American National Standard  
for Emergency Eyewash and Shower Equipment

## MANUFACTURERS STANDARDIZATION SOCIETY OF THE VALVE AND FITTINGS INDUSTRY (MSS)

MSS SP-58 (2018) Pipe Hangers and Supports -  
Materials, Design and Manufacture,  
Selection, Application, and Installation

MSS SP-67 (2022) Butterfly Valves

MSS SP-70 (2011) Gray Iron Gate Valves, Flanged and  
Threaded Ends

MSS SP-71 (2018) Gray Iron Swing Check Valves,  
Flanged and Threaded Ends

MSS SP-72 (2010a) Ball Valves with Flanged or  
Butt-Welding Ends for General Service

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

|            |                                                                                    |
|------------|------------------------------------------------------------------------------------|
| MSS SP-78  | (2011) Cast Iron Plug Valves, Flanged and Threaded Ends                            |
| MSS SP-80  | (2019) Bronze Gate, Globe, Angle and Check Valves                                  |
| MSS SP-85  | (2011) Gray Iron Globe & Angle Valves Flanged and Threaded Ends                    |
| MSS SP-110 | (2010) Ball Valves Threaded, Socket-Welding, Solder Joint, Grooved and Flared Ends |

#### NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

|            |                                                                                     |
|------------|-------------------------------------------------------------------------------------|
| NEMA 250   | (2020) Enclosures for Electrical Equipment (1000 Volts Maximum)                     |
| NEMA MG 1  | (2021) Motors and Generators                                                        |
| NEMA MG 11 | (1977; R 2012) Energy Management Guide for Selection and Use of Single Phase Motors |

#### NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

|          |                                                                                  |
|----------|----------------------------------------------------------------------------------|
| NFPA 31  | (2020) Standard for the Installation of Oil-Burning Equipment                    |
| NFPA 54  | (2021) National Fuel Gas Code                                                    |
| NFPA 90A | (2024) Standard for the Installation of Air Conditioning and Ventilating Systems |

#### NSF INTERNATIONAL (NSF)

|             |                                                                |
|-------------|----------------------------------------------------------------|
| NSF 372     | (2016) Drinking Water System Components - Lead Content         |
| NSF/ANSI 14 | (2022) Plastics Piping System Components and Related Materials |
| NSF/ANSI 61 | (2022) Drinking Water System Components - Health Effects       |

#### PLASTIC PIPE AND FITTINGS ASSOCIATION (PPFA)

|               |                                                                  |
|---------------|------------------------------------------------------------------|
| PPFA Fire Man | (2016) Firestopping: Plastic Pipe in Fire Resistive Construction |
|---------------|------------------------------------------------------------------|

#### PLUMBING AND DRAINAGE INSTITUTE (PDI)

|            |                                        |
|------------|----------------------------------------|
| PDI WH 201 | (2010) Water Hammer Arresters Standard |
|------------|----------------------------------------|

#### SOCIETY OF AUTOMOTIVE ENGINEERS INTERNATIONAL (SAE)

|           |                                  |
|-----------|----------------------------------|
| SAE J1508 | (2023) Hose Clamp Specifications |
|-----------|----------------------------------|

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

U.S. DEPARTMENT OF ENERGY (DOE)

Energy Star (1992; R 2006) Energy Star Energy  
Efficiency Labeling System (FEMP)

U.S. ENVIRONMENTAL PROTECTION AGENCY (EPA)

PL 93-523 (1974; A 1999) Safe Drinking Water Act

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

10 CFR 430 Energy Conservation Program for Consumer  
Products

40 CFR 141.80 National Primary Drinking Water  
Regulations; Control of Lead and Copper;  
General Requirements

UNDERWRITERS LABORATORIES (UL)

UL 174 (2004; Reprint Dec 2021) UL Standard for  
Safety Household Electric Storage Tank  
Water Heaters

## 1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

### SD-03 Product Data

Recycled Content for Steel Pipe; S

Recycled Content for Cast Iron Pipe; S

Backflow Prevention Assemblies; G

Shower Faucets; G

WaterSense Label for Lavatory Faucet; S

### Fixtures

List of installed fixtures with manufacturer, model, and flow rate.

Flush Valve Water Closets

WaterSense Label for Flush Valve Water Closet; S

Flush Valve Urinals

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

WaterSense Label for Urinal; S

Wall Hung Lavatories

Kitchen Sinks

Service Sinks

Drinking-Water Coolers; G

Energy Star Label for Electric Water Cooler; S

Energy Star Label for Wheelchair Electric Water Cooler; S

WaterSense Label for Showerhead; S

Plastic Shower Stalls

Water Heaters; G

Energy Star Label for Gas Storage Water Heater; S

Pumps; G

Welding

A copy of qualified procedures and a list of names and identification symbols of qualified welders and welding operators.

Vibration-Absorbing Features; G

Details of vibration-absorbing features, including arrangement, foundation plan, dimensions and specifications.

Plumbing System

Diagrams, instructions, and other sheets proposed for posting. Manufacturer's recommendations for the installation of bell and spigot and hubless joints for cast iron soil pipe.

SD-06 Test Reports

Tests, Flushing and Disinfection

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

Test reports in booklet form showing all field tests performed to adjust each component and all field tests performed to prove compliance with the specified performance criteria, completion and testing of the installed system. Each test report shall indicate the final position of controls.

Test of Backflow Prevention Assemblies; FIO.

Certification of proper operation shall be as accomplished in accordance with state regulations by an individual certified by the state to perform such tests. If no state requirement exists, the Contractor shall have the manufacturer's representative test the device, to ensure the unit is properly installed and performing as intended. The Contractor shall provide written documentation of the tests performed and signed by the individual performing the tests.

#### SD-07 Certificates

##### Materials and Equipment

Where equipment is specified to conform to requirements of the ASME Boiler and Pressure Vessel Code, the design, fabrication, and installation shall conform to the code.

##### Bolts

Written certification by the bolt manufacturer that the bolts furnished comply with the specified requirements.

#### SD-10 Operation and Maintenance Data

##### Plumbing System; G

Submit in accordance with Section 01 78 23 OPERATION AND MAINTENANCE DATA.

### 1.3 STANDARD PRODUCTS

Specified materials and equipment shall be standard products of a manufacturer regularly engaged in the manufacture of such products. Specified equipment shall essentially duplicate equipment that has performed satisfactorily at least two years prior to bid opening. Standard products shall have been in satisfactory commercial or industrial use for 2 years prior to bid opening. The 2-year use shall include applications of equipment and materials under similar circumstances and of similar size. The product shall have been for sale on the commercial market through advertisements, manufacturers' catalogs, or brochures during the 2 year period.

#### 1.3.1 Alternative Qualifications

Products having less than a two-year field service record will be acceptable if a certified record of satisfactory field operation for not less than 6000 hours, exclusive of the manufacturer's factory or laboratory tests, can be shown.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 1.3.2 Service Support

The equipment items shall be supported by service organizations. Submit a certified list of qualified permanent service organizations for support of the equipment which includes their addresses and qualifications. These service organizations shall be reasonably convenient to the equipment installation and able to render satisfactory service to the equipment on a regular and emergency basis during the warranty period of the contract.

### 1.3.3 Manufacturer's Nameplate

Each item of equipment shall have a nameplate bearing the manufacturer's name, address, model number, and serial number securely affixed in a conspicuous place; the nameplate of the distributing agent will not be acceptable.

### 1.3.4 Modification of References

In each of the publications referred to herein, consider the advisory provisions to be mandatory, as though the word, "shall" had been substituted for "should" wherever it appears. Interpret references in these publications to the "authority having jurisdiction", or words of similar meaning, to mean the Contracting Officer.

#### 1.3.4.1 Definitions

For the International Code Council (ICC) Codes referenced in the contract documents, advisory provisions shall be considered mandatory, the word "should" shall be interpreted as "shall." Reference to the "code official" shall be interpreted to mean the "Contracting Officer." For Navy owned property, references to the "owner" shall be interpreted to mean the "Contracting Officer." For leased facilities, references to the "owner" shall be interpreted to mean the "lessor." References to the "permit holder" shall be interpreted to mean the "Contractor."

#### 1.3.4.2 Administrative Interpretations

For ICC Codes referenced in the contract documents, the provisions of Chapter 1, "Administrator," do not apply. These administrative requirements are covered by the applicable Federal Acquisition Regulations (FAR) included in this contract and by the authority granted to the Officer in Charge of Construction to administer the construction of this project. References in the ICC Codes to sections of Chapter 1, shall be applied appropriately by the Contracting Officer as authorized by his administrative cognizance and the FAR.

### 1.4 DELIVERY, STORAGE, AND HANDLING

Handle, store, and protect equipment and materials to prevent damage before and during installation in accordance with the manufacturer's recommendations, and as approved by the Contracting Officer. Replace damaged or defective items.



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## 1.5 PERFORMANCE REQUIREMENTS

### 1.5.1 Welding

### 1.5.2 Cathodic Protection and Pipe Joint Bonding

Cathodic protection and pipe joint bonding systems shall be in accordance with.

## 1.6 REGULATORY REQUIREMENTS

Unless otherwise required herein, plumbing work shall be in accordance with ICC IPC.

## 1.7 PROJECT/SITE CONDITIONS

The Contractor shall become familiar with details of the work, verify dimensions in the field, and advise the Contracting Officer of any discrepancy before performing any work.

## 1.8 INSTRUCTION TO GOVERNMENT PERSONNEL

When specified in other sections, furnish the services of competent instructors to give full instruction to the designated Government personnel in the adjustment, operation, and maintenance, including pertinent safety requirements, of the specified equipment or system. Instructors shall be thoroughly familiar with all parts of the installation and shall be trained in operating theory as well as practical operation and maintenance work.

Instruction shall be given during the first regular work week after the equipment or system has been accepted and turned over to the Government for regular operation. The number of man-days (8 hours per day) of instruction furnished shall be as specified in the individual section. When more than 4 man-days of instruction are specified, use approximately half of the time for classroom instruction. Use other time for instruction with the equipment or system.

When significant changes or modifications in the equipment or system are made under the terms of the contract, provide additional instruction to acquaint the operating personnel with the changes or modifications.

## 1.9 ACCESSIBILITY OF EQUIPMENT

Install all work so that parts requiring periodic inspection, operation, maintenance, and repair are readily accessible. Install concealed valves, expansion joints, controls, dampers, and equipment requiring access, in locations freely accessible through access doors.

## PART 2 PRODUCTS

### 2.1 MATERIALS

Materials for various services shall be in accordance with TABLES I and II.

Pipe schedules shall be selected based on service requirements. Pipe fittings shall be compatible with the applicable pipe materials. Plastic pipe, fittings, and solvent cement shall meet NSF/ANSI 14 and shall be NSF

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

listed for the service intended. Plastic pipe, fittings, and solvent cement used for potable hot and cold water service shall bear the NSF seal "NSF-PW." Polypropylene pipe and fittings shall conform to dimensional requirements of Schedule 40, Iron Pipe size and shall comply with NSF/ANSI 14, NSF/ANSI 61 and ASTM F2389. Polypropylene piping that will be exposed to UV light shall be provided with a Factory applied UV resistant coating. Pipe threads (except dry seal) shall conform to ASME B1.20.1. Grooved pipe couplings and fittings shall be from the same manufacturer. Material or equipment containing a weighted average of greater than 0.25 percent lead shall not be used in any potable water system intended for human consumption, and shall be certified in accordance with NSF/ANSI 61, Annex G or NSF 372. In line devices such as water meters, building valves, check valves, meter stops, valves, fittings and back flow preventers shall comply with PL 93-523 and NSF/ANSI 61, Section 8. End point devices such as drinking water fountains, lavatory faucets, kitchen and bar faucets, residential ice makers, supply stops and end point control valves used to dispense water for drinking must meet the requirements of NSF/ANSI 61, Section 9. Hubless cast-iron soil pipe shall not be installed underground, under concrete floor slabs, or in crawl spaces below kitchen floors. Plastic pipe shall not be installed in air plenums. Plastic pipe shall not be installed in a pressure piping system in buildings greater than three stories including any basement levels.

#### 2.1.1 Pipe Joint Materials

Grooved pipe and hubless cast-iron soil pipe shall not be used underground. Solder containing lead shall not be used with copper pipe. Cast iron soil pipe and fittings shall be marked with the collective trademark of the Cast Iron Soil Institute. Joints and gasket materials shall conform to the following:

- a. Coupling for Cast-Iron Pipe: for hub and spigot type ASTM A74, AWWA C606. For hubless type: CISPI 310
  
- e. Brazing Material: Brazing material shall conform to AWS A5.8/A5.8M, BCuP-5.
  
- f. Brazing Flux: Flux shall be in paste or liquid form appropriate for use with brazing material. Flux shall be as follows: lead-free; have a 100 percent flushable residue; contain slightly acidic reagents; contain potassium borides; and contain fluorides.
  
- g. Solder Material: Solder metal shall conform to ASTM B32.
  
- h. Solder Flux: Flux shall be liquid form, non-corrosive, and conform to ASTM B813, Standard Test 1.
  
- i. PTFE Tape: PTFE Tape, for use with Threaded Metal or Plastic Pipe.
  
- j. Rubber Gaskets for Cast-Iron Soil-Pipe and Fittings (hub and spigot type and hubless type): ASTM C564.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

- l. Flexible Elastomeric Seals: ASTM D3139, ASTM D3212 or ASTM F477.
  
- p. Plastic Solvent Cement for PVC Plastic Pipe: ASTM D2564 and ASTM D2855.
  
- t. Press fittings for Copper Pipe and Tube: Copper press fittings shall conform to the material and sizing requirements of ASME B16.51 and performance criteria of IAPMO PS 117. Sealing elements for copper press fittings shall be EPDM, FKM or HNBR. Sealing elements shall be factory installed or an alternative supplied fitting manufacturer. Sealing element shall be selected based on manufacturer's approved application guidelines.
- u. Copper tubing shall conform to ASTM B88, Type K, L or M.

#### 2.1.2 Miscellaneous Materials

Miscellaneous materials shall conform to the following:

- a. Water Hammer Arrester: PDI WH 201. Water hammer arrester shall be piston type.
- b. Copper, Sheet and Strip for Building Construction: ASTM B370.
- c. Asphalt Roof Cement: ASTM D2822/D2822M.
- d. Hose Clamps: SAE J1508.
- e. Supports for Off-The-Floor Plumbing Fixtures: ASME A112.6.1M.
- f. Metallic Cleanouts: ASME A112.36.2M.
- g. Plumbing Fixture Setting Compound: A preformed flexible ring seal molded from hydrocarbon wax material. The seal material shall be nonvolatile nonasphaltic and contain germicide and provide watertight, gastight, odorproof and verminproof properties.
- h. Coal-Tar Protective Coatings and Linings for Steel Water Pipelines: AWWA C203.
- i. Hypochlorites: AWWA B300.
- j. Liquid Chlorine: AWWA B301.
- k. Gauges - Pressure and Vacuum Indicating Dial Type - Elastic Element:

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

ASME B40.100.

1. Thermometers: ASTM E1. Mercury shall not be used in thermometers.

### 2.1.3 Pipe Insulation Material

Insulation shall be as specified in Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS.

### 2.2 PIPE HANGERS, INSERTS, AND SUPPORTS

Pipe hangers, inserts, and supports shall conform to MSS SP-58.

### 2.3 VALVES

Valves shall be provided on supplies to equipment and fixtures. Valves 2-1/2 inches and smaller shall be bronze with threaded bodies for pipe and solder-type connections for tubing. Valves 3 inches and larger shall have flanged iron bodies and bronze trim. Pressure ratings shall be based upon the application. Grooved end valves may be provided if the manufacturer certifies that the valves meet the performance requirements of applicable MSS standard. Valves shall conform to the following standards:

| Description                                                                 | Standard       |
|-----------------------------------------------------------------------------|----------------|
| Butterfly Valves                                                            | MSS SP-67      |
| Cast-Iron Gate Valves, Flanged and Threaded Ends                            | MSS SP-70      |
| Cast-Iron Swing Check Valves, Flanged and Threaded Ends                     | MSS SP-71      |
| Ball Valves with Flanged Butt-Welding Ends for General Service              | MSS SP-72      |
| Ball Valves Threaded, Socket-Welding, Solder Joint, Grooved and Flared Ends | MSS SP-110     |
| Cast-Iron Plug Valves, Flanged and Threaded Ends                            | MSS SP-78      |
| Bronze Gate, Globe, Angle, and Check Valves                                 | MSS SP-80      |
| Steel Valves, Socket Welding and Threaded Ends                              | ASME B16.34    |
| Cast-Iron Globe and Angle Valves, Flanged and Threaded Ends                 | MSS SP-85      |
| Backwater Valves                                                            | ASME A112.14.1 |

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

|                                                                                     |                                                                                                 |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| Vacuum Relief Valves                                                                | ANSI Z21.22/CSA 4.4                                                                             |
| Water Pressure Reducing Valves                                                      | ASSE 1003                                                                                       |
| Water Heater Drain Valves                                                           | ASME BPVC SEC IV, Part HLW-810:<br>Requirements for Potable-Water Heaters<br>Bottom Drain Valve |
| Trap Seal Primer Valves                                                             | ASSE 1018                                                                                       |
| Temperature and Pressure Relief Valves<br>for Hot Water Supply Systems              | ANSI Z21.22/CSA 4.4                                                                             |
| Temperature and Pressure Relief Valves<br>for Automatically Fired Hot Water Boilers | ASME CSD-1<br>Safety Code No., Part CW, Article 5                                               |

### 2.3.1 Wall Hydrants (Frostproof)

ASSE 1019 with vacuum-breaker backflow preventer shall have a nickel-brass or nickel-bronze wall plate or flange with nozzle and detachable key handle. A brass or bronze operating rod shall be provided within a galvanized iron casing of sufficient length to extend through the wall so that the valve is inside the building, and the portion of the hydrant between the outlet and valve is self-draining. A brass or bronze valve with coupling and union elbow having metal-to-metal seat shall be provided. Valve rod and seat washer shall be removable through the face of the hydrant. The hydrant shall have 3/4 inch exposed hose thread on spout and 3/4 inch male pipe thread on inlet.

### 2.3.2 Relief Valves

Water heaters and hot water storage tanks shall have a combination pressure and temperature (P&T) relief valve. The pressure relief element of a P&T relief valve shall have adequate capacity to prevent excessive pressure buildup in the system when the system is operating at the maximum rate of heat input. The temperature element of a P&T relief valve shall have a relieving capacity which is at least equal to the total input of the heaters when operating at their maximum capacity. Relief valves shall be rated according to ANSI Z21.22/CSA 4.4. Relief valves for systems where the maximum rate of heat input is less than 200,000 Btuh shall have 3/4 inch minimum inlets, and 3/4 inch outlets. Relief valves for systems where the maximum rate of heat input is greater than 200,000 Btuh shall have 1 inch minimum inlets, and 1 inch outlets. The discharge pipe from the relief valve shall be the size of the valve outlet.

### 2.3.3 Thermostatic Mixing Valves

Provide thermostatic mixing valve for lavatory faucets. Mixing valves, thermostatic type, pressure-balanced or combination thermostatic and pressure-balanced shall be line size and shall be constructed with rough

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

or finish bodies either with or without plating. Each valve shall be constructed to control the mixing of hot and cold water and to deliver water at a desired temperature regardless of pressure or input temperature changes. The control element shall be of an approved type. The body shall be of heavy cast bronze, and interior parts shall be brass, bronze, corrosion-resisting steel or copper. The valve shall be equipped with necessary stops, check valves, unions, and sediment strainers on the inlets. Mixing valves shall maintain water temperature within 5 degrees F of any setting.

## 2.4 FIXTURES

Water closet replacements in major renovations may have a flush valve of up to 1.6 GPF to accommodate existing plumbing capacity. Fixtures for use by the physically handicapped shall be in accordance with ICC A117.1. Vitreous China, nonabsorbent, hard-burned, and vitrified throughout the body shall be provided. Porcelain enameled ware shall have specially selected, clear white, acid-resisting enamel coating evenly applied on surfaces. No fixture will be accepted that shows cracks, crazes, blisters, thin spots, or other flaws. Fixtures shall be equipped with appurtenances such as traps, faucets, stop valves, and drain fittings. Each fixture and piece of equipment requiring connections to the drainage system, except grease interceptors, shall be equipped with a trap. Brass expansion or toggle bolts capped with acorn nuts shall be provided for supports, and polished chromium-plated pipe, valves, and fittings shall be provided where exposed to view. Fixtures with the supply discharge below the rim shall be equipped with backflow preventers. Internal parts of flush valves and flushometer valves, shower mixing valves, shower head face plates, pop-up stoppers of lavatory waste drains, and pop-up stoppers and overflow tees and shoes of bathtub waste drains may contain acetal resin, fluorocarbon, nylon, acrylonitrile-butadiene-styrene (ABS) or other plastic material, if the material has provided satisfactory service under actual commercial or industrial operating conditions for not less than 2 years. Plastic in contact with hot water shall be suitable for 180 degrees F water temperature.

### 2.4.1 Lavatories

Vitreous china lavatories shall be provided with two integral molded lugs on the back-underside of the fixture and drilled for bolting to the wall in a manner similar to the hanger plate. Provide WaterSense labeled faucet with a maximum flow rate of 0.5 gpm at a flowing pressure of 60 psi. Water volume must be limited to 0.25 gal per metering cycle. Provide data identifying WaterSense label for lavatory faucet.

### 2.4.2 Flush Valve Water Closets

ASME A112.19.2/CSA B45.1, white vitreous china, siphon jet, elongated bowl, wall mounted, wall outlet. Top of toilet seat height above floor shall be 14 to 15 inches, except 17 to 19 inches for wheelchair water closets. Provide wax bowl ring including plastic sleeve. Provide white solid plastic elongated open-front seat .

Water flushing volume of the water closet and flush valve combination shall not exceed 1.28 gallons per flush. Water closets must meet the EPA WaterSense product definition specified in [http://www.epa.gov/watersense/partners/product\\_program\\_specs.html](http://www.epa.gov/watersense/partners/product_program_specs.html) and must be EPA WaterSense labeled products. Provide data identifying WaterSense label for flush valve water closet.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

Provide large diameter flush valve including angle control-stop valve, vacuum breaker, tail pieces, slip nuts, and wall plates; exposed to view components shall be chromium-plated or polished stainless steel. Flush valves shall be nonhold-open type. Mount flush valves not less than 11 inches above the fixture. Mounted height of flush valve shall not interfere with the hand rail in ADA stalls.

#### 2.4.3 Flush Valve Urinals

ASME A112.19.2/CSA B45.1, white vitreous china, ,wall-mounted, wall outlet, siphon jet, integral trap, and extended side shields. Provide urinal with the rim 17 inches above the floor. Water flushing volume of the urinal and flush valve combination shall not exceed 0.5 gallons per flush. Urinals must meet the specifications of [http://www.epa.gov/watersense/partners/product\\_program\\_specs.html](http://www.epa.gov/watersense/partners/product_program_specs.html) and must be EPA WaterSense labeled products. Provide data identifying WaterSense label for urinal. Provide ASME A112.6.1M concealed chair carriers with vertical steel pipe supports. Provide large diameter flush valve including angle control-stop valve, vacuum breaker, tail pieces, slip nuts, and wall plates; exposed to view components shall be chromium-plated or polished stainless steel. Flush valves shall be nonhold-open type. Mount flush valves not less than 11 inches above the fixture.

#### 2.4.4 Wall Hung Lavatories

ASME A112.19.2/CSA B45.1, white vitreous china, ,straight back type, minimum dimensions of 19 inches, wide by 17 inches front to rear, with supply openings for use with top mounted centerset faucets, and openings for concealed arm carrier installation. Provide lavatory faucets and accessories meeting the flow rate and product requirements of the paragraph LAVATORIES. Provide ASME A112.6.1M concealed chair carriers with vertical steel pipe supports and concealed arms for the lavatory. Mount lavatory with the front rim 34 inches above floor and with 29 inches minimum clearance from bottom of the front rim to floor. Provide top mounted washerless centerset lavatory faucets.

#### 2.4.5 Kitchen Sinks

ASME A112.19.3/CSA B45.4, 20 gage stainless steel with integral mounting rim for flush installation, minimum dimensions of 33 inches wide by 21 inches front to rear, two compartments, with undersides fully sound deadened, with supply openings for use with top mounted washerless sink faucets with hose spray, and with 3.5 inch drain outlet. Water flow rate shall not exceed 2.2 gpm when measured at a flowing water pressure of 60 psi. Provide stainless steel drain outlets and stainless steel cup strainers. Provide separate 1.5 inch P-trap and drain piping to vertical vent piping from each compartment. Provide top mounted washerless sink faucets with hose spray.

#### 2.4.6 Dual Height (Wheelchair) Drinking-Water Coolers

AHRI 1010 with more than a single thickness of metal between the potable water and the refrigerant in the heat exchanger, wall-hung, bubbler style, air-cooled condensing unit, 4.75 gph minimum capacity, stainless steel splash receptor and basin, and stainless steel cabinet. Bubblers shall be controlled by push levers or push bars, front mounted or side mounted near the front edge of the cabinet. High bubbler spout shall be mounted at maximum of 36 inches above floor and at front of unit basin. Low

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

wheelchair accessible bubbler shall have a minimum of 27 inch minimum knee clearance from front bottom of unit to floor and 36 inch maximum spout height above floor and bottle filler Spouts shall direct water flow at least 4 inches above unit basin and trajectory parallel or nearly parallel to the front of unit. Provide filters for chlorine in supply piping to faucets. Provide ASME A112.6.1M concealed steel pipe chair carriers. Provide electric water cooler that is Energy Star labeled. Provide data identifying Energy Star label for electric water cooler.

#### 2.4.7 Precast Terrazzo Mop Sinks

Terrazzo shall be made of marble chips cast in white portland cement to produce 3000 psi minimum compressive strength 7 days after casting. Provide floor or wall outlet copper alloy body drain cast integral with terrazzo, with polished stainless steel strainers.

#### 2.4.8 Emergency Eyewash and Shower

ANSI/ISEA Z358.1, wall supported frost-proof unit. Provide deluge shower head, stay-open ball valve operated by highly visible push handle. Provide eyewash and stay-open ball valve operated by highly visible push handle. Coordinate wall thickness with manufacturer. Shower piping shall be galvanized steel with safety yellow corrosion resistant coating.

### 2.5 BACKFLOW PREVENTERS

Backflow prevention devices must be approved by the State or local regulatory agencies. If there is no State or local regulatory agency requirements, the backflow prevention devices must be listed by the Foundation for Cross-Connection Control & Hydraulic Research, or any other approved testing laboratory having equivalent capabilities for both laboratory and field evaluation of backflow prevention devices and assemblies.

Reduced pressure principle assemblies, double check valve assemblies, atmospheric (nonpressure) type vacuum breakers, and pressure type vacuum breakers shall be meet the above requirements.

Backflow preventers with intermediate atmospheric vent shall conform to ASSE 1012. Reduced pressure principle backflow preventers shall conform to ASSE 1013. Hose connection vacuum breakers shall conform to ASSE 1011. Pipe applied atmospheric type vacuum breakers shall conform to ASSE 1001. Pressure vacuum breaker assembly shall conform to ASSE 1020. Air gaps in plumbing systems shall conform to ASME A112.1.2.

### 2.6 DRAINS

#### 2.6.1 Floor and Shower Drains

Floor and shower drains shall consist of a galvanized body, integral seepage pan, and adjustable perforated or slotted chromium-plated bronze, nickel-bronze, or nickel-brass strainer, consisting of grate and threaded collar. Floor drains shall be cast iron except where metallic waterproofing membrane is installed. Drains shall be of double drainage pattern for embedding in the floor construction. The seepage pan shall have weep holes or channels for drainage to the drainpipe. The strainer shall be adjustable to floor thickness. A clamping device for attaching flashing or waterproofing membrane to the seepage pan without damaging the flashing or waterproofing membrane shall be provided when required.



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

Drains shall be provided with threaded connection. Between the drain outlet and waste pipe, a neoprene rubber gasket conforming to ASTM C564 may be installed, provided that the drain is specifically designed for the rubber gasket compression type joint. Floor and shower drains shall conform to ASME A112.6.3. Provide drain with trap primer connection, trap primer, and connection piping. Primer shall meet ASSE 1018.

#### 2.6.1.1 Drains and Backwater Valves

Drains and backwater valves installed in connection with waterproofed floors or shower pans shall be equipped with bolted-type device to securely clamp flashing.

#### 2.6.2 Shower Faucets and Drain Fittings

Provide single control pressure equalizing shower faucets with body mounted from behind the wall with threaded connections. Provide ball joint self-cleaning shower heads. Provide WaterSense labeled showerhead with a maximum flow rate of (1.75 gpm). Provide data identifying WaterSense label for showerhead. Provide separate globe valves or angle valves with union connections in each supply to faucet.

#### 2.6.3 Area Drains

Area drains shall be plain pattern with polished stainless steel perforated or slotted grate and bottom outlet. The drain shall be circular or square with a 12 inch nominal overall width or diameter and 10 inch nominal overall depth. Drains shall be cast iron with manufacturer's standard coating. Grate shall be easily lifted out for cleaning. Outlet shall be suitable for inside caulked connection to drain pipe. Drains shall conform to ASME A112.6.3.

#### 2.6.4 Floor Sinks

Floor sinks shall be square, with 12 inch nominal overall width or diameter and 10 inch nominal overall depth. Floor sink shall have an acid-resistant enamel interior finish with cast-iron body, ABS sediment bucket, and perforated grate of cast iron in industrial areas and stainless steel in finished areas. The outlet pipe size shall be as indicated or of the same size as the connecting pipe.

#### 2.6.5 Sight Drains

Sight drains shall consist of body, integral seepage pan, and adjustable strainer with perforated or slotted grate and funnel extension. The strainer shall have a threaded collar to permit adjustment to floor thickness. Drains shall be of double drainage pattern suitable for embedding in the floor construction. A clamping device for attaching flashing or waterproofing membrane to the seepage pan without damaging the flashing or membrane shall be provided for other than concrete construction. Drains shall have a galvanized heavy cast-iron body and seepage pan and chromium-plated bronze, nickel-bronze, or nickel-brass strainer and funnel combination. Drains shall be provided with threaded connection and with a separate cast-iron "P" trap, unless otherwise indicated. Drains shall be circular, unless otherwise indicated. The funnel shall be securely mounted over an opening in the center of the strainer. Minimum dimensions shall be as follows:

Area of strainer and collar: 36 square inches

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

Height of funnel: 3-3/4 inches  
Diameter of lower portion: 2 inches of funnel  
Diameter of upper portion: 4 inches of funnel

#### 2.6.6 Roof Drains and Expansion Joints

Roof drains shall conform to ASME A112.6.4, with dome and integral flange, and shall have a device for making a watertight connection between roofing and flashing. The whole assembly shall be galvanized heavy pattern cast iron. For aggregate surface roofing, the drain shall be provided with a gravel stop. On roofs other than concrete construction, roof drains shall be complete with underdeck clamp, sump receiver, and an extension for the insulation thickness where applicable. A clamping device for attaching flashing or waterproofing membrane to the seepage pan without damaging the flashing or membrane shall be provided when required to suit the building construction. Strainer openings shall have a combined area equal to twice that of the drain outlet. The outlet shall be equipped to make a proper connection to threaded pipe of the same size as the downspout. An expansion joint of proper size to receive the conductor pipe shall be provided. The expansion joint shall consist of a heavy cast-iron housing, brass or bronze sleeve, brass or bronze fastening bolts and nuts, and gaskets or packing. The sleeve shall have a nominal thickness of not less than 0.134 inch. Gaskets and packing shall be close-cell neoprene, O-ring packing shall be close-cell neoprene of 70 durometer. Packing shall be held in place by a packing gland secured with bolts.

#### 2.7 SHOWER PAN

Shower pan may be copper, or nonmetallic material.

##### 2.7.1 Plasticized Polyvinyl Chloride Shower Pan Material

Material shall be sheet form. The material shall be 0.040 inch minimum thickness of plasticized polyvinyl chloride or chlorinated polyethylene and shall be in accordance with ASTM D4551.

#### 2.8 TRAPS

Unless otherwise specified, traps shall be copper-alloy adjustable tube type with slip joint inlet and swivel. Traps shall be without a cleanout.

Provide traps with removable access panels for easy clean-out at sinks and lavatories. Tubes shall be copper alloy with walls not less than 0.032 inch thick within commercial tolerances, except on the outside of bends where the thickness may be reduced slightly in manufacture by usual commercial methods. Inlets shall have rubber washer and copper alloy nuts for slip joints above the discharge level. Swivel joints shall be below the discharge level and shall be of metal-to-metal or metal-to-plastic type as required for the application. Nuts shall have flats for wrench grip. Outlets shall have internal pipe thread, except that when required for the application, the outlets shall have sockets for solder-joint connections. The depth of the water seal shall be not less than 2 inches. The interior diameter shall be not more than 1/8 inch over or under the nominal size, and interior surfaces shall be reasonably smooth throughout. A copper alloy "P" trap assembly consisting of an adjustable "P" trap and threaded trap wall nipple with cast brass wall flange shall be provided for lavatories. The assembly shall be a standard manufactured

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

unit and may have a rubber-gasketed swivel joint.

## 2.9 INTERCEPTORS

### 2.9.1 Combination Oil and Sand Interceptors

Acid resistant coated interior and exterior fabricated steel oil interceptor with bronze cleanout plug and visible double wall trap seal, removable combination pressure equalizing/flow diffusing baffle and sediment bucket, adjustable oil draw-off, vent connections, secured gasketed heavy-duty traffic non-skid cover for flush-with-floor installation, and integral flow control fitting and integral oil storage tank and furnished with inlet and outlet in high position.

### 2.9.2 Lint Interceptors

Acid Resistant Coated interior and exterior fabricated stainless steel lint interceptor, non-skid secured cover with removable lift handle (for 100 gpm and greater units), aluminum lint intercepting secondary screen assembly, and permanent primary straining baffle with 3/8 inch diameter perforated holes. Furnished with high inlet and low outlet.

## 2.10 WATER HEATERS

Water heater types and capacities shall be as indicated. Each water heater shall have replaceable anodes. Each primary water heater shall have controls with an adjustable range that includes 90 to 160 degrees F. Each gas-fired water heater and booster water heater shall have controls with an adjustable range that includes 120 to 180 degrees F. Hot water systems utilizing recirculation systems shall be tied into building off-hour controls. The thermal efficiencies and standby heat losses shall conform to TABLE III in PART 3 of this Section for each type of water heater specified. The only exception is that storage water heaters and hot water storage tanks having more than 500 gallons storage capacity need not meet the standard loss requirement if the tank surface area is insulated to R-12.5 and if a standing light is not used. Plastic materials polyetherimide (PEI) and polyethersulfone (PES) are forbidden to be used for vent piping of combustion gases. A factory pre-charged expansion tank shall be installed on the cold water supply to each water heater. Expansion tanks shall be specifically designed for use on potable water systems and shall be rated for 200 degrees F water temperature and 150 psi working pressure. The expansion tank size and acceptance volume shall be as indicated.

### 2.10.1 Automatic Storage Type

Heaters shall be complete with control system, temperature gauge, and pressure gauge, and shall have ASME rated combination pressure and temperature relief valve.

#### 2.10.1.1 Gas-Fired Type

Gas-fired water heaters shall conform to ANSI Z21.10.1/CSA 4.1 when input is 75,000 BTU per hour or less or ANSI Z21.10.3/CSA 4.3 for heaters with input greater than 75,000 BTU per hour.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### 2.10.1.2 Electric Type

Electric type water heaters shall conform to UL 174 with dual heating elements. Each element shall be 4.5 KW. The elements shall be wired so that only one element can operate at a time.

#### 2.11 HOT-WATER STORAGE TANKS

Hot-water storage tanks shall be constructed by one manufacturer, ASME stamped for the working pressure, and shall have the National Board (ASME) registration. The tank shall be cement-lined or glass-lined steel type in accordance with AWWA D100. The heat loss shall conform to TABLE III in PART 3 of this Section as determined by the requirements of ASHRAE 90.1 - IP. Each tank shall be equipped with a thermometer, conforming to ASTM E1, Type I, Class 3, Range C, style and form as required for the installation, and with 7 inch scale. Thermometer shall have a separable socket suitable for a 3/4 inch tapped opening. Tanks shall be equipped with a pressure gauge 6 inch minimum diameter face. Insulation shall be as specified in Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS. Storage tank capacity shall be as shown.

#### 2.12 PUMPS

##### 2.12.1 Circulating Pumps

Domestic hot water circulating pumps shall be electrically driven, single-stage, centrifugal, with mechanical seals, suitable for the intended service. Pump and motor shall be supported by the piping on which it is installed. The shaft shall be one-piece, heat-treated, corrosion-resisting steel with impeller and smooth-surfaced housing of bronze.

Motor shall be totally enclosed, fan-cooled and shall have sufficient horsepower for the service required. Each pump motor shall be equipped with an across-the-line magnetic controller in a NEMA 250, Type 1 enclosure with "START-STOP" switch in cover.

##### 2.12.2 Flexible Connectors

Flexible connectors shall be provided at the suction and discharge of each pump that is 1 hp or larger. Connectors shall be constructed of neoprene, rubber, or braided bronze, with Class 150 standard flanges. Flexible connectors shall be line size and suitable for the pressure and temperature of the intended service.

#### 2.13 COMPRESSED AIR SYSTEM

##### 2.13.1 Air Compressors

Air compressor unit shall be a factory-packaged assembly, including three phase, 480 volt motor controls, switches, wiring, accessories, and motor controllers, in a NEMA 250, Type 1 enclosure. Tank-mounted air compressors shall be manufactured to comply with UL listing requirements. Air compressors shall have manufacturer's name and address, together with trade name, and catalog number on a nameplate securely attached to the equipment. Each compressor shall start and stop automatically at upper and lower pressure limits of the system. Guards shall shield exposed moving parts. Each duplex compressor system shall be provided with automatic alternation system. Each compressor motor shall be provided

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

with an across-the-line-type magnetic controller, complete with low-voltage release. An intake air filter and silencer shall be provided with each compressor. Aftercooler and moisture separator shall be installed between compressors and air receiver to remove moisture and oil condensate before the air enters the receiver. Aftercoolers shall be either air- or water-cooled, as indicated. The air shall pass through a sufficient number of tubes to affect cooling. Tubes shall be sized to give maximum heat transfer. Water to unit shall be controlled by a solenoid or pneumatic valve, which opens when the compressors start and closes when the compressors shut down. Cooling capacity of the aftercooler shall be sized for the total capacity of the compressors. Means shall be provided for draining condensed moisture from the receiver by an automatic float type trap. Capacities of air compressors and receivers shall be as indicated.

#### 2.13.2 Lubricated Compressors

Compressors shall be two-stage, V-belt drive, capable of operating continuously against their designed discharge pressure, and shall operate at a speed not in excess of 1800 rpm. Compressors shall have the capacity and discharge pressure indicated. Compressors shall be assembled complete on a common subbase. The compressor main bearings shall be either roller or ball. The discharge passage of the high pressure air shall be piped to the air receiver with a copper pipe or tubing. A pressure gauge calibrated to 150 psi and equipped with a gauge cock and pulsation dampener shall be furnished for installation adjacent to pressure switches.

#### 2.13.3 Air Receivers

Receivers shall be designed for 200 psi working pressure. Receivers shall be factory air tested to 1-1/2 times the working pressure. Receivers shall be equipped with safety relief valves and accessories, including pressure gauges and automatic and manual drains. The outside of air receivers may be galvanized or supplied with commercial enamel finish. Receivers shall be designed and constructed in accordance with ASME BPVC SEC VIII D1 and shall have the design working pressures specified herein. A display of the ASME seal on the receiver or a certified test report from an approved independent testing laboratory indicating conformance to the ASME Code shall be provided.

#### 2.13.4 Intake Air Supply Filter

Dry type air filter shall be provided having a collection efficiency of 99 percent of particles larger than 10 microns. Filter body and media shall withstand a maximum 125 psi, capacity as indicated.

#### 2.13.5 Pressure Regulators

The air system shall be provided with the necessary regulator valves to maintain the desired pressure for the installed equipment. Regulators shall be designed for a maximum inlet pressure of 125 psi and a maximum temperature of 200 degrees F. Regulators shall be single-seated, pilot-operated with valve plug, bronze body and trim or equal, and threaded connections. The regulator valve shall include a pressure gauge and shall be provided with an adjustment screw for adjusting the pressure differential from 0 to 125 psi. Regulator shall be sized as indicated.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## 2.14 DOMESTIC WATER SERVICE METER

The requirements for metering and submetering are specified in Section 33 11 00 WATER UTILITY DISTRIBUTION PIPING.

## 2.15 ELECTRICAL WORK

Provide electrical motor driven equipment specified complete with motors, motor starters, and controls as specified herein and in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Provide internal wiring for components of packaged equipment as an integral part of the equipment. Provide high efficiency type, single-phase, fractional-horsepower alternating-current motors, including motors that are part of a system, corresponding to the applications in accordance with NEMA MG 11. In addition to the requirements of Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM, provide polyphase, squirrel-cage medium induction motors with continuous ratings, including motors that are part of a system, that meet the efficiency ratings for premium efficiency motors in accordance with NEMA MG 1. Provide motors in accordance with NEMA MG 1 and of sufficient size to drive the load at the specified capacity without exceeding the nameplate rating of the motor.

Motors shall be rated for continuous duty with the enclosure specified. Motor duty requirements shall allow for maximum frequency start-stop operation and minimum encountered interval between start and stop. Motor torque shall be capable of accelerating the connected load within 20 seconds with 80 percent of the rated voltage maintained at motor terminals during one starting period. Motor bearings shall be fitted with grease supply fittings and grease relief to outside of the enclosure.

Controllers and contactors shall have auxiliary contacts for use with the controls provided. Manual or automatic control and protective or signal devices required for the operation specified and any control wiring required for controls and devices specified, but not shown, shall be provided. For packaged equipment, the manufacturer shall provide controllers, including the required monitors and timed restart.

Power wiring and conduit for field installed equipment shall be provided under and conform to the requirements of Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM.

## 2.16 MISCELLANEOUS PIPING ITEMS

### 2.16.1 Escutcheon Plates

Provide one piece or split hinge metal plates for piping entering floors, walls, and ceilings in exposed spaces. Provide chromium-plated on copper alloy plates or polished stainless steel finish in finished spaces. Provide paint finish on plates in unfinished spaces.

### 2.16.2 Pipe Sleeves

Provide where piping passes entirely through walls, ceilings, roofs, and floors. Sleeves are not required where supply drain, waste, and vent (DWV) piping passes through concrete floor slabs located on grade, except where penetrating a membrane waterproof floor.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### 2.16.2.1 Sleeves in Masonry and Concrete

Provide steel pipe sleeves or schedule 40 PVC plastic pipe sleeves. Sleeves are not required where drain, waste, and vent (DWV) piping passes through concrete floor slabs located on grade. Core drilling of masonry and concrete may be provided in lieu of pipe sleeves when cavities in the core-drilled hole are completely grouted smooth.

#### 2.16.2.2 Sleeves Not in Masonry and Concrete

Provide 26 gage galvanized steel sheet or PVC plastic pipe sleeves.

#### 2.16.3 Pipe Hangers (Supports)

Provide MSS SP-58 Type 1 with adjustable type steel support rods, except as specified or indicated otherwise. Attach to steel joists with Type 19 or 23 clamps and retaining straps. Attach to Steel W or S beams with Type 21, 28, 29, or 30 clamps. Attach to steel angles and vertical web steel channels with Type 20 clamp with beam clamp channel adapter. Attach to horizontal web steel channel and wood with drilled hole on centerline and double nut and washer. Attach to concrete with Type 18 insert or drilled expansion anchor. Provide Type 40 insulation protection shield for insulated piping.

#### 2.16.4 Nameplates

Provide 0.125 inch thick melamine laminated plastic nameplates, black matte finish with white center core, for equipment, gages, thermometers, and valves; valves in supplies to faucets will not require nameplates. Accurately align lettering and engrave minimum of 0.25 inch high normal block lettering into the white core. Minimum size of nameplates shall be 1.0 by 2.5 inches. Key nameplates to a chart and schedule for each system. Frame charts and schedules under glass and place where directed near each system. Furnish two copies of each chart and schedule.

### PART 3 EXECUTION

#### 3.1 GENERAL INSTALLATION REQUIREMENTS

Piping located in air plenums shall conform to NFPA 90A requirements. Piping located in shafts that constitute air ducts or that enclose air ducts shall be noncombustible in accordance with NFPA 90A. Installation of plastic pipe where in compliance with NFPA may be installed in accordance with PPFA Fire Man. The plumbing system shall be installed complete with necessary fixtures, fittings, traps, valves, and accessories. Water and drainage piping shall be extended 5 feet outside the building, unless otherwise indicated. A full port ball valve and drain shall be installed on the water service line inside the building approximately 6 inches above the floor from point of entry. Piping shall be connected to the exterior service lines or capped or plugged if the exterior service is not in place. Sewer and water pipes shall be laid in separate trenches, except when otherwise shown. Exterior underground utilities shall be at least 12 inches below the average local frost depth or as indicated on the drawings. If trenches are closed or the pipes are otherwise covered before being connected to the service lines, the location of the end of each plumbing utility shall be marked with a stake or other acceptable means. Valves shall be installed with control no lower than the valve body.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 3.1.1 Water Pipe, Fittings, and Connections

#### 3.1.1.1 Utilities

The piping shall be extended to fixtures, outlets, and equipment. The hot-water and cold-water piping system shall be arranged and installed to permit draining. The supply line to each item of equipment or fixture, except faucets, flush valves, or other control valves which are supplied with integral stops, shall be equipped with a shutoff valve to enable isolation of the item for repair and maintenance without interfering with operation of other equipment or fixtures. Supply piping to fixtures, faucets, hydrants, shower heads, and flushing devices shall be anchored to prevent movement.

#### 3.1.1.2 Cutting and Repairing

The work shall be carefully laid out in advance, and unnecessary cutting of construction shall be avoided. Damage to building, piping, wiring, or equipment as a result of cutting shall be repaired by mechanics skilled in the trade involved.

#### 3.1.1.3 Protection of Fixtures, Materials, and Equipment

Pipe openings shall be closed with caps or plugs during installation. Fixtures and equipment shall be tightly covered and protected against dirt, water, chemicals, and mechanical injury. Upon completion of the work, the fixtures, materials, and equipment shall be thoroughly cleaned, adjusted, and operated. Safety guards shall be provided for exposed rotating equipment.

#### 3.1.1.4 Mains, Branches, and Runouts

Piping shall be installed as indicated. Pipe shall be accurately cut and worked into place without springing or forcing. Structural portions of the building shall not be weakened. Aboveground piping shall run parallel with the lines of the building, unless otherwise indicated. Branch pipes from service lines may be taken from top, bottom, or side of main, using crossover fittings required by structural or installation conditions. Supply pipes, valves, and fittings shall be kept a sufficient distance from other work and other services to permit not less than 1/2 inch between finished covering on the different services. Bare and insulated water lines shall not bear directly against building structural elements so as to transmit sound to the structure or to prevent flexible movement of the lines. Water pipe shall not be buried in or under floors unless specifically indicated or approved. Changes in pipe sizes shall be made with reducing fittings. Use of bushings will not be permitted except for use in situations in which standard factory fabricated components are furnished to accommodate specific accepted installation practice. Change in direction shall be made with fittings, except that bending of pipe 4 inches and smaller will be permitted, provided a pipe bender is used and wide sweep bends are formed. The center-line radius of bends shall be not less than six diameters of the pipe. Bent pipe showing kinks, wrinkles, flattening, or other malformations will not be acceptable.

#### 3.1.1.5 Pipe Drains

Pipe drains indicated shall consist of 3/4 inch hose bibb with renewable seat and ball valve ahead of hose bibb. At other low points, 3/4 inch brass plugs or caps shall be provided. Disconnection of the supply piping



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

at the fixture is an acceptable drain.

#### 3.1.1.6 Expansion and Contraction of Piping

Allowance shall be made throughout for expansion and contraction of water pipe. Each hot-water and hot-water circulation riser shall have expansion loops or other provisions such as offsets and changes in direction where indicated and required. Risers shall be securely anchored as required or where indicated to force expansion to loops. Branch connections from risers shall be made with ample swing or offset to avoid undue strain on fittings or short pipe lengths. Horizontal runs of pipe over 50 feet in length shall be anchored to the wall or the supporting construction about midway on the run to force expansion, evenly divided, toward the ends. Sufficient flexibility shall be provided on branch runouts from mains and risers to provide for expansion and contraction of piping. Flexibility shall be provided by installing one or more turns in the line so that piping will spring enough to allow for expansion without straining. If mechanical grooved pipe coupling systems are provided, the deviation from design requirements for expansion and contraction may be allowed pending approval of Contracting Officer.

#### 3.1.1.7 Thrust Restraint

Plugs, caps, tees, valves and bends deflecting 11.25 degrees or more, either vertically or horizontally, in waterlines 4 inches in diameter or larger shall be provided with thrust blocks, where indicated, to prevent movement. Thrust blocking shall be concrete of a mix not leaner than: 1 cement, 2-1/2 sand, 5 gravel; and having a compressive strength of not less than 2000 psi after 28 days. Blocking shall be placed between solid ground and the fitting to be anchored. Unless otherwise indicated or directed, the base and thrust bearing sides of the thrust block shall be poured against undisturbed earth. The side of the thrust block not subject to thrust shall be poured against forms. The area of bearing will be as shown. Blocking shall be placed so that the joints of the fitting are accessible for repair. Steel rods and clamps, protected by galvanizing or by coating with bituminous paint, shall be used to anchor vertical down bends into gravity thrust blocks.

#### 3.1.1.8 Commercial-Type Water Hammer Arresters

Commercial-type water hammer arresters shall be provided on hot- and cold-water supplies and shall be located as generally indicated, with precise location and sizing to be in accordance with PDI WH 201. Water hammer arresters, where concealed, shall be accessible by means of access doors or removable panels. Commercial-type water hammer arresters shall conform to ASSE 1010. Vertical capped pipe columns will not be permitted.

#### 3.1.2 Compressed Air Piping (Non-Oil Free)

Compressed air piping shall be installed as specified for water piping and suitable for 125 psig working pressure. Compressed air piping shall have supply lines and discharge terminals legibly and permanently marked at both ends with the name of the system and the direction of flow.

#### 3.1.3 Joints

Installation of pipe and fittings shall be made in accordance with the manufacturer's recommendations. Mitering of joints for elbows and notching of straight runs of pipe for tees will not be permitted. Joints

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

shall be made up with fittings of compatible material and made for the specific purpose intended.

#### 3.1.3.1 Threaded

Threaded joints shall have American Standard taper pipe threads conforming to ASME B1.20.1. Only male pipe threads shall be coated with graphite or with an approved graphite compound, or with an inert filler and oil, or shall have a polytetrafluoroethylene tape applied.

#### 3.1.3.2 Mechanical Couplings

Mechanical couplings may be used in conjunction with grooved pipe for aboveground, ferrous or non-ferrous, domestic hot and cold water systems, in lieu of unions, brazed, soldered, welded, flanged, or threaded joints.

Mechanical couplings are permitted in accessible locations including behind access plates. Flexible grooved joints will not be permitted, except as vibration isolators adjacent to mechanical equipment. Rigid grooved joints shall incorporate an angle bolt pad design which maintains metal-to-metal contact with equal amount of pad offset of housings upon installation to ensure positive rigid clamping of the pipe.

Designs which can only clamp on the bottom of the groove or which utilize gripping teeth or jaws, or which use misaligned housing bolt holes, or which require a torque wrench or torque specifications will not be permitted.

Grooved fittings and couplings, and grooving tools shall be provided from the same manufacturer. Segmentally welded elbows shall not be used. Grooves shall be prepared in accordance with the coupling manufacturer's latest published standards. Grooving shall be performed by qualified grooving operators having demonstrated proper grooving procedures in accordance with the tool manufacturer's recommendations.

The Contracting Officer shall be notified 24 hours in advance of test to demonstrate operator's capability, and the test shall be performed at the work site, if practical, or at a site agreed upon. The operator shall demonstrate the ability to properly adjust the grooving tool, groove the pipe, and to verify the groove dimensions in accordance with the coupling manufacturer's specifications.

#### 3.1.3.3 Unions and Flanges

Unions, flanges and mechanical couplings shall not be concealed in walls, ceilings, or partitions. Unions shall be used on pipe sizes 2-1/2 inches and smaller; flanges shall be used on pipe sizes 3 inches and larger.

#### 3.1.3.4 Grooved Mechanical Joints

Grooves shall be prepared according to the coupling manufacturer's instructions. Grooved fittings, couplings, and grooving tools shall be products of the same manufacturer. Pipe and groove dimensions shall comply with the tolerances specified by the coupling manufacturer. The diameter of grooves made in the field shall be measured using a "go/no-go" gauge, vernier or dial caliper, narrow-land micrometer, or other method specifically approved by the coupling manufacturer for the intended application. Groove width and dimension of groove from end of pipe shall be measured and recorded for each change in grooving tool setup to verify

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

compliance with coupling manufacturer's tolerances. Grooved joints shall not be used in concealed locations.

#### 3.1.3.5 Cast Iron Soil, Waste and Vent Pipe

Bell and spigot compression and hubless gasketed clamp joints for soil, waste and vent piping shall be installed per the manufacturer's recommendations.

#### 3.1.3.6 Copper Tube and Pipe

- a. Brazed. Brazed joints shall be made in conformance with AWS B2.2/B2.2M, ASME B16.50, and CDA A4015 with flux and are acceptable for all pipe sizes. Copper to copper joints shall include the use of copper-phosphorus or copper-phosphorus-silver brazing metal without flux. Brazing of dissimilar metals (copper to bronze or brass) shall include the use of flux with either a copper-phosphorus, copper-phosphorus-silver or a silver brazing filler metal.
- b. Soldered. Soldered joints shall be made with flux and are only acceptable for piping 2 inches and smaller. Soldered joints shall conform to ASME B31.5 and CDA A4015. Soldered joints shall not be used in compressed air piping between the air compressor and the receiver.
- c. Copper Tube Extracted Joint. Mechanically extracted joints shall be made in accordance with ICC IPC.
- d. Press connection. Copper press connections shall be made in **strict** accordance with the manufacturer's installation instructions for manufactured rated size. The joints shall be pressed using the tool(s) approved by the manufacturer **of that joint**. Minimum distance between fittings shall be in accordance with the manufacturer's requirements.

#### 3.1.3.7 Plastic Pipe

Acrylonitrile-Butadiene-Styrene (ABS) pipe shall have joints made with solvent cement. PVC and CPVC pipe shall have joints made with solvent cement elastomeric, threading, (threading of Schedule 80 Pipe is allowed only where required for disconnection and inspection; threading of Schedule 40 Pipe is not allowed), or mated flanged.

#### 3.1.3.8 Glass Pipe

Joints for corrosive waste glass pipe and fittings shall be made with corrosion-resisting steel compression-type couplings with acrylonitrile rubber gaskets lined with polytetrafluoroethylene.

#### 3.1.3.9 Corrosive Waste Plastic Pipe

Joints for polyolefin pipe and fittings shall be made by mechanical joint or electrical fusion coil method in accordance with ASTM D2657 and ASTM F1290. Joints for filament-wound reinforced thermosetting resin pipe shall be made in accordance with manufacturer's instructions. Unions or flanges shall be used where required for disconnection and inspection.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### 3.1.3.10 Polypropylene Pipe

Joints for polypropylene pipe and fittings shall be made by heat fusion welding socket-type or butt-fusion type fittings and shall comply with ASTM F2389.

#### 3.1.3.11 Other Joint Methods

#### 3.1.4 Dissimilar Pipe Materials

Connections between ferrous and non-ferrous copper water pipe shall be made with dielectric unions or flange waterways. Dielectric waterways shall have temperature and pressure rating equal to or greater than that specified for the connecting piping. Waterways shall have metal connections on both ends suited to match connecting piping. Dielectric waterways shall be internally lined with an insulator specifically designed to prevent current flow between dissimilar metals. Dielectric flanges shall meet the performance requirements described herein for dielectric waterways. Connecting joints between plastic and metallic pipe shall be made with transition fitting for the specific purpose.

#### 3.1.5 Pipe Sleeves and Flashing

Pipe sleeves shall be furnished and set in their proper and permanent location.

##### 3.1.5.1 Sleeve Requirements

Unless indicated otherwise, provide pipe sleeves meeting the following requirements:

- a. Secure sleeves in position and location during construction. Provide sleeves of sufficient length to pass through entire thickness of walls, ceilings, roofs, and floors.
- b. A modular mechanical type sealing assembly may be installed in lieu of a waterproofing clamping flange and caulking and sealing of annular space between pipe and sleeve. The seals shall consist of interlocking synthetic rubber links shaped to continuously fill the annular space between the pipe and sleeve using galvanized steel bolts, nuts, and pressure plates. The links shall be loosely assembled with bolts to form a continuous rubber belt around the pipe with a pressure plate under each bolt head and each nut. After the seal assembly is properly positioned in the sleeve, tightening of the bolt shall cause the rubber sealing elements to expand and provide a watertight seal between the pipe and the sleeve. Each seal assembly shall be sized as recommended by the manufacturer to fit the pipe and sleeve involved.
- c. Sleeves shall not be installed in structural members, except where indicated or approved. Rectangular and square openings shall be as detailed. Each sleeve shall extend through its respective floor, or roof, and shall be cut flush with each surface, except for special circumstances. Pipe sleeves passing through floors in wet areas such as mechanical equipment rooms, lavatories, kitchens, and other plumbing fixture areas shall extend a minimum of 4 inches above the finished floor.
- d. Unless otherwise indicated, sleeves shall be of a size to provide a

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

minimum of one inch clearance between bare pipe or insulation and inside of sleeve or between insulation and inside of sleeve. Sleeves in bearing walls and concrete slab on grade floors shall be steel pipe or cast-iron pipe. Sleeves in nonbearing walls or ceilings may be steel pipe, cast-iron pipe, galvanized sheet metal with lock-type longitudinal seam, or plastic.

- e. Except as otherwise specified, the annular space between pipe and sleeve, or between jacket over insulation and sleeve, shall be sealed as indicated with sealants conforming to ASTM C920 and with a primer, backstop material and surface preparation as specified in Section 07 92 00 JOINT SEALANTS. The annular space between pipe and sleeve, between bare insulation and sleeve or between jacket over insulation and sleeve shall not be sealed for interior walls which are not designated as fire rated.
- f. Sleeves through below-grade walls in contact with earth shall be recessed 1/2 inch from wall surfaces on both sides. Annular space between pipe and sleeve shall be filled with backing material and sealants in the joint between the pipe and concrete wall as specified above. Sealant selected for the earth side of the wall shall be compatible with dampproofing/waterproofing materials that are to be applied over the joint sealant. Pipe sleeves in fire-rated walls shall conform to the requirements in Section 07 84 00 FIRESTOPPING.

#### 3.1.5.2 Flashing Requirements

Pipes passing through roof shall be installed through a 16 ounce copper flashing, each within an integral skirt or flange. Flashing shall be suitably formed, and the skirt or flange shall extend not less than 8 inches from the pipe and shall be set over the roof or floor membrane in a solid coating of bituminous cement. The flashing shall extend up the pipe a minimum of 10 inches. For cleanouts, the flashing shall be turned down into the hub and caulked after placing the ferrule. Pipes passing through pitched roofs shall be flashed, using lead or copper flashing, with an adjustable integral flange of adequate size to extend not less than 8 inches from the pipe in all directions and lapped into the roofing to provide a watertight seal. The annular space between the flashing and the bare pipe or between the flashing and the metal-jacket-covered insulation shall be sealed as indicated. Flashing for dry vents shall be turned down into the pipe to form a waterproof joint. Pipes, up to and including 10 inches in diameter, passing through roof or floor waterproofing membrane may be installed through a cast-iron sleeve with caulking recess, anchor lugs, flashing-clamp device, and pressure ring with brass bolts. Flashing shield shall be fitted into the sleeve clamping device. Pipes passing through wall waterproofing membrane shall be sleeved as described above. A waterproofing clamping flange shall be installed.

#### 3.1.5.3 Waterproofing

Waterproofing at floor-mounted water closets shall be accomplished by forming a flashing guard from soft-tempered sheet copper. The center of the sheet shall be perforated and turned down approximately 1-1/2 inches to fit between the outside diameter of the drainpipe and the inside diameter of the cast-iron or steel pipe sleeve. The turned-down portion of the flashing guard shall be embedded in sealant to a depth of approximately 1-1/2 inches; then the sealant shall be finished off flush to floor level between the flashing guard and drainpipe. The flashing guard of sheet copper shall extend not less than 8 inches from the

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

drainpipe and shall be lapped between the floor membrane in a solid coating of bituminous cement. If cast-iron water closet floor flanges are used, the space between the pipe sleeve and drainpipe shall be sealed with sealant and the flashing guard shall be upturned approximately 1-1/2 inches to fit the outside diameter of the drainpipe and the inside diameter of the water closet floor flange. The upturned portion of the sheet fitted into the floor flange shall be sealed.

#### 3.1.5.4 Optional Counterflashing

Instead of turning the flashing down into a dry vent pipe, or caulking and sealing the annular space between the pipe and flashing or metal-jacket-covered insulation and flashing, counterflashing may be accomplished by utilizing the following:

- a. A standard roof coupling for threaded pipe up to 6 inches in diameter.
- b. A tack-welded or banded-metal rain shield around the pipe.

#### 3.1.5.5 Pipe Penetrations of Slab on Grade Floors

Where pipes, fixture drains, floor drains, cleanouts or similar items penetrate slab on grade floors, except at penetrations of floors with waterproofing membrane as specified in paragraphs FLASHING REQUIREMENTS and WATERPROOFING, a groove 1/4 to 1/2 inch wide by 1/4 to 3/8 inch deep shall be formed around the pipe, fitting or drain. The groove shall be filled with a sealant as specified in Section 07 92 00 JOINT SEALANTS.

#### 3.1.5.6 Pipe Penetrations

Provide sealants for all pipe penetrations. All pipe penetrations shall be sealed to prevent infiltration of air, insects, and vermin.

#### 3.1.6 Fire Seal

Where pipes pass through fire walls, fire-partitions, fire-rated pipe chase walls or floors above grade, a fire seal shall be provided as specified in Section 07 84 00 FIRESTOPPING.

#### 3.1.7 Supports

##### 3.1.7.1 General

Hangers used to support piping 2 inches and larger shall be fabricated to permit adequate adjustment after erection while still supporting the load. Pipe guides and anchors shall be installed to keep pipes in accurate alignment, to direct the expansion movement, and to prevent buckling, swaying, and undue strain. Piping subjected to vertical movement when operating temperatures exceed ambient temperatures shall be supported by variable spring hangers and supports or by constant support hangers. In the support of multiple pipe runs on a common base member, a clip or clamp shall be used where each pipe crosses the base support member. Spacing of the base support members shall not exceed the hanger and support spacing required for an individual pipe in the multiple pipe run. Threaded sections of rods shall not be formed or bent.

##### 3.1.7.2 Pipe Hangers, Inserts, and Supports

Installation of pipe hangers, inserts and supports shall conform to

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

MSS SP-58 except as modified herein.

- a. Types 5, 12, and 26 shall not be used.
- b. Type 3 shall not be used on insulated pipe.
- c. Type 18 inserts shall be secured to concrete forms before concrete is placed. Continuous inserts which allow more adjustment may be used if they otherwise meet the requirements for type 18 inserts.
- d. Type 19 and 23 C-clamps shall be torqued per MSS SP-58 and shall have both locknuts and retaining devices furnished by the manufacturer. Field-fabricated C-clamp bodies or retaining devices are not acceptable.
- e. Type 20 attachments used on angles and channels shall be furnished with an added malleable-iron heel plate or adapter.
- f. Type 24 may be used only on trapeze hanger systems or on fabricated frames.
- g. Type 39 saddles shall be used on insulated pipe 4 inches and larger when the temperature of the medium is 60 degrees F or higher. Type 39 saddles shall be welded to the pipe.
- h. Type 40 shields shall:
  - (1) Be used on insulated pipe less than 4 inches.
  - (2) Be used on insulated pipe 4 inches and larger when the temperature of the medium is 60 degrees F or less.
  - (3) Have a high density insert for all pipe sizes. High density inserts shall have a density of 8 pcf or greater.
- i. Horizontal pipe supports shall be spaced as specified in MSS SP-58 and a support shall be installed not over 1 foot from the pipe fitting joint at each change in direction of the piping. Pipe supports shall be spaced not over 5 feet apart at valves. Operating temperatures in determining hanger spacing for PVC or CPVC pipe shall be 120 degrees F for PVC and 180 degrees F for CPVC. Horizontal pipe runs shall include allowances for expansion and contraction.
- j. Vertical pipe shall be supported at each floor, except at slab-on-grade, at intervals of not more than 15 feet nor more than 8 feet from end of risers, and at vent terminations. Vertical pipe risers shall include allowances for expansion and contraction.
- k. Type 35 guides using steel, reinforced polytetrafluoroethylene (PTFE) or graphite slides shall be provided to allow longitudinal pipe movement. Slide materials shall be suitable for the system operating temperatures, atmospheric conditions, and bearing loads encountered. Lateral restraints shall be provided as needed. Where steel slides do not require provisions for lateral restraint the following may be used:
  - (1) On pipe 4 inches and larger when the temperature of the medium is 60 degrees F or higher, a Type 39 saddle, welded to the pipe, may freely rest on a steel plate.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

- (2) On pipe less than 4 inches a Type 40 shield, attached to the pipe or insulation, may freely rest on a steel plate.
- (3) On pipe 4 inches and larger carrying medium less than 60 degrees F a Type 40 shield, attached to the pipe or insulation, may freely rest on a steel plate.
- l. Pipe hangers on horizontal insulated pipe shall be the size of the outside diameter of the insulation. The insulation shall be continuous through the hanger on all pipe sizes and applications.
- m. Where there are high system temperatures and welding to piping is not desirable, the type 35 guide shall include a pipe cradle, welded to the guide structure and strapped securely to the pipe. The pipe shall be separated from the slide material by at least 4 inches or by an amount adequate for the insulation, whichever is greater.
- n. Hangers and supports for plastic pipe shall not compress, distort, cut or abrade the piping, and shall allow free movement of pipe except where otherwise required in the control of expansion/contraction.

### 3.1.7.3 Structural Attachments

Attachment to building structure concrete and masonry shall be by cast-in concrete inserts, built-in anchors, or masonry anchor devices. Inserts and anchors shall be applied with a safety factor not less than 5. Supports shall not be attached to metal decking. Supports shall not be attached to the underside of concrete filled floor or concrete roof decks unless approved by the Contracting Officer. Masonry anchors for overhead applications shall be constructed of ferrous materials only.

### 3.1.8 Welded Installation

Plumbing pipe weldments shall be as indicated. Changes in direction of piping shall be made with welding fittings only; mitering or notching pipe to form elbows and tees or other similar type construction will not be permitted. Branch connection may be made with either welding tees or forged branch outlet fittings. Branch outlet fittings shall be forged, flared for improvement of flow where attached to the run, and reinforced against external strains. Beveling, alignment, heat treatment, and inspection of weld shall conform to ASME B31.1. Weld defects shall be removed and repairs made to the weld, or the weld joints shall be entirely removed and rewelded. After filler metal has been removed from its original package, it shall be protected or stored so that its characteristics or welding properties are not affected. Electrodes that have been wetted or that have lost any of their coating shall not be used.

### 3.1.9 Pipe Cleanouts

Pipe cleanouts shall be the same size as the pipe except that cleanout plugs larger than 4 inches will not be required. A cleanout installed in connection with cast-iron soil pipe shall consist of a long-sweep 1/4 bend or one or two 1/8 bends extended to the place shown. An extra-heavy cast-brass or cast-iron ferrule with countersunk cast-brass head screw plug shall be caulked into the hub of the fitting and shall be flush with the floor. Cleanouts in connection with other pipe, where indicated, shall be T-pattern, 90-degree branch drainage fittings with cast-brass screw plugs, except plastic plugs shall be installed in plastic pipe. Plugs shall be the same size as the pipe up to and including 4 inches.



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

Cleanout tee branches with screw plug shall be installed at the foot of soil and waste stacks, at the foot of interior downspouts, on each connection to building storm drain where interior downspouts are indicated, and on each building drain outside the building. Cleanout tee branches may be omitted on stacks in single story buildings with slab-on-grade construction or where less than 18 inches of crawl space is provided under the floor. Cleanouts on pipe concealed in partitions shall be provided with chromium plated bronze, nickel bronze, nickel brass or stainless steel flush type access cover plates. Round access covers shall be provided and secured to plugs with securing screw. Square access covers may be provided with matching frames, anchoring lugs and cover screws. Cleanouts in finished walls shall have access covers and frames installed flush with the finished wall. Cleanouts installed in finished floors subject to foot traffic shall be provided with a chrome-plated cast brass, nickel brass, or nickel bronze cover secured to the plug or cover frame and set flush with the finished floor. Heads of fastening screws shall not project above the cover surface. Where cleanouts are provided with adjustable heads, the heads shall be cast iron .

### 3.2 WATER HEATERS AND HOT WATER STORAGE TANKS

#### 3.2.1 Relief Valves

No valves shall be installed between a relief valve and its water heater or storage tank. The P&T relief valve shall be installed where the valve actuator comes in contact with the hottest water in the heater. Whenever possible, the relief valve shall be installed directly in a tapping in the tank or heater; otherwise, the P&T valve shall be installed in the hot-water outlet piping. A vacuum relief valve shall be provided on the cold water supply line to the hot-water storage tank or water heater and mounted above and within 6 inches above the top of the tank or water heater.

#### 3.2.2 Installation of Gas- and Oil-Fired Water Heater

Installation shall conform to NFPA 54 for gas fired and NFPA 31 for oil fired. Storage water heaters that are not equipped with integral heat traps and having vertical pipe risers shall be installed with heat traps directly on both the inlet and outlet. Circulating systems need not have heat traps installed. An acceptable heat trap may be a piping arrangement such as elbows connected so that the inlet and outlet piping make vertically upward runs of not less than 24 inches just before turning downward or directly horizontal into the water heater's inlet and outlet fittings. Commercially available heat traps, specifically designed by the manufacturer for the purpose of effectively restricting the natural tendency of hot water to rise through vertical inlet and outlet piping during standby periods may also be approved.

#### 3.2.3 Heat Traps

Piping to and from each water heater and hot water storage tank shall be routed horizontally and downward a minimum of 2 feet before turning in an upward direction.

#### 3.2.4 Connections to Water Heaters

Connections of metallic pipe to water heaters shall be made with dielectric unions or flanges.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 3.2.5 Expansion Tank

A pre-charged expansion tank shall be installed on the cold water supply between the water heater inlet and the cold water supply shut-off valve. The Contractor shall adjust the expansion tank air pressure, as recommended by the tank manufacturer, to match incoming water pressure.

## 3.3 FIXTURES AND FIXTURE TRIMMINGS

Polished chromium-plated pipe, valves, and fittings shall be provided where exposed to view. Angle stops, straight stops, stops integral with the faucets, or concealed type of lock-shield, and loose-key pattern stops for supplies with threaded, sweat or solvent weld inlets shall be furnished and installed with fixtures. Where connections between copper tubing and faucets are made by rubber compression fittings, a beading tool shall be used to mechanically deform the tubing above the compression fitting. Exposed traps and supply pipes for fixtures and equipment shall be connected to the rough piping systems at the wall, unless otherwise specified under the item. Floor and wall escutcheons shall be as specified. Drain lines and hot water lines of fixtures for handicapped personnel shall be insulated and do not require polished chrome finish. Plumbing fixtures and accessories shall be installed within the space shown.

### 3.3.1 Fixture Connections

Where space limitations prohibit standard fittings in conjunction with the cast-iron floor flange, special short-radius fittings shall be provided. Connections between earthenware fixtures and flanges on soil pipe shall be made gastight and watertight with a closet-setting compound or neoprene gasket and seal. Use of natural rubber gaskets or putty will not be permitted. Fixtures with outlet flanges shall be set the proper distance from floor or wall to make a first-class joint with the closet-setting compound or gasket and fixture used.

### 3.3.2 Flushometer Valves

Flushometer valves shall be secured to prevent movement by anchoring the long finished top spud connecting tube to wall adjacent to valve with approved metal bracket. Flushometer valves for water closets shall be installed 39 inches above the floor, except at water closets intended for use by the physically handicapped where flushometer valves shall be mounted at approximately 30 inches above the floor and arranged to avoid interference with grab bars. In addition, for water closets intended for handicap use, the flush valve handle shall be installed on the wide side of the enclosure.

### 3.3.3 Height of Fixture Rims Above Floor

Lavatories shall be mounted with rim 31 inches above finished floor. Wall-hung drinking fountains and water coolers shall be installed with rim 42 inches above floor. Wall-hung service sinks shall be mounted with rim 28 inches above the floor. Installation of fixtures for use by the physically handicapped shall be in accordance with ICC A117.1.

### 3.3.4 Shower Bath Outfits

The area around the water supply piping to the mixing valves and behind the escutcheon plate shall be made watertight by caulking or gasketing.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 3.3.5 Fixture Supports

Fixture supports for off-the-floor lavatories, urinals, water closets, and other fixtures of similar size, design, and use, shall be of the chair-carrier type. The carrier shall provide the necessary means of mounting the fixture, with a foot or feet to anchor the assembly to the floor slab. Adjustability shall be provided to locate the fixture at the desired height and in proper relation to the wall. Support plates, in lieu of chair carrier, shall be fastened to the wall structure only where it is not possible to anchor a floor-mounted chair carrier to the floor slab.

#### 3.3.5.1 Support for Solid Masonry Construction

Chair carrier shall be anchored to the floor slab. Where a floor-anchored chair carrier cannot be used, a suitable wall plate shall be imbedded in the masonry wall.

#### 3.3.5.2 Support for Concrete-Masonry Wall Construction

Chair carrier shall be anchored to floor slab. Where a floor-anchored chair carrier cannot be used, a suitable wall plate shall be fastened to the concrete wall using through bolts and a back-up plate.

#### 3.3.5.3 Support for Steel Stud Frame Partitions

Chair carrier shall be used. The anchor feet and tubular uprights shall be of the heavy duty design; and feet (bases) shall be steel and welded to a square or rectangular steel tube upright. Wall plates, in lieu of floor-anchored chair carriers, shall be used only if adjoining steel partition studs are suitably reinforced to support a wall plate bolted to these studs.

#### 3.3.5.4 Wall-Mounted Water Closet Gaskets

Where wall-mounted water closets are provided, reinforced wax, treated felt, or neoprene gaskets shall be provided. The type of gasket furnished shall be as recommended by the chair-carrier manufacturer.

### 3.3.6 Backflow Prevention Devices

Plumbing fixtures, equipment, and pipe connections shall not cross connect or interconnect between a potable water supply and any source of nonpotable water. Backflow preventers shall be installed where indicated and in accordance with IAPMO UPC at all other locations necessary to preclude a cross-connect or interconnect between a potable water supply and any nonpotable substance. In addition backflow preventers shall be installed at all locations where the potable water outlet is below the flood level of the equipment, or where the potable water outlet will be located below the level of the nonpotable substance. Backflow preventers shall be located so that no part of the device will be submerged. Backflow preventers shall be of sufficient size to allow unrestricted flow of water to the equipment, and preclude the backflow of any nonpotable substance into the potable water system. Bypass piping shall not be provided around backflow preventers. Access shall be provided for maintenance and testing. Each device shall be a standard commercial unit.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 3.3.7 Access Panels

Access panels shall be provided for concealed valves and controls, or any item requiring inspection or maintenance. Access panels shall be of sufficient size and located so that the concealed items may be serviced, maintained, or replaced. Access panels shall be as specified in Section 08 31 00 ACCESS DOORS AND PANELS.

### 3.3.8 Sight Drains

Sight drains shall be installed so that the indirect waste will terminate 2 inches above the flood rim of the funnel to provide an acceptable air gap.

### 3.3.9 Traps

Each trap shall be placed as near the fixture as possible, and no fixture shall be double-trapped. Traps installed on cast-iron soil pipe shall be cast iron. Traps installed on steel pipe or copper tubing shall be recess-drainage pattern, or brass-tube type. Traps installed on plastic pipe may be plastic conforming to ASTM D3311. Traps for acid-resisting waste shall be of the same material as the pipe.

### 3.3.10 Shower Pans

Before installing shower pan, subfloor shall be free of projections such as nail heads or rough edges of aggregate. Drain shall be a bolt-down, clamping-ring type with weepholes, installed so the lip of the subdrain is flush with subfloor.

#### 3.3.10.1 General

The floor of each individual shower, the shower-area portion of combination shower and drying room, and the entire shower and drying room where the two are not separated by curb or partition, shall be made watertight with a shower pan fabricated in place. The shower pan material shall be cut to size and shape of the area indicated, in one piece to the maximum extent practicable, allowing a minimum of 6 inches for turnup on walls or partitions, and shall be folded over the curb with an approximate return of 1/4 of curb height. The upstands shall be placed behind any wall or partition finish. Subflooring shall be smooth and clean, with nailheads driven flush with surface, and shall be sloped to drain. Shower pans shall be clamped to drains with the drain clamping ring.

#### 3.3.10.2 Metal Shower Pans

When a shower pan of required size cannot be furnished in one piece, metal pieces shall be joined with a flintlock seam and soldered or burned. The corners shall be folded, not cut, and the corner seam shall be soldered or burned. Pans, including upstands, shall be coated on all surfaces with one brush coat of asphalt. Asphalt shall be applied evenly at not less than 1 gallon per 50 square feet. A layer of felt covered with building paper shall be placed between shower pans and wood floors. The joining surfaces of metal pan and drain shall be given a brush coat of asphalt after the pan is connected to the drain.

#### 3.3.10.3 Plasticized Chlorinated Polyethylene Shower Pans

Corners of plasticized chlorinated polyethylene shower pans shall be

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

folded against the upstand by making a pig-ear fold. Hot-air gun or heat lamp shall be used in making corner folds. Each pig-ear corner fold shall be nailed or stapled 1/2 inch from the upper edge to hold it in place. Nails shall be galvanized large-head roofing nails. On metal framing or studs, approved duct tape shall be used to secure pig-ear fold and membrane. Where no backing is provided between the studs, the membrane slack shall be taken up by pleating and stapling or nailing to studding 1/2 inch from upper edge. To adhere the membrane to vertical surfaces, the back of the membrane and the surface to which it will be applied shall be coated with adhesive that becomes dry to the touch in 5 to 10 minutes, after which the membrane shall be pressed into place. Surfaces to be solvent-welded shall be clean. Surfaces to be joined with xylene shall be initially sprayed and vigorously cleaned with a cotton cloth, followed by final coating of xylene and the joining of the surfaces by roller or equivalent means. If ambient or membrane temperatures are below 40 degrees F the membrane and the joint shall be heated prior to application of xylene. Heat may be applied with hot-air gun or heat lamp, taking precautions not to scorch the membrane. Adequate ventilation and wearing of gloves are required when working with xylene. Membrane shall be pressed into position on the drain body, and shall be cut and fit to match so that membrane can be properly clamped and an effective gasket-type seal provided. On wood subflooring, two layers of 15 pound dry felt shall be installed prior to installation of shower pan to ensure a smooth surface for installation.

#### 3.3.10.4 Nonplasticized Polyvinyl Chloride (PVC) Shower Pans

Nonplasticized PVC shall be turned up behind walls or wall surfaces a distance of not less than 6 inches in room areas and 3 inches above curb level in curbed spaces with sufficient material to fold over and fasten to outside face of curb. Corners shall be pig-ear type and folded between pan and studs. Only top 1 inch of upstand shall be nailed to hold in place. Nails shall be galvanized large-head roofing type. Approved duct tape shall be used on metal framing or studs to secure pig-ear fold and membrane. Where no backing is provided between studs, the membrane slack shall be taken up by pleating and stapling or nailing to studding at top inch of upstand. To adhere the membrane to vertical surfaces, the back of the membrane and the surface to which it is to be applied shall be coated with adhesive that becomes dry to the touch in 5 to 10 minutes, after which the membrane shall be pressed into place. Trim for drain shall be exactly the size of drain opening. Bolt holes shall be pierced to accommodate bolts with a tight fit. Adhesive shall be used between pan and subdrain. Clamping ring shall be bolted firmly. A small amount of gravel or porous materials shall be placed at weepholes so that holes remain clear when setting bed is poured. Membrane shall be solvent welded with PVC solvent cement. Surfaces to be solvent welded shall be clean (free of grease and grime). Sheets shall be laid on a flat surface with an overlap of about 2 inches. Top edge shall be folded back and surface primed with a PVC primer. PVC cement shall be applied and surfaces immediately placed together, while still wet. Joint shall be lightly rolled with a paint roller, then as the joint sets shall be rolled firmly but not so hard as to distort the material. In long lengths, about 2 or 3 feet at a time shall be welded. On wood subflooring, two layers of 15 pound felt shall be installed prior to installation of shower pan to ensure a smooth surface installation.

#### 3.4 VIBRATION-ABSORBING FEATURES

Mechanical equipment, including compressors and pumps, shall be isolated

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

from the building structure by approved vibration-absorbing features, unless otherwise shown. Each foundation shall include an adequate number of standard isolation units. Each unit shall consist of machine and floor or foundation fastening, together with intermediate isolation material, and shall be a standard product with printed load rating. Piping connected to mechanical equipment shall be provided with flexible connectors. Isolation unit installation shall limit vibration to 10 percent of the lowest equipment rpm.

#### 3.4.1 Tank- or Skid-Mounted Compressors

Floor attachment shall be as recommended by compressor manufacturer.

### 3.5 IDENTIFICATION SYSTEMS

#### 3.5.1 Identification Tags

Identification tags made of brass, engraved laminated plastic, or engraved anodized aluminum, indicating service and valve number shall be installed on valves, except those valves installed on supplies at plumbing fixtures. Tags shall be 1-3/8 inch minimum diameter, and marking shall be stamped or engraved. Indentations shall be black, for reading clarity. Tags shall be attached to valves with No. 12 AWG, copper wire, chrome-plated beaded chain, or plastic straps designed for that purpose.

#### 3.5.2 Pipe Color Code Marking

Color code marking of piping shall be as specified in Section 09 90 00 PAINTS AND COATINGS.

#### 3.5.3 Color Coding Scheme for Locating Hidden Utility Components

Scheme shall be provided in buildings having suspended grid ceilings. The color coding scheme shall identify points of access for maintenance and operation of operable components which are not visible from the finished space and installed in the space directly above the suspended grid ceiling. The operable components shall include valves, dampers, switches, linkages and thermostats. The color coding scheme shall consist of a color code board and colored metal disks. Each colored metal disk shall be approximately 3/8 inch in diameter and secured to removable ceiling panels with fasteners. The fasteners shall be inserted into the ceiling panels so that the fasteners will be concealed from view. The fasteners shall be manually removable without tools and shall not separate from the ceiling panels when panels are dropped from ceiling height. Installation of colored metal disks shall follow completion of the finished surface on which the disks are to be fastened. The color code board shall have the approximate dimensions of 3 foot width, 30 inches height, and 1/2 inch thickness. The board shall be made of wood fiberboard and framed under glass or 1/16 inch transparent plastic cover. Unless otherwise directed, the color code symbols shall be approximately 3/4 inch in diameter and the related lettering in 1/2 inch high capital letters. The color code board shall be mounted and located in the mechanical or equipment room.

### 3.6 ESCUTCHEONS

Escutcheons shall be provided at finished surfaces where bare or insulated piping, exposed to view, passes through floors, walls, or ceilings, except in boiler, utility, or equipment rooms. Escutcheons shall be fastened securely to pipe or pipe covering and shall be satin-finish,

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

corrosion-resisting steel, polished chromium-plated zinc alloy, or polished chromium-plated copper alloy. Escutcheons shall be either one-piece or split-pattern, held in place by internal spring tension or setscrew.

### 3.7 PAINTING

Painting of pipes, hangers, supports, and other iron work, either in concealed spaces or exposed spaces, is specified in Section 09 90 00 PAINTS AND COATINGS.

#### 3.7.1 Painting of New Equipment

New equipment painting shall be factory applied or shop applied, and shall be as specified herein, and provided under each individual section.

##### 3.7.1.1 Factory Painting Systems

Manufacturer's standard factory painting systems may be provided subject to certification that the factory painting system applied will withstand 125 hours in a salt-spray fog test, except that equipment located outdoors shall withstand 500 hours in a salt-spray fog test. Salt-spray fog test shall be in accordance with ASTM B117, and for that test the acceptance criteria shall be as follows: immediately after completion of the test, the paint shall show no signs of blistering, wrinkling, or cracking, and no loss of adhesion; and the specimen shall show no signs of rust creepage beyond 0.125 inch on either side of the scratch mark.

The film thickness of the factory painting system applied on the equipment shall not be less than the film thickness used on the test specimen. If manufacturer's standard factory painting system is being proposed for use on surfaces subject to temperatures above 120 degrees F, the factory painting system shall be designed for the temperature service.

##### 3.7.1.2 Shop Painting Systems for Metal Surfaces

Clean, pretreat, prime and paint metal surfaces; except aluminum surfaces need not be painted. Apply coatings to clean dry surfaces. Clean the surfaces to remove dust, dirt, rust, oil and grease by wire brushing and solvent degreasing prior to application of paint, except metal surfaces subject to temperatures in excess of 120 degrees F shall be cleaned to bare metal.

Where more than one coat of paint is specified, apply the second coat after the preceding coat is thoroughly dry. Lightly sand damaged painting and retouch before applying the succeeding coat. Color of finish coat shall be aluminum or light gray.

- a. Temperatures Less Than 120 Degrees F: Immediately after cleaning, the metal surfaces subject to temperatures less than 120 degrees F shall receive one coat of pretreatment primer applied to a minimum dry film thickness of 0.3 mil, one coat of primer applied to a minimum dry film thickness of one mil; and two coats of enamel applied to a minimum dry film thickness of one mil per coat.
- b. Temperatures Between 120 and 400 Degrees F: Metal surfaces subject to temperatures between 120 and 400 degrees F shall receive two coats of 400 degrees F heat-resisting enamel applied to a total minimum

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

thickness of 2 mils.

- c. Temperatures Greater Than 400 Degrees F: Metal surfaces subject to temperatures greater than 400 degrees F shall receive two coats of 600 degrees F heat-resisting paint applied to a total minimum dry film thickness of 2 mils.

### 3.8 TESTS, FLUSHING AND DISINFECTION

#### 3.8.1 Plumbing System

The following tests shall be performed on the plumbing system in accordance with IAPMO UPC, except that the drainage and vent system final test shall include the smoke test. The Contractor has the option to perform a peppermint test in lieu of the smoke test. If a peppermint test is chosen, the Contractor must submit a testing procedure and reasons for choosing this option in lieu of the smoke test to the Contracting Officer for approval.

- a. Drainage and Vent Systems Test. The final test shall include a smoke test.
- b. Building Sewers Tests.
- c. Water Supply Systems Tests.

##### 3.8.1.1 Test of Backflow Prevention Assemblies

Backflow prevention assembly shall be tested using gauges specifically designed for the testing of backflow prevention assemblies.

Backflow prevention assembly test gauges shall be tested annually for accuracy in accordance with the requirements of State or local regulatory agencies. If there is no State or local regulatory agency requirements, gauges shall be tested annually for accuracy in accordance with the requirements of University of Southern California's Foundation of Cross Connection Control and Hydraulic Research or the American Water Works Association Manual of Cross Connection (Manual M-14), or any other approved testing laboratory having equivalent capabilities for both laboratory and field evaluation of backflow prevention assembly test gauges. Report form for each assembly shall include, as a minimum, the following:

|                  |                      |
|------------------|----------------------|
| Data on Device   | Data on Testing Firm |
| Type of Assembly | Name                 |
| Manufacturer     | Address              |
| Model Number     | Certified Tester     |
| Serial Number    | Certified Tester No. |
| Size             | Date of Test         |



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

|                        |                                       |
|------------------------|---------------------------------------|
| Location               |                                       |
| Test Pressure Readings | Serial Number and Test Data of Gauges |

If the unit fails to meet specified requirements, the unit shall be repaired and retested.

#### 3.8.1.2 Shower Pans

After installation of the pan and finished floor, the drain shall be temporarily plugged below the weep holes. The floor area shall be flooded with water to a minimum depth of 1 inch for a period of 24 hours. Any drop in the water level during test, except for evaporation, will be reason for rejection, repair, and retest.

#### 3.8.1.3 Compressed Air Piping (Nonoil-Free)

Piping systems shall be filled with oil-free dry air or gaseous nitrogen to 150 psig and hold this pressure for 2 hours with no drop in pressure.

#### 3.8.2 Defective Work

If inspection or test shows defects, such defective work or material shall be replaced or repaired as necessary and inspection and tests shall be repeated. Repairs to piping shall be made with new materials. Caulking of screwed joints or holes will not be acceptable.

#### 3.8.3 System Flushing

##### 3.8.3.1 During Flushing

Before operational tests or disinfection, potable water piping system shall be flushed with hot potable water. Sufficient water shall be used to produce a water velocity that is capable of entraining and removing debris in all portions of the piping system. This requires simultaneous operation of all fixtures on a common branch or main in order to produce a flushing velocity of approximately 4 fps through all portions of the piping system. In the event that this is impossible due to size of system, the Contracting Officer (or the designated representative) shall specify the number of fixtures to be operated during flushing. Contractor shall provide adequate personnel to monitor the flushing operation and to ensure that drain lines are unobstructed in order to prevent flooding of the facility. Contractor shall be responsible for any flood damage resulting from flushing of the system. Flushing shall be continued until entrained dirt and other foreign materials have been removed and until discharge water shows no discoloration. All faucets and drinking water fountains, to include any device considered as an end point device by NSF/ANSI 61, Section 9, shall be flushed a minimum of 0.25 gallons per 24 hour period, ten times over a 14 day period.

##### 3.8.3.2 After Flushing

System shall be drained at low points. Strainer screens shall be removed, cleaned, and replaced. After flushing and cleaning, systems shall be prepared for testing by immediately filling water piping with clean, fresh potable water. Any stoppage, discoloration, or other damage to the finish, furnishings, or parts of the building due to the Contractor's failure to properly clean the piping system shall be repaired

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

by the Contractor. When the system flushing is complete, the hot-water system shall be adjusted for uniform circulation. Flushing devices and automatic control systems shall be adjusted for proper operation according to manufacturer's instructions. Flow rates on fixtures must not exceed those stated in PART 2 of this Section. Unless more stringent local requirements exist, lead levels shall not exceed limits established by 40 CFR 141.80 (c)(1). The water supply to the building shall be tested separately to ensure that any lead contamination found during potable water system testing is due to work being performed inside the building.

#### 3.8.4 Operational Test

Upon completion of flushing and prior to disinfection procedures, the Contractor shall subject the plumbing system to operating tests to demonstrate satisfactory installation, connections, adjustments, and functional and operational efficiency. Such operating tests shall cover a period of not less than 8 hours for each system and shall include the following information in a report with conclusion as to the adequacy of the system:

- a. Time, date, and duration of test.
- b. Water pressures at the most remote and the highest fixtures.
- c. Operation of each fixture and fixture trim.
- d. Operation of each valve, hydrant, and faucet.
- e. Pump suction and discharge pressures.
- f. Temperature of each domestic hot-water supply.
- g. Operation of each floor and roof drain by flooding with water.
- h. Operation of each vacuum breaker and backflow preventer.
- i. Complete operation of each water pressure booster system, including pump start pressure and stop pressure.
- j. Compressed air readings at each compressor and at each outlet. Each indicating instrument shall be read at 1/2 hour intervals. The report of the test shall be submitted in quadruplicate. The Contractor shall furnish instruments, equipment, and personnel required for the tests; the Government will furnish the necessary water and electricity.

#### 3.8.5 Disinfection

After all system components are provided and operational tests are complete, the entire domestic hot- and cold-water distribution system shall be disinfected. Before introducing disinfecting chlorination material, entire system shall be flushed with potable water until any entrained dirt and other foreign materials have been removed.

Water chlorination procedure shall be in accordance with AWWA C651 and AWWA C652 as modified and supplemented by this specification. The chlorinating material shall be hypochlorites or liquid chlorine. The chlorinating material shall be fed into the water piping system at a constant rate at a concentration of at least 50 parts per million (ppm). Feed a properly adjusted hypochlorite solution injected into

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

the system with a hypochlorinator, or inject liquid chlorine into the system through a solution-feed chlorinator and booster pump until the entire system is completely filled.

Test the chlorine residual level in the water at 6 hour intervals for a continuous period of 24 hours. If at the end of a 6 hour interval, the chlorine residual has dropped to less than 25 ppm, flush the piping including tanks with potable water, and repeat the above chlorination procedures. During the chlorination period, each valve and faucet shall be opened and closed several times.

After the second 24 hour period, verify that no less than 25 ppm chlorine residual remains in the treated system. The 24 hour chlorination procedure must be repeated until no less than 25 ppm chlorine residual remains in the treated system.

Upon the specified verification, the system including tanks shall then be flushed with potable water until the residual chlorine level is reduced to less than one part per million. During the flushing period, each valve and faucet shall be opened and closed several times.

Take additional samples of water in disinfected containers, for bacterial examination, at locations specified by the Contracting Officer. Test these samples for total coliform organisms (coliform bacteria, fecal coliform, streptococcal, and other bacteria) in accordance with AWWA 10084. The testing method used shall be EPA approved for drinking water systems and shall comply with applicable local and state requirements.

Disinfection shall be repeated until bacterial tests indicate the absence of coliform organisms (zero mean coliform density per 100 milliliters) in the samples for at least 2 full days. The system will not be accepted until satisfactory bacteriological results have been obtained.

### 3.9 POSTED INSTRUCTIONS

Framed instructions under glass or in laminated plastic, including wiring and control diagrams showing the complete layout of the entire system, shall be posted where directed. Condensed operating instructions explaining preventive maintenance procedures, methods of checking the system for normal safe operation, and procedures for safely starting and stopping the system shall be prepared in typed form, framed as specified above for the wiring and control diagrams and posted beside the diagrams. The framed instructions shall be posted before acceptance testing of the systems.

### 3.10 PERFORMANCE OF WATER HEATING EQUIPMENT

Standard rating condition terms are as follows:

EF = Energy factor, minimum overall efficiency.

ET = Minimum thermal efficiency with 70 degrees F delta T.

SL = Standby loss is maximum (Btu/h) based on a 70 degrees F temperature difference between stored water and ambient requirements.

V = Rated volume in gallons

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

Q = Nameplate input rate in kW (Btu/h)

### 3.10.1 Storage Water Heaters

#### 3.10.1.1 Electric

- a. Storage capacity of 60 gallons shall have a minimum energy factor (EF) of 0.93 or higher per FEMP requirements.
- b. Storage capacity of 60 gallons or more shall have a minimum energy factor (EF) of 0.91 or higher per FEMP requirements.

#### 3.10.1.2 Gas

- a. Storage capacity of 50 gallons or less shall have a minimum energy factor (EF) of 0.67 or higher per FEMP requirements.
- b. Storage capacity of 20 gallons - or more and input rating of 75,000 Btu/h or less: minimum EF shall be 0.62 - 0.0019V per 10 CFR 430.
- c. Rating of less than 22980 W: (75,000 Btu/h) ET shall be 80 percent; maximum SL shall be  $(Q/800+110 \times (V^{1/2}))$ , per ANSI Z21.10.3/CSA 4.3

#### 3.10.1.3 Oil

- a. Storage capacity of 20 gallons or more and input rating of 105,000 Btu/h or less: minimum EF shall be 0.59-0.0019V per 10 CFR 430.
- b. Rating of less than 4,000 Btu/h/gallon or input rating more than 105,000 Btu/h: ET shall be 78 percent; maximum SL shall be  $(Q/800+100 \times (V^{1/2}))$ , per ANSI Z21.10.3/CSA 4.3.

### 3.10.2 Unfired Hot Water Storage

All volumes and inputs: shall meet or exceed R-12.5.

### 3.10.3 Instantaneous Water Heater

#### 3.10.3.1 Gas

- a. Rating of 4,000 Btu/h/gal and greater and less than 2 gallons with an input greater than 50,000 Btu/h and less than 200,000 Btu/h shall have a minimum energy factor (EF) of 0.62-0.0019V per 10 CFR 430.
- b. Rating of 4,000 Btu/h/gal and greater and less than 10 gallons with an input of 200,000 Btu/h and greater shall have a minimum thermal efficiency (ET) of 80 percent per ANSI Z21.10.3/CSA 4.3
- c. Rating of 4,000 BTU/h/gal and greater and 10 gallons and greater with an input of 200,000 Btu/h and greater shall have a minimum thermal efficiency (ET) of 80 percent and the maximum SL shall be  $Q/800+110 \times (V^{1/2})$  per ANSI Z21.10.3/CSA 4.3

#### 3.10.3.2 Oil

- a. Rating of 4,000 Btu/h/gal and greater and less than 2 gallons with an input of 210,000 Btu/h and less shall have an energy factor (EF) of 0.59-0.0019V per 10 CFR 430

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

- b. Rating of 4,000 Btu/h/gal and greater and less than 10 gallons with an input greater than 210,000 Btu/h shall have a minimum thermal efficiency (ET) of 80 percent per ANSI Z21.10.3/CSA 4.3
- c. Rating of 4,000 Btu/h/gal and 10 gallons and greater with an input of greater than 210,000 Btu/h shall have a minimum thermal efficiency (ET) of 78 percent and the maximum SL shall be  $Q/800+110 \times (V^{1/2})$  per ANSI Z21.10.3/CSA 4.3

#### 3.10.4 Pool Heaters

- a. Gas/oil fuel, capacities and inputs: ET shall be 78 percent per ASHRAE 146.
- b. Heat Pump, All capacities and inputs shall meet a COP of 4.0 per ASHRAE 146

#### 3.11 TABLES

| TABLE I                                                                                  |                                                                                                                                                                |           |           |           |           |           |           |           |
|------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|
| PIPE AND FITTING MATERIALS FOR DRAINAGE, WASTE, VENT AND CONDENSATE DRAIN PIPING SYSTEMS |                                                                                                                                                                |           |           |           |           |           |           |           |
| Item #                                                                                   | Pipe and Fitting Materials                                                                                                                                     | SERVICE A | SERVICE B | SERVICE C | SERVICE D | SERVICE E | SERVICE F | SERVICE G |
| 2                                                                                        | Cast iron soil pipe and fittings hubless, CISPI 301 and ASTM A888. Pipe and fittings shall be marked with the CISPI trademark.                                 |           | X         |           | X         | X         |           |           |
| 8                                                                                        | Wrought copper grooved joint pressure fittings for non-ferrous pipe ASTM B75/B75M C12200, ASTM B152/B152M, C11000, ASME B16.22 ASME B16.22 for use with Item 5 |           | X         |           | X         |           |           |           |

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

| TABLE I                                                                                  |                                                                                                                |           |           |           |           |           |           |           |
|------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|
| PIPE AND FITTING MATERIALS FOR DRAINAGE, WASTE, VENT AND CONDENSATE DRAIN PIPING SYSTEMS |                                                                                                                |           |           |           |           |           |           |           |
| Item #                                                                                   | Pipe and Fitting Materials                                                                                     | SERVICE A | SERVICE B | SERVICE C | SERVICE D | SERVICE E | SERVICE F | SERVICE G |
| 13                                                                                       | Cast copper alloy solder joint pressure fittings, ASME B16.18 for use with Item 14                             |           |           |           | X         |           |           | X         |
| 14                                                                                       | Seamless copper pipe, ASTM B42                                                                                 |           |           |           | X         |           | X         | X         |
| 15                                                                                       | Cast bronze threaded fittings, ASME B16.15                                                                     |           |           |           | X         |           |           |           |
| 16                                                                                       | Copper drainage tube, (DWV), ASTM B306                                                                         |           | X         |           | X         |           |           | X         |
| 17                                                                                       | Wrought copper and wrought alloy solder-joint drainage fittings. ASME B16.29                                   |           | X         |           | X         |           |           | X         |
| 18                                                                                       | Cast copper alloy solder joint drainage fittings, DWV, ASME B16.23                                             |           | X         |           | X         |           |           | X         |
| 19                                                                                       | Acrylonitrile-Butadiene (ABS) plastic drain, waste, and vent pipe and fittings<br>ASTM D2661, ASTM F628        | X         |           | X         |           |           | X         |           |
| 20                                                                                       | Polyvinyl Chloride plastic drain, waste and vent pipe and fittings, ASTM D2665, ASTM F891, (Sch 40) ASTM F1760 | X         |           | X         |           |           | X         |           |

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

| TABLE I                                                                                                                                                                                                                                                                                                                                                                  |                            |           |           |           |           |           |           |           |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|
| PIPE AND FITTING MATERIALS FOR DRAINAGE, WASTE, VENT AND CONDENSATE DRAIN PIPING SYSTEMS                                                                                                                                                                                                                                                                                 |                            |           |           |           |           |           |           |           |
| Item #                                                                                                                                                                                                                                                                                                                                                                   | Pipe and Fitting Materials | SERVICE A | SERVICE B | SERVICE C | SERVICE D | SERVICE E | SERVICE F | SERVICE G |
| <p>SERVICE:</p> <p>A - Underground Building Soil, Waste and Storm Drain</p> <p>B - Aboveground Soil, Waste, Drain In Buildings</p> <p>C - Underground Vent</p> <p>D - Aboveground Vent</p> <p>E - Interior Rainwater Conductors Aboveground</p> <p>F - Corrosive Waste And Vent Above And Belowground</p> <p>G - Condensate Drain Aboveground</p> <p>* - Hard Temper</p> |                            |           |           |           |           |           |           |           |

| TABLE II                                               |                                                                                                     |           |           |           |           |
|--------------------------------------------------------|-----------------------------------------------------------------------------------------------------|-----------|-----------|-----------|-----------|
| PIPE AND FITTING MATERIALS FOR PRESSURE PIPING SYSTEMS |                                                                                                     |           |           |           |           |
| Item #                                                 | Pipe and Fitting Materials                                                                          | SERVICE A | SERVICE B | SERVICE C | SERVICE D |
| 7                                                      | Seamless copper pipe, ASTM B42                                                                      | X         | X         |           | X         |
| 8                                                      | Seamless copper water tube, ASTM B88, ASTM B88M                                                     | X**       | X**       | X**       | X***      |
| 9                                                      | Cast bronze threaded fittings, ASME B16.15 for use with Items 5 and 7                               | X         | X         |           | X         |
| 10                                                     | Wrought copper and bronze solder-joint pressure fittings, ASME B16.22 for use with Items 5, 7 and 8 | X         | X         |           | X         |
| 11                                                     | Cast copper alloy solder-joint pressure fittings, ASME B16.18 for use with Item 8                   | X         | X         |           | X         |
| 31                                                     | Polypropylene (PP) plastic pipe and fittings; ASTM F2389                                            | X         | X         |           | X         |
| 33                                                     | Fittings: brass or bronze; ASME B16.15, and ASME B16.18 ASTM B828                                   | X         | X         |           |           |

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

| TABLE II                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                |           |           |           |           |
|--------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-----------|-----------|-----------|
| PIPE AND FITTING MATERIALS FOR PRESSURE PIPING SYSTEMS |                                                                                                                                                                                                                                                                                                                                                                                                                                                |           |           |           |           |
| Item #                                                 | Pipe and Fitting Materials                                                                                                                                                                                                                                                                                                                                                                                                                     | SERVICE A | SERVICE B | SERVICE C | SERVICE D |
| 37                                                     | Crosslinked Polyethylene (PEX)<br>Plastic Pipe ASTM F877                                                                                                                                                                                                                                                                                                                                                                                       | X         | X         |           | X         |
| 38                                                     | Press Fittings                                                                                                                                                                                                                                                                                                                                                                                                                                 | X         | X         |           |           |
|                                                        | SERVICE:<br>A - Cold Water Service Aboveground<br>B - Hot and Cold Water Distribution<br>180 degrees F Maximum Aboveground<br>C - Compressed Air Lubricated<br>D - Cold Water Service Belowground<br>Indicated types are minimum wall thicknesses.<br>** - Type L - Hard<br>*** - Type K - Hard temper with brazed joints only or type K-soft<br>temper without joints in or under floors<br>**** - In or under slab floors only brazed joints |           |           |           |           |



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

| TABLE III                                                                                                                                                                                                                                                                                                                                  |                                         |                     |                       |                                                                      |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|---------------------|-----------------------|----------------------------------------------------------------------|
| STANDARD RATING CONDITIONS AND MINIMUM PERFORMANCE RATINGS FOR WATER HEATING EQUIPMENT                                                                                                                                                                                                                                                     |                                         |                     |                       |                                                                      |
| <u>FUEL</u>                                                                                                                                                                                                                                                                                                                                | <u>STORAGE<br/>CAPACITY<br/>GALLONS</u> | <u>INPUT RATING</u> | <u>TEST PROCEDURE</u> | <u>REQUIRED PERFORMANCE</u>                                          |
| A. STORAGE WATER HEATERS                                                                                                                                                                                                                                                                                                                   |                                         |                     |                       |                                                                      |
| Elect.                                                                                                                                                                                                                                                                                                                                     | 20 min.                                 | >12 kW              | 10 CFR 431.106        | $SL = 0.3 + 27/V$                                                    |
| Gas                                                                                                                                                                                                                                                                                                                                        |                                         | > 105,000<br>Btu/h  | 10 CFR 431.106        | $ET = 80\% \text{ min. } SL < Q/800 + 110/(\text{SQRT}(\text{Vol}))$ |
| B. Unfired Hot Water Storage, R-12.5 min.                                                                                                                                                                                                                                                                                                  |                                         |                     |                       |                                                                      |
| TERMS:<br><br>EF = Energy factor, minimum overall efficiency.<br>ET = Minimum thermal efficiency with 70 degrees F delta T.<br>SL = Standby loss is maximum Btu/h based on a 70 degree F temperature difference between stored water and ambient requirements.<br>V = Rated storage volume in gallons<br>Q = Nameplate input rate in Btu/h |                                         |                     |                       |                                                                      |

-- End of Section --

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## SECTION 22 05 48.00 20

## MECHANICAL SOUND, VIBRATION, AND SEISMIC CONTROL

04/06, CHG 1: 05/15

## PART 1 GENERAL

## 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

## AIR-CONDITIONING, HEATING AND REFRIGERATION INSTITUTE (AHRI)

AHRI 575 (2017) Method of Measuring Machinery Sound Within an Equipment Space

ANSI/AHRI 370 (2015; Addendum 1 2016) Sound Rating of Large Outdoor Refrigerating and Air-Conditioning Equipment

## AMERICAN INSTITUTE OF STEEL CONSTRUCTION (AISC)

AISC 360 (2016) Specification for Structural Steel Buildings

## AMERICAN WELDING SOCIETY (AWS)

AWS D1.1/D1.1M (2020; Errata 1 2021) Structural Welding Code - Steel

## ASTM INTERNATIONAL (ASTM)

ASTM A36/A36M (2019) Standard Specification for Carbon Structural Steel

ASTM A123/A123M (2017) Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products

ASTM C94/C94M (2023) Standard Specification for Ready-Mixed Concrete

ASTM D471 (2016a) Standard Test Method for Rubber Property - Effect of Liquids

ASTM D2240 (2015; E 2017) Standard Test Method for Rubber Property - Durometer Hardness

ASTM E84 (2023) Standard Test Method for Surface Burning Characteristics of Building Materials

## SHEET METAL AND AIR CONDITIONING CONTRACTORS' NATIONAL ASSOCIATION (SMACNA)

SMACNA 1793 (2012) Architectural Sheet Metal Manual,

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

7th Edition

SMACNA 1966

(2020) HVAC Duct Construction Standards  
Metal and Flexible, 4th Edition

## 1.2 RELATED REQUIREMENTS

The provisions of Section 23 03 00.00 20 BASIC MECHANICAL MATERIALS AND METHODS apply to this section.

## 1.3 DEFINITIONS

### 1.3.1 Decibels dB

Measure of sound level. Decibels are referenced to either 20 uPa for sound pressure levels or one pW for sound power levels. dBA is the overall "A" weighted sound level.

### 1.3.2 Machinery

The vibration or noise producing equipment that must be isolated.

### 1.3.3 Manufacturer

The fabricator or supplier of vibration-isolation or seismic-protection materials and equipment. For mechanical equipment and machinery the term machinery manufacturer will be used.

### 1.3.4 Micropascal uPa

10 to the minus 6 power newtons per square meter.

### 1.3.5 Picowatt pW

10 to the minus 12 power watts.

## 1.4 SYSTEM DESCRIPTION

### 1.4.1 Spring Isolator Data

For each type and size of spring isolator, submit the spring outside diameter, deflection, operating spring height, unloaded spring height, solid spring height, the ratio of the outside diameter to the operating spring height, the load to deflection ratio of the springs, and weight and sizes of structural steel members.

### 1.4.2 Machinery Manufacturer's Sound Data

For each piece of indicated machinery to be vibration isolated, the calculated sound power test data or sound pressure test data as levels in dB in the eight octave bands between 63 and 8,000 Hz. Refer sound power levels to one pW and sound pressure levels to 20 uPa. Submit the overall "A" weighted scale sound pressure level in dB. Submit the standard test procedure used to obtain the sound power or pressure data for the applicable vibration isolation equipment size.

### 1.4.3 Machinery

For each item of machinery, compare spring static deflections with the

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

specified minimum static deflection, to show that the calculated spring static deflections are not less than the minimum static deflections specified. Rated spring static deflections are not acceptable in lieu of calculated spring static deflections.

#### 1.4.4 Machinery Over 300 Pounds

For machinery items over 300 pounds, provide calculations for shear, pull-up, primary overturning, and secondary overturning.

#### 1.4.5 Machinery Vibration Criteria

| TABLE 1A                                                                |                                |                   |                    |                   |                    |                   |
|-------------------------------------------------------------------------|--------------------------------|-------------------|--------------------|-------------------|--------------------|-------------------|
| Vibration Isolator Types and Minimum Static Deflection                  |                                |                   |                    |                   |                    |                   |
| (MSD, inches) for 4-8 inch slab on grade and column supported.          |                                |                   |                    |                   |                    |                   |
| Column Spacing                                                          | Slab on earth and<br>0-30 feet |                   | 31-40 feet         |                   | 41-50 feet         |                   |
| Equipment                                                               | Type<br>(Note (1))             | MSD (Note<br>(1)) | Type<br>(Note (1)) | MSD (Note<br>(1)) | Type<br>(Note (1)) | MSD (Note<br>(1)) |
| Packaged Boilers                                                        | SV                             | 1.0               | SV                 | 2.5               | SV-R               | 3.5               |
| Closed Coupled Pumps                                                    |                                |                   |                    |                   |                    |                   |
| Up to 7-1/2 hp                                                          | S-I                            | 1.0               | S-I                | 1.0               | S-I                | 1.0               |
| Over 7-1/2 hp                                                           | S-I                            | 1.5               | S-I                | 2.5               | S-I                | 2.5               |
| Base Mounted Pumps                                                      |                                |                   |                    |                   |                    |                   |
| Up to 20 hp                                                             | S-I                            | 1.5               | S-I                | 2.5               | S-I                | 2.5               |
| 20 to 75 hp                                                             | S-I                            | 1.5               | S-I                | 2.5               | S-I                | 3.5               |
| Factory Assembled Air Handling Equipment AH, AC and HV Units (Note (2)) |                                |                   |                    |                   |                    |                   |
| Floor Mounted Units                                                     |                                |                   |                    |                   |                    |                   |
| Up to 5 hp                                                              | S                              | 1.0               | S                  | 1.0               | S                  | 1.0               |
| Over 5 hp                                                               |                                |                   |                    |                   |                    |                   |
| Up to 400 rpm                                                           | S-R                            | 1.75              | S-R                | 1.75              | S-R                | 2.5               |

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

| TABLE 1A                                                                                                             |                                |                   |                    |                   |                    |                   |
|----------------------------------------------------------------------------------------------------------------------|--------------------------------|-------------------|--------------------|-------------------|--------------------|-------------------|
| Vibration Isolator Types and Minimum Static Deflection                                                               |                                |                   |                    |                   |                    |                   |
| (MSD, inches) for 4-8 inch slab on grade and column supported.                                                       |                                |                   |                    |                   |                    |                   |
| Column Spacing                                                                                                       | Slab on earth and<br>0-30 feet |                   | 31-40 feet         |                   | 41-50 feet         |                   |
| Equipment                                                                                                            | Type<br>(Note (1))             | MSD (Note<br>(1)) | Type<br>(Note (1)) | MSD (Note<br>(1)) | Type<br>(Note (1)) | MSD (Note<br>(1)) |
| Over 401 rpm                                                                                                         | S-R                            | 1.0               | S-R                | 1.5               | S-R                | 2.5               |
| Centrifugal Blowers                                                                                                  |                                |                   |                    |                   |                    |                   |
| 175 - 224 rpm                                                                                                        | S-B                            | 4.75              | S-B                | 4.75              | S-B                | 4.75              |
| 225 - 299 rpm                                                                                                        | S-B                            | 3.75              | S-B                | 4.75              | S-B                | 4.75              |
| 300 - 374 rpm                                                                                                        | S-B                            | 2.75              | S-B                | 4.5               | S-B                | 4.75              |
| 375 - 499 rpm                                                                                                        | S-B                            | 2.5               | S-B                | 3.5               | S-B                | 4.5               |
| Over 500 rpm                                                                                                         | S-B                            | 1.75              | S-B                | 2.5               | S-B                | 3.5               |
| Internal Combustion Engines and Engine Driven Equip                                                                  |                                |                   |                    |                   |                    |                   |
| 750 rpm and<br>over                                                                                                  | S                              | 1.5               | S                  | 2.5               | S                  | 3.5               |
| Dimmer Banks and Transformers                                                                                        |                                |                   |                    |                   |                    |                   |
| Up to 1000 lbs.                                                                                                      | NM                             | 0.35              | NM                 | 0.35              | NM                 | 3.5               |
| Over 1000 lbs.                                                                                                       | SV                             | 1.0               | SV                 | 1.0               | SV                 | 1.0               |
| NOTES:                                                                                                               |                                |                   |                    |                   |                    |                   |
| (1) Equipment Vibration Isolation Schedule Designations (Hyphenated designations are combinations of the following:) |                                |                   |                    |                   |                    |                   |
| B - Welded structural steel bases.                                                                                   |                                |                   |                    |                   |                    |                   |
| H - Spring isolators (suspended equipment and piping). Where required, provide with adjustable preloading devices.   |                                |                   |                    |                   |                    |                   |
| HR - Thrust restraints                                                                                               |                                |                   |                    |                   |                    |                   |
| I - Concrete inertia bases with steel forms.                                                                         |                                |                   |                    |                   |                    |                   |

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

| TABLE 1A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                |                   |                    |                   |                    |                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|-------------------|--------------------|-------------------|--------------------|-------------------|
| Vibration Isolator Types and Minimum Static Deflection                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                |                   |                    |                   |                    |                   |
| (MSD, inches) for 4-8 inch slab on grade and column supported.                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                |                   |                    |                   |                    |                   |
| Column Spacing                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Slab on earth and<br>0-30 feet |                   | 31-40 feet         |                   | 41-50 feet         |                   |
| Equipment                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Type<br>(Note (1))             | MSD (Note<br>(1)) | Type<br>(Note (1)) | MSD (Note<br>(1)) | Type<br>(Note (1)) | MSD (Note<br>(1)) |
| NM - Neoprene mounts.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                |                   |                    |                   |                    |                   |
| NP - Neoprene pads.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                |                   |                    |                   |                    |                   |
| R - Structural steel rail for equipment mounts.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                |                   |                    |                   |                    |                   |
| S - Freestanding spring isolators (floor-mounted equipment).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                |                   |                    |                   |                    |                   |
| SV - Freestanding spring isolators (floor-mounted equipment).                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                |                   |                    |                   |                    |                   |
| SX - Freestanding spring isolators with adjustable cushioned vertical stops and cushioned horizontal stops (floor-mounted equipment). Protected spring isolators SX may be substituted wherever S or SV is specified and must meet all requirements.                                                                                                                                                                                                                                                                                        |                                |                   |                    |                   |                    |                   |
| (2) Fans                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                |                   |                    |                   |                    |                   |
| a. When fan motors are 75 hp or larger, use the deflection requirements for the next wider column spacing. Except for building slab on grade a minimum of 2.5 inches should be used unless larger deflections are specified in the centrifugal blower table.                                                                                                                                                                                                                                                                                |                                |                   |                    |                   |                    |                   |
| b. Provide sway brace isolators for tubular centrifugal and axial fans when the fan pressure exceeds 4 inches water column.                                                                                                                                                                                                                                                                                                                                                                                                                 |                                |                   |                    |                   |                    |                   |
| c. Provide inertia bases for all fans in lieu of structural steel bases or rails specified above when the fan pressure exceeds 4 inches water column.                                                                                                                                                                                                                                                                                                                                                                                       |                                |                   |                    |                   |                    |                   |
| d. With attaching brackets, suspension spring isolators bridge between the structure and the thrust-producing machinery such as high-pressure fan. Both types H and HR normally provide reaction in tension, while types S, SV, and SX normally provide reaction in compression. Thrust restraints are low-cost and effective components available from manufacturers. Use thrust restraints to eliminate the need for or reduce the magnitude of inertia mass when the mass is only used to reduce the displacement effects of the thrust. |                                |                   |                    |                   |                    |                   |

Provide vibration isolators for mechanical and electrical machinery and associated piping and ductwork , to minimize transmission of vibrations

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

and structure borne noise to the building structure or spaces or from the building structure to the machinery. Comply with the following vibration schedule.

#### 1.4.6 Machinery Airborne Sound Level Criteria

##### 1.4.6.1 Basic Criteria

For each piece of machinery in the human work environment, do not exceed the maximum airborne sound levels 84 dB A-weighted scale, continuous or intermittent, or 140 dB peak sound pressure-level, impact or impulse, noise.

##### 1.4.6.2 Sound Data Schedule

| TABLE 2A              |                                         |     |     |     |      |      |      |      |
|-----------------------|-----------------------------------------|-----|-----|-----|------|------|------|------|
| Sound Data Schedule   |                                         |     |     |     |      |      |      |      |
| Equipment             | Maximum Sound Power Level (dB)          |     |     |     |      |      |      |      |
|                       | Octave Band Level Center Frequency (Hz) |     |     |     |      |      |      |      |
|                       | 63                                      | 125 | 250 | 500 | 1000 | 2000 | 4000 | 8000 |
| Air Handling Unit     | 94                                      | 90  | 89  | 89  | 89   | 84   | 82   | 79   |
| Make-Up Air Fan       | 91                                      | 91  | 80  | 84  | 82   | 76   | 71   | 65   |
| Air Conditioning Unit | 100                                     | 96  | 90  | 89  | 86   | 80   | 75   | 72   |
| Boiler                | 75                                      | 72  | 72  | 75  | 76   | 63   | 55   | 50   |
| Air Compressor        | 90                                      | 89  | 92  | 93  | 92   | 92   | 90   | 81   |
| Pump                  | 85                                      | 80  | 82  | 82  | 80   | 77   | 74   | 72   |
| Fan                   | 55                                      | 50  | 48  | 47  | 48   | 46   | 42   | 37   |

##### 1.4.7 Welding

AWS D1.1/D1.1M.

#### 1.5 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Inertia Bases

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### SD-03 Product Data

Isolators

Flexible Connectors

Flexible Duct Connectors

Inertia Bases

Machinery Manufacturer's Sound Data

#### SD-05 Design Data

Inertia Bases

Each Item of Machinery

Each Item of Machinery Over 300 Pounds

Submit design calculations for inertia bases, , , , and , either by the machinery manufacturer for the recommended machinery mounting or by the vibration-isolation equipment manufacturer.

#### SD-06 Test Reports

Equipment Vibration Tests

Equipment Sound Level Tests

#### SD-08 Manufacturer's Instructions

Vibration and Noise Isolation Components

### 1.6 QUALITY ASSURANCE

#### 1.6.1 Vibration Isolator Procurement

For each piece of machinery to be isolated from vibration, supply the inertia base, , , , , vibration isolators, , and other associated materials and equipment as a coordinated package by a single manufacturer or by the machinery manufacturer. Select isolators that provide uniform deflection even when machinery weight is not evenly distributed. This requirement does not include the flexible connectors or the hangers for the associated piping and ductwork.

#### 1.6.2 Unitized Machinery Assemblies

Mounting of unitized assemblies directly on vibration isolation springs is acceptable if machinery manufacturer certifies that the end supports of the assemblies have been designed for such installation.



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## PART 2 PRODUCTS

### 2.1 CORROSION PROTECTION FOR STEEL PARTS

ASTM A123/A123M hot-dipped galvanized, or equivalent manufacturer standard coatings. Where steel parts are exposed to the weather, provide galvanized coating of at least 2 ounces of zinc per square foot of surface. Coat springs with neoprene.

### 2.2 NEOPRENE

ASTM D471 and ASTM D2240, Grade Durometer 40, 50, or 60, and oil resistant.

### 2.3 FLOOR-MOUNTED ISOLATORS

#### 2.3.1 Neoprene Isolation Pads

Provide pads at least 1/4 inch thick with cross-ribbed or waffle design. For concentrated loads, provide steel bearing plates bonded or cold cemented to the pads.

#### 2.3.2 Neoprene Isolators

Provide molded neoprene isolators having steel base plates with mounting holes and, at the top, steel mounting plates with mounting holes or threaded inserts. Provide elements of type and size coded with molded letters or color-coded for capacity identification. Embed metal parts completely in neoprene.

### 2.4 SPRING ISOLATORS AND PROTECTED SPRING ISOLATORS

Provide spring isolators or protected spring isolators that are adjustable and laterally stable with free-standing springs of horizontal stiffness at minimum 80 percent of the vertical (axial) stiffness. For machine-attached and floor-attached restraining elements, separate from metal-to-metal contact by neoprene cushions 1/8 inch thick minimum. Provide neoprene acoustic friction pads at least 1/4 inch thick.

#### 2.4.1 Springs

Provide springs with base and compression plates, to keep spring ends parallel during and after deflection to operating height. Provide outside coil diameters at least 0.8 of the operating height. At operating height, springs must have additional travel to complete (solid) compression equal to at least 50 percent of the operating deflection.

#### 2.4.2 Mounting and Adjustment

Provide base and compression plates with mounting holes or threaded fittings. Bolt leveling adjustment bolts to machinery or base.

### 2.5 SUSPENSION ISOLATORS

Provide hangers with suspension isolators encased in open steel brackets. Isolate hanger rods from isolator steel brackets with neoprene-lined opening.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 2.5.1 Suspension Neoprene Isolators

Provide double-deflection elements with minimum 3/8 inch deflection.

### 2.5.2 Suspension Spring Isolators

Provide hangers with springs and molded neoprene elements in series. Provide isolators with adjustable spring-preloading devices where required to maintain constant pipe elevations during installation and when pipe operational loads are transferred to the springs.

## 2.6 MACHINERY BASES

ASTM A36/A36M and AISC 360.

## 2.7 INERTIA BASES

ASTM A36/A36M steel, ASTM C94/C94M ( 3,000 psi) concrete.

## 2.8 FLEXIBLE CONNECTORS FOR PIPING

Straight or elbow flexible connectors rated for temperatures, pressures, and fluids to be conveyed. Provide flexible connectors with the strength 4 times operating pressure at highest system operating temperature. Provide elbow flexible connectors with a permanently set angle.

### 2.8.1 Metal Flexible Connectors

Fabricated of Grade E phosphor bronze, monel or corrugated stainless steel tube covered with comparable bronze or stainless steel braid restraining and pressure cover.

## 2.9 FLEXIBLE DUCT CONNECTORS

Provide flexible duct connectors fabricated in accordance with SMACNA 1966.

## PART 3 EXECUTION

### 3.1 INSTALLATION

#### 3.1.1 Vibration and Noise Isolation Components

Install vibration-and-noise isolation materials and equipment as indicated and in accordance with machinery manufacturer's instructions.

#### 3.1.2 Suspension Vibration Isolators

Provide suspension isolation hangers for piping, suspended equipment, and suspended equipment platforms in mechanical equipment rooms, as indicated and as specified. For operating load static deflections of 1/4 inch or less, provide neoprene pads or single deflection neoprene isolators. For operating load static deflections over 5/16 to 3/8 inch, provide double-deflection neoprene element isolators. For operating load static deflections over 3/8 inch, provide isolators with spring and neoprene elements in series.

#### 3.1.3 Vertical Stops

For machinery affected by wind pressure or having an operational weight

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

different from installed weight, provide resilient vertical limit stops which prevent spring extension when weight is removed. Provide vertical stops for machinery containing liquid, such as boilers. Spring isolated or protected spring isolated machinery must rock and move freely within limits of stops.

#### 3.1.4 Flexible Pipe and Duct Connectors

Install flexible connectors in accordance with the manufacturer's instructions. When liquid pulsation dampening is required, flexible connectors with spherical configuration may be used.

#### 3.1.5 Machinery

Provide vibration isolators, flexible connectors in accordance with manufacturer's recommendations. Machinery with spring isolators or protected spring isolators must rock or move freely within limits of stops.

##### 3.1.5.1 Stability

Isolators must be stable during starting and stopping of machinery without traverse and eccentric movement of machinery that would damage or adversely affect the machinery or attachments.

##### 3.1.5.2 Lateral Motion

The installed vibration isolation system for each piece of floor or ceiling mounted machinery must have a maximum lateral motion under machinery start up and shut down conditions of not more than 1/4 inch. Restrain motions in excess by approved spring mountings.

##### 3.1.5.3 Unbalanced Machinery

Provide foundation suspension systems specifically designed to resist horizontal forces for machinery with large unbalanced horizontal forces. Vibration isolator systems must conform to the machinery manufacturer's recommendations.

##### 3.1.5.4 Nonrotating Machinery

Mount nonrotating machinery in systems which includes rotating or vibrating machinery on isolators having the same deflection as the hangers and supports for the pipe connected to.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 3.1.5.5 Unitized Machinery Assemblies

| TABLE 3A                                                                                                             |                                |                   |                    |                   |                    |                   |
|----------------------------------------------------------------------------------------------------------------------|--------------------------------|-------------------|--------------------|-------------------|--------------------|-------------------|
| Vibration Isolator Types and Minimum Static Deflection                                                               |                                |                   |                    |                   |                    |                   |
| (MSD, inches) for 4-8 inch slab on grade and column supported.                                                       |                                |                   |                    |                   |                    |                   |
| Column Spacing                                                                                                       | Slab on earth and<br>0-30 feet |                   | 31-40 feet         |                   | 41-50 feet         |                   |
| Equipment                                                                                                            | Type<br>(Note (1))             | MSD (Note<br>(1)) | Type<br>(Note (1)) | MSD (Note<br>(1)) | Type<br>(Note (1)) | MSD (Note<br>(1)) |
| Closed Coupled Pumps                                                                                                 |                                |                   |                    |                   |                    |                   |
| Up to 7-1/2 hp                                                                                                       | S-I                            | 1.0               | S-I                | 1.0               | S-I                | 1.0               |
| Over 7-1/2 hp                                                                                                        | S-I                            | 1.5               | S-I                | 2.5               | S-I                | 2.5               |
| Base Mounted Pumps                                                                                                   |                                |                   |                    |                   |                    |                   |
| Up to 20 hp                                                                                                          | S-I                            | 1.5               | S-I                | 2.5               | S-I                | 2.5               |
| 20 to 75 hp                                                                                                          | S-I                            | 1.5               | S-I                | 2.5               | S-I                | 3.5               |
| Internal Combustion Engines and Engine Driven Equip                                                                  |                                |                   |                    |                   |                    |                   |
| 750 rpm and<br>over                                                                                                  | S                              | 1.5               | S                  | 2.5               | S                  | 3.5               |
| Dimmer Banks and Transformers                                                                                        |                                |                   |                    |                   |                    |                   |
| Up to 1000 lbs.                                                                                                      | NM                             | 0.35              | NM                 | 0.35              | NM                 | 3.5               |
| Over 1000 lbs.                                                                                                       | SV                             | 1.0               | SV                 | 1.0               | SV                 | 1.0               |
| NOTES:                                                                                                               |                                |                   |                    |                   |                    |                   |
| (1) Equipment Vibration Isolation Schedule Designations (Hyphenated designations are combinations of the following:) |                                |                   |                    |                   |                    |                   |
| B - Welded structural steel bases.                                                                                   |                                |                   |                    |                   |                    |                   |
| H - Spring isolators (suspended equipment and piping). Where required, provide with adjustable preloading devices.   |                                |                   |                    |                   |                    |                   |
| HR - Thrust restraints                                                                                               |                                |                   |                    |                   |                    |                   |
| I - Concrete inertia bases with steel forms.                                                                         |                                |                   |                    |                   |                    |                   |

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

| TABLE 3A                                                                                                                                                                                                                                            |                                |                   |                    |                   |                    |                   |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|-------------------|--------------------|-------------------|--------------------|-------------------|
| Vibration Isolator Types and Minimum Static Deflection                                                                                                                                                                                              |                                |                   |                    |                   |                    |                   |
| (MSD, inches) for 4-8 inch slab on grade and column supported.                                                                                                                                                                                      |                                |                   |                    |                   |                    |                   |
| Column Spacing                                                                                                                                                                                                                                      | Slab on earth and<br>0-30 feet |                   | 31-40 feet         |                   | 41-50 feet         |                   |
| Equipment                                                                                                                                                                                                                                           | Type<br>(Note (1))             | MSD (Note<br>(1)) | Type<br>(Note (1)) | MSD (Note<br>(1)) | Type<br>(Note (1)) | MSD (Note<br>(1)) |
| NM - Neoprene mounts.                                                                                                                                                                                                                               |                                |                   |                    |                   |                    |                   |
| NP - Neoprene pads.                                                                                                                                                                                                                                 |                                |                   |                    |                   |                    |                   |
| R - Structural steel rail for equipment mounts.                                                                                                                                                                                                     |                                |                   |                    |                   |                    |                   |
| S - Freestanding spring isolators (floor-mounted equipment).                                                                                                                                                                                        |                                |                   |                    |                   |                    |                   |
| SV - Freestanding spring isolators (floor-mounted equipment).                                                                                                                                                                                       |                                |                   |                    |                   |                    |                   |
| SX - Freestanding spring isolators with adjustable cushioned vertical stops and cushioned horizontal stops (floor-mounted equipment. Protected spring isolators SX may be substituted wherever S or SV is specified and must meet all requirements. |                                |                   |                    |                   |                    |                   |

Unitized assemblies such as chillers with evaporator and condenser, and top mounted centrifugal compressor or unitized absorption refrigeration machines, structurally designed with end supports, may be mounted on steel rails and springs in lieu of steel bases and springs. Where the slab or deck is less than 4 inches thick, provide spring isolation units with the deflection double that of the vibration isolation schedule, up to a maximum static deflection of 5 inches.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 3.1.5.6 Roof and Upper Floor Mounted Machinery

| TABLE 3A                                                                                                             |                             |                                                                                          |                 |                |                 |                |
|----------------------------------------------------------------------------------------------------------------------|-----------------------------|------------------------------------------------------------------------------------------|-----------------|----------------|-----------------|----------------|
| Vibration Isolator Types and Minimum Static Deflection                                                               |                             |                                                                                          |                 |                |                 |                |
| (MSD, inches) for 4-8 inch slab on grade and column supported.                                                       |                             |                                                                                          |                 |                |                 |                |
| Column Spacing                                                                                                       | Slab on earth and 0-30 feet |                                                                                          | 31-40 feet      |                | 41-50 feet      |                |
| Equipment                                                                                                            | Type (Note (1))             | MSD (Note (1))                                                                           | Type (Note (1)) | MSD (Note (1)) | Type (Note (1)) | MSD (Note (1)) |
| Factory Assembled Air Handling Equipment AH, AC and HV Units (Note (2))                                              |                             |                                                                                          |                 |                |                 |                |
| Floor Mounted Units                                                                                                  |                             |                                                                                          |                 |                |                 |                |
| Up to 5 hp                                                                                                           | S                           | 1.0                                                                                      | S               | 1.0            | S               | 1.0            |
| Over 5 hp                                                                                                            |                             |                                                                                          |                 |                |                 |                |
| Up to 400 rpm                                                                                                        | S-R                         | 1.75                                                                                     | S-R             | 1.75           | S-R             | 2.5            |
| Over 401 rpm                                                                                                         | S-R                         | 1.0                                                                                      | S-R             | 1.5            | S-R             | 2.5            |
| Centrifugal Blowers                                                                                                  |                             |                                                                                          |                 |                |                 |                |
| 175 - 224 rpm                                                                                                        | S-B                         | 4.75                                                                                     | S-B             | 4.75           | S-B             | 4.75           |
| 225 - 299 rpm                                                                                                        | S-B                         | 3.75                                                                                     | S-B             | 4.75           | S-B             | 4.75           |
| 300 - 374 rpm                                                                                                        | S-B                         | 2.75                                                                                     | S-B             | 4.5            | S-B             | 4.75           |
| 375 - 499 rpm                                                                                                        | S-B                         | 2.5                                                                                      | S-B             | 3.5            | S-B             | 4.5            |
| Over 500 rpm                                                                                                         | S-B                         | 1.75                                                                                     | S-B             | 2.5            | S-B             | 3.5            |
| Utility Fans                                                                                                         |                             |                                                                                          |                 |                |                 |                |
| Floor-Mounted                                                                                                        |                             | S-R with deflections not specified for centrifugal blowers but not to exceed 2.75 inches |                 |                |                 |                |
| NOTES:                                                                                                               |                             |                                                                                          |                 |                |                 |                |
| (1) Equipment Vibration Isolation Schedule Designations (Hyphenated designations are combinations of the following:) |                             |                                                                                          |                 |                |                 |                |

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

| TABLE 3A                                                                                                                                                                                                                                             |                             |                |                 |                |                 |                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|----------------|-----------------|----------------|-----------------|----------------|
| Vibration Isolator Types and Minimum Static Deflection                                                                                                                                                                                               |                             |                |                 |                |                 |                |
| (MSD, inches) for 4-8 inch slab on grade and column supported.                                                                                                                                                                                       |                             |                |                 |                |                 |                |
| Column Spacing                                                                                                                                                                                                                                       | Slab on earth and 0-30 feet |                | 31-40 feet      |                | 41-50 feet      |                |
| Equipment                                                                                                                                                                                                                                            | Type (Note (1))             | MSD (Note (1)) | Type (Note (1)) | MSD (Note (1)) | Type (Note (1)) | MSD (Note (1)) |
| B - Welded structural steel bases.                                                                                                                                                                                                                   |                             |                |                 |                |                 |                |
| H - Spring isolators (suspended equipment and piping). Where required, provide with adjustable preloading devices.                                                                                                                                   |                             |                |                 |                |                 |                |
| HR - Thrust restraints                                                                                                                                                                                                                               |                             |                |                 |                |                 |                |
| I - Concrete inertia bases with steel forms.                                                                                                                                                                                                         |                             |                |                 |                |                 |                |
| NM - Neoprene mounts.                                                                                                                                                                                                                                |                             |                |                 |                |                 |                |
| NP - Neoprene pads.                                                                                                                                                                                                                                  |                             |                |                 |                |                 |                |
| R - Structural steel rail for equipment mounts.                                                                                                                                                                                                      |                             |                |                 |                |                 |                |
| S - Freestanding spring isolators (floor-mounted equipment).                                                                                                                                                                                         |                             |                |                 |                |                 |                |
| SV - Freestanding spring isolators (floor-mounted equipment).                                                                                                                                                                                        |                             |                |                 |                |                 |                |
| SX - Freestanding spring isolators with adjustable cushioned vertical stops and cushioned horizontal stops (floor-mounted equipment). Protected spring isolators SX may be substituted wherever S or SV is specified and must meet all requirements. |                             |                |                 |                |                 |                |

On the roof or upper floors, mount machinery on isolators with vertical stops. Rest isolators on beams or structures designed and installed in accordance with the SMACNA 1793, Plate 61.

### 3.1.6 Piping

Provide vibration isolation for piping . The isolator deflections must be equal to or greater than the static deflection of the vibration isolators provided for the connected machinery as follows:

#### 3.1.6.1 Piping Connected to Vibration Isolated Machinery

For a distance of 50 feet or 50 pipe diameters, whichever is greater.

#### 3.1.6.2 Hot Water Piping

For risers from pumps and for the first 20 feet of the branch connection of the main supply and return piping at each floor.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 3.1.7 Pipe Hanger and Support Installation

#### 3.1.7.1 Pipe Hangers

Provide eye-bolts or swivel joints for pipe hangers to permit pipe thermal or mechanical movement without angular misalignment of hanger vibration isolator.

### 3.1.8 Equipment Room Sound Isolation

Do not allow direct contact between pipe or ducts and walls, floor slabs, roofs, ceilings or partitions of equipment rooms.

#### 3.1.8.1 Pipe Penetrations

Provide galvanized Schedule 40 pipe sleeves and tightly pack annular space between sleeves and pipe with insulation having a flame spread rating not more than 25 and a smoke developed rating not more than 50 when tested in accordance with ASTM E84, maximum effective temperature 1000 degrees F, bulk density 6 pounds/cu. ft. minimum. Provide uninsulated pipe with a one inch thick mineral fiber sleeve the full length of the penetration and seal each end with an interior or exterior and weather resistant non-hardening compound. Provide sealant and mineral-fiber sleeve of a flame spread rating not more than 25 and a smoke developed rating not more than 50 when tested in accordance with ASTM E84.

#### 3.1.8.2 Duct Penetrations

Pack openings around ducts with mineral fiber insulation the full length of the penetration having a flame spread rating not more than 25 and a smoke developed rating not more than 50 when tested in accordance with ASTM E84. At each end of duct opening provide sealing collars and seal with an interior non-hardening compound.

#### 3.1.8.3 Ducts Passing Through Equipment Rooms

Provide with sound insulation equal to the sound attenuation value of the wall, floor, or ceiling penetrated.

### 3.1.9 Machinery Foundations and Subbases

Provide cast in place anchor bolts as recommended by the machinery manufacturer.

#### 3.1.9.1 Machinery Subbases

Provide concrete subbases at least 4 inches high for floor mounted equipment. Rest subbases on structural floor and reinforce with steel rods interconnected with floor reinforcing bars by tie bars hooked at both ends. Provide at least 2 inch clearance between subbases and inertia bases, steel bases, and steel saddles with machinery in operation.

#### 3.1.9.2 Common Machinery Foundations

Mount electrical motors on the same foundations as driven machinery. Support piping connections, strainers, valves, and risers on the same foundation as the pumps.



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### 3.1.9.3 Foundation and Subbase Concrete

Cast concrete foundations and subbases of ASTM C94/C94M 3000 psi concrete reinforced with steel bars as indicated or recommended by machinery manufacturer.

#### 3.1.9.4 Anchor Bolts and Grout

Secure machinery to foundations and inertia bases with anchor bolts. Grout equipment with baseplates, the full area under baseplates with premixed non-shrinking grout. After grout has set, remove wedges, shims, and jack bolts and fill spaces with grout.

#### 3.1.10 Inertia Bases

Install inertia bases in accordance with the recommendations of the machinery manufacturer or inertia base manufacturer, as applicable.

#### 3.1.11 Electrical Connections

Provide flexible conduit or multiple conductor cable connections for machinery with sufficient extra length to permit 2 inch minimum displacement in any direction without damage.

#### 3.1.12 Systems Not To Be Vibration Isolated

Do not provide vibration isolation for electrical raceways and conduits or for fire protection, storm, sanitary, and domestic water piping systems which do not include pumps or other vibrating, rotating, or pulsating equipment including control and pressure reducing valves.

### 3.2 FIELD QUALITY CONTROL

Provide equipment and apparatus required for performing inspections and tests. Notify Contracting Officer 14 days prior to machinery sound and vibration testing. Rebalance, adjust, or replace machinery with noise or vibration levels in excess of those given in the machinery specifications, or machinery manufacturer's data.

#### 3.2.1 Field Inspections

Prior to initial operation, inspect the vibration isolators for conformance to drawings, specifications, and manufacturer's data and instructions. Check for vibration and noise transmission through connections, piping, ductwork, foundations, and walls. Check connector alignment before and after filling of system and during operation. Correct misalignment without damage to connector and in accordance with manufacturer's recommendations.

#### 3.2.2 Spring Isolator Inspection

After installation of spring isolators or protected spring isolators, and seismic restraint devices, the machinery must rock freely on its spring isolators within limits of stops or seismic restraint devices. Eliminate or correct interferences.

#### 3.2.3 Tests

Adjust, repair, or replace isolators as required to reduce vibration and

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

noise transmissions to specified levels.

#### 3.2.3.1 Equipment Vibration Tests

Perform vibration tests to determine conformance with vibration isolation schedule specified indicated.

#### 3.2.3.2 Equipment Sound Level Tests

Measure continuous or intermittent steady state noise with a sound level meter set for low response. Measure impact or impulse noise as dB peak sound pressure level (20 uPa) with an impact noise analyzer. Measure work distance from person to machinery noise center. Perform sound level tests to determine conformance with sound level schedule indicated.

##### a. Interior Machinery Sound

In accordance with AHRI 575, measure the sound data for air conditioning and refrigeration machinery, such as fans, boilers, valves, engines, turbines, or transformers. Measure the sound pressure levels around mechanical and electrical machinery located in equipment spaces, 3 feet horizontally from the edge closest to the acoustical center of the machinery at points 3 feet and 5.5 feet above floor. Take measurements at the center of each side of the machinery. Locate the microphone at least 3 feet from the observer and measuring instruments. Observer must not be between the machinery and the measuring instrument.

##### b. Exterior Machinery Sound

Measure sound data in accordance with ANSI/AHRI 370 for machinery radiating noise outside the building in such applications as grade installations, area-ways, wall and roof installations for cooling towers, refrigerant condensers, engine driven generator sets, fans, air conditioning machinery, heat pumps, evaporative coolers, exhaust silencers, and air intakes.

-- End of Section --

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## SECTION 23 03 00.00 20

### BASIC MECHANICAL MATERIALS AND METHODS

**08/10, CHG 3: 08/18**

#### PART 1 GENERAL

##### 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

##### ASTM INTERNATIONAL (ASTM)

|           |                                                                      |
|-----------|----------------------------------------------------------------------|
| ASTM B117 | (2019) Standard Practice for Operating<br>Salt Spray (Fog) Apparatus |
|-----------|----------------------------------------------------------------------|

##### 1.2 RELATED REQUIREMENTS

This section applies to all sections of Divisions: 21, FIRE SUPPRESSION; 22, PLUMBING; and 23, HEATING, VENTILATING, AND AIR CONDITIONING of this project specification, unless specified otherwise in the individual section.

##### 1.3 QUALITY ASSURANCE

###### 1.3.1 Material and Equipment Qualifications

Provide materials and equipment that are standard products of manufacturers regularly engaged in the manufacture of such products, which are of a similar material, design and workmanship. Standard products must have been in satisfactory commercial or industrial use for 2 years prior to bid opening. The 2-year use must include applications of equipment and materials under similar circumstances and of similar size. The product must have been for sale on the commercial market through advertisements, manufacturers' catalogs, or brochures during the 2 year period.

###### 1.3.2 Alternative Qualifications

Products having less than a two-year field service record will be acceptable if a certified record of satisfactory field operation for not less than 6000 hours, exclusive of the manufacturer's factory or laboratory tests, can be shown.

###### 1.3.3 Service Support

The equipment items must be supported by service organizations. Submit a certified list of qualified permanent service organizations for support of the equipment which includes their addresses and qualifications. These service organizations must be reasonably convenient to the equipment installation and able to render satisfactory service to the equipment on a regular and emergency basis during the warranty period of the contract.

###### 1.3.4 Manufacturer's Nameplate

For each item of equipment, provide a nameplate bearing the manufacturer's name, address, model number, and serial number securely affixed in a

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

conspicuous place; the nameplate of the distributing agent will not be acceptable.

#### 1.3.5 Modification of References

In each of the publications referred to herein, consider the advisory provisions to be mandatory, as though the word, "must" had been substituted for "should" wherever it appears. Interpret references in these publications to the "authority having jurisdiction", or words of similar meaning, to mean the Contracting Officer.

##### 1.3.5.1 Definitions

For the International Code Council (ICC) Codes referenced in the contract documents, advisory provisions must be considered mandatory, the word "should" is interpreted as "must." Reference to the "code official" must be interpreted to mean the "Contracting Officer." For Navy owned property, references to the "owner" must be interpreted to mean the "Contracting Officer." For leased facilities, references to the "owner" must be interpreted to mean the "lessor." References to the "permit holder" must be interpreted to mean the "Contractor."

##### 1.3.5.2 Administrative Interpretations

For ICC Codes referenced in the contract documents, the provisions of Chapter 1, "Administrator," do not apply. These administrative requirements are covered by the applicable Federal Acquisition Regulations (FAR) included in this contract and by the authority granted to the Officer in Charge of Construction to administer the construction of this project. References in the ICC Codes to sections of Chapter 1, must be applied appropriately by the Contracting Officer as authorized by his administrative cognizance and the FAR.

#### 1.4 DELIVERY, STORAGE, AND HANDLING

Handle, store, and protect equipment and materials to prevent damage before and during installation in accordance with the manufacturer's recommendations, and as approved by the Contracting Officer. Replace damaged or defective items.

#### 1.5 ELECTRICAL REQUIREMENTS

Furnish motors, controllers, disconnects and contactors with their respective pieces of equipment. Motors, controllers, disconnects and contactors must conform to and have electrical connections provided under Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Furnish internal wiring for components of packaged equipment as an integral part of the equipment. Extended voltage range motors will not be permitted. Controllers and contactors shall have a maximum of 120 volt control circuits, and must have auxiliary contacts for use with the controls furnished. When motors and equipment furnished are larger than sizes indicated, the cost of additional electrical service and related work must be included under the section that specified that motor or equipment. Power wiring and conduit for field installed equipment must be provided under and conform to the requirements of Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM.

#### 1.6 INSTRUCTION TO GOVERNMENT PERSONNEL

When specified in other sections, furnish the services of competent

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

instructors to give full instruction to the designated Government personnel in the adjustment, operation, and maintenance, including pertinent safety requirements, of the specified equipment or system. Instructors must be thoroughly familiar with all parts of the installation and must be trained in operating theory as well as practical operation and maintenance work.

Instruction must be given during the first regular work week after the equipment or system has been accepted and turned over to the Government for regular operation. The number of man-days (8 hours per day) of instruction furnished must be as specified in the individual section. When more than 4 man-days of instruction are specified, use approximately half of the time for classroom instruction. Use other time for instruction with the equipment or system.

When significant changes or modifications in the equipment or system are made under the terms of the contract, provide additional instruction to acquaint the operating personnel with the changes or modifications.

#### 1.7 ACCESSIBILITY

Install all work so that parts requiring periodic inspection, operation, maintenance, and repair are readily accessible. Install concealed valves, expansion joints, controls, dampers, and equipment requiring access, in locations freely accessible through access doors.

#### PART 2 PRODUCTS

Not Used

#### PART 3 EXECUTION

##### 3.1 PAINTING OF NEW EQUIPMENT

New equipment painting must be factory applied or shop applied, and must be as specified herein, and provided under each individual section.

###### 3.1.1 Factory Painting Systems

Manufacturer's standard factory painting systems may be provided subject to certification that the factory painting system applied will withstand 125 hours in a salt-spray fog test, except that equipment located outdoors must withstand 500 hours in a salt-spray fog test. Salt-spray fog test must be in accordance with ASTM B117, and for that test the acceptance criteria must be as follows: immediately after completion of the test, the paint must show no signs of blistering, wrinkling, or cracking, and no loss of adhesion; and the specimen must show no signs of rust creepage beyond 0.125 inch on either side of the scratch mark.

The film thickness of the factory painting system applied on the equipment must not be less than the film thickness used on the test specimen. If manufacturer's standard factory painting system is being proposed for use on surfaces subject to temperatures above 120 degrees F, the factory painting system must be designed for the temperature service.

###### 3.1.2 Shop Painting Systems for Metal Surfaces

Clean, pretreat, prime and paint metal surfaces; except aluminum surfaces need not be painted. Apply coatings to clean dry surfaces. Clean the

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

surfaces to remove dust, dirt, rust, oil and grease by wire brushing and solvent degreasing prior to application of paint, except metal surfaces subject to temperatures in excess of 120 degrees F must be cleaned to bare metal.

Where more than one coat of paint is specified, apply the second coat after the preceding coat is thoroughly dry. Lightly sand damaged painting and retouch before applying the succeeding coat. Color of finish coat must be aluminum or light gray.

- a. Temperatures Less Than 120 Degrees F: Immediately after cleaning, the metal surfaces subject to temperatures less than 120 degrees F must receive one coat of pretreatment primer applied to a minimum dry film thickness of 0.3 mil, one coat of primer applied to a minimum dry film thickness of 1 mil; and two coats of enamel applied to a minimum dry film thickness of 1 mil per coat.
- b. Temperatures Between 120 and 400 Degrees F: Metal surfaces subject to temperatures between 120 and 400 degrees F must receive two coats of 400 degrees F heat-resisting enamel applied to a total minimum thickness of 2 mils.
- c. Temperatures Greater Than 400 Degrees F: Metal surfaces subject to temperatures greater than 400 degrees F must receive two coats of 600 degrees F heat-resisting paint applied to a total minimum dry film thickness of 2 mils.

-- End of Section --

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## SECTION 23 05 15

### COMMON PIPING FOR HVAC 05/22

#### PART 1 GENERAL

##### 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

#### AMERICAN INSTITUTE OF STEEL CONSTRUCTION (AISC)

AISC 325 (2017) Steel Construction Manual

#### AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

|                           |                                                                                                                          |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------|
| ASME A112.18.1/CSA B125.1 | (2018) Plumbing Supply Fittings                                                                                          |
| ASME A112.19.2/CSA B45.1  | (2018; ERTA 2018) Standard for Vitreous China Plumbing Fixtures and Hydraulic Requirements for Water Closets and Urinals |
| ASME B1.20.7              | (1991; R 2013) Standard for Hose Coupling Screw Threads (Inch)                                                           |
| ASME B16.1                | (2020) Gray Iron Pipe Flanges and Flanged Fittings Classes 25, 125, and 250                                              |
| ASME B16.3                | (2021) Malleable Iron Threaded Fittings, Classes 150 and 300                                                             |
| ASME B16.5                | (2020) Pipe Flanges and Flanged Fittings NPS 1/2 Through NPS 24 Metric/Inch Standard                                     |
| ASME B16.9                | (2018) Factory-Made Wrought Buttwelding Fittings                                                                         |
| ASME B16.22               | (2021) Wrought Copper and Copper Alloy Solder Joint Pressure Fittings                                                    |
| ASME B16.25               | (2022) Buttwelding Ends                                                                                                  |
| ASME B16.34               | (2021) Valves - Flanged, Threaded and Welding End                                                                        |
| ASME B16.39               | (2020) Standard for Malleable Iron Threaded Pipe Unions; Classes 150, 250, and 300                                       |
| ASME B31.3                | (2022; Errata 2023) Process Piping                                                                                       |
| ASME B40.100              | (2022) Pressure Gauges and Gauge Attachments                                                                             |
| ASME BPVC SEC IX          | (2017; Errata 2018) BPVC Section                                                                                         |

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

# IX-Welding, Brazing and Fusing Qualifications

ASME BPVC SEC VIII D1 (2019) BPVC Section VIII-Rules for  
Construction of Pressure Vessels Division 1

## AMERICAN WELDING SOCIETY (AWS)

AWS A5.8/A5.8M (2019) Specification for Filler Metals for  
Brazing and Braze Welding

AWS WHB-2.9 (2004) Welding Handbook; Volume 2, Welding  
Processes, Part 1

## ASTM INTERNATIONAL (ASTM)

ASTM A6/A6M (2024a) Standard Specification for General  
Requirements for Rolled Structural Steel  
Bars, Plates, Shapes, and Sheet Piling

ASTM A53/A53M (2024) Standard Specification for Pipe,  
Steel, Black and Hot-Dipped, Zinc-Coated,  
Welded and Seamless

ASTM A126 (2004; R 2023) Standard Specification for  
Gray Iron Castings for Valves, Flanges,  
and Pipe Fittings

ASTM A183 (2014; R 2020) Standard Specification for  
Carbon Steel Track Bolts and Nuts

ASTM A197/A197M (2020; R 2023; E 2023) Standard  
Specification for Cupola Malleable Iron

ASTM A234/A234M (2024) Standard Specification for Piping  
Fittings of Wrought Carbon Steel and Alloy  
Steel for Moderate and High Temperature  
Service

ASTM A278/A278M (2001; R 2020) Standard Specification for  
Gray Iron Castings for Pressure-Containing  
Parts for Temperatures Up to 650 degrees F  
(350 degrees C)

ASTM A307 (2021) Standard Specification for Carbon  
Steel Bolts, Studs, and Threaded Rod 60  
000 PSI Tensile Strength

ASTM A312/A312M (2022a) Standard Specification for  
Seamless, Welded, and Heavily Cold Worked  
Austenitic Stainless Steel Pipes

ASTM A480/A480M (2023a) Standard Specification for General  
Requirements for Flat-Rolled Stainless and  
Heat-Resisting Steel Plate, Sheet, and  
Strip

ASTM A563 (2021; E 2022a) Standard Specification for  
Carbon and Alloy Steel Nuts



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

|                 |                                                                                                                                     |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------|
| ASTM B62        | (2017) Standard Specification for<br>Composition Bronze or Ounce Metal Castings                                                     |
| ASTM B88        | (2022) Standard Specification for Seamless<br>Copper Water Tube                                                                     |
| ASTM B117       | (2019) Standard Practice for Operating<br>Salt Spray (Fog) Apparatus                                                                |
| ASTM C67/C67M   | (2023a) Standard Test Methods for Sampling<br>and Testing Brick and Structural Clay Tile                                            |
| ASTM C109/C109M | (2023) Standard Test Method for<br>Compressive Strength of Hydraulic Cement<br>Mortars (Using 2-in. or (50-mm) Cube<br>Specimens)   |
| ASTM C404       | (2024) Standard Specification for<br>Aggregates for Masonry Grout                                                                   |
| ASTM C476       | (2023) Standard Specification for Grout<br>for Masonry                                                                              |
| ASTM C553       | (2013; R 2019) Standard Specification for<br>Mineral Fiber Blanket Thermal Insulation<br>for Commercial and Industrial Applications |
| ASTM C920       | (2018) Standard Specification for<br>Elastomeric Joint Sealants                                                                     |
| ASTM D2000      | (2018) Standard Classification System for<br>Rubber Products in Automotive Applications                                             |
| ASTM E1         | (2014) Standard Specification for ASTM<br>Liquid-in-Glass Thermometers                                                              |
| ASTM E814       | (2024) Standard Test Method for Fire Tests<br>of Penetration Firestop Systems                                                       |
| ASTM F104       | (2011; R 2020) Standard Classification<br>System for Nonmetallic Gasket Materials                                                   |
| ASTM F2389      | (2024) Standard Specification for<br>Pressure-rated Polypropylene (PP) Piping<br>Systems                                            |

#### FLUID SEALING ASSOCIATION (FSA)

|          |                                                                                                           |
|----------|-----------------------------------------------------------------------------------------------------------|
| FSA-0017 | (1995e6) Standard for Non-Metallic<br>Expansion Joints and Flexible Pipe<br>Connectors Technical Handbook |
|----------|-----------------------------------------------------------------------------------------------------------|

#### MANUFACTURERS STANDARDIZATION SOCIETY OF THE VALVE AND FITTINGS INDUSTRY (MSS)

|           |                                                                                                                      |
|-----------|----------------------------------------------------------------------------------------------------------------------|
| MSS SP-58 | (2018) Pipe Hangers and Supports -<br>Materials, Design and Manufacture,<br>Selection, Application, and Installation |
|-----------|----------------------------------------------------------------------------------------------------------------------|

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

|            |                                                                                      |
|------------|--------------------------------------------------------------------------------------|
| MSS SP-67  | (2022) Butterfly Valves                                                              |
| MSS SP-72  | (2010a) Ball Valves with Flanged or Butt-Welding Ends for General Service            |
| MSS SP-80  | (2019) Bronze Gate, Globe, Angle and Check Valves                                    |
| MSS SP-110 | (2010) Ball Valves Threaded, Socket-Welding, Solder Joint, Grooved and Flared Ends   |
| MSS SP-125 | (2010) Gray Iron and Ductile Iron In-Line, Spring-Loaded, Center-Guided Check Valves |

#### NSF INTERNATIONAL (NSF)

|             |                                                                |
|-------------|----------------------------------------------------------------|
| NSF/ANSI 14 | (2023) Plastics Piping System Components and Related Materials |
|-------------|----------------------------------------------------------------|

#### U.S. GENERAL SERVICES ADMINISTRATION (GSA)

|               |                                                                                                                             |
|---------------|-----------------------------------------------------------------------------------------------------------------------------|
| CID A-A-1922  | (Rev A; Notice 1; Notice 2; Notice 3; Notice 4) Shield, Expansion (Caulking Anchors, Single Lead)                           |
| CID A-A-1923  | (Rev A; Notice 1; Notice 2; Notice 3; Notice 4) Shield, Expansion (Lag, Machine and Externally Threaded Wedge Bolt Anchors) |
| CID A-A-1924  | (Rev A; Notice 3) Shield, Expansion (Self Drilling Tubular Expansion Shell Bolt Anchors)                                    |
| CID A-A-1925  | (Rev A; Notice 1; Notice 2; Notice 3; Notice 4) Shield Expansion (Nail Anchors)                                             |
| CID A-A-55614 | (Basic; Notice 2) Shield, Expansion (Non-Drilling Expansion Anchors)                                                        |
| CID A-A-55615 | (Basic; Notice 1; Notice 2; Notice 3; Notice 4) Shield, Expansion (Wood Screw and Lag Bolt Self-Threading Anchors)          |

#### UNDERWRITERS LABORATORIES (UL)

|         |                                                                      |
|---------|----------------------------------------------------------------------|
| UL 1479 | (2015; Reprint Apr 2024) Fire Tests of Through-Penetration Firestops |
|---------|----------------------------------------------------------------------|

### 1.2 GENERAL REQUIREMENTS

- a. Section 23 30 00 HVAC AIR DISTRIBUTION applies to work specified in this section
- d. Submit Records of Existing Conditions consisting of the results of Contractor's survey of work area conditions and features of existing structures and facilities within and adjacent to the jobsite. Commencement of work constitutes acceptance of the existing conditions.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

- e. Include with Equipment Foundation Data for piping systems all plan dimensions of foundations and relative elevations, equipment weight and operating loads, horizontal and vertical loads, horizontal and vertical clearances for installation, and size and location of anchor bolts.
- f. Submit Fabrication Drawings for pipes, valves and specialties consisting of fabrication and assembly details to be performed in the factory.
- g. Submit Material, Equipment, and Fixture Lists for pipes, valves and specialties including manufacturer's style or catalog numbers, specification and drawing reference numbers, warranty information, and fabrication site information. Provide a complete list of construction equipment to be used.
- h. Submit Manufacturer's Standard Color Charts for pipes, valves and specialties showing the manufacturer's recommended color and finish selections.
- i. Include with Listing of Product Installations for piping systems identification of at least 5 units, similar to those proposed for use, that have been in successful service for a minimum period of 5 years. Include in the list purchaser, address of installation, service organization, and date of installation.
- j. Submit Record Drawings for pipes, valves and accessories providing current factual information including deviations and amendments to the drawings, and concealed and visible changes in the work.
- k. Submit Connection Diagrams for pipes, valves and specialties indicating the relations and connections of devices and apparatus by showing the general physical layout of all controls, the interconnection of one system (or portion of system) with another, and internal tubing, wiring, and other devices.
- l. Submit Coordination Drawings for pipes, valves and specialties showing coordination of work between different trades and with the structural and architectural elements of work. Detail all drawings sufficiently to show overall dimensions of related items, clearances, and relative locations of work in allotted spaces. Indicate on drawings where conflicts or clearance problems exist between various trades.

### 1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

#### SD-01 Preconstruction Submittals

Material, Equipment, and Fixture Lists; G

#### SD-02 Shop Drawings

Record Drawings; G

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

Connection Diagrams; G

Coordination Drawings; G

Fabrication Drawings; G

Installation Drawings; G

SD-03 Product Data

Pipe and Fittings; G

Piping Specialties; G

Valves; G

Miscellaneous Materials; G

Supporting Elements; G

Equipment Foundation Data; G

SD-04 Samples

Manufacturer's Standard Color Charts; G

SD-05 Design Data

Pipe and Fittings; G

Piping Specialties; G

Valves; G

SD-06 Test Reports

Hydrostatic Tests; G

Air Tests; G

Valve-Operating Tests; G

Drainage Tests; G

Pneumatic Tests; G

Non-Destructive Electric Tests; G

System Operation Tests; G

SD-07 Certificates

Record of Satisfactory Field Operation; FIO

List of Qualified Permanent Service Organizations; FIO

Listing of Product Installations; FIO

Records of Existing Conditions; FIO

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

Surface Resistance; FIO

Shear and Tensile Strengths; FIO

Temperature Ratings; FIO

Bending Tests; FIO

Flattening Tests; FIO

Transverse Guided Weld Bend Tests; FIO

SD-10 Operation and Maintenance Data

Operation and Maintenance Manuals; G

#### 1.4 QUALITY ASSURANCE

##### 1.4.1 Material and Equipment Qualifications

Provide materials and equipment that are standard products of manufacturers regularly engaged in the manufacture of such products, which are of a similar material, design and workmanship. Provide standard products in satisfactory commercial or industrial use for 2 years prior to bid opening. The 2-year use includes applications of equipment and materials under similar circumstances and of similar size. Ensure the product has been for sale on the commercial market through advertisements, manufacturers' catalogs, or brochures during the 2 year period.

##### 1.4.2 Alternative Qualifications

Products having less than a two-year field service record are acceptable if a certified record of satisfactory field operation for not less than 6000 hours, exclusive of the manufacturer's factory or laboratory tests, can be shown.

##### 1.4.3 Service Support

Ensure the equipment items are supported by service organizations. Submit a certified list of qualified permanent service organizations for support of the equipment which includes their addresses and qualifications. Select service organizations that are reasonably convenient to the equipment installation and able to render satisfactory service to the equipment on a regular and emergency basis during the warranty period of the contract.

##### 1.4.4 Manufacturer's Nameplate

Provide a nameplate on each item of equipment bearing the manufacturer's name, address, model number, and serial number securely affixed in a conspicuous place; the nameplate of the distributing agent is not acceptable.

##### 1.4.5 Modification of References

In each of the publications referred to herein, consider the advisory provisions to be mandatory, as though the word, "shall" had been substituted for "should" wherever it appears. Interpret references in

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

these publications to the "authority having jurisdiction," or words of similar meaning, to mean the Contracting Officer.

#### 1.4.5.1 Definitions

For the International Code Council (ICC) Codes referenced in the contract documents, advisory provisions are considered mandatory, the word "should" is interpreted as "shall." Reference to the "code official" is interpreted to mean the "Contracting Officer." For Navy owned property, interpret references to the "owner" to mean the "Contracting Officer." For leased facilities, references to the "owner" is interpreted to mean the "lessor." References to the "permit holder" are interpreted to mean the "Contractor."

#### 1.4.5.2 Administrative Interpretations

For ICC Codes referenced in the contract documents, the provisions of Chapter 1, "Administrator," do not apply. These administrative requirements are covered by the applicable Federal Acquisition Regulations (FAR) included in this contract and by the authority granted to the Officer in Charge of Construction to administer the construction of this project. References in the ICC Codes to sections of Chapter 1, are applied as appropriate by the Contracting Officer and as authorized by his administrative cognizance and the FAR.

### 1.5 DELIVERY, STORAGE, AND HANDLING

Handle, store, and protect equipment and materials to prevent damage before and during installation in accordance with the manufacturer's recommendations, and as approved by the Contracting Officer. Replace damaged or defective items.

### 1.6 ELECTRICAL REQUIREMENTS

Furnish motors, controllers, disconnects and contactors with their respective pieces of equipment. Ensure motors, controllers, disconnects and contactors conform to and have electrical connections provided under Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Furnish internal wiring for components of packaged equipment as an integral part of the equipment. Extended voltage range motors is not permitted. Provide controllers and contactors with a maximum of 120 volt control circuits, and auxiliary contacts for use with the controls furnished. When motors and equipment furnished are larger than sizes indicated, include the cost of additional electrical service and related work under the section that specified that motor or equipment. Provide power wiring and conduit for field installed equipment under and conform to the requirements of Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM.

### 1.7 INSTRUCTION TO GOVERNMENT PERSONNEL

When specified in other sections, furnish the services of competent instructors to give full instruction to the designated Government personnel in the adjustment, operation, and maintenance, including pertinent safety requirements, of the specified equipment or system. Provide instructors thoroughly familiar with all parts of the installation and trained in operating theory as well as practical operation and maintenance work.

Give instruction during the first regular work week after the equipment or

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

system has been accepted and turned over to the Government for regular operation. The number of man-days (8 hours per day) of instruction furnished is as specified in the individual section. When more than four man-days of instruction are specified, use approximately half of the time for classroom instruction. Use other time for instruction with the equipment or system.

When significant changes or modifications in the equipment or system are made under the terms of the contract, provide additional instruction to acquaint the operating personnel with the changes or modifications.

## 1.8 ACCESSIBILITY

Install all work so that parts requiring periodic inspection, operation, maintenance, and repair are readily accessible. Install concealed valves, expansion joints, controls, dampers, and equipment requiring access, in locations freely accessible through access doors.

## PART 2 PRODUCTS

### 2.1 PIPE AND FITTINGS

Submit equipment and performance data for pipe and fittings consisting of corrosion resistance, life expectancy, gage tolerances, and grade line analysis. Submit design analysis and calculations consisting of surface resistance, rates of flow, head losses, inlet and outlet design, required radius of bend, and pressure calculations. Also include in data pipe size, shape, and dimensions, as well as temperature ratings, vibration and thrust limitations minimum burst pressures, shut-off and non-shock pressures and weld characteristics.

#### 2.1.1 Type BCS, Black Carbon Steel

Ensure pipe 1/8 through 12 inches is Schedule 40 black carbon steel, conforming to ASTM A53/A53M.

Ensure pipe 1/8 through 10 inches is Schedule 40 electric-resistance welded black carbon steel, conforming to ASTM A53/A53M, Type E, Grade B (electric-resistance welded). Grade A should be used for permissible field bending, in both cases.

Ensure pipe 12 through 24 inches is 0.375-inch wall seamless black carbon steel, conforming to ASTM A53/A53M, Type E, Grade B (electric-resistance welded).

Ensure fittings 2 inches and under are 150-pounds per square inch, gage (psig) working steam pressure (wsp) banded black malleable iron screwed, conforming to ASTM A197/A197M and ASME B16.3.

Ensure unions 2 inches and under are 250 pounds per square inch, wsp female, screwed, black malleable iron with brass-to-iron seat, and ground joint, conforming to ASME B16.39.

Ensure flanges 2-1/2 inches and over are 150-pound forged-steel conforming to ASME B16.5, welding neck to match pipe wall thickness.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 2.1.2 Type CPR, Copper

#### 2.1.2.1 Type CPR-A, Copper Above Ground

Ensure tubing 2 inches and under is seamless copper tubing, conforming to ASTM B88, Type L (hard-drawn for all horizontal and all exposed vertical lines, annealed for concealed vertical lines).

Ensure fittings 2 inches and under are 150-psig wsp wrought-copper solder joint fittings conforming to ASME B16.22.

Ensure unions 2 inches and under are 150-psig wsp wrought-copper solder joint, conforming to ASME B16.22.

Provide brazing rod with Classification BCuP-5, conforming to AWS A5.8/A5.8M.

#### 2.1.2.2 PEX-a, Radiant Slab Heating

1/2 inch to 1 inch seamless PEX-a tubing NSF/ANSI 14. Designed for hot water applications and rated up 140 deg F at 80 psi in accordance with ASTM F2023

### 2.1.3 Grooved Pipe Couplings and Fittings

Provide housing for all couplings, fabricated in two or more parts, of black, ungalvanized malleable iron castings. Ensure coupling gasket is molded synthetic rubber, conforming to ASTM D2000. Ensure coupling bolts are oval-neck, track-head type, with hexagonal heavy nuts conforming to ASTM A183.

Fabricate all pipe fittings used with couplings of black, ungalvanized malleable iron castings. Where a manufacturer's standard-size malleable iron fitting pattern is not available, approved fabricated fittings may be used.

Fabricate fittings from Schedule 40 or 0.75-inch wall ASTM A53/A53M, Grade B seamless steel pipe; long radius seamless welding fittings with wall thickness to match pipe, conforming to ASTM A234/A234M and ASME B16.9.

## 2.2 PIPING SPECIALTIES

Submit equipment and performance data for piping specialties consisting of corrosion resistance, life expectancy, gage tolerances, and grade line analysis. Submit design analysis and calculations consisting of surface resistance, rates of flow, head losses, inlet and outlet design, required radius of bend, and pressure calculations. Also include in data pipe size, shape, and dimensions, as well as temperature ratings, vibration and thrust limitations minimum burst pressures, shut-off and non-shock pressures and weld characteristics.

### 2.2.1 Air Separator

Air separated from converter discharge water is ejected by a reduced-velocity device vented to the compression tank.

Provide a commercially constructed separator, designed and certified to



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

separate not less than 80 percent of entrained air on the first passage of water and not less than 80 percent of residual on each successive pass. Provide shop drawings detailing all piping connections proposed for this work.

Ensure the air separator is carbon steel, designed, fabricated, tested, and stamped in conformance with ASME BPVC SEC VIII D1 for service pressures not less than 125 psi.

#### 2.2.2 Air Vents

Provide manual air vents using 3/8-inch globe valves.

Provide automatic air vents on pumps, mains, and where indicated using ball-float construction. Ensure the vent inlet is not less than 3/4-inch ips and the outlet not less than 1/4-inch ips. Orifice size is 1/8 inch. Provide corrosion-resistant steel trim conforming to ASTM A480/A480M. Fit vent with try-cock. Ensure vent discharges air at any pressure up to 150 psi. Ensure outlet is copper tube routed.

#### 2.2.3 Compression Tank

Provide compression tank designed, fabricated, tested, and stamped for a working pressure of not less than 125 psi in accordance with ASME BPVC SEC VIII D1. Ensure tank is hot-dip galvanized after fabrication to produce not less than 1.5 ounces of zinc coating per square foot of single-side surface.

Tank accessories include red-lined gage-glass complete with glass protectors and shutoff valves, air charger and drainer, and manual vent.

#### 2.2.4 Dielectric Connections

Electrically isolate dissimilar pipe metals from each other by couplings, unions, or flanges commercially manufactured for that purpose and rated for the service pressure and temperature.

#### 2.2.5 Expansion Vibration Isolation Joints

Construct single or multiple arch-flanged expansion vibration isolation joints of steel-ring reinforced chloroprene-impregnated cloth materials. Design joint to absorb the movement of the pipe sections in which installed with no detrimental effect on the pipe or connected equipment. Back flanges with ferrous-metal backing rings. Provide control rod assemblies to restrict joint movement. Coat all nonmetallic exterior surfaces of the joint with chlorosulphonated polyethylene. Provide grommets in limit bolt hole to absorb noise transmitted through the bolts.

Ensure joints are suitable for continuous-duty working temperature of at least 250 degrees F.

Fill arches with soft chloroprene.

Ensure joint, single-arch, movement limitations and size-related, pressure characteristics conform to FSA-0017.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### 2.2.6 Flexible Metallic Pipe

Ensure flexible pipe is the bellows-type with wire braid cover and designed, constructed, and rated in accordance with the applicable requirements of ASME B31.3.

Minimum working pressure rating is 100 psi at 300 degrees F.

Ensure minimum burst pressure is four times working pressure at 300 degrees F. Bellows material is AISI Type 316L corrosion-resistant steel. Ensure braid is AISI 300 series corrosion-resistant steel wire.

Provide threaded end connections; hex-collared Schedule 40, AISI Type 316L corrosion-resistant steel, conforming to ASTM A312/A312M.

Ensure flanged end connection rating and materials conform to specifications for system primary-pressure rating.

#### 2.2.7 Hose Faucets

Construct hose faucets with 1/2 inch male inlet threads, hexagon shoulder, and 3/4 inch hose connection, conforming to ASME A112.18.1/CSA B125.1. Ensure hose-coupling screw threads conform to ASME B1.20.7.

#### 2.2.8 Pressure Gages

Ensure pressure gages conform to ASME B40.100 and to requirements specified herein. Pressure-gage size is 3-1/2 inches nominal diameter. Ensure case is corrosion-resistant steel, conforming to any of the AISI 300 series of ASTM A6/A6M, with an ASM No. 4 standard commercial polish or better. Equip gages with adjustable red marking pointer and damper-screw adjustment in inlet connection. Align service-pressure reading at midpoint of gage range. Ensure all gages are Grade B or better and be equipped with gage isolators.

#### 2.2.9 Thermometers

Ensure thermometers conform to ASTM E1, except for being filled with a red organic liquid. Provide an industrial pattern armored glass thermometer, (well-threaded and seal-welded). Ensure thermometers installed 6 feet or higher above the floor have an adjustable angle body. Ensure scale is not less than 7 inches long and the case face is manufactured from manufacturer's standard polished aluminum or AISI 300 series polished corrosion-resistant steel. Thermometer range is 20 - 160 deg F. Provide thermometers with nonferrous separable wells. Provide lagging extension to accommodate insulation thickness.

#### 2.2.10 Pump Suction Strainers

Provide a cast iron strainer body, rated for not less than 25 psig at 100 degrees F, with flanges conforming to ASME B16.1, Class 125. Strainer construction is such that there is a machined surface joint between body and basket that is normal to the centerline of the basket.

Ensure minimum ratio of open area of each basket to pipe area is 3 to 1. Provide a basket with AISI 300 series corrosion-resistant steel wire mesh with perforated backing.

Ensure mesh is capable of retaining all particles larger than 1,000

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

micrometer, with a pressure drop across the strainer body of not more than 0.5 psi when the basket is two-thirds dirty at maximum system flow rate. Provide reducing fittings from strainer-flange size to pipe size.

Provide a differential-pressure gage fitted with a two-way brass cock across the strainer.

Provide manual air vent cocks in cap of each strainer.

#### 2.2.11 Line Strainers, Water Service

Provide Y-type strainers with removable basket. Ensure strainers in sizes 2-inch ips and smaller have screwed ends; in sizes 2-1/2-inch ips and larger, strainers have flanged ends. Ensure body working-pressure rating exceeds maximum service pressure of installed system by at least 50 percent. Ensure body has cast-in arrows to indicate direction of flow. Ensure all strainer bodies fitted with screwed screen retainers have straight threads and gasketed with nonferrous metal. For strainer bodies 2-1/2-inches and larger, fitted with bolted-on screen retainers, provide offset blowdown holes. Fit all strainers larger than 2-1/2-inches with manufacturer's standard ball-type blowdown valve. Ensure body material is cast iron conforming to Class 30 ASTM A278/A278M. Where system material is nonferrous, use nonferrous metal for the metal strainer body material.

Ensure minimum free-hole area of strainer element is equal to not less than 3.4 times the internal area of connecting piping. Strainer screens perforation size is not to exceed 0.045-inch. Ensure strainer screens have finished ends fitted to machined screen chamber surfaces to preclude bypass flow. Strainer element material is AISI Type 316 corrosion-resistant steel.

### 2.3 VALVES

Submit equipment and performance data for valves consisting of corrosion resistance and life expectancy. Submit design analysis and calculations consisting of rates of flow, head losses, inlet and outlet design, and pressure calculations. Also include in data, pipe dimensions, as well as temperature ratings, vibration and thrust limitations, minimum burst pressures, shut-off and non-shock pressures and weld characteristics.

Polypropylene valves will comply with the performance requirements of ASTM F2389. Valves shall conform to ASME B16.34.

#### 2.3.1 Ball and Butterfly Valves

Ensure ball valves conform to MSS SP-72 for flanged valves and MSS SP-110 for screwed-end valves for Figure 1A 1 piece body, 1B vertically split body, 1C top entry 1D three piece body and are rated for service at not less than 175 psig at 200 degrees F. For valve bodies in sizes 2 inches and smaller, use screwed-end connection-type constructed of Class A copper alloy. For valve bodies in sizes 2-1/2 inches and larger, use flanged-end connection type, constructed of Class D material. Balls and stems of valves 2 inches and smaller are manufacturer's standard with hard chrome plating finish. Balls and stems of valves 2-1/2 inches and larger are manufacturer's standard Class C corrosion-resistant steel alloy with hard chrome plating. Balls of valves 6 inches and larger may be Class D with 900 Brinell hard chrome plating. Ensure valves are suitable for flow from either direction and seal equally tight in either direction. Valves with ball seals held in place by spring washers are not acceptable. Ensure all

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

valves have adjustable packing glands. Seats and seals are fabricated from tetrafluoroethylene.

Ensure butterfly valves conform to MSS SP-67 and are the wafer type for mounting between specified flanges. Ensure valves are rated for 150-psig shutoff and nonshock working pressure. Select bodies of cast ferrous metal conforming to ASTM A126, Class B, and to ASME B16.1 for body wall thickness. Seats and seals are fabricated from resilient elastomer designed for field removal and replacement.

#### 2.3.2 Drain, Vent, and Gage Cocks

Provide lever handle drain, vent, and gage cocks, ground key type, with washer and screw, constructed of polished ASTM B62 bronze, and rated 125-psi wsp. Ensure end connections are rated for specified service pressure.

Ensure pump vent cocks, and where spray control is required, are UL umbrella-hood type, constructed of manufacturer's standard polished brass. Ensure cocks are 1/2-inch ips male, end threaded, and rated at not less than 125 psi at 225 degrees F.

#### 2.3.3 Globe and Angle Valves (GLV-ANV)

Ensure globe and angle valves 2 inches and smaller, are 125-pound, 125-psi conforming to MSS SP-80 and to requirements specified herein. For valves located in tunnels, equipment rooms, factory-assembled equipment, and where indicated, use union-ring bonnet, screwed-end type. Ensure disc is free to swivel on the stem in all valve sizes. Composition seating-surface disc construction may be substituted for all metal-disc construction. Make packing of non-asbestos type materials. Ensure disk and packing are suitable for pipe service installed.

Ensure globe and angle valves, 2-1/2 inches and larger, are cast iron with bronze trim. Ensure valve bodies are cast iron conforming to ASTM A126, Class A, as specified for Class 1 valves under MSS SP-80. Select flanged valves in conformance with ASME B16.1. Valve construction is outside screw and yoke (OS&Y) type. Make packing of non-asbestos type materials.

#### 2.3.4 Nonslam Check Valves (NSV)

Provide check valves at pump discharges in sizes 2 inches and larger with nonslam or silent-check operation conforming to MSS SP-125. Select a valve disc or plate that closes before line flow can reverse to eliminate slam and water-hammer due to check-valve closure. Ensure valve is Class 125 rated for 200-psi maximum, nonshock pressure at 150 degrees F in sizes to 12 inches. Use valves that are wafer type to fit between flanges conforming to ASME B16.1. Valve body may be cast iron, or equivalent strength ductile iron. Select disks using manufacturer's standard bronze, aluminum bronze, or corrosion-resistant steel. Ensure pins, springs, and miscellaneous trim are manufacturer's standard corrosion-resistant steel. Disk and shaft seals are Buna-N elastomer tetrafluoroethylene.

#### 2.4 MISCELLANEOUS MATERIALS

Submit equipment and performance data for miscellaneous materials consisting of corrosion resistance, life expectancy, gage tolerances, and grade line analysis.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### 2.4.1 Bolting

Ensure flange and general purpose bolting is hex-head and conforms to ASTM A307, Grade B (bolts, for flanged joints in piping systems where one or both flanges are cast iron). Heavy hex-nuts conform to ASTM A563. Square-head bolts and nuts are not acceptable. Ensure threads are coarse-thread series.

#### 2.4.2 Elastomer Caulk

Use two-component polysulfide- or polyurethane-base elastomer caulking material, conforming to ASTM C920.

#### 2.4.3 Escutcheons

Manufacture escutcheons from nonferrous metals and chrome-plated except when AISI 300 series corrosion-resistant steel is provided. Ensure metals and finish conforms to ASME A112.19.2/CSA B45.1.

Use one-piece escutcheons where mounted on chrome-plated pipe or tubing, and one-piece of split-pattern type elsewhere. Ensure all escutcheons have provisions consisting of setscrews for maintaining a fixed position against a surface.

#### 2.4.4 Flange Gaskets

Provide compressed non-asbestos sheets, conforming to ASTM F104, coated on both sides with graphite or similar lubricant, with nitrile composition, binder rated to 750 degrees F.

#### 2.4.5 Grout

Provide shrink-resistant grout as a premixed and packaged metallic-aggregate, mortar-grouting compound conforming to ASTM C404 and ASTM C476.

Ensure shrink-resistant grout is a combination of pre-measured and packaged epoxy polyamide or amine resins and selected aggregate mortar grouting compound conforming to the following requirements:

|                      |                 |                                           |
|----------------------|-----------------|-------------------------------------------|
| Tensile strength     |                 | 1,900 psi, minimum                        |
| Compressive strength | ASTM C109/C109M | 14,000 psi, minimum                       |
| Shrinkage, linear    |                 | 0.00012 inch per inch, maximum            |
| Water absorption     | ASTM C67/C67M   | 0.1 percent, maximum                      |
| Bond strength to     |                 | 1,000 psi, minimum steel in shear minimum |

#### 2.4.6 Pipe Thread Compounds

Use polytetrafluoroethylene tape not less than 2 to 3 mils thick in potable and process water and in chemical systems for pipe sizes to and including 1-inch ips. Use polytetrafluoroethylene dispersions and other suitable compounds for all other applications upon approval by the Contracting Officer; however, do not use lead-containing compounds in potable water systems.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## 2.5 SUPPORTING ELEMENTS

Submit equipment and performance data for the supporting elements consisting of corrosion resistance, life expectancy, gage tolerances, and grade line analysis.

Provide all necessary piping systems and equipment supporting elements, including but not limited to: building structure attachments; supplementary steel; hanger rods, stanchions, and fixtures; vertical pipe attachments; horizontal pipe attachments; anchors; guides; and spring-cushion, variable, or constant supports. Ensure supporting elements are suitable for stresses imposed by systems pressures and temperatures and natural and other external forces normal to this facility without damage to supporting element system or to work being supported.

Ensure supporting elements conform to requirements of ASME B31.3, and MSS SP-58, except as noted.

Ensure attachments welded to pipe are made of materials identical to that of pipe or materials accepted as permissible raw materials by referenced code or standard specification.

Ensure supporting elements exposed to weather are hot-dip galvanized or stainless steel. Select materials of such a nature that their apparent and latent-strength characteristics are not reduced due to galvanizing process. Electroplate supporting elements in contact with copper tubing with copper.

Type designations specified herein are based on MSS SP-58. Ensure masonry anchor group-, type-, and style-combination designations are in accordance with CID A-A-1922, CID A-A-1923, CID A-A-1924, CID A-A-1925, CID A-A-55614, and CID A-A-55615. Provide support elements, except for supplementary steel, that are cataloged, load rated, commercially manufactured products.

### 2.5.1 Building Structure Attachments

#### 2.5.1.1 Anchor Devices, Concrete and Masonry

Ensure anchor devices conform to CID A-A-1922, CID A-A-1923, CID A-A-1924, CID A-A-1925, CID A-A-55614, and CID A-A-55615

For cast-in, floor mounted, equipment anchor devices, provide adjustable positions.

Provide built-in masonry anchor devices.

Do not use powder-actuated anchoring devices to support any mechanical systems components.

#### 2.5.1.2 Beam Clamps

Ensure beam clamps are center-loading MSS SP-58 Type 20.

When it is not possible to use center-loading beam clamps, eccentric-loading beam clamps, MSS SP-58 Type 19 may be used for piping sizes 2 inches and less and for piping sizes 2 through 10 inches provided two counterbalancing clamps are used per point of pipe support. Where more than one rod is used per point of pipe support, determine rod

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

diameter in accordance with referenced standards.

#### 2.5.1.3 C-Clamps

Do not use C-clamps.

#### 2.5.1.4 Inserts, Concrete

Use concrete MSS SP-58 Type 18 inserts. When applied to piping in sizes 2 inches ips and larger and where otherwise required by imposed loads, insert and wire a 1-foot length of 1/2-inch reinforcing rod through wing slots. Submit proprietary-type continuous inserts for approval.

### 2.5.2 Horizontal Pipe Attachments

#### 2.5.2.1 Single Pipes

Support piping in sizes to and including 2-inch ips by MSS SP-58 Type 6 solid malleable iron pipe rings, except that, use split-band-type rings in sizes up to 1-inch ips.

Support piping in sizes through 8-inch ips inclusive by MSS SP-58 Type 1 attachments.

Use MSS SP-58 Type 1 and Type 6 assemblies on vapor-sealed insulated piping and have an inside diameter larger than pipe being supported to provide adequate clearance during pipe movement.

Where thermal movement of a point in a piping system 4 inches and larger would cause a hanger rod to deflect more than 4 degrees from the vertical or where a horizontal point movement exceeds 1/2 inch, use MSS SP-58 Type 44 through 46 pipe rolls.

Use MSS SP-58 Type 40 shields on all insulated piping. Ensure area of the supporting surface is such that compression deformation of insulated surfaces does not occur. Roll away longitudinal and transverse shield edges from the insulation.

Provide insulated piping without vapor barrier on roll supports with MSS SP-58 Type 39 saddles.

Provide spring supports as indicated.

#### 2.5.2.2 Parallel Pipes

Use trapeze hangers fabricated from structural steel shapes, with U-bolts, in congested areas and where multiple pipe runs occur. Ensure structural steel shapes be of commercially available, proprietary design, rolled steel.

### 2.5.3 Vertical Pipe Attachments

Ensure vertical pipe attachments are MSS SP-58 Type 8.

Include complete fabrication and attachment details of any spring supports in shop drawings.

#### 2.5.4 Hanger Rods and Fixtures

Use only circular cross section rod hangers to connect building structure

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

attachments to pipe support devices. Use pipe, straps, or bars of equivalent strength for hangers only where approved by the Contracting Officer.

Provide turnbuckles, swing eyes, and clevises as required by support system to accommodate temperature change, pipe accessibility, and adjustment for load and pitch. Rod couplings are not acceptable.

#### 2.5.5 Supplementary Steel

Where it is necessary to frame structural members between existing members or where structural members are used in lieu of commercially rated supports, design and fabricate such supplementary steel in accordance with AISC 325.

### PART 3 EXECUTION

#### 3.1 PIPE INSTALLATION

Submit certificates for pipes, valves and specialties showing conformance with test requirements as contained in the reference standards contained in this section. Provide certificates verifying Surface Resistance, Shear and Tensile Strengths, Temperature Ratings, Bending Tests, Flattening Tests and Transverse Guided Weld Bend Tests.

Provide test reports for Hydrostatic Tests, Air Tests, Valve-Operating Tests, Drainage Tests, Pneumatic Tests, Non-Destructive Electric Tests and System Operation Tests, in compliance with referenced standards contained within this section.

Fabricate and install piping systems in accordance with ASME B31.3, MSS SP-58, and AWS WHB-2.9.

Submit Installation Drawings for pipes, valves and specialties. Drawings include the manufacturer's design and construction calculations, forces required to obtain rated axial, lateral, or angular movements, installation criteria, anchor and guide requirements for equipment, and equipment room layout and design. Ensure drawings specifically advise on procedures to be followed and provisions required to protect expansion joints during specified hydrostatic testing operations.

Ensure connections between steel piping and copper piping are electrically isolated from each other. Dielectric pipe unions shall be installed to prevent galvanic corrosion. The dielectric unions shall have metal connections on both ends. The ends shall be threaded, flanged, or brazed to match adjacent piping. The metal parts of the union shall be separated so that the electrical current is below one percent of the galvanic current which would exist upon metal-to-metal contact. Gaskets, flanges, and unions shall be installed in accordance with manufacturer's recommendations.

Make final connections to equipment with unions or flanges installed every 100 feet of straight run. Install unions in the line downstream of screwed- and welded-end valves.

Ream all pipe ends before joint connections are made.

Make screwed joints with specified joint compound with not more than three



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

threads showing after joint is made up.

Apply joint compounds to the male thread only and exercise care to prevent compound from reaching the unthreaded interior of the pipe.

Install screwed unions, welded unions, or bolted flanges wherever required to permit convenient removal of equipment, valves, and piping accessories from the piping system for maintenance.

Securely support piping systems with due allowance for thrust forces, thermal expansion and contraction. Do not subject the system to mechanical, chemical, vibrational or other damage as specified in ASME B31.3.

Ensure field welded joints conform to the requirements of the AWS WHB-2.9, ASME B31.3, and ASME BPVC SEC IX.

Make piping systems butt weld joints with backing rings. Use compatible backing ring materials with materials being joined. Ensure joint configuration conforms to ASME B16.25.

Accomplish preheat and postheat treatment of welds in accordance with ASME BPVC SEC IX and ASME B31.3.

### 3.2 VALVES

Install valves in piping mains and all branches and at equipment where indicated and as specified.

Install valves to permit isolation of branch piping and each equipment item from the balance of the system.

Install riser and downcomer drains above piping shutoff valves in piping 2-1/2 inches and larger. Tap and fit shutoff valve body with a 1/2-inch plugged globe valve.

Install valves unavoidably located in furred or other normally inaccessible places with access panels adequately sized for the location and located so that concealed items may be serviced, maintained, or replaced.

### 3.3 SUPPORTING ELEMENTS INSTALLATION

Install supporting elements in accordance with the referenced codes and standards.

Support piping from building structure. Do not support piping from roof deck or from other pipe.

Run piping parallel with the lines of the building. Space and install piping and components so that a threaded pipe fitting may be removed between adjacent pipes and so that there is no less than 1/2 inch of clear space between the finished surface and other work and between the finished surface of parallel adjacent piping. Arrange hangars on different adjacent service lines running parallel with each other in line with each other and parallel to the lines of the building.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

Install piping support elements at intervals specified hereinafter, at locations not more than 3 feet from the ends of each runout, and not over 1 foot from each change in direction of piping.

Base load rating for all pipe-hanger supports on insulated weight of lines filled with water and forces imposed. Deflection per span is not exceed slope gradient of pipe. Ensure supports are in accordance with the following minimum rod size and maximum allowable hanger spacing for specified pipe. For concentrated loads such as valves, reduce the allowable span proportionately:

| <u>PIPE SIZE</u><br><u>INCHES</u> | <u>ROD SIZE</u><br><u>INCHES</u> | <u>STEEL PIPE</u><br><u>FEET</u> | <u>COPPER PIPE</u><br><u>FEET</u> |
|-----------------------------------|----------------------------------|----------------------------------|-----------------------------------|
| 1 and smaller                     | 3/8                              | 8                                | 6                                 |
| 1-1/4 to 1-1/2                    | 3/8                              | 10                               | 8                                 |
| 2                                 | 3/8                              | 10                               | 8                                 |
| 2-1/2 to 3-1/2                    | 1/2                              | 12                               | 12                                |
| 4 to 5                            | 5/8                              | 16                               | 14                                |
| 6                                 | 3/4                              | 16                               | 16                                |
| 8 to 12                           | 7/8                              | 20                               | 20                                |
| 14 to 18                          | 1                                | 20                               | 20                                |
| 20 and over                       | 1-1/4                            | 20                               | 20                                |

Install vibration isolation supports where needed. Refer to Section 23 05 48.00 40 VIBRATION AND SEISMIC CONTROLS FOR HVAC PIPING AND EQUIPMENT where A/C equipment and piping is installed.

Support vertical risers independently of connected horizontal piping, whenever practicable, with fixed or spring supports at the base and at intervals to accommodate system range of thermal conditions. Ensure risers have guides for lateral stability. For risers subject to expansion, install only one rigid support at a point approximately one-third down from the top. Place clamps under fittings unless otherwise specified. Support carbon-steel pipe at each floor and at not more than 15-foot intervals for pipe 2 inches and smaller and at not more than 20-foot intervals for pipe 2-1/2 inches and larger.

### 3.4 PENETRATIONS

Install effective sound stopping and adequate operating clearance to prevent structure contact where piping penetrates walls, floors, or ceilings into occupied spaces adjacent to equipment rooms; where similar penetrations occur between occupied spaces; and where penetrations occur from pipe chases into occupied spaces. Occupied spaces include space above ceilings where no special acoustic treatment of ceiling is provided. Finish penetrations to be compatible with surface being penetrated.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 3.5 SLEEVES

Install sleeves where piping passes through roofs, masonry, concrete walls and floors.

Continuously weld sleeves passing through steel decks to the deck.

Ensure sleeves that extend through floors, roofs, load bearing walls, and fire barriers are continuous and fabricated from Schedule 40 steel pipe, with welded anchor lugs. Form all other sleeves by molded linear polyethylene liners or similar materials that are removable. Ensure diameter of sleeves is large enough to accommodate pipe, insulation, and jacketing without touching the sleeve and provides a minimum 3/8-inch clearance. Install a sleeve size to accommodate mechanical and thermal motion of pipe precluding transmission of vibration to walls and the generation of noise.

Pack the space between a pipe, bare or insulated, and the inside of a pipe sleeve or a construction surface penetration solid with a mineral fiber conforming to ASTM C553 Type V (flexible blanket), to 1,000 degrees F. Install this packing wherever the piping passes through firewalls, equipment room walls, floors, and ceilings connected to occupied spaces, and other locations where sleeves or construction-surface penetrations occur between occupied spaces. Where sleeves or construction surface penetrations occur between conditioned and unconditioned spaces, fill the space between a pipe, bare or insulated, and the inside of a pipe sleeve or construction surface penetration with an elastomer caulk to a depth of 1/2 inch. Ensure all caulked surfaces are oil- and grease-free.

Ensure through-penetration fire stop materials and methods are in accordance with ASTM E814 and UL 1479.

Caulk exterior wall sleeves watertight with lead and oakum or mechanically expandable chloroprene inserts with mastic-sealed metal components.

Ensure sleeve height above roof surface is a minimum of 12 and a maximum of 18-inches.

### 3.6 ESCUTCHEONS

Install escutcheons at all penetrations of piping into finished areas. Where finished areas are separated by partitions through which piping passes, install escutcheons on both sides of the partition. Where suspended ceilings are installed, install plates at the underside only of such ceilings. For insulated pipes, select plates large enough to fit around the insulation. Use chrome-plated escutcheons in all occupied spaces and of size sufficient to effectively conceal openings in building construction. Firmly attach escutcheons with setscrews.

### 3.7 FLASHINGS

Install flashings at penetrations of building boundaries by mechanical systems and related work.

### 3.8 DISINFECTION

Disinfect water piping, including all valves, fittings, and other devices, with a solution of chlorine and water. Ensure the solution contains not less than 50 parts per million (ppm) of available chlorine. Hold solution

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

for a period of not less than 8 hours, after which the solution contains not less than 10 ppm of available chlorine or re-disinfect the piping. After successful sterilization, thoroughly flush the piping before placing into service. Flushing is complete when the flush water contains less than 0.5 ppm of available chlorine. Water for disinfection will be furnished by the Government. Approve disposal of contaminated flush water in accordance with written instructions received from the Environmental authority having jurisdiction through the Contracting Officer and all local, State and Federal Regulations.

### 3.9 OPERATION AND MAINTENANCE

Provide Operation and Maintenance Manuals consistent with manufacturer's standard brochures, schematics, printed instructions, general operating procedures and safety precautions. Submit test data that is clear and readily legible.

### 3.10 PAINTING OF NEW EQUIPMENT

Factory or shop apply new equipment painting, as specified herein, and provided under each individual section.

#### 3.10.1 Factory Painting Systems

Manufacturer's standard factory painting systems may be provided subject to certification that the factory painting system applied withstands 125 hours in a salt-spray fog test, except that equipment located outdoors withstand 500 hours in a salt-spray fog test. Conduct salt-spray fog test in accordance with ASTM B117, and for that test the acceptance criteria is as follows: immediately after completion of the test, the inspected paint shows no signs of blistering, wrinkling, or cracking, and no loss of adhesion; and the specimen shows no signs of rust creepage beyond 0.125 inch on either side of the scratch mark.

Ensure the film thickness of the factory painting system applied on the equipment is not less than the film thickness used on the test specimen. If manufacturer's standard factory painting system is being proposed for use on surfaces subject to temperatures above 120 degrees F, design the factory painting system for the temperature service.

#### 3.10.2 Shop Painting Systems for Metal Surfaces

Clean, pretreat, prime and paint metal surfaces; except aluminum surfaces need not be painted. Apply coatings to clean dry surfaces. Clean the surfaces to remove dust, dirt, rust, oil and grease by wire brushing and solvent degreasing prior to application of paint, except clean to bare metal, surfaces subject to temperatures in excess of 120 degrees F.

Where more than one coat of paint is specified, apply the second coat after the preceding coat is thoroughly dry. Lightly sand damaged painting and retouch before applying the succeeding coat. Selected color of finish coat is aluminum or light gray.

- a. Temperatures Less Than 120 Degrees F: Immediately after cleaning, the metal surfaces subject to temperatures less than 120 degrees F receives one coat of pretreatment primer applied to a minimum dry film thickness of 0.3 mil, one coat of primer applied to a minimum dry film thickness of one mil; and two coats of enamel applied to a minimum dry film thickness of one mil per coat.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

- b. Temperatures Between 120 and 400 Degrees F: Metal surfaces subject to temperatures between 120 and 400 degrees F Receives two coats of 400 degrees F heat-resisting enamel applied to a total minimum thickness of 2 mils.
- c. Temperatures Greater Than 400 Degrees F: Metal surfaces subject to temperatures greater than 400 degrees F receives two coats of 600 degrees F heat-resisting paint applied to a total minimum dry film thickness of 2 mils.

-- End of Section --

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

SECTION 23 05 48.19

BRACING FOR HVAC  
**05/18, CHG 2: 08/20**

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN CONCRETE INSTITUTE (ACI)

- |           |                                                                                                |
|-----------|------------------------------------------------------------------------------------------------|
| ACI 355.2 | (2007) Qualification of Post-Installed Mechanical Anchors in Concrete and Commentary           |
| ACI 355.4 | (2011) Qualification of Post-Installed Adhesive Anchors in Concrete (ACI 355.4) and Commentary |

AMERICAN SOCIETY OF CIVIL ENGINEERS (ASCE)

- |           |                                                                                                                  |
|-----------|------------------------------------------------------------------------------------------------------------------|
| ASCE 7-16 | (2017; Errata 2018; Supp 1 2018) Minimum Design Loads and Associated Criteria for Buildings and Other Structures |
|-----------|------------------------------------------------------------------------------------------------------------------|

AMERICAN WATER WORKS ASSOCIATION (AWWA)

- |                  |                                                                                                                        |
|------------------|------------------------------------------------------------------------------------------------------------------------|
| AWWA C105/A21.5  | (2018) Polyethylene Encasement for Ductile-Iron Pipe Systems                                                           |
| AWWA C116/A21.16 | (2015) Protective Fusion-Bonded Coatings for the Interior and Exterior Surfaces of Ductile-Iron and Gray Iron Fittings |
| AWWA C153/A21.53 | (2019) Ductile-Iron Compact Fittings for Water Service                                                                 |
| AWWA C213        | (2022) Fusion-Bonded Epoxy Coatings and Linings for Steel Water Pipe and Fittings                                      |

ASTM INTERNATIONAL (ASTM)

- |                 |                                                                                                       |
|-----------------|-------------------------------------------------------------------------------------------------------|
| ASTM A36/A36M   | (2019) Standard Specification for Carbon Structural Steel                                             |
| ASTM A53/A53M   | (2022) Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless |
| ASTM A153/A153M | (2023) Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware                   |
| ASTM A325       | (2014) Standard Specification for                                                                     |

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

|                 |                                                                                                                                 |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------|
|                 | Structural Bolts, Steel, Heat Treated,<br>120/105 ksi Minimum Tensile Strength                                                  |
| ASTM A490       | (2014a) Standard Specification for<br>Structural Bolts, Alloy Steel, Heat<br>Treated, 150 ksi Minimum Tensile Strength          |
| ASTM A500/A500M | (2021a) Standard Specification for<br>Cold-Formed Welded and Seamless Carbon<br>Steel Structural Tubing in Rounds and<br>Shapes |
| ASTM A536       | (1984; R 2019; E 2019) Standard<br>Specification for Ductile Iron Castings                                                      |
| ASTM A563       | (2021; E 2022a) Standard Specification for<br>Carbon and Alloy Steel Nuts                                                       |
| ASTM A603       | (2019) Standard Specification for<br>Zinc-Coated Steel Structural Wire Rope                                                     |
| ASTM F1554      | (2020) Standard Specification for Anchor<br>Bolts, Steel, 36, 55, and 105-ksi Yield<br>Strength                                 |

#### FEDERAL EMERGENCY MANAGEMENT AGENCY (FEMA)

|            |                                                                   |
|------------|-------------------------------------------------------------------|
| FEMA P-414 | (January 2004) Installing Seismic<br>Restraints for Duct and Pipe |
|------------|-------------------------------------------------------------------|

#### METAL FRAMING MANUFACTURERS ASSOCIATION (MFMA)

|        |                                            |
|--------|--------------------------------------------|
| MFMA-4 | (2004) Metal Framing Standards Publication |
|--------|--------------------------------------------|

#### NSF INTERNATIONAL (NSF)

|             |                                                             |
|-------------|-------------------------------------------------------------|
| NSF/ANSI 61 | (2022) Drinking Water System Components -<br>Health Effects |
|-------------|-------------------------------------------------------------|

#### VIBRATION ISOLATION AND SEISMIC CONTROL MANUFACTURERS ASSOCIATION (VISCMA)

|            |                                                                  |
|------------|------------------------------------------------------------------|
| VISCMA 412 | (2014) Installing Seismic Restraints for<br>Mechanical Equipment |
|------------|------------------------------------------------------------------|

### 1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

#### SD-02 Shop Drawings

Coupling and Bracing

Flexible Couplings or Joints

Equipment Restraint

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

Contractor Designed Bracing; G

SD-03 Product Data

Coupling and Bracing; G

Flexible Couplings Or Joints; G

Equipment Restraint; G

Contractor Designed Bracing; G

Snubbers

Anchor Bolts

Vibration Isolators

SD-05 Design Data

Design Calculations

SD-06 Test Reports

Anchor Bolts; G

PART 2 PRODUCTS

2.1 EQUIPMENT RESTRAINT

Equipment must be rigidly or flexibly mounted as indicated in the specifications and/or drawings depending on vibration isolation requirements as follows below.

Roof mounted equipment such as condensers, both vibration isolated and nonisolated, must have support members designed and anchored to building structural steel or concrete as required for wind loads.

2.1.1 Rigidly (Base and Suspended) Mounted Equipment

HVAC equipment furnished under this contract must be rigidly mounted that are qualified for earthquake loading in accordance with ACI 355.2 and ACI 355.4. Anchor bolts must conform to ASTM F1554. For any rigid equipment which is rigidly anchored, provide flexible joints for piping, ductwork, electrical conduit, etc., that are capable of accommodating displacements equal to the full width of the joint in both orthogonal directions. Suspended equipment bracing attachments should be located just above the center of gravity to minimize swinging.

2.1.2 Nonrigid or Flexibly-Mounted Equipment

Select vibration isolation devices so that the maximum movement of equipment from the static deflection point is 1/4 inch. Equipment flexibly mounted on vibration isolators must have a bumper restraint or snubber in each horizontal direction and vertical restraints must be provided where required to resist overturning. Isolator housing and



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

restraints must be constructed of ductile materials. A viscoelastic pad or similar material of appropriate thickness must be used between the bumper and components to limit the impact load. Restraints must be designed to resist the calculated horizontal lateral and vertical forces.

Spring vibration isolators must be seismically rated, restrained isolators for equipment subject to load variations and large external forces. The seismically rated housing must be sized to meet or exceed the force requirements applicable to the project and meet the required isolation criteria. Spring vibration isolator manufacturer's will be a member of VISCMA. Design force,  $F_p$ , must be doubled for vibration isolators with an air gap greater than 0.25 inches as specified in ASCE 7-16, Chapter 13.

## 2.2 BOLTS AND NUTS

Hex head bolts, and heavy hexagon nuts must be ASTM A325 or ASTM A490 bolts and ASTM A563 nuts. Provide bolts and nuts galvanized in accordance with ASTM A153/A153M when used underground or exposed to weather.

## 2.3 FLEXIBLE JOINTS

Flexible joints must have same pressure and temperature ratings as adjoining pipe. Braided hoses must not be used where there is torsional or axial movement unless manufacturer allows it.

### 2.3.1 Braided Hose Expansion Joint

Braided hose expansion joint(s) must be installed in the locations indicated on the drawings and as required to accommodate any thermal expansion, contraction or seismic movement of the piping system. Joints must consist of two parallel sections of corrugated metal hose, compatible braid, and 180 degree return bend with inlet and outlet connections. Field fabricated loops are not acceptable. Braided hose expansion joint(s) must be installed in the locations indicated on the drawings and as required to accommodate any thermal expansion, contraction or seismic movement of the piping system. Joints must consist of two parallel sections of corrugated metal hose, compatible braid, and 180 degree return bend with inlet and outlet connections. Field fabricated loops must not be acceptable. Braided hose in a 60 degree flexible V loop arrangement must be used for small diameter pipe connections to coils in variable-air-volume (VAV) terminal units and fan coil units installed in suspended ductwork whether braced or unbraced.

All braided hose expansion joints must be manufactured in accordance with the documented manufacturers weld procedure specifications. The procedure qualification record must be used to document the execution of this procedure and must follow the general "guidelines" of ASME Section IX. Each individual welder must conform to the in-house procedure qualification record and be qualified prior to each production lot. The testing of each individual welder must be documented in a welding procedure qualification record.

#### 2.3.1.1 Flexible Hose Expansion Loops

Flexible hose expansion loops must have a factory supplied, hanger / support lug located at the bottom of the 180deg return. Flexible hose expansion loop(s) must be furnished with a plugged FPT to be used for a drain or air release vent. Flexible hose expansion loop(s) must be rated with an operating pressure which is the same as the adjoining pipe. The

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

operating pressure must be based on burst pressure with a 4 to 1 safety factor.

### 2.3.2 Double Ball Flexible Expansion Joint

Install flexible expansion joints manufactured of ductile iron conforming to the material requirements of ASTM A536 and AWWA C153/A21.53 in the locations indicated on the drawings. Provide foundry certification of material upon request. Each flexible expansion joint must be pressure tested prior to shipment against its own restraint to a minimum of 350 psi (250 psi for flexible expansion joints 2 inch and 30 inches diameter and larger.) A minimum 2:1 safety factor, determined from the published pressure rating, must apply. Factory Mutual Approval for the 3 inch through 12 inch sizes is required. Each flexible expansion joint must consist of an expansion joint designed and cast as an integral part of a ball and socket type flexible joint, having a minimum per ball deflection of: 20°, 2" - 12"; 15°, 14" - 36"; 12°, 42"-48" and 4-inches minimum expansion. Additional expansion sleeves must be available and easily added or removed at the factory or in the field. Both standardized mechanical joint and flange end connections must be available.

#### 2.3.2.1 Internal Surfaces

Line all internal surfaces (wetted parts) with a minimum of 15 mils of fusion bonded epoxy conforming to the applicable requirements of AWWA C213. Sealing gaskets must be constructed of EPDM. The coating must meet NSF/ANSI 61.

#### 2.3.2.2 Exterior Surfaces

Coat exterior surfaces with a minimum of 6 mils of fusion bonded epoxy conforming to the applicable requirements of AWWA C116/A21.16. Include appropriately sized polyethylene sleeves, meeting AWWA C105/A21.5, for direct buried applications.

### 2.4 SWAY BRACING MATERIALS

Material used for members listed in this section , must be structural steel conforming with the following:

- a. Plates, rods, and rolled shapes, ASTM A36/A36M.
- b. Wire rope, ASTM A603 pre-stretched. Class B galv coating Ferrule clamps must be qualified by testing for use in seismic applications per VISCMA 412. A minimum of two clamps are required on each end of wire rope.
- c. Tubes, ASTM A500/A500M, Grade B.
- d. Pipes, ASTM A53/A53M, Grade B.
- e. Angles, ASTM A36/A36M.
- f. Channels (Struts) with in-turned lips and associated hardware for fastening to channels at random points conforming to MFMA-4

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### PART 3 EXECUTION

#### 3.1 COUPLING AND BRACING

- a. Submit detail drawings, as specified here and throughout this specification, along with catalog cuts, templates, and erection and installation details, as appropriate, for the items listed. Submittals must be complete in detail; must indicate thickness, type, grade, class of metal, and dimensions; and must show construction details, reinforcement, anchorage, and installation with relation to the building construction.
- c. Size bracing components as required for the total load carried by the common supports. Bracing rigidly attached to pipe flanges, or similar, must not be used where it would interfere with thermal expansion of piping.
- d. Adjust isolators and restraints after piping systems has been filled and equipment is at its operating weight, following the manufacturer's written instructions.
- e. Install cables at a 45-degree slope. Where interference is present, the slope may be minimum of 30 degrees or a maximum of 60 degrees per VISCMA 412.

#### 3.2 FLEXIBLE COUPLINGS OR JOINTS

##### 3.2.1 Building Piping

Provide flexible couplings or joints in building piping at bottom of all pipe risers for pipe larger than 3-1/2 inches in diameter. Laterally brace flexible couplings or joints without interfering with the action of the flexible coupling or joint. Cast iron waste and vent piping need only comply with these provisions when caulked joints are used. Flexible bell and spigot pipe joints using rubber gaskets may be used at each branch adjacent to tees and elbows for underground waste piping inside of building to satisfy these requirements.

#### 3.3 PIPE SLEEVES

Size pipe sleeves in interior non-fire rated walls as indicated on the drawings to provide clearances that will permit differential movement of piping without the piping striking the pipe sleeve. Pipe sleeves in fire rated walls must conform to the requirements in Section 07 84 00 FIRESTOPPING.

#### 3.4 SWAY BRACES FOR DUCTS

##### 3.4.1 Unbraced Ducts

Attach hangers for unbraced ducts to the duct within 2 inches of the top of the duct with a minimum of two #10 sheet metal screws in accordance with FEMA P-414. Use shortest screws possible when penetrating ductwork to minimize airflow noise inside duct. Install unbraced ducts with a 6 inch minimum clearance to vertical ceiling hanger wires.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 3.5 EQUIPMENT

#### 3.5.1 General

Ensure housekeeping pads have adequate space to mount equipment. Identify position of reinforcing steel and other embedded items prior to drilling holes for anchors. Do not drill holes in concrete or masonry until concrete, mortar, or grout has achieved full design strength. Install neoprene grommet washers or fill the gap with epoxy on equipment anchor bolts where clearance between anchor and equipment support hole exceeds 0.125 inches.

### 3.6 ANCHOR BOLTS

#### 3.6.1 Cast-in-Place Anchor Bolts

Use templates to locate cast-in-place bolts accurately and securely in formwork. Anchor bolts must have an embedded straight length equal to at least 12 times nominal diameter of the bolt. Anchor bolts that exceed the normal depth of equipment foundation piers or pads must either extend into concrete floor or the foundation or be increased in depth to accommodate bolt lengths.

#### 3.6.2 Drilled-In Anchor Bolts

Drill holes with rotary impact hammer drills. Drill bits must be of diameters as specified by the anchor manufacturer. Unless otherwise shown on the Drawings, all holes must be drilled perpendicular to the concrete surface. Where anchors are permitted to be installed in cored holes, use core bits with matched tolerances as specified by the manufacturer. Properly clean cored hole per manufacturer's instructions. Identify position of reinforcing steel and other embedded items prior to drilling holes for anchors. Exercise care in coring or drilling to avoid damaging existing reinforcing or embedded items. Notify the COR if reinforcing steel or other embedded items are encountered during drilling. Take precautions as necessary to avoid damaging prestressing tendons, electrical and telecommunications conduit, and gas lines. Unless otherwise specified, do not drill holes in concrete or masonry until concrete, mortar, or grout has achieved full design strength. Perform anchor installation in accordance with manufacturer instructions.

##### 3.6.2.1 Wedge Anchors, Heavy-Duty Sleeve Anchors, and Undercut Anchors

Protect threads from damage during anchor installation. Heavy-duty sleeve anchors must be installed with sleeve fully engaged in part to be fastened. Set anchors to manufacturer's recommended torque, using a torque wrench. Following attainment of 10% of the specified torque, 100% of the specified torque must be reached within 7 or fewer complete turns of the nut. If the specified torque is not achieved within the required number of turns, the anchor must be removed and replaced unless otherwise directed by the Engineer.

##### 3.6.2.2 Cartridge Injection Adhesive Anchors

Where approved for seismic application, clean all holes per manufacturer instructions to remove loose material and drilling dust prior to installation of adhesive. Inject adhesive into holes proceeding from the bottom of the hole and progressing toward the surface in such a manner as to avoid introduction of air pockets in the adhesive. Follow manufacturer

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

recommendations to ensure proper mixing of adhesive components. Sufficient adhesive must be injected in the hole to ensure that the annular gap is filled to the surface. Remove excess adhesive from the surface. Shim anchors with suitable device to center the anchor in the hole. Do not disturb or load anchors before manufacturer specified cure time has elapsed.

### 3.6.2.3 Capsule Anchors

Where approved for seismic application, perform drilling and setting operations in accordance with manufacturer instructions. Clean all holes to remove loose material and drilling dust prior to installation of adhesive. Remove water from drilled holes in such a manner as to achieve a surface dry condition. Capsule anchors must be installed with equipment conforming to manufacturer recommendations. Do not disturb or load anchors before manufacturer specified cure time has elapsed.

Observe manufacturer recommendations with respect to installation temperatures for cartridge injection adhesive anchors and capsule anchors.

## 3.7 ANCHOR BOLT TESTING

Test in place expansion and chemically bonded anchors not more than 24 hours after installation of the anchor, conducted by an independent testing agency; testing must be performed on random anchor bolts as described below.

### 3.7.1 Torque Wrench Testing

Perform torque wrench testing on not less than 50 percent of the total installed applied torque expansion anchors and at least one anchor for every piece of equipment containing more than two anchors. The test torque must equal the minimum required installation torque as required by the bolt manufacturer. Calibrate torque wrenches at the beginning of each day the torque tests are performed. Recalibrate torque wrenches for each bolt diameter whenever tests are run on bolts of various diameters. Apply torque between 20 and 100 percent of wrench capacity. Reach the test torque within one half turn of the nut, except for 3/8 inch sleeve anchors which must reach their torque by one quarter turn of the nut. If any anchor fails the test, test similar anchors not previously tested until 20 consecutive anchors pass. Failed anchors must be retightened and retested to the specified torque; if the anchor still fails the test it must be replaced.

-- End of Section --

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## SECTION 23 05 93

### TESTING, ADJUSTING, AND BALANCING FOR HVAC **11/15**

#### PART 1 GENERAL

##### 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

#### ACOUSTICAL SOCIETY OF AMERICA (ASA)

ASA S1.4 (1983; Amendment 1985; R 2006)  
Specification for Sound Level Meters (ASA 47)

ASA S1.11 PART 1 (2014) American National Standard  
Electroacoustics - Octave-Band and  
Fractional-Octave-Band Filters - Part 1:  
Specifications

#### AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS (ASHRAE)

ASHRAE HVAC APP IP HDBK (2016) HVAC Applications Handbook, I-P  
Edition

#### ASSOCIATED AIR BALANCE COUNCIL (AABC)

AABC MN-1 (2002; 6th ed) National Standards for  
Total System Balance

AABC MN-4 (1996) Test and Balance Procedures

#### NATIONAL ENVIRONMENTAL BALANCING BUREAU (NEBB)

NEBB MASV (2006) Procedural Standards for  
Measurements and Assessment of Sound and  
Vibration

NEBB PROCEDURAL STANDARDS (2015) Procedural Standards for TAB  
(Testing, Adjusting and Balancing)  
Environmental Systems

#### SHEET METAL AND AIR CONDITIONING CONTRACTORS' NATIONAL ASSOCIATION (SMACNA)

SMACNA 1780 (2002) HVAC Systems - Testing, Adjusting  
and Balancing, 3rd Edition

SMACNA 1858 (2004) HVAC Sound And Vibration Manual -  
First Edition

SMACNA 1972 CD (2012) HVAC Air Duct Leakage Test Manual -  
2nd Edition

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## 1.2 DEFINITIONS

- a. AABC: Associated Air Balance Council
- b. COTR: Contracting Officer's Technical Representative
- c. DALT: Duct air leakage test
- d. DALT'd: Duct air leakage tested
- e. HVAC: Heating, ventilating, and air conditioning; or heating, ventilating, and cooling
- f. NEBB: National Environmental Balancing Bureau
- g. Out-of-tolerance data: Pertains only to field acceptance testing of Final DALT or TAB report. When applied to DALT work, this phase means  
When applied to TAB work this phase means "a measurement taken during TAB field acceptance testing which does not fall within the range of plus 5 to minus 5 percent of the original measurement reported on the TAB Report for a specific parameter."
- h. Season of maximum heating load: The time of year when the outdoor temperature at the project site remains within plus or minus 30 degrees Fahrenheit of the project site's winter outdoor design temperature, throughout the period of TAB data recording.
- i. Season of maximum cooling load: The time of year when the outdoor temperature at the project site remains within plus or minus 5 degrees Fahrenheit of the project site's summer outdoor design temperature, throughout the period of TAB data recording.
- j. Season 1, Season 2: Depending upon when the project HVAC is completed and ready for TAB, Season 1 is defined, thereby defining Season 2. Season 1 could be the season of maximum heating load, or the season of maximum cooling load.
- k. Sound measurements terminology: Defined in AABC MN-1, NEBB MASV, or SMACNA 1858 (TABB).
- l. TAB: Testing, adjusting, and balancing (of HVAC systems)
- m. TAB'd: HVAC Testing/Adjusting/Balancing procedures performed
- n. TAB Agency: TAB Firm
- r. TABB: Testing Adjusting and Balancing Bureau

### 1.2.1 Similar Terms

In some instances, terminology differs between the Contract and the TAB Standard primarily because the intent of this Section is to use the industry standards specified, along with additional requirements listed herein to produce optimal results.

The following table of similar terms is provided for clarification only. Contract requirements take precedent over the corresponding AABC, NEBB, or TABB requirements where differences exist.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

| SIMILAR TERMS           |                                                                                                 |                                                                                    |                                                           |
|-------------------------|-------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-----------------------------------------------------------|
| Contract Term           | AABC Term                                                                                       | NEBB Term                                                                          | TABB Term                                                 |
| TAB Standard            | National Standards for Testing and Balancing Heating, Ventilating, and Air Conditioning Systems | Procedural Standards for Testing, Adjusting and Balancing of Environmental Systems | International Standards for Environmental Systems Balance |
| TAB Specialist          | TAB Engineer                                                                                    | TAB Supervisor                                                                     | TAB Supervisor                                            |
| Systems Readiness Check | Construction Phase Inspection                                                                   | Field Readiness Check & Preliminary Field Procedures                               | Field Readiness Check & Prelim. Field Procedures          |

### 1.3 WORK DESCRIPTION

The work includes duct air leakage testing (DALT) and testing, adjusting, and balancing (TAB) of new heating, ventilating, and cooling (HVAC) air and water distribution systems including equipment and performance data, ducts, and piping which are located within, on, under, between, and adjacent to buildings.

Perform TAB in accordance with the requirements of the TAB procedural standard recommended by the TAB trade association that approved the TAB Firm's qualifications. Comply with requirements of AABC MN-1, NEBB PROCEDURAL STANDARDS, or SMACNA 1780 (TABB) as supplemented and modified by this specification section. All recommendations and suggested practices contained in the TAB procedural standards are considered mandatory.

Conduct DALT and TAB of the indicated existing systems and equipment and submit the specified DALT and TAB reports for approval. Conduct DALT testing in compliance with the requirements specified in SMACNA 1972 CD, except as supplemented and modified by this section. Conduct DALT and TAB work in accordance with the requirements of this section.

#### 1.3.1 Air Distribution Systems

Test, adjust, and balance systems (TAB) in compliance with this section. Obtain Contracting Officer's written approval before applying insulation to exterior of air distribution systems as specified under Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS.

#### 1.3.2 Water Distribution Systems

TAB systems in compliance with this section. Obtain Contracting Officer's written approval before applying insulation to water distribution systems as specified under Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS. At Contractor's option and with Contracting Officer's written approval, the piping systems may be insulated before systems are TAB'd.

Terminate piping insulation immediately adjacent to each flow control



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

valve, automatic control valve, or device. Seal the ends of pipe insulation and the space between ends of pipe insulation and piping, with waterproof vapor barrier coating.

After completion of work under this section, insulate the flow control valves and devices as specified under Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS.

### 1.3.3 TAB SCHEMATIC DRAWINGS

Show the following information on TAB Schematic Drawings:

1. A unique number or mark for each piece of equipment or terminal.
2. Air quantities at air terminals.
3. Air quantities and temperatures in air handling unit schedules.
4. Water quantities and temperatures in thermal energy transfer equipment schedules.
5. Water quantities and heads in pump schedules.
6. Water flow measurement fittings and balancing fittings.
7. Ductwork Construction and Leakage Testing Table that defines the DALT test requirements, including each applicable HVAC duct system ID or mark, duct pressure class, duct seal class, and duct leakage test pressure. This table is included in the file for Graphics for Unified Facilities Guide Specifications:  
<http://www.wbdg.org/ffc/dod/unified-facilities-guide-specifications-ufgs/forms-gra>

Submit three copies of the TAB Schematic Drawings and Report Forms to the Contracting Officer, no later than 21 days prior to the start of TAB field measurements.

### 1.3.4 Related Requirements

## 1.4 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

#### SD-01 Preconstruction Submittals

Records of Existing Conditions; G

#### SD-02 Shop Drawings

TAB Schematic Drawings and Report Forms; G

#### SD-03 Product Data

Equipment and Performance Data; G

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### TAB Related HVAC Submittals; G

A list of the TAB Related HVAC Submittals, no later than 7 days after the approval of the TAB team engineer and assistant.

#### SD-06 Test Reports

Completed Pre-Final DALT Report; G

Certified Final DALT Report; G

#### SD-07 Certificates

Independent TAB Agency and Personnel Qualifications; G

DALT and TAB Submittal and Work Schedule; G

TAB Pre-Field Engineering Report; G

### 1.5 QUALITY ASSURANCE

#### 1.5.1 Independent TAB Agency and Personnel Qualifications

To secure approval for the proposed agency, submit information certifying that the TAB agency is a first tier subcontractor who is not affiliated with any other company participating in work on this contract, including design, furnishing equipment, or construction. Further, submit the following, for the agency, to Contracting Officer for approval:

##### a. Independent AABC or NEBB or TABB TAB agency:

TAB agency: AABC registration number and expiration date of current certification; or NEBB certification number and expiration date of current certification; or TABB certification number and expiration date of current certification.

TAB team supervisor: Name and copy of AABC or NEBB or TABB TAB supervisor certificate and expiration date of current certification.

TAB team field leader: Name and documented evidence that the team field leader has satisfactorily performed full-time supervision of TAB work in the field for not less than 3 years immediately preceding this contract's bid opening date.

TAB team field technicians: Names and documented evidence that each field technician has satisfactorily assisted a TAB team field leader in performance of TAB work in the field for not less than one year immediately preceding this contract's bid opening date.

Current certificates: Registrations and certifications are current, and valid for the duration of this contract. Renew Certifications which expire prior to completion of the TAB work, in a timely manner so that there is no lapse in registration or certification. TAB agency or TAB team personnel without a current registration or current certification are not to perform TAB work on this contract.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

- b. TAB Team Members: TAB team approved to accomplish work on this contract are full-time employees of the TAB agency. No other personnel is allowed to do TAB work on this contract.
- c. Replacement of TAB team members: Replacement of members may occur if each new member complies with the applicable personnel qualifications and each is approved by the Contracting Officer.

## PART 2 PRODUCTS

Not Used

## PART 3 EXECUTION

### 3.1 WORK DESCRIPTIONS OF PARTICIPANTS

Comply with requirements of this section.

### 3.2 PRE-DALT/TAB MEETING

Meet with the Contracting Officer's technical representative (COTR) to develop a mutual understanding relative to the details of the DALT work and TAB work requirements. Ensure that the TAB supervisor is present at this meeting. Requirements to be discussed include required submittals, work schedule, and field quality control.

### 3.3 DALT PROCEDURES

#### 3.3.1 Instruments, Consumables and Personnel

Provide instruments, consumables and personnel required to accomplish the DALT field work. Follow the same basic procedure specified below for TAB Field Work, including maintenance and calibration of instruments, accuracy of measurements, preliminary procedures, field work, workmanship and treatment of deficiencies. Calibrate and maintain instruments in accordance with manufacturer's written procedures.

#### 3.3.2

On completion of the installation of each duct system indicated to be DALT'd, notify the Contracting Officer in writing prior to the COTR's duct selection field visit.

#### 3.3.3 Ductwork To Be DALT'd

From each duct system indicated as subject to DALT, the COTR will randomly select sections of each completed duct system for testing by the Contractor's TAB Firm. The sections selected will not exceed 20 percent of the total measured linear footage of duct systems indicated as subject to DALT. Sections of duct systems subject to DALT will include 20 percent of main ducts, branch main ducts, branch ducts and plenums for supply, return, exhaust, and plenum ductwork.

It is acceptable for an entire duct system to be DALT'd instead of disassembling that system in order to DALT only the 20 percent portion specified above.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### 3.3.4 DALT Testing

Perform DALT on the HVAC duct sections of each system as selected by the COTR. Use the duct class, seal class, leakage class and the leak test pressure data indicated on the drawings, to comply with the procedures specified in SMACNA 1972 CD.

In spite of specifications of SMACNA 1972 CD to the contrary, DALT ductwork of construction class of 3-inch water gauge static pressure and below if indicated to be DALT'd. Complete DALT work on the COTR selected ductwork within 48 hours after the particular ductwork was selected for DALT. Separately conduct DALT work for large duct systems to enable the DALT work to be completed in 48 hours.

#### 3.3.5 Completed Pre-Final DALT Report

#### 3.3.6 Quality Assurance - COTR DALT Field Acceptance Testing

In the presence of the COTR and TAB team field leader, verify for accuracy Pre-final DALT Report data selected by the COTR. For each duct system, this acceptance testing shall be conducted on a maximum of 50 percent of the duct sections DALT'd.

Further, if any data on the Pre-final DALT report form for a given duct section is out-of-tolerance, then field acceptance testing shall be conducted on data for one additional duct section, preferably in the same duct system, in the presence of the COTR.

#### 3.3.7 Additional COTR Field Acceptance Testing

If any of the duct sections checked for a given system are determined to have a leakage rate measured that exceeds the leakage rate allowed by SMACNA Leak Test Manual for an indicated duct construction class and sealant class, terminate data checking for that section. The associated Pre-final DALT Report data for the given duct system will be disapproved. Make the necessary corrections and prepare a revised Pre-final DALT Report. Reschedule a field check of the revised report data with the COTR.

#### 3.3.8 Certified Final DALT Report

#### 3.3.9 Prerequisite for TAB Field Work

Do not commence TAB field work prior to the completion and approval, for all systems, of the Final DALT Report.

### 3.4 TAB PROCEDURES

#### 3.4.1 TAB Field Work

Test, adjust, and balance the HVAC systems until measured flow rates (air and water flow) are within plus or minus 5 percent of the design flow rates as specified or indicated on the contract documents.

That is, comply with the the requirements of AABC MN-1 or SMACNA 1780 (TABB) and SMACNA 1858 (TABB), except as supplemented and modified by this

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

section.

Provide instruments and consumables required to accomplish the TAB work. Calibrate and maintain instruments in accordance with manufacturer's written procedures.

Test, adjust, and balance the HVAC systems until measured flow rates (air and water flow) are within plus or minus 5 percent of the design flow rates as specified or indicated on the contract documents. Conduct TAB work, including measurement accuracy, and sound measurement work in conformance with the AABC MN-1 and AABC MN-4, or NEBB TABES and NEBB MASV, or SMACNA 1780 (used by TABB) and SMACNA 1858 sound measurement procedures, except as supplemented and modified by this section. The only water flow and air flow reporting which can be deferred until the Season 2 is that data which would be affected in terms of accuracy due to outside ambient conditions.

### 3.4.2 Preliminary Procedures

Use the approved pre-field engineering report as instructions and procedures for accomplishing TAB field work. TAB engineer is to locate, in the field, test ports required for testing. It is the responsibility of the sheet metal contractor to provide and install test ports as required by the TAB engineer.

### 3.4.3 TAB Air Distribution Systems

#### 3.4.3.1 Units With Coils

Report heating and cooling performance capacity tests for hot water, chilled water, DX and steam coils for the purpose of verifying that the coils meet the indicated design capacity. Submit the following data and calculations with the coil test reports:

- a. For air handlers with capacities greater than 7.5 tons (90,000 Btu) cooling, such as factory manufactured units, central built-up units and rooftop units, conduct capacity tests in accordance with AABC MN-4, procedure 3.5, "Coil Capacity Testing."

Do not determine entering and leaving wet and dry bulb temperatures by single point measurement, but by the average of multiple readings in compliance with paragraph 3.5-5, "Procedures", (in subparagraph d.) of AABC MN-4, Procedure 3.5, "Coil Capacity Testing."

Submit part-load coil performance data from the coil manufacturer converting test conditions to design conditions; use the data for the purpose of verifying that the coils meet the indicated design capacity in compliance with AABC MN-4, Procedure 3.5, "Coil Capacity Testing," paragraph 3.5.7, "Actual Capacity Vs. Design Capacity" (in subparagraph c.).

- b. For units with capacities of 7.5 tons (90,000 Btu) or less, such as fan coil units, duct mounted reheat coils associated with VAV terminal units, and unitary units, such as through-the-wall heat pumps:

Determine the apparent coil capacity by calculations using single point measurement of entering and leaving wet and dry bulb temperatures; submit the calculations with the coil reports.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 3.4.3.2 Rooftop Air Conditioning

Rooftop air conditioning systems including fans, coils, ducts, plenums, and air distribution devices for supply air, return air, and outside air.

For refrigeration compressors/condensers/condensing units/evaporators, report data as required by NEBB, AABC, and TABB standard procedures, including refrigeration operational data.

### 3.4.3.3 Makeup Air Units

Makeup air unit systems including fans, coils, ducts, plenums, registers, diffusers, grilles, and louvers for supply air, return air, outside air, and mixed air.

### 3.4.3.4 Exhaust Fans

Exhaust fan systems including fans, ducts, plenums, grilles, and hoods for exhaust air.

### 3.4.4 TAB Water Distribution Systems

#### 3.4.4.1 Heating Hot Water

Heating hot water systems including boilers, hot water converters (e.g., heat exchangers), pumps, coils, system balancing valves and flow measuring devices.

### 3.4.5 Sound Measurement Work

#### 3.4.5.1 Areas To Be Sound Measured

In the following spaces, measure and record the sound power level for each octave band listed in ASHRAE HVAC APP IP HDBK Noise Criteria:

- a. All HVAC mechanical rooms, including machinery spaces and other spaces containing HVAC power drivers and power driven equipment.
- b. All spaces sharing a common barrier with each mechanical room, including rooms overhead, rooms on the other side of side walls, and rooms beneath the mechanical room floor.

c. System: MAU-1 Rooms:  
Apparatus Bay

d. System: RTU-1 Rooms: 107, 130, 157, 119, 118, 122

e. System: DOAS-2 Rooms: 107, 130, 157, 119, 118, 122

e. System: MAU-2 Rooms: 120

#### 3.4.5.2 Procedure

Measure sound levels in each room, when unoccupied except for the TAB team, with all HVAC systems that would cause sound readings in the room operating in their noisiest mode. Record the sound level in each octave band. Attempt to mitigate the sound level and bring the level to within the specified ASHRAE HVAC APP IP HDBK noise criteria goals, if such mitigation is within the TAB team's control. State in the report the

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

ASHRAE HVAC APP IP HDBK noise criteria goals. If sound level cannot be brought into compliance, provide written notice of the deficiency to the Contractor for resolution or correction.

#### 3.4.5.3 Timing

Measure sound levels at times prescribed by AABC or NEBB or TABB.

#### 3.4.5.4 Meters

Measure sound levels with a sound meter complying with ASA S1.4, Type 1 or 2, and an octave band filter set complying with ASA S1.11 PART 1. Use measurement methods for overall sound levels and for octave band sound levels as prescribed by NEBB.

#### 3.4.5.5 Calibration

Calibrate sound levels as prescribed by AABC or NEBB or TABB, except that calibrators emitting a sound pressure level tone of 94 dB at 1000 hertz (Hz) are also acceptable.

#### 3.4.5.6 Background Noise Correction

Determine background noise component of room sound (noise) levels for each (of eight) octave bands as prescribed by AABC or NEBB or TABB.

### 3.4.6 TAB Work on Performance Tests Without Seasonal Limitations

#### 3.4.6.1 Performance Tests

In addition to the TAB proportionate balancing work on the air distribution systems and the water distribution systems, accomplish TAB work on the HVAC systems which directly transfer thermal energy. TAB the operational performance of the heating systems and cooling systems.

#### 3.4.6.2 Ambient Temperatures

On each tab report form used for recording data, record the outdoor and indoor ambient dry bulb temperature range and the outdoor and indoor ambient wet bulb temperature range within which the report form's data was recorded. Record these temperatures at beginning and at the end of data taking.

#### 3.4.6.3 Sound Measurements

Comply with the paragraph SOUND MEASUREMENT WORK, specifically, the requirement that a room must be operating in its noisiest mode at the time of sound measurements in the room. The maximum noise level measurements could depend on seasonally related heat or cooling transfer equipment.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 3.4.7 TAB Work on Performance Tests With Seasonal Limitations

#### 3.4.7.1 Performance Tests

#### 3.4.7.2 Season Of Maximum Load

#### 3.4.7.3 Ambient Temperatures

On each tab report form used for recording data, record the outdoor and indoor ambient dry bulb temperature range and the outdoor and indoor ambient wet bulb temperature range within which the report form's data was recorded. Record these temperatures at beginning and at the end of data taking.

#### 3.4.7.4 Sound Measurements

Comply with the paragraph SOUND MEASUREMENT WORK, specifically, the requirement that a room must be operating in its noisiest mode at the time of sound measurements in the room. The maximum noise level measurements could depend on seasonally related heat or cooling transfer equipment.

### 3.4.8 Workmanship

Conduct TAB work on the HVAC systems until measured flow rates are within plus or minus 5 percent of the design flow rates as specified or indicated on the contract documents. This TAB work includes adjustment of balancing valves, balancing dampers, and sheaves. Further, this TAB work includes changing out fan sheaves and pump impellers if required to obtain air and water flow rates specified or indicated. If, with these adjustments and equipment changes, the specified or indicated design flow rates cannot be attained, contact the Contracting Officer for direction.

### 3.4.9 Deficiencies

Strive to meet the intent of this section to maximize the performance of the equipment as designed and installed. However, if deficiencies in equipment design or installation prevent TAB work from being accomplished within the range of design values specified in the paragraph WORKMANSHIP, provide written notice as soon as possible to the Contractor and the Contracting Officer describing the deficiency and recommended correction.

Responsibility for correction of installation deficiencies is the Contractor's. If a deficiency is in equipment design, call the TAB team supervisor for technical assistance. Responsibility for reporting design deficiencies to Contractor is the TAB team supervisor's.

### 3.4.10 TAB Reports

### 3.4.11 Quality Assurance - COTR TAB Field Acceptance Testing

#### 3.4.11.1 TAB Field Acceptance Testing

During the field acceptance testing, verify, in the presence of the COTR, random selections of data (water, air quantities, air motion, sound level readings) recorded in the TAB Report. Points and areas for field



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

acceptance testing are to be selected by the COTR. Measurement and test procedures are the same as approved for TAB work for the TAB Report.

Field acceptance testing includes verification of TAB Report data recorded for the following equipment groups:

Group 1: All chillers, boilers, return fans, computer room units, and air handling units (rooftop and central stations).

Group 2: 25 percent of the VAV terminal boxes and associated diffusers and registers.

Group 3: 25 percent of the supply diffusers, registers, grilles associated with constant volume air handling units.

Group 4: 25 percent of the return grilles, return registers, exhaust grilles and exhaust registers.

Group 5: 25 percent of the supply fans, exhaust fans, and pumps.

Further, if any data on the TAB Report for Groups 2 through 5 is found not to fall within the range of plus 5 to minus 5 percent of the TAB Report data, additional group data verification is required in the presence of the COTR. Verify TAB Report data for one additional piece of equipment in that group. Continue this additional group data verification until out-of-tolerance data ceases to be found.

#### 3.4.11.2 Additional COTR TAB Field Acceptance Testing

If any of the acceptance testing measurements for a given equipment group is found not to fall within the range of plus 5 to minus 5 percent of the TAB Report data, terminate data verification for all affected data for that group. The affected data for the given group will be disapproved. Make the necessary corrections and prepare a revised TAB Report. Reschedule acceptance testing of the revised report data with the COTR.

#### 3.4.11.3 Prerequisite for Approval

Compliance with the field acceptance testing requirements of this section is a prerequisite for the final Contracting Officer approval of the TAB Report submitted.

### 3.5 MARKING OF SETTINGS

Upon the final TAB work approval, permanently mark the settings of HVAC adjustment devices including valves, gauges, splitters, and dampers so that adjustment can be restored if disturbed at any time. Provide permanent markings clearly indicating the settings on the adjustment devices which result in the data reported on the submitted TAB report.

### 3.6 MARKING OF TEST PORTS

The TAB team is to permanently and legibly mark and identify the location points of the duct test ports. If the ducts have exterior insulation, make these markings on the exterior side of the duct insulation. Show the location of test ports on the as-built mechanical drawings with dimensions given where the test port is covered by exterior insulation.

-- End of Section --

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

SECTION 23 07 00

THERMAL INSULATION FOR MECHANICAL SYSTEMS  
**02/13, CHG 7: 05/20**

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only. At the discretion of the Government, the manufacturer of any material supplied will be required to furnish test reports pertaining to any of the tests necessary to assure compliance with the standard or standards referenced in this specification.

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING  
ENGINEERS (ASHRAE)

ASHRAE 90.1 - SI (2019) Energy Standard for Buildings  
Except Low-Rise Residential Buildings

ASTM INTERNATIONAL (ASTM)

ASTM B209 (2014) Standard Specification for Aluminum  
and Aluminum-Alloy Sheet and Plate

ASTM C195 (2007; R 2013) Standard Specification for  
Mineral Fiber Thermal Insulating Cement

ASTM C534/C534M (2023) Standard Specification for  
Preformed Flexible Elastomeric Cellular  
Thermal Insulation in Sheet and Tubular  
Form

ASTM C547 (2022a) Standard Specification for Mineral  
Fiber Pipe Insulation

ASTM C552 (2022) Standard Specification for Cellular  
Glass Thermal Insulation

ASTM C647 (2008; R 2013) Properties and Tests of  
Mastics and Coating Finishes for Thermal  
Insulation

ASTM C755 (2019b) Standard Practice for Selection of  
Water Vapor Retarders for Thermal  
Insulation

ASTM C795 (2008; R 2023) Standard Specification for  
Thermal Insulation for Use in Contact with  
Austenitic Stainless Steel

ASTM C920 (2018) Standard Specification for  
Elastomeric Joint Sealants

ASTM C921 (2010; R 2015) Standard Practice for  
Determining the Properties of Jacketing

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### Materials for Thermal Insulation

|               |                                                                                                                                                                       |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM C1136    | (2023) Standard Specification for Flexible, Low Permeance Vapor Retarders for Thermal Insulation                                                                      |
| ASTM C1710    | (2011) Standard Guide for Installation of Flexible Closed Cell Preformed Insulation in Tube and Sheet Form                                                            |
| ASTM D2863    | (2019) Standard Test Method for Measuring the Minimum Oxygen Concentration to Support Candle-Like Combustion of Plastics (Oxygen Index)                               |
| ASTM D5590    | (2017; R 2021) Standard Test Method for Determining the Resistance of Paint Films and Related Coatings to Fungal Defacement by Accelerated Four-Week Agar Plate Assay |
| ASTM E84      | (2023) Standard Test Method for Surface Burning Characteristics of Building Materials                                                                                 |
| ASTM E96/E96M | (2024) Standard Test Methods for Gravimetric Determination of Water Vapor Transmission Rate of Materials                                                              |
| ASTM E2231    | (2021) Standard Practice for Specimen Preparation and Mounting of Pipe and Duct Insulation Materials to Assess Surface Burning Characteristics                        |

#### CALIFORNIA DEPARTMENT OF PUBLIC HEALTH (CDPH)

|                    |                                                                                                                                                            |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CDPH SECTION 01350 | (2017; Version 1.2) Standard Method for the Testing and Evaluation of Volatile Organic Chemical Emissions from Indoor Sources using Environmental Chambers |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|

#### FM GLOBAL (FM)

|              |                                                                                                                 |
|--------------|-----------------------------------------------------------------------------------------------------------------|
| FM APP GUIDE | (updated on-line) Approval Guide<br><a href="https://www.approvalguide.com/">https://www.approvalguide.com/</a> |
|--------------|-----------------------------------------------------------------------------------------------------------------|

#### GREEN SEAL (GS)

|       |                                     |
|-------|-------------------------------------|
| GS-36 | (2013) Adhesives for Commercial Use |
|-------|-------------------------------------|

#### MANUFACTURERS STANDARDIZATION SOCIETY OF THE VALVE AND FITTINGS INDUSTRY (MSS)

|           |                                                                                                                |
|-----------|----------------------------------------------------------------------------------------------------------------|
| MSS SP-58 | (2018) Pipe Hangers and Supports - Materials, Design and Manufacture, Selection, Application, and Installation |
|-----------|----------------------------------------------------------------------------------------------------------------|

#### MIDWEST INSULATION CONTRACTORS ASSOCIATION (MICA)

|                      |                                           |
|----------------------|-------------------------------------------|
| MICA Insulation Stds | (8th Ed) National Commercial & Industrial |
|----------------------|-------------------------------------------|

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## Insulation Standards

### NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 90A (2024) Standard for the Installation of Air Conditioning and Ventilating Systems

NFPA 90B (2024) Standard for the Installation of Warm Air Heating and Air Conditioning Systems

### SCIENTIFIC CERTIFICATION SYSTEMS (SCS)

SCS SCS Global Services (SCS) Indoor Advantage

### SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT (SCAQMD)

SCAQMD Rule 1168 (2022) Adhesive and Sealant Applications

### TECHNICAL ASSOCIATION OF THE PULP AND PAPER INDUSTRY (TAPPI)

TAPPI T403 OM (2022) Bursting Strength of Paper

### U.S. DEPARTMENT OF DEFENSE (DOD)

MIL-A-24179 (1969; Rev A; Am 2 1980; Notice 1 1987; Notice 2 2020) Adhesive, Flexible Unicellular-Plastic Thermal Insulation

MIL-PRF-19565 (1988; Rev C) Coating Compounds, Thermal Insulation, Fire- and Water-Resistant, Vapor-Barrier

### UNDERWRITERS LABORATORIES (UL)

UL 94 (2023; Reprint Jan 2024) UL Standard for Safety Tests for Flammability of Plastic Materials for Parts in Devices and Appliances

UL 723 (2020) UL Standard for Safety Test for Surface Burning Characteristics of Building Materials

UL 2818 (2022) GREENGUARD Certification Program For Chemical Emissions For Building Materials, Finishes And Furnishings

## 1.2 SYSTEM DESCRIPTION

### 1.2.1 General

Provide field-applied insulation and accessories on mechanical systems as specified herein; factory-applied insulation is specified under the piping, duct or equipment to be insulated. Furnish and install field applied insulation materials required for use on Government-furnished items as listed in the SPECIAL CONTRACT REQUIREMENTS.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

Submit the three SD types, SD-02 Shop Drawings, SD-03 Product Data, and SD-08 Manufacturer's Instructions at the same time for each system.

#### SD-02 Shop Drawings

MICA Plates; G

Pipe Insulation Systems and Associated Accessories

Duct Insulation Systems and Associated Accessories

Recycled content for insulation materials

#### SD-03 Product Data

Pipe Insulation Systems; FIO

Duct Insulation Systems; FIO

#### SD-04 Samples

Thermal Insulation; FIO

Display Samples; FIO

#### SD-07 Certificates

Indoor air quality for adhesives; FIO

#### SD-08 Manufacturer's Instructions

Pipe Insulation Systems; FIO

Duct Insulation Systems; FIO

### 1.4 CERTIFICATIONS

#### 1.4.1 Adhesives and Sealants

Provide products certified to meet indoor air quality requirements by UL 2818 (Greenguard) Gold, SCS Global Services Indoor Advantage Gold or provide certification or validation by other third-party programs that products meet the requirements of this Section. Provide current product certification documentation from certification body. When product does not have certification, provide validation that product meets the indoor air quality product requirements cited herein.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## 1.5 QUALITY ASSURANCE

### 1.5.1 Installer Qualification

Qualified installers will have successfully completed three or more similar type jobs within the last 5 years.

## 1.6 DELIVERY, STORAGE, AND HANDLING

Deliver materials in the manufacturer's unopened containers. Protect materials delivered and placed in storage from weather, humidity, dirt, dust and other contaminants. The Contracting Officer may reject insulation material and supplies that become dirty, dusty, wet, or contaminated by some other means. Attach manufacturer's stamp or label giving the name of the manufacturer and brand, and a description of the material, date codes, and approximate shelf life (if applicable) to packages or standard containers of insulation, jacket material, cements, adhesives, and coatings delivered for use, and samples required for approval. Insulation packages and containers must be asbestos free.

## PART 2 PRODUCTS

### 2.1 STANDARD PRODUCTS

Provide materials which are the standard products of manufacturers regularly engaged in the manufacture of such products and that essentially duplicate items that have been in satisfactory use for at least 2 years prior to bid opening. Submit a complete list of materials, including manufacturer's descriptive technical literature, performance data, catalog cuts, and installation instructions. Include the product number, k-value, thickness and furnished accessories including adhesives, sealants and jackets for each mechanical system requiring insulation. The product data must be copyrighted, have an identifying or publication number, and have been published prior to the issuance date of this solicitation. Submit materials furnished under this section together in a booklet.

#### 2.1.1 Insulation System

Provide insulation systems in accordance with the approved MICA National Insulation Standards plates as supplemented by this specification. Provide field-applied insulation for heating, ventilating, and cooling (HVAC) air distribution systems and piping systems that are located within, on, under, and adjacent to buildings; and for plumbing systems. Provide CFC and HCFC free insulation.

#### 2.1.2 Surface Burning Characteristics

Unless otherwise specified, insulation must have a maximum flame spread index of 25 and a maximum smoke developed index of 50 when tested in accordance with ASTM E84. Determine flame spread, and smoke developed indexes, by ASTM E84 or UL 723. Test insulation in the same density and installed thickness as the material to be used in the actual construction. Prepare and mount test specimens according to ASTM E2231.

### 2.2 MATERIALS

Provide insulation that meets or exceeds the requirements of ASHRAE 90.1. Ensure insulation exterior is cleanable, grease resistant, non-flaking and non-peeling. Provide compatible materials that do not contribute to

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

corrosion, soften, or otherwise attack surfaces to which applied in either wet or dry state. Use materials on stainless steel surfaces meeting ASTM C795 requirements. Do not use calcium silicate on chilled or cold water systems. Use asbestos free materials. Provide product recognized under UL 94 (if containing plastic) and listed in FM APP GUIDE.

#### 2.2.1 Adhesives

Provide non-aerosol adhesive products used on the interior of the building (defined as inside of the weatherproofing system) that meet either emissions requirements of CDPH SECTION 01350 (limit requirements for either office or classroom spaces regardless of space type) or VOC content requirements of SCAQMD Rule 1168 (HVAC duct sealants must meet limit requirements of "Other" category within SCAQMD Rule 1168 sealants table). Provide aerosol adhesives used on the interior of the building that meet either emissions requirements of CDPH SECTION 01350 (use the office or classroom requirements, regardless of space type) or VOC content requirements of GS-36. Provide certification or validation of indoor air quality for adhesives.

##### 2.2.1.1 Mineral Fiber Insulation Cement

Provide cement in accordance with ASTM C195.

##### 2.2.1.2 Contact Adhesive

Adhesives may be any of, but not limited to, the neoprene based, rubber based, or elastomeric type that have a maximum flame spread index of 25 and a maximum smoke developed index of 50 when tested in accordance with ASTM E84. Ensure adhesive does not adversely affect, initially or in service, the insulation to which it is applied, nor cause any corrosive effect on metal to which it is applied. Ensure that any solvent dispersing medium or volatile component of the adhesive has no objectionable odor and does not contain any benzene or carbon tetrachloride. Ensure dried adhesive does not emit nauseous, irritating, or toxic volatile matters or aerosols when the adhesive is heated to any temperature up to 212 degrees F. The dried adhesive must be nonflammable and fire resistant. Flexible Elastomeric Adhesive: Comply with MIL-A-24179, Type II, Class I. Provide product listed in FM APP GUIDE.

#### 2.2.2 Caulking

ASTM C920, Type S, Grade NS, Class 25, Use A.

#### 2.2.3 Corner Angles

Nominal 0.016 inch aluminum 1 by 1 inch with factory applied kraft backing. Aluminum must be ASTM B209, Alloy 3003, 3105, or 5005.

#### 2.2.4 Fittings

Fabricated Fittings are the prefabricated fittings for flexible elastomeric pipe insulation systems in accordance with ASTM C1710. Together with the flexible elastomeric tubes, they provide complete system integrity for retarding heat gain and controlling condensation drip from chilled-water and refrigeration systems. Flexible elastomeric, fabricated fittings provide thermal protection (0.25 k) and condensation resistance (0.05 Water Vapor Transmission factor). For satisfactory performance, use properly installed protective vapor retarder/barriers and vapor stops on



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

high relative humidity and below ambient temperature applications to reduce movement of moisture through or around the insulation to the colder interior surface.

## 2.2.5 Jackets

### 2.2.5.1 Aluminum Jackets

Provide aluminum jackets consisting of embossed sheet, 0.016 inch nominal thickness for pipe < 8 inch diameter, 0.024 inch nominal thickness for exterior ductwork; ASTM B209, Temper H14, Temper H16, Alloy 3003, or 3105. The jacket shall include a polyfilm moisture barrier consisting of three layer of coextruded film of polyethylene and Surlyn (3 mils thick) that is heat bonded to the aluminum outer jacket. Do not use corrugated aluminum jacket outdoors. Aluminum jacket securing bands must be Type 304 stainless steel, 0.015 inch thick, 1/2 inch wide for pipe under 12 inch diameter. Aluminum jacket circumferential seam bands must be 2 by 0.016 inch aluminum matching jacket material. The jacket may, at the option of the Contractor, be provided with a factory fabricated Pittsburgh or "Z" type longitudinal joint. When the "Z" joint is used, use bands at the circumferential joints that are designed by the manufacturer to seal the joints and hold the jacket in place.

### 2.2.5.2 Polyvinyl Chloride (PVC) Jackets

Polyvinyl chloride (PVC) jacket and fitting covers must have high impact strength, ultraviolet (UV) resistant rating or treatment and moderate chemical resistance with minimum thickness 0.030 inch.

### 2.2.5.3 Vapor Barrier/Vapor Retarder

Apply the following criteria to determine which system is required.

- a. On ducts, equip piping and equipment operating below 65 degrees F or located outside with a vapor barrier.
- b. Install ducts, pipes and equipment that are located inside and that always operate above 75 degrees F with a vapor retarder where required as stated in paragraph VAPOR RETARDER REQUIRED.

### 2.2.6 Vapor Retarder Required

ASTM C921, Type I, minimum puncture resistance 50 Beach units on all surfaces except concealed ductwork, where a minimum puncture resistance of 25 Beach units is acceptable. Minimum tensile strength, 35 pounds/inch width. ASTM C921, Type II, minimum puncture resistance 25 Beach units, tensile strength minimum 20 pounds/inch width. Use jackets on insulation exposed in finished areas that have white finish suitable for painting without sizing. Based on the application, insulation materials that require manufacturer or fabricator applied pipe insulation jackets are cellular glass, when all joints are sealed with a vapor barrier mastic, and mineral fiber. Ensure all non-metallic jackets have a maximum flame spread index of 25 and a maximum smoke developed index of 50 when tested in accordance with ASTM E84. Flexible elastomerics require (in addition to vapor barrier skin) vapor retarder jacketing for high relative humidity and below ambient temperature applications.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### 2.2.6.1 White Vapor Retarder All Service Jacket (ASJ)

ASJ is for use on hot/cold pipes, ducts, or equipment indoors or outdoors if covered by a suitable protective jacket. Provide product which meets all physical property and performance requirements of ASTM C1136, Type I, except a minimum burst strength of 85 psi. ASTM D2863 Limited Oxygen Index (LOI) is a minimum of 31.

In addition, do not use paper or other moisture-sensitive material for the outer exposed surface or the inner-most surface contacting the insulation. Ensure the outer exposed surface is white and has an emittance no less than 0.80. Ensure the outer exposed surface is paintable.

#### 2.2.6.2 Vapor Retarder/Vapor Barrier Mastic Coatings

##### 2.2.6.2.1 Vapor Barrier

The vapor barrier must be self adhesive (minimum 2 mils adhesive, 3 mils embossed) greater than 3 plies standard grade, silver, white, black and embossed white jacket for use on hot/cold pipes. Ensure permeability is less than 0.02 when tested in accordance with ASTM E96/E96M. Provide products meeting UL 723 or ASTM E84 flame and smoke requirements and that are UV resistant.

##### 2.2.6.2.2 Vapor Retarder

Provide fire and water resistant vapor retarder coating appropriately selected for either outdoor or indoor service. Color must be white. Ensure the water vapor permeance of the compound is in accordance with ASTM C755, Section 7.2.2, Table 2, for insulation type and service conditions. Provide nonflammable, fire resistant coating. To resist mold/mildew, provide coating meeting ASTM D5590 with 0 growth rating. Ensure coating meets MIL-PRF-19565 Type II (if selected for indoor service) and is Qualified Products Database listed. Determine all other application and service properties pursuant to ASTM C647.

#### 2.2.6.3 Vapor Barrier/Weather Barrier

Ensure the vapor barrier is greater than 3 ply self adhesive laminate -white vapor barrier jacket- superior performance (less than 0.0000 permeability when tested in accordance with ASTM E96/E96M). Provide vapor barrier meeting UL 723 or ASTM E84 25 flame and 50 smoke requirements; and UV resistant. Minimum burst strength 185 psi in accordance with TAPPI T403 OM. Tensile strength 68 lb/inch width (PSTC-1000). Provide tape as specified for laminated film vapor barrier above.

#### 2.2.7 Vapor Retarder Not Required

ASTM C921, Type II, Class D, minimum puncture resistance 50 Beach units on all surfaces except ductwork, where Type IV, maximum moisture vapor transmission 0.10, a minimum puncture resistance of 25 Beach units is acceptable. Provide jacket with a maximum flame spread index of 25 and a maximum smoke developed index of 50 when tested in accordance with ASTM E84.

#### 2.2.8 Insulation Bands

Provide 1/2 inch wide for piping 3/4 inch wide for ductwork; 26 gauge stainless steel insulation bands.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 2.2.9 Sealants

Choose sealants from the butyl polymer type, the styrene-butadiene rubber type, or the butyl type of sealants. Provide sealants with a maximum permeance of 0.02 perms based on Procedure B for ASTM E96/E96M, and a maximum flame spread index of 25 and a maximum smoke developed index of 50 when tested in accordance with ASTM E84.

## 2.3 PIPE INSULATION SYSTEMS

Conform insulation materials to Table 1 and minimum insulation thickness as listed in Table 2 and meet or exceed the requirements of ASHRAE 90.1 - SI. Limit pipe insulation materials to those listed herein and meeting the following requirements:

### 2.3.1 Recycled Materials

Provide insulation materials containing the following minimum percentage of recycled material content by weight:

Rock Wool: 75 percent slag of weight  
Fiberglass: 20 percent glass cullet

Provide data identifying percentage of recycled content for insulation materials.

### 2.3.2 Aboveground Cold Pipeline ( -30 to 60 deg. F)

Provide insulation for outdoor, indoor, exposed or concealed applications, as follows:

#### 2.3.2.1 Flexible Elastomeric Cellular Insulation

Closed-cell, foam- or expanded-rubber materials containing anti-microbial additive, complying with ASTM C534/C534M, Grade 1, Type I or II. Type I, Grade 1 for tubular materials. Type II, Grade 1, for sheet materials. Ensure Type I and II have vapor retarder/vapor barrier skin on one or both sides of the insulation, and require an additional exterior vapor retarder covering for high relative humidity and below ambient temperature applications.

#### 2.3.2.2 Mineral Fiber Insulation with Integral Wicking Material (MFIWM)

ASTM C547. Install in accordance with manufacturer's instructions. Do not use in applications exposed to outdoor ambient conditions in climatic zones 1 through 4.

### 2.3.3 Aboveground Hot Pipeline (Above 60 deg. F)

Provide insulation for outdoor, indoor, exposed or concealed applications meeting the following requirements. Supply the insulation with manufacturer's recommended factory-applied jacket/vapor barrier.

#### 2.3.3.1 Mineral Fiber

ASTM C547, Types I, II or III, supply the insulation with manufacturer's recommended factory-applied jacket.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### 2.3.3.2 Flexible Elastomeric Cellular Insulation

Closed-cell, foam- or expanded-rubber materials containing anti-microbial additive, complying with ASTM C534/C534M, Grade 1, Type I or II to 220 degrees F service. Type I for tubular materials. Type II for sheet materials.

### 2.4 DUCT INSULATION SYSTEMS

#### 2.4.1 Interior Duct Insulation

##### 2.4.1.1 Mineral Fiber

ASTM C547, Types I, II or III, supply the insulation with manufacturer's recommended factory-applied jacket.

##### 2.4.1.2 Rigid Insulation

Calculate the minimum thickness in accordance with ASHRAE 90.1.

##### 2.4.1.3 Blanket Insulation

Calculate minimum thickness in accordance with ASHRAE 90.1.

#### 2.4.2 Kitchen Exhaust Ductwork Insulation

Factory fabricated double-wall round ductwork and coordinated through kitchen hood manufacturer. Inner duct section wall shall be constructed of 0.036 inch thick, 430 type stainless steel and be available in diameters 5 inch through 36 inches. Outer Duct section wall shall be constructed of 430 stainless steel at a minimum of 0.024 inch thickness. Duct shall include 2 layers of Super Wool 607 Plus or Insulfrax Elite Blanket between the inner and outer wall. Duct sections shall be held together by the means of a formed V clamp. V clamps shall be of the hex-head type with flanged stops and tapered "lead in" threads. Duct joints shall be sealed with 3M Fire Barrier 2000+. Duct wall assembly shall be tested and listed at 3/4" or zero inch clearance, according to classifications.

#### 2.4.3 Duct Insulation Jackets

##### 2.4.3.1 All-Purpose Jacket

Provide insulation with insulation manufacturer's standard reinforced fire-retardant jacket with or without integral vapor barrier as required by the service. In exposed locations, provide jacket with a white surface suitable for field painting.

##### 2.4.3.2 Metal Jackets

##### 2.4.3.2.1 Aluminum Jackets

ASTM B209, Temper H14, minimum thickness of 0.024 inch, with factory-applied polyfilm moisture barrier consisting of three layers of coextruded film of polyethylene and Surlyn (3 mils thick) that is heat bonded to the aluminum outer jacket. Surface shall be embossed and painted with an exterior grade white enamel. Provide stainless steel bands, minimum width of 3/4 inch.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### 2.4.4 Weatherproof Duct Insulation

Provide ASTM C552, cellular glass thermal insulation, and weatherproofing as specified in manufacturer's instruction. Multi-ply, Polymeric Blend Laminate Jacketing: Construction of laminate designed to provide UV resistance, high puncture, tear resistance and an excellent WVT rate.

### PART 3 EXECUTION

#### 3.1 APPLICATION - GENERAL

Apply insulation to unheated and uncooled piping and equipment. Do not compress flexible elastomeric cellular insulation at joists, studs, columns, ducts, and hangers. The insulation must not pull apart after a one hour period; replace any insulation found to pull apart after one hour.

##### 3.1.1 Installation

Except as otherwise specified, install material in accordance with the manufacturer's written instructions. Do not apply insulation materials until tests specified in other sections of this specification are completed. Remove material such as rust, scale, dirt and moisture from surfaces to receive insulation. Keep insulation clean and dry. Do not remove insulation from its shipping containers until the day it is ready to use and return to like containers or equally protect from dirt and moisture at the end of each workday. Thoroughly clean insulation that becomes dirty prior to use. If insulation becomes wet or if cleaning does not restore the surfaces to like new condition, reject the insulation, and immediately remove from the jobsite. Stagger joints on multi layer insulation. Mix mineral fiber thermal insulating cement with demineralized water when used on stainless steel surfaces. Install insulation, jacketing and accessories in accordance with MICA Insulation Stds plates except where modified herein or on the drawings.

##### 3.1.2 Firestopping

Where pipes and ducts pass through fire walls, fire partitions, above grade floors, and fire rated chase walls, seal the penetration with fire stopping materials as specified in Section 07 84 00 FIRESTOPPING. The protection of ducts at point of passage through firewalls must be in accordance with NFPA 90A and/or NFPA 90B. All other penetrations, such as piping, conduit, and wiring, through firewalls must be protected with a material or system of the same hourly rating that is listed by UL, FM, or a NRTL.

##### 3.1.3 Painting and Finishing

Paint as specified in Section 09 90 00 PAINTS AND COATINGS.

##### 3.1.4 Installation of Flexible Elastomeric Cellular Insulation

Install flexible elastomeric cellular insulation with seams and joints sealed with rubberized contact adhesive. Do not use flexible elastomeric cellular insulation on surfaces greater than 220 degrees F. Stagger seams when applying multiple layers of insulation. Protect insulation exposed to weather and not shown to have vapor barrier weatherproof jacketing with two coats of UV resistant finish or PVC or metal jacketing as recommended by the manufacturer after the adhesive is dry and cured.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### 3.1.4.1 Adhesive Application

Apply a brush coating of adhesive to both butt ends to be joined and to both slit surfaces to be sealed. Allow the adhesive to set until dry to touch but tacky under slight pressure before joining the surfaces. Ensure insulation seals at seams and joints are not capable of being pulled apart one hour after application. Replace insulation that can be pulled apart one hour after installation.

#### 3.1.4.2 Adhesive Safety Precautions

Use natural cross-ventilation, local (mechanical) pickup, and/or general area (mechanical) ventilation to prevent an accumulation of solvent vapors, keeping in mind the ventilation pattern must remove any heavier-than-air solvent vapors from lower levels of the workspaces. Gloves and spectacle-type safety glasses are recommended in accordance with safe installation practices.

#### 3.1.5 Welding

Welding is not permitted on piping, duct or without written approval of the Contracting Officer. The capacitor discharge welding process may be used for securing metal fasteners to duct.

#### 3.1.6 Pipes/Ducts/ That Require Insulation

Insulation is required on all pipes, ducts, or except for omitted items as specified.

### 3.2 PIPE INSULATION SYSTEMS INSTALLATION

#### 3.2.1 Pipe Insulation

##### 3.2.1.1 General

Install pipe insulation on aboveground hot and cold pipeline systems as specified below to form a continuous thermal retarder/barrier, including straight runs, fittings and appurtenances unless specified otherwise. Install full length units of insulation using a single cut piece to complete a run. Do not use cut pieces or scraps abutting each other. Omit pipe insulation on the following:

- a. Pipe used solely for fire protection.
- b. Chromium plated pipe to plumbing fixtures. However, for fixtures used by the physically handicapped, insulate the hot water supply and drain, including the trap, where exposed.
- c. Sanitary drain lines.
- d. Air chambers.
- e. Adjacent insulation.
- f. ASME stamps.
- g. Access plates of fan housings.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

h. Cleanouts or handholes.

### 3.2.1.2 Pipes Passing Through Walls, Roofs, and Floors

Provide continuous pipe insulation through the sleeve.

Provide an aluminum jacket or vapor barrier/weatherproofing self adhesive jacket (minimum 2 mils adhesive, 3 mils embossed) less than 0.0000 permeability, greater than 3 ply standard grade, silver, white, black and embossed with factory applied moisture retarder over the insulation wherever penetrations require sealing.

#### 3.2.1.2.1 Penetrate Interior Walls

Provide aluminum jacket or vapor barrier/weatherproofing - self adhesive jacket (minimum 2 mils adhesive, 3 mils embossed) less than 0.0000 permeability, greater than 3 plies standard grade, silver, white, black and embossed which extends 2 inches beyond either side of the wall and secure on each end with a band.

#### 3.2.1.2.2 Penetrating Floors

Extend the aluminum jacket from a point below the backup material to a point 10 inches above the floor with one band at the floor and one not more than 1 inch from the end of the aluminum jacket.

#### 3.2.1.2.3 Penetrating Waterproofed Floors

Extend the aluminum jacket from below the backup material to a point 2 inches above the flashing with a band 1 inch from the end of the aluminum jacket.

#### 3.2.1.2.4 Penetrating Exterior Walls

Continue the aluminum jacket required for pipe exposed to weather through the sleeve to a point 2 inches beyond the interior surface of the wall.

#### 3.2.1.2.5 Penetrating Roofs

Insulate pipe as required for interior service to a point flush with the top of the flashing and sealed with flashing sealant. Tightly butt the insulation for exterior application to the top of flashing and interior insulation. Extend the exterior aluminum jacket 2 inches down beyond the end of the insulation to form a counter flashing. Seal the flashing and counter flashing underneath with metal jacketing/flashing sealant.

#### 3.2.1.2.6 Hot Water Pipes Supplying Lavatories or Other Similar Heated Service

Terminate the insulation on the backside of the finished wall. Protect the insulation termination with two coats of vapor barrier coating with a minimum total thickness of 1/16 inch applied with glass tape embedded between coats (if applicable). Extend the coating out onto the insulation 2 inches and seal the end of the insulation. Overlap glass tape seams 1 inch. Caulk the annular space between the pipe and wall penetration with approved fire stop material. Cover the pipe and wall penetration with a properly sized (well fitting) escutcheon plate. Ensure the escutcheon plate overlaps the wall penetration at least 3/8 inches.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### 3.2.1.2.7 Domestic Cold Water Pipes Supplying Lavatories or Other Similar Cooling Service

Terminate the insulation on the finished side of the wall (i.e., insulation must cover the pipe throughout the wall penetration). Protect the insulation with two coats of weather barrier mastic (breather emulsion type weatherproof mastic impermeable to water and permeable to air) with a minimum total thickness of 1/16 inch. Extend the mastic out onto the insulation 2 inches and seal the end of the insulation. The annular space between the outer surface of the pipe insulation and caulk the wall penetration with an approved fire stop material having vapor retarder properties. Cover the pipe and wall penetration with a properly sized (well fitting) escutcheon plate. Ensure the escutcheon plate overlaps the wall penetration by at least 3/8 inches.

#### 3.2.1.3 Pipes Passing Through Hangers

Ensure insulation, whether hot or cold application, is continuous through hangers. Support all horizontal pipes 2 inches and smaller on hangers with the addition of a Type 40 protection shield to protect the insulation in accordance with MSS SP-58. Whenever insulation shows signs of being compressed, or when the insulation or jacket shows visible signs of distortion at or near the support shield, install insulation inserts as specified below for piping larger than 2 inches, or factory insulated hangers (designed with a load bearing core) can be used.

##### 3.2.1.3.1 Horizontal Pipes Larger Than 2 Inches at 60 Degrees F and Above

Supported on hangers in accordance with MSS SP-58, and Section 22 00 00 PLUMBING, GENERAL PURPOSE.

##### 3.2.1.3.2 Vertical Pipes

Supported with either Type 8 or Type 42 riser clamps with the addition of two Type 40 protection shields in accordance with MSS SP-58 covering the 360-degree arc of the insulation. Install an insulation insert of cellular glass or calcium silicate between each shield and the pipe. Ensure the insert covers the 360-degree arc of the pipe. Provide inserts that are the same thickness as the insulation, and extend 2 inches on each end beyond the protection shield. When insulation inserts are required in accordance with the above, and the insulation thickness is less than 1 inch, wooden or cork dowels or blocks may be installed between the pipe and the shield to prevent the hanger from crushing the insulation, as an option instead of installing insulation inserts. Ensure the insulation jacket is continuous over the wooden dowel, wooden block, or insulation insert. Support the vertical weight of the pipe with hangers located in a horizontal section of the pipe. When the pipe riser is longer than 30 feet, support the weight of the pipe additionally with hangers in the vertical run of the pipe that are directly clamped to the pipe, penetrating the pipe insulation. Use insulated hangers and seal the insulation jacket as indicated herein for anchors in a similar service.

##### 3.2.1.3.3 Inserts

Covered with a jacket material of the same appearance and quality as the adjoining pipe insulation jacket, overlap the adjoining pipe jacket 1-1/2 inches, and seal as required for the pipe jacket. Use jacket material to cover inserts in flexible elastomeric cellular insulation conforming to



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

ASTM C1136, Type 1, and is allowed to be of a different material than the adjoining insulation material.

#### 3.2.1.4 Flexible Elastomeric Cellular Pipe Insulation

Use tubular form flexible elastomeric cellular pipe insulation for pipe sizes 6 inches and less. Grade 1, Do not stretch Type II sheet insulation used on pipes larger than 6 inches around the pipe. On pipes larger than 12 inches, adhere the insulation directly to the pipe on the lower 1/3 of the pipe. Stagger seams when applying multiple layers of insulation. Insulate sweat fittings with miter-cut pieces the same size as on adjacent piping. Insulate screwed fittings with sleeved fitting covers fabricated from miter-cut pieces and overlap and seal to the adjacent pipe insulation. Type II requires an additional exterior vapor retarder/barrier covering for high relative humidity and below ambient temperature applications.

#### 3.2.1.5 Pipes in high abuse areas.

In high abuse areas such as janitor closets and traffic areas in equipment rooms, kitchens, and mechanical rooms, utilize welded PVC , aluminum or flexible laminate cladding (comprised of elastomeric, plastic or metal foil laminate) laminated self-adhesive (minimum 2 mils adhesive, 3 mils embossed) vapor barrier/weatherproofing jacket, - less than 0.0000 permeability; (greater than 3 ply, standard grade, silver, white, black and embossed) aluminum jackets. Protect pipe insulation to the 8 foot level. Other areas that specifically require protection to the 8 foot level are Apparatus Bay.

#### 3.2.1.6 Pipe Insulation Material and Thickness

Pipe insulation materials must be as listed in Table 1 and must meet or exceed the requirements of ASHRAE 90.1.

| TABLE 1                                                                   |                               |                 |      |       |             |
|---------------------------------------------------------------------------|-------------------------------|-----------------|------|-------|-------------|
| Insulation Material for Piping                                            |                               |                 |      |       |             |
| Service                                                                   |                               |                 |      |       |             |
|                                                                           | Material                      | Specification   | Type | Class | VR/VB Req'd |
| Heating Hot Water Supply & Return, Heated Oil (Max 250 F)                 |                               |                 |      |       |             |
|                                                                           | Mineral Fiber                 | ASTM C547       | I    | 1     | No          |
| Cold Domestic Water Piping, Makeup Water & Drinking Fountain Drain Piping |                               |                 |      |       |             |
|                                                                           | Mineral Fiber                 | ASTM C547       | I    | 1     | Yes         |
| Hot Domestic Water Supply & Recirculating Piping (Max 200 F)              |                               |                 |      |       |             |
|                                                                           | Mineral Fiber                 | ASTM C547       | I    | 1     | No          |
|                                                                           | Flexible Elastomeric Cellular | ASTM C534/C534M | I    |       | No          |
| Refrigerant Piping (35 degrees F nominal)                                 |                               |                 |      |       |             |

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

| TABLE 1                                                                                            |                               |                 |      |       |             |
|----------------------------------------------------------------------------------------------------|-------------------------------|-----------------|------|-------|-------------|
| Insulation Material for Piping                                                                     |                               |                 |      |       |             |
| Service                                                                                            |                               |                 |      |       |             |
|                                                                                                    | Material                      | Specification   | Type | Class | VR/VB Req'd |
|                                                                                                    | Flexible Elastomeric Cellular | ASTM C534/C534M | I    |       | Yes         |
| Exposed Lavatory Drains, Exposed Domestic Water Piping & Drains to Areas for Handicapped Personnel |                               |                 |      |       |             |
|                                                                                                    | Mineral Fiber                 | ASTM C547       | I    | 1     | No          |
| Horizontal Roof Drain Leaders (Including Underside of Roof Drain Fittings)                         |                               |                 |      |       |             |
|                                                                                                    | Mineral Fiber                 | ASTM C547       | I    | 1     | Yes         |
| Condensate Drain Located Inside Building                                                           |                               |                 |      |       |             |
|                                                                                                    | Mineral Fiber                 | ASTM C547       | I    | 1     | Yes         |
| Note: VR/VB = Vapor Retarder/Vapor Barrier                                                         |                               |                 |      |       |             |

| TABLE 2                                                                                                                                                                       |               |                           |        |        |      |           |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|---------------------------|--------|--------|------|-----------|
| Piping Insulation Thickness (inch)<br>Do not use integral wicking material in Chilled water applications exposed to outdoor ambient conditions in climatic zones 1 through 4. |               |                           |        |        |      |           |
| Service                                                                                                                                                                       |               |                           |        |        |      |           |
|                                                                                                                                                                               | Material      | Tube And Pipe Size (inch) |        |        |      |           |
|                                                                                                                                                                               |               | <1                        | 1-<1.5 | 1.5-<4 | 4-<8 | > or = >8 |
| Heating Hot Water Supply & Return, Heated Oil (Max 250 F)                                                                                                                     |               |                           |        |        |      |           |
|                                                                                                                                                                               | Mineral Fiber | 1.5                       | 1.5    | 2      | 2    | 2         |
| Cold Domestic Water Piping, Makeup Water & Drinking Fountain Drain Piping                                                                                                     |               |                           |        |        |      |           |
|                                                                                                                                                                               | Mineral Fiber | 1.5                       | 1.5    | 1.5    | 1.5  | 1.5       |
| Hot Domestic Water Supply & Recirculating Piping (Max 200 F)                                                                                                                  |               |                           |        |        |      |           |
|                                                                                                                                                                               | Mineral Fiber | 1                         | 1      | 1      | 1.5  | 1.5       |
| Refrigerant Piping )                                                                                                                                                          |               |                           |        |        |      |           |

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

| TABLE 2                                                                                                                                                                       |                               |                           |        |        |      |           |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|---------------------------|--------|--------|------|-----------|
| Piping Insulation Thickness (inch)<br>Do not use integral wicking material in Chilled water applications exposed to outdoor ambient conditions in climatic zones 1 through 4. |                               |                           |        |        |      |           |
| Service                                                                                                                                                                       |                               |                           |        |        |      |           |
|                                                                                                                                                                               | Material                      | Tube And Pipe Size (inch) |        |        |      |           |
|                                                                                                                                                                               |                               | <1                        | 1-<1.5 | 1.5-<4 | 4-<8 | > or = >8 |
|                                                                                                                                                                               | Flexible Elastomeric Cellular | 1                         | 1      | 1      | N/A  | N/A       |
| Exposed Lavatory Drains, Exposed Domestic Water Piping & Drains to Areas for Handicapped Personnel                                                                            |                               |                           |        |        |      |           |
|                                                                                                                                                                               | Mineral Fiber                 | 0.5                       | 0.5    | 0.5    | 0.5  | 0.5       |
| Horizontal Roof Drain Leaders (Including Underside of Roof Drain Fittings)                                                                                                    |                               |                           |        |        |      |           |
|                                                                                                                                                                               | Mineral Fiber                 | 1.5                       | 1.5    | 1.5    | 1.5  | 1.5       |
| Condensate Drain Located Inside Building                                                                                                                                      |                               |                           |        |        |      |           |
|                                                                                                                                                                               | Mineral Fiber                 | 1                         | 1      | 1      | N/A  | N/A       |

### 3.2.2 Aboveground Cold Pipelines

Insulate the following cold pipelines for minus 30 to plus 60 degrees F in accordance with Table 2 except those piping listed in subparagraph Pipe Insulation in PART 3 as to be omitted. This includes but is not limited to the following:

- a. Make-up water.
- b. Horizontal and vertical portions of interior roof drains.
- c. Refrigerant lines.
- f. Air conditioner condensate drains.
- h. Exposed lavatory drains and domestic water lines serving plumbing fixtures for handicap persons.
- i. Domestic cold and chilled drinking water.

#### 3.2.2.1 Insulation Material and Thickness

Determine insulation thickness for cold pipelines using Table 2.

#### 3.2.2.2 Field applied Jacket

Where indicated cover interior insulation with a field applied seal welded

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

PVC jacket, white, for use with Mineral Fiber Insulated Pipe. For insulation outside the building, to be protected with an aluminum jacket, 3 mils embossed product, less than 0.0000 permeability, standard grade, Embossed White, install the insulation and vapor retarder jacket as specified herein. In high abuse areas such as janitor closets and traffic areas in equipment rooms, kitchens, and mechanical rooms, provide PVC jackets, less than 0.0000 permeability, standard grade, for pipe insulation to the 8 ft level. Other areas that specifically require protection to the 8 ft level are Apparatus Bay.

#### 3.2.2.3 Installing Insulation for Straight Runs Hot and Cold Pipe

Apply insulation to the pipe with tight butt joints. Seal all butted joints and ends with joint sealant and seal with a vapor retarder coating, greater than 3 ply laminate jacket - less than 0.0000 perm adhesive tape or PVDC adhesive tape.

##### 3.2.2.3.1 Longitudinal Laps of the Jacket Material

Overlap not less than 1-1/2 inches. Provide butt strips 3 inches wide for circumferential joints.

##### 3.2.2.3.2 Laps and Butt Strips

Secure with adhesive and staple on 4 inch centers if not factory self-sealing. If staples are used, seal in accordance with paragraph STAPLES below. Note that staples are not required with cellular glass systems.

##### 3.2.2.3.3 Factory Self-Sealing Lap Systems

May be used when the ambient temperature is between 40 and 120 degrees F during installation. Install the lap system in accordance with manufacturer's recommendations. Use a stapler only if specifically recommended by the manufacturer. Where gaps occur, replace the section or repair the gap by applying adhesive under the lap and then stapling.

##### 3.2.2.3.4 Staples

Coat all staples, including those used to repair factory self-seal lap systems, with a vapor retarder coating or PVDC adhesive tape or greater than 3 ply laminate jacket - 0.0000 perm adhesive tape. Coat all seams, except those on factory self-seal systems, with vapor retarder coating or PVDC adhesive tape or greater than 3 ply laminate jacket - less than 0.0000 perm adhesive tape.

##### 3.2.2.3.5 Breaks and Punctures in the Jacket Material

Patch by wrapping a strip of jacket material around the pipe and secure it with adhesive, staple, and coat with vapor retarder coating or PVDC adhesive tape or greater than 3 ply laminate jacket - less than 0.0000 perm adhesive tape. Extend the patch not less than 1-1/2 inches past the break.

##### 3.2.2.3.6 Penetrations Such as Thermometers

Fill the voids in the insulation and seal with vapor retarder coating or PVDC adhesive tape or greater than 3 ply laminate jacket - less than 0.0000 perm adhesive tape.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 3.2.2.3.7 Flexible Elastomeric Cellular Pipe Insulation

Install by slitting the tubular sections and applying them onto the piping or tubing. Alternately, whenever possible slide un-slit sections over the open ends of piping or tubing. Secure all seams and butt joints and seal with adhesive. When using self seal products, secure only the butt joints with adhesive. Push insulation on the pipe, never pulled. Stretching of insulation may result in open seams and joints. Clean cut all edges. Rough or jagged edges of the insulation are not be permitted. Use proper tools such as sharp knives. Do not stretch Grade 1, Type II sheet insulation around the pipe when used on pipe larger than 6 inches. On pipes larger than 12 inches, adhere sheet insulation directly to the pipe on the lower 1/3 of the pipe.

### 3.2.2.4 Insulation for Fittings and Accessories

- a. Butt pipe insulation tightly to the insulation of the fittings and accessories. Seal the butted joints and ends with joint sealant and seal with a vapor retarder coating or PVDC adhesive tape or greater than 3 ply laminate jacket - less than 0.0000 perm adhesive tape.
- b. Place precut or preformed insulation around all fittings and accessories and conform to MICA plates except as modified herein: 5 for anchors; 10, 11, and 13 for fittings; 14 for valves; and 17 for flanges and unions. Insulation must be the same insulation as the pipe insulation, including same density, thickness, and thermal conductivity. Where precut/preformed is unavailable, rigid preformed pipe insulation sections may be segmented into the shape required. Use insulation of the same thickness and conductivity as the adjoining pipe insulation. If nesting size insulation is used, overlap the insulation 2 inches or one pipe diameter. Elbows insulated using segments must conform to MICA Tables 12.20 "Mitered Insulation Elbow". Submit a booklet containing completed MICA Insulation Stds plates detailing each insulating system for each pipe, duct, insulating system, after approval of materials and prior to applying insulation.
  - (1) Ensure MICA plates detail the materials to be installed and the specific insulation application. Submit all MICA plates required showing the entire insulating system, including plates required to show insulation penetrations, vessel bottom and top heads, legs, and skirt insulation as applicable. Present all variations of insulation systems including locations, materials, vaporproofing, jackets and insulation accessories.
  - (2) If the Contractor elects to submit detailed drawings instead of edited MICA Plates, ensure the detail drawings are technically equivalent to the edited MICA Plate submittal.
- c. Upon completion of insulation installation on flanges, unions, valves, anchors, fittings and accessories, terminations, seams, joints and insulation not protected by factory vapor retarder jackets or PVC fitting covers must be protected with PVDC or greater than 3 ply laminate jacket - less than 0.0000 perm adhesive tape or two coats of vapor retarder coating with a minimum total thickness of 1/16 inch, applied with glass tape embedded between coats. Overlap tap seams 1 inch. Extend the coating out onto the adjoining pipe insulation 2 inches. Protect fabricated insulation with a factory vapor retarder

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

jacket with either greater than 3 ply laminate jacket - less than 0.0000 perm adhesive tape, standard grade, silver, white, black and embossed or PVDC adhesive tape or two coats of vapor retarder coating with a minimum thickness of 1/16 inch and with a 2 inch wide glass tape embedded between coats. Where fitting insulation butts to pipe insulation, seal the joints with a vapor retarder coating and a 4 inch wide ASJ tape which matches the jacket of the pipe insulation.

- d. Insulate anchors attached directly to the pipe for a sufficient distance to prevent condensation but no less than 6 inches from the insulation surface.
- e. Mark insulation to show the location of unions, strainers, and check valves.

#### 3.2.2.5 PVC Fitting Covers

Provide factory precut or premolded insulation segments under the PVC fitting covers for elbows. Use insulation segments which are the same insulation as the pipe insulation including same density, thickness, and thermal conductivity. Secure the covers by PVC vapor retarder tape, adhesive, seal welding or with tacks made for securing PVC covers. Seal seams in the cover, and tacks and laps to adjoining pipe insulation jacket, with vapor retarder tape to ensure that the assembly has a continuous vapor seal.

#### 3.2.3 Aboveground Hot Pipelines

##### 3.2.3.1 General Requirements

Insulate all hot pipe lines above 60 degrees F, except those piping listed in subparagraph Pipe Insulation in PART 3 as to be omitted, in accordance with Table 2. This includes but is not limited to the following:

- a. Domestic hot water supply & re-circulating system.
- d. Hot water heating.

Cover insulation, in accordance with manufacturer's recommendations, with a factory applied Type I jacket or field applied seal welded PVC.

##### 3.2.3.2 Insulation for Fittings and Accessories

Butt pipe insulation tightly to the insulation of the fittings and accessories. Seal butted joints and ends with joint sealant. Mark insulation to show the location of unions, strainers, check valves and other components that would otherwise be hidden from view by the insulation.

###### 3.2.3.2.1 Precut or Preformed

Place precut or preformed insulation around all fittings and accessories. Use the same insulation as the pipe insulation, including same density, thickness, and thermal conductivity.

###### 3.2.3.2.2 Rigid Preformed

Where precut/preformed is unavailable, rigid preformed pipe insulation sections may be segmented into the shape required. Use insulation that is

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

the same thickness and conductivity as the adjoining pipe insulation. If nesting size insulation is used, do not overlap insulation 2 inches or one pipe diameter. Elbows insulated using segments must conform to MICA Tables 12.20 "Mitered Insulation Elbow".

#### 3.2.4 Piping Exposed to Weather

Insulate and jacket piping exposed to weather as specified for the applicable service inside the building. After this procedure, apply a laminated self-adhesive (minimum 2 mils adhesive, 3 mils embossed) vapor barrier/weatherproofing jacket - less than 0.0000 permeability (greater than 3 ply, standard grade, silver, white, black and embossed aluminum jacket, stainless steel).

Treat flexible elastomeric cellular insulation exposed to weather in accordance with paragraph INSTALLATION OF FLEXIBLE ELASTOMERIC CELLULAR INSULATION in PART 3.

##### 3.2.4.1 Aluminum Jacket

The jacket for hot piping may be factory applied. Overlap the jacket no less than 2 inches at longitudinal and circumferential joints and secure with bands at no more than 12 inch centers. Overlap longitudinal joints down to shed water and locate at 4 or 8 o'clock positions. Seal joints on piping 60 degrees F and below with metal jacketing/flashing sealant while overlapping to prevent moisture penetration. Where jacketing on piping 60 degrees F and below abuts an un-insulated surface, caulk joints to prevent moisture penetration. Seal joints on piping above 60 degrees F with a moisture retarder.

##### 3.2.4.2 Insulation for Fittings

Insulate and finish flanges, unions, valves, fittings, and accessories as specified for the applicable service. Apply two coats of breather emulsion type weatherproof mastic (impermeable to water, permeable to air) recommended by the insulation manufacturer with glass tape embedded between coats. Overlap tap no less than 1 inch and the adjoining aluminum jacket no less than 2 inches. Factory preformed aluminum jackets may be used in lieu of the above. Provide molded PVC fitting covers when PVC jackets are used for straight runs of pipe. Provide PVC fitting covers that have adhesive welded joints and are weatherproof laminated self-adhesive (minimum 2 mils adhesive, 3 mils embossed) vapor barrier/weatherproofing jacket - less than 0.0000 permeability, (greater than 3 ply, standard grade, silver, white, black and embossed, and UV resistant).

##### 3.2.4.3 PVC Jacket

Provide ultraviolet resistant PVC jacket that is adhesive welded weather tight with manufacturer's recommended adhesive. Include provision for thermal expansion.

#### 3.3 DUCT INSULATION SYSTEMS INSTALLATION

Except for oven hood exhaust duct insulation, install corner angles on external corners of insulation on ductwork in exposed finished spaces before covering with jacket. Air conditioned spaces are defined as those spaces directly supplied with cooled conditioned air (or provided with a

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

cooling device such as a fan-coil unit) and heated conditioned air (or provided with a heating device such as a unit heater, radiator or convector).

### 3.3.1 Duct Insulation Minimum Thickness

Duct insulation minimum thickness in accordance with Table 4.

| Table 4 - Minimum Duct Insulation (inches) |     |
|--------------------------------------------|-----|
| Cold Air Ducts                             | 2.0 |
| Fresh Air Intake Ducts                     | 1.5 |
|                                            |     |
| Warm Air Ducts                             | 2.0 |
| Fresh Air Intake Ducts                     | 1.5 |

### 3.3.2 Insulation and Vapor Retarder/Vapor Barrier for Cold Air Duct

Provide insulation and vapor retarder/vapor barrier for the following cold air ducts and associated equipment.

- a. Supply ducts.
- e. Plenums.
- n. Ducts exposed to weather.
- o. Combustion air intake ducts.

Use insulation for rectangular ducts that is flexible type where concealed, minimum density 3/4 pcf, and rigid type where exposed, minimum density 3 pcf. Provide insulation for both concealed or exposed round/oval ducts that is flexible type, minimum density 3/4 pcf or a semi rigid board, minimum density 3 pcf, formed or fabricated to a tight fit, edges beveled and joints tightly butted and staggered. Provide insulation for all exposed ducts with either a white, paint-able, factory-applied Type I jacket or a field applied vapor retarder/vapor barrier jacket coating finish as specified. Ensure the total field applied dry film thickness is approximately 1/16 inch. Provide insulation on all concealed duct with a factory-applied Type I or II vapor retarder/vapor barrier jacket. Continue duct insulation through sleeves and prepare openings except firewall penetrations. Duct insulation terminating at fire dampers, must be continuous over the damper collar and retaining angle of fire dampers, which are exposed to unconditioned air and which may be prone to condensate formation. Provide duct insulation and vapor retarder/vapor barrier to cover the collar, neck, and un-insulated surfaces of diffusers, registers and grills. Apply vapor retarder/vapor barrier materials to form a complete unbroken vapor seal over the insulation. Seal sheet metal duct in accordance with Section 23 30 00 HVAC AIR DISTRIBUTION.

#### 3.3.2.1 Installation on Concealed Duct

- a. For rectangular, oval or round ducts, attach flexible insulation by applying adhesive around the entire perimeter of the duct in 6 inch wide strips on 12 inch centers.



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

- b. For rectangular and oval ducts, 24 inches and larger, additionally secure insulation to bottom of ducts using mechanical fasteners. Space fasteners on 16 inch centers and no more than 16 inches from duct corners.
- c. For rectangular, oval and round ducts, provide mechanical fasteners on sides of duct risers for all duct sizes. Space fasteners on 16 inch centers and no more than 16 inches from duct corners.
- d. Impale insulation on the mechanical fasteners (self stick pins) where used and press thoroughly into the adhesive. Take care to ensure vapor retarder/vapor barrier jacket joints overlap 2 inches. Do not compress insulation to a thickness less than that specified. Carry insulation over standing seams and trapeze-type duct hangers.
- e. Where mechanical fasteners are used, install self-locking washers and trim and bend the pin over.
- f. Secure jacket overlaps with staples and tape as necessary to ensure a secure seal. Coat staples, tape and seams with a brush coat of vapor retarder coating or PVDC adhesive tape or greater than 3 ply laminate (minimum 2 mils adhesive, 3 mils embossed) - less than 0.0000 perm adhesive tape.
- g. Cover breaks in the jacket material with patches of the same material as the vapor retarder jacket. Do not extend patches less than 2 inches beyond the break or penetration in all directions and secure with tape and staples. Seal staples and tape joints with a brush coat of vapor retarder coating or PVDC adhesive tape or greater than 3 ply laminate (minimum 2 mils adhesive, 3 mils embossed) - less than 0.0000 perm adhesive tape.
- h. At jacket penetrations such as hangers, thermometers, and damper operating rods, fill voids in the insulation and seal the penetration with a brush coat of vapor retarder coating or PVDC adhesive tape greater than 3 ply laminate (minimum 2 mils adhesive, 3 mils embossed) - less than 0.0000 perm adhesive tape.
- i. Seal insulation terminations and pin punctures and flash with a reinforced vapor retarder coating finish or tape with a brush coat of vapor retarder coating.. Ensure the coating overlaps the adjoining insulation and un-insulated surface 2 inches. Extend pin puncture coatings 2 inches from the puncture in all directions.
- j. Where insulation standoff brackets occur, extend insulation under the bracket and terminate the jacket at the bracket.

#### 3.3.2.2 Installation on Exposed Duct Work

- a. For rectangular ducts, secure rigid insulation to the duct by mechanical fasteners on all four sides of the duct, space no more than 12 inches apart and no more than 3 inches from the edges of the insulation joints. Provide a minimum of two rows of fasteners for each side of duct 12 inches and larger. Provide one row for each side of duct less than 12 inches. Provide mechanical fasteners that are corrosion resistant as G60 coated galvanized steel, and indefinitely sustain a 50 lb tensile dead load test perpendicular to the duct wall.
- b. Form duct insulation with minimum jacket seams. Fasten each piece of

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

rigid insulation to the duct using mechanical fasteners. When the height of projections is less than the insulation thickness, bring insulation up to standing seams, reinforcing, and other vertical projections and do not carry over. Continue vapor retarder/barrier jacket across seams, reinforcing, and projections. When height of projections is greater than the insulation thickness, carry over insulation and jacket. Apply insulation with joints tightly butted. Neatly bevel insulation around name plates and access plates and doors.

- c. Impale insulation on the fasteners; install self-locking washers and trim and bend the pin over.
- d. Seal joints in the insulation jacket with a 4 inch wide strip of tape. Seal taped seams with a brush coat of vapor retarder coating.
- e. Cover breaks and ribs or standing seam penetrations in the jacket material with a patch of the same material as the jacket. Do not extend patches less than 2 inches beyond the break or penetration and secure with tape and staple. Seal staples and joints with a brush coat of vapor retarder coating.
- f. At jacket penetrations such as hangers, thermometers, and damper operating rods, fill the voids in the insulation and seal the penetrations with a flashing sealant.
- g. Seal and flash insulation terminations and pin punctures with a reinforced vapor retarder coating finish. Ensure coating overlaps the adjoining insulation and un-insulated surface 2 inches. Extend pin puncture coatings 2 inches from the puncture in all directions.
- h. Insulate oval and round ducts, flexible type, with factory Type I jacket insulation with minimum density of 3/4 pcf, attach in accordance with MICA standards.

### 3.3.3 Insulation for Warm Air Duct

Provide insulation and vapor barrier for the following warm air ducts and associated equipment:.

- a. Supply ducts.
- e. Plenums.
- n. Ducts exposed to weather.
- o. Exhaust ducts passing through concealed spaces exhausting conditioned air.

Provide insulation for rectangular ducts that is flexible type where concealed, and rigid type where exposed. Provide insulation on exposed ducts with a white, paint-able, factory-applied Type II jacket, or finish with adhesive finish. Use flexible type insulation for round ducts, with a factory-applied Type II jacket. Provide insulation on concealed duct with a factory-applied Type II jacket. Accomplish adhesive finish where indicated to be used by applying two coats of adhesive with a layer of glass cloth embedded between the coats. Ensure total dry film thickness is approximately 1/16 inch. Continue duct insulation through sleeves and prepare openings. Terminate duct insulation at fire dampers and flexible connections.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 3.3.3.1 Installation on Concealed Duct

- a. For rectangular, oval and round ducts, attach insulation by applying adhesive around the entire perimeter of the duct in 6 inch wide strips on 12 inch centers.
- b. For rectangular and oval ducts 24 inches and larger, secure insulation to the bottom of ducts using mechanical fasteners. Space fasteners on 18 inch centers and no more than 18 inches from duct corner.
- c. For rectangular, oval and round ducts, provide mechanical fasteners on sides of duct risers for all duct sizes. Space fasteners on 18 inch centers and no more than 18 inches from duct corners.
- d. Impale insulation on the mechanical fasteners where used. Do not compress insulation to a thickness less than that specified. Carry insulation over standing seams and trapeze-type hangers.
- e. Install self-locking washers where mechanical fasteners are used and trim and bend the pin over.
- f. Do not overlap insulation jacket less than 2 inches at joints and secure the lap and staple on 4 inch centers.

### 3.3.3.2 Installation on Exposed Duct

- a. For rectangular ducts, secure the rigid insulation to the duct using mechanical fasteners on all four sides of the duct, space no more than 16 inches apart and no more than 6 inches from the edges of the insulation joints. Provide a minimum of two rows of fasteners for each side of duct 12 inches and larger and a minimum of one row for each side of duct less than 12 inches.
- b. Form duct insulation with factory-applied jacket with minimum jacket seams, and fasten each piece of rigid insulation to the duct using mechanical fasteners. When the height of projection is less than the insulation thickness, bring insulation up to standing seams, reinforcing, and other vertical projections and do not carry over the projection. Continue jacket across seams, reinforcing, and projections. Where the height of projections is greater than the insulation thickness, carry insulation and jacket over the projection.
- c. Impale insulation on the fasteners; install self-locking washers and trim and bend the pin over.
- d. Seal joints on jacketed insulation with a 4 inch wide strip of tape and brush with vapor retarder coating.
- e. Cover breaks and penetrations in the jacket material with a patch of the same material as the jacket. Extend patches no less than 2 inches beyond the break or penetration and secure with adhesive and staple.
- f. Seal insulation terminations and pin punctures with tape and brush with vapor retarder coating.
- g. Insulate oval and round ducts, flexible type, with factory Type I jacket insulation, minimum density of 3/4 pcf attach by staples spaced no more than 16 inches and no more than 6 inches from the degrees of

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

joints. Seal joints in accordance with item "d." above.

#### 3.3.4 Ducts Handling Air for Dual Purpose

For air handling ducts for dual purpose below and above 60 degrees F, insulate ducts as specified for cold air duct.

#### 3.3.5 Duct Test Holes

After duct systems have been tested, adjusted, and balanced, repair breaks in the insulation and jacket in accordance with the applicable section of this specification for the type of duct insulation to be repaired.

#### 3.3.6 Duct Exposed to Weather

##### 3.3.6.1 Installation

Insulate and finish ducts exposed to weather as specified for the applicable service for exposed duct inside the building. After the above is accomplished, further finish the insulation as detailed in the following subparagraphs.

##### 3.3.6.2 Round Duct

Laminated self-adhesive (minimum 2 mils adhesive, 3 mils embossed) vapor barrier/weatherproofing jacket - Less than 0.0000 permeability, (greater than 3 ply, standard grade, silver, white, black and embossed or greater than 8 ply, heavy duty, white and natural) membrane must be applied overlapping material by 3 inches no bands or caulking needed - see manufacturer's recommended installation instructions. Aluminum jacket with factory applied moisture retarder must be applied with the joints lapped no less than 3 inches and secured with bands located at circumferential laps and at no more than 12 inch intervals throughout. Lap horizontal joints down to shed water and located at 4 or 8 o'clock position. Seal joints with metal jacketing sealant to prevent moisture penetration. Where jacketing abuts an un-insulated surface, seal joints with metal jacketing sealant.

##### 3.3.6.3 Fittings

Finish fittings and other irregular shapes as specified for rectangular ducts.

##### 3.3.6.4 Rectangular Ducts

Apply two coats of weather barrier mastic reinforced with fabric or mesh for outdoor application to the entire surface. Ensure each coat of weatherproof mastic has a minimum thickness of 1/16 inch. Ensure exterior is a metal jacketing applied for mechanical abuse and weather protection, and secure with screws or vapor barrier/weatherproofing jacket less than 0.0000 permeability greater than 3 ply, standard grade, silver, white, black, and embossed or greater than 8 ply, heavy duty white and natural. Apply membrane overlapping material by 3 inches. No bands or caulking needed-see manufacturing recommend installation instructions.

-- End of Section --

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## SECTION 23 08 00

### COMMISSIONING OF MECHANICAL AND PLUMBING SYSTEMS 05/23

#### PART 1 GENERAL

Building Commissioning is a systematic, quality-focused process for enhancing the delivery of a project that focuses on verifying and documenting that all of the commissioned systems and assemblies are planned, designed, installed, tested, operated, and maintained to meet the project requirements. The purpose is to reduce the cost and performance risks associated with delivering facilities projects, and to increase value to owners, occupants, and users. Comply with specification Section 01 91 00.15 BUILDING COMMISSIONING.

#### 1.1 SEQUENCING AND SCHEDULING

Complete the following prior to starting Functional Performance Tests of mechanical systems:

- a. All equipment and systems completed, cleaned, flushed, disinfected, calibrated, tested, and operate in accordance with contract documents and construction plans and specifications
- b. Final DALT Report submitted and approved in accordance with Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC
- c. Pre-final Testing, Adjusting, and Balancing Report submitted in accordance with Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC
- e. The Certificate of Readiness submitted and approved in accordance with Section 01 46 00.00 06 TOTAL BUILDING COMMISSIONING
- f. Air Leakage Test Reports and Diagnostic Test Reports submitted and approved in accordance with Section 01 46 00.00 06 TOTAL BUILDING COMMISSIONING
- g. Tests, Flushing, and Disinfection in accordance with Section 22 00 00 PLUMBING, GENERAL PURPOSE

#### 1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Test Equipment; G

SD-06 Test Reports

Pipe Flushing, Testing, And Water Treatment Reports; G

Completed Pre-Functional Checklists; G

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

Seasonal Test Report; G

Full-Load Test Report; G

Post-Construction Trend Log Report; G

SD-07 Certificates

Certificate Of Readiness; G

### 1.3 ACCESSIBILITY REQUIREMENTS

Equipment, systems, and devices for commissioned systems must be accessible. Make necessary modifications if systems and devices are not accessible for inspections and testing.

Assist commissioning team in testing by removing equipment covers, opening access panels, and other required activities that assist with visual oversight. Furnish ladders, flashlights, meters, gauges, or other inspection equipment as necessary.

### 1.4 COORDINATION

Refer to Section 01 46 00.00 06 TOTAL BUILDING COMMISSIONING for requirements pertaining to coordination during the commissioning process. Coordinate with the in accordance with Section 01 46 00.00 06 TOTAL BUILDING COMMISSIONING and in accordance with the Commissioning Plan to schedule inspections as required to support the commissioning process. Furnish additional information requested by the. Coordinate scheduling of Functional Performance Testing with the commissioning team. Provide plans, reports, notes, and other documentation to the as specified in the commissioning plan, as it is completed.

### 1.5 PIPE FLUSHING, TESTING, AND WATER TREATMENT REPORTS

Test requirements are specified in Division 22 and 23 piping Sections. Prepare a pipe system cleaning, flushing, and hydrostatic testing log. Provide cleaning, flushing, testing, and water treatment log and final reports.

Include the following in the pipe system cleaning, flushing, and hydrostatic testing log:

- a. Minimum flushing water velocity.
- b. Water treatment reports.
- c. Tracking checklist for managing and ensuring that all pipe sections have been cleaned, flushed, hydrostatically tested, and chemically treated.

### 1.6 CERTIFICATE OF READINESS

Submit Certificate of Readiness documentation in accordance with Section 01 91 00.15 BUILDING COMMISSIONING for all equipment and systems including start-up reports; completed Pre-Functional Checklists; Testing, Adjusting, and Balancing (TAB) Report; Issues Log; HVAC Controls Start-Up Reports. Do not schedule Functional Performance Tests for the system until the

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

Certificate of Readiness for that system receives approval by the Government. The and the Mechanical, Electrical, Controls, and TAB subcontractor representatives must sign and date the Certificate of Readiness.

## PART 2 PRODUCTS

### 2.1 TEST EQUIPMENT

Provide all testing equipment required to perform testing for the systems to be commissioned, except for equipment specific to and used by TAB as required by Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC. Provide a sufficient quantity of two-way radios for each subcontractor. Submit list of Test Equipment and instrumentation to be used for testing including equipment/instrument identification number, equipment application or planned use, manufacturer, make, model, and serial number, and calibration history with certificates. Also list special equipment and proprietary tools specific to a piece of equipment required for testing.

#### 2.1.1 Proprietary Equipment

Provide manufacturer's proprietary test equipment and software required by any equipment manufacturer for programming and start-up, whether specified or not. Provide manufacturer test equipment, demonstrate its use, and assist in the commissioning process as needed. Provide data logging equipment and software required to test equipment.

#### 2.1.2 Calibration and Accuracy

Comply with equipment manufacturer's test equipment calibration procedures and intervals. Recalibrate test instruments immediately after instruments have been repaired resulting from being dropped or damaged. Affix calibration tags to test instruments. Furnish calibration records to Contracting Officer upon request.

Provide all testing equipment of sufficient quality and accuracy to test and measure system performance with the tolerances specified. Unless otherwise noted, the following minimum requirements apply: Provide temperature sensors and digital thermometers with a certified calibration within the past year to an accuracy of 0.5 degrees F and a resolution of plus or minus 0.1 degrees F. Provide pressure sensors with an accuracy of plus or minus 2.0 percent of the value range being measured (not full range of meter) and calibrated within the last year.

## PART 3 EXECUTION

### 3.1 MEETINGS

Attend all meetings in accordance with Section 01 91 00.15 BUILDING COMMISSIONING.

Provide timely updates on construction schedule changes to allow the to execute commissioning process efficiently. Notify Contracting Officer of anticipated construction delays to commissioning activities not yet performed or not yet scheduled.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 3.2 COMMISSIONING CONSTRUCTION OBSERVATION CHECKLISTS

Commissioning construction observation checklists include Pre-Functional Checklists, Integrated Systems Test Checklists, and Functional Performance Test Checklists. Provide commissioning construction observation checklists for the Interim and Final Construction Phase Commissioning Plan in accordance with Section 01 46 00.00 06 TOTAL BUILDING COMMISSIONING.

#### 3.2.1 Pre-Functional Checklists

#### 3.2.2 Functional Performance Test Checklists

#### 3.2.3 Integrated Systems Test Checklists

### 3.3 PRE-FUNCTIONAL CHECKS

Pre-Functional Checks are a type of Commissioning Inspection in accordance with Section 01 46 00.00 06 TOTAL BUILDING COMMISSIONING. Complete one Pre-Functional Checklist for each individual item of equipment or system for each system required to be commissioned including, but not limited to, ductwork, piping, equipment, fixtures, and controls. Include manufacturer start-up checklists associated with equipment with the submission of the Pre-Functional Checklists. Provide manufacturer's installation manual for each type of unit. Indicate commissioning team member inspection and validation of each Pre-Functional Checklist item by initials. Validation of each Pre-Functional Checklist item by each team member indicates that item conforms to the contract documents and validated design in their area of responsibility. Commissioning Specialist validation of each Pre-Functional Checklist item indicates that each item has been installed correctly and in accordance with contract documents. Required commissioning team members for Pre-Functional Checks includes the CxC, Mechanical Commissioning Specialist, sub-contractor representative for each trade responsible for construction/installation of the checklist item, and the TAB representative (for items impacting TAB). Submit the initialed and Completed Pre-Functional Checklists no later than 7 calendar days after completion of inspection of all checklists items for each system.

### 3.4 STARTUP AND INITIAL CHECKOUT

Document start-up and initial testing procedures, and include in the Completed Pre-Functional Checklists submittal, including:

- a. Startup tests and factory testing reports.
- b. Manufacturer's representative start-up, operating, troubleshooting and maintenance procedures.
- c. Additional documentation necessary for third party certification programs.
- d. Perform and clearly document system operational checks and quality control checks as they are completed, and providing a copy to the commissioning team.
- e. Correct deficiencies and sign the Certificate of Readiness for each system before functional performance testing



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 3.5 DUCT AIR LEAKAGE TEST (DALT) REPORT REVIEW AND VERIFICATION

#### 3.5.1 Duct Air Leakage Test (DALT) Report Review

All deficiencies must be resolved prior to DALT Report approval.

#### 3.5.2 Duct Air Leakage Test (DALT) Report Verification

All deficiencies must be resolved prior to DALT Report approval.

### 3.6 TESTING, ADJUSTING, AND BALANCING REPORT REVIEW AND VERIFICATION

#### 3.6.1 Testing, Adjusting, and Balancing (TAB) Report Review

All deficiencies must be resolved prior to TAB Report approval.

#### 3.6.2 Testing, Adjusting, and Balancing (TAB) Report Verification

All deficiencies must be resolved prior to TAB Report approval.

### 3.7 HVAC CONTROLS START-UP AND PERFORMANCE VERIFICATION TEST REVIEW

All deficiencies must be resolved prior to final acceptance.

### 3.8 FUNCTIONAL PERFORMANCE AND INTEGRATED SYSTEMS TESTING

Functional Performance and Integrated Systems Testing are a type of Commissioning Testing in accordance with Section 01 46 00.00 06 TOTAL BUILDING COMMISSIONING. Conduct Functional Performance Testing in accordance with Section 01 46 00.00 06 TOTAL BUILDING COMMISSIONING and requirements in this Section. Prior to Functional Performance Testing, complete all prerequisites in accordance with paragraph SEQUENCING AND SCHEDULING. Demonstrate that all system components have been installed, that each control device and item of equipment operates, and that the systems operate and perform, including interactive operation between systems, in accordance with contract documents.

Perform Integrated Systems Tests only after the Functional Performance Tests for each associated system are completed with all deficiencies resolved and after the related Functional Performance Test Checklists have been signed by each commissioning team member.

#### 3.8.1 Test Scheduling and Coordination

Schedule and conduct Initial Functional Performance Tests. Develop and implement means of artificial loading to demonstrate, to a reasonable level of confidence, the ability of the HVAC systems to handle peak seasonal loads. Schedule Functional Performance Tests for each system only after the Certificate of Readiness has been approved by the Government for the system. Correct all deficiencies identified through any prior review, inspection, or test activity before the start of Functional Performance Tests.

Functional Performance Tests and Integrated Systems Tests must be performed with the CxC present. Government reserves the right to witness all tests. Coordinate test schedule with Government representatives.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 3.8.2 Preparation

Put equipment and systems into operation and continue operation during each working day of functional performance and integrated systems testing, as required. Verify temperature and pressure taps in accordance with Contract Documents. Provide a pressure/temperature plug at each water sensor which is an input point to control system.

Perform minor adjustments to equipment and systems during Functional Performance Tests as deemed necessary by the commissioning team. Where calibrated DDC sensors cannot be used to record test data, provide measuring instruments, logging devices, and data acquisition equipment to record data for the complete range of test data for the required test period.

### 3.8.3 Testing Procedures

Provide all necessary materials and system modifications to produce the necessary flows, pressures, temperatures, and other conditions necessary to execute the test according to the specified conditions. At completion of the test, return the affected building equipment and systems to their pre-test condition.

Follow the Functional Performance Test from the approved Final Construction Phase Commissioning Plan. Perform Functional Performance Tests for each item of equipment and each system required to be commissioned. Verify all sensor calibrations, control responses, safeties, interlocks, operating modes, sequences of operation, capacities, and all other performance requirements comply with contract, regardless of the specific items listed within the checklists provided. In general, testing must progress from equipment or components to subsystems to systems to interlocks and connections between systems. Commissioning Specialists are responsible for determining the order of components and systems to be tested.

Indicate validation of each item of equipment and systems tested by signature of each commissioning team member for each test. The Quality Control Representative, Commissioning Specialists, and Contracting Officer's Representative, if present, must indicate validation after the equipment and systems are free of deficiencies.

### 3.8.4 Simulating Conditions

Functional performance testing is conducted by simulating conditions at control devices to initiate a control system response. Before testing, calibrate all sensors, transducers and devices. Over-writing control input values through the control system is not acceptable unless approved by the Contracting Officer. Perform each test under conditions that simulate actual conditions as close as is practically possible. Specific examples of simulating conditions are provided below. Do not simulate conditions when damage to the system or building may result.

- a. When varying static pressures inside ductwork cannot be simulated within the duct, and where a sensor signals the controls system to initiate sequences at various duct static pressures, it is acceptable to simulate the various pressures with a Pneumatic Squeeze-Bulb Type Signaling Device with gauge temporarily attached to the sensing tube leading to the transmitter. It is not acceptable to reset the various set-points, nor to simulate an electric analog signal (unless approved

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

as noted above).

- b. Dirty filter pressure drops can be simulated by partially blocking filter face.
- c. Freeze-stat safeties can be simulated by packing portion of sensor with ice.
- d. High outside air temperatures can be simulated with a hair blower.
- e. Raising entering cooling coil temperatures by activating a heating/preheat coil can be used to simulate entering cooling coil conditions.
- f. Do not use signal generators to simulate sensor signals unless approved by the Contracting Officer, as noted above, for special cases.
- g. Control set points can be altered. For example, to see the air conditioning compressor lockout work at an outside air temperature below 55 degrees F, when the outside air temperature is above 55 degrees F, temporarily change the lockout set point to be 0 degrees F above the current outside air temperature. Caution: Set points are not to be raised or lowered to a point to cause damage to the components, systems, or the building structure and contents.
- h. Test duct mounted smoke detectors in accordance with the manufacturer's recommendations. Perform the tests with air system at minimum airflow condition.
- i. Test current sensing relays used for fan and pump status signals to control system to indicate unit failure and run status by resetting the set point on the relay to simulate a lost belt or unit failure while the unit is running. Confirm that the failure alarm was generated and received at the control system. After the test is conducted, return the set point to its original set-point or a set-point as indicated by the Contracting Officer.

### 3.8.5 Manufacturer's Representative

Provide a factory trained representative authorized by the equipment manufacturer to perform Functional Performance Testing for the following equipment:

Boilers

Packaged Direct-Expansion Refrigeration Equipment.

Packaged and Split Computer Room Air Conditioners (CRAC)

Packaged Air Compressors

Water Quality and Chemical Treatment Systems

Ensure the test representative reviews, approves, and signs the completed field test report. Include person's name with signatures.

### 3.8.6 Integrated Systems Tests

Follow the Integrated Systems Test Checklists from the approved Final

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

Construction Phase Commissioning Plan. Integrated Systems Tests must be performed for the interactive operation between HVAC systems and other systems such as fire protection systems, smoke control systems, or back-up electrical supply. Verify correct interactive operation, acceptable speed of response, and other contract requirements for both normal and failure modes. Examples of Integrated Systems Tests include the correct operation of HVAC systems during emergency system activation or correct operation of uninterruptible power supplies or energy generators and connected systems.

### 3.8.7 Sample Strategy

Complete a Functional Performance Test Checklist for each item of equipment or system to be tested. During testing, Government representatives may select the specific equipment or system to be tested for sample sizes less than 100 percent.

#### 3.8.7.1 100 Percent Sample Procedures

Systems or equipment for which 100 percent sample size are tested fail if one or more of the test procedures results in discovery of a deficiency and the deficiency cannot be resolved within 5 minutes during the test.

Re-test to the extent necessary to confirm that the deficiencies have been corrected without negatively impacting the performance of the rest of the system.

#### 3.8.7.2 Less than 100 Percent Sample Procedures

Randomly test each sample group of identical equipment. If 10 percent of the units in the first sample fail the functional performance tests, test a second sample group, the same size as the first sample group. The second sample must not include any units from the first sample group.

If 10 percent of the units in the second sample fail, test all remaining units. If at any point frequent failures occur the CxC may stop the testing and require the contractor to perform and document a checkout of the remaining units prior to continuing functional testing.

### 3.9 INTEGRATED SYSTEMS TEST

Perform Integrated Systems Tests only after the Functional Performance Tests for each associated system are completed with all deficiencies resolved and after the related Functional Performance Test Checklists have been signed by each commissioning team member.

### 3.10 RETESTING REQUIREMENTS

Abort tests if any deficiency prevents successful completion of the test or if any required commissioning team member is not present for the test. Re-test only after all deficiencies identified during the original tests have been corrected.

Contracting Officer may withhold payment equivalent to lost time, re-testing, and aborted tests. These costs may include salary, travel costs, and per diem for Government team members. Correct deficiencies as identified by the commissioning team and retest the systems to be commissioned.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 3.11 SYSTEM ACCEPTANCE

Systems may be partially accepted prior to seasonal testing if they comply with all construction contract and accepted design requirements that can be tested during initial Functional Performance Tests. All test procedures must be successful completed prior to full systems acceptance.

### 3.12 SEASONAL TESTS

Perform Initial Functional Performance Tests as soon as all contract work is completed, but prior to facility turnover, regardless of the season.

In addition to the Initial Functional Performance Tests, perform Functional Performance Tests of HVAC systems during peak heating and cooling seasons during outdoor air condition design extremes. Schedule Seasonal Functional Performance Tests in coordination with the Contracting Officer. Submit Seasonal Test Report within 14 days of test completion. Include seasonal test report in the Updated Final Commissioning Report as specified in Section 01 91 00.15 BUILDING COMMISSIONING

Execute seasonal functional performance testing, witnessed by the Contracting Officer. Correct deficiencies and make adjustments to O&M manuals and as-built drawings for applicable issues identified in any seasonal testing.

### 3.13 TRAINING

### 3.14 COMMISSIONING REPORT

Include all completed Pre-Functional Checklists and Functional Performance and Integrated Systems Test Checklists in the Commissioning Report as specified in Section 01 91 00.15 BUILDING COMMISSIONING. Include the approved TAB Report.

### 3.15 POST-CONSTRUCTION ENDURANCE TEST

Perform an Endurance Test in accordance with the paragraph in Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC once during the peak heating season and once during the peak cooling season during outdoor air condition extremes with the exception that network bandwidth usage measurement and recording is not required. Use the Temporary Trending Hardware, if necessary, in accordance with Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC.

Include the Post-Construction Trend Log Reports in the Updated Final Commissioning Report as specified in Section 01 91 00.15 BUILDING COMMISSIONING

-- End of Section --

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## SECTION 23 09 00

### INSTRUMENTATION AND CONTROL FOR HVAC

**02/19, CHG 3: 05/21**

#### PART 1 GENERAL

##### 1.1 SUMMARY

Provide a complete Direct Digital Control (DDC) system, except for the Front End which is specified in Section 25 10 10 UTILITY MONITORING AND CONTROL (UMCS) FRONT END AND INTEGRATION, suitable for the control of the heating, ventilating and air conditioning (HVAC) and other building-level systems as indicated and shown and in accordance with Section 23 09 13 INSTRUMENTATION AND CONTROL DEVICES FOR HVAC, Section 23 09 93 SEQUENCES OF OPERATION FOR HVAC CONTROL, Section 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS for BACnet or Niagara BACnet systems, and other referenced Sections.

##### 1.1.1 Proprietary Systems

##### 1.1.1.1 Proprietary Systems Exempted From Open Protocol Requirements

The following systems are specifically exempted from the open protocol requirements of Section 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS:

- a. A simple split (DX) system consisting of a single indoor unit and a single outdoor unit from the same manufacturer.
- b. Systems in Table I (previously approved by the designer in accordance with UFC 3-410-02).

| TABLE I: Systems Approved to Use Proprietary Communications |                                                |                                                                                       |
|-------------------------------------------------------------|------------------------------------------------|---------------------------------------------------------------------------------------|
| System                                                      | Type (Multi-Split/VRF or Chiller/Boiler Plant) | Proprietary Multi-Split Engineering Tool Software Required (for Multi-Split/VRF only) |
|                                                             |                                                |                                                                                       |
|                                                             |                                                |                                                                                       |

- c. A system (not already shown Table I) of multiple boilers or multiple chillers communicating with a proprietary network for which an approved request has been obtained and for which: all units are from the same manufacturer, they are all co-located in the same room, the network connecting them is fully contained in that room, and the units are operating using a common "plant" sequence of operation which stages the units in a manner that requires operational parameters be shared between them and which cannot be accomplished with a single lead-lag command from a third-party controller.

##### 1.1.1.2 Implementation of Proprietary Systems

For proprietary systems exempted from open protocol requirements, a

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

proprietary network and DDC hardware communicating via proprietary protocol are permitted. For these systems a building control network meeting the requirements of Section 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS must also be provided, along with a gateway or interface to connect the proprietary system to the open building control network.

The proprietary system gateway or interface must provide the required functionality as shown on the points schedule. Scheduling, alarming, trending, overrides, network inputs, network outputs and other protocol related requirements must be met on the open protocol control system as specified in Section 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS.

#### 1.1.1.3 Proprietary Multi-Split Engineering Tool Software

For each permitted proprietary systems in Table 1 shown as requiring Proprietary Multi-Split Engineering Tool Software, provide the software needed to replace a unit and configure the replacement. Submit hard copies of the software user manuals with the software submittal.

Submit Proprietary Multi-Split Engineering Tool Software on CD-ROM as a Technical Data Package. Submit two hard copies of the software user manual for each piece of software.

#### 1.1.2 System Requirements

Provide systems meeting the requirements this Section and other Sections referenced by this Section, and which have the following characteristics:

- a. The system implements the control sequences of operation shown in the Contract Drawings using DDC hardware to control mechanical and electrical equipment
- b. The system meet the requirements of this specification as a stand-alone system and does not require connection to any other system.
- c. Control sequences reside in DDC hardware in the building. The building control network is not dependent upon connection to a Utility Monitoring and Control System (UMCS) Front End or to any other system for performance of control sequences. To the greatest extent practical, the hardware performs control sequences without reliance on the building network, unless otherwise pre-approved by the Contracting Officer.
- d. The hardware is installed such that individual control equipment can be replaced by similar control equipment from other equipment manufacturers with no loss of system functionality.
- e. All necessary documentation, configuration information, programming tools, programs, drivers, and other software are licensed to and otherwise remain with the Government such that the Government or their agents are able to perform repair, replacement, upgrades, and expansions of the system without subsequent or future dependence on the Contractor, Vendor or Manufacturer.
- f. Sufficient documentation and data, including rights to documentation and data, are provided such that the Government or their agents can execute work to perform repair, replacement, upgrades, and expansions

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

of the system without subsequent or future dependence on the Contractor, Vendor or Manufacturer.

- g. Hardware is installed and configured such that the Government or their agents are able to perform repair, replacement, and upgrades of individual hardware without further interaction with the Contractor, Vendor or Manufacturer.
- h. All Niagara Framework components have an unrestricted interoperability license with a Niagara Compatibility Statement (NiCS) following the Tridium Open NiCS Specification and have a value of "ALL" for "Station Compatibility In", "Station Compatibility Out", "Tool Compatibility In" and "Tool Compatibility Out". Note that this will result in the following entries in the license file:
 

```
accept.station.in="*"
accept.station.out="*"
accept.wb.in="*"
accept.wb.out="*"

```

#### 1.1.3 End to End Accuracy

Select products, install and configure the system such that the maximum error of a measured value as read from the DDC Hardware over the network is less than the maximum allowable error specified for the sensor or instrumentation.

#### 1.1.4 Verification of Dimensions

After becoming familiar with all details of the work, verify all dimensions in the field, and advise the Contracting Officer of any discrepancy before performing any work.

#### 1.1.5 Drawings

The Government will not indicate all offsets, fittings, and accessories that may be required on the drawings. Carefully investigate the mechanical, electrical, and finish conditions that could affect the work to be performed, arrange such work accordingly, and provide all work necessary to meet such conditions.

### 1.2 RELATED SECTIONS

Related work specified elsewhere:

- a. Section 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS for BACnet systems with or without Niagara Framework.
- b. Section 23 09 13 INSTRUMENTATION AND CONTROL DEVICES FOR HVAC
- d. Section 25 08 10 UTILITY MONITORING AND CONTROL SYSTEMS TESTING
- e. Section 25 10 10 UTILITY MONITORING AND CONTROL SYSTEMS (UMCS) FRONT END AND INTEGRATION
- f. Section 25 05 11 CYBERSECURITY FOR FACILITY-RELATED CONTROL SYSTEMS
- g.



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 1.3 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

#### AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS (ASHRAE)

ASHRAE 135 (2020; Interpretation 1-8 2021; Errata 1-2 2021; Addenda CD 2021; Addenda BV-CE 2022; Interpretation 9-12 2022; Interpretation 13-17 2023; Addenda BV-CF 2023; Errata 3 2023) BACnet—A Data Communication Protocol for Building Automation and Control Networks

ASHRAE FUN IP (2021) Fundamentals Handbook, I-P Edition

#### INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE C62.41 (1991; R 1995) Recommended Practice on Surge Voltages in Low-Voltage AC Power Circuits

#### NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA 250 (2020) Enclosures for Electrical Equipment (1000 Volts Maximum)

#### NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 (2023; ERTA 4 2023) National Electrical Code

NFPA 90A (2024) Standard for the Installation of Air Conditioning and Ventilating Systems

#### TRIDIUM, INC (TRIDIUM)

Niagara Framework (2012) NiagaraAX User's Guide

Tridium Open NiCS (2005) Understanding the NiagaraAX Compatibility Statement (NiCS)

#### U.S. DEPARTMENT OF DEFENSE (DOD)

UFC 3-410-02 (2018; with Change 2, 2021) Direct Digital Control for HVAC and Other Building Control Systems

#### UNDERWRITERS LABORATORIES (UL)

UL 5085-3 (2006; Reprint Jan 2022) UL Standard for Safety Low Voltage Transformers - Part 3: Class 2 and Class 3 Transformers

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### 1.4 DEFINITIONS

The following list of definitions includes terms used in Sections referenced by this Section and are included here for completeness. The definitions contained in this Section may disagree with how terms are defined or used in other documents, including documents referenced by this Section. The definitions included here are the authoritative definitions for this Section and all Sections referenced by this Section.

After each term the protocol related to that term is included in parenthesis.

##### 1.4.1 Alarm Generation (All protocols)

Alarm Generation is the monitoring of a value, comparison of the value to alarm conditions and the creation of an alarm when the conditions set for the alarm are met. Note that this does NOT include delivery of the alarm to the final destination (such as a user interface) - see paragraph ALARM ROUTING in Section 25 10 10 UTILITY MONITORING AND CONTROL SYSTEM (UMCS) FRONT END AND INTEGRATION.

##### 1.4.2 Building Automation and Control Network (BACnet) (BACnet)

The term BACnet is used in two ways. First meaning the BACnet Protocol Standard - the communication requirements as defined by ASHRAE 135 including all annexes and addenda. The second to refer to the overall technology related to the ASHRAE 135 protocol.

##### 1.4.3 BACnet Advanced Application Controller (B-AAC) (BACnet)

A hardware device BTL Listed as a B-AAC, which is required to support BACnet Interoperability Building Blocks (BIBBs) for scheduling and alarming, but is not required to support as many BIBBs as a B-BC.

##### 1.4.4 BACnet Application Specific Controller (B-ASC) (BACnet)

A hardware device BTL Listed as a B-ASC, with fewer BIBB requirements than a B-AAC. It is intended for use in a specific application.

##### 1.4.5 BACnet Building Controller (B-BC) (BACnet)

A hardware device BTL Listed as a B-BC. A general-purpose, field-programmable device capable of carrying out a variety of building automation and control tasks including control and monitoring via direct digital control (DDC) of specific systems and data storage for trend information, time schedules, and alarm data. Like the other BTL Listed controller types (B-AAC, B-ASC etc.) a B-BC device is required to support the server ("B") side of the ReadProperty and WriteProperty services, but unlike the other controller types it is also required to support the client ("A") side of these services. Communication between controllers requires that one of them support the client side and the other support the server side, so a B-BC is often used when communication between controllers is needed.

##### 1.4.6 BACnet Broadcast Management Device (BBMD) (BACnet)

A communications device, typically combined with a BACnet router. A BBMD forwards BACnet broadcast messages to BACnet/IP devices and other BBMDs connected to the same BACnet/IP network. Each IP subnet that is part of a

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

BACnet/IP network must have at least one BBMD. Note there are additional restrictions when multiple BBMDs share an IP subnet.

#### 1.4.7 BACnet/IP (BACnet)

An extension of BACnet, Annex J, defines the use of a reserved UDP socket to transmit BACnet messages over IP networks. A BACnet/IP network is a collection of one or more IP subnets that share the same BACnet network number. See also paragraph BACNET BROADCAST MANAGEMENT DEVICE.

#### 1.4.8 BACnet Internetwork (BACnet)

Two or more BACnet networks, connected with BACnet routers. In a BACnet Internetwork, there exists only one message path between devices.

#### 1.4.9 BACnet Interoperability Building Blocks (BIBBs) (BACnet)

A BIBB is a collection of one or more ASHRAE 135 Services intended to define a higher level of interoperability. BIBBs are combined to build the BACnet functional requirements for a device in a specification. Some BIBBs define additional requirements (beyond requiring support for specific services) in order to achieve a level of interoperability. For example, the BIBB DS-V-A (Data Sharing-View-A), which would typically be used by a front-end, not only requires the client to support the ReadProperty Service, but also provides a list of data types (Object / Properties) which the client must be able to interpret and display for the user.

In the BIBB shorthand notation, -A is the client side and -B is the server side.

| The following is a list of some BIBBs used by this or referenced Sections: |                                                     |
|----------------------------------------------------------------------------|-----------------------------------------------------|
| DS-COV-A                                                                   | Data Sharing-Change of Value (A side)               |
| DS-COV-B                                                                   | Data Sharing-Change of Value (B side)               |
| NM-RC-B                                                                    | Network Management-Router Configuration (B side)    |
| DS-RP-A                                                                    | Data Sharing-Read Property (A side)                 |
| DS-RP-B                                                                    | Data Sharing-Read Property (B side)                 |
| DS-RPM-A                                                                   | Data Sharing-Read Property Multiple (A Side)        |
| DS-RPM-B                                                                   | Data Sharing-Read Property Multiple (B Side)        |
| DS-WP-A                                                                    | Data Sharing-Write Property (A Side)                |
| DM-TS-B                                                                    | Device Management-Time Synchronization (B Side)     |
| DM-UTC-B                                                                   | Device Management-UTC Time Synchronization (B Side) |
| DS-WP-B                                                                    | Data Sharing-Write Property (B side)                |

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

|                                                                            |                                                         |
|----------------------------------------------------------------------------|---------------------------------------------------------|
| The following is a list of some BIBBs used by this or referenced Sections: |                                                         |
| SCHED-E-B                                                                  | Scheduling-External (B side)                            |
| DM-OCD-B                                                                   | Device Management-Object Creation and Deletion (B side) |
| AE-N-I-B                                                                   | Alarm and Event-Notification Internal (B Side)          |
| AE-N-E-B                                                                   | Alarm and Event-Notification External (B Side)          |
| T-VMT-I-B                                                                  | Trending-Viewing and Modifying Trends Internal (B Side) |
| T-VMT-E-B                                                                  | Trending-Viewing and Modifying Trends External (B Side) |

#### 1.4.10 BACnet Network (BACnet)

In BACnet, a portion of the control Internetwork consisting of one or more segments connected by repeaters. Networks are separated by routers.

#### 1.4.11 BACnet Operator Display (B-OD) (BACnet)

A basic operator interface with limited capabilities relative to a B-OWS. It is not intended to perform direct digital control. A B-OD profile could be used for LCD devices, displays affixed to BACnet devices, handheld terminals or other very simple user interfaces.

#### 1.4.12 BACnet Segment (BACnet)

One or more physical segments interconnected by repeaters (ASHRAE 135).

#### 1.4.13 BACnet Smart Actuator (B-SA) (BACnet)

A simple actuator device with limited resources intended for specific applications.

#### 1.4.14 BACnet Smart Sensor (B-SS) (BACnet)

A simple sensing device with limited resources.

#### 1.4.15 BACnet Testing Laboratories (BTL) (BACnet)

Established by BACnet International to support compliance testing and interoperability testing activities and consists of BTL Manager and the BTL Working Group (BTL-WG). BTL also publishes Implementation Guidelines.

#### 1.4.16 BACnet Testing Laboratories (BTL) Listed (BACnet)

A device that has been listed by BACnet Testing Laboratory. Devices may be certified to a specific device profile, in which case the listing indicates that the device supports the required capabilities for that profile, or may be listed as "other".

#### 1.4.17 Binary (All protocols)

A two-state system where an "ON" condition is represented by a high signal level and an "OFF" condition is represented by a low signal level.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

'Digital' is sometimes used interchangeably with 'binary'.

#### 1.4.18 Broadcast (BACnet)

Unlike most messages, which are intended for a specific recipient device, a broadcast message is intended for all devices on the network.

#### 1.4.19 Building Control Network (BCN) (All protocols)

The network connecting all DDC Hardware within a building (or specific group of buildings).

#### 1.4.20 Building Point of Connection (BPOC) (All protocols)

A FPOC for a Building Control System. (This term is being phased out of use in preference for FPOC but is still used in some specifications and criteria. When it was used, it typically referred to a piece of control hardware. The current FPOC definition typically refers instead to IT hardware.)

#### 1.4.21 Commandable (All protocols)

See Overridable.

#### 1.4.22 Commandable Objects (BACnet)

Commandable Objects have a Commandable Property, Priority\_Array, and Relinquish\_Default Property as defined in ASHRAE 135, Clause 19.2, Command Prioritization.

#### 1.4.23 Configurable (All protocols)

A property, setting, or value is configurable if it can be changed via hardware settings on the device, via the use of engineering software or over the control network from the front end, and is retained through (after) loss of power.

In a non-Niagara Framework BACnet system, a property, setting, or value is configurable if it can be changed via one or more of:

- 1) via BACnet services (including proprietary BACnet services)
- 2) via hardware settings on the device

In a Niagara Framework BACnet system, a property, setting, or value is configurable if it can be changed via one or more of:

- 1) via BACnet services (including proprietary BACnet services)
- 2) via hardware settings on the device
- 3) via the Niagara Framework

Note this is more stringent than the ASHRAE 135 definition.

#### 1.4.24 Control Logic Diagram (All protocols)

A graphical representation of control logic for multiple processes that make up a system.

#### 1.4.25 Device (BACnet)

A Digital Controller that contains a BACnet Device Object and uses BACnet to communicate with other devices.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### 1.4.26 Device Object (BACnet)

Every BACnet device requires one Device Object, whose properties represent the network visible properties of that device. Every Device Object requires a unique Object Identifier number on the BACnet Internetwork. This number is often referred to as the device instance or device ID.

#### 1.4.27 Device Profile (BACnet)

A collection of BIBBs determining minimum BACnet capabilities of a device, defined in ASHRAE 135. Standard device profiles include BACnet Advanced Workstations (B-AWS), BACnet Building Controllers (B-BC), BACnet Advanced Application Controllers (B-AAC), BACnet Application Specific Controllers (B-ASC), BACnet Smart Actuator (B-SA), and BACnet Smart Sensor (B-SS).

#### 1.4.28 Digital Controller (All protocols)

An electronic controller, usually with internal programming logic and digital and analog input/output capability, which performs control functions.

#### 1.4.29 Direct Digital Control (DDC) (All protocols)

Digital controllers performing control logic. Usually the controller directly senses physical values, makes control decisions with internal programs, and outputs control signals to directly operate switches, valves, dampers, and motor controllers.

#### 1.4.30 Field Point of Connection (FPOC) (All protocols)

The FPOC is the point of connection between the UMCS IP Network and the field control network (either an IP network, a non-IP network, or a combination of both). The hardware at this location which provides the connection is generally an IT device such as a switch, IP router, or firewall.

In general, the term "FPOC Location" means the place where this connection occurs, and "FPOC Hardware" means the device that provides the connection. Sometimes the term "FPOC" is used to mean either and its actual meaning (i.e. location or hardware) is determined by the context in which it is used.

#### 1.4.31 Fox Protocol (Niagara Framework)

The protocol used for communication between components in the Niagara Framework. By default, Fox uses TCP port 1911.

#### 1.4.32 Gateway (All protocols)

A device that translates from one protocol application data format to another. Devices that change only the transport mechanism of the protocol - "translating" from TP/FT-10 to Ethernet/IP or from BACnet MS/TP to BACnet over IP for example - are not gateways as the underlying data format does not change. Gateways are also called Communications Bridges or Protocol Translators.

A Niagara Framework Supervisory Gateway is one type of Gateway.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### 1.4.33 IEEE 802.3 Ethernet (All protocols)

A family of local-area-network technologies providing high-speed networking features over various media, typically Cat 5, 5e or Cat 6 twisted pair copper or fiber optic cable.

#### 1.4.34 Internet Protocol (IP, TCP/IP, UDP/IP) (All protocols)

A communication method, the most common use is the World Wide Web. At the lowest level, it is based on Internet Protocol (IP), a method for conveying and routing packets of information over various LAN media. Two common protocols using IP are User Datagram Protocol (UDP) and Transmission Control Protocol (TCP). UDP conveys information to well-known "sockets" without confirmation of receipt. TCP establishes connections, also known as "sessions", which have end-to-end confirmation and guaranteed sequence of delivery.

#### 1.4.35 Input/Output (I/O) (All protocols)

Physical inputs and outputs to and from a device, although the term sometimes describes network or "virtual" inputs or outputs. See also "Points".

#### 1.4.36 I/O Expansion Unit (All protocols)

An I/O expansion unit provides additional point capacity to a digital controller

#### 1.4.37 IP subnet (All protocols)

A group of devices which share a defined range IP addresses. Devices on a common IP subnet can share data (including broadcasts) directly without the need for the traffic to traverse an IP router.

#### 1.4.38 JACE (Niagara Framework)

Java Application Control Engine. See paragraph NIAGARA FRAMEWORK SUPERVISORY GATEWAY

#### 1.4.39 Local-Area Network (LAN) (All protocols)

A communication network that spans a limited geographic area and uses the same basic communication technology throughout.

#### 1.4.40 Local Display Panels (LDPs) (All protocols)

A DDC Hardware with a display and navigation buttons, and must provide display and adjustment of points as shown on the Points Schedule and as indicated.

#### 1.4.41 MAC Address (All protocols)

Media Access Control address. The physical device address that identifies a device on a Local Area Network.

#### 1.4.42 Master-Slave/Token-Passing (MS/TP) (BACnet)

Data link protocol as defined by the BACnet standard. Multiple speeds (data rates) are permitted by the BACnet MS/TP standard.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### 1.4.43 Monitoring and Control (M&C) Software (All protocols)

The UMCS 'front end' software which performs supervisory functions such as alarm handling, scheduling and data logging and provides a user interface for monitoring the system and configuring these functions.

#### 1.4.44 Network Number (BACnet)

A site-specific number assigned to each network. This network number must be unique throughout the BACnet Internetwork.

#### 1.4.45 Niagara Framework (Niagara Framework)

A set of hardware and software specifications for building and utility control owned by Tridium Inc. and licensed to multiple vendors. The Framework consists of front end (M&C) software, web based clients, field level control hardware, and engineering tools. While the Niagara Framework is not adopted by a recognized standards body and does not use an open licensing model, it is sufficiently well-supported by multiple HVAC vendors to be considered a de-facto Open Standard.

#### 1.4.46 Niagara Framework Supervisory Gateway (Niagara Framework)

DDC Hardware component of the Niagara Framework. A typical Niagara architecture has Niagara specific supervisory gateways at the IP level and other (non-Niagara specific) controllers on field networks (TP/FT-10, MS/TP, etc.) beneath the Niagara supervisory gateways. The Niagara specific controllers function as a gateway between the Niagara framework protocol (Fox) and the field network beneath. These supervisory gateways may also be used as general purpose controllers and also have the capability to provide a web-based user interface.

Note that different vendors refer to this component by different names. The most common name is "JACE"; other names include (but are not limited to) "EC-BOS", "FX-40", "TMN", "SLX" and "UNC".

#### 1.4.47 Object (BACnet)

An ASHRAE 135 Object. The concept of organizing BACnet information into standard components with various associated Properties. Examples include Analog Input objects and Binary Output objects.

#### 1.4.48 Object Identifier (BACnet)

A grouping of two Object properties: Object Type (e.g. Analog Value, Schedule, etc.) and Object Instance (in this case, a number). Object Identifiers must be unique within a device.

#### 1.4.49 Object Instance (BACnet)

See paragraph OBJECT IDENTIFIER

#### 1.4.50 Object Properties (BACnet)

Attributes of an object. Examples include present value and high limit properties of an analog input object. Properties are defined in ASHRAE 135; some are optional and some are required. Objects are controlled by reading from and writing to object properties.



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### 1.4.51 Operator Configurable (All protocols)

Operator configurable values are values that can be changed from a single common front end user interface across multiple vendor systems.

For Niagara Framework Systems, a property, setting, or value is Operator Configurable when it is configurable from a Niagara Framework Front End.

For non Niagara-based BACnet systems, a property, setting, or value in a device is Operator Configurable when it is Configurable and is either:

- a. a Writable Property of a Standard BACnet Object; or
- b. a Property of a Standard BACnet Object that is Writable when Out\_Of\_Service is TRUE and Out\_Of\_Service is Writable.

#### 1.4.52 Override (All protocols)

Changing the value of a point outside of the normal sequence of operation where the change has priority over the sequence and where there is a mechanism for releasing the change such that the point returns to the normal value. Overrides persist until released or overridden at the same or higher priority but are not required to persist through a loss of power. Overrides are often used by operators to change values, and generally originate at a user interface (workstation or local display panel).

#### 1.4.53 Packaged Equipment (All protocols)

Packaged equipment is a single piece of equipment provided by a manufacturer in a substantially complete and operable condition, where the controls (DDC Hardware) are factory installed, and the equipment is sold and shipped from the manufacturer as a single entity. Disassembly and reassembly of a large piece of equipment for shipping does not prevent it from being packaged equipment. Package units may require field installation of remote sensors. Packaged equipment is also called a "packaged unit".

Note industry may use the term "Packaged System" to mean a collection of equipment that is designed to work together where each piece of equipment is packaged equipment and there is a network that connects the equipment together. A "packaged system" of this type is NOT packaged equipment; it is a collection of packaged equipment, and each piece of equipment must individually meet specification requirements.

#### 1.4.54 Packaged Unit (All protocols)

See packaged equipment.

#### 1.4.55 Performance Verification Test (PVT) (All protocols)

The procedure for determining if the installed BAS meets design criteria prior to final acceptance. The PVT is performed after installation, testing, and balancing of mechanical systems. Typically the PVT is performed by the Contractor in the presence of the Government.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### 1.4.56 Physical Segment (BACnet)

A single contiguous medium to which BACnet devices are attached (ASHRAE 135).

#### 1.4.57 Polling (All protocols)

A device periodically requesting data from another device.

#### 1.4.58 Points (All protocols)

Physical and virtual inputs and outputs. See also paragraph INPUT/OUTPUT (I/O).

#### 1.4.59 Proportional, Integral, and Derivative (PID) Control Loop (All protocols)

Three parameters used to control modulating equipment to maintain a setpoint. Derivative control is often not required for HVAC systems (leaving "PI" control).

#### 1.4.60 Proprietary (BACnet)

Within the context of BACnet, any extension of or addition to object types, properties, PrivateTransfer services, or enumerations specified in ASHRAE 135. Objects with Object\_Type values of 128 and above are Proprietary Objects. Properties with Property\_Identifier of 512 and above are proprietary Properties.

#### 1.4.61 Protocol Implementation Conformance Statement (PICS) (BACnet)

A document, created by the manufacturer of a device, which describes which portions of the BACnet standard may be implemented by a given device. ASHRAE 135 requires that all ASHRAE 135 devices have a PICS, and also defines a minimum set of information that must be in it. A device as installed for a specific project may not implement everything in its PICS.

#### 1.4.62 Repeater (All protocols)

A device that connects two control network segments and retransmits all information received on one side onto the other.

#### 1.4.63 Router (All protocols)

A device that connects two ASHRAE 135 networks and controls traffic between the two by retransmitting signals received from one side onto the other based on the signal destination. Routers are used to subdivide a BACnet internetwork and to limit network traffic.

#### 1.4.64 Segment (All protocols)

A 'single' section of a control network that contains no repeaters or routers. There is generally a limit on the number of devices on a segment, and this limit is dependent on the topology/media and device type.

#### 1.4.65 Standard BACnet Objects (BACnet)

Objects with Object\_Type values below 128 and specifically enumerated in

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

Clause 21 of ASHRAE 135. Objects which are not proprietary. See paragraph PROPRIETARY.

#### 1.4.66 Standard BACnet Properties (BACnet)

Properties with Property\_Identifier values below 512 and specifically enumerated in Clause 21 of ASHRAE 135. Properties which are not proprietary. See PROPRIETARY.

#### 1.4.67 Standard BACnet Services (BACnet)

ASHRAE 135 services other than ConfirmedPrivateTransfer or UnconfirmedPrivateTransfer. See paragraph PROPRIETARY.

#### 1.4.68 UMCS (All protocols)

UMCS stands for Utility Monitoring and Control System. The term refers to all components by which a project site monitors, manages, and controls real-time operation of HVAC and other building systems. These components include the UMCS "front-end" and all field building control systems connected to the front-end. The front-end consists of Monitoring and Control Software (user interface software), browser-based user interfaces and network infrastructure.

The network infrastructure (the "UMCS Network"), is an IP network connecting multiple building or facility control networks to the Monitoring and Control Software.

#### 1.4.69 UMCS Network (All protocols)

The UMCS Network connects multiple building or facility control networks to the Monitoring and Control Software.

#### 1.4.70 Writable Property (BACnet)

A Property is Writable when it can be changed through the use of one or more of the WriteProperty services defined in ASHRAE 135, Clause 15 regardless of the value of any other Property. Note that in the ASHRAE 135 standard, some Properties may be writable when the Out of Service Property is TRUE; for purposes of this Section, Properties that are only writable when the Out of Service Property is TRUE are not considered to be Writable.

### 1.5 PROJECT SEQUENCING

TABLE II: PROJECT SEQUENCING lists the sequencing of submittals as specified in paragraph SUBMITTALS (denoted by an 'S' in the 'TYPE' column) and activities as specified in PART 3 EXECUTION (denoted by an 'E' in the 'TYPE' column). TABLE II does not specify overall project milestone and completion dates; these dates are specified in the contract documents.

- a. Sequencing for Submittals: The sequencing specified for submittals is the deadline by which the submittal must be initially submitted to the Government. Following submission there will be a Government review period as specified in Section 01 33 00 SUBMITTAL PROCEDURES. If the submittal is not accepted by the Government, revise the submittal and resubmit it to the Government within 14 days of notification that the submittal has been rejected. Upon resubmittal there will be an additional Government review period. If the submittal is not accepted

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

the process repeats until the submittal is accepted by the Government.

- b. Sequencing for Activities: The sequencing specified for activities indicates the earliest the activity may begin.
- c. Abbreviations: In TABLE II the abbreviation AAO is used for 'after approval of' and 'ACO' is used for 'after completion of'.

| TABLE II. PROJECT SEQUENCING |      |                                                                                                                       |                                                          |
|------------------------------|------|-----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| ITEM #                       | TYPE | DESCRIPTION                                                                                                           | SEQUENCING (START OF ACTIVITY OR DEADLINE FOR SUBMITTAL) |
| 1                            | S    | Existing Conditions Report                                                                                            |                                                          |
| 2                            | S    | DDC Contractor Design Drawings                                                                                        |                                                          |
| 3                            | S    | Manufacturer's Product Data                                                                                           |                                                          |
| 4                            | S    | Pre-construction QC Checklist                                                                                         |                                                          |
| 5                            | E    | Install Building Control System                                                                                       | AAO #1 thru #4                                           |
| 6                            | E    | Start-Up and Start-Up Testing                                                                                         | ACO #5                                                   |
| 7                            | S    | Post-Construction QC Checklist                                                                                        | ACO #6                                                   |
| 8                            | S    | Programming Software<br>Configuration Software<br><br>Niagara Framework Engineering Tool<br>Niagara Framework Wizards | ACO #6                                                   |
| 9                            | S    | Draft As-Built Drawings                                                                                               | ACO #6                                                   |
| 10                           | S    | Start-Up Testing Report                                                                                               | ACO #6                                                   |
| 11                           | S    | PVT Procedures                                                                                                        | before schedule start of #12 and AAO #10                 |

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

| TABLE II. PROJECT SEQUENCING |      |                                                                                                                           |                                                          |
|------------------------------|------|---------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| ITEM #                       | TYPE | DESCRIPTION                                                                                                               | SEQUENCING (START OF ACTIVITY OR DEADLINE FOR SUBMITTAL) |
| 12                           | E    | Execute PVT                                                                                                               | AAO #9 and #11                                           |
| 13                           | S    | PVT Report                                                                                                                | ACO #12                                                  |
| 14                           | S    | Controller Application Programs<br>Controller Configuration Settings<br><br>Niagara Framework Supervisory Gateway Backups | AAO #13                                                  |
| 15                           | S    | Final As-Built Drawings                                                                                                   | AAO #13                                                  |
| 16                           | S    | O&M Instructions                                                                                                          | AAO #15                                                  |
| 17                           | S    | Training Documentation                                                                                                    | AAO #10 and before scheduled start of #18                |
| 18                           | E    | Training                                                                                                                  | AAO #16 and #17                                          |
| 19                           | S    | Closeout QC Checklist                                                                                                     |                                                          |

#### 1.6 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

##### SD-02 Shop Drawings

DDC Contractor Design Drawings; G

Draft As-Built Drawings; G

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

Final As-Built Drawings; G

SD-03 Product Data

Programming Software; G

Controller Application Programs; G

Configuration Software; G

Controller Configuration Settings; G

Proprietary Multi-Split Engineering Tool Software; G

Manufacturer's Product Data; G

Niagara Framework Supervisory Gateway Backups; G

Niagara Framework Engineering Tool; G

Niagara Framework Wizards; G

SD-05 Design Data

Boiler Or Chiller Plant Gateway Request

SD-06 Test Reports

Existing Conditions Report

Pre-Construction Quality Control (QC) Checklist; G

Post-Construction Quality Control (QC) Checklist; G

Start-Up Testing Report; G

PVT Procedures; G

PVT Report; G

SD-10 Operation and Maintenance Data

Operation and Maintenance (O&M) Instructions; G

Training Documentation; G

SD-11 Closeout Submittals

Enclosure Keys; G

Password Summary Report; G

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## Closeout Quality Control (QC) Checklist; G

### 1.7 DATA PACKAGE AND SUBMITTAL REQUIREMENTS

Technical data packages consisting of technical data and computer software (meaning technical data which relates to computer software) which are specifically identified in this project and which may be defined/required in other specifications must be delivered strictly in accordance with the CONTRACT CLAUSES and in accordance with the Contract Data Requirements List, DD Form 1423. Data delivered must be identified by reference to the particular specification paragraph against which it is furnished. All submittals not specified as technical data packages are considered 'shop drawings' under the Federal Acquisition Regulation Supplement (FARS) and must contain no proprietary information and be delivered with unrestricted rights.

### 1.8 SOFTWARE FOR DDC HARDWARE AND GATEWAYS

Provide all software related to the programming and configuration of DDC Hardware and Gateways as indicated. License all Software to the project site. The term "controller" as used in these requirements means both DDC Hardware and Gateways.

#### 1.8.1 Configuration Software

For each type of controller, provide the configuration tool software in accordance with Section 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS. Submit hard copies of the software user manuals for each software with the software submittal.

Submit Configuration Software on CD-ROM as a Technical Data Package. Submit two hard copies of the software user manual for each piece of software.

#### 1.8.2 Controller Configuration Settings

For each controller, provide copies of the installed configuration settings as source code compatible with the configuration tool software for that controller in accordance with Section 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS.

Submit Controller Configuration Settings on CD-ROM as a Technical Data Package. Include on the CD-ROM a list or table of contents clearly indicating which files are associated with each device. Submit 2 copies of the Controller Configuration Settings CD-ROM.

#### 1.8.3 Programming Software

For each type of programmable controller, provide the programming software in accordance with Section 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS. Submit hard copies of software user manuals for each software with the software submittal.

Submit Programming Software on CD-ROM as a Technical Data Package. Submit two hard copies of the software user manual for each piece of software.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### 1.8.4 Controller Application Programs

For each programmable controller, provide copies of the application program as source code compatible with the programming software for that controller in accordance with Section 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS.

Submit Controller Application Programs on CD-ROM as a Technical Data Package. Include on the CD-ROM a list or table of contents clearly indicating which application program is associated with each device. Submit 2 copies of the Controller Application Programs CD-ROM.

#### 1.8.5 Niagara Framework Supervisory Gateway Backups

For each Niagara Framework Supervisory Gateway, provide a backup of all software within the Niagara Framework Supervisory Gateway, including configuration settings. This backup must be sufficient to allow the restoration of the Niagara Framework Supervisory Gateway or the replacement of the Niagara Framework Supervisory Gateway.

Submit backups for each Niagara Framework Supervisory Gateway on CD-ROM as a Technical Data Package. Mark each backup indicating clearly the source Niagara Framework Supervisory Gateway.

#### 1.8.6 Niagara Framework Engineering Tool(for all Niagara Framework system)

Provide a Niagara Framework Engineering Tool in accordance with Section 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS. Submit software user manuals with the Niagara Framework Engineering Tool submittal.

Submit the Niagara Framework Engineering Tool on CD-ROM as a Technical Data Package. Submit two hard copies of the software user manual for the Niagara Framework Engineering Tool.

#### 1.9 BOILER OR CHILLER PLANT GATEWAY REQUEST

If requesting the use of a gateway to a boiler or chiller plant as indicated in paragraph Proprietary Systems Exempted From Open Protocol Requirements, submit a Boiler or Chiller Plant Gateway Request describing the configuration of the boilers or chillers including model numbers for equipment and controllers, the sequence of operation for the units, and a justification for the need to operate the units on a shared non-BACnet network.

#### 1.10 QUALITY CONTROL CHECKLISTS

The QC Checklist for BACnet Systems in APPENDIX A of this Section must be completed by the Contractor's Chief Quality Control (QC) Representative and submitted as indicated.

The QC Checklist for Niagara Framework Based BACnet Systems in APPENDIX A of this Section must be completed by the Contractor's Chief Quality Control (QC) Representative and submitted as indicated.

The QC Representative must verify each item indicated and initial in the



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

space provided to indicate that the requirement has been met. The QC Representative must sign and date the Checklist prior to submission to the Government.

#### 1.10.1 Pre-Construction Quality Control (QC) Checklist

Complete items indicated as Pre-Construction QC Checklist items in the QC Checklist. Submit four copies of the Pre-Construction QC Checklist.

#### 1.10.2 Post-Construction Quality Control (QC) Checklist

Complete items indicated as Post-Construction QC Checklist items in the QC Checklist. Submit four copies of the Post-Construction QC Checklist.

#### 1.10.3 Closeout Quality Control (QC) Checklist

Complete items indicated as Closeout QC Checklist items in the QC Checklist. Submit four copies of the Closeout QC Checklist.

### PART 2 PRODUCTS

Provide products meeting the requirements of Section 23 09 13 INSTRUMENTATION AND CONTROL DEVICES FOR HVAC, Section 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS for BACnet or Niagara BACnet systems, other referenced Sections, and this Section.

#### 2.1 GENERAL PRODUCT REQUIREMENTS

Units of the same type of equipment must be products of a single manufacturer. Each major component of equipment must have the manufacturer's name and address, and the model and serial number in a conspicuous place. Materials and equipment must be standard products of a manufacturer regularly engaged in the manufacturing of these and similar products. The standard products must have been in a satisfactory commercial or industrial use for two years prior to use on this project. The two year use must include applications of equipment and materials under similar circumstances and of similar size. DDC Hardware not meeting the two-year field service requirement is acceptable provided it has been successfully used by the Contractor in a minimum of two previous projects. The equipment items must be supported by a service organization. Items of the same type and purpose must be identical, including equipment, assemblies, parts and components.

#### 2.2 PRODUCT DATA

Provide manufacturer's product data sheets documenting compliance with product specifications for each product provided under Section 23 09 13 INSTRUMENTATION AND CONTROL DEVICES FOR HVAC, Section 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS, or this Section. Provide product data for all products in a single indexed compendium, organized by product type.

For all BACnet hardware: for each manufacturer, model and version (revision) of DDC Hardware provide the Protocol Implementation Conformance Statement (PICS) in accordance with Section 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS.

Submit Manufacturer's Product Data on CD-ROM.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## 2.3 OPERATION ENVIRONMENT

Unless otherwise specified, provide products rated for continuous operation under the following conditions:

- a. Pressure: Pressure conditions normally encountered in the installed location.
- b. Vibration: Vibration conditions normally encountered in the installed location.
- c. Temperature:
  - (1) Products installed indoors: Ambient temperatures in the range of 32 to 112 degrees F and temperature conditions outside this range normally encountered at the installed location.
  - (2) Products installed outdoors or in unconditioned indoor spaces: Ambient temperatures in the range of -35 to +151 degrees F and temperature conditions outside this range normally encountered at the installed location.
- d. Humidity: 10 to 95 percent relative humidity, noncondensing and humidity conditions outside this range normally encountered at the installed location.

## 2.4 WIRELESS CAPABILITY

For products incorporating any wireless capability (including but not limited to radio frequency (RF), infrared and optical), provide products for which wireless capability can be permanently disabled at the device. Optical and infrared capabilities may be disabled via a permanently affixed opaque cover plate.

## 2.5 ENCLOSURES

Enclosures supplied as an integral (pre-packaged) part of another product are acceptable. Provide two Enclosure Keys for each lockable enclosure on a single ring per enclosure with a tag identifying the enclosure the keys operate. Provide enclosures meeting the following minimum requirements:

### 2.5.1 Outdoors

For enclosures located outdoors, provide enclosures meeting NEMA 250 Type 4 requirements.

### 2.5.2 Mechanical and Electrical Rooms

For enclosures located in mechanical or electrical rooms, provide enclosures meeting NEMA 250 Type 2 requirements.

### 2.5.3 Other Locations

For enclosures in other locations including but not limited to occupied spaces, above ceilings, and in plenum returns, provide enclosures meeting NEMA 250 Type 1 requirements.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## 2.6 WIRE AND CABLE

Provide wire and cable meeting the requirements of NFPA 70 and NFPA 90A in addition to the requirements of this specification and referenced specifications.

### 2.6.1 Terminal Blocks

For terminal blocks which are not integral to other equipment, provide terminal blocks which are insulated, modular, feed-through, clamp style with recessed captive screw-type clamping mechanism, suitable for DIN rail mounting, and which have enclosed sides or end plates and partition plates for separation.

### 2.6.2 Control Wiring for Binary Signals

For Control Wiring for Binary Signals, provide 18 AWG copper or thicker wire rated for 300-volt service.

### 2.6.3 Control Wiring for Analog Signals

For Control Wiring for Analog Signals, provide 18 AWG or thicker, copper, single- or multiple-twisted wire meeting the following requirements:

- a. minimum 2 inch lay of twist
- b. 100 percent shielded pairs
- c. at least 300-volt insulation
- d. each pair has a 20 AWG tinned-copper drain wire and individual overall pair insulation
- e. cables have an overall aluminum-polyester or tinned-copper cable-shield tape, overall 20 AWG tinned-copper cable drain wire, and overall cable insulation.

### 2.6.4 Power Wiring for Control Devices

For 24-volt circuits, provide insulated copper 18 AWG or thicker wire rated for 300 VAC service. For 120-volt circuits, provide 14 AWG or thicker stranded copper wire rated for 600-volt service.

### 2.6.5 Transformers

Provide UL 5085-3 approved transformers. Select transformers sized so that the connected load is no greater than 80 percent of the transformer rated capacity.

## PART 3 EXECUTION

### 3.1 INSTALLATION

Fully install and test the control system in accordance Section 23 09 13 INSTRUMENTATION AND CONTROL DEVICES FOR HVAC, Section 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS for BACnet or Niagara BACnet systems, and this Section.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 3.1.1 Dielectric Isolation

Provide dielectric isolation where dissimilar metals are used for connection and support. Install control system in a manner that provides clearance for control system maintenance by maintaining access space required to calibrate, remove, repair, or replace control system devices. Install control system such that it does not interfere with the clearance requirements for mechanical and electrical system maintenance.

### 3.1.2 Penetrations in Building Exterior

Make all penetrations through and mounting holes in the building exterior watertight.

### 3.1.3 Device Mounting Criteria

Install devices in accordance with the manufacturer's recommendations and as indicated and shown. Provide a weathershield for all devices installed outdoors. Provide clearance for control system maintenance by maintaining access space required to calibrate, remove, repair, or replace control system devices. Provide clearance for mechanical and electrical system maintenance; do not not interfere with the clearance requirements for mechanical and electrical system maintenance.

### 3.1.4 Labels and Tags

Key all labels and tags to the unique identifiers shown on the As-Built drawings. For labels exterior to protective enclosures provide engraved plastic labels mechanically attached to the enclosure or DDC Hardware. Labels inside protective enclosures may be attached using adhesive, but must not be hand written. For tags, provide plastic or metal tags mechanically attached directly to each device or attached by a metal chain or wire.

- a. Label all Enclosures and DDC Hardware.
- b. Tag Airflow measurement arrays (AFMA) with flow rate range for signal output range, duct size, and pitot tube AFMA flow coefficient.
- c. Tag duct static pressure taps at the location of the pressure tap

### 3.1.5 Surge Protection

#### 3.1.5.1 Power-Line Surge Protection

Protect equipment connected to AC circuits to withstand power-line surges in accordance with IEEE C62.41. Do not use fuses for surge protection.

#### 3.1.5.2 Surge Protection for Transmitter and Control Wiring

Protect DDC hardware against or provided DDC hardware capable of withstanding surges induced on control and transmitter wiring installed outdoors and as shown. Protect equipment against the following two waveforms:

- a. A waveform with a 10-microsecond rise time, a 1000-microsecond decay time and a peak current of 60 amps.
- b. A waveform with an 8-microsecond rise time, a 20-microsecond decay

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

time and a peak current of 500 amperes.

### 3.1.6 Basic Cybersecurity Requirements

#### 3.1.6.1 Passwords

For all devices with a password, change the password from the default password. Do not use the same password for more than one device. Coordinate selection of passwords with Owner. Provide a Password Summary Report documenting the password for each device and describing the procedure to change the password for each device.

Provide two hardcopies of the Password Summary Report, each copy in its own sealed envelope.

#### 3.1.6.2 Wireless Capability

Unless otherwise indicated, disable wireless capability (including but not limited to radio frequency (RF), infrared and optical) for all devices with wireless capability. Optical and infrared capabilities may be disabled via a permanently affixed opaque cover plate. Password protecting a wireless connections does not meet this requirement; the wireless capability must be disabled.

#### 3.1.6.3 IP Network Physical Security

Install all IP Network media in conduit. Install all IP devices including but not limited to IP-enabled DDC hardware and IP Network Hardware in lockable enclosures.

### 3.2 DRAWINGS AND CALCULATIONS

Provide drawings in the form and arrangement indicated and shown. Use the same abbreviations, symbols, nomenclature and identifiers shown. Assign a unique identifier as shown to each control system element on a drawing. When packaging drawings, group schedules by system. When space allows, it is permissible to include multiple schedules for the same system on a single sheet. Except for drawings covering all systems, do not put information for different systems on the same sheet.

Submit hardcopy drawings on A3 17 by 11 inches sheets, and electronic drawings in PDF and in AutoCAD format. In addition, submit electronic drawings in editable Excel format for all drawings that are tabular, including but not limited to the Point Schedule and Equipment Schedule.

- a. Submit DDC Contractor Design Drawings consisting of each drawing indicated with pre-construction information depicting the intended control system design and plans. Submit DDC Contractor Design Drawings as a single complete package: two hard copies and two copies on CD-ROM.
- b. Submit Draft As-Built Drawings consisting of each drawing indicated updated with as-built data for the system prior to PVT. Submit Draft As-Built Drawings as a single complete package: two hard copies and two copies on CD-ROM.
- c. Submit Final As-Built Drawings consisting of each drawing indicated updated with all final as-built data. Final As-Built Drawings as a single complete package: two hard copies and two copies on CD-ROM.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 3.2.1 Sample Drawings

Sample drawings in electronic format are available at the Whole Building Design Guide page for this section:

<http://www.wbdg.org/ffc/dod/unified-facilities-guide-specifications-ufgs/ufgs-23-09-00>

These drawings may prove useful in demonstrating expected drawing formatting and example content and are provided for illustrative purposes only. Note that these drawings do not meet the content requirements of this Section and must be completed to meet project requirements.

### 3.2.2 Drawing Index and Legend

Provide an HVAC Control System Drawing Index showing the name and number of the building, military site, State or other similar designation, and Country. In the Drawing Index, list all Contractor Design Drawings, including the drawing number, sheet number, drawing title, and computer filename when used. In the Design Drawing Legend, show and describe all symbols, abbreviations and acronyms used on the Design Drawings. Provide a single Index and Legend for the entire drawing package.

### 3.2.3 Thermostat and Occupancy Sensor Schedule

Provide a thermostat and occupancy sensor schedule containing each thermostat's unique identifier, room identifier and control features and functions as shown. Provide a single thermostat and occupancy sensor schedule for the entire project.

### 3.2.4 Valve Schedule

Provide a valve schedule containing each valve's unique identifier, size, flow coefficient Kv (Cv), pressure drop at specified flow rate, spring range, positive positioner range, actuator size, close-off pressure to torque data, dimensions, and access and clearance requirements data. In the valve schedule include actuator selection data supported by calculations of the force required to move and seal the valve, access and clearance requirements. Provide a single valve schedule for the entire project.

### 3.2.5 Damper Schedule

Provide a damper schedule containing each damper's unique identifier, type (opposed or parallel blade), nominal and actual sizes, orientation of axis and frame, direction of blade rotation, actuator size and spring ranges, operation rate, positive positioner range, location of actuators and damper end switches, arrangement of sections in multi-section dampers, and methods of connecting dampers, actuators, and linkages. Include the AMCA 511 maximum leakage rate at the operating static-pressure differential for each damper in the Damper Schedule. Provide a single damper schedule for the entire project.

### 3.2.6 Project Summary Equipment Schedule

Provide a project summary equipment schedule containing the manufacturer, model number, part number and descriptive name for each control device, hardware and component provided under this specification. Provide a single project equipment schedule for the entire project.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 3.2.7 Equipment Schedule

Provide system equipment schedules containing the unique identifier, manufacturer, model number, part number and descriptive name for each control device, hardware and component provided under this specification. Provide a separate equipment schedule for each HVAC system.

### 3.2.8 Occupancy Schedule

Provide an occupancy schedule drawing containing the same fields as the occupancy schedule Contract Drawing with Contractor updated information. Provide a single occupancy schedule for the entire project.

### 3.2.9 DDC Hardware Schedule

Provide a single DDC Hardware Schedule for the entire project and including following information for each device.

#### 3.2.9.1 DDC Hardware Identifier

The Unique DDC Hardware Identifier for the device.

#### 3.2.9.2 HVAC System

The system "name" used to identify a specific system (the name used on the system schematic drawing for that system).

#### 3.2.9.3 BACnet Device Information

##### 3.2.9.3.1 Device Object Identifier

The Device Object Identifier: The Object\_Identifier of the Device Object

##### 3.2.9.3.2 Network Number

The Network Number for the device.

##### 3.2.9.3.3 MAC Address

The MAC Address for the device

##### 3.2.9.3.4 BTL Listing

The BTL Listing of the device. If the device is listed under multiple BTL Profiles, indicate the profile that matches the use and configuration of the device as installed.

##### 3.2.9.3.5 Proprietary Services Information

If the device uses non-standard ASHRAE 135 services as defined and permitted in Section 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS, indicate that the device uses non-standard services and include a description of all non-standard services used. Describe usage and content such that a device from another vendor can interoperate with the device using the non-standard service. Provide descriptions with sufficient detail to allow a device from a different manufacturer to be programmed to both read and write the non-standard service request:

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

- a. read: interpret the data contained in the non-standard service and;
- b. write: given similar data, generate the appropriate non-standard service request.

#### 3.2.9.3.6 Alarming Information

Indicate whether the device is used for alarm generation, and which types of alarm generation the device implements: intrinsic, local algorithmic, remote algorithmic.

#### 3.2.9.3.7 Scheduling Information

Indicate whether the device is used for scheduling.

#### 3.2.9.3.8 Trending Information

Indicate whether the device is used for trending, and indicate if the device is used to trend local values, remote values, or both.

#### 3.2.9.4 Niagara Station ID

The Niagara Station ID for each Niagara Framework Supervisory Gateway

#### 3.2.10 Points Schedule

Provide a Points Schedule in tabular form for each HVAC system, with the indicated columns and with each row representing a hardware point, network point or configuration point in the system.

- a. When a Points Schedule was included in the Contract Drawing package, use the same fields as the Contract Drawing with updated information in addition to the indicated fields.
- b. When Point Schedules are included in the contract package, items requiring contractor verification or input have been shown in angle brackets ("<" and ">"), such as <\_\_\_> for a required entry or <value> for a value requiring confirmation. Complete all items in brackets as well as any blank cells. Do not modify values which are not in brackets without approval.

Points Schedule Columns must include:

##### 3.2.10.1 Point Name

The abbreviated name for the point using the indicated naming convention.

##### 3.2.10.2 Description

A brief functional description of the point such as "Supply Air Temperature".

##### 3.2.10.3 DDC Hardware Identifier

The Unique DDC Hardware Identifier shown on the DDC Hardware Schedule and used across all drawings for the DDC Hardware containing the point.



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### 3.2.10.4 Settings

The value and units of any setpoints, configured setpoints, configuration parameters, and settings related to each point.

#### 3.2.10.5 Range

The range of values, including units, associated with the point, including but not limited to a zone temperature setpoint adjustment range, a sensor measurement range, occupancy values for an occupancy input, or the status of a safety.

#### 3.2.10.6 Input or Output (I/O) Type

The type of input or output signal associated with the point. Use the following abbreviations for entries in this column:

- a. AI: The value comes from a hardware (physical) Analog Input
- b. AO: The value is output as a hardware (physical) Analog Output
- c. BI: The value comes from a hardware (physical) Binary Input
- d. BO: The value is output as a hardware (physical) Binary Output
- e. PULSE: The value comes from a hardware (physical) Pulse Accumulator Input
- f. NET-IN: The value is provided from the network (generally from another device). Use this entry only when the value is received from another device as part of scheduling or as part of a sequence of operation, not when the value is received on the network for supervisory functions such as trending, alarming, override or display at a user interface.
- g. NET-OUT: The value is provided to another controller over the network. Use this entry only when the value is transmitted to another device as part of scheduling or as part of a sequence of operation, not when the value is transmitted on the network for supervisory functions such as trending, alarming, override or display at a user interface.

#### 3.2.10.7 Object and Property Information

The Object Type and Instance Number for the Object associated with the point. If the value of the point is not in the Present\_Value Property, then also provide the Property ID for the Property containing the value of the point. Any point that is displayed at the front end or on an LDP, is trended, is used by another device on the network, or has an alarm condition must be documented here.

#### 3.2.10.8 Niagara Station ID

The Niagara Station ID of the Niagara Framework Supervisory Gateway the point is mapped into.

#### 3.2.10.9 Network Data Exchange Information (Gets Data From, Sends Data To)

Provide the DDC Hardware Identifier of other DDC Hardware the point is

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

shared with.

#### 3.2.10.10 Override Information (Object Type and Instance Number)

For each point requiring an Override and not residing in a Niagara Framework Supervisory Gateway, indicate if the Object for the point is Commandable or, if the use of a separate Object was specifically approved by the Contracting Officer, provide the Object Type and Instance Number of the Object to be used in overriding the point.

#### 3.2.10.11 Trend Object Information

For each point requiring a trend, indicate if the trend is Local or Remote, the trend Object type and the trend Object instance number. For remote trends provide the DDC Hardware Identifier for the device containing the trend Object in the Points Schedule notes.

#### 3.2.10.12 Alarm Information

Indicate the Alarm Generation Type, Event Enrollment Object Instance Number, and Notification Class Object Instance Number for each point requiring an alarm. (Note that not all alarms will have Event Enrollment Objects.)

For Niagara BACnet systems: Indicate the Alarm Generation Type and Notification Class Object Instance Number for each point requiring an alarm. (Note that not all alarms will have a Notification Class Object.)

#### 3.2.10.13 Configuration Information

Indicate the means of configuration associated with each point. For points in a Niagara Framework Supervisory Gateway, indicate the point within the Niagara Framework Supervisory Gateway used to configure the value. For other points:

- a. For Operator Configurable Points indicate BACnet Object and Property information (Name, Type, Identifiers) containing the configurable value. Indicate whether the property is writable always, or only when Out\_Of\_Service is TRUE.
- b. For Configurable Points indicate the BACnet Object and Property information as for Operator Configurable points, or identification of the configurable settings from within the engineering software for the device or identification of the hardware settings on the device.

#### 3.2.11 Riser Diagram

The Riser Diagram of the Building Control Network may be in tabular form, and must show all DDC Hardware and all Network Hardware, including network terminators. For each item, provide the unique identifier, common descriptive name, physical sequential order (previous and next device on the network), room identifier and location within room. A single riser diagram must be submitted for the entire system.

#### 3.2.12 Control System Schematics

Provide control system schematics in the same form as the control system

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

schematic Contract Drawing with Contractor updated information. Provide a control system schematic for each HVAC system.

### 3.2.13 Sequences of Operation Including Control Logic Diagrams

Provide HVAC control system sequence of operation and control logic diagrams in the same format as the Contract Drawings. Within these drawings, refer to devices by their unique identifiers. Submit sequences of operation and control logic diagrams for each HVAC system

### 3.2.14 Controller, Motor Starter and Relay Wiring Diagram

Provide controller wiring diagrams as functional wiring diagrams which show the interconnection of conductors and cables to each controller and to the identified terminals of input and output devices, starters and package equipment. Show necessary jumpers and ground connections and the labels of all conductors. Identify sources of power required for control systems and for packaged equipment control systems back to the panel board circuit breaker number, controller enclosures, magnetic starter, or packaged equipment control circuit. Show each power supply and transformer not integral to a controller, starter, or packaged equipment. Show the connected volt-ampere load and the power supply volt-ampere rating. Provide wiring diagrams for each HVAC system.

## 3.3 CONTROLLER TUNING

Tune each controller in a manner consistent with that described in the ASHRAE FUN IP and in the manufacturer's instruction manual. Tuning must consist of adjustment of the proportional, integral, and where applicable, the derivative (PID) settings to provide stable closed-loop control. Each loop must be tuned while the system or plant is operating at a high gain (worst case) condition, where high gain can generally be defined as a low-flow or low-load condition. Upon final adjustment of the PID settings, in response to a change in controller setpoint, the controlled variable must settle out at the new setpoint with no more than two (2) oscillations above and below setpoint. Upon settling out at the new setpoint the controller output must be steady. With the exception of naturally slow processes such as zone temperature control, the controller must settle out at the new setpoint within five (5) minutes. Set the controller to its correct setpoint and record and submit the final PID configuration settings with the O&M Instructions and on the associated Points Schedule.

## 3.4 START-UP

### 3.4.1 Start-Up Test

Perform the following startup tests for each control system to ensure that the described control system components are installed and functioning per this specification.

Adjust, calibrate, measure, program, configure, set the time schedules, and otherwise perform all necessary actions to ensure that the systems function as indicated and shown in the sequence of operation and other contract documents.

#### 3.4.1.1 Systems Check

An item-by-item check must be performed for each HVAC system

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### 3.4.1.1.1 Step 1 - System Inspection

With the system in unoccupied mode and with fan hand-off-auto switches in the OFF position, verify that power and main air are available where required and that all output devices are in their failsafe and normal positions. Inspect each local display panel and each M&C Client to verify that all displays indicate shutdown conditions.

#### 3.4.1.1.2 Step 2 - Calibration Accuracy Check

Perform a two-point accuracy check of the calibration of each HVAC control system sensing element and transmitter by comparing the value from the test instrument to the network value provided by the DDC Hardware. Use digital indicating test instruments, such as digital thermometers, motor-driven psychrometers, and tachometers. Use test instruments with accuracy at least twice as accurate as the specified sensor accuracy and with calibration traceable to National Institute of Standards and Technology standards. Check one the first check point in the bottom one-third of the sensor range, and the second in the top one-third of the sensor range. Verify that the sensing element-to-DDC readout accuracies at two points are within the specified product accuracy tolerances, and if not recalibrate or replace the device and repeat the calibration check.

#### 3.4.1.1.3 Step 3 - Actuator Range Check

With the system running, apply a signal to each actuator through the DDC Hardware controller. Verify proper operation of the actuators and positioners for all actuated devices and record the signal levels for the extreme positions of each device. Vary the signal over its full range, and verify that the actuators travel from zero stroke to full stroke within the signal range. Where applicable, verify that all sequenced actuators move from zero stroke to full stroke in the proper direction, and move the connected device in the proper direction from one extreme position to the other. For valve actuators and damper actuators, perform the actuator range check under normal system pressures.

#### 3.4.1.2 Weather Dependent Test

Perform weather dependent test procedures in the appropriate climatic season.

#### 3.4.2 Start-Up Testing Report

Submit 4 copies of the Start-Up Testing Report. The report may be submitted as a Technical Data Package documenting the results of the tests performed and certifying that the system is installed and functioning per this specification, and is ready for the Performance Verification Test (PVT).

### 3.5 PERFORMANCE VERIFICATION TEST (PVT)

#### 3.5.1 PVT Procedures

Prepare PVT Procedures based on Section 25 08 10 UTILITY MONITORING AND CONTROL SYSTEM TESTING explaining step-by-step, the actions and expected results that will demonstrate that the control system performs in

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

accordance with the sequences of operation, and other contract documents. Submit 4 copies of the PVT Procedures. The PVT Procedures may be submitted as a Technical Data Package.

#### 3.5.1.1 Sensor Accuracy Checks

Include a one-point accuracy check of each sensor in the PVT procedures.

#### 3.5.1.2 Endurance Test

Include a one-week endurance test as part of the PVT during which the system is operated continuously.

Use the building control system BACnet Trend Log or Trend Log Multiple Objects to trend all points shown as requiring a trend on the Point Schedule for the entire endurance test. If insufficient buffer capacity exists to trend the entire endurance test, upload trend logs during the course of the endurance test to ensure that no trend data is lost.

Use the building control system Niagara Trend Log Objects to trend all points shown as requiring a trend on the Point Schedule for the entire endurance test. If insufficient buffer capacity exists to trend the entire endurance test, upload trend logs during the course of the endurance test to ensure that no trend data is lost.

#### 3.5.1.3 PVT Equipment List

Include in the PVT procedures a control system performance verification test equipment list that lists the equipment to be used during performance verification testing. For each piece of equipment, include manufacturer name, model number, equipment function, the date of the latest calibration, and the results of the latest calibration

#### 3.5.2 PVT Execution

Demonstrate compliance of the control system with the contract documents. Using test plans and procedures approved by the Government, software capable of reading and writing COV Notification Subscriptions, Notification Class Recipient List Properties, event enrollments, demonstrate all physical and functional requirements of the project. Show, step-by-step, the actions and results demonstrating that the control systems perform in accordance with the sequences of operation. Do not start the performance verification test until after receipt of written permission by the Government, based on Government approval of the PVT Plan and Draft As-Builts and completion of balancing. UNLESS GOVERNMENT WITNESSING OF A TEST IS SPECIFICALLY WAIVED BY THE GOVERNMENT, PERFORM ALL TESTS WITH A GOVERNMENT WITNESS. Do not conduct tests during scheduled seasonal off periods of base heating and cooling systems. If the system experiences any failures during the endurance test portion of the PVT, repair the system repeat the endurance test portion of the PVT until the system operates continuously and without failure for the specified endurance test period.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 3.5.3 PVT Report

Prepare and submit a PVT report documenting all tests performed during the PVT and their results. Include all tests in the PVT procedures and any additional tests performed during PVT. Document test failures and repairs conducted with the test results.

Submit four copies of the PVT Report. The PVT Report may be submitted as a Technical Data Package.

### 3.6 OPERATION AND MAINTENANCE (O&M) INSTRUCTIONS

Provide HVAC control System Operation and Maintenance Instructions which include:

- a. "Data Package 3" as indicated in Section 01 78 23 OPERATION AND MAINTENANCE DATA for each piece of control equipment.
- b. "Data Package 4" as described in Section 01 78 23 OPERATION AND MAINTENANCE DATA for all air compressors.
- c. HVAC control system sequences of operation formatted as indicated.
- d. Procedures for the HVAC system start-up, operation and shut-down including the manufacturer's supplied procedures for each piece of equipment, and procedures for the overall HVAC system.
- e. As-built HVAC control system detail drawings formatted as indicated.
- f. Routine maintenance checklist. Provide the routine maintenance checklist arranged in a columnar format, where the first column lists all installed devices, the second column states the maintenance activity or that no maintenance required, the third column states the frequency of the maintenance activity, and the fourth column is used for additional comments or reference.
- g. Qualified service organization list, including at a minimum company name, contact name and phone number.
- h. Start-Up Testing Report.
- i. Performance Verification Test (PVT) Procedures and Report.

Submit 2 copies of the Operation and Maintenance Instructions, indexed and in booklet form. The Operation and Maintenance Instructions may be submitted as a Technical Data Package.

### 3.7 MAINTENANCE AND SERVICE

Provide services, materials and equipment as necessary to maintain the entire system in an operational state as indicated for a period of one year from the date of final acceptance of the project. Minimize impacts on facility operations.

- a. The integration of the system specified in this section into a Utility Monitoring and Control System must not, of itself, void the warranty or otherwise alter the requirement for the one year maintenance and service period. Integration into a UMCS includes but is not limited

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

to establishing communication between devices in the control system and the front end or devices in another system.

- b. The changing of configuration properties must not, of itself, void the warranty or otherwise alter the requirement for the one year maintenance and service period.

#### 3.7.1 Description of Work

Provide adjustment and repair of the system including the manufacturer's required sensor and actuator (including transducer) calibration, span and range adjustment.

#### 3.7.2 Personnel

Use only service personnel qualified to accomplish work promptly and satisfactorily. Advise the Government in writing of the name of the designated service representative, and of any changes in personnel.

#### 3.7.3 Scheduled Inspections

Perform two inspections at six-month intervals and provide work required. Perform inspections in June and December. During each inspection perform the indicated tasks:

- a. Perform visual checks and operational tests of equipment.
- b. Clean control system equipment including interior and exterior surfaces.
- c. Check and calibrate each field device. Check and calibrate 50 percent of the total analog inputs and outputs during the first inspection. Check and calibrate the remaining 50 percent of the analog inputs and outputs during the second major inspection. Certify analog test instrumentation accuracy to be twice the specified accuracy of the device being calibrated. Randomly check at least 25 percent of all binary inputs and outputs for proper operation during the first inspection. Randomly check at least 25 percent of the remaining binary inputs and outputs during the second inspection. If more than 20 percent of checked inputs or outputs failed the calibration check during any inspection, check and recalibrate all inputs and outputs during that inspection.
- d. Run system software diagnostics and correct diagnosed problems.
- e. Resolve any previous outstanding problems.

#### 3.7.4 Scheduled Work

This work must be performed during regular working hours, Monday through Friday, excluding Federal holidays.

#### 3.7.5 Emergency Service

The Government will initiate service calls when the system is not functioning properly. Qualified personnel must be available to provide service to the system. A telephone number where the service supervisor can be reached at all times must be provided. Service personnel must be at the site within 24 hours after receiving a request for service. The

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

control system must be restored to proper operating condition as required per Section 01 78 00 CLOSEOUT SUBMITTALS.

### 3.7.6 Operation

After performing scheduled adjustments and repairs, verify control system operation as demonstrated by the applicable tests of the performance verification test.

### 3.7.7 Records and Logs

Keep dated records and logs of each task, with cumulative records for each major component, and for the complete system chronologically. Maintain a continuous log for all devices, including initial analog span and zero calibration values and digital points. Keep complete logs and provide logs for inspection onsite, demonstrating that planned and systematic adjustments and repairs have been accomplished for the control system.

### 3.7.8 Work Requests

Record each service call request as received and include its location, date and time the call was received, nature of trouble, names of the service personnel assigned to the task, instructions describing what has to be done, the amount and nature of the materials to be used, the time and date work started, and the time and date of completion. Submit a record of the work performed within 5 days after work is accomplished.

### 3.7.9 System Modifications

Submit recommendations for system modification in writing. Do not make system modifications, including operating parameters and control settings, without prior approval of the Government.

## 3.8 TRAINING

Conduct a training course for 8 operating staff members designated by the Government in the maintenance and operation of the system, including specified hardware and software. Conduct 32 hours of training at the project site within 30 days after successful completion of the performance verification test. The Government reserves the right to make audio and visual recordings (using Government supplied equipment) of the training sessions for later use. Provide audiovisual equipment and other training materials and supplies required to conduct training. A training day is defined as 8 hours of classroom instruction, including two 15 minute breaks and excluding lunchtime, Monday through Friday, during the daytime shift in effect at the training facility.

### 3.8.1 Training Documentation

Prepare training documentation consisting of:

- a. Course Attendee List: Develop the list of course attendees in coordination with and signed by the HVAC shop supervisor.
- b. Training Manuals: Provide training manuals which include an agenda, defined objectives for each lesson, and a detailed description of the subject matter for each lesson. When presenting portions of the course material by audiovisuals, deliver copies of those audiovisuals as a part of the printed training manuals.



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 3.8.2 Training Course Content

For guidance in planning the required instruction, assume that attendees will have a high school education, and are familiar with HVAC systems. During the training course, cover all of the material contained in the Operating and Maintenance Instructions, the layout and location of each controller enclosure, the layout of one of each type of equipment and the locations of each, the location of each control device external to the panels, the location of the compressed air station, preventive maintenance, troubleshooting, diagnostics, calibration, adjustment, commissioning, tuning, and repair procedures. Typical systems and similar systems may be treated as a group, with instruction on the physical layout of one such system. Present the results of the performance verification test and the Start-Up Testing Report as benchmarks of HVAC control system performance by which to measure operation and maintenance effectiveness.

### 3.8.3 Training Documentation Submittal Requirements

Submit hardcopy training manuals and all training materials on CD-ROM. Provide one hardcopy manual for each trainee on the Course Attendee List and 2 additional copies for archive at the project site. Provide 2 copies of the Course Attendee List with the archival copies. Training Documentation may be submitted as a Technical Data Package.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## APPENDIX A

| <u>QC CHECKLIST FOR BACNET SYSTEMS</u>                                                                                                                                                                                                         |                                                                                                                   |      |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|------|
| <p>This checklist is not all-inclusive of the requirements of this specification and should not be interpreted as such.</p> <p>Instructions: Initial each item in the space provided ( ____ ) verifying that the requirement has been met.</p> |                                                                                                                   |      |
| <p>This checklist is for (circle one:)</p> <p>Pre-Construction QC Checklist Submittal</p> <p>Post-Construction QC Checklist Submittal</p> <p>Close-out QC Checklist Submittal</p>                                                              |                                                                                                                   |      |
| Items verified for Pre-Construction, Post-Construction and Closeout QC Checklist Submittals:                                                                                                                                                   |                                                                                                                   |      |
| 1                                                                                                                                                                                                                                              | All DDC Hardware is numbered on Control System Schematic Drawings.                                                | ____ |
| 2                                                                                                                                                                                                                                              | Signal lines on Control System Schematic are labeled with the signal type.                                        | ____ |
| 3                                                                                                                                                                                                                                              | Local Display Panel (LDP) Locations are shown on Control System Schematic drawings.                               | ____ |
| Items verified for Post-Construction and Closeout QC Checklist Submittals:                                                                                                                                                                     |                                                                                                                   |      |
| 4                                                                                                                                                                                                                                              | All sequences are performed as specified using DDC Hardware.                                                      | ____ |
| 5                                                                                                                                                                                                                                              | Training schedule and course attendee list has been developed and coordinated with shops and submitted.           | ____ |
| Items verified for Closeout QC Checklist Submittal:                                                                                                                                                                                            |                                                                                                                   |      |
| 6                                                                                                                                                                                                                                              | Final As-built Drawings, including all Points Schedule drawings, accurately represent the final installed system. | ____ |
| 7                                                                                                                                                                                                                                              | Programming software has been submitted for all programmable controllers.                                         | ____ |
| 8                                                                                                                                                                                                                                              | All software has been licensed to the Government.                                                                 | ____ |

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

| <b><u>QC CHECKLIST FOR BACNET SYSTEMS</u></b>                                  |                                                                                                                                                                                                                      |        |
|--------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| 9                                                                              | O&M Instructions have been completed and submitted.                                                                                                                                                                  | ____   |
| 10                                                                             | Training course has been completed.                                                                                                                                                                                  | ____   |
| 11                                                                             | All DDC Hardware is installed on a BACnet ASHRAE 135 network using either MS/TP in accordance with Clause 9 or IP in accordance with Annex J.                                                                        | ____   |
| 12                                                                             | All DDC Hardware is BTL listed.                                                                                                                                                                                      | ____   |
| 13                                                                             | Communication between DDC Hardware is only via BACnet using standard services, except as specifically permitted by the specification. Non-standard services have been fully documented in the DDC Hardware Schedule. | ____   |
| 14                                                                             | Scheduling, Alarming, and Trending have been implemented using the standard BACnet Objects for these functions.                                                                                                      | ____   |
| 15                                                                             | All Properties indicated as required to be Writable are Writable and Overrides have been provided as indicated                                                                                                       | ____   |
| <div style="border-bottom: 1px solid black; height: 20px; width: 100%;"></div> |                                                                                                                                                                                                                      |        |
|                                                                                | (QC Representative Signature)                                                                                                                                                                                        | (Date) |

| <b><u>QC CHECKLIST FOR NIAGARA FRAMEWORK BASED BACNET SYSTEMS</u></b>                                                                                                                                                                                                 |                                                                    |      |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|------|
| <p>This checklist is not all-inclusive of the requirements of this specification and should not be interpreted as such.</p> <p>Instructions: Initial each item in the space provided ( ____ ) verifying that the requirement has been met.</p>                        |                                                                    |      |
| <p>This checklist is for (circle one:)</p> <p style="padding-left: 40px;">Pre-Construction QC Checklist Submittal</p> <p style="padding-left: 40px;">Post-Construction QC Checklist Submittal</p> <p style="padding-left: 40px;">Close-out QC Checklist Submittal</p> |                                                                    |      |
| <p>Items verified for Pre-Construction, Post-Construction and Closeout QC Checklist Submittals:</p>                                                                                                                                                                   |                                                                    |      |
| 1                                                                                                                                                                                                                                                                     | All DDC Hardware is numbered on Control System Schematic Drawings. | ____ |

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

| <b><u>QC CHECKLIST FOR NIAGARA FRAMEWORK BASED BACNET SYSTEMS</u></b>                                                                                                                                                 |                                                                                                                                                                                                                      |      |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 2                                                                                                                                                                                                                     | Signal lines on Control System Schematic are labeled with the signal type.                                                                                                                                           | ____ |
| 3                                                                                                                                                                                                                     | Local Display Panel (LDP) Locations are shown on Control System Schematic drawings.                                                                                                                                  | ____ |
| Items verified for Post-Construction and Closeout QC Checklist Submittals:                                                                                                                                            |                                                                                                                                                                                                                      |      |
| 4                                                                                                                                                                                                                     | All sequences are performed as specified using DDC Hardware.                                                                                                                                                         | ____ |
| 5                                                                                                                                                                                                                     | Training schedule and course attendee list has been developed and coordinated with shops and submitted.                                                                                                              | ____ |
| Items verified for Closeout QC Checklist Submittal:                                                                                                                                                                   |                                                                                                                                                                                                                      |      |
| 6                                                                                                                                                                                                                     | Final As-built Drawings, including all Points Schedule drawings, accurately represent the final installed system.                                                                                                    | ____ |
| 7                                                                                                                                                                                                                     | Programming software has been submitted for all programmable controllers.                                                                                                                                            | ____ |
| 8                                                                                                                                                                                                                     | All software has been licensed to the Government.                                                                                                                                                                    | ____ |
| 9                                                                                                                                                                                                                     | O&M Instructions have been completed and submitted.                                                                                                                                                                  | ____ |
| 10                                                                                                                                                                                                                    | Training course has been completed.                                                                                                                                                                                  | ____ |
| 11                                                                                                                                                                                                                    | All DDC Hardware is installed on a BACnet ASHRAE 135 network using either MS/TP in accordance with Clause 9 or IP in accordance with Annex J.                                                                        | ____ |
| 12                                                                                                                                                                                                                    | All DDC Hardware is BTL listed.                                                                                                                                                                                      | ____ |
| 13                                                                                                                                                                                                                    | Communication between DDC Hardware is only via BACnet using standard services, except as specifically permitted by the specification. Non-standard services have been fully documented in the DDC Hardware Schedule. | ____ |
| 14                                                                                                                                                                                                                    | Scheduling, Alarming, and Trending have been implemented using Niagara Framework objects and services, and BACnet Intrinsic Alarming as indicated.                                                                   | ____ |
| 15                                                                                                                                                                                                                    | All Properties indicated as required to be Writable are Writable and Overrides have been provided as indicated                                                                                                       | ____ |
| <div style="display: flex; justify-content: space-between; margin-top: 20px;"> <div style="width: 45%; border-bottom: 1px solid black;"></div> <div style="width: 45%; border-bottom: 1px solid black;"></div> </div> |                                                                                                                                                                                                                      |      |
| <div style="display: flex; justify-content: space-between;"> <span>(QC Representative Signature)</span> <span>(Date)</span> </div>                                                                                    |                                                                                                                                                                                                                      |      |

-- End of Section --

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## SECTION 23 09 13

### INSTRUMENTATION AND CONTROL DEVICES FOR HVAC

**11/15, CHG 2: 05/21**

#### PART 1 GENERAL

##### 1.1 SUMMARY

This section provides for the instrumentation control system components excluding direct digital controllers, network controllers, gateways etc. that are necessary for a completely functional automatic control system. When combined with a Direct Digital Control (DDC) system, the Instrumentation and Control Devices covered under this section must be a complete system suitable for the control of the heating, ventilating and air conditioning (HVAC) and other building-level systems as specified and indicated.

- a. Install hardware to perform the control sequences as specified and indicated and to provide control of the equipment as specified and indicated.
- b. Install hardware such that individual control equipment can be replaced by similar control equipment from other equipment manufacturers with no loss of system functionality.
- c. Install and configure hardware such that the Government or their agents are able to perform repair, replacement, and upgrades of individual hardware without further interaction with the installing Contractor.

##### 1.1.1 Verification of Dimensions

After becoming familiar with all details of the work, verify all dimensions in the field, and advise the Contracting Officer of any discrepancy before performing any work.

##### 1.1.2 Drawings

The Government will not indicate all offsets, fittings, and accessories that may be required on the drawings. Carefully investigate the mechanical, electrical, and finish conditions that could affect the work to be performed, arrange such work accordingly, and provide all work necessary to meet such conditions.

##### 1.2 RELATED SECTIONS

Related work specified elsewhere.

Section 01 30 00 ADMINISTRATIVE REQUIREMENTS

Section 23 30 00 HVAC AIR DISTRIBUTION

Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC

Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 1.3 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

#### AIR MOVEMENT AND CONTROL ASSOCIATION INTERNATIONAL, INC. (AMCA)

AMCA 500-D (2018) Laboratory Methods of Testing  
Dampers for Rating

#### AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI C12.1 (2014; Errata 2016) Electric Meters - Code  
for Electricity Metering

#### AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B16.15 (2018) Cast Copper Alloy Threaded Fittings  
Classes 125 and 250

ASME B16.34 (2021) Valves - Flanged, Threaded and  
Welding End

ASME BPVC SEC VIII D1 (2019) BPVC Section VIII-Rules for  
Construction of Pressure Vessels Division 1

#### ASTM INTERNATIONAL (ASTM)

ASTM A536 (1984; R 2019; E 2019) Standard  
Specification for Ductile Iron Castings

#### FLUID CONTROLS INSTITUTE (FCI)

FCI 70-2 (2021) Control Valve Seat Leakage

#### INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 142 (2007; Errata 2014) Recommended Practice  
for Grounding of Industrial and Commercial  
Power Systems - IEEE Green Book

#### INTERNATIONAL SOCIETY OF AUTOMATION (ISA)

ISA 7.0.01 (1996) Quality Standard for Instrument Air

#### NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

ANSI C12.20 (2015; E 2018) Electricity Meters - 0.1,  
0.2, and 0.5 Accuracy Classes

NEMA/ANSI C12.10 (2011; R 2021) Physical Aspects of  
Watt-hour Meters - Safety Standard

#### NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 (2023; ERTA 4 2023) National Electrical  
Code

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

NFPA 90A (2024) Standard for the Installation of  
Air Conditioning and Ventilating Systems

UNDERWRITERS LABORATORIES (UL)

UL 5085-3 (2006; Reprint Jan 2022) UL Standard for  
Safety Low Voltage Transformers - Part 3:  
Class 2 and Class 3 Transformers

#### 1.4 SUBMITTALS

Submittal requirements are specified in Section 23 09 00 INSTRUMENTATION  
AND CONTROL FOR HVAC.

#### 1.5 DELIVERY AND STORAGE

Store and protect products from the weather, humidity, and temperature  
variations, dirt and dust, and other contaminants, within the storage  
condition limits published by the equipment manufacturer.

#### 1.6 INPUT MEASUREMENT ACCURACY

Select, install and configure sensors, transmitters and DDC Hardware such  
that the maximum error of the measured value at the input of the DDC  
hardware is less than the maximum allowable error specified for the sensor  
or instrumentation.

#### 1.7 SUBCONTRACTOR SPECIAL REQUIREMENTS

Perform all work in this section in accordance with the paragraph entitled  
CONTRACTOR SPECIAL REQUIREMENTS in Section 01 30 00 ADMINISTRATIVE  
REQUIREMENTS.

### PART 2 PRODUCTS

#### 2.1 EQUIPMENT

##### 2.1.1 General Requirements

All products used to meet this specification must meet the indicated  
requirements, but not all products specified here will be required by  
every project. All products must meet the requirements both Section  
23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC and this Section.

##### 2.1.2 Operation Environment Requirements

Unless otherwise specified, provide products rated for continuous  
operation under the following conditions:

###### 2.1.2.1 Pressure

Pressure conditions normally encountered in the installed location.

###### 2.1.2.2 Vibration

Vibration conditions normally encountered in the installed location.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### 2.1.2.3 Temperature

- a. Products installed indoors: Ambient temperatures in the range of 32 to 112 degrees F and temperature conditions outside this range normally encountered at the installed location.
- b. Products installed outdoors or in unconditioned indoor spaces: Ambient temperatures in the range of -35 to +151 degrees F and temperature conditions outside this range normally encountered at the installed location.

#### 2.1.2.4 Humidity

10 to 95 percent relative humidity, non-condensing and also humidity conditions outside this range normally encountered at the installed location.

### 2.2 WEATHERSHIELDS

Provide weathershields constructed of galvanized steel painted white, unpainted aluminum, aluminum painted white, or white PVC.

### 2.3 WIRE AND CABLE

Provide wire and cable meeting the requirements of NFPA 70 and NFPA 90A in addition to the requirements of this specification and referenced specifications.

#### 2.3.1 Terminal Blocks

For terminal blocks which are not integral to other equipment, provide terminal blocks which are insulated, modular, feed-through, clamp style with recessed captive screw-type clamping mechanism, suitable for DIN rail mounting, and which have enclosed sides or end plates and partition plates for separation.

#### 2.3.2 Control Wiring for Binary Signals

For Control Wiring for Binary Signals, provide 18 AWG copper or thicker wire rated for 300-volt service.

#### 2.3.3 Control Wiring for Analog Signals

For Control Wiring for Analog Signals, provide 18 AWG or thicker, copper, single- or multiple-twisted wire meeting the following requirements:

- a. minimum 2 inch lay of twist
- b. 100 percent shielded pairs
- c. at least 300-volt insulation
- d. each pair has a 20 AWG tinned-copper drain wire and individual overall pair insulation
- e. cables have an overall aluminum-polyester or tinned-copper cable-shield tape, overall 20 AWG tinned-copper cable drain wire, and overall cable insulation.



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### 2.3.4 Power Wiring for Control Devices

For 24-volt circuits, provide insulated copper 18 AWG or thicker wire rated for 300 VAC service. For 120-volt circuits, provide 14 AWG or thicker stranded copper wire rated for 600-volt service.

#### 2.3.5 Transformers

Provide UL 5085-3 approved transformers. Select transformers sized so that the connected load is no greater than 80 percent of the transformer rated capacity.

### 2.4 AUTOMATIC CONTROL VALVES

Provide valves with stainless-steel stems and stuffing boxes with extended necks to clear the piping insulation. Provide valves with bodies meeting ASME B16.34 or ASME B16.15 pressure and temperature class ratings based on the design operating temperature and 150 percent of the system design operating pressure. Unless otherwise specified or indicated, provide valves meeting FCI 70-2 Class IV leakage rating. Provide valves rated for modulating or two-position service as indicated, which close against a differential pressure indicated as the Close-Off pressure and which are Normally-Open, Normally-Closed, or Fail-In-Last-Position as indicated.

#### 2.4.1 Valve Type

##### 2.4.1.1 Liquid Service 150 Degrees F or Less

Use either globe valves or ball valves except that butterfly valves may be used for sizes 4 inch and larger.

##### 2.4.1.2 Liquid Service Above 150 Degrees F

- a. Two-position valves: Use either globe valves or ball valves except that butterfly valves may be used for sizes 4 inch and larger.
- b. Modulating valves: Use globe valves except that butterfly valves may be used for sizes 4 inch and larger.

##### 2.4.1.3 Steam Service

Use globe valves except that butterfly valves may be used for sizes 4 inch and larger.

#### 2.4.2 Valve Flow Coefficient and Flow Characteristic

##### 2.4.2.1 Two-Way Modulating Valves

Provide the valve coefficient (Cv) indicated. Provide equal-percentage flow characteristic for liquid service except for butterfly valves. Provide linear flow characteristic for steam service except for butterfly valves.

##### 2.4.2.2 Three-Way Modulating Valves

Provide the valve coefficient (Cv) indicated. Provide linear flow characteristic with constant total flow throughout full plug travel.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### 2.4.3 Two-Position Valves

Use full line size full port valves with maximum available (Cv).

#### 2.4.4 Globe Valves

##### 2.4.4.1 Liquid Service Not Exceeding 150 Degrees F

a. Valve body and body connections:

- (1) valves 1-1/2 inches and smaller: brass or bronze body, with threaded or union ends
- (2) valves from 2 inches to 3 inches inclusive: brass, bronze, or iron bodies. 2 inch valves with threaded connections; 2-1/2 to 3 inches valves with flanged connections

b. Internal valve trim: Brass or bronze.

c. Stems: Stainless steel.

d. Provide valves compatible with a solution of 50 percent ethylene or propylene glycol.

#### 2.4.5 Ball Valves

##### 2.4.5.1 Liquid Service Not Exceeding 150 Degrees F

a. Valve body and connections:

(1) valves 1-1/2 inches and smaller: bodies of brass or bronze, with threaded or union ends

(2) valves from 2 inches to 3 inches inclusive: bodies of brass, bronze, or iron. 2 inch valves with threaded connections; valves from 2-1/2 to 3 inches with flanged connections.

b. Ball: Stainless steel or nickel-plated brass or chrome-plated brass.

c. Seals: Reinforced Teflon seals and EPDM O-rings.

d. Stem: Stainless steel, blow-out proof.

e. Provide valves compatible with a solution of 50 percent ethylene or propylene glycol.

#### 2.4.6 Butterfly Valves

Provide butterfly valves which are threaded lug type suitable for dead-end service and modulation to the fully-closed position, with carbon-steel bodies or with ductile iron bodies in accordance with ASTM A536. Provide butterfly valves with non-corrosive discs, stainless steel shafts supported by bearings, and EPDM seats suitable for temperatures from -20 to +250 degrees F. Provide valves with rated Cv of the Cv at 70 percent (60 degrees) open position. Provide valves meeting FCI 70-2 Class VI leakage rating.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### 2.4.7 Pressure Independent Control Valves (PICV)

Provide pressure independent control valves which include a regulator valve which maintains the differential pressure across a flow control valve. Pressure independent control valves must accurately control the flow from 0-100 percent full rated flow regardless of changes in the piping pressure and not vary the flow more than plus or minus 5 percent at any given flow control valve position when the PICV differential pressure lies between the manufacturer's stated minimum and maximum. The rated minimum differential pressure for steady flow must not exceed 5 psid across the PICV. Provide either globe or ball type valves meeting the indicated requirements for globe and ball valves. Provide valves with a flow tag listing full rated flow and minimum required pressure drop. Provide valves with factory installed Pressure/Temperature ports ("Pete's Plugs") to measure the pressure drop to determine the valve flow rate.

#### 2.4.8 Duct-Coil and Terminal-Unit-Coil Valves

For duct or terminal-unit coils provide control valves with screw type ends.

### 2.5 DAMPERS

#### 2.5.1 Damper Assembly

Provide single damper sections with blades no longer than 48 inches and which are no higher than 72 inches and damper blade width of 8 inches or less. When larger sizes are required, combine damper sections. Provide dampers made of steel, or other materials where indicated and with assembly frames constructed of 0.07 inch minimum thickness galvanized steel channels with mitered and welded corners. Steel channel frames constructed of 0.06 inch minimum thickness are acceptable provided the corners are reinforced.

- a. Flat blades must be made rigid by folding the edges. Blade-operating linkages must be within the frame so that blade-connecting devices within the same damper section must not be located directly in the air stream.
- b. Damper axles must be 1/2 inch minimum, plated steel rods supported in the damper frame by stainless steel or bronze bearings. Blades mounted vertically must be supported by thrust bearings.
- c. Provide dampers which do not exceed a pressure drop through the damper of 0.04 inches water gauge at 1000 ft/min in the wide-open position. Provide dampers with frames not less than 2 inch in width. Provide dampers which have been tested in accordance with AMCA 500-D.

#### 2.5.2 Operating Linkages

For operating links external to dampers, such as crank arms, connecting rods, and line shafting for transmitting motion from damper actuators to dampers, provide links able to withstand a load equal to at least 300 percent of the maximum required damper-operating force without deforming. Rod lengths must be adjustable. Links must be brass, bronze, zinc-coated steel, or stainless steel. Working parts of joints and clevises must be brass, bronze, or stainless steel. Adjustments of crank arms must control the open and closed positions of dampers.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 2.5.3 Damper Types

## 2.6 SENSORS AND INSTRUMENTATION

Unless otherwise specified, provide sensors and instrumentation which incorporate an integral transmitter. Sensors and instrumentation, including their transmitters, must meet the specified accuracy and drift requirements at the input of the connected DDC Hardware's analog-to-digital conversion.

### 2.6.1 Analog and Binary Transmitters

Provide transmitters which match the characteristics of the sensor. Transmitters providing analog values must produce a linear 4-20 mAdc, 0-10 Vdc signal corresponding to the required operating range and must have zero and span adjustment. Transmitters providing binary values must have dry contacts rated at 1A at 24 Volts AC.

### 2.6.2 Network Transmitters

Sensors and Instrumentation incorporating an integral network connection are considered DDC Hardware and must meet the DDC Hardware requirements of Section 23 09 23.01 LONWORKS DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS when used in a Lonworks network, or the requirements of 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS when used in a BACnet network.

### 2.6.3 Temperature Sensors

Provide the same sensor type throughout the project. Temperature sensors may be provided without transmitters. Where transmitters are used, the range must be the smallest available from the manufacturer and suitable for the application such that the range encompasses the expected range of temperatures to be measured. The end to end accuracy includes the combined effect of sensitivity, hysteresis, linearity and repeatability between the measured variable and the end user interface (graphic presentation) including transmitters if used.

#### 2.6.3.1 Sensor Accuracy and Stability of Control

##### 2.6.3.1.1 Conditioned Space Temperature

Plus or minus 0.5 degree F over the operating range.

##### 2.6.3.1.2 Duct Temperature

Plus or minus 0.5 degree F

##### 2.6.3.1.3 Outside Air Temperature

- a. Plus or minus 2 degrees F over the range of -30 to +130 degrees F AND
- b. Plus or minus 1 degree F over the range of 30 to 130 degrees F.

##### 2.6.3.1.4 Heating Hot Water

Plus or minus 2 degrees F.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### 2.6.3.2 Transmitter Drift

The maximum allowable transmitter drift: 0.25 degrees F per year.

#### 2.6.3.3 Point Temperature Sensors

Point Sensors must be encapsulated in epoxy, series 300 stainless steel, anodized aluminum, or copper.

#### 2.6.3.4 Temperature Sensor Details

##### 2.6.3.4.1 Room Type

Provide the sensing element components within a decorative protective cover suitable for surrounding decor.

##### 2.6.3.4.2 Duct Probe Type

Ensure the probe is long enough to properly sense the air stream temperature.

##### 2.6.3.4.3 Duct Averaging Type

Continuous averaging sensors must be one foot in length for each 1 square foot of duct cross-sectional area, and a minimum length of 5 feet.

##### 2.6.3.4.4 Pipe Immersion Type

For pipes with larger than 3 inch diameter, provide minimum 3 inch immersion. For pipes with less than 3 inch diameter, provide immersion at least half the diameter of the pipe. Provide each sensor with a corresponding pipe-mounted sensor well, unless indicated otherwise. Sensor wells must be stainless steel when used in steel piping, and brass when used in copper piping.

##### 2.6.3.4.5 Outside Air Type

Provide the sensing element rated for outdoor use

#### 2.6.4 Differential Pressure Instrumentation

##### 2.6.4.1 Differential Pressure Sensors

Provide Differential Pressure Sensors with ranges as indicated or as required for the application. Pressure sensor ranges must not exceed the high end range indicated on the Points Schedule by more than 50 percent. The over pressure rating must be a minimum of 150 percent of the highest design pressure of either input to the sensor. The accuracy must be plus or minus 1 percent of full scale. The sensor must have a maximum drift of 2 percent per year

##### 2.6.4.2 Differential Pressure Switch

Provide differential pressure switches with a user-adjustable setpoint which are sized for the application such that the setpoint is between 25 percent and 75 percent of the full range. The over pressure rating must be a minimum of 150 percent of the highest design pressure of either input to the sensor. The switch must have two sets of contacts and each contact must have a rating greater than it's connected load. Contacts must open

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

or close upon rise of pressure above the setpoint or drop of pressure below the setpoint as indicated.

## 2.6.5 Flow Sensors

### 2.6.5.1 Insertion Turbine Flowmeter

Provide dual axial turbine flowmeter with all installation hardware necessary to enable insertion and removal of the meter without system shutdown. All parts must meet or exceed the pressure classification of the pipe system it is installed in. Insertion Turbine Flowmeter accuracy must be plus or minus 0.5 percent of rate at calibrated velocity., within plus or minus of rate over a 10:1 turndown and within plus or minus 2 percent of rate over a 50:1 turndown. Repeatability must be plus or minus 0.25 percent of reading. The meter flow sensing element must operate over a range suitable for the installed location with a pressure loss limited to 1 percent of operating pressure at maximum flow rate. The flowmeter ,must include either dry contact pulse outputs, 4-20mA, 0-10Vdc or 0-5Vdc outputs. The turbine rotor assembly must be constructed of Series 300 stainless steel and use Teflon seals.

### 2.6.5.2 Ultrasonic Flow Meter

Provide Ultrasonic Flow Meters complete with matched transducers, self aligning installation hardware and transducer cables. Ultrasonic transducers must be optimized for the specific pipe and process conditions for the application. The flow meter accuracy must plus or minus 1 percent of rate from 0 to 40 ft/sec. The flowmeter must include either dry contact pulse outputs, 4-20mA, 0-10Vdc or 0-5Vdc output.

### 2.6.5.3 Flow Switch

Flow switch must have a repetitive accuracy of plus or minus 10 percent of actual flow setting. Switch actuation must be adjustable over the operating flow range, and must be sized for the application such that the setpoint is between 25 percent and 75 percent of the full range.. The switch must have Form C snap-action contacts, rated for the application. The flow switch must have non flexible paddle with magnetically actuated contacts and be rated for service at a pressure greater than the installed conditions. Flow switch for use in sewage system must be rated for use in corrosive environments encountered.

## 2.6.6 Electrical Instruments

Provide Electrical Instruments with an input range as indicated or sized for the application. Unless otherwise specified, AC instrumentation must be suitable for 60 Hz operation.

### 2.6.6.1 Current Transducers

Current transducers must accept an AC current input and must have an accuracy of plus or minus 0.5 percent of full scale. The device must have a means for calibration. Current transducers for variable frequency applications must be rated for variable frequency operation.

### 2.6.6.2 Current Sensing Relays (CSRs)

Current sensing relays (CSRs) must provide a normally-open contact with a voltage and amperage rating greater than its connected load. Current

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

sensing relays must be of split-core design. The CSR must be rated for operation at 200 percent of the connected load. Voltage isolation must be a minimum of 600 volts. The CSR must auto-calibrate to the connected load or be adjustable and field calibrated. Current sensors for variable frequency applications must be rated for variable frequency operation.

#### 2.6.6.3 Voltage Transducers

Voltage transducers must accept an AC voltage input and have an accuracy of plus or minus 0.25 percent of full scale. The device must have a means for calibration. Line side fuses for transducer protection must be provided.

#### 2.6.6.4 Energy Metering

##### 2.6.6.4.1 Watt or Watthour Transducers

Watt transducers must measure voltage and current and must output kW or kWh or both kW and kWh as indicated. kW outputs must have an accuracy of plus or minus 0.5 percent over a power factor range of 0.1 to 1. kWh outputs must have an accuracy of plus or minus 0.5 percent over a power factor range of 0.1 to 1.

##### 2.6.6.4.2 Watthour Revenue Meter (with and without Demand Register)

All Watthour revenue meters must measure voltage and current and must be in accordance with ANSI C12.1 with an ANSI C12.20 Accuracy class of 0.2 and must have pulse initiators for remote monitoring of Watthour consumption. Pulse initiators must consist of form C contacts with a current rating not to exceed two amperes and voltage not to exceed 500 V, with combinations of VA not to exceed 100 VA, and a life rating of one billion operations. Meter sockets must be in accordance with NEMA/ANSI C12.10. Watthour revenue meters with demand registers must output instantaneous demand in addition to the pulse initiators.

##### 2.6.6.4.3 Steam Meters

Steam meters must be the vortex type, with pressure compensation, a minimum turndown ratio of 10 to 1, and an output signal compatible with the DDC system.

##### 2.6.6.4.4 Hydronic BTU Meters

The BTU meter is to be supplied with wall mount hardware and be capable of being installed remote from the flow meter. The BTU meter must include an LCD display for local indication of energy rate and for display of parameters and settings during configuration. Each BTU meter must be factory configured for its specific application and be completely field configurable by the user via a front panel keypad (no special interface device or computer required). The unit must output Energy Rate, Energy Total, Flow Rate, Supply Temperature, and Return Temperature. An integral transmitter is to provide a linear analog or configurable pulse output signal representing the energy rate; and the signal must be compatible with building automation system DDC Hardware to which the output is connected.

#### 2.6.7 Damper End Switches

Each end switch must be a hermetically sealed switch with a trip lever and

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

over-travel mechanism. The switch enclosure must be suitable for mounting on the duct exterior and must permit setting the position of the trip lever that actuates the switch. The trip lever must be aligned with the damper blade.

End switches integral to an electric damper actuator are allowed as long as at least one is adjustable over the travel of the actuator.

#### 2.6.8 Air Quality Sensors

Provide full spectrum air quality sensors using a hot wire element based on the Taguchi principle. The sensor must monitor a wide range of gaseous volatile organic components common in indoor air contaminants like paint fumes, solvents, cigarette smoke, and vehicle exhaust. The sensor must automatically compensate for temperature and humidity, have span and calibration potentiometers, operate on 24 VDC power with output of 0-10 VDC, and have a service rating of 32 to 140 degrees F and 5 to 95 percent relative humidity.

### 2.7 OUTPUT DEVICES

#### 2.7.1 Actuators

Actuators must be electric (electronic) . All actuators must be normally open (NO), normally closed (NC) or fail-in-last-position (FILP) as indicated. Normally open and normally closed actuators must be of mechanical spring return type. Electric actuators must have an electronic cut off or other means to provide burnout protection if stalled. Actuators must have a visible position indicator. Electric actuators must provide position feedback to the controller as indicated. Actuators must smoothly and fully open or close the devices to which they are applied. Electric actuators must have a full stroke response time in both directions of 90 seconds or less at rated load. Electric actuators must be of the foot-mounted type with an oil-immersed gear train or the direct-coupled type. Where multiple electric actuators operate from a common signal, the actuators must provide an output signal identical to its input signal to the additional devices. All actuators must be rated for their operating environment. Actuators used outdoors must be designed and rated for outdoor use. Actuators under continuous exposure to water, such as those used in sumps, must be submersible.

Actuators incorporating an integral network connection are considered DDC Hardware and must meet the DDC Hardware requirements of Section 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS.

##### 2.7.1.1 Valve Actuators

Valve actuators must provide shutoff pressures and torques as indicated on the Valve Schedule.

##### 2.7.1.2 Damper Actuators

Damper actuators must provide the torque necessary per damper manufacturer's instructions to modulate the dampers smoothly over its full range of operation and torque must be at least 6 inch-pounds/1 square foot of damper area for opposed blade dampers and 9 inch-pounds/1 square foot of damper area for parallel blade dampers.



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 2.7.1.3 Electric Actuators

Each actuator must have distinct markings indicating the full-open and full-closed position. Each actuator must deliver the torque required for continuous uniform motion and must have internal end switches to limit the travel, or be capable of withstanding continuous stalling without damage. Actuators must function properly within 85 to 110 percent of rated line voltage. Provide actuators with hardened steel running shafts and gears of steel or copper alloy. Fiber or reinforced nylon gears may be used for torques less than 16 inch-pounds..

- a. Two-position actuators must be single direction, spring return, or reversing type. Two position actuator signals may either be the control power voltage or line voltage as needed for torque or appropriate interlock circuits.
- b. Modulating actuators must be capable of stopping at any point in the cycle, and starting in either direction from any point. Actuators must be equipped with a switch for reversing direction, and a button to disengage the clutch to allow manual adjustments. Provide the actuator with a hand crank for manual adjustments, as applicable. Modulating actuator input signals can either be a 4 to 20 mA<sub>dc</sub> or a 0-10 VDC signal.
- c. Floating or pulse width modulation actuators are acceptable for non-fail safe applications unless indicated otherwise provided that the floating point control (timed actuation) must have a scheduled re-calibration of span and position no more than once a day and no less than once a week. The schedule for the re-calibration should not affect occupied conditions and be staggered between equipment to prevent falsely loading or unloading central plant equipment.

### 2.7.2 Solenoid-Operated Electric to Pneumatic Switch (EPS)

Solenoid-Operated Electric to Pneumatic Switches (EPS) must accept a voltage input to actuate its air valve. Each valve must have three-port operation: common, normally open, and normally closed. Each valve must have an outer cast aluminum body and internal parts of brass, bronze, or stainless steel. The air connection must be a 0.38 inch NPT threaded connection. Valves must be rated for 50 psig.

### 2.7.3 Relays

Relays must have contacts rated for the intended application, indicator light, and dust proof enclosure. The indicator light must be lit when the coil is energized and off when coil is not energized.

Control relay contacts must have utilization category and ratings selected for the application. Each set of contacts must incorporate a normally open (NO), normally closed (NC) and common contact. Relays must be rated for a minimum life of one million operations.

## 2.8 USER INPUT DEVICES

User Input Devices, including potentiometers, switches and momentary contact push-buttons. Potentiometers must be of the thumb wheel or sliding bar type. Momentary Contact Push-Buttons may include an adjustable timer for their output. User input devices must be labeled for their function.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## 2.9 MULTIFUNCTION DEVICES

Multifunction devices are products which combine the functions of multiple sensor, user input or output devices into a single product. Unless otherwise specified, the multifunction device must meet all requirements of each component device. Where the requirements for the component devices conflict, the multifunction device must meet the most stringent of the requirements.

### 2.9.1 Current Sensing Relay Command Switch

The Current Sensing Relay portion must meet all requirements of the Current Sensing Relay input device. The Command Switch portion must meet all requirements of the Relay output device except that it must have at least one normally-open (NO) contact.

Current Sensing Relays used for Variable Frequency Drives must be rated for Variable Frequency applications unless installed on the source side of the drive. If used in this situation, the threshold for showing status must be set to allow for the VFD's control power when the drive is not enabled and provide indication of operation when the drive is enabled at minimum speed.

### 2.9.2 Space Sensor Module

Space Sensor Modules must be multifunction devices incorporating a temperature sensor and one or more of the following as specified and indicated on the Space Sensor Module Schedule:

- a. A temperature indicating device.
- b. A User Input Device which must adjust a temperature setpoint output.
- c. A User Input Momentary Contact Button and an output to the control system indicating zone occupancy.
- d. A three position User Input Switch labeled to indicate heating, cooling and off positions ('HEAT-COOL-OFF' switch) and providing corresponding outputs to the control system.
- e. A two position User Input Switch labeled with 'AUTO' and 'ON' positions and providing corresponding output to the control system..
- f. A multi-position User Input Switch with 'OFF' and at least two fan speed positions and providing corresponding outputs to the control system.

Space Sensor Modules cannot contain mercury (Hg).

## PART 3 EXECUTION

### 3.1 INSTALLATION

#### 3.1.1 General Installation Requirements

Perform the installation under the supervision of competent technicians regularly employed in the installation of DDC systems.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### 3.1.1.1 Device Mounting Criteria

All devices must be installed in accordance with manufacturer's recommendations and as specified and indicated. Control devices to be installed in piping and ductwork must be provided with required gaskets, flanges, thermal compounds, insulation, piping, fittings, and manual valves for shutoff, equalization, purging, and calibration. Strap-on temperature sensing elements must not be used except as specified. Spare thermowells must be installed adjacent to each thermowell containing a sensor and as indicated. Devices located outdoors must have a weathershield.

#### 3.1.1.2 Labels and Tags

Match labels and tags to the unique identifiers indicated on the As-Built drawings. Label all enclosures and instrumentation. Tag all sensors and actuators in mechanical rooms. Tag airflow measurement arrays to show flow rate range for signal output range, duct size, and pitot tube AFMA flow coefficient. Tag duct static pressure taps at the location of the pressure tap. Provide plastic or metal tags, mechanically attached directly to each device or attached by a metal chain or wire. Labels exterior to protective enclosures must be engraved plastic and mechanically attached to the enclosure or instrumentation. Labels inside protective enclosures may be attached using adhesive, but must not be hand written.

#### 3.1.2 Weathershield

Provide weathershields for sensors located outdoors. Install weathershields such that they prevent the sun from directly striking the sensor and prevent rain from directly striking or dripping onto the sensor. Install weather shields with adequate ventilation so that the sensing element responds to the ambient conditions of the surroundings. When installing weathershields near outside air intake ducts, install them such that normal outside air flow does not cause rainwater to strike the sensor.

#### 3.1.3 Room Instrument Mounting

Mount room instruments, including but not limited to wall mounted non-adjustable space sensor modules and sensors located in occupied spaces, 48 inches above the floor unless otherwise indicated. Install adjustable devices to be ADA compliant unless otherwise indicated on the Room Sensor Schedule:

- a. Space Sensor Modules for Fan Coil Units may be either unit or wall mounted but not mounted on an exterior wall.
- b. Wall mount all other Space Sensor Modules.

#### 3.1.4 Indication Devices Installed in Piping and Liquid Systems

Provide snubbers for gauges in piping systems subject to pulsation. For gauges for steam service use pigtail fittings with cock. Install thermometers and temperature sensing elements in liquid systems in thermowells. Provide spare Pressure/Temperature Ports (Pete's Plug) for all temperature and pressure sensing elements installed in liquid systems for calibration/testing.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 3.1.5 Switches

#### 3.1.5.1 Temperature Limit Switch

Provide a temperature limit switch (freezestat) to sense the temperature at the location indicated. Provide a sufficient number of temperature limit switches (freezestats) to provide complete coverage of the duct section but no less than 1 foot in length per square foot of cross sectional area. Install manual reset limit switches in approved, accessible locations where they can be reset easily. Install temperature limit switch (freezestat) sensing elements in a side-to-side (not top-to-bottom) serpentine pattern with the relay section at the highest point and in accordance with the manufacturer's installation instructions.

#### 3.1.5.2 Hand-Off Auto Switches

Wire safety controls such as smoke detectors and freeze protection thermostats to protect the equipment during both hand and auto operation.

### 3.1.6 Temperature Sensors

Install temperature sensors in locations that are accessible and provide a good representation of sensed media. Installations in dead spaces are not acceptable. Calibrate and install sensors according to manufacturer's instructions. Select sensors only for intended application as designated or recommended by manufacturer.

#### 3.1.6.1 Room Temperature Sensors

Mount the sensors on interior walls to sense the average room temperature at the locations indicated. Avoid locations near heat sources such as copy machines or locations by supply air outlet drafts. Mount the center of all user-adjustable sensors 48 inches above the floor to meet ADA requirements. Non user-adjustable sensors can be mounted as indicated in paragraph ROOM INSTRUMENT MOUNTING.

#### 3.1.6.2 Duct Temperature Sensors

##### 3.1.6.2.1 Probe Type

Place tip of the sensor in the middle of the airstream or in accordance with manufacturer's recommendations or instructions. Provide a gasket between the sensor housing and the duct wall. Seal the duct penetration air tight. When installed in insulated duct, provide enclosure or stand off fitting to accommodate the thickness of duct insulation to allow for maintenance or replacement of the sensor and wiring terminations. Seal the duct insulation penetration vapor tight.

##### 3.1.6.2.2 Averaging Type

Weave the sensing element in a serpentine fashion from side to side perpendicular to the flow, across the duct or air handler cross-section, using durable non-metal supports in accordance with manufacturer's installation instructions. Avoid tight radius bends or kinking of the sensing element. Prevent contact between the sensing element and the duct or air handler internals. Provide a duct access door at the sensor location. The access door must be hinged on the side, factory insulated, have cam type locks, and be as large as the duct will permit, maximum 18 by 18 inches. For sensors inside air handlers, the sensors must be fully

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

accessible through the air handler's access doors without removing any of the air handler's internals.

#### 3.1.6.3 Immersion Temperature Sensors

Provide thermowells for sensors measuring piping, tank, or pressure vessel temperatures. Locate wells to sense continuous flow conditions. Do not install wells using extension couplings. When installed on insulated piping, provide stand enclosure or stand off fitting to accommodate the thickness of the pipe insulation and allow for maintenance or replacement of the sensor or wiring terminations. Where piping diameters are smaller than the length of the wells, provide wells in piping at elbows to sense flow across entire area of well. Wells must not restrict flow area to less than 70 percent of pipe area. Increase piping size as required to avoid restriction. Provide the sensor well with a heat-sensitive transfer agent between the sensor and the well interior ensuring contact between the sensor and the well.

#### 3.1.6.4 Outside Air Temperature Sensors

Provide outside air temperature sensors on the building's north side with a protective weather shade that does not inhibit free air flow across the sensing element, and protects the sensor from snow, ice, and rain. Location must not be near exhaust hoods and other areas such that it is not influenced by radiation or convection sources which may affect the reading. Provide a shield to shade the sensor from direct sunlight.

#### 3.1.7 Duct Static Pressure Sensors

Locate the duct static pressure sensing tap at 75 percent of the distance between the first and last air terminal units as indicated on the design documents. If the transmitter output is a 0-10Vdc signal, locate the transmitter in the same enclosure as the air handling unit (AHU) controller for the AHU serving the terminal units. If a remote duct static pressure sensor is to be used, run the signal wire back to the controller for the air handling unit.

#### 3.1.8 Meters

##### 3.1.8.1 Flowmeters

Install flowmeters to ensure minimum straight unobstructed piping for at least 10 pipe diameters upstream and at least 5 pipe diameters downstream of the flowmeter, and in accordance with the manufacturer's installation instructions.

##### 3.1.8.2 Energy Meters

Locate energy meters as indicated. Connect each meter output to the DDC system, to measure both instantaneous demand/energy and other variables as indicated.

#### 3.1.9 Dampers

##### 3.1.9.1 Damper Actuators

Provide spring return actuators which fail to a position that protects the served equipment and space on all control dampers related to freeze protection or force protection. For all outside, makeup and relief

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

dampers provide dampers which fail closed. Terminal fan coil units, terminal VAV units, convectors, and unit heaters may be non-spring return unless indicated otherwise. Do not mount actuators in the air stream. Do not connect multiple actuators to a common drive shaft. Install actuators so that their action seal the damper to the extent required to maintain leakage at or below the specified rate and so that they move the blades smoothly throughout the full range of motion.

#### 3.1.9.2 Damper Installation

Install dampers straight and true, level in all planes, and square in all dimensions. Dampers must move freely without undue stress due to twisting, racking (parallelogramming), bowing, or other installation error. External linkages must operate smoothly over the entire range of motion, without deformation or slipping of any connecting rods, joints or brackets that will prevent a return to it's normal position. Blades must close completely and leakage must not exceed that specified at the rated static pressure. Provide structural support for multi-section dampers. Acceptable methods of structural support include but are not limited to U-channel, angle iron, corner angles and bolts, bent galvanized steel stiffeners, sleeve attachments, braces, and building structure. Where multi-section dampers are installed in ducts or sleeves, they must not sag due to lack of support. Do not use jackshafts to link more than three damper sections. Do not use blade to blade linkages. Install outside and return air dampers such that their blades direct their respective air streams towards each other to provide for maximum mixing of air streams.

#### 3.1.10 Valves

Install the valves in accordance with the manufacturer's instructions.

##### 3.1.10.1 Valve Actuators

Provide spring return actuators on all control valves where freeze protection is required. Spring return actuators for terminal fan coil units, terminal VAV units, convectors, and unit heaters are not required unless indicated otherwise.

#### 3.1.11 Thermometers and Gauges

##### 3.1.11.1 Thermometers

Mount devices to allow reading while standing on the floor or ground, as applicable.

#### 3.1.12 Wire and Cable

Provide complete electrical wiring for the Control System, including wiring to transformer primaries. Wire and Cable must be installed without splices between control devices and in accordance with NFPA 70 and NFPA 90A. Instrumentation grounding must be installed per the device manufacturer's instructions and as necessary to prevent ground loops, noise, and surges from adversely affecting operation of the system. Test installed ground rods as specified in IEEE 142. Cables and conductor wires must be tagged at both ends, with the identifier indicated on the shop drawings. Electrical work must be as specified in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM and as indicated. Wiring external to enclosures must be run in raceways, except low-voltage control and low-voltage network wiring may be installed as follows:

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

- a. plenum rated cable in suspended ceilings over occupied spaces may be run without raceways
- b. nonmetallic-sheathed cables or metallic-armored cables may be installed as permitted by NFPA 70.

Install control circuit wiring not in raceways in a neat and safe manner. Wiring must not use the suspended ceiling system (including tiles, frames or hangers) for support. Where conduit or raceways are required, control circuit wiring must not run in the same conduit/raceway as power wiring over 50 volts. Run all circuits over 50 volts in conduit, metallic tubing, covered metal raceways, or armored cable.

#### 3.1.13 Copper Tubing

Provide hard-drawn copper tubing in exposed areas and either hard-drawn or annealed copper tubing in concealed areas. Use only tool-made bends. Use only brass or copper solder joint type fittings, except for connections to apparatus. For connections to apparatus use brass compression type fittings.

#### 3.1.14 Plastic Tubing

Install plastic tubing within covered raceways or conduit except when otherwise specified. Do not use plastic tubing for applications where the tubing could be subjected to a temperature exceeding 130 degrees F. For fittings, use brass or acetal resin of the compression or barbed push-on type for instrument service. Except in walls and exposed locations, plastic multitube instrument tubing bundle without conduit or raceway protection may be used where a number of air lines run to the same points, provided the multitube bundle is enclosed in a protective sheath, is run parallel to the building lines and is adequately supported as specified.

#### 3.1.15 Compressed Air Stations

Mount the air compressor assembly on vibration eliminators, in accordance with ASME BPVC SEC VIII D1 for tank clearance. Connect the air line to the tank with a flexible pipe connector. Provide compressed air station specialties with required tubing, including condensate tubing to a floor drain. Compressed air stations must deliver control air meeting the requirements of ISA 7.0.01. Provide foundations and housekeeping pads for the HVAC control system air compressors .

-- End of Section --

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## SECTION 23 09 23.02

### BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS 02/19, CHG 1: 02/20

#### PART 1 GENERAL

##### 1.1 SUMMARY

Provide a complete Direct Digital Control (DDC) system, suitable for the control of the heating, ventilating and air conditioning (HVAC) and other building-level systems as specified and shown and in accordance with Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC.

##### 1.1.1 System Requirements

Provide a system meeting the requirements of both Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC and this Section and with the following characteristics:

- a. Except for Gateways, the control system must be an open implementation of BACnet technology using ASHRAE 135 and Fox as the communications protocols. The system must use standard ASHRAE 135 Objects and Properties and the Niagara Framework. The system must use standard ASHRAE 135 Services and the Niagara Framework exclusively for communication over the network. Gateways to packaged units must communicate with other DDC hardware using ASHRAE 135 or the Fox protocol exclusively and may communicate with packaged equipment using other protocols. The control system must be installed such that any two ASHRAE 135 devices on the Internetwork can communicate using standard ASHRAE 135 Services.
- b. Install and configure control hardware to provide ASHRAE 135 Objects and Properties or Niagara Framework Objects as indicated and as needed to meet the requirements of this specification.
- c. Use Niagara Framework hardware and software exclusively for scheduling, trending, and communication with a front end (UMCS). Use Niagara Framework or standard BACnet Objects and services for alarming. Use the Fox protocol for all communication between Niagara Framework Supervisory Gateways; use the ASHRAE 135 protocol for all other building communication. Niagara Framework Supervisory Gateway must serve web pages as specified.
- d. Use Niagara Framework Version 4.0 or later.

##### 1.1.2 Verification of Specification Requirements

Review all specifications related to the control system installation and advise the Contracting Officer of any discrepancies before performing any work. If Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC or any other Section referenced in this specification is not included in the project specifications advise the Contracting Officer and either obtain the missing Section or obtain Contracting Officer approval before performing any work.



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## 1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

### AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS (ASHRAE)

ASHRAE 135 (2020; Interpretation 1-8 2021; Errata 1-2 2021; Addenda CD 2021; Addenda BV-CE 2022; Interpretation 9-12 2022; Interpretation 13-17 2023; Addenda BV-CF 2023; Errata 3 2023) BACnet—A Data Communication Protocol for Building Automation and Control Networks

### BACNET INTERNATIONAL (BTL)

BTL Guide (v.49; 2017) BACnet Testing Laboratory Implementation Guidelines

### INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 802.3 (2022) Ethernet

### TELECOMMUNICATIONS INDUSTRY ASSOCIATION (TIA)

TIA-485 (1998a; R 2012) Electrical Characteristics of Generators and Receivers for Use in Balanced Digital Multipoint Systems

### TRIDIUM, INC (TRIDIUM)

Niagara Framework (2012) NiagaraAX User's Guide

Tridium Open NiCS (2005) Understanding the NiagaraAX Compatibility Statement (NiCS)

### U.S. FEDERAL COMMUNICATIONS COMMISSION (FCC)

FCC Part 15 Radio Frequency Devices (47 CFR 15)

### UNDERWRITERS LABORATORIES (UL)

UL 916 (2015; Reprint Oct 2021) UL Standard for Safety Energy Management Equipment

## 1.3 DEFINITIONS

For definitions related to this section, see Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC.

## 1.4 SUBMITTALS

Submittal requirements related to this Section are specified in Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## PART 2 PRODUCTS

All products used to meet this specification must meet the indicated requirements, but not all products specified here will be required by every project. All products must meet the requirements both Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC and this Section.

### 2.1 NETWORK HARDWARE

#### 2.1.1 BACnet Router

All BACnet Routers must be BACnet/IP Routers and must perform layer 3 routing of ASHRAE 135 packets over an IP network in accordance with ASHRAE 135 Annex J and Clause 6. The router must provide the appropriate connection to the IP network and connections to one or more ASHRAE 135 MS/TP networks. Devices used as BACnet Routers must meet the requirements for DDC Hardware, and except for Niagara Framework Supervisory Gateways, devices used as BACnet routers must support the NM-RC-B BIBB.

#### 2.1.2 BACnet Gateways

In addition to the requirements for DDC Hardware, the BACnet Gateway must be a Niagara Framework Supervisory Gateway or must meet the following requirements:

- a. It must perform bi-directional protocol translation from one non-ASHRAE 135 protocol to ASHRAE 135. BACnet Gateways must incorporate a network connection to an ASHRAE 135 network (either BACnet over IP in accordance with Annex J or MS/TP) and a separate connection appropriate for the non-ASHRAE 135 protocol and media.
- b. It must retain its configuration after a power loss of an indefinite time, and must automatically return to their pre-power loss state once power is restored.
- c. It must allow bi-directional mapping of data between the non-ASHRAE 135 protocol and Standard Objects as defined in ASHRAE 135. It must support the DS-RP-B BIBB for Objects requiring read access and the DS-WP-B BIBB for Objects requiring write access.
- d. It must support the DS-COV-B BIBB.

Although Gateways must meet DDC Hardware requirements, except for Niagara Framework Supervisory Gateways, they are not DDC Hardware and must not be used when DDC Hardware is required. (Niagara Framework Supervisory Gateways are both Gateways and DDC Hardware.)

#### 2.1.3 Ethernet Switch

Ethernet Switches must autoconfigure between 10,100 and 1000 megabits per second (MBPS).

### 2.2 CONTROL NETWORK WIRING

- a. BACnet MS/TP communications wiring must be in accordance with ASHRAE 135. The wiring must use shielded, three wire (twisted-pair with reference) cable with characteristic impedance between 100 and 120 ohms. Distributed capacitance between conductors must be less than

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

30 pF per foot.

- b. Building Control Network Backbone IP Network must use Ethernet media. Ethernet cables must be CAT-5e at a minimum and meet all requirements of IEEE 802.3 .

## 2.3 DIRECT DIGITAL CONTROL (DDC) HARDWARE

### 2.3.1 General Requirements

All DDC Hardware must meet the following requirements:

- a. It must be locally powered and must incorporate a light to indicate the device is receiving power.
- b. It must conform to the BTL Guide
- c. It must be BACnet Testing Laboratory (BTL) Listed.
- d. The Manufacturer's Product Data submittal for each piece of DDC Hardware must include the Protocol Implementation Conformance Statement (PICS) for that hardware as specified in Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC.
- e. It must communicate and be interoperable in accordance with ASHRAE 135 and have connections for BACnet IP or MS/TP control network wiring.
- f. Other than devices controlling terminal units or functioning solely as a BACnet Router, it must support DS-COV-B, DS-RPM-A and DS-RPM-B BIBBs.
- g. Devices supporting the DS-RP-A BIBB must also support the DS-COV-A BIBB.
- h. Application programs, configuration settings and communication information must be stored in a manner such that they persist through loss of power:
  - (1) Application programs must persist regardless of the length of time power is lost.
  - (2) Configured settings must persist for any loss of power less than 2,500 hours.
  - (3) Communication information, including but not limited to COV subscriptions, event reporting destinations, Notification Class Object settings, and internal communication settings, must persist for any loss of power less than 2,500 hours.
- i. Internal Clocks:
  - (1) Clocks in DDC Hardware incorporating a Clock must continue to function for 120 hours upon loss of power to the DDC Hardware.
  - (2) DDC Hardware incorporating a Clock must support the DM-TS-B or DM-UTC-B BIBB.
- j. It must have all functionality indicated and required to support the application (Sequence of Operation or portion thereof) in which it is used, including but not limited to providing Objects or Niagara

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

Framework Points as specified and as indicated on the Points Schedule.

- k. In addition to these general requirements and the DDC Hardware Input-Output (I/O) Function requirements, all DDC Hardware must also meet any additional requirements for the application in which it is used (e.g. scheduling, alarming, trending, etc.).
- l. It must meet FCC Part 15 requirements and have UL 916 or equivalent safety listing.
- m. Except for Niagara Framework Supervisory Gateways, Device must support Commandable Objects to support Override requirements as detailed in PART 3 EXECUTION
- n. User interfaces which allow for modification of Properties or settings must be password-protected.
- o. Devices communicating BACnet MS/TP must meet the following requirements:
  - (1) Must have a configurable Max\_Master Property.
  - (2) DDC Hardware other than hardware controlling a single terminal unit must have a configurable Max\_Info\_Frames Property.
  - (3) Must respond to any valid request within 50 msec with either the appropriate response or with a response of "Reply Postponed".
  - (4) Must use twisted pair with reference and shield (3-wire media) wiring, or twisted pair with shield (2-wire media) wiring and use half-wave rectification.
- p. Devices communicating BACnet/IP must use UDP Port 0xBAC0. Devices with configurable UDP Ports must default to 0xBAC0.
- q. All Device IDs, Network Numbers, and BACnet MAC addresses of devices must be fully configurable without limitation, except MS/TP MAC addresses may be limited by ASHRAE 135 requirements.
- r. Except for Niagara Framework Supervisory Gateways, DDC Hardware controlling a single terminal unit must have:
  - (1) Objects (including the Device Object) with an Object Name Property of at least 8 characters in length.
  - (2) A configurable Device Object Name.
  - (3) A configurable Device Object Description Property at least 16 characters in length.
- s. Except for Objects in either Niagara Framework Supervisory Gateways or DDC Hardware controlling a single terminal unit, all Objects (including Device Objects) must:
  - (1) Have a configurable Object Name Property of at least 12 characters in length.
  - (2) Have a configurable Object Description Property of at least 24 characters in length.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

- t. For programmable DDC Hardware, provide and license to the project site all programming software required to program the Hardware in accordance with Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC.
- u. For programmable DDC Hardware, provide copies of the installed application programs (all software that is not common to every controller of the same manufacturer and model) as source code compatible with the supplied programming software in accordance with Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC. The submitted application program must be the complete application necessary for controller to function as installed and be sufficient to allow replacement of the installed controller with another controller of the same type.

### 2.3.2 Hardware Input-Output (I/O) Functions

DDC Hardware incorporating hardware input-output (I/O) functions must meet the following requirements:

#### 2.3.2.1 Analog Inputs

DC Hardware analog inputs (AIs) must be implemented using ASHRAE 135 Analog Input Objects and perform analog to digital (A-to-D) conversion with a minimum resolution of 8 bits plus sign or better as needed to meet the accuracy requirements specified in Section 23 09 00. Signal conditioning including transient rejection must be provided for each analog input. Analog inputs must be capable of being individually calibrated for zero and span. Calibration via software scaling performed as part of point configuration is acceptable. The AI must incorporate common mode noise rejection of at least 50 dB from 0 to 100 Hz for differential inputs, and normal mode noise rejection of at least 20 dB at 60 Hz from a source impedance of 10,000 ohms.

#### 2.3.2.2 Analog Outputs

DDC Hardware analog outputs (AOs) must be implemented using ASHRAE 135 Analog Output Objects and perform digital to analog (D-to-A) conversion with a minimum resolution of 8 bits plus sign, and output a signal with a range of 4-20 mA<sub>dc</sub> or 0-10 V<sub>dc</sub>. Analog outputs must be capable of being individually calibrated for zero and span. Calibration via software scaling performed as part of point configuration is acceptable. DDC Hardware with Hand-Off-Auto (H-O-A) switches for analog outputs must provide for overriding the output to 0 percent and to 100 percent

#### 2.3.2.3 Binary Inputs

DDC Hardware binary inputs (BIs) must be implemented using ASHRAE 135 Binary Input Objects and accept contact closures and must ignore transients of less than 5 milli-second duration. Protection against a transient 50VAC must be provided.

#### 2.3.2.4 Binary Outputs

DDC Hardware binary outputs (BOs) must be implemented using ASHRAE 135 Binary Output Objects and provide relay contact closures or triac outputs for momentary and maintained operation of output devices. DDC Hardware with H-O-A switches for binary outputs must provide for overriding the output open or closed.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### 2.3.2.4.1 Relay Contact Closures

Closures must have a minimum duration of 0.1 second. Relays must provide at least 180V of isolation. Electromagnetic interference suppression must be provided on all output lines to limit transients to 50 Vac. Minimum contact rating must be 0.5 amperes at 24 Vac.

#### 2.3.2.4.2 Triac Outputs

Triac outputs must provide at least 180 V of isolation. Minimum contact rating must be 0.5 amperes at 24 Vac.

#### 2.3.2.5 Pulse Accumulator

DDC Hardware pulse accumulators must be implemented using either an ASHRAE 135 Accumulator Object or an ASHRAE 135 Analog Value Object where the Present\_Value is the totalized pulse count. Pulse accumulators must accept contact closures, ignore transients less than 5 msec duration, protect against transients of 50 VAC, and accept rates of at least 20 pulses per second.

#### 2.3.2.6 ASHRAE 135 Objects for Hardware Inputs and Outputs

The requirements for use of ASHRAE 135 objects for hardware input and outputs includes devices where the hardware sensor or actuator is integral to the controller (e.g. a VAV box with integral damper actuator, a smart sensor, a VFD, etc.)

#### 2.3.3 Local Display Panel (LDP)

The Local Display Panels (LDPs) must be DDC Hardware with a display and navigation buttons or a touch screen display, and must provide display and adjustment of Niagara Framework points or ASHRAE 135 Properties as indicated on the Points Schedule and as specified. LDPs must be either BTL Listed as a B-OD, B-OWS, B-AWS, or be an integral part of another piece of DDC Hardware listed as a B-BC. For LDPs listed as B-OWS or B-AWS, the hardware must be BTL listed and the product must come factory installed with all applications necessary for the device to function as an LDP.

The adjustment of values using display and navigation buttons must be password protected.

#### 2.3.4 Expansion Modules and Tethered Hardware

A single piece of DDC Hardware may consist of a base unit and also:

- a. An unlimited number of hardware expansion modules, where the individual hardware expansion modules are designed to directly connect, both mechanically and electrically, to the base unit hardware. The expansion modules must be commercially available as an optional add-on to the base unit.
- b. A single piece of hardware connected (tethered) to a base unit by a single cable where the cable carries a proprietary protocol between the base unit and tethered hardware. The tethered hardware must not contain control logic and be commercially available as an optional add-on to the base unit as a single package.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

Note that this restriction on tethered hardware does not apply to sensors or actuators using standard binary or analog signals (not a communications protocol); sensors or actuators using standard binary or analog signals are not considered part of the DDC Hardware.

Hardware capable of being installed stand-alone, or without a separate base unit, is DDC Hardware and must not be used as expansion modules or tethered hardware.

### 2.3.5 Supervisory Control Requirements

#### 2.3.5.1 Alarm Generation Hardware

Non-Niagara Framework DDC Hardware used for alarm generation must meet the following requirements:

- a. Device must support the AE-N-I-B BIBB
- b. The Recipient\_List Property must be Writable for all Notification Class Objects used for alarm generation.
- c. For all Objects implementing Intrinsic Alarming, the following Properties must be Writable:
  - (1) Time\_Delay
  - (2) High\_Limit
  - (3) Low\_Limit
  - (4) Deadband
  - (5) Event\_Enable
  - (6) If the issue date of this project specification is after 1 January 2016, Time\_Delay\_Normal must be writable.
- d. It is preferred, but not required, that devices support the DM-OCD-B BIBB on all Notification Class Objects. It is also preferred, but not required that devices supporting the DM-OCD-B BIBB accept any valid value as an initial value for properties of Notification Class Objects.

#### 2.3.6 Niagara Framework Supervisory Gateway

Any device implementing the Niagara Framework is a Niagara Framework Supervisory Gateway and must meet these requirements. In addition to the general requirements for all DDC Hardware, Niagara Framework Supervisory Gateway Hardware must:

- a. Be direct digital control hardware.
- b. Have an unrestricted interoperability license and its Niagara Compatibility Statement (NiCS) must follow the Tridium Open NiCS Specification.
- c. Manage communications between a field control network and the Niagara Framework Monitoring and Control Software, and between itself and other Niagara Framework Supervisory Gateways. Niagara Framework Supervisory Gateway Hardware must use Fox protocol for communication with other Niagara Framework Components, regardless of the manufacturer of the other components.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

- d. Be fully programmable using the Niagara Framework Engineering Tool and must support the following:
  - (1) Time synchronization, Calendar, and Scheduling using Niagara Scheduling Objects
  - (2) Alarm generation and routing using the Niagara Alarm Service
  - (3) Trending using the Niagara History Service and Niagara Trend Log Objects
  - (4) Integration of field control networks using the Niagara Framework Engineering Tool
  - (5) Configuration of integrated field control system using the Niagara Framework Engineering Tool when supported by the field control system
- e. Meet the following minimum hardware requirements:
  - (1) One 10/100 Mbps Ethernet Port
  - (2) One or more MS/TP ports.
- f. Provide access to field control network data and supervisory functions via web interface and support a minimum of 16 simultaneous users.  
Note: implementation of this capability may not be required on all projects.
- g. Submit a backup of each Niagara Framework Supervisory Gateway as specified in Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC. The backup must be sufficient to restore a Niagara Framework Supervisory Gateway to the final as-built condition such that a new Niagara Framework Supervisory Gateway loaded with the backup is indistinguishable in functionality from the original.

#### 2.4 NIAGARA FRAMEWORK ENGINEERING TOOL

The Niagara Framework Engineering Tool must be Niagara Workbench or an equivalent Niagara Framework engineering tool software must:

- a. Have an unrestricted interoperability license and its Niagara Compatibility Statement (NiCS) must follow the Tridium Open NiCS Specification.
- b. Be capable of performing network configuration for Niagara Framework Supervisory Gateways and Niagara Framework Monitoring and Control Software.
- c. Be capable of programming and configuring of Niagara Framework Supervisory Gateways and Niagara Framework Monitoring and Control Software.
- d. Be capable of discovery of Niagara Framework Supervisory Gateways and all points mapped into each Niagara Framework Supervisory Gateway and making these points accessible to Niagara Framework Monitoring and Control Software.



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## PART 3 EXECUTION

### 3.1 CONTROL SYSTEM INSTALLATION 3.1.1 Niagara Framework Engineering Tool

Provide a Niagara Framework Engineering Tool.

### 3.1.2 Building Control Network (BCN)

Install the Building Control Network (BCN) as a single BACnet Internetwork consisting of a single IP network as the BCN Backbone and zero or more BACnet MS/TP networks. Note that in some cases there may only be a single device on the BCN Backbone.

Except for the IP Network and as permitted for the non-BACnet side of Gateways, use exclusively ASHRAE 135 networks.

#### 3.1.2.1 Building Control Network IP Backbone

Install IP Network Cabling in conduit. Install Ethernet Switches in lockable enclosures. Install the Building Control Network (BCN) IP Backbone such that it is available at the Facility Point of Connection (FPOC) location as indicated. When the FPOC location is a room number, provide sufficient additional media to ensure that the Building Control Network (BCN) IP Backbone can be extended to any location in the room.

Use UDP port 0xBAC0 for all BACnet traffic on the IP network. (Note that in a Niagara Framework system there may not be BACnet traffic on the IP Network)

#### 3.1.2.2 BACnet MS/TP Networks

When using MS/TP, provide MS/TP networks in accordance with ASHRAE 135 and in accordance with the ASHRAE 135 figure "Mixed Devices on 3-Conductor Cable with Shield" (Figure 9-1.4 in the 2012 version of ASHRAE 135). Ground the shield at the BACnet Router and at no other point. Ground the reference wire at the BACnet Router through a 100 ohm resistor and do not ground it at any other point. In addition:

- a. Provide each segment in a doubly terminated bus topology in accordance with TIA-485.
- b. Provide each segment with 2 sets of network bias resistors in accordance with ASHRAE 135, with one set of resistors at each end of the MS/TP network.
- c. Use 3 wire (twisted pair and reference) with shield media for all MS/TP media installed inside. Use fiber optic isolation in accordance with ASHRAE 135 for all MS/TP media installed outside buildings, or between multiple buildings.
- d. For 18 AWG cable, use segments with a maximum length of 4000 ft. When using greater distances or different wire gauges comply with the electrical specifications of TIA-485.
- e. For each controller that does not use the reference wire provide transient suppression at the network connection of the controller if the controller itself does not incorporate transient suppression.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

- f. Install no more than 32 devices on each MS/TP segment. Do not use MS/TP to MS/TP routers.
- g. Connect each MS/TP network to the BCN backbone via a Niagara Framework Supervisory Gateway configured as a BACnet Router.
- h. For BACnet Routers, configure the MS/TP MAC address to 0. Assign MAC Addresses to other devices consecutively beginning at 1, with no gaps.
- i. Configure the Max\_Master Property of all devices to be 31.

#### 3.1.2.3 Building Control Network (BCN) Installation

Provide a building control network meeting the following requirements:

- a. Install all DDC Hardware connected to the Building Control Network.
- b. Where multiple pieces of DDC Hardware are used to execute one sequence, install all DDC Hardware executing that sequence on a single MS/TP network dedicated to that sequence.
- c. Traffic between BACnet networks must be exclusively via BACnet routers.
- d. Use the Fox protocol for all traffic both originating and terminating at Niagara Framework components. Use the Fox protocol for all traffic originating or terminating at a Niagara Framework UMCS (including traffic to or from a future UMCS). All other traffic, including traffic between ASHRAE 135 devices and traffic between Niagara Framework Supervisory Gateways and ASHRAE 135 devices must be in accordance with ASHRAE 135.

#### 3.1.3 DDC Hardware

Install all DDC Hardware that connects to an IP network in lockable enclosure. Install other DDC Hardware that is not in suspended ceilings in lockable enclosures. For all DDC hardware with a user interface, coordinate with site to determine proper passwords and configure passwords into device.

- a. Except for zone sensors (thermostats), install all Tethered Hardware within 6 feet of its base unit.
- b. Install and configure all BTL-Listed devices in a manner consistent with their BTL Listing such that the device as provided still meets all requirements necessary for its BTL Listing.
- c. Install and configure all BTL-Listed devices in a manner consistent with the BTL Device Implementation Guidelines such that the device as provided meets all those Guidelines.

##### 3.1.3.1 Device Identifiers, Network Addresses, and IP addresses

- a. Do not use any Device Identifier or Network Number already used by another BACnet system at the project site. Coordinate Device IDs and Network Numbers with the installation.
- b. Coordinate device IP addresses with installation.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 3.1.3.2 ASHRAE 135 Object Name Property and Object Description Property

Configure the Object\_Names and Object\_Descriptions properties of all ASHRAE 135 Objects (including Device Objects) as indicated on the Points Schedule (Point Name and Point Description) and as specified. At a minimum:

- a. Except for DDC Hardware controlling a single terminal unit, configure the Object\_Name and Object\_Description properties of all Objects (including Device Objects) as indicated on the Points Schedule and as specified.
- b. In DDC Hardware controlling a single terminal unit, configure the Device Object\_Name and Device Object\_Description as indicated on the Points Schedule and as specified.

When Points Schedule entries exceed the length limitations in the device, provide recommended alternatives for approval.

### 3.1.3.3 Niagara Framework Point Names and Descriptions

Configure the names and descriptions of all Points in Niagara Framework Supervisory Gateways as indicated on the Points Schedule and as specified.

### 3.1.3.4 Niagara Station IDs

Ensure that Niagara Station IDs of new Niagara Framework Supervisory Gateways are maintained as unique within UMCS front-end, including ensuring they do not conflict with any existing Niagara Station ID.

### 3.1.3.5 Hand-Off-Auto (H-O-A) Switches

Provide Hand-Off-Auto (H-O-A) switches for all DDC Hardware analog outputs and binary outputs used for control of systems other than terminal units, as specified and as indicated on the Points Schedule. Provide H-O-A switches that are integral to the controller hardware, an external device co-located with (in the same enclosure as) the controller, integral to the controlled equipment, or an external device co-located with (in the same enclosure as) the controlled equipment.

- a. For H-O-A switches integral to DDC Hardware, meet the requirements specified in paragraph DIRECT DIGITAL CONTROL (DDC) HARDWARE.
- b. For external H-O-A switches used for binary outputs, provide for overriding the output open or closed.
- c. For external H-O-A switches used for analog outputs, provide for overriding to 0 percent or 100 percent.

### 3.1.3.6 Local Display Panels

Provide LDPs to display and override values of points in a Niagara Framework Supervisory Gateway or ASHRAE 135 Object Properties as indicated on the Points Schedule. Install LDPs displaying points for anything other than a terminal unit in the same room as the equipment. Install LDPs displaying points for only terminal units in a mechanical room central to the group of terminal units it serves. For LDPs using WriteProperty to commandable objects to implement an override, write values with priority 9.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 3.1.3.7 MS/TP Slave Devices

Configure all MS/TP devices as Master devices. Do not configure any devices to act as slave devices.

### 3.1.3.8 Change of Value (COV) and Read Property

- a. To the greatest extent possible, configure all devices to support the SubscribeCOV service (the DS-COV-B BIBB). At a minimum, all devices supporting the DS-RP-B BIBB, other than devices controlling only a single terminal unit, must be configured to support the DS-COV-B BIBB.
- b. Whenever supported by the server side, configure client devices to use the DS-COV-A BIBB.

### 3.1.3.9 Engineering Units

Configure devices to use English (Inch-Pound) engineering units as follows:

- a. Temperature in degrees F
- b. Air or natural gas flows in cubic feet per minute (CFM)
- c. Water in gallons per minute (GPM)
- d. Steam flow in pounds per hour (pph)
- e. Differential Air pressures in inches of water column (IWC)
- f. Water, steam, and natural gas pressures in PSI
- g. Enthalpy in BTU/lb
- h. Heating and cooling energy in MBTU (1MBTU = 1,000,000 BTU))
- i. Cooling load in tons (1 ton = 12,000 BTU/hour)
- j. Heating load in MBTU/hour (1MBTU = 1,000,000 BTU)
- k. Electrical Power: kilowatts (kW)
- l. Electrical Energy: kilowatt-hours (kWh)

### 3.1.3.10 Occupancy Modes

Use the following correspondence between value and occupancy mode whenever an occupancy state or value is required:

- a. OCCUPIED mode: a value of one
- b. UNOCCUPIED mode: a value of two
- c. WARM-UP/COOL-DOWN (PRE-OCCUPANCY) mode: a value of three

Note that elsewhere in this Section the Schedule Object is required to also support a value of four, which is reserved for future use. Also note that the behavior of a system in each of these occupancy modes is indicated in the sequence of operation for the system.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 3.1.3.11 Use of BACnet Objects

Except as specifically indicated for Niagara Framework Objects, Use only standard non-proprietary ASHRAE 135 Objects and services to accomplish the project scope of work as follows:

- a. Use Analog Input or Analog Output Objects for all analog hardware I/O. Do not use Analog Value Object for analog hardware I/O) .
- b. Use Binary Input or Binary Output Objects for all binary hardware I/O. Do not use Binary Value Objects for binary hardware I/O.
- c. Use Analog Value Objects for analog setpoints.
- d. Use Accumulator Objects or Analog Value Objects for pulse inputs.
- e. For occupancy modes, use Multistate Value Objects and the correspondence between value and occupancy mode specified in paragraph OCCUPANCY MODES.
- f. Use a combination of Niagara Framework Alarm Extensions and Alarm Services, Intrinsic Alarming, and Notification Class Objects for alarm generation.
- g. For all other points shown on the Points Schedule as requiring an ASHRAE 135 Object, use the Object type shown on the Points Schedule or, if no Object Type is shown, use a standard Object appropriate to the point.

#### 3.1.3.11.1 Niagara Framework Objects

Points in the Niagara Framework Supervisory Gateway, even if used in a sequence or are shown on the Points Schedule, are not required to be exposed as BACnet Objects unless they are required to be available on the network by another device or sequence of operation (i.e. there is some other reason they are needed).

Use a Niagara Framework Supervisory Gateway as specified for all scheduling and trending. Use a Niagara Framework Supervisory Gateway as specified for all alarming except for intrinsic alarming.

#### 3.1.3.12 Use of Standard BACnet Services

Except as noted in this paragraph, for all DDC Hardware (including Niagara Frameworks Supervisory Gateways when communicating with non-Niagara Framework DDC Hardware) use Standard BACnet Services as defined in this specification (which excludes some ASHRAE 135 services) exclusively for application control functionality and communication.

DDC Hardware that cannot meet this requirement may use non-standard services provided they can provide identical functionality using Standard BACnet Services when communicating with BACnet devices from a different vendor. When implementing non-standard services, document all non-standard services in the DDC Hardware Schedule as specified and as specified in Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC.

#### 3.1.3.13 Device Application Configuration

- a. For every property, setting or value shown on the Points Schedule or

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

otherwise indicated as Configurable, provide a value that is retained through loss of power and can be changed via one or more of:

- (1) BACnet services (including proprietary services)
  - (2) Hardware settings on the device
  - (3) The Niagara Framework
- b. For every property, setting or value in non-Niagara Framework Hardware shown on the Points Schedule or otherwise indicated as Operator Configurable, provide a value that is retained through loss of power and can be changed via one or more of:
- (1) A Writable Property of a standard BACnet Object
  - (2) A Property of a standard BACnet Object that is Writable when Out\_Of\_Service is TRUE and Out\_Of\_Service is Writable.
  - (3) Using some other method supported by a Niagara Framework Supervisory Gateway
- c. Configure Niagara Framework Supervisory Gateways such that the property, setting or value is configurable from a Niagara Framework Front End.
- d. For every property, setting or value in a Niagara Framework Supervisory Gateway which is shown on the Points Schedule or otherwise indicated as Operator Configurable, configure the value to be configurable from within the Niagara Framework such that it can be configured from a system graphic page at a Niagara Framework Front End.

#### 3.1.3.14 Niagara Framework Engineering Tool

Use the Niagara Framework Engineering Tool to fully discover the field control system and make all field control system information available to the Niagara Framework Supervisory Gateway. Ensure that all points on the points schedule are available to the front end via the Fox protocol.

#### 3.1.3.15 Graphics and Web Pages

Configure Niagara Framework Supervisory Gateways to use web pages to provide a graphical user interface including System Displays using the project site sample displays, including overrides, as indicated on the Points Schedule and as specified. Label all points on displays with full English language descriptions. Configure user permissions for access to and executions of action using graphic pages. Coordinate user permissions with the HVAC shop supervisor. Configure the web server to use HTTPS based on the Transport Layer Security (TLS) protocol in accordance with RFC 5246 using a Government furnished certificate.

#### 3.1.4 Scheduling, Alarming, Trending, and Overrides

##### 3.1.4.1 Scheduling

Configure schedules in Niagara Framework Supervisory Gateway using Niagara Schedule Objects as indicated on the Points Schedule and as specified. When the schedule is controlling occupancy modes in DDC Hardware other than a Niagara Framework Supervisory Gateway use the indicated correspondence between value and occupancy mode.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

Provide a separate schedule for each AHU including it's associated Terminal Units and for each stand-alone Terminal Unit (those not dependent upon AHU service) or group of stand-alone Terminal Units acting according to a common schedule as indicated.

#### 3.1.4.2 Alarm Configuration

Configure alarm generation and management as indicated on the Points Schedule and as specified. Configure alarm generation in Niagara Framework Supervisory Gateways using Niagara Framework Alarm Extensions and Alarm Services or in other DDC Hardware (not Niagara Framework Supervisory Gateways) using ASHRAE 135 Intrinsic Alarming. Configure alarm management and routing for all alarms, including those generated via intrinsic alarming in other devices, in the Niagara Framework Supervisory Gateway such that the alarms are able to be accessed from the Niagara Framework Front End.

Where Intrinsic Alarming is used, configure intrinsic alarming as specified in paragraph "Configuration of ASHRAE 135 Intrinsic Alarm Generation". Configure a Niagara Framework Supervisory Gateway to provide a means to configure the intrinsic alarm parameters such that the Intrinsic Alarm is configurable from the front end via the Niagara Framework.

#### 3.1.4.3 Configuration of ASHRAE 135 Intrinsic Alarm Generation

Intrinsic alarm generation must meet the following requirements:

Configure alarm generation as indicated on the Points Schedule and as specified using Intrinsic Alarming in accordance with ASHRAE 135 or Algorithmic Alarming in accordance with ASHRAE 135. Alarm generation must meet the following requirements:

- a. Send alarm events as Alarms (not Events).
- b. Use the ConfirmedNotification Service for alarm events.
- c. For alarm generation, support two priority levels for alarms: critical and non-critical. Configure the Priority of Notification Class Objects to use Priority 112 for critical and 224 for non-critical alarms.
- d. Number of Notification Class Objects for Alarm Generation:
  - (1) If the device implements non-critical alarms, or if any Object in the device supports Intrinsic Alarms, then provide a single Notification Class Object specifically for (shared by) all non-critical alarms.
  - (2) If the device implements critical alarms, provide a single Notification Class Object specifically for (shared by) all critical alarms.
  - (3) If the device implements both critical and non-critical alarms, provide both Notification Class Objects (one for critical, one for non-critical).
  - (4) If the device controls equipment other than a single terminal

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

unit, provide both Notification Class Objects (one for critical, one for non-critical) even if no alarm generation is required at time of installation.

- e. For all intrinsic alarms configure the Limit\_Enable Property to set both HighLimitEnable and LowLimitEnable to TRUE. If the specified alarm conditions are for a single-sided alarm (only High\_Limit used or only Low\_Limit used) assign a value to the unused limit such that the unused alarm condition will not occur.
- f. For all objects supporting intrinsic alarming, even if no alarm generation is required during installation, configure the following Properties as follows:
  - (1) Notification\_Class to point to the non-Critical Notification Class Object in that device.
  - (2) Limit\_Enable to enable both the HighLimitEnable and LowLimitEnable
  - (3) Notify\_Type to Alarm
- g. Configure the Recipient\_List Property of the Notification Class Object to point to the Niagara Framework Supervisory Gateway managing the alarm.

#### 3.1.4.4 Trending

Perform all trending using a Niagara Framework Supervisory Gateway using Niagara Framework History Extensions and Niagara Framework History Service exclusively.

#### 3.1.4.5 Overrides

Provide an override for each point shown on the Points Schedule as requiring an override. Use the Niagara Framework for all overrides to points in Niagara Framework Supervisory Gateways. For overrides to other points, provide an override to a point in a Niagara Framework Supervisory Gateway via the Niagara Framework where the Niagara Framework Supervisory Gateway overrides the other point as specified.

Unless otherwise approved, provide Commandable Objects to support all Overrides in non-Niagara Framework Supervisory Gateway DDC Hardware. With specific approval from the Contracting Officer, Overrides for points which are not hardware outputs and which are in DDC hardware controlling a single terminal unit may support overrides via an additional Object provided for the override. No other means of implementing Overrides may be used.

- a. Where Commandable Objects are used, ensure that WriteProperty service requests with a Priority of 10 or less take precedence over the SEQUENCE VALUE and that WriteProperty service request with a priority of 11 or more have a lower precedence than the SEQUENCE VALUE.
- b. For devices implementing overrides via additional Objects, provide Objects which are NOT Written to as part of the normal Sequence of Operations and are Writable when Out\_Of\_Service is TRUE and Out\_Of\_Service is Writable. Use this point as an Override of the



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

normal value when Out\_Of\_Service is TRUE and the normal value otherwise. Note these Objects may be modified as part of the sequence via local processes, but must not be modified by local processes when Out\_Of\_Service is TRUE.

### 3.1.5 BACnet Gateways

The requirements in this paragraph do not themselves permit the installation of hardware not meeting the other requirements of this section. Except for proprietary systems specifically indicated in Section 23 09 00, all control hardware installed under this project must meet the requirements of this specification, including the control hardware providing the network interface for a package unit or split system specified under another section. Only use gateways to connect to pre-existing control devices, and to proprietary systems specifically permitted by Section 23 09 00.

#### 3.1.5.1 General Gateway Requirements

Provide BACnet Gateways to connect non-BACnet control hardware in accordance with the following:

- a. Configure gateways to map writable data points in the controlled equipment to Writable Properties of Standard Objects, or to Niagara Framework points, as indicated in the Points Schedule and as specified.
- b. Configure gateway to map readable data points in the controlled equipment to Readable Properties of Standard Objects, or to Niagara Framework points, as indicated in the Points Schedule and as specified.
- c. Configure gateway to support the DS-COV-B BIBB for all points mapped to BACnet Objects.
- d. Do not use non-BACnet control hardware for controlling built-up units or any other equipment that was not furnished with factory-installed controls. (Note: A Niagara Framework Supervisory Gateway is BACnet control hardware.)
- e. Do not use non-BACnet control hardware for system scheduling functions.
- f. Each gateway must communicate with and perform protocol translation for non-BACnet control hardware controlling one and only one package unit or a single non-BACnet system specifically permitted by Section 23 09 00.
- g. Connect one network port on the gateway to the Building Control Backbone IP Network or to a BACnet MS/TP network and the other port to the single piece of controlled equipment or the non-BACnet system specifically permitted by Section 23 09 00..
- h. For gateways to existing package units or simple split systems, non-BACnet network wiring connecting the gateway to the package unit must not exceed 10 feet in length and must connect to exactly two devices: the controlled equipment (packaged unit) or split system interface and the gateway.

-- End of Section --

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## SECTION 23 11 20

### FACILITY GAS PIPING 05/20

#### PART 1 GENERAL

##### 1.1 SUMMARY

This specification section applies to gas piping installed within buildings incidental underground piping under building, above ground steel piping and corrugated stainless steel tubing (CSST) both outside (up to 5 feet beyond exterior walls) and within buildings in compliance with NFPA 54 /AGA Z223.1, "National Fuel Gas Code" , "Fuel Gas Piping".

##### 1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

#### AMERICAN GAS ASSOCIATION (AGA)

AGA XR0603 (2006; 8th Ed) AGA Plastic Pipe Manual for Gas Service

AGA Z223.1 (2012) National Fuel Gas Code

#### AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI Z21.1/CSA 1.1 (2023) Household Cooking Gas Appliances

ANSI Z21.15/CSA 9.1 (2021) Manually Operated Gas Valves for Appliances, Appliance Connector Valves and Hose End Valves

ANSI Z21.18/CSA 6.3 (2007; R 2017) Gas Appliance Pressure Regulators

ANSI Z21.24/CSA 6.10 (2022) Connectors for Gas Appliances

ANSI Z21.41/CSA 6.9 (2023) Quick-Disconnect Devices for Use with Gas Fuel Appliances

ANSI Z21.69/CSA 6.16 (2015; R 2020) Connectors for Movable Gas Appliances

ANSI Z21.80/CSA 6.22 (2019) Line Pressure Regulators

ANSI Z21.93/CSA 6.30 (2017) Excess Flow Valves for Natural Gas and Propane Gas with Pressures up to 5 psig

#### AMERICAN PETROLEUM INSTITUTE (API)

API 570 (2016; Addendum 1 2017; Addendum 2 2018; ERTA 1 2018) Piping Inspection Code: In-Service Inspection, Rating, Repair, and Alteration of Piping Systems

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

|              |                                                                                                                               |
|--------------|-------------------------------------------------------------------------------------------------------------------------------|
| API RP 2009  | (2002; R 2007; 7th Ed) Safe Welding, Cutting, and Hot Work Practices in Refineries, Gasoline Plants, and Petrochemical Plants |
| API Spec 5CT | (2018) Casing and Tubing                                                                                                      |
| API Spec 6D  | (June 2018, 4th Ed; Errata 1 July 2018; Errata 2 August 2018) Specification for Pipeline and Piping Valves                    |
| API Std 598  | (2009) Valve Inspecting and Testing                                                                                           |
| API Std 607  | (2016) Fire Test for Quarter-turn Valves and Valves Equipped with Non-metallic Seats                                          |

#### AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

|                  |                                                                                                                           |
|------------------|---------------------------------------------------------------------------------------------------------------------------|
| ASME A13.1       | (2020) Scheme for the Identification of Piping Systems                                                                    |
| ASME B1.1        | (2003; R 2018) Unified Inch Screw Threads (UN and UNR Thread Form)                                                        |
| ASME B1.20.1     | (2013; R 2018) Pipe Threads, General Purpose (Inch)                                                                       |
| ASME B16.21      | (2021) Nonmetallic Flat Gaskets for Pipe Flanges                                                                          |
| ASME B16.33      | (2012; R 2017) Manually Operated Metallic Gas Valves for Use in Gas Piping Systems Up to 125 psi, (Sizes NPS 1/2 - NPS 2) |
| ASME B18.2.1     | (2012; Errata 2013) Square and Hex Bolts and Screws (Inch Series)                                                         |
| ASME B18.2.2     | (2022) Nuts for General Applications: Machine Screw Nuts, and Hex, Square, Hex Flange, and Coupling Nuts (Inch Series)    |
| ASME B31.8       | (2022; Supplement 2022) Gas Transmission and Distribution Piping Systems                                                  |
| ASME B31.9       | (2020) Building Services Piping                                                                                           |
| ASME BPVC SEC IX | (2017; Errata 2018) BPVC Section IX-Welding, Brazing and Fusing Qualifications                                            |

#### ASTM INTERNATIONAL (ASTM)

|                 |                                                                                                                                                         |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM A193/A193M | (2023) Standard Specification for Alloy-Steel and Stainless Steel Bolting Materials for High-Temperature Service and Other Special Purpose Applications |
| ASTM A194/A194M | (2023) Standard Specification for Carbon                                                                                                                |

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

Steel, Alloy Steel, and Stainless Steel  
Nuts for Bolts for High-Pressure or  
High-Temperature Service, or Both

ASTM B88 (2022) Standard Specification for Seamless  
Copper Water Tube

ASTM B280 (2020) Standard Specification for Seamless  
Copper Tube for Air Conditioning and  
Refrigeration Field Service

ASTM D2513 (2018a) Standard Specification for  
Polyethylene (PE) Gas Pressure Pipe,  
Tubing, and Fittings

ASTM F2015 (2000; R 2013) Standard Specification for  
Lap Joint Flange Pipe End Applications

#### CSA GROUP (CSA)

ANSI LC 1/CSA 6.26 (2023) Fuel Gas Piping Systems Using  
Corrugated Stainless Steel Tubing (CSST)

CGA 3.11-M88 (2015) Lever Operated Pressure Lubricated  
Plug Type Gas Shut-Off Valves

#### MANUFACTURERS STANDARDIZATION SOCIETY OF THE VALVE AND FITTINGS INDUSTRY (MSS)

MSS SP-25 (2018) Standard Marking System for Valves,  
Fittings, Flanges and Unions

MSS SP-58 (2018) Pipe Hangers and Supports -  
Materials, Design and Manufacture,  
Selection, Application, and Installation

#### NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 54 (2021) National Fuel Gas Code

NFPA 58 (2020; TIA 20-1; TIA 20-2; TIA 20-3)  
Liquefied Petroleum Gas Code

NFPA 70 (2023; ERTA 4 2023) National Electrical  
Code

#### SOCIETY FOR PROTECTIVE COATINGS (SSPC)

SSPC SP 6/NACE No.3 (2007) Commercial Blast Cleaning

#### U.S. DEPARTMENT OF DEFENSE (DOD)

MIL-STD-101 (2014; Rev C) Color Code for Pipelines and  
for Compressed Gas Cylinders

#### U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

49 CFR 192 Transportation of Natural and Other Gas by  
Pipeline: Minimum Federal Safety Standards

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## UNDERWRITERS LABORATORIES (UL)

UL FLAMMABLE & COMBUSTIBLE (2012) Flammable and Combustible Liquids  
and Gases Equipment Directory

### 1.3 SYSTEM DESCRIPTION

The gas piping system includes natural gas piping and appurtenances from point of connection with supply system, as indicated, to gas operated equipment within the facility. Submit operation and maintenance data in accordance with Section 01 78 23 OPERATION AND MAINTENANCE DATA, in three separate packages.

#### 1.3.1 Gas Facility System and Equipment Operation

Include shop drawings showing piping layout, locations of system valves, gas line markers; step-by-step procedures for system start up, operation and shutdown (index system components and equipment to the system drawings); isolation procedures including valve operation to shutdown or isolate each section of the system (index valves to the system maps and provide separate procedures for normal operation and emergency shutdown if required to be different). Submit Data package No. 4.

#### 1.3.2 Gas Facility System Maintenance

Include maintenance procedures and frequency for system and equipment; identification of pipe materials and manufacturer by locations, pipe repair procedures, and jointing procedures at transitions to other piping material or material from a different manufacturer. Submit Data Package No.4.

#### 1.3.3 Gas Facility Equipment Maintenance

Include identification of valves, shut-offs, disconnects, and other equipment by materials, manufacturer, vendor identification and location; maintenance procedures and recommended tool kits for valves and equipment; recommended repair methods (i.e., field repair, factory repair, or replacement) for each valve and piece of equipment; and preventive maintenance procedures, possible failure modes and troubleshooting guide. Submit Data Package No. 3.

### 1.4 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Gas Piping System; G

SD-03 Product Data

Pipe and Fittings; G

Gas Equipment Connectors; G

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

Gas Piping System; G

Pipe Coating Materials; G

Pressure Regulators; G

Risers; G

Transition Fittings; G

Valves; G

Warning and Identification Tape; G

#### SD-06 Test Reports

Testing; FIO

Pressure Tests; FIO

Test with Gas; FIO

#### SD-07 Certificates

Welders Procedures and Qualifications; FIO

Assigned Number, Letter, or Symbol; FIO

#### SD-08 Manufacturer's Instructions

PE Pipe and Fittings; FIO

Pipe Coating Materials; FIO

#### SD-10 Operation and Maintenance Data

Gas Facility System and Equipment Operation; G

Gas Facility System Maintenance; G

Gas Facility Equipment Maintenance; G

### 1.5 QUALITY ASSURANCE

Submit manufacturer's descriptive data and installation instructions for approval for compression-type mechanical joints used in joining dissimilar materials and for insulating joints. Mark all valves, flanges and fittings in accordance with MSS SP-25.

#### 1.5.1 Welding Qualifications

- a. Weld piping in accordance with qualified procedures using performance qualified welders and welding operators in accordance with API RP 2009, ASME BPVC SEC IX, and ASME B31.9. Welding procedures qualified by others, and welders and welding operators qualified by another

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

employer may be accepted as permitted by ASME B31.9. Notify the Contracting Officer at least 24 hours in advance of tests, and perform at the work site if practicable.

- b. Submit a certified copy of welders procedures and qualifications metal and PE in conformance with ASME B31.9 for each welder and welding operator. Submit the assigned number, letter, or symbol that will be used in identifying the work of each welder to the Contracting Officer.

#### 1.5.2 Jointing Thermoplastic and Fiberglass Piping

Perform all jointing of piping using qualified joiners and qualified procedures in accordance with AGA XR0603. Furnish the Contracting Officer with a copy of qualified procedures and list of and identification symbols of qualified joiners. Submit manufacturer's installation instructions and manufacturer's visual joint appearance chart, including all PE pipe and fittings.

#### 1.5.3 Shop Drawings

Submit drawings for complete Gas Piping System, within 30 days of contract award, showing location, size and all branches of pipeline; location of all required shutoff valves; and instructions necessary for the installation of gas equipment connectors and supports.

### 1.6 DELIVERY, STORAGE, AND HANDLING

#### 1.6.1 Plastic Pipe

Handle, transport, and store plastic pipe and fittings carefully. Plug or cap pipe and fittings ends during transportation or storage to minimize dirt and moisture entry. Do not subject piping to abrasion or concentrated external loads. Discard PE pipe sections and fittings that have been damaged.

#### 1.6.2 CSST Tubing

Handle, transport and store CSST tubing on the wooden spool or shipping container provided by the manufacturer. Insure tubing ends are capped during transportation and storage to minimize dirt and moisture entry. Discard any tubing segment and fitting that has been damaged.

## PART 2 PRODUCTS

### 2.1 MATERIALS AND EQUIPMENT

Provide materials and equipment which are the standard products of a manufacturer regularly engaged in the manufacture of the products and that essentially duplicate items that have been in satisfactory use for at least 2 years prior to bid opening. Asbestos or products containing asbestos are not allowed. Submit catalog data and installation instructions for pipe, valves, all related system components, pipe coating materials and application procedures. Conform to NFPA 54 and with requirements specified herein. Provide supply piping to appliances or equipment at least as large as the inlets thereof.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## 2.2 GAS PIPING SYSTEM AND FITTINGS

### 2.2.1 Steel Pipe, Joints, and Fittings

### 2.2.2 Copper Tubing, Joints and Fittings

Provide copper tubing conforming to ASTM B88, Type K or L, or ASTM B280, with tubing joints made up with tubing fittings recommended by the tubing manufacturer. Provide copper and copper alloy press fittings, with sealing elements of Hydrogenated Nitrile Butadiene Rubber (HNBR), factory installed, or an alternative supplied by the fitting manufacturer. Press fittings are not a permitted connection fitting for natural and LP gas on Army and Navy projects.

### 2.2.3 Thermoplastic Pipe, Tubing, Joints, and Fittings

Provide thermoplastic pipe, tubing, casing and joints and fittings conforming to ASTM D2513 and API Spec 5CT.

### 2.2.4 Corrugated Stainless Steel Tubing, Fittings and Accessories

Provide corrugated stainless steel tubing conforming to ANSI LC 1/CSA 6.26 (austenitic stainless steel of series 300) with tubing joints made with special mechanical fittings as supplied by the tubing manufacturer.

#### 2.2.4.1 Tubing

Austenitic stainless alloy of series 300 with polyethylene jacket/coating in accordance with ANSI LC 1/CSA 6.26 for sizes 3/8-inch through 2-inch

#### 2.2.4.2 Mechanical Fittings

Copper alloy with one end matched to the corrugated tubing and one end with NPT threads in accordance with ASME B1.20.1

#### 2.2.4.3 Striker Plates

Hardened steel designed to protect tubing from mechanical damage in accordance with ANSI LC 1/CSA 6.26

#### 2.2.4.4 Manifolds

Malleable iron, steel or copper alloy with threaded connections/ports in accordance with ASME B1.20.1

### 2.2.5 Sealants for Steel Pipe Threaded Joints

Provide joint sealing compound as listed in UL FLAMMABLE & COMBUSTIBLE, Class 20 or less. For taping, use tetrafluoroethylene tape conforming to UL FLAMMABLE & COMBUSTIBLE.

### 2.2.6 Warning and Identification

Provide pipe flow markings, warning and identification tape, and metal tags as required.

### 2.2.7 Flange Gaskets

Provide gaskets of nonasbestos compressed material in accordance with ASME B16.21, 1/16 inch thickness, full face or self-centering flat ring



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

type, containing aramid fibers bonded with styrene butadiene rubber (SBR) or nitrile butadiene rubber (NBR) suitable for a maximum 600 degree F service, to be used for hydrocarbon service.

#### 2.2.8 Pipe Threads

Provide pipe threads conforming to ASME B1.20.1.

#### 2.2.9 Escutcheons

Provide chromium-plated steel or chromium-plated brass escutcheons, either one piece or split pattern, held in place by internal spring tension or set screw.

#### 2.2.10 Gas Transition Fittings

- a. Provide steel to plastic (PE) designed for steel-to-plastic with tapping tee or sleeve conforming to AGA XR0603 requirements for transitions fittings.. Coat or wrap exposed steel pipe with heavy plastic coating.
- c. Provide lever operated pressure lubricated plug type gas shut-off valve conforming to CGA 3.11-M88.

#### 2.2.11 Insulating Pipe Joints

##### 2.2.11.1 Insulating Joint Material

Provide insulating joint material between flanged or threaded metallic pipe systems where shown to control galvanic or electrical action.

##### 2.2.11.2 Threaded Pipe Joints

Provide threaded pipe joints of steel body nut type dielectric unions with insulating gaskets.

##### 2.2.11.3 Flanged Pipe Joints

Provide joints for flanged pipe consisting of full face sandwich-type flange insulating gasket of the dielectric type, insulating sleeves for flange bolts, and insulating washers for flange nuts. Provide lap joint flange pipe ends conforming to ASTM F2015.

#### 2.2.12 Flexible Connectors

- a. Provide flexible connectors for connecting gas utilization equipment to building gas piping conforming to ANSI Z21.24/CSA 6.10 or ANSI Z21.41/CSA 6.9 for quick disconnect devices, and flexible connectors for movable food service equipment conforming to ANSI Z21.69/CSA 6.16.
- b. Do not install the flexible connector through the appliance cabinet face. Provide rigid metallic pipe and fittings to extend the final connection beyond the cabinet, except when appliance is provided with an external connection point.

#### 2.3 VALVES

Provide lockable shutoff or service isolation valves conforming to the

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

following:

#### 2.3.1 Valves 2 Inches and Smaller

Provide valves 2 inches and smaller conforming to ASME B16.33 of materials and manufacture compatible with system materials used. Provide manually operated household cooking gas appliance valves conforming to ANSI Z21.1/CSA 1.1 and ANSI Z21.15/CSA 9.1.

#### 2.3.2 Valves 2-1/2 Inches and Larger

Provide valves 2-1/2 inches and larger of carbon steel conforming to API Spec 6D, Class 150.

#### 2.3.3 Valve Support on PE Piping

Provide valve support assembly in accordance with the PE piping manufacturer's requirements at valve terminations points.

#### 2.4 RISERS

Provide manufacturer's standard riser, transition from plastic to steel pipe with 7 to 12 mil thick epoxy coating. Use swaged gas-tight construction with O-ring seals, metal insert, and protective sleeve. Provide remote bolt-on or bracket or wall-mounted riser supports .

#### 2.5 PIPE HANGERS AND SUPPORTS

Provide pipe hangers and supports conforming to MSS SP-58.

#### 2.6 LINE AND APPLIANCE REGULATORS AND SHUTOFF VALVES

Provide regulators conforming to ANSI Z21.18/CSA 6.3 for appliances , and ANSI Z21.80/CSA 6.22 for line pressure regulators. Provide shutoff valves conforming to ANSI Z21.15/CSA 9.1 for manually controlled gas shutoff valves .

#### 2.7 AUTOMATIC GAS SHUT-OFF

Provide low pressure automatic gas shutoff or excess flow valve (EFV) downstream of the point of delivery after the meter/regulator conforming to ANSI Z21.93/CSA 6.30 and UL listed or CSA listed or International Association of Plumbing and Mechanical Officials (IAPMO) listed. The EFV may be either a bypass (automatic reset) or a non-bypass type (manual reset).

#### 2.8 BOLTING (BOLTS AND NUTS)

Stainless steel bolting; ASTM A193/A193M, Grade B8M or B8MA, Type 316, for bolts; and ASTM A194/A194M, Grade 8M, Type 316, for nuts. Dimensions of bolts, studs, and nuts must conform with ASME B18.2.1 and ASME B18.2.2 with coarse threads conforming to ASME B1.1, with Class 2A fit for bolts and studs and Class 2B fit for nuts. Bolts or bolt-studs must extend through the nuts and may have reduced shanks of a diameter not less than the diameter at root of threads. Bolts must have American Standard regular square or heavy hexagon heads; nuts must be American Standard heavy semifinished hexagonal.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## 2.9 GASKETS

Fluorinated elastomer, compatible with flange faces.

## 2.10 IDENTIFICATION FOR ABOVEGROUND PIPING

MIL-STD-101 for legends and type and size of characters. For pipes 3/4 inch od and larger, provide printed legends to identify contents of pipes and arrows to show direction of flow. Color code label backgrounds to signify levels of hazard. Make labels of plastic sheet with pressure-sensitive adhesive suitable for the intended application. For pipes smaller than 3/4 inch od, provide brass identification tags 1 1/2 inches in diameter with legends in depressed black-filled characters.

## PART 3 EXECUTION

### 3.1 EXAMINATION

After becoming familiar with all details of the work, verify all dimensions in the field, and advise the Contracting Officer of any discrepancy or areas of conflict before performing the work.

### 3.2 EXCAVATION AND BACKFILLING

Provide required excavation, backfilling, and compaction as specified in Section 31 00 00 EARTHWORK.

### 3.3 GAS PIPING SYSTEM

Provide a gas piping system from the point of delivery, defined as the outlet of the meter set assembly, as specified in Section 33 51 15 NATURAL-GAS / LIQUEFIED PETROLEUM GAS DISTRIBUTION PIPELINES, to the connections to each gas utilization device that is in compliance with NFPA 54..

#### 3.3.1 Protection and Cleaning of Materials and Components

Protect equipment, pipe, and tube openings by closing with caps or plugs during installation. At the completion of all work, thoroughly clean the entire system.

#### 3.3.2 Workmanship and Defects

Piping, tubing and fittings must be clear and free of cutting burrs and defects in structure or threading and must be thoroughly brushed and chip-and scale-blown. Repair of defects in piping, tubing or fittings is not allowed; replace defective items when found.

### 3.4 PROTECTIVE COVERING

#### 3.4.1 Underground Metallic Pipe

Protect buried metallic piping and tubing from corrosion by either: (1) applying protective coatings as specified in Section 33 51 15 NATURAL-GAS / LIQUEFIED PETROLEUM GAS DISTRIBUTION PIPELINES; (2) encasement in a water tight plastic conduit; or (3) encasement in a protective system designed and listed by the manufacturer for this application. When dissimilar metals are joined underground, use gastight insulating fittings.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 3.4.2 Aboveground Metallic Piping Systems

#### 3.4.2.1 Ferrous Surfaces

Touch up shop primed surfaces with ferrous metal primer. Solvent clean surfaces that have not been shop primed. Mechanically clean surfaces that contain loose rust, loose mill scale and other foreign substances by power wire brushing or commercial sand blasted conforming to SSPC SP 6/NACE No.3 and prime with ferrous metal primer. Finish primed surfaces with two coats of exterior oil paint.

#### 3.4.2.2 Nonferrous Surfaces

Except for aluminum alloy pipe, do not paint nonferrous surfaces. Paint surfaces of aluminum alloy pipe and fittings to protect against external corrosion where they contact masonry, plaster, insulation, or are subject to repeated wettings by such liquids as water, detergents or sewage. Solvent-clean the surfaces and treat with vinyl type wash coat. Apply a first coat of aluminum paint and a second coat of alkyd gloss enamel or silicone alkyd copolymer enamel.

### 3.5 INSTALLATION

Install the gas system in conformance with the manufacturer's recommendations and applicable provisions of NFPA 54 and AGA XR0603, and as indicated. Perform all pipe cutting without damage to the pipe, with an approved type of mechanical cutter, unless otherwise authorized. Use wheel cutters where practicable. On steel pipe 6 inches and larger, an approved gas cutting and beveling machine may be used. Cut thermoplastic and fiberglass pipe in accordance with AGA XR0603.

#### 3.5.1 Metallic Piping Installation

Bury underground piping a minimum of 18 inches below grade. Make changes in direction of piping with fittings only; mitering or notching pipe to form elbows and tees or other similar type construction is not permitted. Branch connection may be made with either tees or forged branch outlet fittings. Provide branch outlet fittings which are forged, flared for improvement of flow where attached to the run, and reinforced against external strains. Do not use aluminum alloy pipe in exterior locations or underground.

#### 3.5.2 Metallic Tubing Installation

Install metallic tubing using gas tubing fittings approved by the tubing manufacturer. CSST gas piping systems must be installed by contractors who have completed the manufacturer's training program as indicated on a certification card. Make branch connections with tees. Prepare all tubing ends with tools designed for that purpose. Do not use aluminum alloy tubing in exterior locations or underground. Maintain electrical continuity of gas piping system in accordance with NFPA 54, paragraph entitled 'Electrical Bonding and Grounding'.

#### 3.5.3 Thermoplastic and Fiberglass Piping, Tubing, and Fittings

Installation of thermoplastic and fiberglass piping, tubing, and fittings is permitted only outside and underground. Bury piping a minimum of 18 inches below grade. Install the piping to avoid excessive stresses due to thermal contraction, and use only where indicated. Installations must be

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

made using qualified procedures, by qualified installers, and in compliance with AGA XR0603 and NFPA 54 , and must be inspected by a qualified inspector.

#### 3.5.4 Connections Between Metallic and Plastic Piping

Connections between metallic and plastic piping are only allowed outside, underground, and with approved transition fittings.

#### 3.5.5 Piping and Tubing Buried Under Buildings

Run underground piping and tubing installed beneath buildings in a steel pipe casing protected from corrosion with protective coatings as specified in Section 33 51 15 NATURAL-GAS / LIQUEFIED PETROLEUM GAS DISTRIBUTION PIPELINES or installed within a water tight plastic conduit or as part of a listed encasement system. Extend casing or encasement system at least 4 inches outside the building, and provide the pipe with spacers and end bushings to seal at both ends to prevent the entrance of water and/or the escape of gas. Extend a vent line from the annular space above grade outside to a point where gas will not be a hazard, and terminate in a rain/insect-resistant fitting.

#### 3.5.6 Concealed Piping in Buildings

Do not use combinations of fittings ( unions, tubing fittings, running threads, right- and left-hand couplings, bushings, and swing joints) to conceal piping within buildings.

##### 3.5.6.1 Piping and Tubing in Partitions

Locate concealed piping and tubing in hollow, rather than solid, partitions. Protect tubing passing through walls or partitions against physical damage both during and after construction, and provide appropriate safety markings and labels. Provide protection of concealed pipe and tubing in accordance with ANSI LC 1/CSA 6.26.

#### 3.5.7 Aboveground Piping

Run aboveground piping as straight as practicable along the alignment and elevation indicated, with a minimum of joints, and separately supported from other piping system and equipment. Install exposed horizontal piping no farther than 6 inches from nearest parallel wall and at an elevation which prevents standing, sitting, or placement of objects on the piping.

#### 3.5.8 Final Gas Connections

Unless otherwise specified, make final connections with rigid metallic pipe and fittings. Make final connections to kitchen ranges using flexible connectors not less than 40 inch long, to afford access to coupling and to permit movement of equipment for cleaning. Flexible connectors may be used for final connections to residential dryers. Flexible connectors may be used for final connections to gas utilization equipment. In addition to cautions listed in instructions required by ANSI standards for flexible connectors, insure that flexible connectors do not pass through equipment cabinet. Provide accessible gas shutoff valve and coupling for each gas equipment item.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 3.6 PIPE JOINTS

Design and install pipe joints to effectively sustain the longitudinal pull-out forces caused by contraction of the piping or superimposed loads.

#### 3.6.1 Threaded Metallic Joints

Provide threaded joints in metallic pipe with tapered threads evenly cut and made with UL approved graphite joint sealing compound for gas service or tetrafluoroethylene tape applied to the male threads only. Threaded joints up to 1-1/2 inches in diameter may be made with approved tetrafluoroethylene tape. Threaded joints up to 2 inches in diameter may be made with approved joint sealing compound. After cutting and before threading, ream pipe and remove all burrs. Caulking of threaded joints to stop or prevent leaks is not permitted.

#### 3.6.2 Welded Metallic Joints

Conform beveling, alignment, heat treatment, and inspection of welds to NFPA 54. Remove weld defects and make repairs to the weld, or remove the weld joints entirely and reweld. After filler metal has been removed from its original package, protect and store so that its characteristics or welding properties are not affected adversely. Do not use electrodes that have been wetted or have lost any of their coating.

#### 3.6.3 Thermoplastic and Fiberglass Joints

##### 3.6.3.1 Thermoplastic and Fiberglass

Conform jointing procedures to AGA XR0603. Do not make joints with solvent cement or heat of fusion between different kinds of plastics.

##### 3.6.3.2 PE Fusion Welding Inspection

Visually inspect butt joints by comparing with, manufacturer's visual joint appearance chart. Inspect fusion joints for proper fused connection. Replace defective joints by cutting out defective joints or replacing fittings. Inspect, in conformance with API 570, 100 percent of all joints and re-inspect all corrections. Arrange with the pipe manufacturer's representative in the presence of the Contracting Officer to make first time inspection.

#### 3.6.4 Flared Metallic Tubing Joints

Make flared joints in metallic tubing with special tools recommended by the tubing manufacturer. Use flared joints only in systems constructed from nonferrous pipe and tubing, when experience or tests have demonstrated that the joint is suitable for the conditions, and when adequate provisions are made in the design to prevent separation of the joints. Do not use metallic ball sleeve compression-type tubing fittings for tubing joints.

#### 3.6.5 Solder or Brazed Joints

Make all joints in metallic tubing and fittings with materials and procedures recommended by the tubing supplier. Braze joints with material having a melting point above 1000 degrees F, containing no phosphorous.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 3.6.6 Joining Thermoplastic or Fiberglass to Metallic Piping or Tubing

When compression type mechanical joints are used, provide gasket material in the fittings compatible with the plastic piping and with the gas in the system. Use an internal tubular rigid stiffener in conjunction with the fitting, flush with end of the pipe or tubing, extending at least to the outside end of the compression fitting when installed. Remove all rough or sharp edges from stiffener. Do not force fit stiffener in the plastic. Split tubular stiffeners are not allowed.

### 3.6.7 Press Connections

Make press connections in accordance with manufacturer's installation instructions using tools approved by the manufacturer. Fully insert the tubing into the fitting and then mark at the shoulder of the fitting. Check the fitting alignment against the mark on the tubing to assure the tubing is fully inserted before the joint is pressed.

## 3.7 PIPE SLEEVES

Provide pipes passing through concrete or masonry walls or concrete floors or roofs with pipe sleeves fitted into place at the time of construction. Do not install sleeves in structural members except where indicated or approved. Make all rectangular and square openings as detailed. Extend each sleeve through its respective wall, floor or roof, and cut flush with each surface, except in mechanical room floors not located on grade where clamping flanges or riser pipe clamps are used. Extend sleeves in mechanical room floors above grade at least 4 inches above finish floor. Unless otherwise indicated, use sleeves large enough to provide a minimum clearance of 1/4 inch all around the pipe. Provide steel pipe for sleeves in bearing walls, waterproofing membrane floors, and wet areas. Provide sleeves in nonbearing walls, floors, or ceilings of steel pipe, galvanized sheet metal with lock-type longitudinal seam, or moisture-resistant fiber or plastic. For penetrations of fire walls, fire partitions and floors which are not on grade, seal the annular space between the pipe and sleeve with fire-stopping material and sealant that meet the requirement of Section 07 84 00 FIRESTOPPING.

## 3.8 PIPES PENETRATING WATERPROOFING MEMBRANES

Install pipes penetrating waterproofing membranes as specified in Section 22 00 00 PLUMBING, GENERAL PURPOSE.

## 3.9 FIRE SEAL

Fire seal all penetrations of fire rated partitions, walls and floors in accordance with Section 07 84 00 FIRESTOPPING.

## 3.10 ESCUTCHEONS

Provide escutcheons for all finished surfaces where gas piping passes through floors, walls, or ceilings except in boiler, utility, or equipment rooms.

## 3.11 SPECIAL REQUIREMENTS

Provide drips, grading of the lines, freeze protection, and branch outlet locations as shown and conforming to the requirements of NFPA 54.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 3.12 BUILDING STRUCTURE

Do not weaken any building structure by the installation of any gas piping. Do not cut or notch beams, joists or columns. Attach piping supports to metal decking. Do not attach supports to the underside of concrete filled floors or concrete roof decks unless approved by the Contracting Officer.

### 3.13 PIPING SYSTEM SUPPORTS

Support gas piping systems in buildings with pipe hooks, metal pipe straps, bands or hangers suitable for the size of piping or tubing. Do not support any gas piping system by other piping. Conform spacing of supports in gas piping and tubing installations to the requirements of NFPA 54. Conform the selection and application of supports in gas piping and tubing installations to the requirements of MSS SP-58. In the support of multiple pipe runs on a common base member, use a clip or clamp where each pipe crosses the base support member. Spacing of the base support members is not to exceed the hanger and support spacing required for any of the individual pipes in the multiple pipe run. Rigidly connect the clips or clamps to the common base member. Provide a clearance of 1/8 inch between the pipe and clip or clamp for all piping which may be subjected to thermal expansion.

### 3.14 ELECTRICAL BONDING AND GROUNDING

Provide a gas piping system within the building that is electrically continuous and bonded to a grounding electrode as required by NFPA 54, NFPA 58, and NFPA 70.

### 3.15 SHUTOFF VALVE

Install the main gas shutoff valve controlling the gas piping system to be easily accessible for operation, as indicated, protected from physical damage, and marked with a metal tag to clearly identify the piping system controlled. Install valves approximately at locations indicated. Orient stems vertically, with operators on top, or horizontally. Provide PE piping manufacturer bracket support assembly securely fastened to structure for valve connections to resist operating torque applied to PE pipes. Provide stop valve on service branch at connection to main and shut-off valve on riser outside of building.

### 3.16 LINE AND APPLIANCE PRESSURE REGULATORS

Install line pressure regulators and appliance regulators in accordance with the manufacturer's requirements and in accordance with NFPA 54. Install each regulator in an accessible location and install shutoff valves ahead of each line and appliance regulator to allow for maintenance. Where vent limiting devices are not included in the regulators, install a vent pipe to the exterior of the building. Terminate all service regulator vents and relief vents in the outside air in rain and insect resistant fittings. Locate the open end of the vent where gas can escape freely into the atmosphere, away from any openings into the building and above areas subject to flooding.

### 3.17 GAS SERVICE INSTALLATION

Gas service line, service regulator and gas company meter must be installed in accordance with Section 33 51 15 NATURAL-GAS / LIQUEFIED



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

PETROLEUM GAS DISTRIBUTION PIPELINES. Installations must be in accordance with 49 CFR 192 and ASME B31.8. Contractor must submit and use only tested and approved work procedures. Contractor must use only welders and jointers who have been recently qualified by training and test for joining and installing the gas pipe material used on this job. The finished product must be inspected by a person qualified to inspect joints made by the particular procedures used to make joints.

### 3.18 TESTING

Submit test procedures and reports in booklet form tabulating test and measurements performed; dated after award of this contract, and stating the Contractor's name and address, the project name and location, and a list of the specific requirements which are being certified. Test entire gas piping system to ensure that it is gastight prior to putting into service. Prior to testing, purge the system, clean, and clear all foreign material. Test each joint with an approved gas detector, soap and water, or an equivalent nonflammable solution. Inspect and test each valve in conformance with API Std 598 and API Std 607. Complete testing before any work is covered, enclosed, or concealed, and perform with due regard for the safety of employees and the public during the test. Install bulkheads, anchorage and bracing suitably designed to resist test pressures if necessary, and as directed and or approved by the Contracting Officer. Do not use oxygen as a testing medium.

#### 3.18.1 Pressure Tests

Submit test procedures and reports in booklet form tabulating test and measurements performed; dated after award of this contract, and stating the Contractor's name and address, the project name and location, and a list of the specific requirements which are being certified. Before appliances are connected, test by filling the piping systems with air or an inert gas to withstand a minimum pressure of 3 pounds gauge for a period of not less than 10 minutes as specified in NFPA 54 without showing any drop in pressure. Do not use Oxygen for test. Measure pressure with a mercury manometer, slope gauge, or an equivalent device calibrated to be read in increments of not greater than 0.1 pound. Isolate the source of pressure before the pressure tests are made.

#### 3.18.2 Test With Gas

Before turning on gas under pressure into any piping, close all openings from which gas can escape. Immediately after turning on the gas, check the piping system for leakage by using a laboratory-certified gas meter, an appliance orifice, a manometer, or equivalent device. Conform all testing to the requirements of NFPA 54. If leakage is recorded, shut off the gas supply, repair the leak, and repeat the tests until all leaks have been stopped.

#### 3.18.3 Purging

After testing is completed, and before connecting any appliances, fully purge all gas piping. Do not purge piping into the combustion chamber of an appliance. Do not purge the open end of piping systems into confined spaces or areas where there are ignition sources unless the safety precautions recommended in NFPA 54 are followed.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

3.18.4 Labor, Materials and Equipment

Furnish all labor, materials and equipment necessary for conducting the testing and purging.

3.19 PIPE COLOR CODE MARKING

Provide color code marking of piping as specified in Section 09 90 00  
PAINTS AND COATINGS, conforming to ASME A13.1.

-- End of Section --

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## SECTION 23 21 13.00 20

## LOW TEMPERATURE WATER (LTW) HEATING SYSTEM

04/06, CHG 2: 11/19

## PART 1 GENERAL

## 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

## AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

|              |                                                                                                                        |
|--------------|------------------------------------------------------------------------------------------------------------------------|
| ASME B1.1    | (2024) Unified Inch Screw Threads (UN, UNR, and UNJ Thread Form)                                                       |
| ASME B1.20.1 | (2013; R 2018) Pipe Threads, General Purpose (Inch)                                                                    |
| ASME B16.1   | (2020) Gray Iron Pipe Flanges and Flanged Fittings Classes 25, 125, and 250                                            |
| ASME B16.3   | (2021) Malleable Iron Threaded Fittings, Classes 150 and 300                                                           |
| ASME B16.5   | (2020) Pipe Flanges and Flanged Fittings NPS 1/2 Through NPS 24 Metric/Inch Standard                                   |
| ASME B16.9   | (2018) Factory-Made Wrought Buttwelding Fittings                                                                       |
| ASME B16.11  | (2022) Forged Fittings, Socket-Welding and Threaded                                                                    |
| ASME B16.18  | (2021) Cast Copper Alloy Solder Joint Pressure Fittings                                                                |
| ASME B16.21  | (2021) Nonmetallic Flat Gaskets for Pipe Flanges                                                                       |
| ASME B16.22  | (2021) Wrought Copper and Copper Alloy Solder Joint Pressure Fittings                                                  |
| ASME B16.24  | (2022) Cast Copper Alloy Pipe Flanges, Flanged Fittings, and Valves Classes 150, 300, 600, 900, 1500, and 2500         |
| ASME B16.39  | (2020) Standard for Malleable Iron Threaded Pipe Unions; Classes 150, 250, and 300                                     |
| ASME B18.2.2 | (2022) Nuts for General Applications: Machine Screw Nuts, and Hex, Square, Hex Flange, and Coupling Nuts (Inch Series) |
| ASME B31.9   | (2020) Building Services Piping                                                                                        |

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

ASME B40.100 (2022) Pressure Gauges and Gauge Attachments

AMERICAN WELDING SOCIETY (AWS)

AWS Z49.1 (2021) Safety in Welding and Cutting and Allied Processes

ASTM INTERNATIONAL (ASTM)

ASTM A53/A53M (2024) Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless

ASTM A194/A194M (2024) Standard Specification for Carbon Steel, Alloy Steel, and Stainless Steel Nuts for Bolts for High-Pressure or High-Temperature Service, or Both

ASTM A307 (2021) Standard Specification for Carbon Steel Bolts, Studs, and Threaded Rod 60 000 PSI Tensile Strength

ASTM B32 (2020) Standard Specification for Solder Metal

ASTM B88 (2022) Standard Specification for Seamless Copper Water Tube

ASTM F1007 (2018; R 2022) Standard Specification for Pipeline Expansion Joints of the Packed Slip Type for Marine Application

ASTM F1120 (1987; R 2019) Standard Specification for Circular Metallic Bellows Type Expansion Joints for Piping Applications

COPPER DEVELOPMENT ASSOCIATION (CDA)

CDA A4015 (2016; 14/17) Copper Tube Handbook

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA MG 1 (2021) Motors and Generators

U.S. GENERAL SERVICES ADMINISTRATION (GSA)

CID A-A-1689 (Rev B) Tape, Pressure-Sensitive Adhesive, (Plastic Film)

CID A-A-50560 (Basic) Pumps, Centrifugal, Water Circulating, Electric-Motor-Driven

CID A-A-59617 (Basic, Notice 1) Unions, Brass or Bronze, Threaded Pipe Connections and Solder-Joint Tube Connections

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1910.144                      Safety Color Code for Marking Physical Hazards

29 CFR 1910.219                      Mechanical Power Transmission Apparatus

1.2 RELATED REQUIREMENTS

Section 23 03 00.00 20 BASIC MECHANICAL MATERIALS AND METHODS applies to this section with additions and modifications specified herein.

1.3 SYSTEM DESCRIPTION

Except as specified otherwise, equipment and piping components shall be suitable for use in low temperature water heating system. Except as modified herein, the pressure temperature limitations shall be as specified in the referenced standards and specifications. Pressures in this specification are pressures in pounds per square inch above atmospheric pressure, and temperatures are in degrees Fahrenheit (F).

1.3.1 Hot Water Heating System

Submit plan, elevations, dimensions, capacities, and ratings. Include the following:

- a. Unit heaters
- b. Convectors
- c. Finned tube radiators
- d. Pumps
- e. Valves
- f. Expansion tanks
- g. Flow measuring equipment
- h. Backflow preventer
- i. Air separating tank
- j. Boilers

1.4 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Hot water heating system; G

SD-03 Product Data

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

Convectors

Finned tube radiators

Pumps; G

Include pump speed and characteristic curve for performance of impeller selected for each pump. Curves shall indicate capacity vs head, efficiency, and brake power for full range, from shut-off to free delivery.

Expansion tanks; G

Flow measuring equipment

Backflow preventers

External air separation tanks

Hot water heating pipe

#### SD-06 Test Reports

Hydrostatic test of piping system

Auxiliary equipment and accessory tests

Submit test reports in accordance with the paragraph entitled "Field Quality Control."

#### SD-07 Certificates

Backflow preventer certification

Report of prior installations

Welding procedures

Welder's qualifications

#### SD-10 Operation and Maintenance Data

Convectors, Data Package 3; G

Finned tube radiators, Data Package 3; G

Submit in accordance with Section 01 78 23 OPERATION AND MAINTENANCE DATA. Submit a list of qualified service organizations which includes addresses and qualifications.

### 1.5 QUALITY ASSURANCE

#### 1.5.1 Standard Commercial Product for Terminal Units

Terminal units provided shall comply with features called out in this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not prohibited by this specification but which are a part of the manufacturer's standard

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

commercial product, shall be included in the terminal units being furnished. A standard commercial product is a product which has been sold or is currently being offered for sale, on the commercial market through advertisements or manufacturer's catalogs, or brochures. Provide Institute of Boiler and Radiator Manufacturer (IBR) or Steel Boiler Institute (SBI) rating for required capacity.

#### 1.5.2 Welding

##### 1.5.2.1 Report of Prior Installations

Submit a Certificate of Full Approval or a current Certificate of Approval for each design, size, and make of backflow preventer being provided for the project. Certificate shall be from the Foundation for Cross-Connection Control and Hydraulic Research, University of Southern California, and shall attest that this design, size, and make of backflow preventer has satisfactorily passed the complete sequence of performance testing and evaluation for the respective level of approval. A Certificate of Provisional Approval is not acceptable in lieu of the above.

##### 1.5.2.2 Welding Procedures

Before performing welding, submit three copies of welding procedure specification for all metals to be used in the work, together with proof of welder's qualification as outlines in ASME B31.9.

##### 1.5.2.3 Welder's Qualifications

Before welder or operator performs welding, submit three copies of Welder's Performance Qualification Record in conformance with ASME B31.9 showing that the welder was tested under the approved procedure specification submitted by the Contractor. In addition, submit each welder's assigned number, letter, or symbol used to identify the work of the welder.

##### 1.5.2.4 Identification of Welder's Work

Ensure that each welder's assigned number, letter or symbol is affixed immediately upon completion of the weld. To welders making defective welds after passing a qualification test, give a requalification test. Upon failing to pass the test, do not permit welder to work in this contract.

##### 1.5.2.5 Previous Qualifications

Welding procedures, welders, and welding operators previously qualified by test may be accepted for this contract without requalification subject to the approval and provided that all the conditions specified in ASME B31.9 are met before a procedure can be used.

#### 1.5.3 Brazing and Soldering

##### 1.5.3.1 Brazing Procedure

ASME B31.9. Brazing procedure for joints shall be as outlined in CDA A4015.

##### 1.5.3.2 Soldering, Soldering Preparation, and Procedures for Joints

ASME B31.9 and as outlined in CDA A4015.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### 1.5.4 Backflow Preventer Certification

Submit a Certificate of Full Approval or a current Certificate of Approval for backflow preventers.

### 1.6 SAFETY STANDARDS

#### 1.6.1 Welding

Safety in welding and cutting of pipe shall conform to AWS Z49.1.

#### 1.6.2 Guards

Couplings, motor shafts, gears and other moving parts shall be guarded, in accordance with OSHA 29 CFR 1910.219. Guards shall be cast iron or expanded metal. Guard parts shall be rigid and removable without disassembling the guarded unit.

## PART 2 PRODUCTS

### 2.1 PIPE AND FITTINGS

#### 2.1.1 Hot Water Heating Pipe (Supply and Return)

ASTM A53/A53M electric resistance welded or seamless Schedule 40 steel pipe or ASTM B88 Type L hard drawn Copper tubing.

#### 2.1.2 Fittings

Provide fittings compatible with the pipe being provided and shall conform to the following requirements.

##### 2.1.2.1 Steel or Malleable Iron Pipe

Sizes 1/8 to 2 inches. ASME B16.11 steel socket welding or screwed type or ASME B16.3 for screwed type malleable iron fittings.

##### 2.1.2.2 Steel, Cast Iron, or Bronze

Sizes 2 1/2 inches and above. Steel fitting butt welding type ASME B16.9 or ASME B16.5 flanged type. Cast iron fittings flanged type ASME B16.1. Bronze fittings up to 8 inch size flanged type ASME B16.24.

##### 2.1.2.3 Fittings for Copper Tubing

ASME B16.18 cast bronze solder joint type or ASME B16.22 wrought copper solder joint type. Fittings may be flared or compression joint type.

#### 2.1.3 Unions

##### 2.1.3.1 Steel Pipe

Provide ASME B16.39, malleable iron unions, threaded connections.

##### 2.1.3.2 Copper Tubing

Provide CID A-A-59617, bronze unions, solder joint end.



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### 2.1.3.3 Dielectric Union

Provide insulated union with galvanized steel female pipe-threaded end and a copper solder joint end conforming with ASME B16.39, Class 1, dimensional, strength and pressure requirements. Union shall have a water-impervious insulation barrier capable of limiting galvanic current to one percent of the short-circuit current in a corresponding bimetallic joint. When dry, insulation barrier shall be able to withstand a 600-volt breakdown test.

#### 2.1.4 Flanges

Remove raised faces when used with flanges having a flat face.

##### 2.1.4.1 Steel Flanges

ASME B16.5 forged steel, welding type.

##### 2.1.4.2 Cast Iron Screwed Flanges

ASME B16.1.

##### 2.1.4.3 Bronze Screwed Flanges

ASME B16.24.

#### 2.1.5 Drains and Overflows

##### 2.1.5.1 Steel Pipe

ASTM A53/A53M, Seamless Schedule 40, Malleable iron or forged steel fittings, screwed or welded joints.

##### 2.1.5.2 Copper Tubing

ASTM B88, Type L, hard drawn, cast brass or wrought copper fittings, Grade Sb5 solder joints.

#### 2.1.6 Valves

Valves shall have rising stems and shall open when turned counterclockwise.

##### 2.1.6.1 Globe and Angle Valves

Provide as specified in Section 23 05 15 COMMON PIPING FOR HVAC.

##### 2.1.6.2 Check Valves

Provide as specified in Section 23 05 15 COMMON PIPING FOR HVAC.

##### 2.1.6.3 Ball Valves

Provide as specified in Section 23 05 15 COMMON PIPING FOR HVAC.

##### 2.1.6.4 Relief Valves

Bronze body, teflon seat, stainless steel stem and springs, automatic, direct pressure actuated, capacities ASME certified and labelled.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### 2.1.6.5 Balancing Valves

Balancing valves shall be calibrated bronze body balancing valves with integral ball valve and venturi or valve orifice and valve body pressure taps for flow measurement based on differential pressure readings. Valve pressure taps and meter connections shall have seals and built-in check valves with threaded connections for a portable meter. Meter shall be provided by the same manufacturer and be capable of reading system pressures and shall meet the requirements of the paragraph entitled "Flow Measuring Equipment." Valves shall have internal seals to prevent leakage around rotating element and be suitable for full shut-off rated pressure. Valves shall have an operator with integral pointer and memory stop. Balancing valves shall be selected for the required flows as indicated on the plans.

#### 2.1.7 End Connections

##### 2.1.7.1 Flexible Connectors

Provide flexible pipe connectors on piping connected to equipment. Flexible section shall consist of rubber, tetrafluoroethylene resin, corrosion-resistant steel, bronze, monel, or galvanized steel. Material provided and configuration shall be suitable for pressure, temperature, and circulating medium. Flexible section shall have threaded, or flanged ends and shall be suitable for service intended. Flexible section may be reinforced with metal retaining rings, with built-in braided wire reinforcement and restriction bolts or with wire braid cover suitable for service intended.

##### 2.1.7.2 Steel Piping

Screwed or socket welded for 2 inches and smaller and flanged or butt welded for 2 1/2 inches and larger.

- a. Screwed Joints With Taper Threads: ASME B1.20.1.
- b. Flanged Joints: Bolting and gaskets shall be as follows:
  - (1) Bolting: Bolt and stud material ASTM A307, Grade B, and nut material ASTM A194/A194M, Grade 2. Bolt, stud, and nut dimensions ASME B18.2.2 threads ASME B1.1coarse type with Class 2A fit for bolts and studs, and Class 2B fit for nuts. Bolts or bolt studs shall extend completely through the nuts and may have reduced shanks of a diameter not less than the diameter at root of threads. Carbon steel bolts shall have American Standard regular square or heavy hexagon heads and shall have American Standard heavy semifinished hexagonal nuts conforming to ASME B18.2.2.
  - (2) Gaskets: ASME B16.21, Nonasbestos compressed material 1/16 inch thickness full face or self-centering flat ring type and suitable for pressure and temperature of the piping system.
- c. Butt Weld Joints: ASME B31.9. Backing rings shall conform to ASME B31.9. Ferrous rings shall not exceed 0.05 percent sulfur; for alloy pipe, backing rings shall be of material compatible with the chemical composition of the parts to be welded and preferably of the same composition. Provide continuous machined or split band backing rings.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

d. Socket Weld Joints: ASME B31.9.

#### 2.1.7.3 Joints for Copper Tubing

- a. Solder conforming to ASTM B32 alloy grade Sb5 or Sn96. Solder and flux shall be lead free (less than 0.2 percent of lead).
- b. Copper Tube Extracted Joint: An extracted mechanical tee joint may be made in copper tube. Make joint with an appropriate tool by drilling a pilot hole and drawing out the tube surface to form a collar having a minimum height of three times the thickness of the tube wall. To prevent the branch tube from being inserted beyond the depth of the extracted joint, provide dimpled depth stops. Notch the branch tube for proper penetration into fitting to assure a free flow joint. Braze extracted joints using a copper phosphorous classification brazing filler metal. Soldered joints shall not be permitted.

#### 2.1.8 Expansion Joints

##### 2.1.8.1 Packless Type

Provide ASTM F1120, Type III with fabricated corrosion-resistant steel bellows.

##### 2.1.8.2 Guided Slip-Tube Type

Provide ASTM F1007, Type IV internally-externally guided, injected semiplastic type packing.

#### 2.1.9 Instrumentation

##### 2.1.9.1 Pressure and Vacuum Gauges

Provide ASME B40.100 with restrictor.

##### 2.1.9.2 Indicating Thermometers

Provide as specified in Section 23 05 15 COMMON PIPING FOR HVAC.

##### 2.1.9.3 Pressure/Temperature Test Ports

Pressure/Temperature Test Ports shall have brass body and EPDM and/or Neoprene valve seals. Ports shall be rated for service between 35 and 275 degrees F and up to 500 psig. Ports shall be provided in lengths appropriate for the insulation thickness specified in Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS and installed to allow a minimum of 12 inches of access for probe insertion. Provide with screw-on cap attached with a strap or chain to prevent loss when removed. Ports shall be 1/4 inch NPT and accept 1/8 inch diameter probes.

##### 2.1.10 Miscellaneous Pipeline Components

Provide as specified in Section 23 05 15 COMMON PIPING FOR HVAC.

#### 2.2 CENTRAL MECHANICAL EQUIPMENT

##### 2.2.1 Boilers

Provide as specified in Section 23 52 43.00 20 LOW PRESSURE WATER HEATING

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

BOILERS UNDER 800,000 BTU/HR OUTPUT.

## 2.3 PIPING SYSTEM EQUIPMENT

### 2.3.1 Pumps

Provide hot water circulating pumps, CID A-A-50560, Service A. Pump casing and flange shall be made of close-grained cast iron. Shaft shall be carbon or alloy steel with lubricated bearings and impeller shall be bronze. Select pumps so that the operating point on selected impeller-curve will lie at or to the left of shutoff side of, and not more than 5 percent below, point of maximum efficiency for impeller. Provide motors of totally enclosed type conforming to NEMA MG 1 and suitable for electrical characteristic as indicated. Provide a BACnet compatible, variable speed drive with NEMA type 12 enclosure, with automatic bypass.

### 2.3.2 Expansion Tanks

Provide as specified in Section 23 05 15 COMMON PIPING FOR HVAC.

### 2.3.3 External Air Separation Tanks

Provide as specified in Section 23 05 15 COMMON PIPING FOR HVAC.

## 2.4 TERMINAL UNITS

### 2.4.1 Hydronic Radiant Slab Heating

<...specifications to reference PEX-a piping, control valves, loop dimensions, etc....>

## 2.5 ELECTRICAL EQUIPMENT

Provide complete with motors, motor starters, thermal overload protection, and controls. Equipment and wiring shall be in accordance with Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM.

## 2.6 CONTROLS

Provide controls as specified in Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC

## 2.7 INSULATION

Provide shop and field applied insulation as specified in Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS.

## 2.8 ASBESTOS PROHIBITION

Asbestos and asbestos containing products are prohibited.

## PART 3 EXECUTION

### 3.1 PREPARATION

Provide storage for equipment and material at the project site. All parts shall be readily accessible for inspection, repair, and renewal. Protect material and equipment from the weather.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 3.2 INSTALLATION

Piping fabrication, assembly, welding, soldering, and brazing shall conform to ASME B31.9. Piping shall follow the general arrangement shown. Route piping and equipment within buildings out of the way of lighting fixtures and doors, windows, and other openings. Run overhead piping in buildings in inconspicuous positions. Provide adequate clearances from walls, ceilings, and floors to permit welding of joints and application of insulation. Make provision for expansion and contraction of pipe lines. Make changes in size of water lines with reducing fittings. Do not bury, conceal, or insulate until piping has been inspected, tested, and approved. Do not run piping concealed in walls, partitions, underground, or under the floor except as otherwise indicated. Where pipe passes through building structure, locate pipe joints and expansion joints where they may be inspected. Provide flanged joints where necessary for normal maintenance and where required to match valves and equipment. Furnish gaskets, packing, and thread compounds suitable for the service. Provide long radius ells where possible to reduce pressure drops. Pipe bends in lieu of welding fittings may be used where space permits. Pipe bends shall have a uniform radius of at least five times the pipe diameter and shall be free from appreciable flattening, wrinkling, or thinning of the pipe. Do not use mitering of pipe to form elbows, notching straight runs to form full sized tees, or any similar construction. Make branch connections over 2 inches with welding tees except factory made forged welding branch outlets or nozzles having integral reinforcements conforming to ASME B31.9 may be used, provided the nominal diameter of the branch is at least one pipe size less than the nominal diameter of the run. Branch connections 2 inches and under can be threaded or welded. Run vertical piping plumb and straight and parallel to walls. Provide sleeves for lines passing through building structure. Provide a fire seal where pipes pass through fire wall, fire partitions, fire rated pipe chase walls, or floors above grade. Install piping connected to equipment with flexibility for thermal stresses and for vibration, and support and anchor so that strain from weight and thermal movement of piping is not imposed on the equipment.

#### 3.2.1 Hangers and Supports

Provide as specified in Section 23 05 15 COMMON PIPING FOR HVAC.

#### 3.2.2 Grading of Pipe Lines

Unless otherwise indicated, install horizontal lines of hot water piping to grade down in the direction of flow with a pitch of not less than one inch in 30 feet, except in loop mains and main headers where the flow may be in either direction.

#### 3.2.3 Pipe Sleeves

Provide as specified in Section 23 05 15 COMMON PIPING FOR HVAC.

#### 3.2.4 Unions and Flanges

Provide unions and flanges to permit easy disconnection of piping and apparatus. Each connection having a screwed-end valve shall have a union. Place unions and flanges no farther apart than 100 feet. Install unions downstream of valves and at equipment or apparatus connections. Provide unions on piping under 2 inches in diameter, and provide flanges on piping 2 inches and over in diameter. Provide dielectric unions or

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

flanges between ferrous and non-ferrous piping, equipment, and fittings; except that bronze valves and fittings may be used without dielectric couplings for ferrous-to-ferrous or non-ferrous-to-non-ferrous connections.

### 3.2.5 Changes in Pipe Size

Provide reducing fittings for changes in pipe size; reducing bushings are not permitted. In horizontal lines, provide eccentric reducing fittings to maintain the top of the lines in the same plane.

### 3.2.6 Cleaning of Pipe

Thoroughly clean each section of pipe, fittings, and valves free of foreign matter before erection. Prior to erection, hold each piece of pipe in an inclined position and tap along its full length to loosen sand, mill scale and other foreign matter. For pipe 2 inches and larger, draw wire brush, of a diameter larger than that of the inside of the pipe, several times through the entire length of pipe. Before making final connections to apparatus, wash out interior of piping thoroughly with water. Plug or cap open ends of mains during shutdown periods. Do not leave lines open where foreign matter might enter the pipe.

### 3.2.7 Valves

Install valves in conformance with ASME B31.9. Provide gate valves unless otherwise directed. Install valves with stems horizontal or above. Locate or equip stop valves to permit operation from floor level, or provide with safe access in the form of walkways or ladders. Install valves in positions accessible for operation and repair.

#### 3.2.7.1 Globe Valves

Install globe valves so that the pressure is below the disk and the stem horizontal.

#### 3.2.7.2 Radiators Valves

Provide radiator valves on water inlet and balancing valves on water outlet of terminal heating units such as radiation, unit heaters, and fan coil unit.

#### 3.2.7.3 Relief Valves

Provide valves on pressure tanks, low pressure side of reducing valves, heat exchangers, and expansion tanks. Select system relief valve so that capacity is greater than make-up pressure reducing valve capacity. Select equipment relief valve capacity to exceed rating of connected equipment. Pipe relief valve outlet to the nearest floor drain.

### 3.2.8 Pressure Gage

Provide a shut-off valve or pet cock between pressure gages and the line.

### 3.2.9 Thermometers

Provide thermometers and thermal sensing elements of control valves with a separable socket. Install separable sockets in pipe lines in such a manner to sense the temperature of flowing the fluid and minimize obstruction to flow.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 3.2.10 Strainers

Provide strainers, with meshes suitable for the services, where indicated, or where dirt might interfere with the proper operation of valve parts, orifices, or moving parts of equipment.

### 3.2.11 Pumps

Select pumps for specified fluid temperatures, are non-overloading in parallel or individual operation, and operate within 25 percent of midpoint of published maximum efficiency curve. Support piping adjacent to pump such that no weight is carried on pump casings. Install close coupled and base mounted pumps on concrete base, with anchor bolts, set and level, and grout in place and provide supports under elbows on pump suction and discharge line sizes 4 inches and over. Lubricate pump before start-up.

### 3.2.12 Equipment Foundations

Locate equipment foundations as shown on the drawings. Size, weight, and design shall preclude shifting of equipment under operating conditions. Foundations shall meet the requirements of the equipment manufacturer. Concrete shall conform to Section 03 30 00 CAST-IN-PLACE CONCRETE, and grout shall be approved non-shrinking.

### 3.2.13 Equipment Installation

Install equipment in accordance with installation instructions of the manufacturers. Grout equipment mounted on concrete foundations before installing piping. Install piping in such a manner as not to place a strain on the equipment. Do not bolt flanged joints tight unless they match. Grade, anchor, guide, and support piping without low pockets.

### 3.2.14 Cleaning of Systems

As installation of the various system components is completed, fill, start, and vent prior to cleaning. Place terminal control valves in open position. Add cleaner to closed system at concentration as recommended by manufacturer. Apply heat while circulating, slowly raising temperature to 160 degrees F and maintain for 12 hours minimum. Remove heat and circulate to 100 degrees F or less; drain systems as quickly as possible and refill with clean water. Circulate for 6 hours at design temperatures, then drain. Refill with clean water and repeat until system cleaner is removed. Use neutralizer agents on recommendation of system cleaner supplier and approval of Contracting Officer. Remove, clean, and replace strainer screens. Inspect, remove sludge, and flush low points with clean water after cleaning process is completed. Include disassembly of components as required. Preliminary or final tests are not permitted until cleaning is approved.

### 3.2.15 Painting of Piping and Equipment

Provide in accordance with Section 09 90 00 PAINTS AND COATINGS.

### 3.2.16 Identification of Piping

Identify piping in accordance with OSHA 29 CFR 1910.144, except that labels or tapes may be used in lieu of painting or stenciling. Spacing of

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

identification marking on runs shall not exceed 50 feet. Materials for labels and tapes shall conform to CID A-A-1689, and shall be general purpose type and color class. Painting and stenciling shall conform to Section 09 90 00 PAINTS AND COATINGS.

### 3.3 FIELD QUALITY CONTROL

Perform inspections and tests as specified herein to demonstrate that piping and equipment, as installed, is in compliance with contract requirements. Start up and operate the system. During this time, periodically clean the various strainers until no further accumulation of foreign material occurs. Exercise care so that minimum loss of water occurs when strainers are cleaned. Adjust safety and automatic control instruments to place them in proper operation and sequence.

#### 3.3.1 Hydrostatic Test of Piping System

Test piping system hydrostatically using water not exceeding 100 degrees F. Conduct tests in accordance with the requirements of ASME B31.9 and as follows. Test piping system after all lines have been cleaned and before applying insulation covering. Remove or valve off from the system, gages, and other apparatus which may be damaged by the test before the tests are made. Install calibrated test pressure gage in the system to observe any loss in pressure. Maintain test pressure for a sufficient length of time to enable an inspection of each joint and connection. Perform tests after installation and prior to acceptance. Notify the Contracting Officer in writing 30 days prior to the time scheduled for the tests.

#### 3.3.2 Auxiliary Equipment and Accessory Tests

Observe and check pumps, accessories, and equipment during operational and capacity tests for leakage, malfunctions, defects, noncompliance with referenced standards, or overloading.

### 3.4 TESTING, ADJUSTING, AND BALANCING

Test, adjust, and balance the hydronic system in accordance with Section 23 05 93 TESTING, ADJUSTING AND BALANCING FOR HVAC.

#### 3.4.1 Markings of Settings

Following final acceptance of the balancing report, the settings of all valves, splitters, dampers, and other adjustment devices shall be permanently marked so that adjustment can be restored if disturbed at anytime.

-- End of Section --



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## SECTION 23 23 00

### REFRIGERANT PIPING

**08/21**

#### PART 1 GENERAL

##### 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

#### AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS (ASHRAE)

|                |                                                                                                                                                                                |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASHRAE 15 & 34 | (2013) ASHRAE Standard 34-2016 Safety Standard for Refrigeration Systems/ASHRAE Standard 34-2016 Designation and Safety Classification of Refrigerants-ASHRAE Standard 34-2016 |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

#### AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

|             |                                                                        |
|-------------|------------------------------------------------------------------------|
| ASME B16.22 | (2021) Wrought Copper and Copper Alloy Solder Joint Pressure Fittings  |
| ASME B16.26 | (2018) Standard for Cast Copper Alloy Fittings for Flared Copper Tubes |
| ASME B31.1  | (2022) Power Piping                                                    |
| ASME B31.5  | (2022) Refrigeration Piping and Heat Transfer Components               |

#### AMERICAN WELDING SOCIETY (AWS)

|                  |                                                                      |
|------------------|----------------------------------------------------------------------|
| AWS A5.8/A5.8M   | (2019) Specification for Filler Metals for Brazing and Braze Welding |
| AWS A5.31/A5.31M | (2012) Specification for Fluxes for Brazing and Braze Welding        |
| AWS BRH          | (2007; 5th Ed) Brazing Handbook                                      |
| AWS Z49.1        | (2021) Safety in Welding and Cutting and Allied Processes            |

#### ASTM INTERNATIONAL (ASTM)

|                 |                                                                                                                                         |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| ASTM A53/A53M   | (2022) Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless                                   |
| ASTM A653/A653M | (2023) Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process |

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

|                                                                                   |                                                                                                                      |
|-----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| ASTM B62                                                                          | (2017) Standard Specification for<br>Composition Bronze or Ounce Metal Castings                                      |
| ASTM B75/B75M                                                                     | (2020) Standard Specification for Seamless<br>Copper Tube                                                            |
| ASTM B117                                                                         | (2019) Standard Practice for Operating<br>Salt Spray (Fog) Apparatus                                                 |
| ASTM B280                                                                         | (2020) Standard Specification for Seamless<br>Copper Tube for Air Conditioning and<br>Refrigeration Field Service    |
| ASTM D520                                                                         | (2000; R 2011) Zinc Dust Pigment                                                                                     |
| ASTM D3308                                                                        | (2012; R 2017) Standard Specification for<br>PTFE Resin Skived Tape                                                  |
| MANUFACTURERS STANDARDIZATION SOCIETY OF THE VALVE AND FITTINGS<br>INDUSTRY (MSS) |                                                                                                                      |
| MSS SP-58                                                                         | (2018) Pipe Hangers and Supports -<br>Materials, Design and Manufacture,<br>Selection, Application, and Installation |

## 1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

### SD-02 Shop Drawings

Refrigerant Piping System; G

### SD-03 Product Data

Refrigerant Piping System

Spare Parts

Qualifications

Refrigerant Piping Tests

Verification of Dimensions

### SD-06 Test Reports

Refrigerant Piping Tests

### SD-07 Certificates

Service Organization

### SD-10 Operation and Maintenance Data

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

Maintenance; G

Operation and Maintenance Manuals; G

Demonstrations; G

### 1.3 QUALITY ASSURANCE

#### 1.3.1 Contract Drawings

Because of the small scale of the drawings, it is not possible to indicate all offsets, fittings, and accessories that may be required. Carefully investigate the plumbing, fire protection, electrical, structural and finish conditions that would affect the work to be performed and arrange such work accordingly, furnishing required offsets, fittings, and accessories to meet such conditions.

### 1.4 DELIVERY, STORAGE, AND HANDLING

Protect stored items from the weather, humidity and temperature variations, dirt and dust, or other contaminants. Proper protection and care of all material both before and during installation is the Contractor's responsibility. Replace any materials found to be damaged at the Contractor's expense. During installation, cap piping and similar openings to keep out dirt and other foreign matter.

### 1.5 MAINTENANCE

#### 1.5.1 General

Submit Data Package 2 plus operation and maintenance data complying with the requirements of Section 01 78 23 OPERATION AND MAINTENANCE DATA and as specified herein.

## PART 2 PRODUCTS

### 2.1 STANDARD COMMERCIAL PRODUCTS

- a. Provide materials and equipment which are standard products of a manufacturer regularly engaged in the manufacturing of such products, that are of a similar material, design and workmanship and that have been in satisfactory commercial or industrial use for 2 years prior to bid opening.
- b. The 2 year use must include applications of equipment and materials under similar circumstances and of similar size. The 2 years' experience must be satisfactorily completed by a product which has been sold or is offered for sale on the commercial market through advertisements, manufacturer's catalogs, or brochures. Products having less than a 2-year field service record will be acceptable if a certified record of satisfactory field operation, for not less than 6000 hours exclusive of the manufacturer's factory tests, can be shown.
- c. Products must be supported by a service organization. System components must be environmentally suitable for the indicated locations. Submit a certified list of qualified permanent service organizations for support of the equipment which includes their addresses and qualifications. The service organizations must be reasonably convenient to the equipment installation and be able to

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

render satisfactory service to the equipment on a regular and emergency basis during the warranty period of the contract.

- d. Exposed equipment moving parts, parts that produce high operating temperature, parts which may be electrically energized, and parts that may be a hazard to operating personnel must be insulated, fully enclosed, guarded, or fitted with other types of safety devices. Install safety devices so that proper operation of equipment is not impaired. Welding and cutting safety requirements must be in accordance with AWS Z49.1.
- e. Provide the manufacturer's standard catalog data, at least 5 weeks prior to the purchase or installation of a particular component. Highlight the data to show information such as, but not limited to, material, size, options, performance charts, and curves in adequate detail to demonstrate compliance with contract requirements. Include the manufacturer's recommended installation instructions and procedures in the data provided. Provide data for the following components as a minimum:
  - (1) Piping and Fittings
  - (2) Valves
  - (3) Piping Accessories
  - (4) Pipe Hangers, Inserts, and Supports

## 2.2 REFRIGERANT PIPING SYSTEM

Provide refrigerant piping, valves, fittings, and accessories in accordance with ASHRAE 15 & 34 and ASME B31.5, except as specified herein. Refrigerant piping, valves, fittings, and accessories must be compatible with the fluids used and capable of withstanding the pressures and temperatures of the service. Refrigerant piping, valves, and accessories used for refrigerant service must be cleaned, dehydrated, and sealed (capped or plugged) prior to shipment from the manufacturer's plant. Submit drawings, at least 5 weeks prior to beginning construction, provided in adequate detail to demonstrate compliance with contract requirements. Drawings must consist of:

- a. Piping layouts which identify all valves and fittings.
- b. Plans and elevations which identify clearances required for maintenance and operation.

## 2.3 PIPE, FITTINGS AND END CONNECTIONS (JOINTS)

### 2.3.1 Copper Tubing

Provide copper tubing conforming to ASTM B280 annealed or hard drawn as required. Copper tubing must bear the product identification markings in accordance with ASTM B280, "ACR" must be present on copper tubing. Copper tubing must be soft annealed where bending is required and hard drawn where no bending is required. Soft annealed copper tubing must not be used in sizes larger than 1-3/8 inches. Joints must be brazed except that joints on lines 7/8 inch and smaller may be flared. Cast copper alloy fittings for flared copper tube must conform to ASME B16.26 and ASTM B62. Wrought copper and bronze solder-joint pressure fittings must conform to ASME B16.22 and ASTM B75/B75M. Joints and fittings for brazed joint must be wrought-copper or forged-brass sweat fittings. Cast sweat-type joints and fittings are not allowed for brazed joints. Brass or bronze adapters

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

for brazed tubing may be used for connecting tubing to flanges and to threaded ends of valves and equipment.

#### 2.3.2 Brazing Filler Metal

Filler metal must conform to AWS A5.8/A5.8M, Type BAg-5 with AWS Type FB3-A or Type FB3-C flux, except Type BCuP-3, BCuP-4, or BCuP-5 may be used for brazing copper-to-copper joints. BAlSi-4 with AWS Type FB1-A flux may be used when joining copper piping to aluminum components.

#### 2.3.3 Brazing Flux

Brazing flux must conform to AWS A5.31/A5.31M, Type FB3-A or Type FB3-C when using Type BAg-5 filler metal. Type FB1-A is to be used with Type BAlSi-4 filler metal.

#### 2.3.4 Press Fittings

Press fittings are not acceptable for use in refrigerant piping systems.

### 2.4 VALVES

Valves must be designed, manufactured, and tested specifically for refrigerant service. The valve material and all internal components must be compatible with the specific refrigerant and lubricant used. Valve bodies must be of bronze construction. Valves 1 inch and smaller must have brazed or socket welded connections. Valves larger than 1 inch must have butt welded end connections. Do not use threaded end connections, except in pilot pressure or gauge lines where maintenance disassembly is required and welded flanges cannot be used. Internal parts must be removable for inspection or replacement without applying heat or breaking pipe connections. Valve stems exposed to the atmosphere must be stainless steel or corrosion resistant metal plated carbon steel. Direction of flow must be legibly and permanently indicated on the valve body.

#### 2.4.1 Refrigerant Stop Valves

Valve must be the globe or full-port ball type with a back-seating stem especially packed for refrigerant service. Valve packing must be replaceable under line pressure. Valve must be the straight or angle pattern design as indicated.

### 2.5 PIPING ACCESSORIES

#### 2.5.1 Flexible Pipe Connectors

Connector must be a composite of interior corrugated phosphor bronze or Type 300 Series stainless steel, as required for fluid service, with exterior reinforcement of bronze, stainless steel or monel wire braid. Assembly must be constructed with a safety factor of not less than 4 at 300 degrees F. Unless otherwise indicated, the length of a flexible connector must be as recommended by the manufacturer for the service intended.

#### 2.5.2 Pipe Hangers, Inserts, and Supports

Provide pipe hangers, inserts, guides, and supports conforming to MSS SP-58. and as specified in Section 23 05 15 COMMON PIPING FOR HVAC.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 2.5.3 Escutcheons

Provide specified in Section 23 05 15 COMMON PIPING FOR HVAC.

## 2.6 FABRICATION

### 2.6.1 Factory Coating

Unless otherwise specified, equipment and component items, when fabricated from ferrous metal, must be factory finished with the manufacturer's standard finish, except that items located outside of buildings must have weather resistant finishes that will withstand 500 hours exposure to the salt spray test specified in ASTM B117 using a 5 percent sodium chloride solution. Immediately after completion of the test, the specimen must show no signs of blistering, wrinkling, cracking, or loss of adhesion and no sign of rust creepage beyond 1/8 inch on either side of the scratch mark. Cut edges of galvanized surfaces where hot-dip galvanized sheet steel is used must be coated with a zinc-rich coating conforming to ASTM D520, Type I.

## PART 3 EXECUTION

### 3.1 EXAMINATION

After becoming familiar with all details of the work, perform a verification of dimensions in the field. Submit a letter, at least 2 weeks prior to beginning construction, including the date the site was visited, conformation of existing conditions, and any discrepancies found before performing any work.

### 3.2 INSTALLATION

Pipe and fitting installation must conform to the requirements of ASME B31.1. Cut pipe accurately to measurements established at the jobsite, and work into place without springing or forcing, completely clearing all windows, doors, and other openings. Cutting or other weakening of the building structure to facilitate piping installation is not permitted without written approval. Cut pipe or tubing square, remove by reaming, and permit free expansion and contraction without causing damage to the building structure, pipe, joints, or hangers.

#### 3.2.1 Directional Changes

Make changes in direction with fittings, except that bending of pipe 4 inches and smaller is permitted, provided a pipe bender is used and wide weep bends are formed. Mitering or notching pipe or other similar construction to form elbows or tees is not permitted. The centerline radius of bends must not be less than 6 diameters of the pipe. Bent pipe showing kinks, wrinkles, flattening, or other malformations will not be accepted.

#### 3.2.2 Functional Requirements

Install piping 1/2 inch/10 feet of pipe in the direction of flow to ensure adequate oil drainage. Properly cap or plug open ends of refrigerant lines or equipment during installation to keep moisture, dirt, or other foreign material out of the system. Piping must remain capped until installation. Equipment piping must be in accordance with the equipment manufacturer's recommendations and the contract drawings.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

Equipment and piping arrangements must fit into space allotted and allow adequate acceptable clearances for installation, replacement, entry, servicing, and maintenance.

### 3.2.3 Fittings and End Connections

#### 3.2.3.1 Threaded Connections

Make threaded connections with tapered threads and make tight with PTFE tape complying with ASTM D3308 or equivalent thread-joint compound applied to the male threads only. Do not show more than three threads after the joint is made.

#### 3.2.3.2 Brazed Connections

Perform brazing in accordance with AWS BRH, except as modified herein. During brazing, fill the pipe and fittings with a pressure regulated inert gas, such as nitrogen, to prevent the formation of scale. Before brazing copper joints, clean both the outside of the tube and the inside of the fitting with a wire fitting brush until the entire joint surface is bright and clean. Do not use brazing flux on copper-to-copper connections. Remove surplus brazing material at all joints. Make steel tubing joints in accordance with the manufacturer's recommendations. Paint joints in steel tubing with the same material as the baked-on coating within 8 hours after joints are made. Protect tubing against oxidation during brazing by continuous purging of the inside of the piping using nitrogen. Support piping prior to brazing and do not spring or force.

### 3.2.4 Valves

#### 3.2.4.1 General

Install refrigerant stop valves on each side of each piece of equipment such as compressors condensers, evaporators, receivers, and other similar items in multiple-unit installation, to provide partial system isolation as required for maintenance or repair. Install stop valves with stems horizontal unless otherwise indicated. Install ball valves must be installed with stems positioned to facilitate operation and maintenance. Isolating valves for pressure gauges and switches must be external to thermal insulation. Safety switches must not be fitted with isolation valves. Filter dryers having access ports may be considered a point of isolation. Purge valves must be provided at all points of systems where accumulated non-condensable gases would prevent proper system operation. Valves must be furnished to match line size, unless otherwise indicated or approved.

#### 3.2.5 Flexible Pipe Connectors

Install connectors perpendicular to line of motion being isolated. Fit piping for equipment with bidirectional motion with two flexible connectors, in perpendicular planes. Install reinforced elastomer flexible connectors in accordance with manufacturer's instructions. Provide piping guides and restraints related to flexible connectors as required.

#### 3.2.6 Pipe Hangers, Inserts, and Supports

Pipe hangers, inserts, and supports must conform to MSS SP-58, except as modified herein. Do not use pipe hanger types 5, 12, and 26. Fabricate

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

hangers used to support piping 2 inches and larger to permit adequate adjustment after erection while still supporting the load. Support piping subjected to vertical movement, when operating temperatures exceed ambient temperatures, by variable spring hangers and supports or by constant support hangers.

#### 3.2.6.1 Hangers

Do not use Type 3 on insulated piping. Type 24 may be used only on trapeze hanger systems or on fabricated frames.

#### 3.2.6.2 Inserts

Secure Type 18 inserts to concrete forms before concrete is placed. Continuous inserts which allow more adjustments may be used if they otherwise meet the requirements for Type 18 inserts.

#### 3.2.6.3 C-Clamps

Torque Type 19 and 23 C-clamps in accordance with MSS SP-58 and have both locknuts and retaining devices, furnished by the manufacturer. Field-fabricated C-clamp bodies or retaining devices are not acceptable.

#### 3.2.6.4 Angle Attachments

Furnish Type 20 attachments used on angles and channels with an added malleable-iron heel plate or adapter.

#### 3.2.6.5 Saddles and Shields

Where Type 39 saddle or Type 40 shield are permitted for a particular pipe attachment application, the Type 39 saddle, connected to the pipe, must be used on all pipe 4 inches and larger when the temperature of the medium is 60 degrees F or higher. Use Type 40 shields on all piping less than 4 inches and all piping 4 inches and larger carrying medium less than 60 degrees F. Use a high-density insulation insert of cellular glass under the Type 40 shield for piping 2 inches and larger.

#### 3.2.6.6 Horizontal Pipe Supports

Space horizontal pipe supports as specified in MSS SP-58 and install a support no more than 1 foot from the pipe fitting joint at each change in direction of the piping. Space pipe supports no more than 5 feet apart at valves. Pipe hanger loads suspended from steel joist with hanger loads between panel points in excess of 50 pounds must have the excess hanger loads suspended from panel points.

#### 3.2.6.7 Vertical Pipe Supports

Support vertical pipe at each floor, except at slab-on-grade, and at intervals of not more than 15 feet not more than 8 feet from end of risers, and at vent terminations.

#### 3.2.6.8 Multiple Pipe Runs

In the support of multiple pipe runs on a common base member, use a clip or clamp where each pipe crosses the base support member. Spacing of the base support members must not exceed the hanger and support spacing required for an individual pipe in the multiple pipe run.



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 3.2.6.9 Structural Attachments

Attachment to building structure concrete and masonry must be by cast-in concrete inserts, built-in anchors, or masonry anchor devices. Inserts and anchors must be applied with a safety factor not less than 5. Do not attach supports to metal decking. Construct masonry anchors for overhead applications of ferrous materials only. Provide structural steel brackets required to support piping, headers, and equipment, but not shown, under this section. Specify material used for support under Section 05 12 00 STRUCTURAL STEEL.

### 3.2.7 Building Surface Penetrations

Do not install sleeves in structural members except where indicated or approved. Provide galvanized sheet metal sleeves in non-load bearing surfaces conforming to ASTM A653/A653M, Coating Class G-90, 20 gauge. Provide uncoated carbon steel pipe sleeves in load bearing surfaces conforming to ASTM A53/A53M, Schedule 30. Apply sealants to moisture and oil-free surfaces and elastomers to not less than 1/2 inch depth. Do not install sleeves in structural members.

#### 3.2.7.1 General Service Areas

Extend each sleeve through its respective wall, floor, or roof, and cut flush with each surface. Provide pipes passing through concrete or masonry wall or concrete floors or roofs with pipe sleeves fitted into place at the time of construction. Provide sleeves that allow a minimum of 1/4 inch all-around clearance between bare pipe and sleeves or between jacketed-insulation and sleeves. Except in pipe chases or interior walls, seal the annular space between pipe and sleeve or between jacket over-insulation and sleeve in accordance with Section 07 92 00 JOINT SEALANTS.

#### 3.2.7.2 Waterproof Penetrations

Install pipes passing through roof or floor waterproofing membrane through a 17 ounce copper sleeve, or a 0.032 inch thick aluminum sleeve, each within an integral skirt or flange. Form flashing sleeve, and extend skirt or flange greater than 8 inches from the pipe and set over the roof or floor membrane in a troweled coating of bituminous cement. Extend the flashing sleeve up the pipe a minimum of 2 inches above the roof or floor penetration. Seal the annular space between the flashing sleeve and the bare pipe or between the flashing sleeve and the metal-jacket-covered insulation as indicated. Seal penetrations by either one of the following methods.

##### 3.2.7.2.1 Waterproofing Clamping Flange

Pipes up to and including 10 inches in diameter passing through roof or floor waterproofing membrane may be installed through a cast iron sleeve with caulking recess, anchor lugs, flashing clamp device, and pressure ring with brass bolts. Clamp waterproofing membrane into place and place sealant in the caulking recess.

##### 3.2.7.2.2 Modular Mechanical Type Sealing Assembly

In lieu of a waterproofing clamping flange and caulking and sealing of annular space between pipe and sleeve or conduit and sleeve, a modular

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

mechanical type sealing assembly may be installed. Provide seals consisting of interlocking synthetic rubber links shaped to continuously fill the annular space between the pipe/conduit and sleeve with corrosion protected carbon steel bolts, nuts, and pressure plates. Loosely assemble links with bolts to form a continuous rubber belt around the pipe with a pressure plate under each bolt head and each nut. After the seal assembly is properly positioned in the sleeve, tighten the bolt to cause the rubber sealing elements to expand and provide a watertight seal between the pipe/conduit and the sleeve. Size each seal assembly as recommended by the manufacturer to fit the pipe/conduit and sleeve involved. The Contractor electing to use the modular mechanical type seals must provide sleeves of the proper diameters.

#### 3.2.7.3 Fire-Rated Penetrations

Seal penetration of fire-rated walls, partitions, and floors as specified in Section 07 84 00 FIRESTOPPING.

#### 3.2.7.4 Escutcheons

Provide escutcheons for finished surfaces where exposed piping, bare or insulated, pass through floors, walls, or ceilings, except in boiler, utility, or equipment rooms. Where sleeves project slightly from floors, use special deep-type escutcheons. Secure escutcheon to pipe or pipe covering.

#### 3.2.8 Access Panels

Provide access panels for all concealed valves, vents, controls, and items requiring inspection or maintenance. Provide access panels of sufficient size and locate so that the concealed items may be serviced and maintained or completely removed and replaced. Provide access panels as specified in Section 08 31 00 ACCESS DOORS AND PANELS.

#### 3.2.9 Field Applied Insulation

Field installed insulation is specified in Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS, except as defined differently herein.

#### 3.2.10 Field Painting

Painting required for surfaces not otherwise specified, and finish painting of items only primed at the factory are specified in Section 09 90 00 PAINTS AND COATINGS.

##### 3.2.10.1 Color Coding

Color coding for piping identification is specified in Section 09 90 00 PAINTS AND COATINGS.

##### 3.2.10.2 Color Coding Scheme

Provide a color coding scheme for locating hidden piping in accordance with Section 22 00 00 PLUMBING, GENERAL PURPOSE.

#### 3.2.11 Identification Tags

Provide identification tags made of brass, engraved laminated plastic or engraved anodized aluminum indicating service and item number on all

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

valves and dampers. Tags must be 1-3/8 inch minimum diameter and marking must be stamped or engraved. Indentations must be black for reading clarity. Attach tags to valves with No. 12 AWG copper wire, chrome-plated beaded chain or plastic straps designed for that purpose.

### 3.3 CLEANING AND ADJUSTING

Clean uncontaminated system(s) by evacuation and purging procedures currently recommended by refrigerant and refrigerant equipment manufacturers, and as specified herein, to remove small amounts of air and moisture. Systems containing moderate amounts of air, moisture, contaminated refrigerant, or any foreign matter are considered contaminated systems. Restore contaminated systems to clean condition including disassembly, component replacement, evacuation, flushing, purging, and re-charging, using currently approved refrigerant and refrigeration manufacturer's procedures. Restore contaminated systems at no additional cost to the Government as determined by the Contracting Officer. Do not use water in any procedure or test.

### 3.4 TRAINING COURSE

- a. Submit a schedule, at least 2 weeks prior to the date of the proposed training course, which identifies the date, time, and location for the training. Conduct a training course for 8 members of the operating staff as designated by the Contracting Officer. The training period must consist of a total 8 hours of normal working time and start after the system is functionally completed but prior to final acceptance tests.
- b. Cover all of the items contained in the approved operation and maintenance manuals as well as demonstrations of routine maintenance operations in the field posted instructions..
- c. Submit 6 complete copies of an operation manual in bound 8 1/2 by 11 inch booklets listing step-by-step procedures required for system startup, operation, abnormal shutdown, emergency shutdown, and normal shutdown at least 4 weeks prior to the first training course. Include the manufacturer's name, model number, and parts list in the booklets. Include the manufacturer's name, model number, service manual, and a brief description of all equipment and their basic operating features in the manuals.
- d. Submit 6 complete copies of maintenance manual in bound 8 1/2 x 11 inch booklets listing routine maintenance procedures, possible breakdowns and repairs, and a trouble shooting guide. Include piping layouts and simplified wiring and control diagrams of the system as installed in the manuals.

### 3.5 REFRIGERANT PIPING TESTS

After all components of the refrigerant system have been installed and connected, subject the entire refrigeration system to pneumatic, evacuation, and startup tests as described herein. Submit a schedule, at least 2 weeks prior to the start of related testing, for each test. Identify the proposed date, time, and location for each test. Conduct tests in the presence of the Contracting Officer. Water and electricity required for the tests will be furnished by the Government. Provide all material, equipment, instruments, and personnel required for the test. Provide the services of a qualified technician, as required, to perform

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

all tests and procedures indicated herein. Coordinate field tests with Section 23 05 93 TESTING, ADJUSTING, AND BALANCING OF HVAC SYSTEMS. Submit 6 copies of the tests report in bound 8 1/2 by 11 inch booklets documenting all phases of the tests performed. Include initial test summaries, all repairs/adjustments made, and the final test results in the report.

### 3.5.1 Preliminary Procedures

Prior to pneumatic testing, isolate equipment which has been factory tested and refrigerant charged as well as equipment which could be damaged or cause personnel injury by imposed test pressure, positive or negative, from the test pressure, or remove from the system. Remove safety relief valves and rupture discs that are not part of factory sealed systems, and cap or plug openings.

### 3.5.2 Pneumatic Test

Provide pressure control and excess pressure protection at the source of test pressure. Valves must be wide open, except those leading to the atmosphere. Test gas must be dry nitrogen, with minus 70 degree F dewpoint and less than 5 ppm oil. Apply test pressure in two stages before any refrigerant pipe is insulated or covered. In accordance with ASME B31.5, a preliminary test not to exceed 25 psi must be applied as a means of locating major leaks. Every joint being tested must be coated with a thick soap or color indicating solution. The second stage test pressure must be at least 110 percent of the design pressure, but cannot exceed 130 percent of the design pressure of any component in the system. For large systems that are not completely visible, the pressure in the system must be gradually increased to one-half of the test pressure after which the pressure must be increased in steps of one-tenth of the test pressure, until the required test pressure has been reached. The test pressure must be continuously maintained for at least 24 hours, after which it can be reduced to the leak test pressure. A correction factor of 0.3 psi will be allowed for each degree F change between test space initial and final ambient temperature, plus for increase and minus for a decrease. The leak test pressure must be the design pressure, or a pressure specified in the engineering design. To repair leaks, the joint must be taken apart, thoroughly cleaned, and reconstructed as a new joint. Joints repaired by caulking, re-melting, or back-welding/brazing are not acceptable. Following repair, the entire system must be retested using the pneumatic tests described above. Reassemble the entire system once the pneumatic tests are satisfactorily completed.

### 3.5.3 Evacuation Test

Following satisfactory completion of the pneumatic tests, relieve the pressure and evacuate the entire system to an absolute pressure of 300 micrometers. During evacuation of the system, the ambient temperature must be higher than 35 degrees F. Do not evacuate no more than one system at one time by one vacuum pump. Once the desired vacuum has been reached, close the vacuum line and allow the system to stand for 1 hour. If the pressure rises over 500 micrometers after the 1 hour period, evacuate the system again down to 300 micrometers and let set for another 1 hour period. Do not charge the system until a vacuum of at least 500 micrometers is maintained for a period of 1 hour without the assistance of a vacuum line. If during the testing the pressure rises above 500 micrometers, continue to repeat the evacuation procedures until all residual moisture has been removed. During evacuation, record pressures

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

by a thermocouple-type, electronic-type, or a calibrated-micrometer type gauge.

#### 3.5.4 System Charging and Startup Test

Following satisfactory completion of the evacuation tests, charge the system with the required amount of refrigerant by raising pressure to normal operating pressure and in accordance with manufacturer's procedures. Following charging, the system must operate with high-side and low-side pressures and corresponding refrigerant temperatures, at design or improved values. Test the entire system tested for leaks. Test fluorocarbon systems with halide torch or electronic leak detectors.

#### 3.5.5 Refrigerant Leakage

If a refrigerant leak is discovered after the system has been charged, the leaking portion of the system must be immediately isolated from the remainder of the system and the refrigerant pumped into the system receiver or other suitable container. The refrigerant must not be discharged into the atmosphere.

#### 3.5.6 Contractor's Responsibility

At all times during the installation and testing of the refrigeration system, take steps to prevent the release of refrigerants into the atmosphere. The steps must include, but not be limited to, procedures which will minimize the release of refrigerants to the atmosphere and the use of refrigerant recovery devices to remove refrigerant from the system and store the refrigerant for reuse or reclaim. At no time will the allowable leak rate exceed the leak rates allowed in Section 608 of the Clean Air Act: 30 percent of the full charge per year for industrial refrigeration, 20 percent of the full charge per year for commercial refrigeration, and 10 percent of the full charge per year for comfort cooling. Any system leaks within the first year must be repaired in accordance with the requirements herein at no cost to the Government including material, labor, and refrigerant if the leak is the result of defective equipment, material, or installation.

-- End of Section --

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

SECTION 23 30 00

HVAC AIR DISTRIBUTION  
**05/20, CHG 1: 02/22**

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AIR MOVEMENT AND CONTROL ASSOCIATION INTERNATIONAL, INC. (AMCA)

|            |                                                                              |
|------------|------------------------------------------------------------------------------|
| AMCA 201   | (2002; R 2011) Fans and Systems                                              |
| AMCA 210   | (2016) Laboratory Methods of Testing Fans for Aerodynamic Performance Rating |
| AMCA 300   | (2014) Reverberant Room Method for Sound Testing of Fans                     |
| AMCA 301   | (2014) Methods for Calculating Fan Sound Ratings from Laboratory Test Data   |
| AMCA 500-D | (2018) Laboratory Methods of Testing Dampers for Rating                      |

AIR-CONDITIONING, HEATING AND REFRIGERATION INSTITUTE (AHRI)

|              |                                                                                                                                |
|--------------|--------------------------------------------------------------------------------------------------------------------------------|
| AHRI 880 I-P | (2011) Performance Rating of Air Terminals                                                                                     |
| AHRI 885     | (2008; Addendum 2011) Procedure for Estimating Occupied Space Sound Levels in the Application of Air Terminals and Air Outlets |

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS (ASHRAE)

|                  |                                                                                                                                 |
|------------------|---------------------------------------------------------------------------------------------------------------------------------|
| ASHRAE 62.1      | (2022; Addenda E-X 2022; Errata 1-2 2022; Errata 3 2023; Interpretation 1-3 2023) Ventilation for Acceptable Indoor Air Quality |
| ASHRAE 70        | (2023) Method of Testing the Performance of Air Outlets and Inlets                                                              |
| ASHRAE 90.1 - IP | (2019) Energy Standard for Buildings Except Low-Rise Residential Buildings                                                      |

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

|            |                                                        |
|------------|--------------------------------------------------------|
| ASME A13.1 | (2020) Scheme for the Identification of Piping Systems |
|------------|--------------------------------------------------------|

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### ASTM INTERNATIONAL (ASTM)

|                 |                                                                                                                               |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------|
| ASTM A53/A53M   | (2022) Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless                         |
| ASTM A123/A123M | (2017) Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products                               |
| ASTM A924/A924M | (2022a) Standard Specification for General Requirements for Steel Sheet, Metallic-Coated by the Hot-Dip Process               |
| ASTM B117       | (2019) Standard Practice for Operating Salt Spray (Fog) Apparatus                                                             |
| ASTM B766       | (2023) Standard Specification for Electrodeposited Coatings of Cadmium                                                        |
| ASTM C553       | (2013; R 2019) Standard Specification for Mineral Fiber Blanket Thermal Insulation for Commercial and Industrial Applications |
| ASTM D520       | (2000; R 2011) Zinc Dust Pigment                                                                                              |
| ASTM D1654      | (2008; R 2016; E 2017) Standard Test Method for Evaluation of Painted or Coated Specimens Subjected to Corrosive Environments |
| ASTM D1785      | (2015; E 2018) Standard Specification for Poly(Vinyl Chloride) (PVC), Plastic Pipe, Schedules 40, 80, and 120                 |
| ASTM D2466      | (2017) Standard Specification for Poly(Vinyl Chloride) (PVC) Plastic Pipe Fittings, Schedule 40                               |
| ASTM D2564      | (2020) Standard Specification for Solvent Cements for Poly(Vinyl Chloride) (PVC) Plastic Piping Systems                       |
| ASTM D2855      | (2015) Standard Practice for Making Solvent-Cemented Joints with Poly(Vinyl Chloride) (PVC) Pipe and Fittings                 |
| ASTM D3359      | (2017) Standard Test Methods for Rating Adhesion by Tape Test                                                                 |
| ASTM E2016      | (2022) Standard Specification for Industrial Woven Wire Cloth                                                                 |

#### CALIFORNIA DEPARTMENT OF PUBLIC HEALTH (CDPH)

|                    |                                                                                                                                                            |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CDPH SECTION 01350 | (2017; Version 1.2) Standard Method for the Testing and Evaluation of Volatile Organic Chemical Emissions from Indoor Sources using Environmental Chambers |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

|          |                                                                                                |
|----------|------------------------------------------------------------------------------------------------|
| NFPA 70  | (2023; ERTA 4 2023) National Electrical Code                                                   |
| NFPA 90A | (2024) Standard for the Installation of Air Conditioning and Ventilating Systems               |
| NFPA 96  | (2024) Standard for Ventilation Control and Fire Protection of Commercial Cooking Operations   |
| NFPA 701 | (2023; ERTA 1 2023) Standard Methods of Fire Tests for Flame Propagation of Textiles and Films |

#### SHEET METAL AND AIR CONDITIONING CONTRACTORS' NATIONAL ASSOCIATION (SMACNA)

|                |                                                                                          |
|----------------|------------------------------------------------------------------------------------------|
| SMACNA 1819    | (2002) Fire, Smoke and Radiation Damper Installation Guide for HVAC Systems, 5th Edition |
| SMACNA 1966    | (2020) HVAC Duct Construction Standards Metal and Flexible, 4th Edition                  |
| SMACNA 1972 CD | (2012) HVAC Air Duct Leakage Test Manual - 2nd Edition                                   |

#### SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT (SCAQMD)

|                  |                                          |
|------------------|------------------------------------------|
| SCAQMD Rule 1168 | (2017) Adhesive and Sealant Applications |
|------------------|------------------------------------------|

#### U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

|           |                                   |
|-----------|-----------------------------------|
| 40 CFR 82 | Protection of Stratospheric Ozone |
|-----------|-----------------------------------|

#### UNDERWRITERS LABORATORIES (UL)

|                |                                                                                                    |
|----------------|----------------------------------------------------------------------------------------------------|
| UL 6           | (2022) UL Standard for Safety Electrical Rigid Metal Conduit-Steel                                 |
| UL 181         | (2013; Reprint Dec 2021) UL Standard for Safety Factory-Made Air Ducts and Air Connectors          |
| UL 555         | (2006; Reprint Aug 2016) UL Standard for Safety Fire Dampers                                       |
| UL 705         | (2017; Reprint Aug 2022) UL Standard for Safety Power Ventilators                                  |
| UL 2021        | (2015; Reprint Dec 2016) UL Standard for Safety Fixed and Location-Dedicated Electric Room Heaters |
| UL Bld Mat Dir | (updated continuously online) Building Materials Directory                                         |



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

UL Electrical Construction                      (2012) Electrical Construction Equipment Directory

UL Fire Resistance                                (2014) Fire Resistance Directory

## 1.2 SYSTEM DESCRIPTION

Furnish ductwork, piping offsets, fittings, and accessories as required to provide a complete installation. Coordinate the work of the different trades to avoid interference between piping, equipment, structural, and electrical work. Provide complete, in place, all necessary offsets in piping and ductwork, and all fittings, and other components, required to install the work as indicated and specified.

### 1.2.1 Mechanical Equipment Identification

The number of charts and diagrams must be equal to or greater than the number of mechanical equipment rooms. Where more than one chart or diagram per space is required, mount these in edge pivoted, swinging leaf, extruded aluminum frame holders which open to 170 degrees.

#### 1.2.1.1 Charts

Provide chart listing of equipment by designation numbers and capacities such as flow rates, pressure and temperature differences, heating and cooling capacities, horsepower, pipe sizes, and voltage and current characteristics.

#### 1.2.1.2 Diagrams

Submit proposed diagrams, at least 2 weeks prior to start of related testing. provide neat mechanical drawings provided with extruded aluminum frame under 1/8-inch glass or laminated plastic, system diagrams that show the layout of equipment, piping, and ductwork, and typed condensed operation manuals explaining preventative maintenance procedures, methods of checking the system for normal, safe operation, and procedures for safely starting and stopping the system. After approval, post these items where directed.

### 1.2.2 Service Labeling

Label equipment, including fans, air handlers, terminal units, etc. with labels made of self-sticking, plastic film designed for permanent installation. Provide labels consistent with the contract documents.

Identify similar services with different temperatures or pressures. Where pressures could exceed 125 pounds per square inch, gage, include the maximum system pressure in the label. Label and arrow piping in accordance with the following:

- a. Each point of entry and exit of pipe passing through walls.
- b. Each change in direction, i.e., elbows, tees.
- c. In congested or hidden areas and at all access panels at each point required to clarify service or indicated hazard.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

- d. In long straight runs, locate labels at distances within eyesight of each other not to exceed 75 feet. All labels must be visible and legible from the primary service and operating area.

| For Bare or Insulated Pipes |            |
|-----------------------------|------------|
| for Outside Diameters of    | Lettering  |
| 1/2 thru 1-3/8 inch         | 1/2 inch   |
| 1-1/2 thru 2-3/8 inch       | 3/4 inch   |
| 2-1/2 inch and larger       | 1-1/4 inch |

### 1.2.3 Color Coding

Color coding of all piping systems must be in accordance with ASME A13.1 .

### 1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

#### SD-02 Shop Drawings

Detail Drawings; G

#### SD-03 Product Data

Insulated Nonmetallic Flexible Duct Runouts

Duct Connectors

Duct Access Doors; G

Fire Dampers

Manual Balancing Dampers; G

Diffusers

Registers and Grilles

Louvers

Centrifugal Type Power Roof Ventilators

Variable Volume, Single Duct Terminal Units; G

Prefabricated Radiant-Heating Electric Panels; G

Test Procedures

Diagrams; G

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

Indoor Air Quality for Duct Sealants; S

SD-06 Test Reports

Performance Tests; FIO

Damper Acceptance Test; FIO

SD-07 Certificates

Bolts

Ozone Depleting Substances Technician Certification

SD-08 Manufacturer's Instructions

Manufacturer's Installation Instructions

Operation and Maintenance Training

SD-10 Operation and Maintenance Data

Operation and Maintenance Manuals; G

Fire Dampers; G

Manual Balancing Dampers; G

Centrifugal Type Power Roof Ventilators; G

Variable Volume, Single Duct Terminal Units; G

Prefabricated Radiant-Heating Electric Panels; G

SD-11 Closeout Submittals

Indoor Air Quality During Construction; S

1.4 QUALITY ASSURANCE

Except as otherwise specified, approval of materials and equipment is based on manufacturer's published data.

- a. Where materials and equipment are specified to conform to the standards of the Underwriters Laboratories, the label of or listing with reexamination in UL Bld Mat Dir, and UL 6 is acceptable as sufficient evidence that the items conform to Underwriters Laboratories requirements. In lieu of such label or listing, submit a written certificate from any nationally recognized testing agency, adequately equipped and competent to perform such services, stating that the items have been tested and that the units conform to the specified requirements. Outline methods of testing used by the specified agencies.
- b. Where materials or equipment are specified to be constructed or tested, or both, in accordance with the standards of the ASTM International (ASTM), the ASME International (ASME), or other standards, a manufacturer's certificate of compliance of each item is

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

acceptable as proof of compliance.

- c. Conformance to such agency requirements does not relieve the item from compliance with other requirements of these specifications.
- d. Where products are specified to meet or exceed the specified energy efficiency requirement of FEMP-designated or ENERGY STAR covered product categories, equipment selected must have as a minimum the efficiency rating identified under "Energy-Efficient Products" at <http://femp.energy.gov/procurement>.

#### 1.4.1 Prevention of Corrosion

Protect metallic materials against corrosion. Provide rust-inhibiting treatment and standard finish for the equipment enclosures. Do not use aluminum in contact with earth, and where connected to dissimilar metal. Protect aluminum by approved fittings, barrier material, or treatment. Provide hot-dip galvanized ferrous parts such as anchors, bolts, braces, boxes, bodies, clamps, fittings, guards, nuts, pins, rods, shims, thimbles, washers, and miscellaneous parts not of corrosion-resistant steel or nonferrous materials in accordance with ASTM A123/A123M for exterior locations and cadmium-plated in conformance with ASTM B766 for interior locations.

#### 1.4.2 Asbestos Prohibition

Do not use asbestos and asbestos-containing products.

#### 1.4.3 Ozone Depleting Substances Technician Certification

All technicians working on equipment that contain ozone depleting refrigerants must be certified as a Section 608 Technician to meet requirements in 40 CFR 82, Subpart F. Provide copies of technician certifications to the Contracting Officer at least 14 calendar days prior to work on any equipment containing these refrigerants.

#### 1.4.4 Detail Drawings

Submit detail drawings showing equipment layout, including assembly and installation details and electrical connection diagrams; ductwork layout showing the location of all supports and hangers, typical hanger details, gauge reinforcement, reinforcement spacing rigidity classification, and static pressure and seal classifications. Include any information required to demonstrate that the system has been coordinated and functions properly as a unit on the drawings and show equipment relationship to other parts of the work, including clearances required for operation and maintenance. Submit drawings showing bolt-setting information, and foundation bolts prior to concrete foundation construction for all equipment indicated or required to have concrete foundations. Submit function designation of the equipment and any other requirements specified throughout this Section with the shop drawings.

#### 1.4.5 Test Procedures

Conduct performance tests as required in Section 23 05 93 Testing, Adjusting and Balancing for HVAC and Section 23 09 00 Instrumentation and Control for HVAC.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 1.5 DELIVERY, STORAGE, AND HANDLING

Protect stored equipment at the jobsite from the weather, humidity and temperature variations, dirt and dust, or other contaminants. Additionally, cap or plug all pipes until installed.

## PART 2 PRODUCTS

### 2.1 STANDARD PRODUCTS

Provide components and equipment that are "standard products" of a manufacturer regularly engaged in the manufacturing of products that are of a similar material, design and workmanship. "Standard products" is defined as being in satisfactory commercial or industrial use for 2 years before bid opening, including applications of components and equipment under similar circumstances and of similar size, satisfactorily completed by a product that is sold on the commercial market through advertisements, manufacturers' catalogs, or brochures. Products having less than a 2-year field service record are acceptable if a certified record of satisfactory field operation, for not less than 6000 hours exclusive of the manufacturer's factory tests, can be shown. Provide equipment items that are supported by a service organization. In product categories covered by ENERGY STAR or the Federal Energy Management Program, provide equipment that is listed on the ENERGY STAR Qualified Products List or that meets or exceeds the FEMP-designated Efficiency Requirements.

### 2.2 IDENTIFICATION PLATES

In addition to standard manufacturer's identification plates, provide engraved laminated phenolic identification plates for each piece of mechanical equipment. Identification plates are to designate the function of the equipment. Submit designation with the shop drawings. Provide identification plates that are layers, black-white-black, engraved to show white letters on black background. Letters must be upper case. Identification plates that are 1-1/2-inches high and smaller must be 1/16-inch thick, with engraved lettering 1/8-inch high; identification plates larger than 1-1/2-inches high must be 1/8-inch thick, with engraved lettering of suitable height. Identification plates 1-1/2-inches high and larger must have beveled edges. Install identification plates using a compatible adhesive.

### 2.3 EQUIPMENT GUARDS AND ACCESS

Fully enclose or guard belts, pulleys, chains, gears, couplings, projecting setscrews, keys, and other rotating parts exposed to personnel contact according to OSHA requirements. Properly guard or cover with insulation of a type specified, high temperature equipment and piping exposed to contact by personnel or where it creates a potential fire hazard. The requirements for catwalks, operating platforms, ladders, and guardrails are specified in Section 08 31 00 ACCESS DOORS AND PANELS.

### 2.4 ELECTRICAL WORK

- a. Provide motors, controllers, integral disconnects, contactors, and controls with their respective pieces of equipment, except controllers indicated as part of motor control centers. Provide electrical equipment, including motors and wiring, as specified in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Provide manual or automatic control and protective or signal devices required for the operation

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

specified and control wiring required for controls and devices specified, but not shown. For packaged equipment, include manufacturer provided controllers with the required monitors and timed restart.

## 2.5 ANCHOR BOLTS

Provide anchor bolts for equipment placed on concrete equipment pads or on concrete slabs. Bolts to be of the size and number recommended by the equipment manufacturer and located by means of suitable templates. Installation of anchor bolts must not degrade the surrounding concrete.

## 2.6 PAINTING

Paint equipment units in accordance with approved equipment manufacturer's standards unless specified otherwise. Field retouch only if approved. Otherwise, return equipment to the factory for refinishing. Paint in accordance with Section 09 96 00 HIGH-PERFORMANCE COATINGS.

## 2.7 INDOOR AIR QUALITY

Provide equipment and components that comply with the requirements of ASHRAE 62.1 unless more stringent requirements are specified herein.

## 2.8 DUCT SYSTEMS

### 2.8.1 Metal Ductwork

Provide metal ductwork construction, including all fittings and components, that complies with SMACNA 1966, as supplemented and modified by this specification .

- b. Provide radius type elbows with a centerline radius of 1.5 times the width or diameter of the duct where space permits. Otherwise, elbows having a minimum radius equal to the width or diameter of the duct or square elbows with factory fabricated turning vanes are allowed.
- c. Provide ductwork that meets the requirements of Seal Class A. Provide ductwork in VAV systems upstream of the VAV boxes that meets the requirements of Seal Class A.
- e. Provide sealants that conform to fire hazard classification specified in Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS and are suitable for the range of air distribution and ambient temperatures to which it is exposed. Do not use pressure sensitive tape as a sealant. Provide duct sealant products that meet either emissions requirements of CDPH SECTION 01350 (limit requirements for either office or classroom spaces regardless of space type) or VOC content requirements of SCAQMD Rule 1168 (HVAC duct sealants are classified as "Other" within the SCAQMD Rule 1168 sealants table). Provide validation of indoor air quality for duct sealants.
- f. Make spiral lock seam duct, and flat oval with duct sealant and lock with not less than 3 equally spaced drive screws or other approved methods indicated in SMACNA 1966. Apply the sealant to the exposed

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

male part of the fitting collar so that the sealer is on the inside of the joint and fully protected by the metal of the duct fitting. Apply one brush coat of the sealant over the outside of the joint to at least 2 inch band width covering all screw heads and joint gap. Dents in the male portion of the slip fitting collar are not acceptable.

- g. Fabricate outdoor air intake ducts and plenums with watertight soldered or brazed joints and seams.

#### 2.8.1.1 Insulated Nonmetallic Flexible Duct Runouts

Use flexible duct runouts only where indicated. Runout length is indicated on the drawings, and is not to exceed 5 feet. Provide runouts that are preinsulated, factory fabricated, and that comply with NFPA 90A and UL 181. Provide either field or factory applied vapor barrier. Provide not less than 20 ounce glass fabric duct connectors coated on both sides with neoprene. Where coil induction or high velocity units are supplied with vertical air inlets, use a streamlined, vaned and mitered elbow transition piece for connection to the flexible duct or hose. Provide a die-stamped elbow and not a flexible connector as the last elbow to these units other than the vertical air inlet type. Insulated flexible connectors are allowed as runouts. Provide insulated material and vapor barrier that conform to the requirements of Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS. Do not expose the insulation material surface to the air stream.

#### 2.8.1.2 General Service Duct Connectors

Provide a flexible duct connector approximately 6 inches in width where sheet metal connections are made to fans or where ducts of dissimilar metals are connected. For round/oval ducts, secure the flexible material by stainless steel or zinc-coated, iron clinch-type draw bands. For rectangular ducts, install the flexible material locked to metal collars using normal duct construction methods. Provide a composite connector system that complies with NFPA 701 and is classified as "flame-retardent fabrics" in UL Bld Mat Dir.

#### 2.8.2 Duct Access Doors

Provide hinged access doors conforming to SMACNA 1966 in ductwork and plenums where indicated and at all air flow measuring primaries, automatic dampers, fire dampers, coils, thermostats, and other apparatus requiring service and inspection in the duct system. Provide access doors upstream and downstream of air flow measuring primaries and heating and cooling coils. Provide doors that are a minimum 15 by 18 inches, unless otherwise shown. Where duct size does not accommodate this size door, make the doors as large as practicable. Equip doors 24 by 24 inches or larger with fasteners operable from inside and outside the duct. Use insulated type doors in insulated ducts.

#### 2.8.3 Fire Dampers

Use 1.5 hour rated fire dampers unless otherwise indicated. Provide fire dampers that conform to the requirements of NFPA 90A and UL 555. Perform the fire damper test as outlined in NFPA 90A. Provide a pressure relief door upstream of the fire damper. If the ductwork connected to the fire damper is to be insulated then provide a factory installed pressure relief damper. Provide automatic operating fire dampers with a dynamic rating suitable for the maximum air velocity and pressure differential to which

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

it is subjected. Provide fire dampers approved for the specific application, and install according to their listing. Equip fire dampers with a steel sleeve or adequately sized frame installed in such a manner that disruption of the attached ductwork, if any, does not impair the operation of the damper. Equip sleeves or frames with perimeter mounting angles attached on both sides of the wall or floor opening. Construct ductwork in fire-rated floor-ceiling or roof-ceiling assembly systems with air ducts that pierce the ceiling of the assemblies in conformance with UL Fire Resistance. Provide curtain type with damper blades out of the air stream fire dampers. Install dampers that do not reduce the duct or the air transfer opening cross-sectional area. Install dampers so that the centerline of the damper depth or thickness is located in the centerline of the wall, partition or floor slab depth or thickness. Unless otherwise indicated, comply with the installation details given in SMACNA 1819 and in manufacturer's instructions for fire dampers. Perform acceptance testing of fire dampers according to paragraph DAMPER ACCEPTANCE TEST and NFPA 90A.

#### 2.8.4 Manual Balancing Dampers

Furnish manual balancing dampers with accessible operating mechanisms. Use chromium plated operators (with all exposed edges rounded) in finished portions of the building. Provide manual volume control dampers that are operated by locking-type quadrant operators. Install dampers that are 2 gauges heavier than the duct in which installed. Unless otherwise indicated, provide opposed blade type multileaf dampers with maximum blade width of 12 inches. Provide access doors or panels for all concealed damper operators and locking setscrews. Provide stand-off mounting brackets, bases, or adapters not less than the thickness of the insulation when the locking-type quadrant operators for dampers are installed on ducts to be thermally insulated, to provide clearance between the duct surface and the operator. Provide stand-off mounting items that are integral with the operator or standard accessory of the damper manufacturer.

#### 2.8.5 Air Supply And Exhaust Air Dampers

Provide outdoor air supply and exhaust air dampers that have a maximum leakage rate when tested in accordance with AMCA 500-D as required by ASHRAE 90.1 - IP, including maximum Damper Leakage for:

- a. Climate Zones 1,2,6,7,8 the maximum damper leakage at 1.0 inch w.g. for motorized dampers is 4 cfm per square foot of damper area and non-motorized dampers are not allowed.
- b. All other Climate Zones the maximum damper leakage at 1.0 inch w.g. is 10 cfm per square foot and for non-motorized dampers is 20 cfm per square foot of damper area.

Dampers smaller than 24 inches in either direction may have leakage of 40 cfm per square foot.

#### 2.8.6 Air Deflectors (Volume Extractors) and Branch Connections

Provide air deflectors (volume extractors) at all duct mounted supply outlets, at takeoff or extension collars to supply outlets, at duct branch takeoff connections, and at 90 degree elbows, as well as at locations as indicated on the drawings or otherwise specified. Conical branch connections or 45 degree entry connections are allowed in lieu of



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

deflectors for branch connections. Furnish all air deflectors (volume extractors), except those installed in 90 degree elbows, with an approved means of adjustment. Provide easily accessible means for adjustment inside the duct or from an adjustment with sturdy lock on the face of the duct. When installed on ducts to be thermally insulated, provide external adjustments with stand-off mounting brackets, integral with the adjustment device, to provide clearance between the duct surface and the adjustment device not less than the thickness of the thermal insulation. Provide factory-fabricated air deflectors consisting of curved turning vanes or louver blades designed to provide uniform air distribution and change of direction with minimum turbulence or pressure loss. Provide factory or field assembled air deflectors (volume extractors). Make adjustment from the face of the diffuser or by position adjustment and lock external to the duct. Provide stand-off brackets on insulated ducts as described herein. Provide fixed air deflectors (volume extractors), also called turning vanes, in 90 degree elbows.

#### 2.8.7 Diffusers, Registers, and Grilles

Provide factory-fabricated units of steel or aluminum that distribute the specified quantity of air evenly over space intended without causing noticeable drafts, air movement faster than 50 fpm in occupied zone, or dead spots anywhere in the conditioned area. Provide outlets for diffusion, spread, throw, and noise level as required for specified performance. Certify performance according to ASHRAE 70. Provide sound rated and certified inlets and outlets according to ASHRAE 70. Provide sound power level as indicated. Provide diffusers and registers with volume damper with accessible operator, unless otherwise indicated; or if standard with the manufacturer, an automatically controlled device is acceptable. Provide opposed blade type volume dampers for all diffusers and registers, except linear slot diffusers. Provide linear slot diffusers with round or elliptical balancing dampers. Where the inlet and outlet openings are located less than 7 feet above the floor, protect them by a grille or screen according to NFPA 90A.

##### 2.8.7.1 Diffusers

Provide diffuser types indicated. Furnish ceiling mounted units with anti-smudge devices, unless the diffuser unit minimizes ceiling smudging through design features. Provide diffusers with air deflectors of the type indicated. Provide air handling troffers or combination light and ceiling diffusers conforming to the requirements of UL Electrical Construction for the interchangeable use as cooled or heated air supply diffusers or return air units. Install ceiling mounted units with rims tight against ceiling. Provide sponge rubber gaskets between ceiling and surface mounted diffusers for air leakage control. Provide suitable trim for flush mounted diffusers. For connecting the duct to diffuser, provide duct collar that is airtight and does not interfere with volume controller. Provide return or exhaust units that are similar to supply diffusers.

##### 2.8.7.2 Registers and Grilles

Provide units that are four-way directional-control type, except provide return and exhaust registers that are fixed horizontal or vertical louver type similar in appearance to the supply register face. Furnish registers with sponge-rubber gasket between flanges and wall or ceiling. Install wall supply registers at least 6 inches below the ceiling unless otherwise indicated. Locate return and exhaust registers 6 inches above the floor

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

unless otherwise indicated. Achieve four-way directional control by a grille face which can be rotated in 4 positions or by adjustment of horizontal and vertical vanes. Provide grilles as specified for registers, without volume control damper.

#### 2.8.8 Louvers

Provide louvers for installation in exterior walls that are associated with the air supply and distribution system as specified in Section 08 91 00 METAL WALL LOUVERS.

#### 2.8.9 Bird Screens and Frames

Provide bird screens that conform to ASTM E2016, No. 2 mesh, aluminum or stainless steel. Provide "medium-light" rated aluminum screens. Provide "light" rated stainless steel screens. Provide removable type frames fabricated from either stainless steel or extruded aluminum.

#### 2.8.10 Radon Exhaust Ductwork

Fabricate radon exhaust ductwork installed in or beneath slabs from Schedule 40 PVC pipe that conforms to ASTM D1785. Provide fittings that conform to ASTM D2466. Use solvent cement conforming to ASTM D2564 to make joints. Otherwise provide metal radon exhaust ductwork as specified herein.

### 2.9 AIR SYSTEMS EQUIPMENT 2.9.1 Fans

Test and rate fans according to AMCA 210. Calculate system effect on air moving devices in accordance with AMCA 201 where installed ductwork differs from that indicated on drawings. Install air moving devices to minimize fan system effect. Where system effect is unavoidable, determine the most effective way to accommodate the inefficiencies caused by system effect on the installed air moving device. The sound power level of the fans must not exceed 85 dBA when tested according to AMCA 300 and rated in accordance with AMCA 301. Provide all fans with an AMCA seal. Connect fans to the motors directly. Select each fan to produce the capacity required at the fan static pressure indicated. Provide sound power level as indicated. Obtain the sound power level values according to AMCA 300. Provide standard AMCA arrangement, rotation, and discharge as indicated. Provide power ventilators that conform to UL 705 and have a UL label.

#### 2.9.1.1 Centrifugal Type Power Roof Ventilators

Provide direct driven centrifugal type fans with backward inclined, non-overloading wheel. Provide hinged or removable and weatherproof motor compartment housing, constructed of heavy gauge aluminum. Provide fans with disconnect switch, motorized dampers, roof curb. Provide dripproof type motor enclosure. Provide centrifugal type kitchen exhaust fans according to UL 705 and NFPA 96, fitted with a round hood, and windband upblast discharge configuration, integral residue trough and collection device, with motor and power transmission components located in outside positively air ventilated compartment. Use only lubricated bearings. If there is a conflict between NFPA 96 and UL 705 the most stringent wording must be adhered to.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## 2.10 TERMINAL UNITS

### 2.10.1 Variable Air Volume (VAV) Terminal Units

- a. Provide VAV terminal units that are the type, size, and capacity shown, mounted in the ceiling that are suitable for single system applications. Provide actuators and controls as specified in paragraph SUPPLEMENTAL COMPONENTS/SERVICES, subparagraph CONTROLS.
- b. Provide unit enclosures that are constructed of galvanized steel not lighter than 22 gauge or aluminum sheet not lighter than 18 gauge. Provide single or multiple discharge outlets as required. Units with flow limiters are not acceptable. Provide unit air volume that is factory preset and readily field adjustable without special tools. Provide reheat coils as indicated.
- c. Attach a flow chart to each unit. Base acoustic performance of the terminal units upon units tested according to AHRI 880 I-P with the calculations prepared in accordance with AHRI 885. Provide sound power level as indicated. Show discharge sound power for minimum and 1-1/2 inches water gauge inlet static pressure. Provide acoustical lining according to NFPA 90A.

#### 2.10.1.1 Variable Volume, Single Duct Terminal Units

Provide variable volume, single duct, terminal units with a calibrated air volume sensing device, air valve or damper, actuator, and accessory relays. Provide units that control air volume to within plus or minus 5 percent of each air set point volume as determined by the thermostat with variations in inlet pressures from 3/4 to 6 inch water gauge. Provide units with an internal resistance not exceeding 0.4 inch water gauge at maximum flow range. Provide external differential pressure taps separate from the control pressure taps for air flow measurement with a 0 to 1 inch water gauge range.

#### 2.10.1.2 Reheat Units

##### 2.10.1.2.1 Electric Resistance Heaters

Provide the duct-mounting type electric resistance heaters consisting of a nickel-chromium resistor mounted on refractory material and a steel or aluminum frame for attachment to ductwork. Provide electric duct heater that meets the requirement of Underwriters Laboratories and NFPA 70 and is provided with a built-in or surface-mounted high-limit thermostat. Interlock electric duct heaters electrically so that they cannot be energized unless the fan is running.

## 2.11 FACTORY PAINTING

Factory paint new equipment, which are not of galvanized construction. Paint with a corrosion resisting paint finish according to ASTM A123/A123M or ASTM A924/A924M. Clean, phosphatize and coat internal and external ferrous metal surfaces with a paint finish which has been tested according to ASTM B117, ASTM D1654, and ASTM D3359. Submit evidence of satisfactory paint performance for a minimum of 125 hours for units to be installed indoors and 500 hours for units to be installed outdoors. Provide rating of failure at the scribe mark that is not less than 6, average creepage not greater than 1/8 inch. Provide rating of the inscribed area that is not less than 10, no failure. On units constructed of galvanized steel

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

that have been welded, provide a final shop docket of zinc-rich protective paint on exterior surfaces of welds or welds that have burned through from the interior according to ASTM D520 Type I.

Field paint factory painting that has been damaged prior to acceptance by the Contracting Officer in compliance with the requirements of paragraph FIELD PAINTING OF MECHANICAL EQUIPMENT.

## 2.12 SUPPLEMENTAL COMPONENTS/SERVICES

### 2.12.1 Insulation

The requirements for shop and field applied insulation are specified in Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS.

### 2.12.2 Controls

The requirements for controls are specified in Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC.

## 2.13 RADIANT PANELS

### 2.13.1 Prefabricated Radiant-Heating Electric Panels

#### 2.13.1.1 Description

Sheet metal enclosed panel with heating element suitable for recessed mounting. Comply with UL 2021

#### 2.13.1.2 Panel

Minimum 0.027 inch thick, galvanized steel sheet back panel riveted to minimum 0.040 inch thick, galvanized steel sheet front panel with fused-on crystalline surface.

#### 2.13.1.3 Heating Element

Powdered graphite sandwiched between sheets of electric insulation.

#### 2.13.1.4 Electrical Connections

Nonheating, high-temperature, insulated-copper leads, factory connected to heating element.

#### 2.13.1.5 Wall Thermostat

Bimetal, sensing elements; with contacts suitable for low -voltage circuit, and manually operated on-off switch with contactors, relays, and control transformers.

## PART 3 EXECUTION

### 3.1 EXAMINATION

After becoming familiar with all details of the work, verify all dimensions in the field, and advise the Contracting Officer of any discrepancy before performing the work.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 3.2 INSTALLATION

- a. Install materials and equipment in accordance with the requirements of the contract drawings and approved manufacturer's installation instructions. Accomplish installation by workers skilled in this type of work. Perform installation so that there is no degradation of the designed fire ratings of walls, partitions, ceilings, and floors.
- b. No installation is permitted to block or otherwise impede access to any existing machine or system. Install all hinged doors to swing open a minimum of 120 degrees. Provide an area in front of all access doors that clears a minimum of 3 feet. In front of all access doors to electrical circuits, clear the area the minimum distance to energized circuits as specified in OSHA Standards, part 1910.333 (Electrical-Safety Related work practices) and an additional 3 feet.
- c. Except as otherwise indicated, install emergency switches and alarms in conspicuous locations. Mount all indicators, to include gauges, meters, and alarms in order to be easily visible by people in the area.

#### 3.2.1 Access Panels

Install access panels for concealed valves, vents, controls, dampers, and items requiring inspection or maintenance of sufficient size, and locate them so that the concealed items are easily serviced and maintained or completely removed and replaced. Provide access panels as specified in Section 08 31 00 ACCESS DOORS AND PANELS.

#### 3.2.2 Flexible Duct

Install pre-insulated flexible duct in accordance with the latest printed instructions of the manufacturer to ensure a vapor tight joint. Provide hangers, when required to suspend the duct, of the type recommended by the duct manufacturer and set at the intervals recommended.

#### 3.2.3 Metal Ductwork

Install according to SMACNA 1966 unless otherwise indicated. Install duct supports for sheet metal ductwork according to SMACNA 1966, unless otherwise specified. Do not use friction beam clamps indicated in SMACNA 1966. Anchor risers on high velocity ducts in the center of the vertical run to allow ends of riser to move due to thermal expansion. Erect supports on the risers that allow free vertical movement of the duct. Attach supports only to structural framing members and concrete slabs. Do not anchor supports to metal decking unless a means is provided and approved for preventing the anchor from puncturing the metal decking. Where supports are required between structural framing members, provide suitable intermediate metal framing. Where C-clamps are used, provide retainer clips.

##### 3.2.3.1 Radon Exhaust Ductwork

Perforate subslab suction piping where indicated. Install PVC joints as specified in ASTM D2855.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 3.2.4 Kitchen Exhaust Ductwork

#### 3.2.4.1 Ducts Conveying Smoke and Grease Laden Vapors

Provide ducts conveying smoke and grease laden vapors that conform to requirements of NFPA 96. Make seams, joints, penetrations, and duct-to-hood collar connections with a liquid tight continuous external weld. Provide duct material that is a minimum 18 gauge, Type 304L or 316L, stainless steel.

#### 3.2.4.2 Exposed Ductwork

Provide exposed ductwork that is fabricated from minimum 18 gauge, Type 304L or 316L, stainless steel with continuously welded joints and seams. Pitch ducts to drain at hoods and low points indicated. Match surface finish to hoods.

### 3.2.5 Dust Control

To prevent the accumulation of dust, debris and foreign material during construction, perform temporary dust control protection. Protect the distribution system (supply and return) with temporary seal-offs at all inlets and outlets at the end of each day's work. Keep temporary protection in place until system is ready for startup.

### 3.2.6 Insulation

Provide thickness and application of insulation materials for ductwork, piping, and equipment according to Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS.

### 3.2.7 Duct Test Holes

Provide holes with closures or threaded holes with plugs in ducts and plenums as indicated or where necessary for the use of pitot tube in balancing the air system. Plug insulated duct at the duct surface, patched over with insulation and then marked to indicate location of test hole if needed for future use.

## 3.3 CUTTING AND PATCHING

Install work in such a manner and at such time that a minimum of cutting and patching of the building structure is required. Make holes in exposed locations, in or through existing floors, by drilling and smooth by sanding. Use of a jackhammer is permitted only where specifically approved. Make holes through masonry walls to accommodate sleeves with an iron pipe masonry core saw.

## 3.4 CLEANING

Thoroughly clean surfaces of piping and equipment that have become covered with dirt, plaster, or other material during handling and construction before such surfaces are prepared for final finish painting or are enclosed within the building structure. Before final acceptance, clean mechanical equipment, including piping, ducting, and fixtures, and free from dirt, grease, and finger marks. When the work area is in an occupied space such as office, laboratory or warehouse protect all furniture and equipment from dirt and debris. Incorporate housekeeping for field construction work which leaves all furniture and equipment in the affected

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

area free of construction generated dust and debris; and, all floor surfaces vacuum-swept clean.

### 3.5 PENETRATIONS

Provide sleeves and prepared openings for duct mains, branches, and other penetrating items, and install during the construction of the surface to be penetrated. Cut sleeves flush with each surface. Place sleeves for round duct 15 inches and smaller. Build framed, prepared openings for round duct larger than 15 inches and square, rectangular or oval ducts. Sleeves and framed openings are also required where grilles, registers, and diffusers are installed at the openings. Provide one inch clearance between penetrating and penetrated surfaces except at grilles, registers, and diffusers. Pack spaces between sleeve or opening and duct or duct insulation with mineral fiber conforming with ASTM C553, Type 1, Class B-2.

#### 3.5.1 Sleeves

Fabricate sleeves, except as otherwise specified or indicated, from 20 gauge thick mill galvanized sheet metal. Where sleeves are installed in bearing walls or partitions, provide black steel pipe conforming with ASTM A53/A53M, Schedule 20.

#### 3.5.2 Framed Prepared Openings

Fabricate framed prepared openings from 20 gauge galvanized steel, unless otherwise indicated.

#### 3.5.3 Insulation

Provide duct insulation in accordance with Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS continuous through sleeves and prepared openings except firewall penetrations. Terminate duct insulation at fire dampers and flexible connections. For duct handling air at or below 60 degrees F, provide insulation continuous over the damper collar and retaining angle of fire dampers, which are exposed to unconditioned air.

#### 3.5.4 Closure Collars

Provide closure collars of a minimum 4 inches wide, unless otherwise indicated, for exposed ducts and items on each side of penetrated surface, except where equipment is installed. Install collar tight against the surface and fit snugly around the duct or insulation. Grind sharp edges smooth to prevent damage to penetrating surface. Fabricate collars for round ducts 15 inches in diameter or less from 20 gauge galvanized steel. Fabricate collars for square and rectangular ducts, or round ducts with minimum dimension over 15 inches from 18 gauge galvanized steel. Fabricate collars for square and rectangular ducts with a maximum side of 15 inches or less from 20 gauge galvanized steel. Install collars with fasteners a maximum of 6 inches on center. Attach to collars a minimum of 4 fasteners where the opening is 12 inches in diameter or less, and a minimum of 8 fasteners where the opening is 20 inches in diameter or less.

#### 3.5.5 Firestopping

Where ducts pass through fire-rated walls, fire partitions, and fire rated chase walls, seal the penetration with fire stopping materials as specified in Section 07 84 00 FIRESTOPPING.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 3.6 IDENTIFICATION SYSTEMS

Provide identification tags made of brass, engraved laminated plastic, or engraved anodized aluminum, indicating service and item number on all valves and dampers. Provide tags that are 1-3/8 inch minimum diameter with stamped or engraved markings. Make indentations black for reading clarity. Attach tags to valves with No. 12 AWG 0.0808-inch diameter corrosion-resistant steel wire, copper wire, chrome-plated beaded chain or plastic straps designed for that purpose.

### 3.7 DUCTWORK LEAK TEST

Perform ductwork leak test for the entire air distribution and exhaust system, including fans, coils, filters, etc. Provide test procedure, apparatus, and report that conform to SMACNA 1972 CD. The maximum allowable leakage rate is  $=(\text{Duct Leakage Class in CFM}/100 \text{ SF}) \times (\text{Pressure Class} - \text{in-wg})^{0.65} \text{ cfm}$ . Complete ductwork leak test with satisfactory results prior to applying insulation to ductwork exterior or concealing ductwork.

### 3.8 DAMPER ACCEPTANCE TEST

Submit the proposed schedule, at least 2 weeks prior to the start of test. Operate all fire dampers and smoke dampers under normal operating conditions, prior to the occupancy of a building to determine that they function properly. Test each fire damper equipped with fusible link by having the fusible link cut in place. Test dynamic fire dampers with the air handling and distribution system running. Reset all fire dampers with the fusible links replaced after acceptance testing. To ensure optimum operation and performance, install the damper so it is square and free from racking.

### 3.9 TESTING, ADJUSTING, AND BALANCING

The requirements for testing, adjusting, and balancing are specified in Section 23 05 93 TESTING, ADJUSTING AND BALANCING FOR HVAC. Begin testing, adjusting, and balancing only when the air supply and distribution, including controls, has been completed, with the exception of performance tests.

### 3.10 PERFORMANCE TESTS

Conduct performance tests as required in Section 23 05 93 Testing, Adjusting and Balancing for HVAC and Section 23 09 00 Instrumentation and Control for HVAC.

### 3.11 CLEANING AND ADJUSTING

Thoroughly clean ducts, plenums, and casing of debris and blow free of small particles of rubbish and dust and then vacuum clean before installing outlet faces. Wipe equipment clean, with no traces of oil, dust, dirt, or paint spots. Provide temporary filters prior to startup of all fans that are operated during construction, and provide new filters after all construction dirt has been removed from the building, and the ducts, plenums, casings, and other items specified have been vacuum cleaned. Perform and document that proper "Indoor Air Quality During Construction" procedures have been followed; provide documentation showing that after construction ends, and prior to occupancy, new filters were provided and installed. Maintain system in this clean condition until



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

final acceptance. Properly lubricate bearings with oil or grease as recommended by the manufacturer. Tighten belts to proper tension. Adjust control valves and other miscellaneous equipment requiring adjustment to setting indicated or directed. Adjust fans to the speed indicated by the manufacturer to meet specified conditions. Maintain all equipment installed under the contract until close out documentation is received, the project is completed and the building has been documented as beneficially occupied.

### 3.12 RADIANT PANELS

#### 3.12.1 Installation

Install radiant panels level and plumb, maintaining sufficient clearance for normal services and maintenance.

### 3.13 OPERATION AND MAINTENANCE

#### 3.13.1 Operation and Maintenance Manuals

Submit six manuals at least 2 weeks prior to field training. Submit data complying with the requirements specified in Section 01 78 23 OPERATION AND MAINTENANCE DATA. Submit Data Package 3 for the items/units listed under SD-10 Operation and Maintenance Data

#### 3.13.2 Operation And Maintenance Training

Conduct a training course for the members of the operating staff as designated by the Contracting Officer. Make the training period consist of a total of 8 hours of normal working time and start it after all work specified herein is functionally completed and the Performance Tests have been approved. Conduct field instruction that covers all of the items contained in the Operation and Maintenance Manuals as well as demonstrations of routine maintenance operations. Submit the proposed On-site Training schedule concurrently with the Operation and Maintenance Manuals and at least 14 days prior to conducting the training course.

-- End of Section --

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## SECTION 23 35 16.17 10

### MECHANICAL ENGINE EXHAUST SYSTEMS

**05/20**

#### PART 1 GENERAL

##### 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

#### AIR MOVEMENT AND CONTROL ASSOCIATION INTERNATIONAL, INC. (AMCA)

AMCA 210 (2016) Laboratory Methods of Testing Fans for Aerodynamic Performance Rating

AMCA 300 (2014) Reverberant Room Method for Sound Testing of Fans

#### AMERICAN BEARING MANUFACTURERS ASSOCIATION (ABMA)

ABMA 9 (2015) Load Ratings and Fatigue Life for Ball Bearings

ABMA 11 (2014) Load Ratings and Fatigue Life for Roller Bearings

#### AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B16.21 (2021) Nonmetallic Flat Gaskets for Pipe Flanges

ASME BPVC SEC IX (2017; Errata 2018) BPVC Section IX-Welding, Brazing and Fusing Qualifications

#### AMERICAN WELDING SOCIETY (AWS)

AWS A5.8/A5.8M (2019) Specification for Filler Metals for Brazing and Braze Welding

#### ASTM INTERNATIONAL (ASTM)

ASTM A36/A36M (2019) Standard Specification for Carbon Structural Steel

ASTM A53/A53M (2022) Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless

ASTM A167 (2011) Standard Specification for Stainless and Heat-Resisting Chromium-Nickel Steel Plate, Sheet, and Strip

ASTM A193/A193M (2023) Standard Specification for

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

Alloy-Steel and Stainless Steel Bolting  
Materials for High-Temperature Service and  
Other Special Purpose Applications

ASTM A307 (2021) Standard Specification for Carbon  
Steel Bolts, Studs, and Threaded Rod 60  
000 PSI Tensile Strength

ASTM A924/A924M (2022a) Standard Specification for General  
Requirements for Steel Sheet,  
Metallic-Coated by the Hot-Dip Process

ASTM B32 (2020) Standard Specification for Solder  
Metal

ASTM B117 (2019) Standard Practice for Operating  
Salt Spray (Fog) Apparatus

#### NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA MG 1 (2021) Motors and Generators

#### SHEET METAL AND AIR CONDITIONING CONTRACTORS' NATIONAL ASSOCIATION (SMACNA)

SMACNA 1520 (1999) Round Industrial Duct Construction  
Standards, 3rd Edition

### 1.2 SYSTEM DESCRIPTION

Construct, complete and operational, an exhaust system as specified herein. Provide adequate air exhaust quantities and velocities. Properly size all duct for pressure loss and adequate velocity including locating intakes, ductwork size, layout, equipment and controls. Base construction of the exhaust system on the referenced publications, and other provisions as specified herein. Furnish ductwork offsets, fittings, and any other accessories required, as specified, to provide a complete exhaust system installation and to eliminate interference with other construction. Provide controls as specified in Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC.

### 1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

#### SD-02 Shop Drawings

Detail Drawings; G

Exhaust System Installation; G

#### SD-03 Product Data

Related Submittals

Ductwork Components; G

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

Materials and Equipment

Spare Parts

Field Instructions

Final Acceptance Tests

Onsite Training; G

Exhaust System Specialist; G

SD-06 Test Reports

Final Acceptance Tests

SD-07 Certificates

Inspection; G

SD-10 Operation and Maintenance Data

Exhaust System; G

Operation and Maintenance Manuals; G

#### 1.4 QUALITY ASSURANCE

##### 1.4.1 Detail Drawings

Submit 3 copies of the Exhaust System Drawings, no later than 21 days prior to the start of exhaust system installation. Include a complete list of equipment and materials, including manufacturer's descriptive and technical literature, performance charts and curves, catalog cuts, installation instructions, complete duct, wiring, and schematic diagrams and any other details to demonstrate that the system has been coordinated and will properly function as a unit. Also show proposed layout and anchorage of equipment and appurtenances, and equipment in relation to other parts of the work including clearances required for maintenance and operation.

##### 1.4.2 Exhaust System Specialist

Submit the name and documentation of certification of the proposed Exhaust System Specialists, no later than 14 days after the Notice to Proceed and prior to the submittal of the exhaust system drawings and hydraulic calculations. Prepare a list of the submittals from the Contract Submittal Register that relate to the successful installation of the exhaust systems(s). Submit the list no later than 7 days after the approval of the Exhaust System Specialist. The related submittals identified on this list must be accompanied by a letter of approval signed and dated by the Exhaust System Specialist when submitted to the Government. The Exhaust System Specialist must be regularly engaged in the installation of the type and complexity of system specified in the Contract documents, and will have served in a similar capacity for at least three systems that have performed in the manner intended for a period of not less than 6 months.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 1.5 DELIVERY, STORAGE, AND HANDLING

House all equipment delivered and placed in storage in a manner to preclude any damage from the weather, humidity and temperature variations, dirt and dust, or other contaminants. Additionally, cap or plug all ductwork, flexible connections and pipes until installed.

### 1.6 EXTRA MATERIALS

Submit spare parts data for each item of equipment and material specified. Include a complete list of parts and supplies, with current unit prices and source of supply, and a list of parts recommended by the manufacturer to be replaced after 1 year and 3 years of service. Include a list of special tools and test equipment required for maintenance and testing of the products supplied.

## PART 2 PRODUCTS

### 2.1 STANDARD PRODUCTS

- a. Provide materials and equipment which are standard products of a manufacturer regularly engaged in the manufacture of the product and that essentially duplicate items that have been in satisfactory use for at least 2 years prior to bid opening. Submit manufacturer's catalog data included with the Exhaust System Drawings for all items specified herein. Highlight data to show model, size, options, etc., that are intended for consideration. Demonstrate compliance with all contract requirements. In addition, provide a complete equipment list that includes equipment description, model number and quantity.
- b. Where an integrated, packaged exhaust system is furnished, all items will be the product of the system manufacturer. System component parts may be by other manufacturers. Provide equipment by a service organization that is capable of responding to service calls within four hours.
- c. Asbestos and asbestos-containing products are not acceptable.

### 2.2 NAMEPLATES

Furnish a nameplate that identifies the manufacturer's name, address, type or style, model or serial number, and catalog number on all equipment.

### 2.3 EQUIPMENT GUARDS AND ACCESS

Fully enclose or guard belts, pulleys, chains, gears, couplings, projecting setscrews, keys, and other rotating parts exposed to personnel contact according to OSHA requirements. Properly guard or cover high temperature equipment and piping exposed to contact by personnel or where it creates a potential fire hazard with insulation of a type specified.

### 2.4 DUCTWORK COMPONENTS

#### 2.4.1 General

Provide duct constructed of galvanized sheets of the minimum gauge thickness for ducts as required in SMACNA 1520. Construct and seal ducts in accordance with SMACNA 1520 for a negative pressure of 5.0 inch water gauge static pressure. Unless otherwise approved, provide ducts that are

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

round with longitudinal lock seam and conform to the dimensions indicated. Provide ducts that are straight and smooth on the inside with airtight joints. Where ducts with crimped ends are used to make up joints, the joints must have crimp and bead. The bead must provide a rigid stop for the mating open end to seat against. Steel spiral wound duct is not acceptable.

#### 2.4.2 Fittings

Furnish reducing fittings that have a minimum of 1 inch increase in diameter per 8 inches in length. Elbows that have a centerline radius less than 1.5 times the diameter are not permitted. Stub branches into mains at main expansion points at an angle of not more than 30 degrees with the centerline of the main duct in the direction of air flow, unless otherwise indicated or approved. Where riser ducts with single or multiple inlets are indicated, connect the riser duct into the bottom of the main duct at an angle as specified for branches. Where flexible connections connect to the main duct, brace the duct branch takeoff or stub with approved metal straps or members.

#### 2.4.3 Cleanout

Provide cleanout on the end of the main ductwork opposite the end of the fan suction connection. Size the cleanout opening to the approximate inside area of the duct. Provide removable airtight caps or flange type covers of minimum gauge thickness as the main duct. Provide other cleanout openings where indicated.

#### 2.4.4 Apparatus Connections

Where sheet metal connections are made to fan suction and discharge, or where ducts of dissimilar metals are connected, install an approved noncombustible flexible connection approximately 6 inches and securely fasten by zinc-coated steel clinch-type draw bands for round ducts. For rectangular ducts, install the flexible connections locked to metal collars using normal duct construction methods.

#### 2.4.5 Duct Test Holes

Provide test holes with covers where indicated, directed, or where necessary in ducts and plenums for using Pitot tubes for taking air measurements to balance the air systems.

#### 2.4.6 Duct Sleeves and Framed Openings

Provide duct sleeves for all round ducts 15 inch diameter or less passing through floors, walls, ceilings, or roofs. Provide sleeves in non-load bearing walls fabricated of 20 gauge steel sheets conforming to ASTM A924/A924M. Provide sleeves in load-bearing walls fabricated of standard-weight galvanized steel pipe conforming to ASTM A53/A53M. Install round ducts larger than 15 inch diameter and all square and rectangular ducts passing through floors, walls, ceilings, or roofs through framed openings. Furnish structural steel members for framed openings conforming to ASTM A36/A36M. Provide 1 inch clearance between the duct and the opening. Provide closure collars of galvanized steel no less than 4 inches wide on each side of walls or floors where sleeves or framed openings are provided. Fabricate collars for round ducts 15 inch diameter or less from 20 gauge galvanized steel. Fabricate collars for round, square or rectangular ducts with minimum dimension over 15 inches

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

from 18 gauge galvanized steel.

## 2.5 EXHAUST HOSE SYSTEM

### 2.5.1 Tailpipe Adapters

Provide tapered-cone adapters with magnetic connectors or other suitable devices for exhaust pipe attachment. The adapter may vary between 3 and 5 inch nominal diameter exhaust pipe.

### 2.5.2 Flexible Exhaust Hose

Flexible exhaust hose must be approved heat-resistant wire-reinforced glass fiber and neoprene tubing. Provide flexible tubing inside diameter and length as shown. Connect the tubing to the bottom of the ductwork. Provide a flanged connection where the flexible tubing and overhead ductwork are joined. The flanged connection must consist of steel flanges not less than 0.078 inch thick, 1/8 inch gasket. Provide gasket suitable for the system design temperature shown, in accordance with ASME B16.21, full face or self-centering flat ring type. It must contain aramid fibers bonded with styrene butadiene rubber (SBR) or nitrile butadiene rubber (NBR). Size or design the flange to suit the hose as approved. The connection of the neoprene hose may be installed with an approved hose clamp or as recommended by the manufacturer.

### 2.5.3 Exhaust Hose Suspension System

Suspend the flexible tubing overhead when not in use; allowing it to be lowered to the operating level, when required. Furnish the suspension system complete with cable, and operating mechanism. The suspension system must be manually operated balancer type with safety ratchet lock or automatic brake having slip resistant hand grip.

## 2.6 DAMPERS

Provide dampers of the type indicated and install where shown. Provide circular disk dampers with quadrant locking device or blast gate type. Damper blades less than 16 gauge thickness of stainless steel are not permitted. Provide blast gate dampers consisting of two piece construction with adjustable sliding gate and setscrew.

## 2.7 MATERIALS

Provide materials conforming to the following requirements.

### 2.7.1 Iron and Steel Sheets

#### 2.7.1.1 Galvanized Iron and Steel

ASTM A924/A924M, Coating Designation G90.

#### 2.7.1.2 Stainless Steel

ASTM A167, Type 304.

### 2.7.2 Steel Structural Shapes

ASTM A36/A36M.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### 2.7.3 Solder Silver

AWS A5.8/A5.8M, brazing alloy; grade to suit application.

### 2.7.4 Solder

ASTM B32, composition to suit application.

### 2.7.5 Bolts and Nuts

Furnish bolts and nuts, except as required for high temperature exhaust applications, in accordance with ASTM A307. Use bolts and nuts for exhaust applications where the temperature of the bolt may rise above 400 degrees F or use as flange bolts in corrosion resistant material in accordance with ASTM A193/A193M Class 2. Mark the bolt head to identify the manufacturer and the standard with which the bolt complies in accordance with ASTM A307 or ASTM A193/A193M as applicable.

## 2.8 ELECTRICAL WORK

Provide specified electrical motor-driven equipment complete with motor, motor starter, and controls. Unless otherwise specified, provide electric equipment, including wiring and motor efficiencies, according to Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Provide electrical characteristics and enclosure type as shown. Unless otherwise indicated, motors of 1 hp and above must be high efficiency type. Provide motor starters complete with thermal overload protection and other appurtenances necessary. Provide each motor according to NEMA MG 1 and of sufficient size to drive the equipment at the specified capacity without exceeding the nameplate rating of the motor. Manual or automatic control and protective or signal devices required for the operation specified, and any control wiring required for controls and devices, but not shown, must be provided. Where two-speed or variable-speed motors are indicated, solid-state variable-speed controller may be provided to accomplish the same function. Utilize solid-state variable-speed controllers for motors rated 10 hp or less. Use adjustable frequency drives for larger motors.

## 2.9 AIR MOVING DEVICES

### 2.9.1 General

Test and rate fans in accordance with the standards of AMCA 210, Type "D" Ducted Inlet, Ducted Outlet Configuration. Fans will be directly connected to the motor shaft. Provide fans with personnel screens or guards on both suction and supply ends except where ducts or dampers are connected to the fan. Provide fans and motors with vibration isolation supports or mountings. Vibration isolation units must be standard products with published load ratings, and must be single rubber-in-shear, neoprene coated fiberglass, double rubber-in-shear springs, or springs under inertia base. Select each fan to produce the capacity required at the fan total pressure indicated. Provide standard AMCA arrangements unless otherwise indicated and provide the rotation and discharge as indicated. Provide fans with nonoverloading characteristics. Construct fan housing with no less than 16 gauge thick steel. Construct fan impellers to meet AMCA Spark Resistance "B" Classification and accurately balance both statically and dynamically when installed in the assembled fan unit. Coat impeller and housing in the air stream with neoprene, epoxy, phenolic resins, or otherwise be suitable to resist the corrosive gases and temperatures produced. Fans must be free of objectionable



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

vibration or noise. Certified performance curves indicating that the fan supplied will operate in its most efficient operating range will be provided. In addition, furnish "sound power" ratings with each fan. Provide fans indicated to be mounted on exterior of building with weatherproof covers for the motor drive unit or other weatherproofing as recommended by the manufacturer. Select each fan to produce the capacity required at the fan total pressure indicated. Provide weather hoods, flashing, and bird screens where indicated.

## 2.9.2 Fans

Provide sound power level as indicated and obtain values according to AMCA 300. Indicate Standard AMCA arrangement, rotation, and discharge. Test and rate fans according to AMCA 210. Select each fan to produce the capacity required at the fan static pressure indicated. Fans shall be connected to the motors directly.

### 2.9.2.1 Protective Devices

Provide fan and motor assemblies with vibration-isolation supports or mountings as indicated. Vibration-isolation units must be standard products with published loading ratings.

### 2.9.2.2 Centrifugal Fans

Centrifugal fans must be fully enclosed, single-width single-inlet, or double-width double-inlet, AMCA Pressure Class I, II, or III as required or indicated for the design system pressure. Provide rigidly constructed impellar wheels, accurately balance both statically and dynamically. Fan blades may be forward curved, backward-inclined or airfoil design in wheel sizes up to 30 inches. Fan blades for wheels over 30 inches in diameter must be backward-inclined or airfoil design. These fans must be suitable for the temperatures encountered. Provide fan shaft with a heat slinger to dissipate heat buildup along the shaft. Supply an access (service) door to facilitate maintenance with these fans. Fan wheels over 36 inches in diameter must have overhung pulleys and a bearing on each side of the wheel. Indirect drive fan wheels 36 inches or less in diameter may have one or more extra long bearings between the fan wheel and the drive. Provide bearings that are sleeve type, self-aligning and self-oiling with oil reservoirs, or precision self-aligning roller or ball-type with accessible grease fittings or permanently lubricated type. Connect grease fittings to tubing and service from a single accessible point. Bearing life must be L50 rated at not less than 200,000 hours as defined by ABMA 9 and ABMA 11. Provide steel fan shafts, accurately finish, and provide key seats and keys for impeller hubs and fan pulleys. Provide each fan outlet of ample proportions and design for the attachment of angles and bolts for attaching flexible connections. Provide motors, unless otherwise indicated, that do not exceed 1800 rpm and have totally enclosed enclosures. Provide motor starters that are magnetic type with general-purpose enclosure. Provide remote manual switch with pilot indicating light where indicated.

## 2.10 FACTORY COATING

Provide factory finished equipment and component items, when fabricated from ferrous metal as defined by ASTM (or similar) standard, with the manufacturers standard finish except provide weather-resistant finishes that will withstand 500 hours exposure to the salt spray test specified in ASTM B117 to items located outside of building.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### PART 3 EXECUTION

#### 3.1 EXAMINATION

After becoming familiar with all details of the work, verify all dimensions in the field, and advise the Contracting Officer of any discrepancy before performing the work.

#### 3.2 INSTALLATION

Install work as shown and according to the manufacturer's diagrams and recommendations.

#### 3.3 INSPECTION

The Exhaust System Specialist must (1) Inspect the exhaust system periodically during the installation. (2) Witness the final tests, and sign approval of the test results. (3) Certify in writing that the system has been installed in accordance with the contract requirements. Bring any discrepancy to the attention of the Contracting Officer in writing, no later than three working days after the discrepancy is discovered.

#### 3.4 EXHAUST SYSTEM INSTALLATION

##### 3.4.1 General Requirements

Perform welding and brazing conforming to ASME BPVC SEC IX. Install horizontal sections of the main duct with the longitudinal lock seam on the top. Seal slip joints in accordance with SMACNA 1520. Support and anchor riser duct to the structure as indicated. Attach main duct to the structural members of the building as recommended by SMACNA 1520.

##### 3.4.2 Building Surface Penetrations

Utilize sleeves or framed openings where duct penetrates building surfaces. Penetrations must be sealed, and fireproofed in accordance with Section 07 84 00 FIRESTOPPING. Pack the space between the sleeve or framed opening and the duct with mineral wool or other approved material. Install closure collars around the duct on both sides of the penetrated surface. Collars must fit tight against the building surfaces and snugly around the duct.

#### 3.5 PIPE COLOR CODE MARKING

Provide color code marking of piping as specified in Section 09 90 00 PAINTS AND COATINGS.

#### 3.6 ONSITE TRAINING

Submit proposed Onsite Training schedule, at least 14 days prior to the start of related training for the operating staff as designated by the Contracting Officer. The training period must consist of a total 8 hours of normal working time and must start after the system is functionally completed but prior to final acceptance tests. The field instructions must cover all of the items contained in the approved operation and maintenance manuals, as well as demonstrations of routine maintenance operations. Submit 6 manuals listing step-by-step procedures required for system startup, operation, shutdown, and routine maintenance, at least 14

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

days prior to on-site training. Include the manufacturer's name, model number, list of parts and tools that should be kept in stock by the owner for routine maintenance including the name of a local supplier, simplified wiring and controls diagrams, troubleshooting guide, and recommended service organization (including address and telephone number) for each item of equipment. Each service organization submitted must be capable of providing 4 hour on-site response to a service call on an emergency basis.

Notify the Contracting Officer at least 14 days prior to date of proposed conduction of the training course.

### 3.7 FINAL ACCEPTANCE TESTS

Balance each exhaust system and inlet to produce the indicated air quantities within 10 percent at the conditions shown. Set control devices to control at the points indicated or directed. Lubricate bearings, and check the speed, direction or rotation of each fan. Check the running current of each motor. Upon completion, and prior to acceptance of the installation, test the exhaust system at operating conditions to demonstrate satisfactory functional and operating efficiency.

- a. Operating tests must cover a period of not less than 2 hours for each system, and conduct all tests in the presence of the Contracting Officer. If tests do not demonstrate satisfactory operation of the exhaust system, correct deficiencies and retest. Provide all instruments, facilities, and labor required to properly conduct the tests. The electricity required for testing will be furnished by the Government.
- b. Submit 3 copies of the completed Final Acceptance Tests Reports, no later than 7 days after the completion of the Tests. Sign all items in the Final Acceptance Report. Submit proposed diagrams, instructions, and other sheets, concurrent with the Final Acceptance Test Procedures. Post framed instructions under glass or in laminated plastic where directed, including wiring and control diagrams showing the complete layout of the entire system. Prepare condensed operating instructions explaining preventive maintenance procedures, methods of checking the system for normal safe operation, and procedures for safely starting and stopping the system in typed form, frame as specified above for the wiring and control diagrams and post beside the diagrams. Post the framed instructions before acceptance testing of the systems.
- c. Submit proposed procedures for Final Acceptance Tests, no later than 14 days prior to the proposed start of the tests.
- d. Submit proposed date and time to begin Final Acceptance Tests, with the Final Acceptance Test Procedures. Provide notification at least 14 days prior to the proposed start of the test.

-- End of Section --

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## SECTION 23 52 43.00 20

### LOW PRESSURE WATER HEATING BOILERS (UNDER 800,000 BTU/HR OUTPUT)

**05/15, CHG 2: 08/18**

#### PART 1 GENERAL

##### 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

#### AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI Z21.13/CSA 4.9 (2022) Gas-Fired Low Pressure Steam and Hot Water Boilers

#### AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME BPVC SEC IV (2017) BPVC Section IV-Rules for Construction of Heating Boilers

ASME CSD-1 (2021) Control and Safety Devices for Automatically Fired Boilers

#### ASTM INTERNATIONAL (ASTM)

ASTM A53/A53M (2022) Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless

#### NATIONAL BOARD OF BOILER AND PRESSURE VESSEL INSPECTORS (NBBI)

NBBI NB-23 PART 1 (2013) National Board Inspection Code - Part 1 Installation

#### NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA MG 1 (2021) Motors and Generators

#### NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 31 (2020; TIA 23-1) Standard for the Installation of Oil-Burning Equipment

NFPA 54 (2024) National Fuel Gas Code

NFPA 70 (2023; ERTA 4 2023; ERTA 5 2023; ERTA 6 2023) National Electrical Code

NFPA 211 (2019) Standard for Chimneys, Fireplaces, Vents, and Solid Fuel-Burning Appliances

#### U.S. DEPARTMENT OF ENERGY (DOE)

Energy Star (1992; R 2006) Energy Star Energy Efficiency Labeling System (FEMP)

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## UNDERWRITERS LABORATORIES (UL)

UL 795

(2016; Reprint May 2022) UL Standard for  
Safety Commercial-Industrial Gas Heating  
Equipment

### 1.2 RELATED REQUIREMENTS

Section 23 03 00.00 20 BASIC MECHANICAL MATERIALS AND METHODS, applies to this section, with the additions and modifications specified herein.

### 1.3 DESIGN REQUIREMENTS

Boiler must be suitable for installation in the space shown with ample room for opening doors and cleaning and removal and replacement of heat exchanger. Boiler must have an output of 395,000 BTU/hr at max fire at AHRI conditions. Boiler must be designed and tested in accordance with ASME BPVC SEC IV, and ASME CSD-1. Boiler must be installed in accordance with manufacturer instructions. Boiler shall have standard manufacturer's enclosure and corrosion resistant finish. Boiler design working pressure must be 160 psig with ASME stamp. Boiler operating temperature must be 102.5 degrees F. Boiler return water temperature must be 82.5 degrees F.

#### 1.3.1 Boiler Installation Requirements

##### 1.3.1.1 Location

Install Boiler(s) and associated hot water pumps in a mechanical room inside the facility in accordance with NBBI NB-23 PART 1. Provide ample clearance around boilers to allow access for inspection, maintenance and repair. Passageways around all sides of boilers must have an unobstructed minimum width of 36 inches or the clearances recommended by the boiler manufacturer whichever is greater.

##### 1.3.1.2 Combustion Air

Provide supply of air for combustion and ventilation. In accordance with NFPA 54, NFPA 211 and manufacturer's installation manual, calculate the amount of combustion air necessary to operate the boiler. Install and locate properly sized combustion air ductwork.

##### 1.3.1.3 Sequence of Operation

Local, manual starting of boilers is required. Remote starting and stopping of the boiler by the HVAC control system is not permitted. This is to ensure that an operator witness the initial firing of the boiler at the beginning of each heating season to verify proper operation of the boiler and to promote proper maintenance.

#### 1.3.2 Detail Drawings

##### 1.3.2.1 Drawings

Show boiler hot water isolation valves, emergency disconnect switch, and complete boiler gas train on the contract drawings.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### 1.3.2.2 Fuel Train / Wiring Diagram

Submit fuel train and wiring diagram.

#### 1.3.3 Water Analysis

Provide test reports of water analysis. UFC 3-240-13FN Industrial Water Treatment must be followed for all boiler installations.

#### 1.4 SAFETY STANDARDS

Hot water boilers, burners and supplementary control devices, safety interlocks, or limit controls required under this specification, must meet requirements of the following standards as applicable:

- b. Gas-Fired Units: ASME CSD-1, NFPA 54, NFPA 70, ANSI Z21.13/CSA 4.9 or UL 795.
- d. All Units: ASME BPVC SEC IV, NFPA 70 and ASME CSD-1.

Controls not covered by the above must have a UL label, UL listing mark, or must be listed in the Factory Mutual Approval Guide.

#### 1.5 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

Submittals for this Section must be delivered to the project Contracting Officer, who will forward two complete sets of copies to the appropriate approving official for review and approval.

##### SD-02 Shop Drawings

Fuel Train; G

Wiring Diagram; G

##### SD-03 Product Data

Boilers; G

Energy Star Label for Gas Fired Residential Boiler Product

Boiler Trim and Control Equipment

Burners and Control Equipment

Stack, Breeching, and Supports

##### SD-06 Test Reports

Operational Tests

Water Analysis

##### SD-07 Certificates

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## Boilers

### Burners and Control Equipment

#### Boiler Trim and Control Equipment

Boiler manufacturer's certificate of boiler performance including evidence that the burners provided must be a make, model, and type certified and approved by the manufacturer of the boiler being provided.

#### SD-08 Manufacturer's Instructions

##### Boilers

##### Feedwater Treatment Feeder

#### SD-10 Operation and Maintenance Data

##### Boilers, Data Package 4; G

Submit in accordance with Section 01 78 23 OPERATION AND MAINTENANCE DATA.

#### SD-11 Closeout Submittals

##### Posted Operating Instructions for Heating Water Boilers; G

## PART 2 PRODUCTS

### 2.1 BOILERS

Provide hot water heating boiler complete with firing equipment, combustion chamber, insulation with steel jacket, safety and operating controls, integral electrical wiring and other appurtenances, to make the boiler a complete, self-contained, fully-automatic unit, ready for service upon completion of utility connections. Condensing gas fired boilers greater than 300,000 Btuh input must have a AHRI thermal efficiency of at least 93.8 percent when fired at the maximum and minimum capacities which are provided and allowed by the controls.

#### 2.1.1 General Requirements

Design, construction, installation, testing, and operation of boiler and appurtenances shall comply with NBBI NB-23 PART 1, ASME BPVC SEC IV, ASME CSD-1, NFPA 54, NFPA 31, ANSI Z21.13/CSA 4.9, and the manufacturer's instructions.

### 2.2 BURNERS AND CONTROL EQUIPMENT

#### 2.2.1 Gas-Fired Power Burner

Gas-fired power burner. Interrupted pilot type ignition system, and pilot must be the electrode-ignited natural gas type. Design burner and combustion control equipment for firing natural gas having a specific gravity of 0.6 and a heating value of approximately 1000 BTU per cubic foot and be an integral part of the boiler. Burner controls and safety equipment must conform to applicable requirements of ASME CSD-1, NFPA 54,

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

ANSI Z21.13/CSA 4.9 and UL 795. Mount controls; including operating switches, indicating lights, gages, alarms, motor starters, fuses, and circuit elements of control systems on a single control panel or cabinet designed for separate mounting not on the burner. The combustion control system must be the modulating 8:1 turndown type. Locate flame scanner such that testing and cleaning of scanner can be accomplished without disassembly of burner. Provide fuel train as indicated. Gas pressure available: 3 - 13 inches of water gage.

## 2.3 BOILER TRIM AND CONTROL EQUIPMENT

Provide in accordance with ASME CSD-1 and ASME BPVC SEC IV and additional requirements specified below.

### 2.3.1 Emergency Disconnect Switch

Provide and locate on wall outside boiler room entrance or just inside door, when boiler room door is on building exterior as required by ASME CSD-1 to allow rapid and complete shutdown of the boiler in the event of an emergency. Emergency switch must be a 20-amp. fuse-type safety switch. Switch must be red and furnished with a label indicating function of switch.

### 2.3.2 Relief Valves

Provide relieving capacity for the full output of boiler installed. Safety relief-valve piping must conform to ASTM A53/A53M, schedule 40 steel pipe and be piped full-size floor sink.

### 2.3.3 Pressure and Altitude Gage or Combination Pressure/Altitude Gage

Provide one located on supply water piping and one on return water piping.

### 2.3.4 Thermometer

Provide thermometer with a scale equivalent to 1.5 times outlet water temperature. Provide one located on supply water piping and one on return water piping.

### 2.3.5 Drain Tapping

Provide drain valve and piping floor sink.

### 2.3.6 Make-up Water Station

#### 2.3.6.1 Pressure Reducing Station

Provide a water pressure-reducing valve and relief valve, or a combination of the two in the makeup water line to the boiler to maintain a water pressure of 12 psig in the hot water system. Provide a 3/4 inch globe valve by-pass around this valve.

#### 2.3.6.2 Backflow Preventers

Section 22 00 00 PLUMBING, GENERAL PURPOSE. Locate upstream of by-pass.

### 2.3.7 Feedwater Treatment Feeder

Provide floor mounted, Type II - Shot-Type Feeder (manual, intermittent



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

feed), as indicated for use with pressures up to 200 psig maximum.

#### 2.3.8 Combustion Regulator

Provide adjustable temperature, thermostatic immersion type that must limit boiler water temperature to a maximum of 250 degrees F. Control must actuate burner through an electric relay system to maintain boiler water temperature within normal prescribed limits at loads within rated capacity of boiler.

#### 2.3.9 Air Vent Valve

Provide with screwed connection, stainless steel disk, and stainless steel seats to vent entrapped air.

#### 2.3.10 High Temperature Limit Switch

Provide adjustable immersible aquastat type with a temperature setting above that of the combustion regulator and below that of the lowest relief valve setting. Aquastat must function to cause a safety shutdown by closing fuel valves, and shutting down burner equipment, activating a red indicating light, and sounding an alarm in the event that boiler water temperature rises to the high temperature limit setting. A safety shutdown due to high temperature must require manual reset before operation can resume and prevent recycling of burner equipment. Pre-set high temperature limit devices that cannot be easily tested are not allowed.

#### 2.3.11 Low Water Level Cutoff Switch

Low water level cutoff must cause a safety shutdown by closing fuel valves, and shutting down burner equipment, activating a red indicating light, and sounding an alarm in the event that water level drops below the lowest safe permissible water level established by the boiler manufacturer and ASME BPVC SEC IV. A safety shutdown due to low water must require manual reset before operation can resume and prevent recycling of burner equipment.

#### 2.3.12 Boiler Safety Control Circuits

Provide boiler safety control circuits, including control circuits for burner, must be single-phase, two-wire one-side grounded, and not over 120 volts. Provide safety control switching in ungrounded conductors. Provide overcurrent protection. In addition to circuit grounds, ground metal parts which do not carry current to a grounding conductor.

#### 2.3.13 Draft

Comply with boiler manufacturer's recommendations.

#### 2.3.14 Stack, Breeching, and Supports

NFPA 211, Type B - gas, prefabricated multi-wall type, flashed to the roof, and complete with rain cap. Stack diameter and height must be in accordance with manufacturer's recommendations and conform to NFPA 211.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### 2.3.15 Hot Water Coils

Provide coils capable of heating 40 GPM of water with 20 degree F rise conforming to ASME BPVC SEC IV.

#### 2.3.16 Stack Thermometer

Provide flue gas dial type thermometer with scale calibrated from 150 to 750 degrees F and mounted in flue gas outlet.

### 2.4 ELECTRIC MOTORS

Electric motors must meet requirements of NEMA MG 1. Motors less than 1 hp must meet NEMA High Efficiency requirements. Motors 1 hp and larger must meet NEMA Premium Efficiency requirements. Motors which are an integral part of the packaged boiler system must be the highest efficiency available by the manufacturer of the packaged boiler. Motors must be variable speed.

## PART 3 EXECUTION

### 3.1 EQUIPMENT INSTALLATION

Install equipment in accordance with manufacturer's installation instructions and NBBI NB-23 PART 1. Grout equipment mounted on concrete foundations before installing piping. Install piping in such a manner as not to place a strain on equipment. Do not bolt flanged joints tight unless they match. Grade, anchor, guide, and support piping without low pockets. Feedwater treatment feeders must be mounted so that the top of the feeder is no higher than 48 inches above the finished floor.

### 3.2 EQUIPMENT FOUNDATIONS

Locate equipment foundations as indicated, designed, and made of sufficient size and weight to preclude shifting of equipment under operating conditions or under abnormal conditions that could be imposed upon the equipment. Foundations must meet requirements of the equipment manufacturer. Concrete and grout must conform to Section 03 30 00 CAST-IN-PLACE CONCRETE.

### 3.3 BOILER CLEANING

Before being placed in service, boiler must be boiled out for a period of 24 hours at a pressure not exceeding 12 psig. Solution to be used in the boiler for the boiling out process must consist of two pounds of trisodium phosphate per 100 gallons of water. Upon completion of boiling out, flush out boiler with potable water, drain, and charge with chemically treated water. Protect boiler and appurtenances against internal corrosion until testing is completed and boiler is accepted. Professional services are required for cleaning/treatment process.

### 3.4 FIELD QUALITY CONTROL

Perform and furnish everything required for inspections and tests as specified herein to demonstrate that boiler and auxiliary equipment, as installed, are in compliance with contract requirements. Start-up and operate the system. During this time, clean strainers until no further accumulation of foreign material occurs. Exercise care to minimize loss of water when strainers are cleaned. Adjust safety and automatic control

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

instruments as necessary to place them in proper operation and sequence. Test instrumentation must be calibrated and have full scale readings from 1.5 to 2 times test values.

#### 3.4.1 Operational Tests

Furnish the services of an engineer or technician approved by the boiler manufacturer of installation, startup, operational and safety testing. This person must remain on the job until each boiler has been successfully operated. Furnish and perform everything required for inspections and tests of the boiler and auxiliary equipment. Test instrumentation must be calibrated and have full-scale reading from 1.5 to 2 times test values. Demonstrate proper operability of combustion control, flame safeguard control and safety interlocks. Provide a detailed description of all boiler startup and operational tests in the Commissioning Plan.

##### 3.4.1.1 Preliminary Operational Test

Operate the boilers continuously for a period of at least 8 hours to demonstrate proper operability of the combustion control, flame safeguard control, and safety interlocks.

##### 3.4.1.2 Acceptance Operational Test and Inspection

Prior to requesting an acceptance test, conduct a satisfactory operational test for at least 8 hours, and provide a certified statement that the equipment is installed per all requirements of this guide. Contracting Officer, upon receipt of the notice from the Contractor, will request a boiler inspection by a Naval Facilities Engineering and Expeditionary Warfare Center (EXWC) NAVFAC boiler inspector. Fifteen days advance notice is required for scheduling inspector to conduct acceptance operational test and inspection.

-- End of Section --

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## SECTION 23 80 20.00 10

## GAS-FIRED HEATING EQUIPMENT

05/20

## PART 1 GENERAL

## 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

## CSA GROUP (CSA)

CSA Directory (updated continuously online) Product Index

## NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA MG 1 (2021) Motors and Generators

## NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 54 (2021) National Fuel Gas Code

## UNDERWRITERS LABORATORIES (UL)

UL FLAMMABLE & COMBUSTIBLE (2012) Flammable and Combustible Liquids and Gases Equipment Directory

## 1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

## SD-02 Shop Drawings

Detail Drawings; G

Installation

## SD-03 Product Data

Spare Parts

## SD-06 Test Reports

Testing, Adjusting, and Balancing

## SD-10 Operation and Maintenance Data

Operation and Maintenance Instructions; G

## 1.3 QUALITY ASSURANCE

Submit detail drawings consisting of illustrations, schedules, performance

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

charts, instructions, brochures, diagrams, and other information to illustrate the requirements and operation of the system. Detail drawings for space heating equipment, controls, associated equipment, and for piping and wiring. Show proposed layout and anchorage of equipment and appurtenances, and equipment relationship to other parts of the work including clearances for maintenance and operation.

#### 1.4 DELIVERY, STORAGE, AND HANDLING

Protect all equipment delivered and placed in storage from weather, humidity and temperature variations, dirt and dust, or other contaminants.

#### 1.5 EXTRA MATERIALS

Submit spare parts data for each different item of material and equipment specified, after approval of the detail drawings, and not later than one month prior to the date of beneficial occupancy. Include in the data a complete list of parts and supplies, with current unit prices and source of supply.

### PART 2 PRODUCTS

#### 2.1 MATERIALS AND EQUIPMENT

##### 2.1.1 General

Provide materials and equipment which are standard products of a manufacturer regularly engaged in manufacturing of the products and that essentially duplicate equipment that has been in satisfactory use at least 2 years prior to bid opening. All gas fired appliances must meet the requirements of NFPA 54.

##### 2.1.2 Nameplates

Secure a plate to each major component of equipment containing the manufacturer's name, address, type or style, model or serial number, and catalog number. Also, affix an ENERGY STAR label as applicable.

##### 2.1.3 Equipment Guards

Completely enclose or guard belts, pulleys, chains, gears, couplings, projecting setscrews, keys, and other rotating parts so located that any person may come in close proximity thereto. Guard or cover high-temperature equipment and piping so located as to endanger personnel or create a fire hazard with insulation of type specified for service.

#### 2.2 ELECTRICAL WORK

Provide electrical motor driven equipment complete with motors, motor starters, and controls. Provide motors conforming to NEMA MG 1. Provide electrical equipment and wiring accordance with Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Provide electrical characteristics as specified or indicated. Provide premium efficiency integral size motors in accordance with NEMA MG 1. Provide motor starters complete with thermal overload protection and other appurtenances necessary for the motor control specified. Each motor must be of sufficient size to drive the equipment at the specified capacity without exceeding the nameplate rating of the motor. Manual or automatic control and protective or signal devices required for the operation specified and any control wiring required for

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

controls and devices specified, but not shown, must be provided.

## 2.3 HEATERS

Equip heaters for and adjust to burn natural gas. Provide each heater with a gas pressure regulator that will satisfactorily limit the main gas burner supply pressure. Heaters must have an intermittent or interrupted electrically ignited pilot or a direct electric ignition system. Provide safety controls that conform to the ANSI standard specified for each heater. Furnish mounting brackets and hardware by the heater manufacturer and factory finish to match the supported equipment.

### 2.3.1 Indirect Fired Make-Up Heaters

Provide heaters in accordance with ANSI and CSA Standards. Equip heaters with motorized inlet dampers, duct collar, and air filters. Provide modulating gas control valve. Maximum air temperature rise during minimum burner fire must be 7 degrees F. Supply and exhaust fans must be variable speed. Provide motorized inlet dampers that close when the unit is shut down. Dampers must be interlocked to prevent burner operation when dampers are closed. Provide heaters with a space thermostat, a low limit air stream thermostat, and an ambient air thermostat. Provide low limit air stream thermostat to shut down the entire unit if the discharge air temperature drops below the low-limit setpoint. Provide ambient air thermostat to shut down the burner if the outside air exceeds the supply air temperature setpoint.

#### 2.3.1.1 General Construction

Unit shall be of double-wall construction. The units outer casing shall be constructed of a minimum 18 gauge steel. The exterior finish shall be the manufacturer's standard corrosion resistant coating and color. Inner liner shall be 22 gauge perforated galvanized steel. The floor of the unit shall be constructed of 16 gauge galvanized steel over C channel base frame. The inner and outer layers shall sandwich 2 inches of 1.5 lb density insulation. Duct connections shall be 1-1/2" flanges. The intake hood shall be equipped with a combination bird screen/moisture eliminator. Intake filters shall be 2" MERV 8 throw-away pleated filters. All major components of the unit shall include insulated access doors with 1/4 turn handles. Additional access sections shall be provided as necessary to service the unit.

#### 2.3.1.2 Supply Fan

Belt driven, backward incline steel in galvanized steel housing. Heavy welded steel L angle frames with T-bar bearing supports for 4-way mounting. Detachable inlet veterri. Shafts shall have corrosion inhibiting coating and are keywayed both ends. Provide high quality cast iron pillow block bearings with grease fittings for lubrication.

#### 2.3.1.3 Exhaust Fan

Direct Drive, backward inclined, aluminum welded air foil wheel. Bolted 4-piece galvanized frame construction. Suitable for Class III air.

#### 2.3.1.4 Energy Recovery

AHRI 1060 rated. Constructed of 0.005 inch thick 99.9% pure aluminum plates (alloys or other materials shall not be exceptable). Frame of heat

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

exchanger shall be of 20 gauge galvanized steel. The heat exchanger shall withstand with degradation of performance pressure differential of at least 6 in-wg. It shall withstand a pressure drop of 10 in-wg without permanent deformation.

#### 2.3.1.5 Vibration Isolation Roof Curb

Prefabricated Vibration isolation curb to be manufactured of prime galvanized steel construction, 18 or 14 gauge as required, meeting ASTM A653/653M, with welded corners and with seams joined by continuous water and air tight welds. Vibration isolation curb shall be internally reinforced with bulkheads 48" on center and factory installed wood nailer. Top of all vibration isolation curb shall be level, with pitch built into curb when deck slopes. Vibration isolation curb shall be designed to provide a minimum of 90% isolation efficiency with 1" deflection. 9" continuous rubber cover around perimeter of Vibration isolation curb over spring isolators.

#### 2.4 FACTORY FINISHES

Equipment and component items, when fabricated from ferrous metal, must be factory finished with the manufacturer's standard finish.

### PART 3 EXECUTION

#### 3.1 EXAMINATION

After becoming thoroughly familiar with all details of the work, verify all dimensions in the field, and advise the Contracting Officer of any discrepancy before performing any work.

#### 3.2 INSTALLATION

install equipment as indicated and in accordance with the recommendations of the equipment manufacturer and the listing agency, except as otherwise specified.

##### 3.2.1 Heating Equipment

Install heaters with clearance to combustibles, complying with minimum distances as determined by CSA Directory, UL FLAMMABLE & COMBUSTIBLE and as indicated on each heater approval and listing plate. Support heaters independently from the building structure, as indicated, but not relying on suspended ceiling systems for support.

##### 3.2.2 Gas Piping

Connect gas piping as indicated, complying with the applicable requirements at Section 23 11 20 FACILITY GAS PIPING.

#### 3.3 TRAINING

Conduct a training course for the maintenance and operating staff. The training period of 8 hours normal working time must start after the system is functionally complete but before the final acceptance tests. Give the Contracting Officer at least two weeks advance notice of such training. Include all of the items contained in the approved operation and maintenance instructions as well as demonstrations of routine maintenance

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

operations. Submit 6 complete copies of operating instructions outlining the step-by-step procedures required for system startup, operation and shutdown. Include the manufacturer's name, model number, service manual, parts list, and brief description of all equipment and basic operating features. Submit 6 complete copies of maintenance instructions listing routine maintenance, possible breakdowns, repairs and troubleshooting guide. Include simplified piping, wiring, and control diagrams for the system as installed.

#### 3.4 TESTING, ADJUSTING, AND BALANCING

Perform testing, adjusting, and balancing as specified in Section 23 05 93 TESTING, ADJUSTING, AND BALANCING OF HVAC SYSTEMS. Submit test reports in booklet form showing all field tests performed to adjust each component and all field tests performed to prove compliance with the specified performance criteria, upon completion and testing of the installed system. Indicate the final position of controls.

-- End of Section --



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## SECTION 23 81 00

## DECENTRALIZED UNITARY HVAC EQUIPMENT

05/18, CHG 1: 02/21

## PART 1 GENERAL

## 1.1 RELATED REQUIREMENTS

Section 23 03 00.00 20 BASIC MECHANICAL MATERIALS AND METHODS, applies to this section with the additions and modifications specified herein.

## 1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

## AIR-CONDITIONING, HEATING AND REFRIGERATION INSTITUTE (AHRI)

|                  |                                                                                                         |
|------------------|---------------------------------------------------------------------------------------------------------|
| AHRI 340/360 I-P | (2015) Performance Rating of Commercial and Industrial Unitary Air-Conditioning and Heat Pump Equipment |
|------------------|---------------------------------------------------------------------------------------------------------|

|          |                                                     |
|----------|-----------------------------------------------------|
| AHRI 700 | (2016) Specifications for Fluorocarbon Refrigerants |
|----------|-----------------------------------------------------|

## AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS (ASHRAE)

|                     |                                                                                                                                                            |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ANSI/ASHRAE 15 & 34 | (2013) ANSI/ASHRAE Standard 15-Safety Standard for Refrigeration Systems and ANSI/ASHRAE Standard 34-Designation and Safety Classification of Refrigerants |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                |                                                                                                                                                                                |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASHRAE 15 & 34 | (2013) ASHRAE Standard 34-2016 Safety Standard for Refrigeration Systems/ASHRAE Standard 34-2016 Designation and Safety Classification of Refrigerants-ASHRAE Standard 34-2016 |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|             |                                                                                                           |
|-------------|-----------------------------------------------------------------------------------------------------------|
| ASHRAE 52.2 | (2012) Method of Testing General Ventilation Air-Cleaning Devices for Removal Efficiency by Particle Size |
|-------------|-----------------------------------------------------------------------------------------------------------|

|           |                                                                                                                               |
|-----------|-------------------------------------------------------------------------------------------------------------------------------|
| ASHRAE 55 | (2020; Errata 1 2021; Addenda A-D 2022; Addenda G-I 2023; Errata 2 2022) Thermal Environmental Conditions for Human Occupancy |
|-----------|-------------------------------------------------------------------------------------------------------------------------------|

|             |                                                                                                                                 |
|-------------|---------------------------------------------------------------------------------------------------------------------------------|
| ASHRAE 62.1 | (2022; Addenda E-X 2022; Errata 1-2 2022; Errata 3 2023; Interpretation 1-3 2023) Ventilation for Acceptable Indoor Air Quality |
|-------------|---------------------------------------------------------------------------------------------------------------------------------|

|                  |                                      |
|------------------|--------------------------------------|
| ASHRAE 90.1 - IP | (2019) Energy Standard for Buildings |
|------------------|--------------------------------------|

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

### Except Low-Rise Residential Buildings

#### AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

|                       |                                                                                |
|-----------------------|--------------------------------------------------------------------------------|
| ASME BPVC SEC IX      | (2017; Errata 2018) BPVC Section IX-Welding, Brazing and Fusing Qualifications |
| ASME BPVC SEC VIII D1 | (2019) BPVC Section VIII-Rules for Construction of Pressure Vessels Division 1 |

#### AMERICAN WELDING SOCIETY (AWS)

|           |                                                           |
|-----------|-----------------------------------------------------------|
| AWS Z49.1 | (2021) Safety in Welding and Cutting and Allied Processes |
|-----------|-----------------------------------------------------------|

#### ASTM INTERNATIONAL (ASTM)

|           |                                                                   |
|-----------|-------------------------------------------------------------------|
| ASTM B117 | (2019) Standard Practice for Operating Salt Spray (Fog) Apparatus |
|-----------|-------------------------------------------------------------------|

#### NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

|           |                                                                                                                         |
|-----------|-------------------------------------------------------------------------------------------------------------------------|
| NEMA MG 1 | (2021) Motors and Generators                                                                                            |
| NEMA MG 2 | (2014) Safety Standard for Construction and Guide for Selection, Installation and Use of Electric Motors and Generators |

#### UNDERWRITERS LABORATORIES (UL)

|         |                                                                                               |
|---------|-----------------------------------------------------------------------------------------------|
| UL 586  | (2009; Reprint Sep 2022) UL Standard for Safety High-Efficiency Particulate, Air Filter Units |
| UL 900  | (2015; Reprint Aug 2022) UL Standard for Safety Standard for Air Filter Units                 |
| UL 1995 | (2015; Reprint Aug 2022) UL Standard for Safety Heating and Cooling Equipment                 |

### 1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Spare Parts

Posted Instructions

Coil Corrosion Protection

System Performance Tests

Training; G

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

Inventory

Environmental Data

Supplied Products

Manufacturer's Standard Catalog Data

Humidifier

SD-06 Test Reports

Refrigerant Tests, Charging, and Start-Up; FIO

System Performance Tests; FIO

SD-07 Certificates

Service Organizations

SD-10 Operation and Maintenance Data

Operation and Maintenance Manuals; G

SD-11 Closeout Submittals

Ozone Depleting Substances; S

#### 1.4 QUALITY ASSURANCE

Carefully investigate the plumbing, fire protection, electrical, structural and finish conditions that would affect the work to be performed and arrange such work accordingly, furnishing required offsets, fittings, and accessories to meet such conditions. Submit drawings consisting of:

- a. Equipment layouts which identify assembly and installation details.
- b. Plans and elevations which identify clearances required for maintenance and operation.
- c. Wiring diagrams which identify each component individually and interconnected or interlocked relationships between components.
- d. Foundation drawings, bolt-setting information, and foundation bolts prior to concrete foundation construction for equipment indicated or required to have concrete foundations.
- e. Details, if piping and equipment are to be supported other than as indicated, which include loadings and type of frames, brackets, stanchions, or other supports.
- f. Automatic temperature control diagrams and control sequences.
- g. Installation details which includes the amount of factory set superheat and corresponding refrigerant pressure/temperature.
- h. Equipment schedules

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### 1.5 DELIVERY, STORAGE, AND HANDLING

Protect stored items from the weather, humidity and temperature variations, dirt and dust, or other contaminants. Properly protect and care for all material both before and during installation. Submit an inventory of all the stored items. Replace any materials found to be damaged, at no additional cost to the Government. During installation, cap piping and similar openings capped to keep out dirt and other foreign matter.

#### 1.6 ENVIRONMENTAL REQUIREMENTS

For proper Indoor Environmental Quality, maintain pressure within the building as indicated. Ventilation must meet or exceed ASHRAE 62.1 and all published addenda. Meet or exceed filter media efficiency as tested in accordance with ASHRAE 52.2. Thermal comfort must meet or exceed ASHRAE 55.

#### 1.7 WARRANTY

Provide equipment with the Manufacturer's Standard Warranty.

### PART 2 PRODUCTS

#### 2.1 MATERIALS

Provide Manufacturer's standard catalog data, at least 5 weeks prior to the purchase or installation of a particular component, highlighted to show material, size, options, performance charts and curves, etc. in adequate detail to demonstrate compliance with contract requirements. Data includes manufacturer's recommended installation instructions and procedures. If vibration isolation is specified for a unit, include vibration isolator literature containing catalog cuts and certification that the isolation characteristics of the isolators provided meet the manufacturer's recommendations. Submit data for each specified component. Minimum efficiency requirements must be in accordance with ASHRAE 90.1 - IP.

##### 2.1.1 Standard Products

Provide materials and equipment that are standard products of a manufacturer regularly engaged in the manufacturing of such products, which are of a similar material, design and workmanship. The standard products must have been in satisfactory commercial or industrial use for 2 years prior to bid opening. The 2 year use includes applications of equipment and materials under similar circumstances and of similar size. The 2 years' experience must be satisfactorily completed by a product which has been sold or is offered for sale on the commercial market through advertisements, manufacturer's catalogs, or brochures. Products having less than a 2 year field service record will be acceptable if a certified record of satisfactory field operation, for not less than 6000 hours exclusive of the manufacturer's factory tests, can be shown. Products must be supported by a service organization. Ensure system components are environmentally suitable for the indicated geographic locations.

##### 2.1.2 Product Sustainability Criteria

###### 2.1.2.1 Energy Efficient Equipment

Provide equipment meeting the efficiency requirements as stated within this section and provide documentation in conformance with Section 01 33 29 SUSTAINABILITY REQUIREMENTS AND REPORTING paragraph ENERGY EFFICIENT

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### EQUIPMENT.

##### 2.1.2.2 Electrical Equipment / Motors

Provide electrical equipment, motors, motor efficiencies, and wiring which are in accordance with Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Electrical motor driven equipment specified must be provided complete with motors, motor starters, and controls. Electrical characteristics must be as shown, and unless otherwise indicated, all motors of 1 horsepower and above with open, dripproof, totally enclosed, or explosion proof fan cooled enclosures, must be the premium efficiency type in accordance with NEMA MG 1. Field wiring must be in accordance with manufacturer's instructions. Each motor must conform to NEMA MG 1 and NEMA MG 2 and be of sufficient size to drive the equipment at the specified capacity without exceeding the nameplate rating of the motor. Motors must be continuous duty with the enclosure specified. Motor starters must be provided complete with thermal overload protection and other appurtenances necessary for the motor control indicated. Motors must be furnished with a magnetic across-the-line or reduced voltage type starter as required by the manufacturer. Motor duty requirements must allow for maximum frequency start-stop operation and minimum encountered interval between start and stop. Motors must be sized for the applicable loads. Motor torque must be capable of accelerating the connected load within 20 seconds with 80 percent of the rated voltage maintained at motor terminals during one starting period. Motor bearings must be fitted with grease supply fittings and grease relief to outside of enclosure. Manual or automatic control and protective or signal devices required for the operation specified and any control wiring required for controls and devices specified, but not shown, must be provided.

##### 2.1.2.3 Ozone Depleting Substances

Unitary air conditioning equipment must not use CFC-based refrigerants. Refrigerant may be an approved alternative refrigerant in accordance with EPA's Significant New Alternative Policy (SNAP) listing. Provide documentation in conformance with Section 01 33 29 SUSTAINABILITY REQUIREMENTS AND REPORTING paragraph OZONE DEPLETING SUBSTANCES.

##### 2.1.2.4 Local/Regional Materials

Use materials or products extracted, harvested, or recovered, as well as manufactured, within a 500 mile radius from the project site, if available from a minimum of three sources.

##### 2.1.3 Nameplates

Major equipment including compressors, condensers, receivers, heat exchanges, fans, and motors must have the manufacturer's name, address, type or style, model or serial number, and catalog number on a plate secured to the item of equipment. Plates must be durable and legible throughout equipment life and made of stainless steel. Fix plates in prominent locations with nonferrous screws or bolts.

##### 2.1.4 Safety Devices

Exposed moving parts, parts that produce high operating temperature, parts which may be electrically energized, and parts that may be a hazard to operating personnel must be insulated, fully enclosed, guarded, or fitted with other types of safety devices. Safety devices must be installed so that proper operation of equipment is not impaired. Welding and cutting

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

safety requirements must be in accordance with AWS Z49.1.

## 2.2 EQUIPMENT

### 2.2.1 Mini-Split-System Heat Pumps

#### 2.2.1.1 Small-Capacity Split-System Air-Conditioners (Not Exceeding 65,000 Btu/hr)

Provide an air-cooled, split system which employs a remote condensing unit, a separate wall mounted ceiling mounted indoor unit, and interconnecting refrigerant piping. Provide the heat pump type unit conforming to applicable Underwriters Laboratories (UL) standards including UL 1995. Unit must be rated in accordance with AHRI 340/360 I-P. Provide indoor unit with necessary fans, air filters, and galvanized steel cabinet construction. The remote unit must be as specified in paragraph CONDENSING UNIT. Provide double-width, double inlet, forward curved backward inclined, or airfoil blade, centrifugal scroll type evaporator or supply fans. Provide the manufacturer's standard condenser or outdoor fans for the unit specified and may be propeller type. Fan and condenser motors must have totally enclosed enclosures. Design unit to operate at outdoor ambient temperatures up to 115 degrees F.

##### 2.2.1.1.1 Energy Efficiency

Provide unit with an Energy Star label. Air Conditioners must have a minimum seasonal energy efficiency ratio (SEER) of 17.0, and a heating seasonal performance factor (HSPF) of 8.0

##### 2.2.1.1.2 Air-to-Refrigerant Coil

Provide condensing coils with copper tubes of 3/8 inch minimum diameter with aluminum fins that are mechanically bonded or soldered to the tubes. Casing must be galvanized steel. Avoid contact of dissimilar metals. Test coils in accordance with ASHRAE 15 & 34 at the factory and ensure suitability for the working pressure of the installed system. Dehydrate and seal each coil testing and prior to evaluation and charging.

Coat condenser coil with a uniformly applied anti-corrosion acrylic resin and hydrophilic film type E1 type coating to all coil surface areas without material bridging between fins. Apply coating at either the coil or coating manufacturer's factory. Coating process must ensure complete coil encapsulation and be capable of withstanding a minimum 1,000 hours exposure to the salt spray test specified in ASTM B117 using a 5 percent sodium chloride solution.

##### 2.2.1.1.3 Compressor

Provide direct drive variable speed type compressor. Provide compressor with internal over temperature and pressure protector; sump heater; oil pump; high pressure and low pressure controls; and liquid line dryer.

##### 2.2.1.1.4 Refrigeration Circuit

Refrigerant-containing components must comply with ASHRAE 15 & 34 and be factory tested, cleaned, dehydrated, charged, and sealed. Provide each unit with a factory operating charge of refrigerant and oil or a holding charge. Field charge unit shipped with a holding charge. Provide refrigerant charging valves. Provide filter-drier in liquid line.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

prevent freeze-up in event of loss of water flow during heating cycle.

#### 2.2.1.1.5 Unit Controls

Provide unit internally prewired with a 16 volt control circuit powered by an internal transformer. Provide terminal blocks for power wiring and external control wiring. Internally protect unit by fuses or a circuit breaker in accordance with UL 1995. Equip units with three-phase power with phase monitoring protection to protect against problems caused by phase loss, phase imbalance and phase reversal. Provide unit with microprocessor controls to provide all 24V control functions. Control unit by a programmable electronic thermostat with heating setback and cooling setup with 7-day programming capability.

Controls must include a control system interface to a BACnet Control system. The control system interface, as well as any network between physically separate units, must meet the requirements of Section 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS.

#### 2.2.1.1.6 Evaporator Coil

Provide coils with copper tubes of 3/8 inch minimum diameter with aluminum fins that are mechanically bonded or soldered to the tubes. Provide galvanized steel or aluminum casing. Avoid contact of dissimilar metals. Test coils in accordance with ANSI/ASHRAE 15 & 34 at the factory and ensure suitability for the working pressure of the installed system. Dehydrate and seal each coil after testing and prior to evaluation and charging. Provide separate expansion devices for each compressor circuit.

#### 2.2.1.1.7 Air Filters

Provide filters of the cleanable type that are capable of filtering the entire air supply. Mount filter(s) integral within the unit and make accessible by hinged access panel(s).

#### 2.2.1.1.8 Indoor Unit Fans

Provide direct driven, statically and dynamically balanced, centrifugal type fans. Design the outdoor fan so that condensate will evaporate without drip, splash, or spray on building exterior. Provide indoor fan with a minimum two-speed motor with built-in overload protection. Fan motors must be the inherently protected, permanent split-capacitor type.

### 2.2.2 Packaged Rooftop Dedicated Outdoor Air System (DOAS) Units

#### 2.2.2.1 Cabinet Casing and Frame

Panel construction shall be double-wall construction for all panels. All floor panels shall have a solid galvanized steel inner liner on the air stream side of the unit to protect insulation during service and maintenance. Insulation shall be a minimum of 2" thick with an R-value of 13.0, and shall be 2 part injected foam. Panel design shall include no exposed insulation edges. Unit cabinet shall be designed to operate at total static pressures up to 5.0 inches w.g.

Exterior surfaces shall be constructed of painted galvanized steel, for

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

aesthetics and long-term durability. Paint finish will include a base primer with a high-quality polyester resin topcoat. Finished, unabraded panel surfaces shall be exposed to an ASTM B117 salt spray environment and exhibit no visible red rust at a minimum of 3,000 hours exposure. Finished, abraded surfaces shall be tested per ASTM D1654, having a mean scribe creepage not exceeding 1/16" at 1,000 hours minimum exposure to an ASTM B117 salt spray environment. Measurements of results shall be quantified using ASTM D1654 in conjunction with ASTM D610 and ASTM D714 to evaluate blister and rust ratings.

Service doors shall be provided on the fan section, filter section, control panel section, and heating vestibule in order to provide user access to unit components. All service access doors shall be mounted on multiple, stainless steel hinges and shall be secured by a latch system. Removable service panels secured by multiple mechanical fasteners are not acceptable.

The unit base shall overhang the roof curb for positive water runoff and shall seat on the roof curb gasket to provide a positive, weathertight seal. Lifting brackets shall be provided on the unit base to accept cable or chain hooks for rigging the equipment.

#### 2.2.2.2 Outdoor Air Section

Unit shall be provided with a 100% outdoor air hood. The 100% outdoor air hood shall allow outdoor air to enter from the back of the unit, at the draw-through filter section. The outdoor air hood shall be factory installed and constructed from galvanized steel finished with the same durable paint finish as the main unit. The hood shall include a bird screen to prevent infiltration of foreign materials and a rain lip to drain water away from the entering air stream.

#### 2.2.2.3 Plate and Frame Energy Recovery

A. The fixed plate energy recovery (ER) core is equipped with a bypass damper on the outside air path. If the RTU has an economizer internal to it then the bypass damper will open when the unit enters the economizer operating state and close when the unit leaves the economizer operating state.

When the outside air is below 32°F (adjustable) the bypass damper will open for 5 minutes (adjustable) every 60 minute period (adjustable). Exhaust air continues to run through the core during this time to remove frost buildup.

The ER core shall transfer both sensible and latent energy between the incoming fresh air stream and the exhaust stale air stream

The ER core shall be in either a cross-flow or counter cross-flow orientation and have no moving parts.

The ER core shall be certified by AHRI under its Standard 1060 for Energy Recovery Ventilators. Products not currently AHRI certified will not be accepted.

The ER core shall achieve the minimum effectiveness value as indicated in the schedule.



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

The fresh air stream must have complete separation from the stale air stream to prevent cross contamination.

The ER core shall have Exhaust Air Transport Ratio of 0.5% as tested to AHRI 1060 (EATR) to prevent cross-over of gases, contaminants or odors.

The ER core's Outdoor Air Correction Factor (OACF) shall not exceed 1.0 as tested to AHRI 1060 (OACF) Standard.

The ER core shall not be degraded or promote the growth of mold and bacteria with a rating of zero in testing according to ISO846 A and C.

The ER core must be able to tolerate freezing temperatures of -30°C (-22°F) and not have an increase in EATR or decrease in performance after being frozen.

The ER core must be able to tolerate high temperatures of +60°C and not have an increase in EATR or decrease in performance at these elevated temperatures.

The ERV core must be freeze tolerant tested to 40 freeze thaw cycles from -20°C to +20°C while maintaining the energy recovery effectiveness and EATR rating of 0.5%.

The ER core must be water washable to remove dust and contaminants.

The ER core must be flame proof and comply with UL 723 with a flame spread index that shall not be over 25 and a smoke index that shall not be over 50.

The ER cores should have particulate filters positioned before the incoming air streams.

#### 2.2.2.4 Exhaust Fan

Exhaust fan shall be a single width, single inlet (SWSI) airfoil centrifugal fan. The fan wheel shall be Class II construction with aluminum fan blades that are continuously welded to the hub plate and end rim. The exhaust fan shall be a direct drive fan mounted to the motor shaft. Belts and sheaves are not acceptable due to the additional maintenance.

The fan motor shall be a totally enclosed EC motor that is speed controlled by the rooftop unit controller. The motor shall include thermal overload protection and protect the motor in the case of excessive motor temperatures. The motor shall have phase failure protection and prevent the motor from operation in the event of a loss of phase. Motors shall be premium efficiency. The unit DDC controller shall provide building static pressure control. The unit controller shall provide proportional control of the exhaust fans from 25% to 100% of the supply air fan designed airflow to maintain the adjustable building pressure setpoint. The field shall mount the required sensing tubing from the building to the factory mounted building static pressure sensor.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### 2.2.2.5 Cooling Coil

The indoor coil section shall be installed in a draw through configuration, upstream of the supply air fan. The coil section shall be complete with a factory piped cooling coil and an ASHRAE 62.1 compliant double sloped drain pan.

The direct expansion (DX) cooling coils shall be fabricated of seamless high efficiency copper tubing that is mechanically expanded into high efficiency aluminum plate fins. Coils shall be a multi-row, staggered tube design with a minimum of 3 rows. All cooling coils shall have an interlaced coil circuiting that keeps the full coil face active at all load conditions. All coils shall be factory leak tested with high pressure air under water.

The cooling coil shall have an electronic controlled expansion valve. The unit controller shall control the expansion valve to maintain liquid subcooling and the superheat of the refrigerant system.

The refrigerant suction lines shall be fully insulated from the expansion valve to the compressors.

The drain pan shall be stainless steel and positively sloped. The slope of the drain pan shall be in two directions and comply with ASHRAE Standard 62.1. The drain pan shall have a minimum slope of 1/8" per foot to provide positive draining. The drain pan shall extend beyond the leaving side of the coil. The drain pan shall have a threaded drain connection extending through the base of the unit base.

#### 2.2.2.6 Hot Gas Reheat

Unit shall be equipped with a fully modulating hot gas reheat coil with hot gas coming from the unit condenser

Hot gas reheat coil shall be a microchannel design. The aluminum tube shall be a microchannel design with high efficiency aluminum fins. Fins shall be brazed to the tubing for a direct bond. The capacity of the reheat coil shall allow for a 20°F temperature rise at all operating conditions.

The modulating hot gas reheat systems shall allow for independent control of the cooling coil leaving air temperature and the reheat coil leaving air temperature. The cooling coil and reheat coil leaving air temperature setpoints shall be adjustable through the unit controller. During the dehumidification cycle the unit shall be capable of 100% of the cooling capacity. The hot gas reheat coil shall provide discharge temperature control within +/- 2°F.

Each coil shall be factory leak tested with high-pressure air under water.

#### 2.2.2.7 Supply Fan

Supply fan shall be a single width, single inlet (SWSI) airfoil centrifugal fan. The fan wheel shall be Class II construction with fan blades that are continuously welded to the hub plate and end rim. The supply fan shall be a direct drive fan mounted to the motor shaft. Belts and sheaves are not acceptable due to the additional maintenance.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

All fan assemblies shall be statically and dynamically balanced at the factory, including a final trim balance, prior to shipment. The fan motor shall be a totally enclosed EC motor that is speed controlled by the rooftop unit controller. The motor shall include thermal overload protection and protect the motor in the case of excessive motor temperatures. The motor shall have phase failure protection and prevent the motor from operation in the event of a loss of phase. Motors shall be premium efficiency.

The supply fan shall be capable of airflow modulation from 30% to 100% of the scheduled designed airflow. The fan shall not operate in a state of surge at any point within the modulation range.

#### 2.2.2.8 Gas Heater

The rooftop unit shall include a natural gas heating section. The gas furnace design shall be one natural gas fired heating module factory installed downstream of the supply air fan in the heat section. The heating module shall be a tubular design with in-shot gas burners.

Each module shall have modulating heating control.

The heat exchanger tubes shall be constructed of stainless steel. The module shall have an induced draft fan that will maintain a negative pressure in the heat exchanger tubes for the removal of the flue gases.

Each burner module shall have two flame roll-out safety protection switches and a high temperature limit switch that will shut the gas valve off upon detection of improper burner manifold operation. The induced draft fan shall have an airflow safety switch that will prevent the heating module from turning on in the event of no airflow in the flue chamber. The factory-installed DDC unit control system shall control the gas heat module. The manufacturer's rooftop unit ETL certification shall cover the complete unit including the gas heating modules.

#### 2.2.2.9 Condensing Section

Outdoor coils shall be cast aluminum, micro-channel coils. Plate fins shall be protected and brazed between adjoining flat tubes such that they shall not extend outside the tubes. A sub-cooling coil shall be an integral part of the main outdoor air coil. Each outdoor air coil shall be factory leak tested with high-pressure air under water.

Fan motors shall be an ECM type motor for proportional control. The unit controller shall proportionally control the speed of the condenser fan motors to maintain the head pressure of the refrigerant circuit from ambient condition of 25 - 120°F. Mechanical cooling shall be provided to 25 ° F. The motor shall include thermal overload protection and protect the motor in the case of excessive motor temperatures. The motor shall have phase failure protection and prevent the motor from operation in the event of a loss of phase.

The condenser fan shall be low noise blade design. Fan blade design shall be a dynamic profile for low tip speed. Fan blade shall be of a composite

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

material.

The unit shall have scroll compressors. One of the compressors shall be an inverter compressor providing proportional control. The unit controller shall control the speed of the compressor to maintain the discharge air temperature. The inverter compressor shall have a separate oil pump and low oil safety protection.

Pressure transducers shall be provided for the suction pressure and head pressure. Temperature sensor shall be provided for the suction temperature and the refrigerant discharge temperature of the compressors. All of the above devices shall be an input to the unit controller and the values be displayed at the unit controller. Each circuit shall be dehydrated and factory charged with R-32 Refrigerant and oil.

#### 2.2.2.10 Electrical

Unit wiring shall comply with NEC requirements and with all applicable UL standards. All electrical components shall be UL recognized where applicable. All wiring and electrical components provided with the unit shall be number and color-coded and labeled according to the electrical diagram provided for easy identification. The unit shall be provided with a factory wired weatherproof control panel. Unit shall have a single point power terminal block for main power connection. A terminal board shall be provided for low voltage control wiring. Branch short circuit protection, 120-volt control circuit transformer and fuse, system switches, and a high temperature sensor shall also be provided with the unit. Each compressor and condenser fan motor shall be furnished with contactors and inherent thermal overload protection. Supply fan motors shall have contactors and external overload protection. Knockouts shall be provided in the bottom of the main control panels for field wiring entrance.

A single non-fused disconnect switch shall be provided for disconnecting electrical power at the unit. Disconnect switches shall be mounted internally to the control panel and operated by an externally mounted handle.

#### 2.2.2.11 Controls

Provide a complete integrated microprocessor based Direct Digital Control (DDC) system to control all unit functions including temperature control, scheduling, monitoring, unit safety protection, including compressor minimum run and minimum off times, and diagnostics. This system shall consist of all required temperature sensors, pressure sensors, controller and keypad/display operator interface. All MCBs and sensors shall be factory mounted, wired and tested.

The stand-alone DDC controllers shall not be dependent on communications with any on-site or remote PC or master control panel for proper unit operation. The microprocessor shall maintain existing set points and operate stand alone if the unit loses either direct connect or network communications. The microprocessor memory shall be protected from voltage fluctuations as well as any extended power failures. All factory and user set schedules and control points shall be maintained in nonvolatile memory. No settings shall be lost, even during extended power shutdowns.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

The DDC control system shall permit starting and stopping of the unit locally or remotely. The control system shall be capable of providing a remote alarm indication. The unit control system shall provide for outside air damper actuation, emergency shutdown, remote heat enable/disable, remote cool enable/disable, heat indication, cool indication, and fan operation.

All digital inputs and outputs shall be protected against damage from transients or incorrect voltages. All field wiring shall be terminated at a separate, clearly marked terminal strip.

The DDC controller shall have a built-in time schedule. The schedule shall be programmable from the unit keypad interface. The schedule shall be maintained in nonvolatile memory to insure that it is not lost during a power failure. There shall be one start/stop per day and a separate holiday schedule. The controller shall accept up to sixteen holidays each with up to a 5-day duration. Each unit shall also have the ability to accept a time schedule via BAS network communications.

The keypad interface shall allow convenient navigation and access to all control functions. The unit keypad/display character format shall be 4 lines x 20 characters. All control settings shall be password protected against unauthorized changes. For ease of service, the display format shall be English language readout. Coded formats with look-up tables will not be accepted. The user interaction with the display shall provide the following information as a minimum:

1. Return air temperature.
2. Discharge air temperature.
3. Outdoor air temperature.
4. Space air temperature.
5. Outdoor enthalpy, high/low.
6. Compressor suction temperature and pressure
7. Compressor head pressure and temperature
8. Expansion valve position
9. Condenser fan speed
10. Inverter compressor speed
11. Dirty filter indication.
12. Airflow verification.
13. Cooling status.
14. Control temperature (Changeover).
15. VAV box output status.
16. Cooling status/capacity.
17. Unit status.
18. All time schedules.
19. Active alarms with time and date.
20. Previous alarms with time and date.
21. Optimal start
22. Supply fan and exhaust fan speed.
23. System operating hours.
  - a. Fan
  - b. Exhaust fan
  - c. Cooling
  - d. Individual compressor
  - e. Heating
  - f. Economizer
  - g. Tenant override

G. The user interaction with the keypad shall provide the following:

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

1. Controls mode
  - a. Off manual
  - b. Auto
  - c. Heat/Cool
  - d. Cool only
  - e. Heat only
  - f. Fan only
2. Occupancy mode
  - a. Auto
  - b. Occupied
  - c. Unoccupied
  - d. Tenant override
3. Unit operation changeover control
  - a. Return air temperature
  - b. Space temperature
  - c. Network signal
4. Cooling and heating change-over temperature with deadband
5. Cooling discharge air temperature (DAT)
6. Supply reset options
  - a. Return air temperature
  - b. Outdoor air temperature
  - c. Space temperature
  - d. Airflow (VAV)
  - e. Network signal
  - f. External (0-10 vdc)
  - g. External (0-20 mA)
7. Temperature alarm limits
  - a. High supply air temperature
  - b. Low supply air temperature
  - c. High return air temperature
8. Lockout control for compressors.
9. Compressor interstage timers
10. Night setback and setup space temperature.
11. Building static pressure.
12. Economizer changeover
  - a. Enthalpy
  - b. Drybulb temperature
13. Currently time and date
14. Tenant override time
15. Occupied/unoccupied time schedule
16. One event schedule
17. Holiday dates and duration
18. Adjustable set points
19. Service mode
  - a. Timers normal (all time delays normal)
  - b. Timers fast (all time delays 20 sec)
- I. To increase the efficiency of the cooling system the DDC controller shall include a discharge air temperature reset program for part load operating conditions. The discharge air temperature shall be controlled between a minimum and a maximum

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

discharge air temperature (DAT) based on one of the following inputs:

1. Airflow
2. Outside air temperature
3. Space temperature
4. Return air temperature
5. External signal of 1-5 vdc
6. External signal of 0-20 mA
7. Network signal

#### 2.2.2.12 Roof Curb

Prefabricated Vibration isolation curb to be manufactured of prime galvanized steel construction, 18 or 14 gauge as required, meeting ASTM A653/653M, with welded corners and with seams joined by continuous water and air tight welds. Vibration isolation curb shall be internally reinforced with bulkheads 48" on center and factory installed wood nailer. Top of all vibration isolation curb shall be level, with pitch built into curb when deck slopes. Vibration isolation curb shall be designed to provide a minimum of 90% isolation efficiency with 1" deflection. 9" continuous rubber cover around perimeter of Vibration isolation curb over spring isolators.

#### 2.2.3 Packaged Rooftop Variable Air Volume (VAV) Units

##### 2.2.3.1 Cabinet Casing and Frame

Panel construction shall be double-wall construction for all panels. All floor panels shall have a solid galvanized steel inner liner on the air stream side of the unit to protect insulation during service and maintenance. Insulation shall be a minimum of 2" thick with an R-value of 13.0, and shall be 2 part injected foam. Panel design shall include no exposed insulation edges. Unit cabinet shall be designed to operate at total static pressures up to 5.0 inches w.g.

Exterior surfaces shall be constructed of painted galvanized steel, for aesthetics and long-term durability. Paint finish will include a base primer with a high-quality polyester resin topcoat. Finished, unabraded panel surfaces shall be exposed to an ASTM B117 salt spray environment and exhibit no visible red rust at a minimum of 3,000 hours exposure. Finished, abraded surfaces shall be tested per ASTM D1654, having a mean scribe creepage not exceeding 1/16" at 1,000 hours minimum exposure to an ASTM B117 salt spray environment. Measurements of results shall be quantified using ASTM D1654 in conjunction with ASTM D610 and ASTM D714 to evaluate blister and rust ratings.

Service doors shall be provided on the fan section, filter section, control panel section, and heating vestibule in order to provide user access to unit components. All service access doors shall be mounted on multiple, stainless steel hinges and shall be secured by a latch system. Removable service panels secured by multiple mechanical fasteners are not acceptable.

The unit base shall overhang the roof curb for positive water runoff and shall seat on the roof curb gasket to provide a positive, weathertight seal. Lifting brackets shall be provided on the unit base to accept cable or chain hooks for rigging the equipment.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### 2.2.3.2 Outdoor/Return Air Section

Unit shall be provided with an outdoor air economizer section. The economizer section shall include outdoor, return, and exhaust air dampers. The economizer operation shall be fully integral to the mechanical cooling and allow up to 100% of mechanical cooling if needed to maintain the cooling discharge air temperature. The outdoor air hood shall be factory installed and constructed from galvanized steel finished with the same durable paint finish as the main unit. The hood shall include moisture eliminator filters to drain water away from the entering air stream. The outside and return air dampers shall be sized to handle 100% of the supply air volume. The dampers shall be parallel blade design. Damper blades shall be gasketed with side seals to provide an air leakage rate of 1.5 cfm /square foot of damper area at 1" differential pressure in according with testing defined in AMCA 500. A barometric exhaust damper shall be provided to exhaust air out of the back of the unit. A bird screen shall be provided to prevent infiltration of rain and foreign materials. Exhaust damper blades shall be lined with vinyl gasketing on contact edges.

Control of the dampers shall be by a factory installed direct coupled actuator. Damper actuator shall be of the modulating, spring return type. A comparative enthalpy control shall be provided to sense and compare enthalpy in both the outdoor and return air streams to determine if outdoor air is suitable for "free" cooling. If outdoor air is suitable for "free" cooling, the outdoor air dampers shall modulate in response to the unit's temperature control system.

#### 2.2.3.3 Exhaust Fan

Exhaust fan shall be a single width, single inlet (SWSI) airfoil centrifugal fan. The fan wheel shall be Class II construction with aluminum fan blades that are continuously welded to the hub plate and end rim. The exhaust fan shall be a direct drive fan mounted to the motor shaft. Belts and sheaves are not acceptable due to the additional maintenance.

The fan motor shall be a totally enclosed EC motor that is speed controlled by the rooftop unit controller. The motor shall include thermal overload protection and protect the motor in the case of excessive motor temperatures. The motor shall have phase failure protection and prevent the motor from operation in the event of a loss of phase. Motors shall be premium efficiency. The unit DDC controller shall provide building static pressure control. The unit controller shall provide proportional control of the exhaust fans from 25% to 100% of the supply air fan designed airflow to maintain the adjustable building pressure setpoint. The field shall mount the required sensing tubing from the building to the factory mounted building static pressure sensor.

#### 2.2.3.4 Cooling Coil

The indoor coil section shall be installed in a draw through configuration, upstream of the supply air fan. The coil section shall be complete with a factory piped cooling coil and an ASHRAE 62.1 compliant double sloped drain pan.

The direct expansion (DX) cooling coils shall be fabricated of seamless high efficiency copper tubing that is mechanically expanded into high



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

efficiency aluminum plate fins. Coils shall be a multi-row, staggered tube design with a minimum of 3 rows. All cooling coils shall have an interlaced coil circuiting that keeps the full coil face active at all load conditions. All coils shall be factory leak tested with high pressure air under water.

The cooling coil shall have an electronic controlled expansion valve. The unit controller shall control the expansion valve to maintain liquid subcooling and the superheat of the refrigerant system.

The refrigerant suction lines shall be fully insulated from the expansion valve to the compressors.

The drain pan shall be stainless steel and positively sloped. The slope of the drain pan shall be in two directions and comply with ASHRAE Standard 62.1. The drain pan shall have a minimum slope of 1/8" per foot to provide positive draining. The drain pan shall extend beyond the leaving side of the coil. The drain pan shall have a threaded drain connection extending through the base of the unit base.

#### 2.2.3.5 Supply Fan

Supply fan shall be a single width, single inlet (SWSI) airfoil centrifugal fan. The fan wheel shall be Class II construction with fan blades that are continuously welded to the hub plate and end rim. The supply fan shall be a direct drive fan mounted to the motor shaft. Belts and sheaves are not acceptable due to the additional maintenance.

All fan assemblies shall be statically and dynamically balanced at the factory, including a final trim balance, prior to shipment. The fan motor shall be a totally enclosed EC motor that is speed controlled by the rooftop unit controller. The motor shall include thermal overload protection and protect the motor in the case of excessive motor temperatures. The motor shall have phase failure protection and prevent the motor from operation in the event of a loss of phase. Motors shall be premium efficiency.

The supply fan shall be capable of airflow modulation from 30% to 100% of the scheduled designed airflow. The fan shall not operate in a state of surge at any point within the modulation range.

#### 2.2.3.6 Condensing Section

Outdoor coils shall be cast aluminum, micro-channel coils. Plate fins shall be protected and brazed between adjoining flat tubes such that they shall not extend outside the tubes. A sub-cooling coil shall be an integral part of the main outdoor air coil. Each outdoor air coil shall be factory leak tested with high-pressure air under water.

Fan motors shall be an ECM type motor for proportional control. The unit controller shall proportionally control the speed of the condenser fan motors to maintain the head pressure of the refrigerant circuit from ambient condition of 25 - 120°F. Mechanical cooling shall be provided to 25 ° F. The motor shall include thermal overload protection and protect the motor in the case of excessive motor temperatures. The motor shall have

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

phase failure protection and prevent the motor from operation in the event of a loss of phase.

The condenser fan shall be low noise blade design. Fan blade design shall be a dynamic profile for low tip speed. Fan blade shall be of a composite material.

The unit shall have scroll compressors. One of the compressors shall be an inverter compressor providing proportional control. The unit controller shall control the speed of the compressor to maintain the discharge air temperature. The inverter compressor shall have a separate oil pump and low oil safety protection.

Pressure transducers shall be provided for the suction pressure and head pressure. Temperature sensor shall be provided for the suction temperature and the refrigerant discharge temperature of the compressors. All of the above devices shall be an input to the unit controller and the values be displayed at the unit controller. Each circuit shall be dehydrated and factory charged with R-32 Refrigerant and oil.

#### 2.2.3.7 Electrical

Unit wiring shall comply with NEC requirements and with all applicable UL standards. All electrical components shall be UL recognized where applicable. All wiring and electrical components provided with the unit shall be number and color-coded and labeled according to the electrical diagram provided for easy identification. The unit shall be provided with a factory wired weatherproof control panel. Unit shall have a single point power terminal block for main power connection. A terminal board shall be provided for low voltage control wiring. Branch short circuit protection, 120-volt control circuit transformer and fuse, system switches, and a high temperature sensor shall also be provided with the unit. Each compressor and condenser fan motor shall be furnished with contactors and inherent thermal overload protection. Supply fan motors shall have contactors and external overload protection. Knockouts shall be provided in the bottom of the main control panels for field wiring entrance.

A single non-fused disconnect switch shall be provided for disconnecting electrical power at the unit. Disconnect switches shall be mounted internally to the control panel and operated by an externally mounted handle.

#### 2.2.3.8 Controls

Provide a complete integrated microprocessor based Direct Digital Control (DDC) system to control all unit functions including temperature control, scheduling, monitoring, unit safety protection, including compressor minimum run and minimum off times, and diagnostics. This system shall consist of all required temperature sensors, pressure sensors, controller and keypad/display operator interface. All MCBs and sensors shall be factory mounted, wired and tested.

The stand-alone DDC controllers shall not be dependent on communications with any on-site or remote PC or master control panel for proper unit operation. The microprocessor shall maintain existing set points and

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

operate stand alone if the unit loses either direct connect or network communications. The microprocessor memory shall be protected from voltage fluctuations as well as any extended power failures. All factory and user set schedules and control points shall be maintained in nonvolatile memory. No settings shall be lost, even during extended power shutdowns.

The DDC control system shall permit starting and stopping of the unit locally or remotely. The control system shall be capable of providing a remote alarm indication. The unit control system shall provide for outside air damper actuation, emergency shutdown, remote heat enable/disable, remote cool enable/disable, heat indication, cool indication, and fan operation.

All digital inputs and outputs shall be protected against damage from transients or incorrect voltages. All field wiring shall be terminated at a separate, clearly marked terminal strip.

The DDC controller shall have a built-in time schedule. The schedule shall be programmable from the unit keypad interface. The schedule shall be maintained in nonvolatile memory to insure that it is not lost during a power failure. There shall be one start/stop per day and a separate holiday schedule. The controller shall accept up to sixteen holidays each with up to a 5-day duration. Each unit shall also have the ability to accept a time schedule via BAS network communications.

The keypad interface shall allow convenient navigation and access to all control functions. The unit keypad/display character format shall be 4 lines x 20 characters. All control settings shall be password protected against unauthorized changes. For ease of service, the display format shall be English language readout. Coded formats with look-up tables will not be accepted. The user interaction with the display shall provide the following information as a minimum:

1. Return air temperature.
2. Discharge air temperature.
3. Outdoor air temperature.
4. Space air temperature.
5. Outdoor enthalpy, high/low.
6. Compressor suction temperature and pressure
7. Compressor head pressure and temperature
8. Expansion valve position
9. Condenser fan speed
10. Inverter compressor speed
11. Dirty filter indication.
12. Airflow verification.
13. Cooling status.
14. Control temperature (Changeover).
15. VAV box output status.
16. Cooling status/capacity.
17. Unit status.
18. All time schedules.
19. Active alarms with time and date.
20. Previous alarms with time and date.
21. Optimal start
22. Supply fan and exhaust fan speed.
23. System operating hours.
  - a. Fan
  - b. Exhaust fan
  - c. Cooling
  - d. Individual compressor

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

- e. Heating
- f. Economizer
- g. Tenant override

G. The user interaction with the keypad shall provide the following:

1. Controls mode
  - a. Off manual
  - b. Auto
  - c. Heat/Cool
  - d. Cool only
  - e. Heat only
  - f. Fan only
2. Occupancy mode
  - a. Auto
  - b. Occupied
  - c. Unoccupied
  - d. Tenant override
3. Unit operation changeover control
  - a. Return air temperature
  - b. Space temperature
  - c. Network signal
4. Cooling and heating change-over temperature with deadband
5. Cooling discharge air temperature (DAT)
6. Supply reset options
  - a. Return air temperature
  - b. Outdoor air temperature
  - c. Space temperature
  - d. Airflow (VAV)
  - e. Network signal
  - f. External (0-10 vdc)
  - g. External (0-20 mA)
7. Temperature alarm limits
  - a. High supply air temperature
  - b. Low supply air temperature
  - c. High return air temperature
8. Lockout control for compressors.
9. Compressor interstage timers
10. Night setback and setup space temperature.
11. Building static pressure.
12. Economizer changeover
  - a. Enthalpy
  - b. Drybulb temperature
13. Currently time and date
14. Tenant override time
15. Occupied/unoccupied time schedule
16. One event schedule
17. Holiday dates and duration
18. Adjustable set points
19. Service mode
  - a. Timers normal (all time delays normal)
  - b. Timers fast (all time delays 20 sec)

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

I. To increase the efficiency of the cooling system the DDC controller shall include a discharge air temperature reset program for part load operating conditions. The discharge air temperature shall be controlled between a minimum and a maximum discharge air temperature (DAT) based on one of the following inputs:

1. Airflow
2. Outside air temperature
3. Space temperature
4. Return air temperature
5. External signal of 1-5 vdc
6. External signal of 0-20 mA
7. Network signal

#### 2.2.3.9 Roof Curb

Prefabricated Vibration isolation curb to be manufactured of prime galvanized steel construction, 18 or 14 gauge as required, meeting ASTM A653/653M, with welded corners and with seams joined by continuous water and air tight welds. Vibration isolation curb shall be internally reinforced with bulkheads 48" on center and factory installed wood nailer. Top of all vibration isolation curb shall be level, with pitch built into curb when deck slopes. Vibration isolation curb shall be designed to provide a minimum of 90% isolation efficiency with 1" deflection. 9" continuous rubber cover around perimeter of Vibration isolation curb over spring isolators.

### 2.3 COMPONENTS

#### 2.3.1 Refrigerant and Oil

Refrigerant must be one of the fluorocarbon gases. Refrigerants must have number designations and safety classifications in accordance with ASHRAE 15 & 34. Refrigerants must meet the requirements of AHRI 700 as a minimum. Provide a complete charge of refrigerant for the installed system as recommended by the manufacturer. Lubricating oil must be of a type and grade recommended by the manufacturer for each compressor. Where color leak indicator dye is incorporated, charge must be in accordance with manufacturer's recommendation.

#### 2.3.2 Air Filters

Provide filters to filter outside air and return air and locate as indicated. Provide replaceable (throw-away) type. Filters must conform to UL 900, Class 1 or Class 2. Polyurethane filters cannot be used on units with multiframe filters.

Air filters must be listed in accordance with requirements of UL 900, except high efficiency particulate air filters of 99.97 percent efficiency by the DOP Test Method must be as listed under the label service and must meet the requirements of UL 586.

##### 2.3.2.1 Extended Surface Pleated Panel Filters

Filters must be 2 inch depth sectional type of the size indicated and must

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

have an average efficiency of 25 to 30 percent when tested in accordance with ASHRAE 52.2. Initial resistance at 500 feet/minute must not exceed 0.36 inches water gauge. Filters must be UL Class 2. Media must be nonwoven cotton and synthetic fiber mat. A wire support grid bonded to the media must be attached to a moisture resistant fiberboard frame. Four edges of the filter media must be bonded to the inside of the frame to prevent air bypass and increase rigidity.

#### 2.3.3 Refrigerant Piping

Provide refrigerant piping in accordance with Section 23 23 00 REFRIGERANT PIPING.

#### 2.3.4 Condensate Drain Piping

provide condensate drain piping in accordance with Section 23 05 15 COMMON PIPING FOR HVAC.

#### 2.3.5 Ductwork

Provide ductwork in accordance with Section 23 30 00 HVAC AIR DISTRIBUTION.

#### 2.3.6 Temperature Controls

Provide temperature controls in accordance with Section 23 09 23.02 BACnet DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS.

### 2.4 TESTS, INSPECTIONS, AND VERIFICATIONS

All manufactured units must be inspected and tested, and documentation provided to demonstrate that each unit is in compliance with ANSI/AHRI and UL requirements and that the minimum efficiency requirements of ASHRAE 90.1 - IP have been met.

## PART 3 EXECUTION

### 3.1 EXAMINATION

After becoming familiar with all details of the work, perform Verification of Dimensions in the field, and advise the Contracting Officer of any discrepancy before performing any work.

### 3.2 INSTALLATION

Perform work in accordance with the manufacturer's published diagrams, recommendations, and equipment warranty requirements. Where equipment is specified to conform to the requirements of ASME BPVC SEC VIII D1 and ASME BPVC SEC IX, the design, fabrication, and installation of the system must conform to ASME BPVC SEC VIII D1 and ASME BPVC SEC IX.

#### 3.2.1 Equipment

Provide refrigeration equipment conforming to ASHRAE 15 & 34. Provide necessary supports for all equipment, appurtenances, and pipe as required, including frames or supports for compressors, pumps, cooling towers, condensers, and similar items. Isolate compressors from the building structure. If mechanical vibration isolators are not provided, provide

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

vibration absorbing foundations. Other floor-mounted equipment must be set on not less than a 6 inch concrete pad doweled in place. Isolators must be selected and sized based on load-bearing requirements and the lowest frequency of vibration to be isolated. Isolators must conform to the requirements of Section 22 05 48.00 20 MECHANICAL SOUND, VIBRATION, AND SEISMIC CONTROL. Equipment must be properly leveled, aligned, and secured in place in accordance with manufacturer's instructions.

### 3.2.2 Field Applied Insulation

Apply field applied insulation as specified in Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS, except as defined differently herein.

### 3.2.3 Field Painting

Painting required for surfaces not otherwise specified, and finish painting of items only primed at the factory are specified in Section 09 90 00 PAINTS AND COATINGS.

### 3.3 CLEANING AND ADJUSTING

Equipment must be wiped clean, with all traces of oil, dust, dirt, or paint spots removed. Temporary filters must be provided for all fans that are operated during construction, and new filters must be installed after all construction dirt has been removed from the building. System must be maintained in this clean condition until final acceptance. Bearings must be properly lubricated with oil or grease as recommended by the manufacturer. Belts must be tightened to proper tension. Control valves and other miscellaneous equipment requiring adjustment must be adjusted to setting indicated or directed. Fans must be adjusted to the speed indicated by the manufacturer to meet specified conditions. Testing, adjusting, and balancing must be as specified in Section 23 05 93 TESTING, ADJUSTING, AND BALANCING OF HVAC SYSTEMS.

### 3.4 TRAINING

Conduct a training course for the operating staff as designated by the Contracting Officer. The training period must consist of a total 8 hours of normal working time and start after the system is functionally completed but prior to final acceptance tests.

- a. Submit a schedule, at least 2 weeks prior to the date of the proposed training course, which identifies the date, time, and location for the training.
- b. Submit the field posted instructions, at least 2 weeks prior to construction completion, including equipment layout, wiring and control diagrams, piping, valves and control sequences, and typed condensed operation instructions. The condensed operation instructions must include preventative maintenance procedures, methods of checking the system for normal and safe operation, and procedures for safely starting and stopping the system. The posted instructions must be framed under glass or laminated plastic and be posted where indicated by the Contracting Officer.
- c. The posted instructions must cover all of the items contained in the approved operation and maintenance manuals as well as demonstrations of routine maintenance operations. Submit 2 complete copies of an operation manual in bound 8-1/2 by 11 inch booklets listing

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

step-by-step procedures required for system startup, operation, abnormal shutdown, emergency shutdown, and normal shutdown at least 4 weeks prior to the first training course. The booklets must include the manufacturer's name, model number, and parts list. The manuals must include the manufacturer's name, model number, service manual, and a brief description of all equipment and their basic operating features.

- d. Submit 2 complete copies of maintenance manual in bound 8-1/2 by 11 inch booklets listing routine maintenance procedures, possible breakdowns and repairs, and a trouble shooting guide. The manuals must include piping and equipment layouts and simplified wiring and control diagrams of the system as installed.

### 3.5 REFRIGERANT TESTS, CHARGING, AND START-UP

Split-system refrigerant piping systems must be tested and charged as specified in Section 23 23 00 REFRIGERANT PIPING. Packaged refrigerant systems which are factory charged must be checked for refrigerant and oil capacity to verify proper refrigerant levels in accordance with manufacturer's recommendations. Following charging, packaged systems must be tested for leaks with a halide torch or an electronic leak detector. Submit 2 copies of each test containing the information described below in bound 8-1/2 by 11 inch booklets. Individual reports must be submitted for the refrigerant system tests.

- a. The date the tests were performed.
- b. A list of equipment used, with calibration certifications.
- c. Initial test summaries.
- d. Repairs/adjustments performed.
- e. Final test results.

#### 3.5.1 Refrigerant Leakage

If a refrigerant leak is discovered after the system has been charged, the leaking portion of the system must immediately be isolated from the remainder of the system and the refrigerant pumped into the system receiver or other suitable container. Under no circumstances must the refrigerant be discharged into the atmosphere.

#### 3.5.2 Contractor's Responsibility

Take steps, at all times during the installation and testing of the refrigeration system, to prevent the release of refrigerants into the atmosphere. The steps must include, but not be limited to, procedures which will minimize the release of refrigerants to the atmosphere and the use of refrigerant recovery devices to remove refrigerant from the system and store the refrigerant for reuse or reclaim. At no time must more than 3 ounces of refrigerant be released to the atmosphere in any one occurrence. Any system leaks within the first year must be repaired in accordance with the requirements herein at no cost to the Government including material, labor, and refrigerant if the leak is the result of defective equipment, material, or installation.

### 3.6 SYSTEM PERFORMANCE TESTS

Before each refrigeration system is accepted, conduct tests to demonstrate the general operating characteristics of all equipment by a registered professional engineer or an approved manufacturer's start-up representative experienced in system start-up and testing, at such times as directed. Two



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

copies of the report provided in bound 8-1/2 by 11 inch booklets. The report must document compliance with the specified performance criteria upon completion and testing of the system. The report must indicate the number of days covered by the tests and any conclusions as to the adequacy of the system.

For equipment providing heating and cooling the system performance tests must be performed during the heating and cooling seasons.

- a. Submit a schedule, at least 2 weeks prior to the start of related testing, for the system performance tests. The schedules must identify the proposed date, time, and location for each test. Tests must cover a period of not less than 48 hours for each system and must demonstrate that the entire system is functioning in accordance with the drawings and specifications.
- b. Make corrections and adjustments, as necessary, tests must be re-conducted to demonstrate that the entire system is functioning as specified. Prior to acceptance, install and tighten service valve seal caps and blanks over gauge points. Replace any refrigerant lost during the system startup.
- c. If tests do not demonstrate satisfactory system performance, correct deficiencies and retest the system. Conduct tests in the presence of the Contracting Officer. Water and electricity required for the tests will be furnished by the Government. Provide all material, equipment, instruments, and personnel required for the test.
- d. Coordinate field tests with Section 23 05 93 TESTING, ADJUSTING, AND BALANCING OF HVAC SYSTEMS. Submit 6 copies of the report provided in bound 8-1/2 by 11 inch booklets. The report must document compliance with the specified performance criteria upon completion and testing of the system. The report must indicate the number of days covered by the tests and any conclusions as to the adequacy of the system. Submit the report including the following information (where values are taken at least three different times at outside dry-bulb temperatures that are at least 5 degrees F apart):
  - (1) Date and outside weather conditions.
  - (2) The load on the system based on the following:
    - (a) The refrigerant used in the system.
    - (b) Condensing temperature and pressure.
    - (c) Suction temperature and pressure.
    - (d) Ambient, condensing and coolant temperatures.
    - (e) Running current, voltage and proper phase sequence for each phase of all motors.
  - (3) The actual on-site setting of operating and safety controls.
  - (4) Thermostatic expansion valve superheat - value as determined by field test.
  - (5) Subcooling.
  - (6) High and low refrigerant temperature switch set-points
  - (7) Low oil pressure switch set-point.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

- (8) Defrost system timer and thermostat set-points.
- (9) Moisture content.
- (10) Capacity control set-points.
- (11) Field data and adjustments which affect unit performance and energy consumption.
- (12) Field adjustments and settings which were not permanently marked as an integral part of a device.

### 3.7 MAINTENANCE

#### 3.7.1 EXTRA MATERIALS

Submit spare parts data for each different item of equipment specified, after approval of detail drawings and not later than 2 months prior to the date of beneficial occupancy. Include in the data a complete list of parts and supplies, with current unit prices and source of supply, a recommended spare parts list for 1 year of operation, and a list of the parts recommended by the manufacturer to be replaced on a routine basis.

#### 3.7.2 Maintenance Service

Submit a certified list of qualified permanent service organizations, which includes their addresses and qualifications, for support of the equipment. The service organizations must be reasonably convenient to the equipment installation and be able to render satisfactory service to the equipment on a regular and emergency basis during the warranty period of the contract.

-- End of Section --

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## SECTION 23 81 23

### COMPUTER ROOM AIR CONDITIONING UNITS

**11/20**

#### PART 1 GENERAL

##### 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

#### AIR-CONDITIONING, HEATING AND REFRIGERATION INSTITUTE (AHRI)

AHRI 1360 (2017) Performance Rating of Computer and Data Processing Room Air Conditioners

ANSI/AHRI 520 (2004) Performance Rating of Positive Displacement Condensing Units

#### AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS (ASHRAE)

ASHRAE 52.2 (2012) Method of Testing General Ventilation Air-Cleaning Devices for Removal Efficiency by Particle Size

ASHRAE 62.1 (2016) Ventilation for Acceptable Indoor Air Quality

ASHRAE 90.1 - IP (2019) Energy Standard for Buildings Except Low-Rise Residential Buildings

#### AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B31.1 (2022) Power Piping

ASME B31.5 (2022) Refrigeration Piping and Heat Transfer Components

#### ASTM INTERNATIONAL (ASTM)

ASTM B117 (2019) Standard Practice for Operating Salt Spray (Fog) Apparatus

ASTM C1338 (2019) Standard Test Method for Determining Fungi Resistance of Insulation Materials and Facings

#### ETL TESTING LABORATORIES (ETL)

ETL DLP (updated continuously) ETL Listed Mark Directory

#### NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 (2023; ERTA 4 2023; ERTA 5 2023; ERTA 6

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

## 2023) National Electrical Code

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

40 CFR 82

Protection of Stratospheric Ozone

UNDERWRITERS LABORATORIES (UL)

UL 181

(2013; Reprint Dec 2021) UL Standard for  
Safety Factory-Made Air Ducts and Air  
Connectors

UL Elec Equip Dir

(2011) Electrical Appliance and  
Utilization Equipment Directory

### 1.2 DEFINITIONS

Computer Room Air Conditioner (CRAC): A single, self-contained unit or split-system unit designed and manufactured specifically for temperature and humidity control of data processing environments.

Cold Aisle: The aisle between or adjacent to rows of racks from which the computing equipment draws cool air.

Hot Aisle: The aisle between or adjacent to rows of racks to which the computing equipment ejects hot air.

Rack: Telecommunications support frame that can consist of post-and-frame or full cabinet construction. Racks are provided under Section 27 10 00 BUILDING TELECOMMUNICATIONS CABLING SYSTEM.

### 1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

#### SD-03 Product Data

Small Computer Room Air Conditioners; G

Space Temperature Control System Drawings; G

Filters

Refrigerants; S

#### SD-06 Test Reports

Field Test Schedule; G

Manufacturer's Field Test Plans; G

Field Test Reports; G

#### SD-07 Certificates

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

Certificate of Specification Compliance; FIO

Credentials of the Manufacturer's Field Test Representative; FIO  
Ozone Depleting Substances Technician Certification  
Certified List Of Qualified Permanent Service Organizations  
SD-08 Manufacturer's Instructions

Installation Manual for Each Type of CRAC

SD-10 Operation and Maintenance Data

Computer Room Air Conditioner Operation and Maintenance Data, Data  
Package 4; G

SD-11 Closeout Submittals

Indoor Air Quality During Construction

#### 1.4 REFRIGERANTS

Refrigerants must have an Ozone Depletion Potential (ODP) no greater than 0.0. CFC-based refrigerants are prohibited. HCFCs and Halons are not permitted. Provide SDS sheets for all refrigerants.

#### 1.5 QUALIFICATIONS

##### 1.5.1 Ozone Depleting Substances Technician Certification

All technicians working on equipment that contain ozone depleting refrigerants must be certified as a Section 608 Technician to meet requirements in 40 CFR 82, Subpart F. Provide copies of technician certifications to the Contracting Officer at least 14 calendar days prior to work on any equipment containing these refrigerants.

#### 1.6 QUALIFICATIONS

##### 1.6.1 Material and Equipment Qualifications

Provide materials and equipment that are standard products of manufacturers regularly engaged in the manufacture of such products, which are of a similar material, design, and workmanship. Standard products must have been in satisfactory commercial or industrial use for two years prior to bid opening. The two-year use must include applications of equipment and materials under similar circumstances and of similar size. The product must have been for sale on the commercial market through advertisements, manufacturers' catalogs, or brochures during the two-year period.

##### 1.6.2 Alternative Equipment Qualifications

Products having less than a two-year field service record will be acceptable if a certified record of satisfactory field operation for not less than 6000 hours, exclusive of the manufacturer's factory or laboratory tests, can be shown.

##### 1.6.3 Service Support

The equipment items must be supported by service organizations. Submit a certified list of qualified permanent service organizations for support of

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

the equipment which includes their addresses and qualifications. These service organizations must be reasonably convenient to the equipment installation and able to render satisfactory service to the equipment on a regular and emergency basis during the warranty period of the contract.

#### 1.6.4 Manufacturer's Nameplate

For each item of equipment, provide a nameplate bearing the manufacturer's name, address, model number, and serial number securely affixed in a conspicuous place; the nameplate of the distributing agent will not be acceptable.

#### 1.6.5 Modification of References

In each of the publications referred to herein, consider the advisory provisions to be mandatory, as though the word, "must" had been substituted for "should" wherever it appears. Interpret references in these publications to the "authority having jurisdiction", or words of similar meaning, to mean the Contracting Officer.

##### 1.6.5.1 Definitions

For the International Code Council (ICC) Codes referenced in the contract documents, advisory provisions must be considered mandatory, the word "should" is interpreted as "must." Reference to the "code official" must be interpreted to mean the "Contracting Officer." For Navy owned property, references to the "owner" must be interpreted to mean the "Contracting Officer." For leased facilities, references to the "owner" must be interpreted to mean the "lessor." References to the "permit holder" must be interpreted to mean the "Contractor."

##### 1.6.5.2 Administrative Interpretations

For ICC Codes referenced in the contract documents, the provisions of Chapter 1, "Administrator," do not apply. These administrative requirements are covered by the applicable Federal Acquisition Regulations (FAR) included in this contract and by the authority granted to the Officer in Charge of Construction to administer the construction of this project. References in the ICC Codes to sections of Chapter 1, must be applied appropriately by the Contracting Officer as authorized by his administrative cognizance and the FAR.

#### 1.7 PROJECT REQUIREMENTS

##### 1.7.1 Verification of Dimensions

Become familiar with the details of the work, verify all dimensions in the field, and provide adequate clearance for all connections and service access. Notify the Contracting Officer of any discrepancy before performing any work.

##### 1.7.2 Energy Efficiency

Provide equipment with minimum efficiencies as required by ASHRAE 90.1 - IP.

#### 1.8 DELIVERY, STORAGE, AND HANDLING

Handle, store, and protect equipment and materials to prevent damage before and during installation in accordance with the manufacturer's

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

recommendations, and as approved by the Contracting Officer. Replace damaged or defective items.

## PART 2 PRODUCTS

### 2.1 SMALL COMPUTER ROOM AIR CONDITIONERS

Provide complete working CRACs, designed and factory assembled. Equipment must be listed in UL Elec Equip Dir or ETL DLP for computer room application. CRACs must include room cabinet and frame, fan, filter, cooling coil, compressor, condenser, controls, and, interconnecting piping internal to the CRAC. Provide units rated in accordance with AHRI 1360. Provide all refrigerant piping in accordance with Section 23 23 00 REFRIGERANT PIPING.

#### 2.1.1 System Configuration

#### 2.1.2 Evaporator Cabinet Construction

Provide cabinet and chassis constructed of heavy gauge galvanized steel with all service access from a single side of the unit. Mounting brackets must be integral to the cabinet. Internal cabinet insulation must meet ASHRAE 62.1 requirements for Mold Growth, Humidity & Erosion, tested per UL 181 and ASTM C1338 standards.

#### 2.1.3 Air Distribution Components

Provide direct-drive fan assembly equipped with double-inlet blower, self-aligning ball bearings and lifetime lubrication. Fan motor must be permanent-split capacitor, high-efficiency type, equipped with two speeds for airflow modulation. The microprocessor controller must use the lower fan speed for precise dehumidification control. Fan speed must also be user selectable from the wall controller. System must be suitable for supply and return air plenum or ducted supply and return air distribution. Provide filter rack designed to accept 4 inch thick filters. Provide pleated filters with a MERV 8 rating in accordance with ASHRAE 52.2.

#### 2.1.4 Direct Expansion System Evaporator Components

The evaporator section must include evaporator coil, thermostatic expansion valve and filter drier. The evaporator coil must be constructed of copper tubes and aluminum fins. Provide an externally equalized thermostatic expansion valve to control refrigerant flow. The refrigerant piping must be spun-closed and filled with a nitrogen holding charge. Evaporator and condensing unit must be field piped using copper lines, brazed, evacuated and field charged with R-407C refrigerant. The evaporator unit can be coupled directly with the condensing unit or mounted remote to the condensing unit. The coil assembly must be mounted in a condensate drain pan with an internally trapped drain line. The evaporator drain pan must include a factory-installed float switch to shut down the evaporator upon high water condition.

#### 2.1.5 Outdoor, Air-Cooled Condensing Unit

Provide condensing unit rated in accordance with ANSI/AHRI 520 and designed for permanent outdoor installation. Provide removable panels for access to all components. Condensing unit components must include condenser coil, direct drive propeller fan, variable capacity scroll

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

compressor, high-pressure switch, refrigerant receiver, head pressure control valve, , and liquid line solenoid valve. Unit casing and chassis must be constructed of heavy gauge galvanized steel.

Provide a high pressure switch to protect the unit from abnormal refrigerant pressure conditions and deactivate the compressor and annunciate an alarm at the wall controller. The blower must continue to circulate air. The wall controller must be used to manually restart the compressor function after the automatic pressure switch resets. Three high head pressure alarms in a rolling 12-hour period must lock out the manual restart feature until power is cycled to the evaporator unit. A pressure balancing valve must be factory installed to reduce the chance of high pressure cut-out due to excessive refrigerant migration to the receiver due to changing outdoor temperatures during off-cycles. The refrigerant piping must be spun-closed and filled with a nitrogen holding charge. Evaporator and condensing unit must be field piped using copper lines, brazed, evacuated and field charged with R-407C refrigerant. Condensing unit must be designed for 95 degrees F ambient and be capable of operation to minus 30 degrees F. The condensing coil must be constructed of copper tubes and aluminum fins.

#### 2.1.6 Controls

Provide remote mounted color touchscreen display for each unit. Provide remote mounted temperature sensor for each unit. Controls must be organized by menus with minimum menu selection of: Alarms, Event Log, Graphics, and Status Overview. The Graphics menu must display a minimum of the following: zone temperature, zone setpoints, fan status. Controls must include a control system interface. The control system interface must meet DDC Hardware requirements of Section 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS.

Integrate CRAC control into the HVAC control system defined in Section 23 09 23.02 BACNET DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS. HVAC control system interface point is located in 121 MECH/ELEC. Refer to controls drawings for minimum points required to interface with the HVAC control system.

#### 2.2 FACTORY PAINTING SYSTEMS

Provide manufacturer's standard factory painting. Certify that the factory painting system applied will withstand 125 hours in a salt-spray fog test, except that equipment located outdoors must withstand 500 hours in a salt-spray fog test. Salt-spray fog test must be in accordance with ASTM B117, and for that test the acceptance criteria must be as follows: immediately after completion of the test, the paint must show no signs of blistering, wrinkling, or cracking, and no loss of adhesion; and the specimen must show no signs of rust creepage beyond 0.125 inch on either side of the scratch mark.

The film thickness of the factory painting system applied on the equipment must not be less than the film thickness used on the test specimen. The factory painting system must be designed for the anticipated temperature service.

#### 2.3 ELECTRICAL

Provide an integral electrical panel of similar construction to the unit cabinet. Within the electrical panel, provide a single point power



P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

connection terminal block and disconnect switch, . The electrical panel must provide at least 65,000 (outdoor condensing unit) and 5000 (indoor evaporator unit) amp Short Circuit Current Rating (SCCR).

#### 2.3.1 Electrical Motors, Controllers, Contactors, and Disconnects

Provide motors, controllers, disconnects and contactors with their respective pieces of equipment. Motors, controllers, disconnects and contactors must conform to and have electrical connections provided under Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Provide internal wiring for components of packaged equipment as an integral part of the equipment. Extended voltage range motors will not be permitted. Controllers and contactors must have a maximum of 120 volt control circuits, and must have auxiliary contacts for use with the controls provided. When motors and equipment provided are larger than sizes indicated, the cost of additional electrical service and related work must be included under the section that specified that motor or equipment. Power wiring and conduit for field installed equipment must be provided under and conform to the requirements of Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM.

#### 2.3.2 Electrical Control Wiring

Provide control wiring under Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC. Provide Space temperature control system drawings which include point-to-point electrical wiring diagrams.

### PART 3 EXECUTION

#### 3.1 INSTALLATION

##### 3.1.1 CRAC System

Installation of each CRAC system including equipment, materials, installation, workmanship, fabrication, assembly, erection, examination, inspection, and testing, must be in accordance with ASME B31.1, ASME B31.5, NFPA 70, as modified and supplemented by the requirements of this section and the CRAC manufacturer's written installation instructions.

Install all work so that parts requiring periodic inspection, operation, maintenance, and repair are readily accessible. Install concealed valves, expansion joints, controls, dampers, and equipment requiring access, in locations freely accessible through access doors.

##### 3.1.2 Installation Instructions

Provide a manufacturer's installation manual for each type of CRAC.

##### 3.1.3 Operation and Maintenance Data

Submit Computer Room Air Conditioner Operation and Maintenance Data in accordance with Section 01 78 23 OPERATION AND MAINTENANCE DATA.

##### 3.1.4 Connections to Existing Systems

Notify the Contracting Officer in writing at least 15 calendar days prior to the date the connections are required. Obtain approval before interrupting service. Provide materials required to make connections into

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

existing systems and perform excavating, backfilling, compacting, and other incidental labor as required. Provide labor and tools for making actual connections to existing systems.

### 3.2 FIELD QUALITY CONTROL

Upon completion and before final acceptance of work, test each CRAC subsystem in service to demonstrate compliance with the contract requirements, including field testing specified below. Adjust controls and balance systems prior to final acceptance of completed systems. Test controls through every cycle of operation. Test safety controls to demonstrate performance of required function. Correct defects in work provided and repeat tests. Provide steam, fuel, water, electricity, instruments, connecting devices, and personnel for tests. Flush and clean piping before placing in operation. Clean equipment, piping, strainers, and ducts. Prior to commencement of field testing, remove all filters and provide new filters. Perform and document that proper Indoor Air Quality During Construction procedures have been followed; this includes providing documentation showing that after construction ends, and prior to occupancy, new filters were provided.

### 3.3 FIELD TESTING

Provide field test plan, field test schedule, field test and field test report on each of the CRAC. Field test each CRAC for Contracting Officer acceptance in accordance with the CRAC manufacturer's approved field test plan.

#### 3.3.1 Manufacturer's Field Test Plans

Submit field test plans developed by the manufacturer for each CRAC; submit the field test plans at least 90 calendar days prior to planned date of the field test. Field test plans developed by the installing Contractor, or the equipment sales agency furnishing the CRAC, will not be acceptable.

The Contracting Officer will review and approve the field test plan for each of the listed CRACs prior to commencement of field testing of the equipment. The approved field test plans must be followed for the field tests of the CRAC and test reporting.

##### 3.3.1.1 Coordinated Testing

Indicate in each field test plan when work required by this section requires coordination with test work required by other specification sections. Provide test procedures for the simultaneous or integrated testing of: CRAC controls which interlock and interface with controls factory prewired; and external controls for the CRAC provided under Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC.

##### 3.3.1.2 Prerequisite Testing

Each CRAC for which performance testing is dependent upon the completion of the work covered by Section 23 05 93 TESTING, ADJUSTING AND BALANCING FOR HVAC must have that work completed as a prerequisite to testing work under this section. Indicate in each field test plan when such prerequisite work is required.

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

#### 3.3.1.3 Test Procedure

Indicate in each field test plan the CRAC manufacturer's published start-up, and field acceptance test procedures. Include in each test plan a detailed step-by-step procedure for testing automatic controls provided by the manufacturer.

Procedures must be structured to test the controls through all modes of control to confirm that the controls are performing with the intended sequence of control.

Controllers must be verified to be properly calibrated and have the proper set point to provide stable control of their respective equipment.

#### 3.3.1.4 Performance Variables

Each test plan must list performance variables that are required to be measured or tested as part of the field test.

Include, in the listed performance variables, requirements indicated on the CRAC schedules on the design drawings. Manufacturer must provide, with each test procedure, a description of acceptable results that have been verified.

Manufacturer must identify the acceptable limits or tolerances within which each tested performance variable must acceptably operate.

#### 3.3.1.5 Test Configuration

Plans must indicate that tests are to be performed for a minimum of four continuous hours in a wet coil condition. If test period is interrupted, the four hour test period must be started over. Each test plan must be job specific and must address the particular CRAC and particular conditions which exist with this contract. Generic or general preprinted test procedures are not acceptable.

#### 3.3.1.6 Tested Variables

Plans must provide for air side testing which includes verification of the airflow, total static pressure; fan drive motor KW, amperage and RPM; and fan RPM. Provide entering air temperatures equal to those indicated on the CRAC schedules.

#### 3.3.1.7 Field Test Reporting Forms

Each test plan must include the required test reporting forms to be completed by the Contractor's testing representatives.

#### 3.3.2 Field Test Schedule

Notify the Contracting Officer in writing at least 30 calendar days prior to the testing. Within 30 calendar days after acceptable completion of testing, submit each test report for the review and approval of the Contracting Officer.

#### 3.3.3 Manufacturer's Test Representative

Provide a factory trained field test representative authorized by the CRAC manufacturer to oversee the complete execution of the field testing. This

P2#: 513392, YARS FIRE STATION  
AIR FORCE RESERVE BASE, YOUNGSTOWN, OH

test representative must also review, approve, and sign the completed field test report. Signatures must be accompanied by the person's name typed.

Submit credentials of the manufacturer's field test representative proposed, including current telephone number, to the Contracting Officer for review and approval. Submit these credentials with the written advance notice of the field tests.

#### 3.3.4 Field Tests

Conduct the field testing in compliance with the Contracting Officer approved manufacturer's field test plan, and in accordance with additional field testing requirements specified herein. Record the required data using the test reporting forms approved of the approved field test plan. Conduct the test for each CRAC for a continuous 24-hour test period. A CRAC shutdown before the continuous 24-hour test period is completed must result in the 24-hour test period being started again and run for the required duration.

#### 3.3.5 Deficiency Resolution

The test requirements must be acceptably met; deficiencies identified during the tests must be corrected in compliance with the manufacturer's recommendations. Corrections must be tested again in compliance with the requirements specified in the paragraph FIELD TEST PLANS.

#### 3.3.6 Field Test Reports

Use the test reporting forms approved in the field test plan. Final test report forms must be typed, including data entries and remarks. Completed test report forms for each CRAC must be reviewed, approved, and signed by the Contractor's test director and the QC manager.

### 3.4 INSTRUCTION TO GOVERNMENT PERSONNEL

Provide the services of competent instructors to give full instruction to the designated Government personnel in the adjustment, operation, and maintenance, including pertinent safety requirements, of the specified equipment or system. Instructors must be thoroughly familiar with all parts of the installation and must be trained in operating theory as well as practical operation and maintenance work.

Instruction must be given during the first regular work week after the equipment or system has been accepted and turned over to the Government for regular operation. Provide 4 hours of training for each type of CRAC specified.

-- End of Section --